



CANDIDATE
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9702/41

May/June 2022

2 hours

You must answer on the question paper.

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [].

This document has **24** pages. Any blank pages are indicated.

Data

acceleration of free fall	$g = 9.81 \text{ m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
unified atomic mass unit	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27} \text{ kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$ ($\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ m F}^{-1}$)
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_o = \frac{f_s v}{v \pm v_s}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \dots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$

gravitational potential	$\phi = -\frac{GM}{r}$
gravitational potential energy	$E_P = -\frac{GMm}{r}$
pressure of an ideal gas	$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$
simple harmonic motion	$a = -\omega^2 x$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t$ $v = \omega \sqrt{(x_0^2 - x^2)}$
electric potential	$V = \frac{Q}{4\pi\epsilon_0 r}$
electrical potential energy	$E_P = \frac{Qq}{4\pi\epsilon_0 r}$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$
capacitors in parallel	$C = C_1 + C_2 + \dots$
discharge of a capacitor	$x = x_0 e^{-\frac{t}{RC}}$
Hall voltage	$V_H = \frac{BI}{ntq}$
alternating current/voltage	$x = x_0 \sin \omega t$
radioactive decay	$x = x_0 e^{-\lambda t}$
decay constant	$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$
intensity reflection coefficient	$\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$
Stefan–Boltzmann law	$L = 4\pi\sigma r^2 T^4$
Doppler redshift	$\frac{\Delta\lambda}{\lambda} \approx \frac{Df}{f} \approx \frac{v}{c}$

- 1 (a) (i) State Newton's law of gravitation.

.....

.....

..... [2]

- (ii) Use Newton's law of gravitation to show that the gravitational field strength g at a distance r away from a point mass M is given by

$$g = \frac{GM}{r^2}.$$

[2]

- (b) The Earth has a mass of 5.98×10^{24} kg and a radius of 6.37×10^6 m.
 The Moon has a mass of 7.35×10^{22} kg and a radius of 1.74×10^6 m.
 The Earth and the Moon can both be considered as point masses at their centres. Their centres are a distance of 3.84×10^8 m apart.

- (i) Show that the gravitational field strength at the surface of the Moon due to the mass of the Moon is 1.62 N kg^{-1} .

[1]

- (ii) Explain why there is a point X on the line between the centres of the Earth and the Moon where the resultant gravitational field strength due to the Earth and the Moon is zero.

.....

.....

..... [2]

- (iii) Calculate the distance x of point X from the centre of the Moon.

$x = \dots\dots\dots$ m [3]

[Total: 10]

- 2 A sphere of mass $1.6 \times 10^{-10} \text{ kg}$ has a charge of $+0.27 \text{ nC}$. The sphere is in a uniform electric field that acts vertically upwards, as shown in the side view in Fig. 2.1.

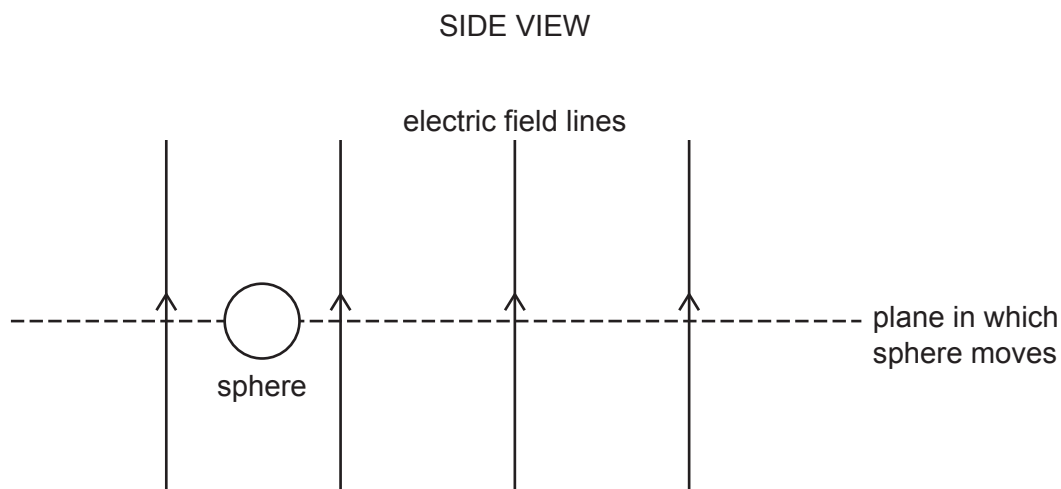


Fig. 2.1

The force exerted on the sphere by the electric field causes the sphere to remain at a fixed vertical height in a horizontal plane.

There is a uniform magnetic field in the region of the electric field. The sphere moves at a speed of 0.78 m s^{-1} in the horizontal plane. The magnetic field causes the sphere to move in a circular path of radius 3.4 m , as shown in the view from above in Fig. 2.2.

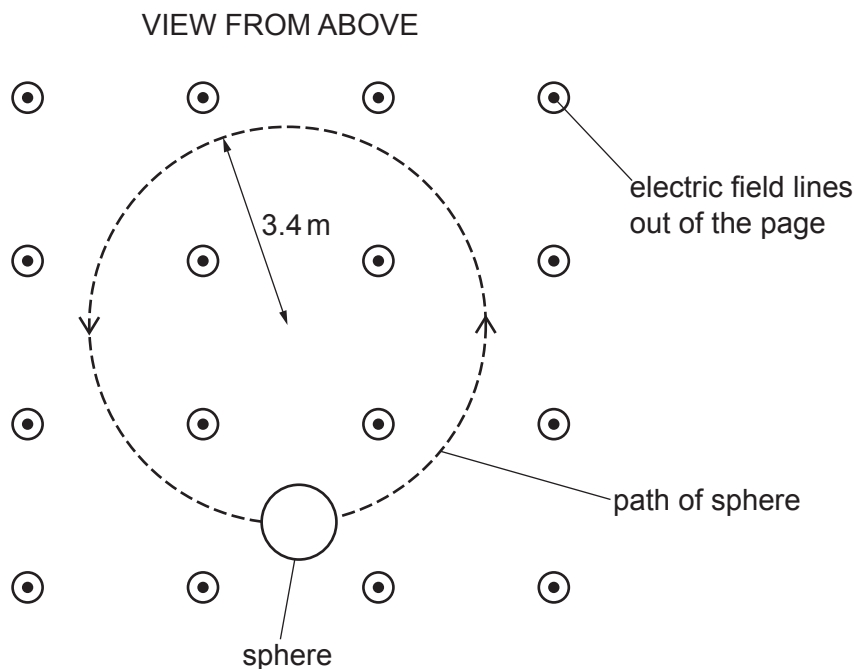


Fig. 2.2

- (a) (i) Determine the direction of the uniform magnetic field.

..... [1]

- (ii) Explain why the motion of the sphere in the horizontal plane is circular.

.....

.....

..... [2]

- (b) Calculate the strength of the uniform electric field.

electric field strength = NC^{-1} [2]

- (c) By considering the magnetic force on the sphere, show that the flux density of the uniform magnetic field is 0.14 T.

[3]

[Total: 8]

- 3 A fixed mass of an ideal gas is initially at a temperature of 17°C . The gas has a volume of 0.24 m^3 and a pressure of $1.2 \times 10^5\text{ Pa}$.

(a) (i) State what is meant by an ideal gas.

.....

 [2]

(ii) Calculate the amount n of gas.

$n = \dots\dots\dots$ mol [2]

(b) The gas undergoes three successive changes, as shown in Fig. 3.1.

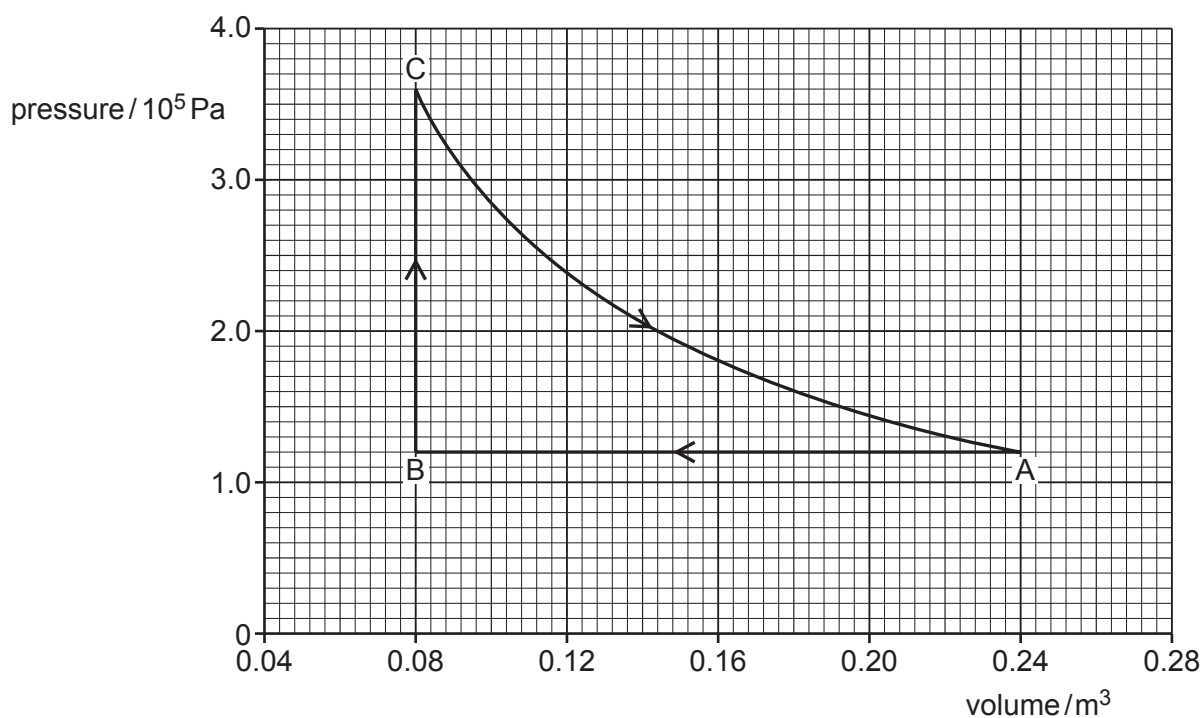


Fig. 3.1

The initial state is represented by point A. The gas is cooled at constant pressure to point B by the removal of 48.0 kJ of thermal energy.

The gas is then heated at constant volume to point C.

Finally, the gas expands at constant temperature back to its original pressure and volume at point A. During this expansion, the gas does 31.6 kJ of work.

- (i) Show that the magnitude of the work done during the change AB is 19.2 kJ.

[2]

- (ii) Complete Table 3.1 to show the work done on the gas, the thermal energy supplied to the gas and the increase in internal energy of the gas, for each of the changes AB, BC and CA.

Table 3.1

change	work done on gas / kJ	thermal energy supplied to gas / kJ	increase in internal energy of gas / kJ
AB		-48.0	
BC			
CA	-31.6		

[5]

[Total: 11]

- 4 A pendulum consists of a bob (small metal sphere) attached to the end of a piece of string. The other end of the string is attached to a fixed point. The bob oscillates with small oscillations about its equilibrium position, as shown in Fig. 4.1.

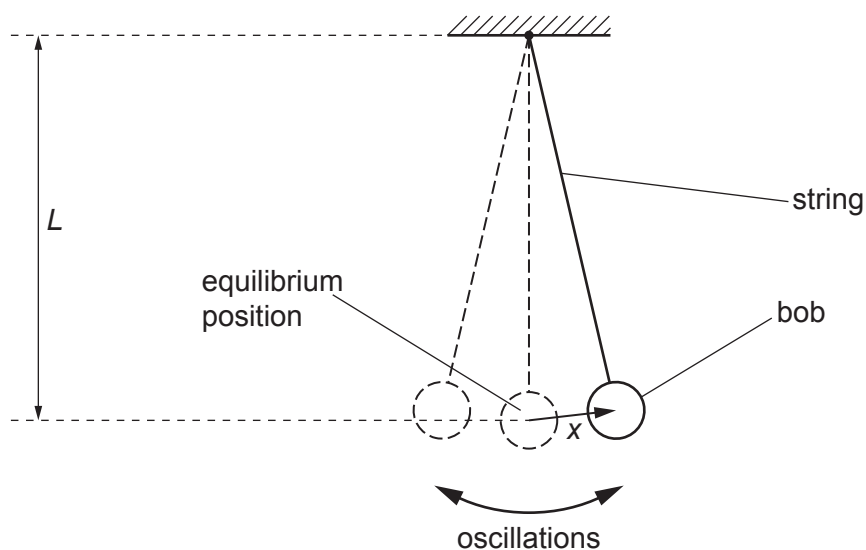


Fig. 4.1 (not to scale)

The length L of the pendulum, measured from the fixed point to the centre of the bob, is 1.24 m.

The acceleration a of the bob varies with its displacement x from the equilibrium position as shown in Fig. 4.2.

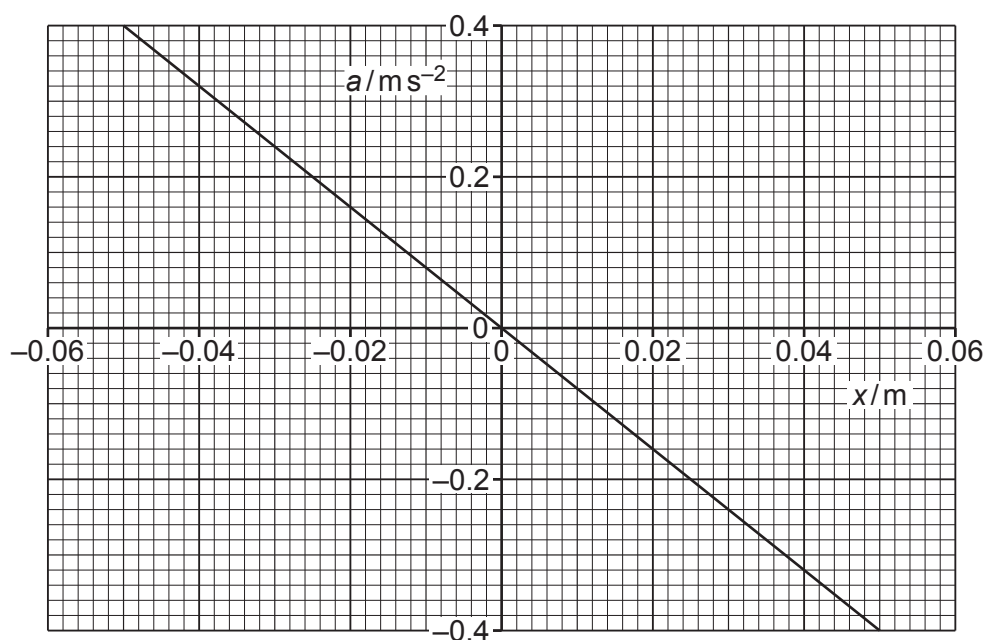


Fig. 4.2

- (a) State how Fig. 4.2 shows that the motion of the pendulum is simple harmonic.

.....

.....

..... [2]

- (b) (i) Use Fig. 4.2 to determine the angular frequency ω of the oscillations.

$$\omega = \dots\dots\dots \text{rad s}^{-1} \quad [2]$$

- (ii) The angular frequency ω is related to the length L of the pendulum by

$$\omega = \sqrt{\frac{k}{L}}$$

where k is a constant.

Use your answer in (b)(i) to determine k . Give a unit with your answer.

$$k = \dots\dots\dots \text{unit} \dots\dots\dots [2]$$

- (c) While the pendulum is oscillating, the length of the string is increased in such a way that the total energy of the oscillations remains constant.

Suggest and explain the qualitative effect of this change on the amplitude of the oscillations.

.....

.....

..... [2]

[Total: 8]

- 5 Fig. 5.1 shows four diodes and a load resistor of resistance $1.2\text{ k}\Omega$, connected in a circuit that is used to produce rectification of an alternating voltage.

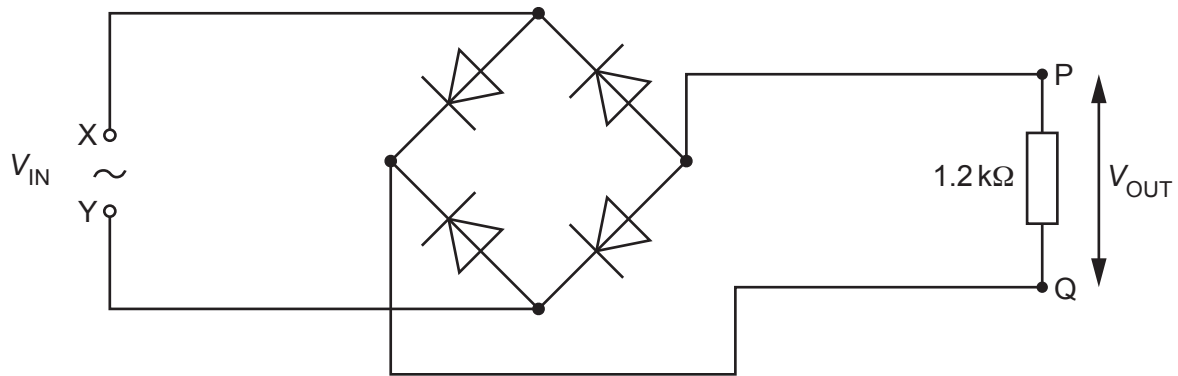


Fig. 5.1

- (a) (i) State what is meant by rectification.

.....
 [1]

- (ii) State the type of rectification produced by the circuit in Fig. 5.1.

..... [1]

- (b) A sinusoidal alternating voltage V_{IN} is applied across the input terminals X and Y. The variation with time t of V_{IN} is given by the equation

$$V_{IN} = 6.0 \sin 25\pi t$$

where V_{IN} is in volts and t is in seconds.

- (i) On Fig. 5.1, label the output terminals P and Q with the appropriate symbols to indicate the polarity of the output voltage V_{OUT} . [1]

- (ii) The magnitude of the output voltage V_{OUT} varies with t as shown in Fig. 5.2.

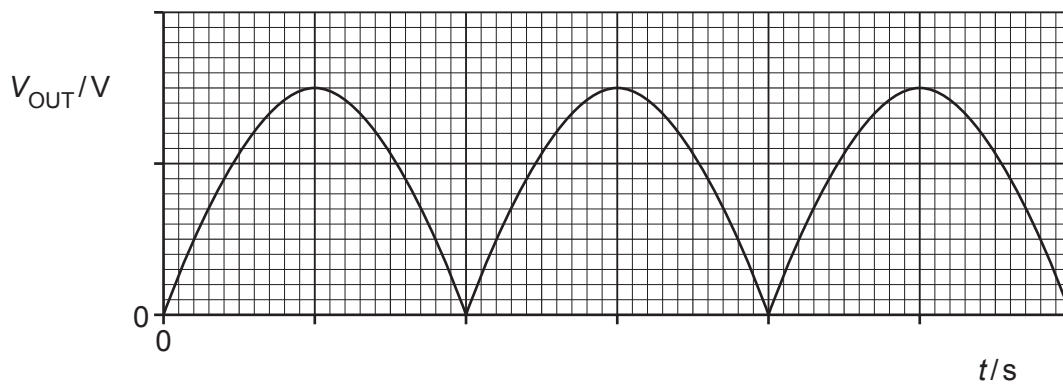


Fig. 5.2

On Fig. 5.2, label both of the axes with the correct scales. Use the space below for any working that you need.

[3]

- (c) The output voltage in (b) is smoothed by adding a capacitor to the circuit in Fig. 5.1. The difference between the maximum and minimum values of the smoothed output voltage is 10% of the peak voltage.

- (i) On Fig. 5.1, draw the circuit symbol for a capacitor showing the capacitor correctly connected into the circuit. [1]

- (ii) On Fig. 5.2, sketch the variation with t of the smoothed output voltage. [2]

- (iii) Calculate the capacitance C of the capacitor.

$C = \dots\dots\dots$ F [3]

[Total: 12]

- 6 (a) Define magnetic flux.

.....

.....

..... [2]

- (b) A square coil of wire of side length 12 cm consists of 8 insulated turns. The coil is stationary in a uniform magnetic field. The plane of the coil is perpendicular to the magnetic field, as shown in Fig. 6.1.

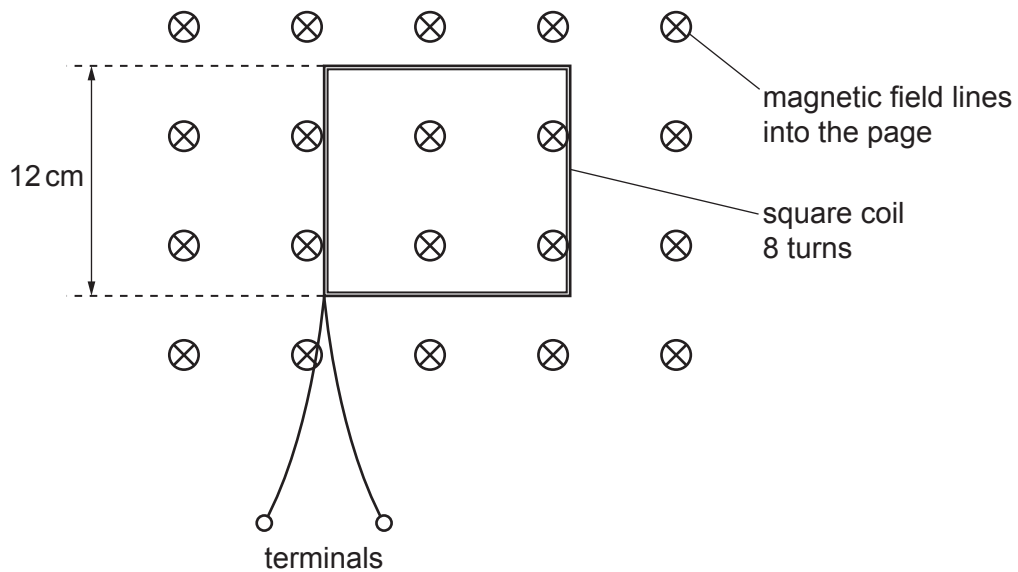


Fig. 6.1

The flux density B of the magnetic field varies with time t as shown in Fig. 6.2.

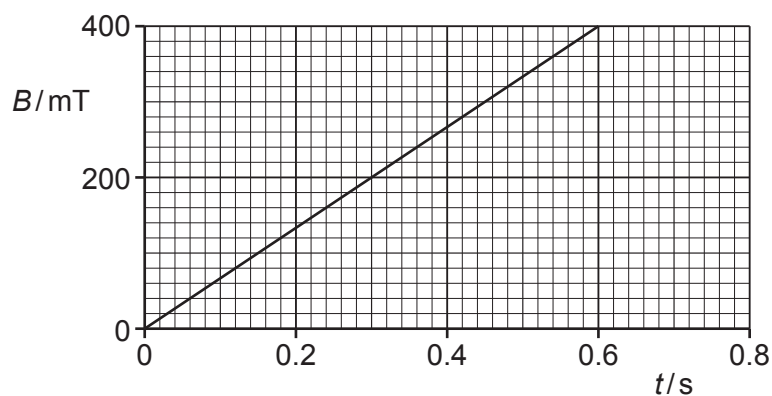


Fig. 6.2

- (i) Determine the magnetic flux linkage inside the coil at time $t = 0.60\text{ s}$. Give a unit with your answer.

magnetic flux linkage = unit [3]

- (ii) State how Fig. 6.2 shows that the electromotive force (e.m.f.) E induced across the terminals between $t = 0$ and $t = 0.60\text{ s}$ is constant.

..... [1]

- (iii) Calculate the magnitude of E .

$E = \dots\dots\dots \text{ V}$ [2]

- (c) The procedure in (b) is repeated, but this time the terminals of the coil are connected together.

State and explain the effect on the coil of connecting the terminals together during the change of magnetic flux density shown in Fig. 6.2.

.....

 [3]

[Total: 11]

- 7 (a) State what is meant by a photon.

.....

 [2]

- (b) Electromagnetic radiation of a varying frequency f and constant intensity I is used to illuminate a metal surface. At certain frequencies, electrons are emitted from the surface of the metal. The variation with f of the maximum kinetic energy E_{MAX} of the emitted electrons is shown in Fig. 7.1.

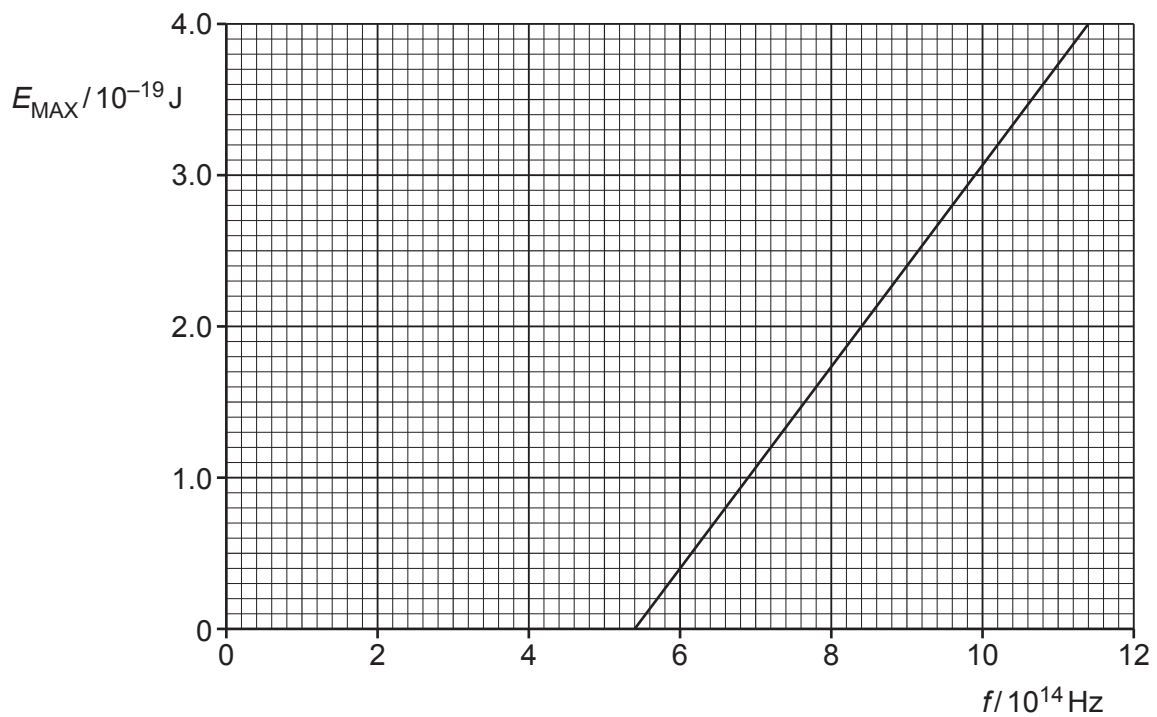


Fig. 7.1

- (i) State the name of this phenomenon.

..... [1]

- (ii) Describe **three** conclusions that can be drawn from the graph in Fig. 7.1. The conclusions may be qualitative or quantitative.

1

.....

2

.....

3

.....

[3]

- (c) The experiment in (b) is repeated twice, each time making one change.

State, with a reason, how the graph obtained would compare with Fig. 7.1 when:

- (i) a different metal is used, but keeping the intensity I of the radiation the same

.....

.....

..... [2]

- (ii) the same metal is used, but with electromagnetic radiation of intensity $2I$.

.....

.....

..... [2]

[Total: 10]

- 8 (a) (i) State what is meant by nuclear binding energy.

.....

 [2]

- (ii) On Fig. 8.1, sketch a line to show the variation with nucleon number A of the binding energy per nucleon E of a nucleus.

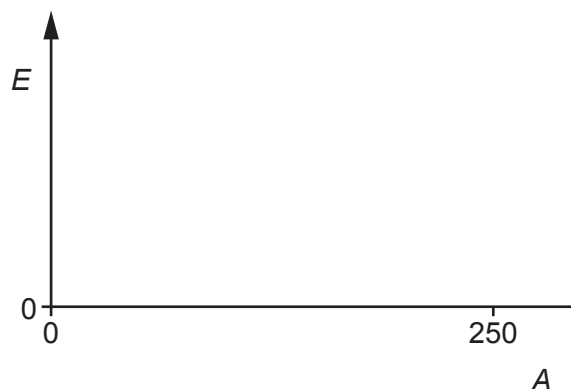


Fig. 8.1

[2]

- (b) In one type of nuclear process, deuterium (${}^2_1\text{H}$) undergoes the reaction



- (i) State the name of this type of nuclear process.

..... [1]

- (ii) Explain, with reference to your line in (a)(ii), why this reaction results in the release of energy.

.....

 [2]

(c) Table 8.1 shows the masses of the particles involved in the reaction in (b).

Table 8.1

particle	mass/u
${}^1_0\text{n}$	1.008 665
${}^2_1\text{H}$	2.014 102
${}^3_2\text{He}$	3.016 029

Calculate the energy released when 1.00 mol of deuterium undergoes the reaction.

energy = J [5]

[Total: 12]

- 9 (a) (i) Explain how X-rays are produced for use in medical diagnosis.

.....

 [3]

- (ii) State why X-ray images are taken of multiple sections of the body during computed tomography (CT) scanning.

.....
 [1]

- (b) An X-ray image is taken of the structure shown in Fig. 9.1.

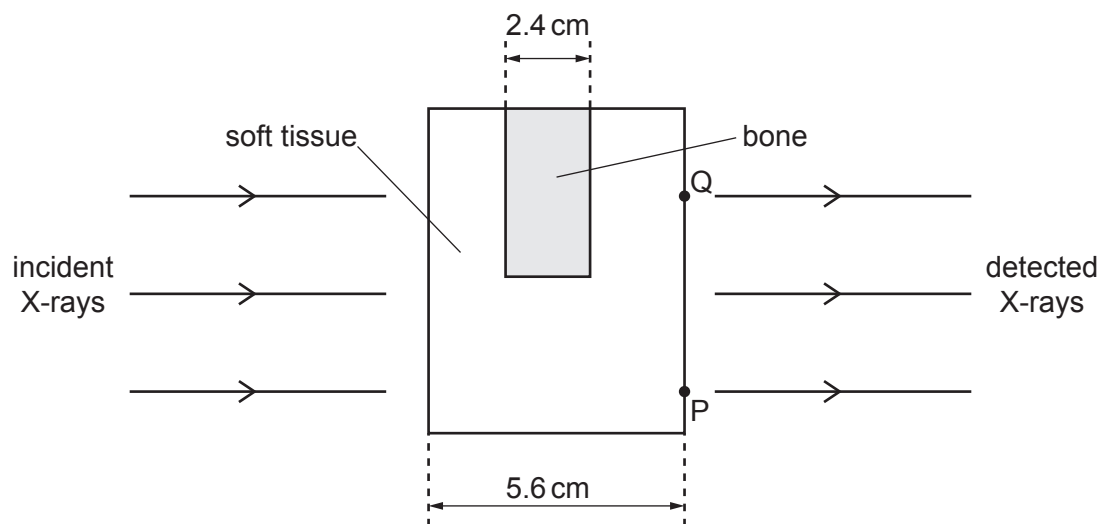


Fig. 9.1

The linear attenuation coefficient of bone is 3.4 cm^{-1} .

The linear attenuation coefficient of soft tissue is 0.89 cm^{-1} .

The incident X-rays are parallel and have a uniform intensity I_0 across the structure.

Determine, in terms of I_0 , the intensity of the detected X-rays from:

(i) point P

detected intensity = I_0 [2]

(ii) point Q.

detected intensity = I_0 [2]

(c) Explain, with reference to your answers in (b), whether the X-ray image of the structure in Fig. 9.1 has good contrast.

.....

 [1]

[Total: 9]

10 (a) State Wien's displacement law.

.....
 [1]

(b) Fig. 10.1 shows the wavelength distributions of electromagnetic radiation emitted by two stars A and B.

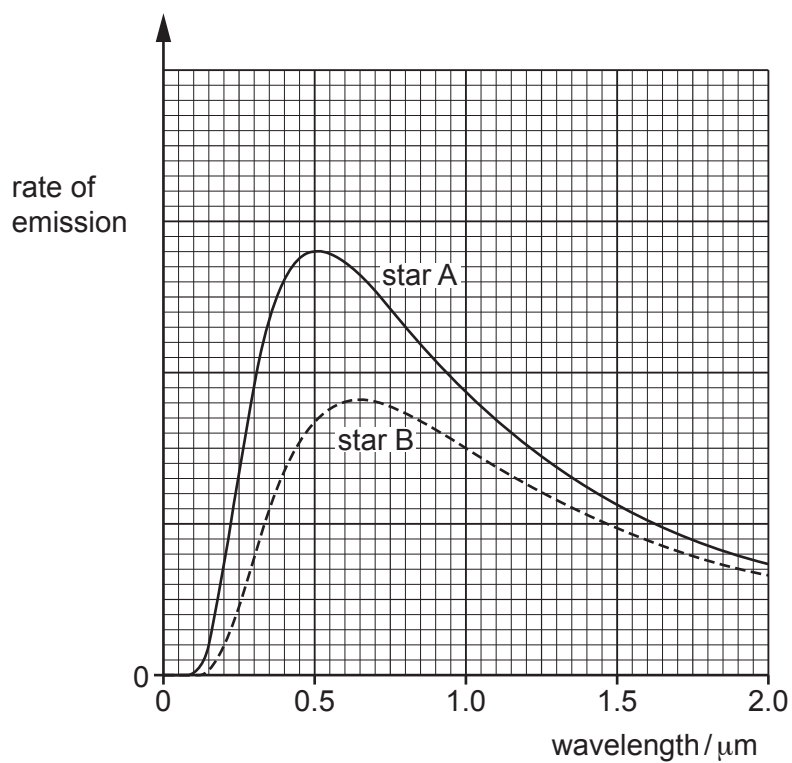


Fig. 10.1

The surface temperature of star A is known to be 5800 K.

(i) Determine the surface temperature of star B.

surface temperature = K [2]

- (ii) Star B appears less bright than star A when viewed from the Earth.

Use Fig. 10.1 to suggest, with a reason, how else the physical appearance of star B compares with that of star A.

.....
.....
..... [2]

- (c) The lines in Fig. 10.1 have been corrected for redshift.

- (i) State what is meant by redshift.

.....
.....
..... [2]

- (ii) Explain how cosmologists are able to determine that light from a distant star has undergone redshift.

.....
.....
..... [2]

[Total: 9]

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Stefan–Boltzmann law	$L = 4\pi\sigma r^2 T^4$
Doppler redshift	$\frac{\Delta\lambda}{\lambda} \hat{=} \frac{Df}{f} \hat{=} \frac{v}{c}$

- 1 (a) (i) Define gravitational potential at a point.

.....

 [2]

- (ii) Starting from the equation for the gravitational potential due to a point mass, show that the gravitational potential energy E_P of a point mass m at a distance r from another point mass M is given by

$$E_P = -\frac{GMm}{r}$$

where G is the gravitational constant.

[1]

- (b) Fig. 1.1 shows the path of a comet of mass $2.20 \times 10^{14} \text{ kg}$ as it passes around a star of mass $1.99 \times 10^{30} \text{ kg}$.

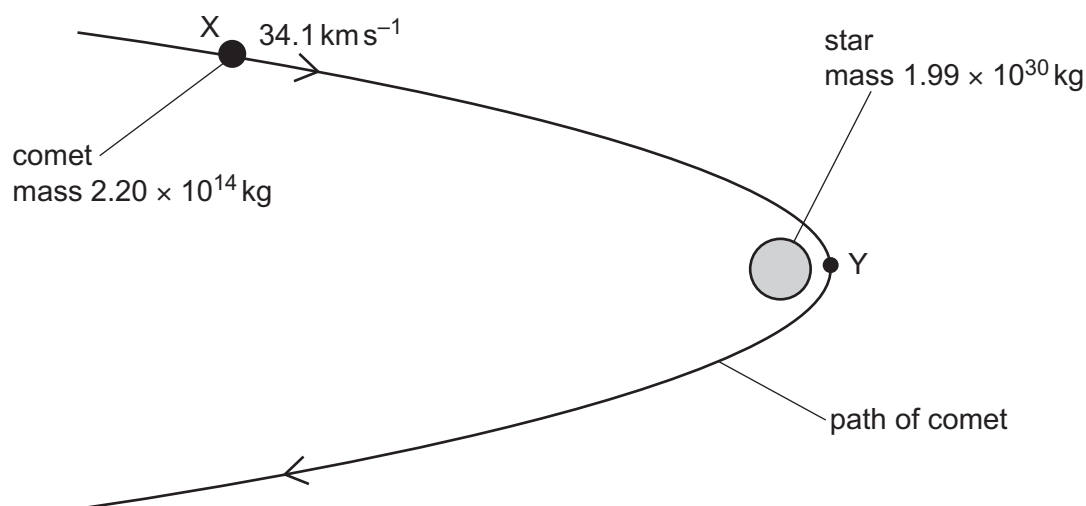


Fig. 1.1 (not to scale)

At point X, the comet is $8.44 \times 10^{11} \text{ m}$ from the centre of the star and is moving at a speed of 34.1 km s^{-1} .

At point Y, the comet passes its point of closest approach to the star. At this point, the comet is a distance of $6.38 \times 10^{10} \text{ m}$ from the centre of the star.

Both the comet and the star can be considered as point masses at their centres.

- (i) Calculate the magnitude of the change in the gravitational potential energy ΔE_p of the comet as it moves from position X to position Y.

$$\Delta E_p = \dots\dots\dots \text{ J [2]}$$

- (ii) State, with a reason, whether the change in gravitational potential energy in (b)(i) is an increase or a decrease.

.....
 [1]

- (iii) Use your answer in (b)(i) to determine the speed, in km s^{-1} , of the comet at point Y.

$$\text{speed} = \dots\dots\dots \text{ km s}^{-1} [3]$$

- (c) A second comet passes point X with the same speed as the comet in (b) and travelling in the same direction. This comet is gradually losing mass. The mass of this comet when it passes point X is the same as the mass of the comet in (b).

Suggest, with a reason, how the path of the second comet compares with the path shown in Fig. 1.1.

.....
 [1]

[Total: 10]

- 2 (a) State Coulomb's law.

.....

.....

..... [2]

- (b) Positronium is a system in which an electron and a positron orbit, with the same period, around their common centre of mass, as shown in Fig. 2.1.

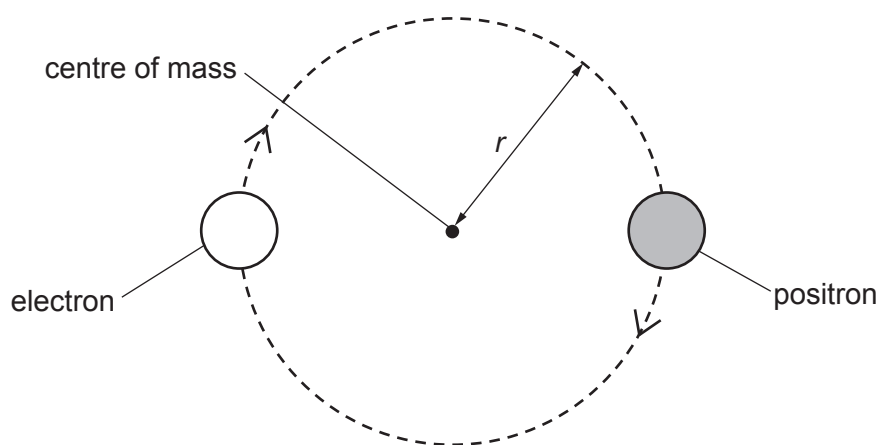


Fig. 2.1 (not to scale)

The radius r of the orbit of both particles is $1.59 \times 10^{-10} \text{ m}$.

- (i) Explain how the electric force between the electron and the positron causes the path of the moving particles to be circular.

.....

.....

..... [2]

- (ii) Show that the magnitude of the electric force between the electron and the positron is $2.28 \times 10^{-9} \text{ N}$.

[2]

- (iii) Use the information in (b)(ii) to determine the period of the circular orbit of the two particles.

period = s [3]

- (c) Positronium is highly unstable, and after a very short period of time it becomes gamma radiation.

- (i) Describe how gamma radiation is formed from the two particles in positronium.

.....
.....
.....
..... [3]

- (ii) State **one** medical application of the process described in (c)(i).

..... [1]

[Total: 13]

- 3 (a) Define specific latent heat of vaporisation.

.....

.....

..... [2]

- (b) The specific latent heat of vaporisation of water at atmospheric pressure of 1.0×10^5 Pa is $2.3 \times 10^6 \text{ J kg}^{-1}$. A mass of 0.37 kg of liquid water at 100°C is provided with the thermal energy needed to vaporise all of the water at atmospheric pressure.

- (i) Calculate the thermal energy q supplied to the water.

$$q = \dots\dots\dots \text{ J [1]}$$

- (ii) The mass of 1.0 mol of water is 18g. Assume that water vapour can be considered to behave as an ideal gas.

Show that the volume of water vapour produced is 0.64 m^3 .

[3]

- (iii) Assume that the initial volume of the liquid water is negligible compared with the volume of water vapour produced.

Determine the magnitude of the work done by the water in expanding against the atmosphere when it vaporises.

$$\text{work done} = \dots\dots\dots \text{ J [2]}$$

- (iv) Use your answers in (b)(i) and (b)(iii) to determine the increase in internal energy of the water when it vaporises at 100 °C. Explain your reasoning.

increase in internal energy = J [2]

- (c) Use the first law of thermodynamics to suggest, with a reason, how the specific latent heat of vaporisation of water at a pressure greater than atmospheric pressure compares with its value at atmospheric pressure.

.....
.....
..... [2]

[Total: 12]

- 4 (a) State what is meant by resonance.

.....

.....

..... [2]

- (b) Fig. 4.1 shows a heavy pendulum and a light pendulum, both suspended from the same piece of string. This string is secured at each end to fixed points.

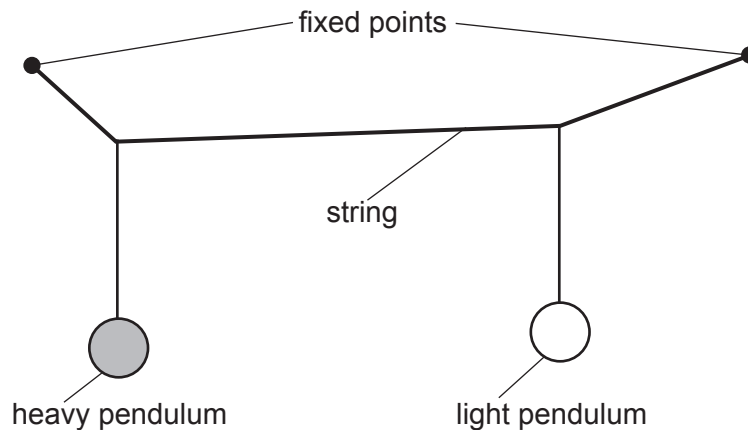


Fig. 4.1

Both pendulums have the same natural frequency.

The heavy pendulum is set oscillating perpendicular to the plane of the diagram. As it oscillates, it causes the light pendulum to oscillate.

Fig. 4.2 shows the variation with time t of the displacements of the two pendulums for three oscillations.

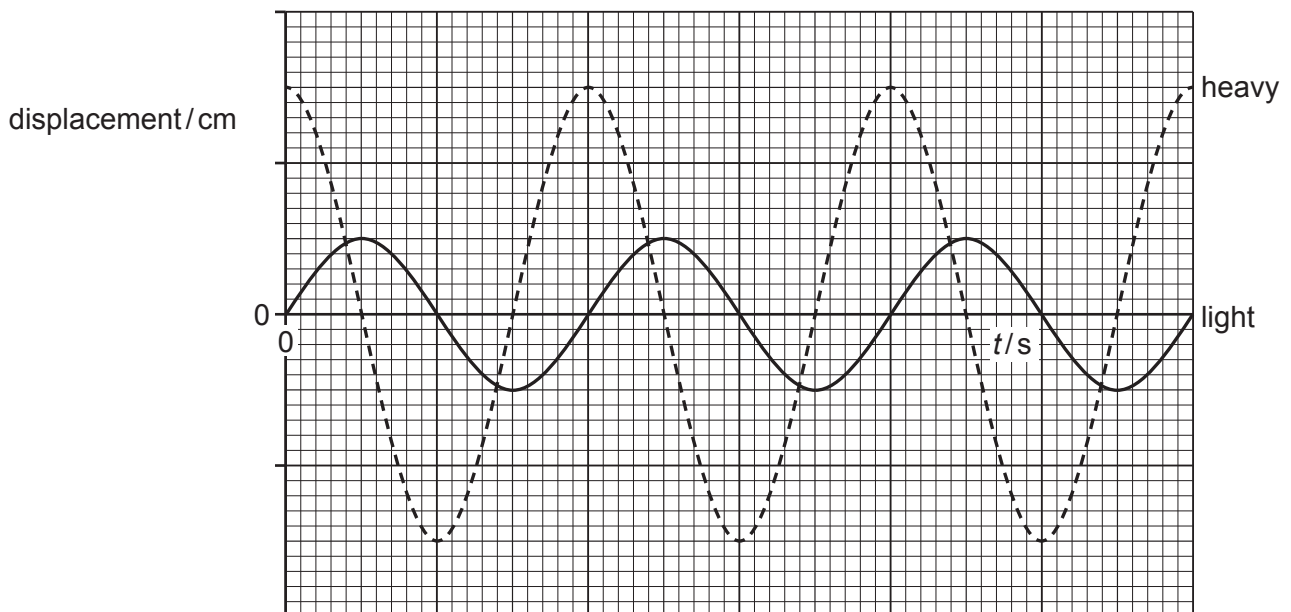


Fig. 4.2

The variation with t of the displacement x of the light pendulum is given by

$$x = 0.25 \sin 5.0\pi t$$

where x is in centimetres and t is in seconds.

- (i) Calculate the period T of the oscillations.

$$T = \dots\dots\dots \text{ s [2]}$$

- (ii) On Fig. 4.2, label both of the axes with the correct scales. Use the space below for any additional working that you need.

[2]

- (iii) Determine the magnitude of the phase difference ϕ between the oscillations of the light and heavy pendulums. Give a unit with your answer.

$$\phi = \dots\dots\dots \text{ unit } \dots\dots\dots [2]$$

[Total: 8]

- 5 (a) Define the capacitance of a parallel plate capacitor.

.....

.....

..... [2]

- (b) Two capacitors, of capacitances C_1 and C_2 , are connected in parallel to a power supply of electromotive force (e.m.f.) E , as shown in Fig. 5.1.

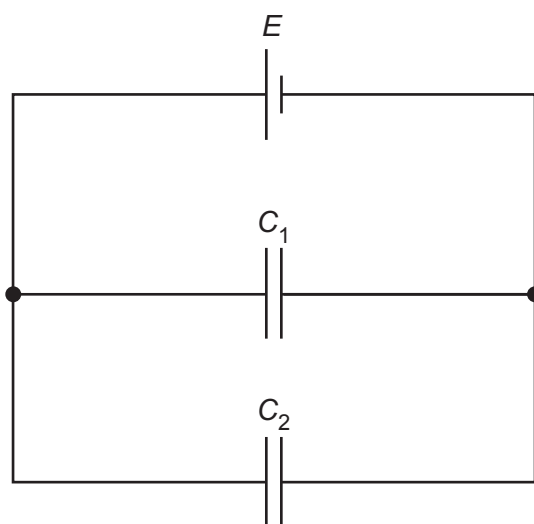


Fig. 5.1

Show that the combined capacitance C_T of the two capacitors is given by

$$C_T = C_1 + C_2.$$

Explain your reasoning. You may draw on Fig. 5.1 if you wish.

[3]

- (c) Two capacitors of capacitances $22\mu\text{F}$ and $47\mu\text{F}$, and a resistor of resistance $2.7\text{M}\Omega$, are connected into the circuit of Fig. 5.2.

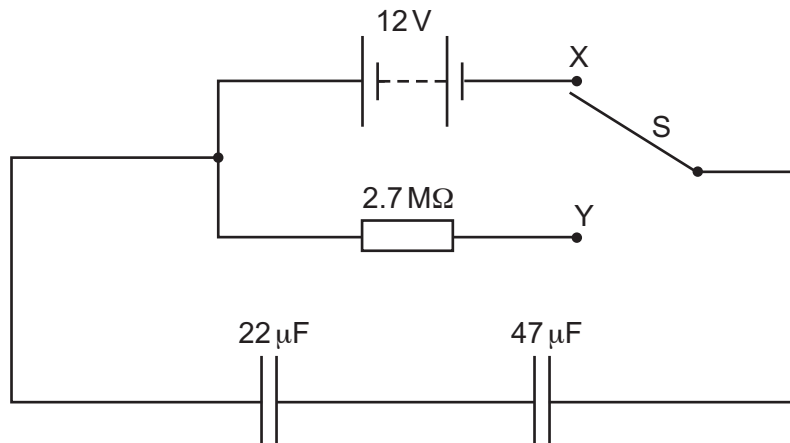


Fig. 5.2

The battery has an e.m.f. of 12 V.

- (i) Show that the combined capacitance of the two capacitors is $15\mu\text{F}$.

[1]

- (ii) The two-way switch S is initially at position X, so that the capacitors are fully charged.

Use the information in (c)(i) to calculate the total energy stored in the two capacitors.

total energy = J [2]

- (iii) The two-way switch is now moved to position Y.

Determine the time taken for the potential difference (p.d.) across the $22\mu\text{F}$ capacitor to become 6.0 V.

time = s [3]

[Total: 11]

[Turn over]

- 6 (a) State the **two** conditions that must be satisfied for a copper wire, placed in a magnetic field, to experience a magnetic force.

1

.....

2

.....

[2]

- (b) A long air-cored solenoid is connected to a power supply, so that the solenoid creates a magnetic field. Fig. 6.1 shows a cross-section through the middle of the solenoid.

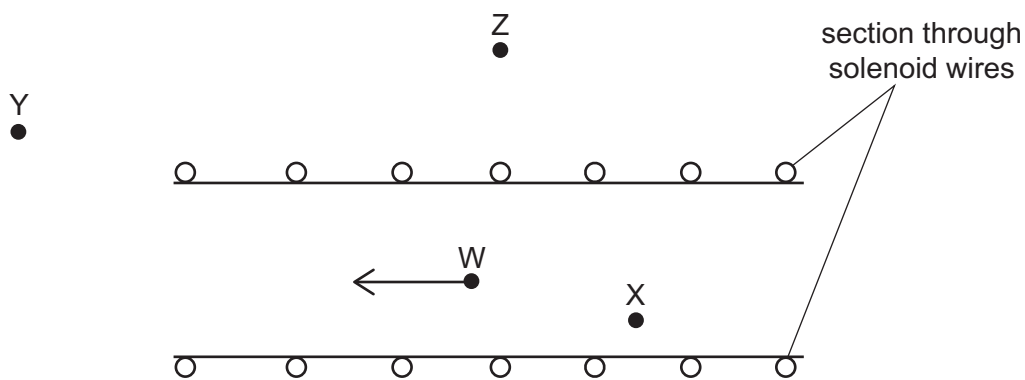


Fig. 6.1

The direction of the magnetic field at point W is indicated by the arrow. Three other points are labelled X, Y and Z.

- (i) On Fig. 6.1, draw arrows to indicate the direction of the magnetic field at each of the points X, Y and Z. [3]

- (ii) Compare the magnitude of the flux density of the magnetic field:

• at X and at W

.....

• at Y and at Z.

.....

[2]

- (c) Two long parallel current-carrying wires are placed near to each other in a vacuum.

Explain why these wires exert a magnetic force on each other. You may draw a labelled diagram if you wish.

.....

.....

.....

..... [3]

[Total: 10]

- 7 (a) State Faraday's law of electromagnetic induction.

.....

.....

..... [2]

- (b) Two coils are wound on an iron bar, as shown in Fig. 7.1.

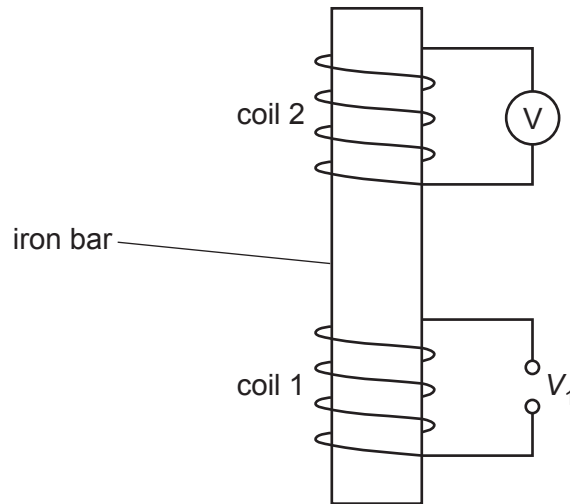


Fig. 7.1

Coil 1 is connected to a potential difference (p.d.) V_1 that gives rise to a magnetic field in the iron bar.

Fig. 7.2 shows the variation with time t of the magnetic flux density B in the iron bar.

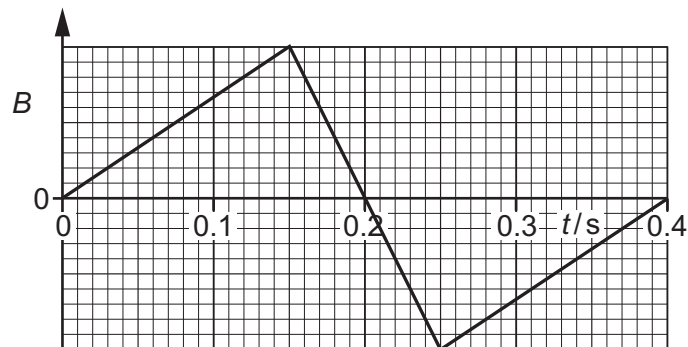


Fig. 7.2

A voltmeter measures the electromotive force (e.m.f.) V_2 that is induced across coil 2.

On Fig. 7.3, sketch the variation with t of V_2 between $t = 0$ and $t = 0.40$ s.

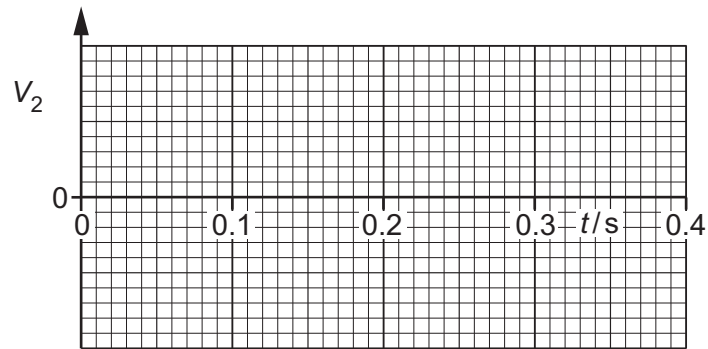


Fig. 7.3

[4]

- (c) Coil 2 in (b) is now replaced with a copper ring that rests loosely on top of coil 1. The supply to coil 1 is replaced with a cell and a switch that is initially open, as shown in Fig. 7.4.

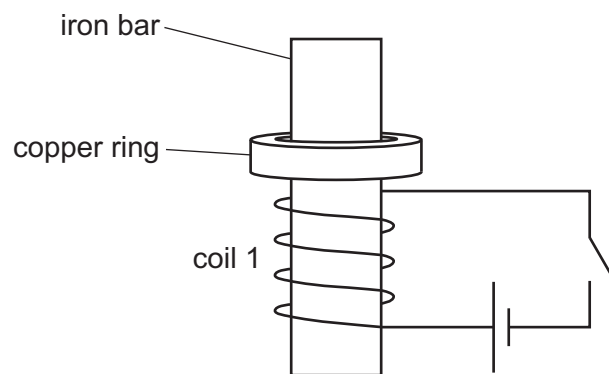


Fig. 7.4

- (i) The switch is now closed. As it is closed, the copper ring is observed to jump upwards.

Explain why this happens.

.....

 [3]

- (ii) Suggest, with a reason, what would be the effect of repeating the procedure in (c)(i) with the terminals of the cell reversed.

.....
 [1]

[Total: 10]

8 (a) State **one** piece of experimental evidence for:

(i) the particulate nature of electromagnetic radiation

..... [1]

(ii) the wave nature of matter.

..... [1]

(b) (i) Calculate the de Broglie wavelength λ of an alpha-particle moving at a speed of $6.2 \times 10^7 \text{ m s}^{-1}$.

$\lambda =$ m [3]

(ii) The speed v of the alpha-particle in (b)(i) is gradually reduced to zero.

On Fig. 8.1, sketch the variation with v of λ .

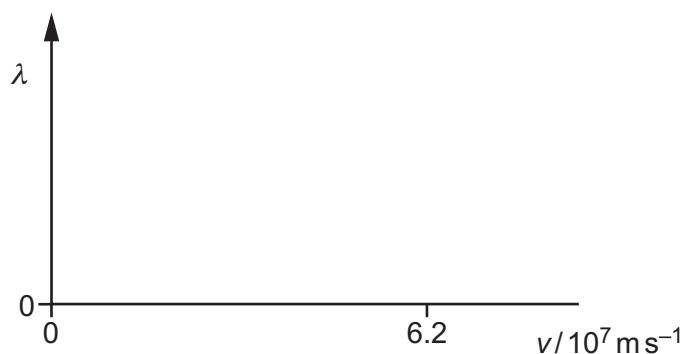


Fig. 8.1

[2]

(c) Suggest an explanation for why people are not observed to diffract when they walk through a doorway.

.....

 [1]

[Total: 8]

9 (a) (i) State Hubble's law.

.....

.....

..... [2]

(ii) Explain how cosmologists use observations of emission spectra from stars in distant galaxies to determine that the Universe is expanding.

.....

.....

.....

..... [2]

(b) Explain how Hubble's law and the idea of the expanding Universe lead to the Big Bang theory of the origin of the Universe.

.....

.....

.....

..... [3]

[Total: 7]

- 10 (a) State what is meant by radioactive decay.

.....

.....

..... [2]

- (b) A radioactive sample consists of an isotope X of half-life T that decays to form a stable product. Only X and the stable product are present in the sample.

At time $t = 0$, the sample has an activity of A_0 and contains N_0 nuclei of X.

- (i) On Fig. 10.1, sketch the variation with t of the number N of nuclei of X present in the sample. Your line should extend from time $t = 0$ to time $t = 3T$.

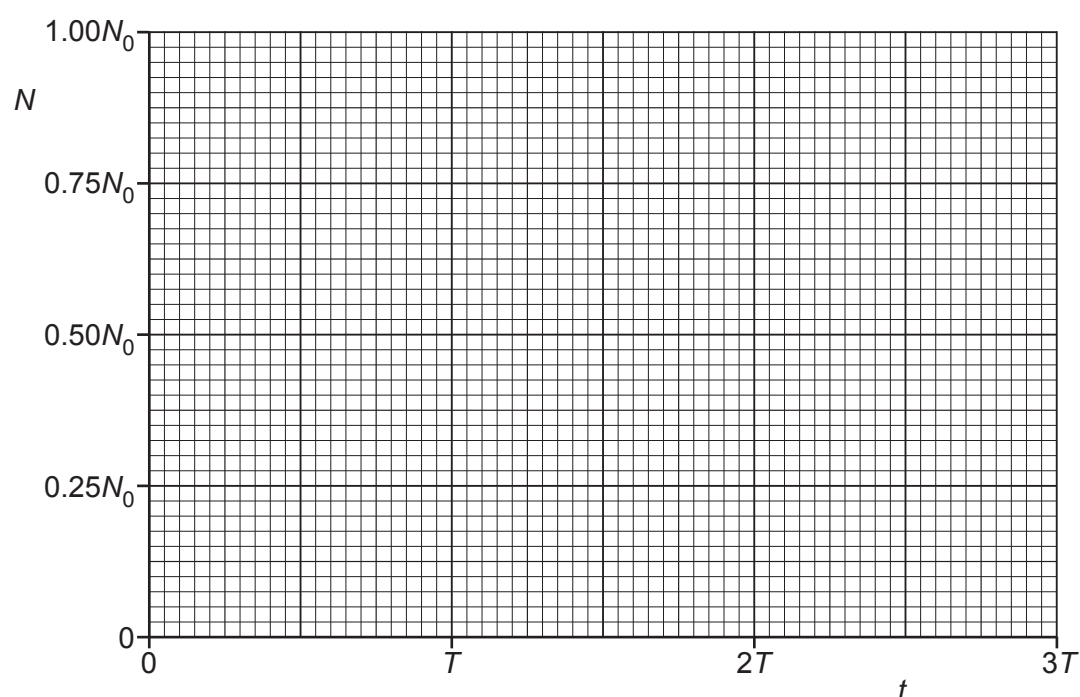


Fig. 10.1

[3]

- (ii) On Fig. 10.2, sketch the variation with N of the activity A of the sample for values of N between $N = 0$ and $N = N_0$.

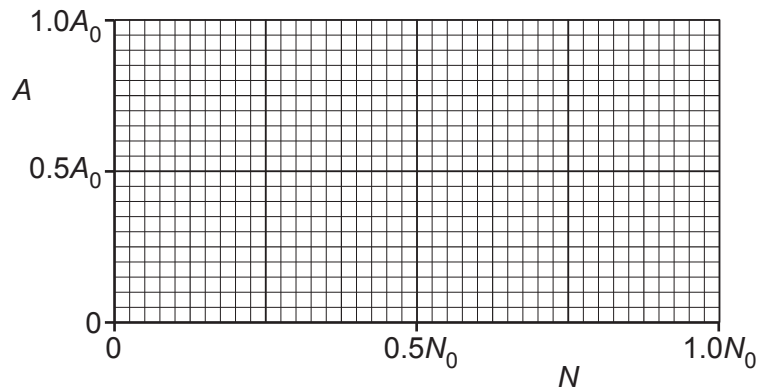


Fig. 10.2

[2]

- (c) State the name of the quantity represented by the gradient of your line in:

- (i) Fig. 10.1

..... [1]

- (ii) Fig. 10.2.

..... [1]

- (d) For the sample in (b), calculate the fraction $\frac{N}{N_0}$ at time $t = 1.70T$.

$$\frac{N}{N_0} = \text{.....} [2]$$

[Total: 11]

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9702/43

May/June 2022

2 hours

You must answer on the question paper.

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [].

This document has **24** pages. Any blank pages are indicated.

Data

acceleration of free fall	$g = 9.81 \text{ m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
unified atomic mass unit	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27} \text{ kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$ ($\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ m F}^{-1}$)
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_o = \frac{f_s v}{v \pm v_s}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \dots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$

gravitational potential	$\phi = -\frac{GM}{r}$
gravitational potential energy	$E_P = -\frac{GMm}{r}$
pressure of an ideal gas	$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$
simple harmonic motion	$a = -\omega^2 x$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t$ $v = \omega \sqrt{(x_0^2 - x^2)}$
electric potential	$V = \frac{Q}{4\pi\epsilon_0 r}$
electrical potential energy	$E_P = \frac{Qq}{4\pi\epsilon_0 r}$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$
capacitors in parallel	$C = C_1 + C_2 + \dots$
discharge of a capacitor	$x = x_0 e^{-\frac{t}{RC}}$
Hall voltage	$V_H = \frac{BI}{ntq}$
alternating current/voltage	$x = x_0 \sin \omega t$
radioactive decay	$x = x_0 e^{-\lambda t}$
decay constant	$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$
intensity reflection coefficient	$\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$
Stefan–Boltzmann law	$L = 4\pi\sigma r^2 T^4$
Doppler redshift	$\frac{\Delta\lambda}{\lambda} \approx \frac{Df}{f} \approx \frac{v}{c}$

- 1 (a) (i) State Newton's law of gravitation.

.....

.....

..... [2]

- (ii) Use Newton's law of gravitation to show that the gravitational field strength g at a distance r away from a point mass M is given by

$$g = \frac{GM}{r^2}.$$

[2]

- (b) The Earth has a mass of 5.98×10^{24} kg and a radius of 6.37×10^6 m.
 The Moon has a mass of 7.35×10^{22} kg and a radius of 1.74×10^6 m.
 The Earth and the Moon can both be considered as point masses at their centres. Their centres are a distance of 3.84×10^8 m apart.

- (i) Show that the gravitational field strength at the surface of the Moon due to the mass of the Moon is 1.62 N kg^{-1} .

[1]

- (ii) Explain why there is a point X on the line between the centres of the Earth and the Moon where the resultant gravitational field strength due to the Earth and the Moon is zero.

.....

.....

..... [2]

- (iii) Calculate the distance x of point X from the centre of the Moon.

$x =$ m [3]

[Total: 10]

- 2 A sphere of mass $1.6 \times 10^{-10} \text{ kg}$ has a charge of $+0.27 \text{ nC}$. The sphere is in a uniform electric field that acts vertically upwards, as shown in the side view in Fig. 2.1.

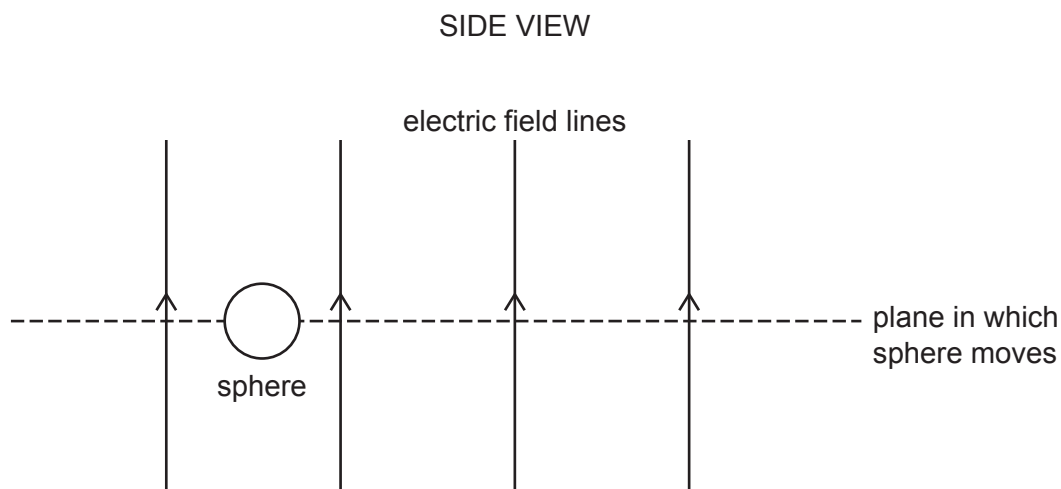


Fig. 2.1

The force exerted on the sphere by the electric field causes the sphere to remain at a fixed vertical height in a horizontal plane.

There is a uniform magnetic field in the region of the electric field. The sphere moves at a speed of 0.78 m s^{-1} in the horizontal plane. The magnetic field causes the sphere to move in a circular path of radius 3.4 m , as shown in the view from above in Fig. 2.2.

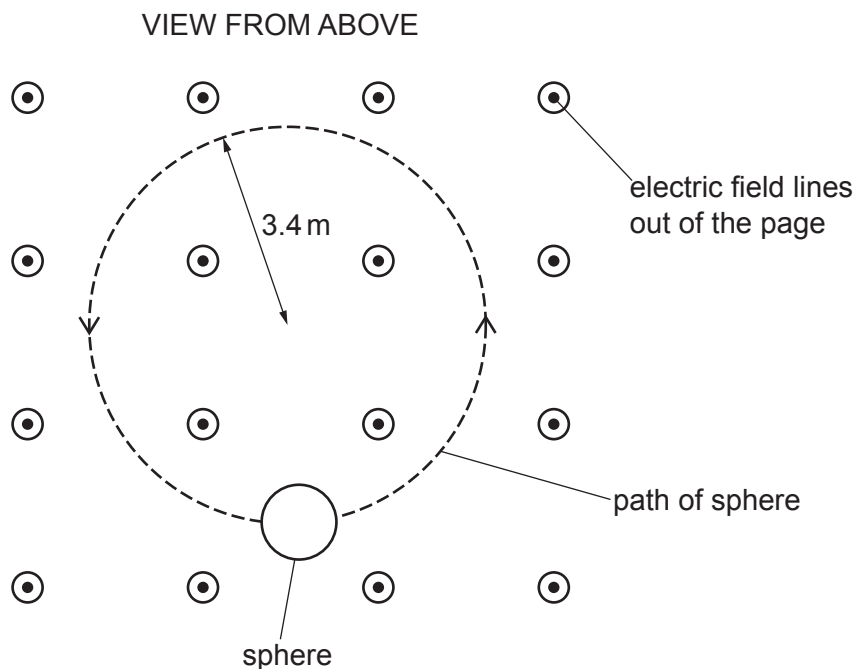


Fig. 2.2

- (a) (i) Determine the direction of the uniform magnetic field.

..... [1]

- (ii) Explain why the motion of the sphere in the horizontal plane is circular.

.....

.....

..... [2]

- (b) Calculate the strength of the uniform electric field.

electric field strength = NC^{-1} [2]

- (c) By considering the magnetic force on the sphere, show that the flux density of the uniform magnetic field is 0.14 T.

[3]

[Total: 8]

- 3 A fixed mass of an ideal gas is initially at a temperature of 17°C . The gas has a volume of 0.24 m^3 and a pressure of $1.2 \times 10^5\text{ Pa}$.

(a) (i) State what is meant by an ideal gas.

.....

 [2]

(ii) Calculate the amount n of gas.

$n = \dots\dots\dots$ mol [2]

(b) The gas undergoes three successive changes, as shown in Fig. 3.1.

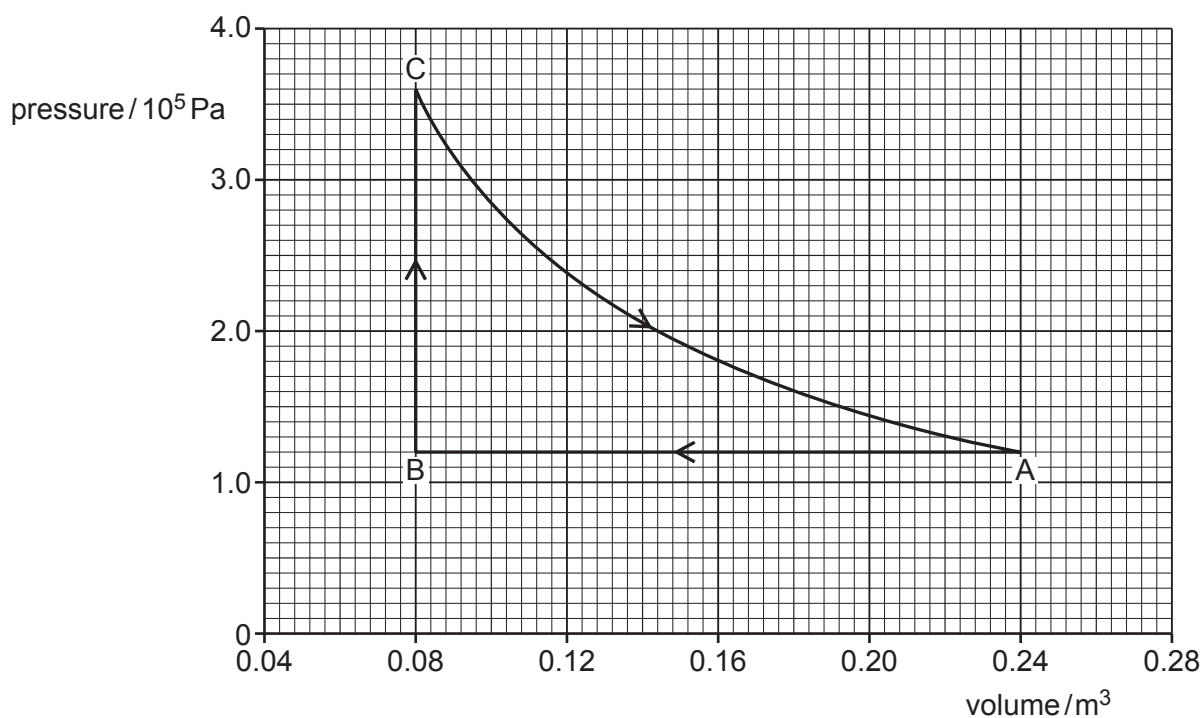


Fig. 3.1

The initial state is represented by point A. The gas is cooled at constant pressure to point B by the removal of 48.0 kJ of thermal energy.

The gas is then heated at constant volume to point C.

Finally, the gas expands at constant temperature back to its original pressure and volume at point A. During this expansion, the gas does 31.6 kJ of work.

- (i) Show that the magnitude of the work done during the change AB is 19.2 kJ.

[2]

- (ii) Complete Table 3.1 to show the work done on the gas, the thermal energy supplied to the gas and the increase in internal energy of the gas, for each of the changes AB, BC and CA.

Table 3.1

change	work done on gas / kJ	thermal energy supplied to gas / kJ	increase in internal energy of gas / kJ
AB		-48.0	
BC			
CA	-31.6		

[5]

[Total: 11]

- 4 A pendulum consists of a bob (small metal sphere) attached to the end of a piece of string. The other end of the string is attached to a fixed point. The bob oscillates with small oscillations about its equilibrium position, as shown in Fig. 4.1.

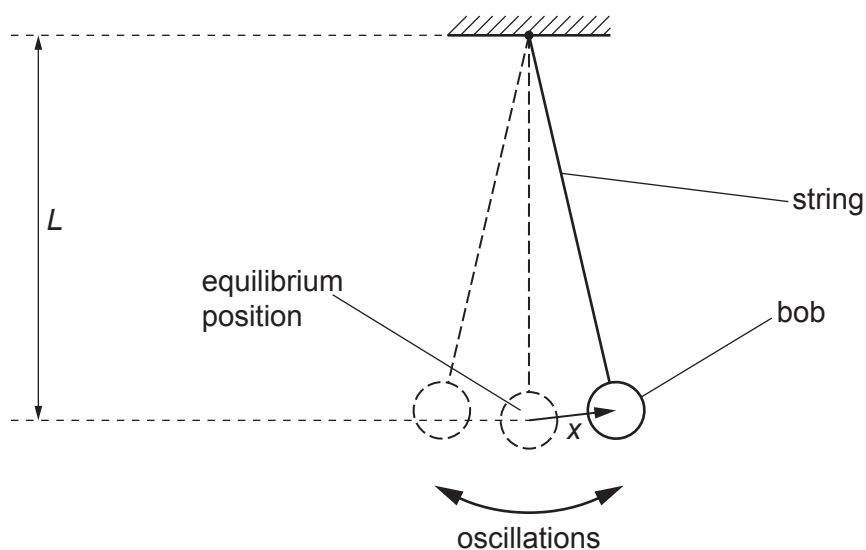


Fig. 4.1 (not to scale)

The length L of the pendulum, measured from the fixed point to the centre of the bob, is 1.24 m.

The acceleration a of the bob varies with its displacement x from the equilibrium position as shown in Fig. 4.2.

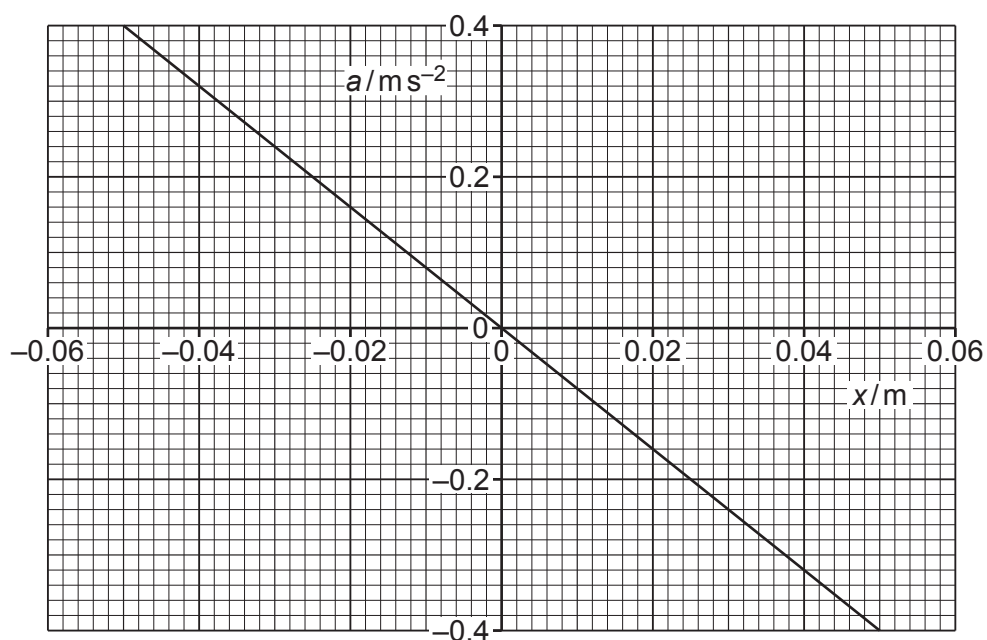


Fig. 4.2

- (a) State how Fig. 4.2 shows that the motion of the pendulum is simple harmonic.

.....

.....

..... [2]

- (b) (i) Use Fig. 4.2 to determine the angular frequency ω of the oscillations.

$$\omega = \dots\dots\dots \text{rad s}^{-1} \quad [2]$$

- (ii) The angular frequency ω is related to the length L of the pendulum by

$$\omega = \sqrt{\frac{k}{L}}$$

where k is a constant.

Use your answer in (b)(i) to determine k . Give a unit with your answer.

$$k = \dots\dots\dots \text{unit} \dots\dots\dots [2]$$

- (c) While the pendulum is oscillating, the length of the string is increased in such a way that the total energy of the oscillations remains constant.

Suggest and explain the qualitative effect of this change on the amplitude of the oscillations.

.....

.....

..... [2]

[Total: 8]

- 5 Fig. 5.1 shows four diodes and a load resistor of resistance $1.2\text{ k}\Omega$, connected in a circuit that is used to produce rectification of an alternating voltage.

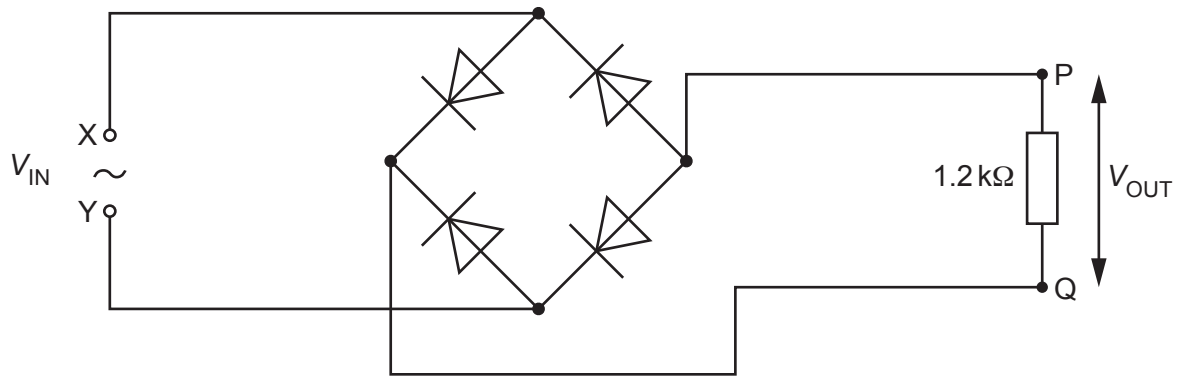


Fig. 5.1

- (a) (i) State what is meant by rectification.

.....
 [1]

- (ii) State the type of rectification produced by the circuit in Fig. 5.1.

..... [1]

- (b) A sinusoidal alternating voltage V_{IN} is applied across the input terminals X and Y. The variation with time t of V_{IN} is given by the equation

$$V_{IN} = 6.0 \sin 25\pi t$$

where V_{IN} is in volts and t is in seconds.

- (i) On Fig. 5.1, label the output terminals P and Q with the appropriate symbols to indicate the polarity of the output voltage V_{OUT} . [1]

- (ii) The magnitude of the output voltage V_{OUT} varies with t as shown in Fig. 5.2.

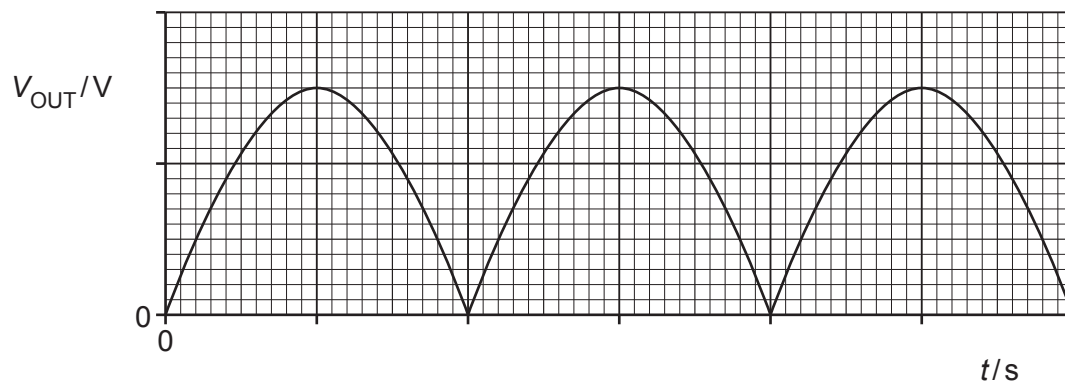


Fig. 5.2

On Fig. 5.2, label both of the axes with the correct scales. Use the space below for any working that you need.

[3]

- (c) The output voltage in (b) is smoothed by adding a capacitor to the circuit in Fig. 5.1. The difference between the maximum and minimum values of the smoothed output voltage is 10% of the peak voltage.

- (i) On Fig. 5.1, draw the circuit symbol for a capacitor showing the capacitor correctly connected into the circuit. [1]

- (ii) On Fig. 5.2, sketch the variation with t of the smoothed output voltage. [2]

- (iii) Calculate the capacitance C of the capacitor.

$C = \dots\dots\dots$ F [3]

[Total: 12]

- 6 (a) Define magnetic flux.

.....

.....

..... [2]

- (b) A square coil of wire of side length 12 cm consists of 8 insulated turns. The coil is stationary in a uniform magnetic field. The plane of the coil is perpendicular to the magnetic field, as shown in Fig. 6.1.

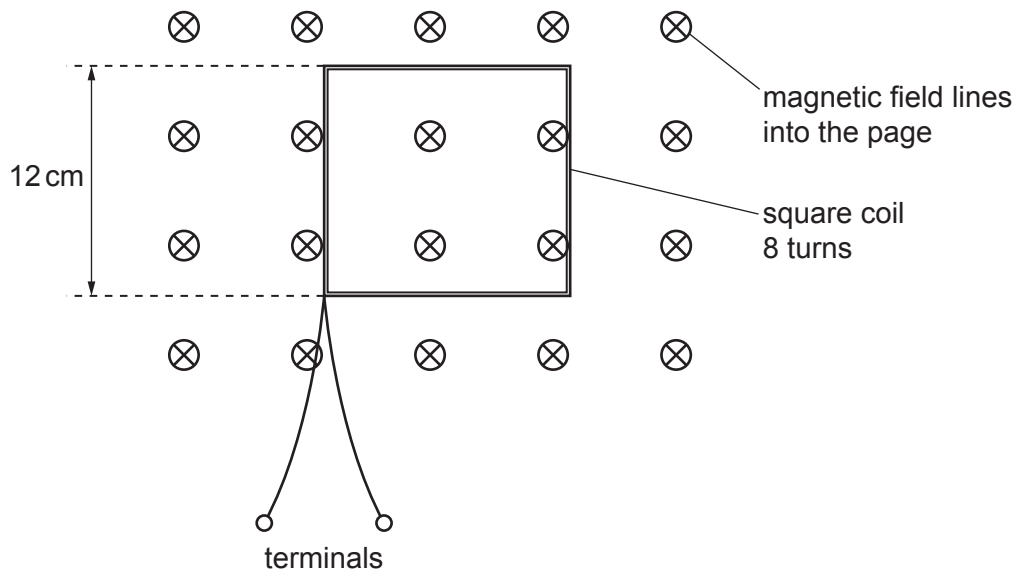


Fig. 6.1

The flux density B of the magnetic field varies with time t as shown in Fig. 6.2.

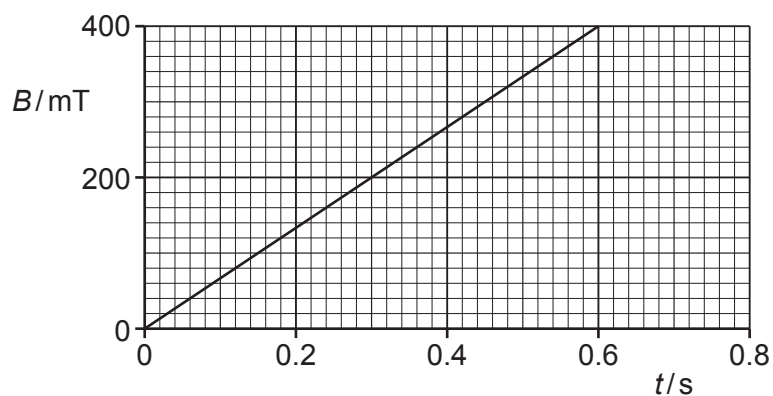


Fig. 6.2

- (i) Determine the magnetic flux linkage inside the coil at time $t = 0.60\text{ s}$. Give a unit with your answer.

magnetic flux linkage = unit [3]

- (ii) State how Fig. 6.2 shows that the electromotive force (e.m.f.) E induced across the terminals between $t = 0$ and $t = 0.60\text{ s}$ is constant.

..... [1]

- (iii) Calculate the magnitude of E .

$E = \dots\dots\dots \text{ V}$ [2]

- (c) The procedure in (b) is repeated, but this time the terminals of the coil are connected together.

State and explain the effect on the coil of connecting the terminals together during the change of magnetic flux density shown in Fig. 6.2.

.....

 [3]

[Total: 11]

- 7 (a) State what is meant by a photon.

.....

 [2]

- (b) Electromagnetic radiation of a varying frequency f and constant intensity I is used to illuminate a metal surface. At certain frequencies, electrons are emitted from the surface of the metal. The variation with f of the maximum kinetic energy E_{MAX} of the emitted electrons is shown in Fig. 7.1.

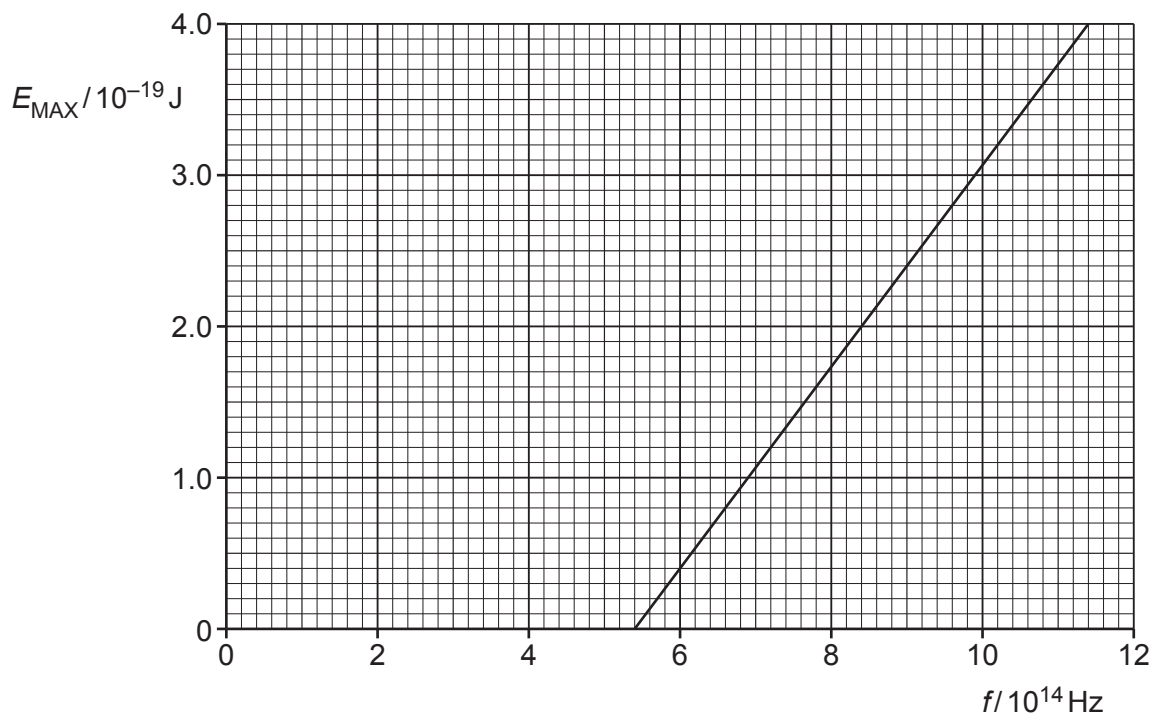


Fig. 7.1

- (i) State the name of this phenomenon.

..... [1]

- (ii) Describe **three** conclusions that can be drawn from the graph in Fig. 7.1. The conclusions may be qualitative or quantitative.

1

.....

2

.....

3

.....

[3]

- (c) The experiment in (b) is repeated twice, each time making one change.

State, with a reason, how the graph obtained would compare with Fig. 7.1 when:

- (i) a different metal is used, but keeping the intensity I of the radiation the same

.....

.....

..... [2]

- (ii) the same metal is used, but with electromagnetic radiation of intensity $2I$.

.....

.....

..... [2]

[Total: 10]

- 8 (a) (i) State what is meant by nuclear binding energy.

.....

 [2]

- (ii) On Fig. 8.1, sketch a line to show the variation with nucleon number A of the binding energy per nucleon E of a nucleus.

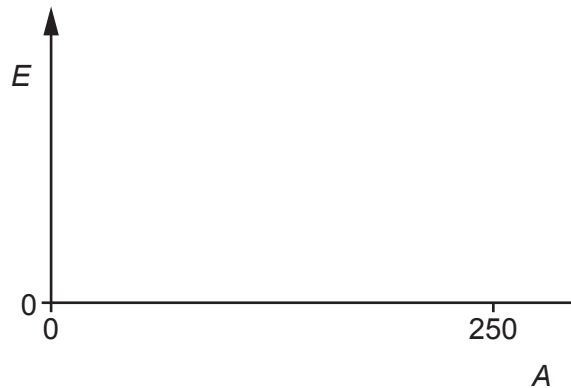


Fig. 8.1

[2]

- (b) In one type of nuclear process, deuterium (${}^2_1\text{H}$) undergoes the reaction



- (i) State the name of this type of nuclear process.

..... [1]

- (ii) Explain, with reference to your line in (a)(ii), why this reaction results in the release of energy.

.....

 [2]

(c) Table 8.1 shows the masses of the particles involved in the reaction in (b).

Table 8.1

particle	mass/u
${}_0^1\text{n}$	1.008 665
${}_1^2\text{H}$	2.014 102
${}_2^3\text{He}$	3.016 029

Calculate the energy released when 1.00 mol of deuterium undergoes the reaction.

energy = J [5]

[Total: 12]

- 9 (a) (i) Explain how X-rays are produced for use in medical diagnosis.

.....

 [3]

- (ii) State why X-ray images are taken of multiple sections of the body during computed tomography (CT) scanning.

.....
 [1]

- (b) An X-ray image is taken of the structure shown in Fig. 9.1.

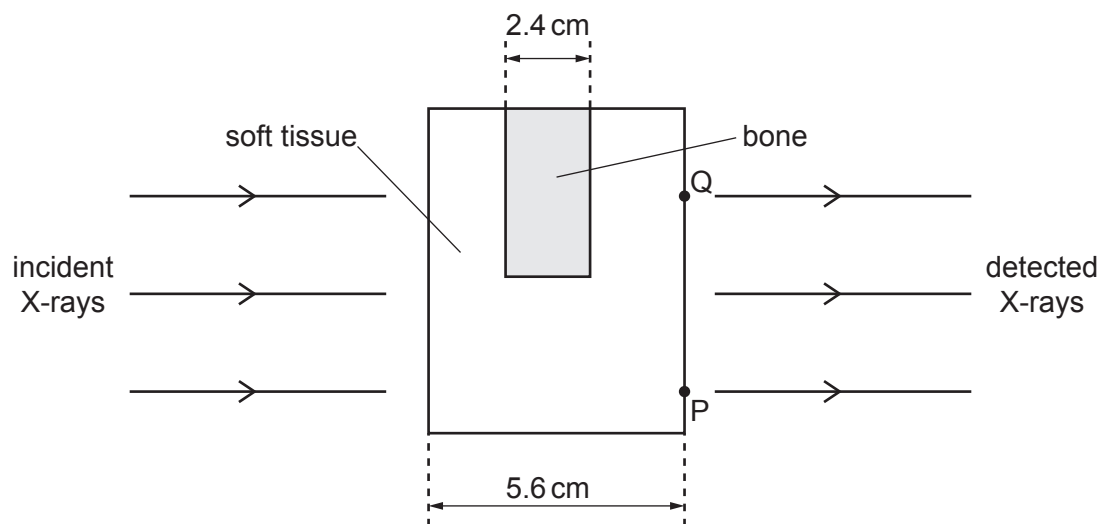


Fig. 9.1

The linear attenuation coefficient of bone is 3.4 cm^{-1} .

The linear attenuation coefficient of soft tissue is 0.89 cm^{-1} .

The incident X-rays are parallel and have a uniform intensity I_0 across the structure.

Determine, in terms of I_0 , the intensity of the detected X-rays from:

(i) point P

detected intensity = I_0 [2]

(ii) point Q.

detected intensity = I_0 [2]

(c) Explain, with reference to your answers in (b), whether the X-ray image of the structure in Fig. 9.1 has good contrast.

.....

 [1]

[Total: 9]

10 (a) State Wien's displacement law.

.....
 [1]

(b) Fig. 10.1 shows the wavelength distributions of electromagnetic radiation emitted by two stars A and B.

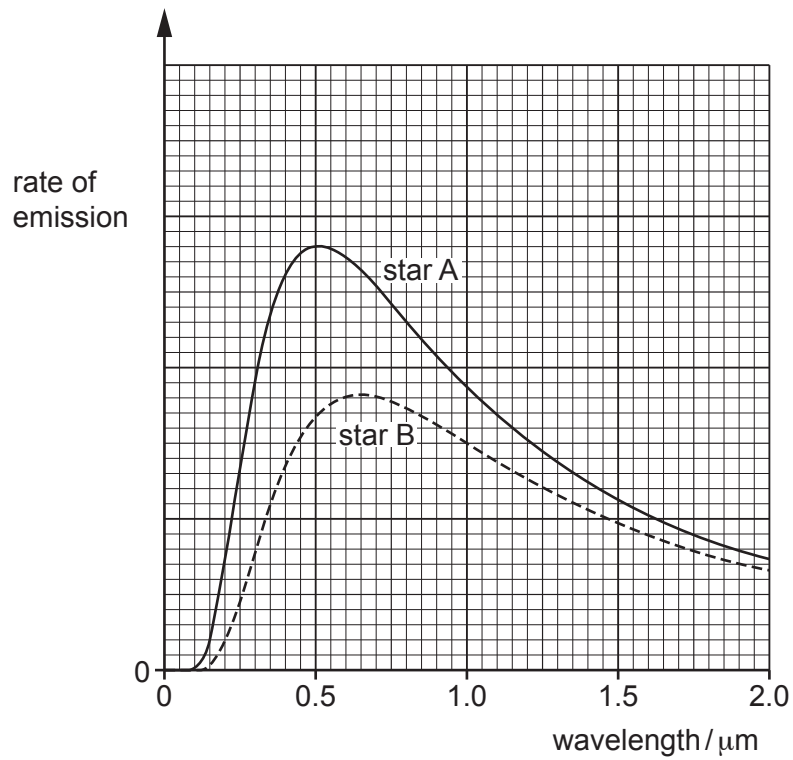


Fig. 10.1

The surface temperature of star A is known to be 5800 K.

(i) Determine the surface temperature of star B.

surface temperature = K [2]

- (ii) Star B appears less bright than star A when viewed from the Earth.

Use Fig. 10.1 to suggest, with a reason, how else the physical appearance of star B compares with that of star A.

.....
.....
..... [2]

- (c) The lines in Fig. 10.1 have been corrected for redshift.

- (i) State what is meant by redshift.

.....
.....
..... [2]

- (ii) Explain how cosmologists are able to determine that light from a distant star has undergone redshift.

.....
.....
..... [2]

[Total: 9]

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9702/51

May/June 2022

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 30.
- The number of marks for each question or part question is shown in brackets [].

This document has 8 pages.

- 1 Two parallel metal plates, each of area A , are separated by a small distance d , as shown in Fig. 1.1.

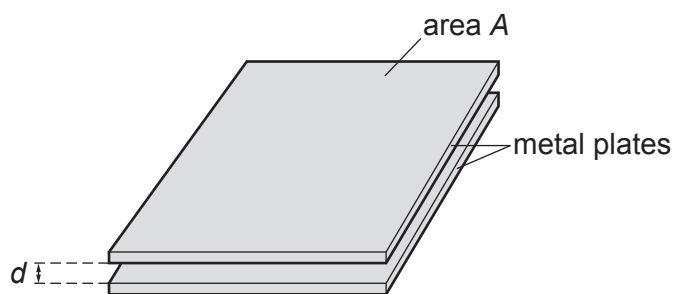


Fig. 1.1 (not to scale)

The plates are initially charged using a power supply.

The plates are then connected to an uncharged capacitor. The potential difference V across the capacitor is measured.

It is suggested that V is related to d by the relationship

$$\frac{W}{V} = 1 + \frac{Cd}{KA}$$

where C is the capacitance of the capacitor, and K and W are constants.

Plan a laboratory experiment to test the relationship between V and d .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for K and W .

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

Diagram

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- 2 A student investigates the relationship between the luminosity L of a star and its mass M for a set of stars known as main-sequence stars.

It is suggested that L and M are related by the equation

$$L = SZM^n$$

where S is the luminosity of the Sun, and Z and n are constants.

- (a) A graph is plotted of $\lg L$ on the y -axis against $\lg M$ on the x -axis.

Determine expressions for the gradient and y -intercept.

gradient =

y -intercept =

[1]

(b) Values of M and L are given in Table 2.1.

Table 2.1

$M/10^{30} \text{ kg}$	$L/10^{28} \text{ W}$	$\lg (M/10^{30} \text{ kg})$	$\lg (L/10^{28} \text{ W})$
4.8 ± 0.4	1.4		
6.4 ± 0.4	3.1		
12 ± 2	32		
23 ± 2	350		
43 ± 4	3600		
91 ± 4	66 000		

Calculate and record values of $\lg (M/10^{30} \text{ kg})$ and $\lg (L/10^{28} \text{ W})$ in Table 2.1.
Include the absolute uncertainties in $\lg (M/10^{30} \text{ kg})$.

[2]

(c) (i) Plot a graph of $\lg (L/10^{28} \text{ W})$ against $\lg (M/10^{30} \text{ kg})$.
Include error bars for $\lg (M/10^{30} \text{ kg})$.

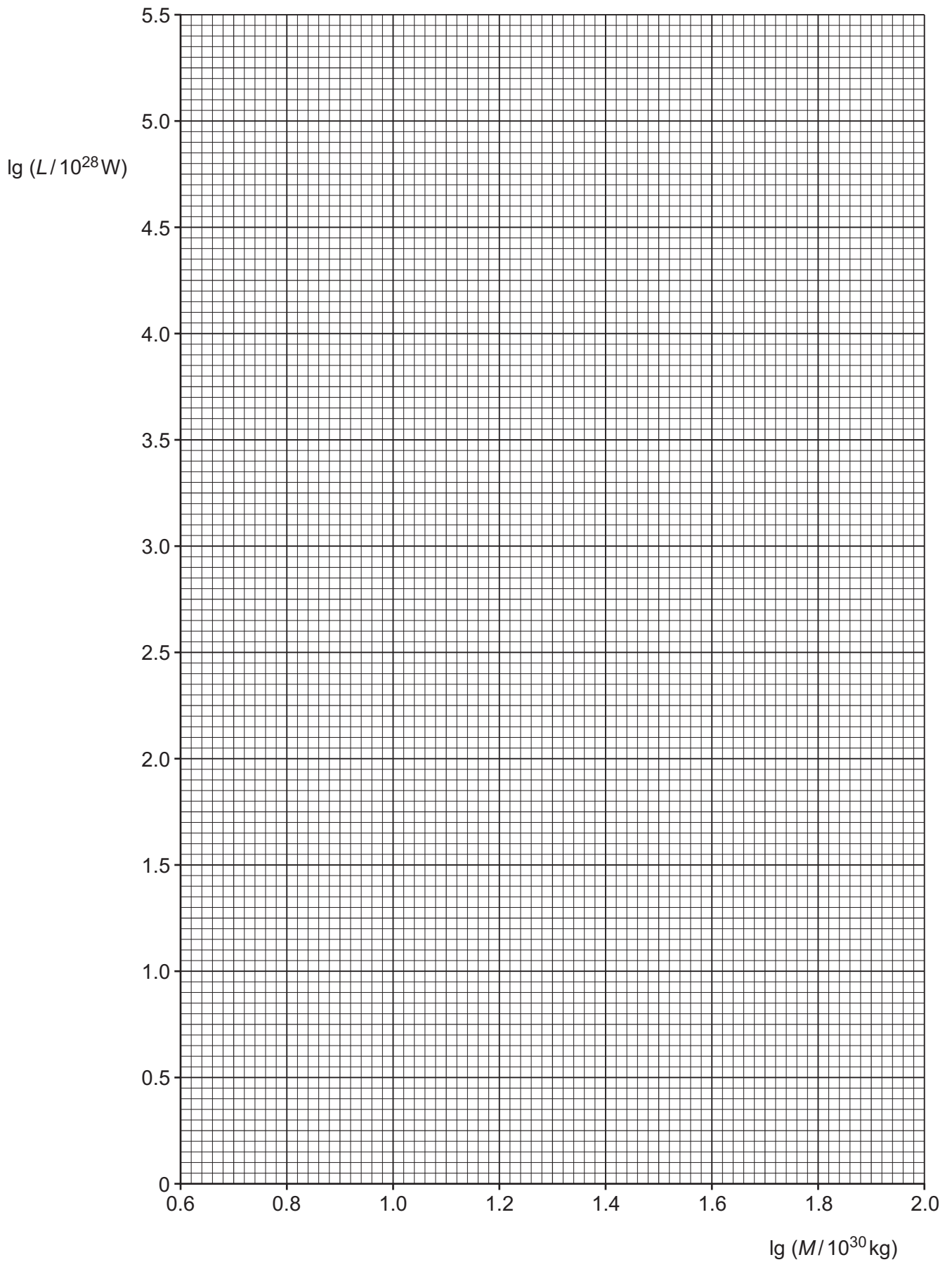
[2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines.

[2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]



- (iv) Determine the y -intercept of the line of best fit. Include the absolute uncertainty in your answer.

y -intercept = [2]

- (d) Using your answers to (a), (c)(iii) and (c)(iv), determine the values of n and Z . Include the absolute uncertainties in your values. You need not be concerned with units.

Data: $S = 3.85 \times 10^{26} \text{ W}$

$n = \dots\dots\dots$

$Z = \dots\dots\dots$ [3]

- (e) Another main-sequence star has a mass of $3.0 \times 10^{30} \text{ kg}$.

Determine the luminosity L of this star.

$L = \dots\dots\dots \text{ W}$ [1]

[Total: 15]



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9702/52

May/June 2022

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 30.
- The number of marks for each question or part question is shown in brackets [].

This document has 8 pages.

- 1 Two parallel cylindrical conductors each have a small cross-sectional area A . A thin metal bar connects the two conductors, as shown in Fig. 1.1.

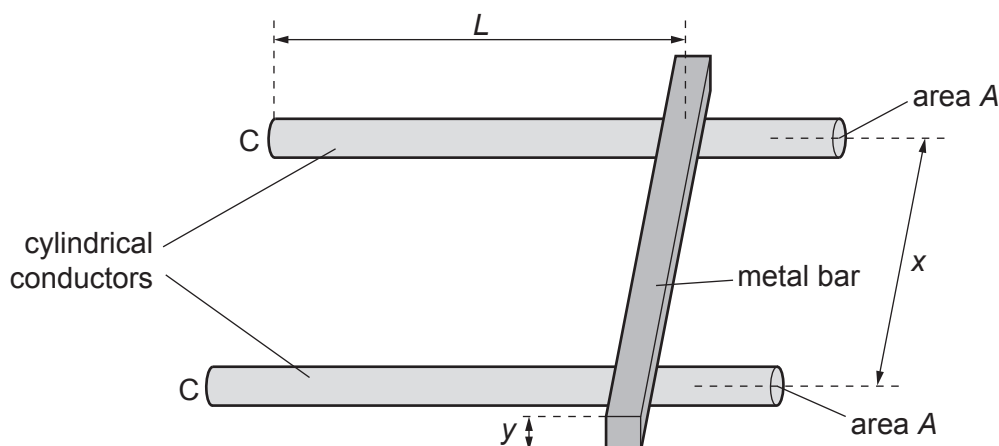


Fig. 1.1 (not to scale)

The metal bar has a square cross-section with sides of length y . For each conductor, the distance between its end C and the centre of the metal bar is L . The distance between the centres of the conductors is x .

The ends C are connected to a power supply and the current I in the conductors is measured.

It is suggested that I is related to L by the relationship

$$\frac{E}{I} = \frac{2PL}{A} + \frac{Qx}{y^2}$$

where E is the electromotive force (e.m.f.) of the power supply, and P and Q are constants.

Plan a laboratory experiment to test the relationship between I and L .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for P and Q .

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

Diagram

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[illegible]

- 2 The brightness of some stars varies regularly. These stars are called variable stars.

Fig. 2.1 shows the variation of luminosity with time for a variable star.

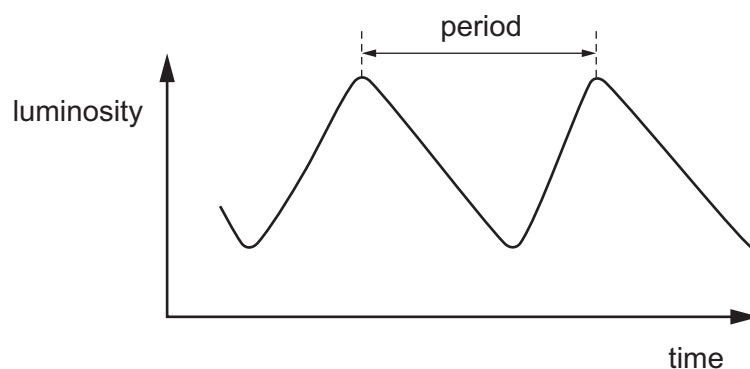


Fig. 2.1

A student determines the period T and mean luminosity L of the star.

The student repeats the process for different variable stars.

It is suggested that L and T are related by the equation

$$L = SKT^a$$

where S is the luminosity of the Sun, and a and K are constants.

- (a) A graph is plotted of $\lg L$ on the y -axis against $\lg T$ on the x -axis.

Determine expressions for the gradient and y -intercept.

gradient =

y -intercept =

[1]

(b) Values of T and L are given in Table 2.1.

Table 2.1

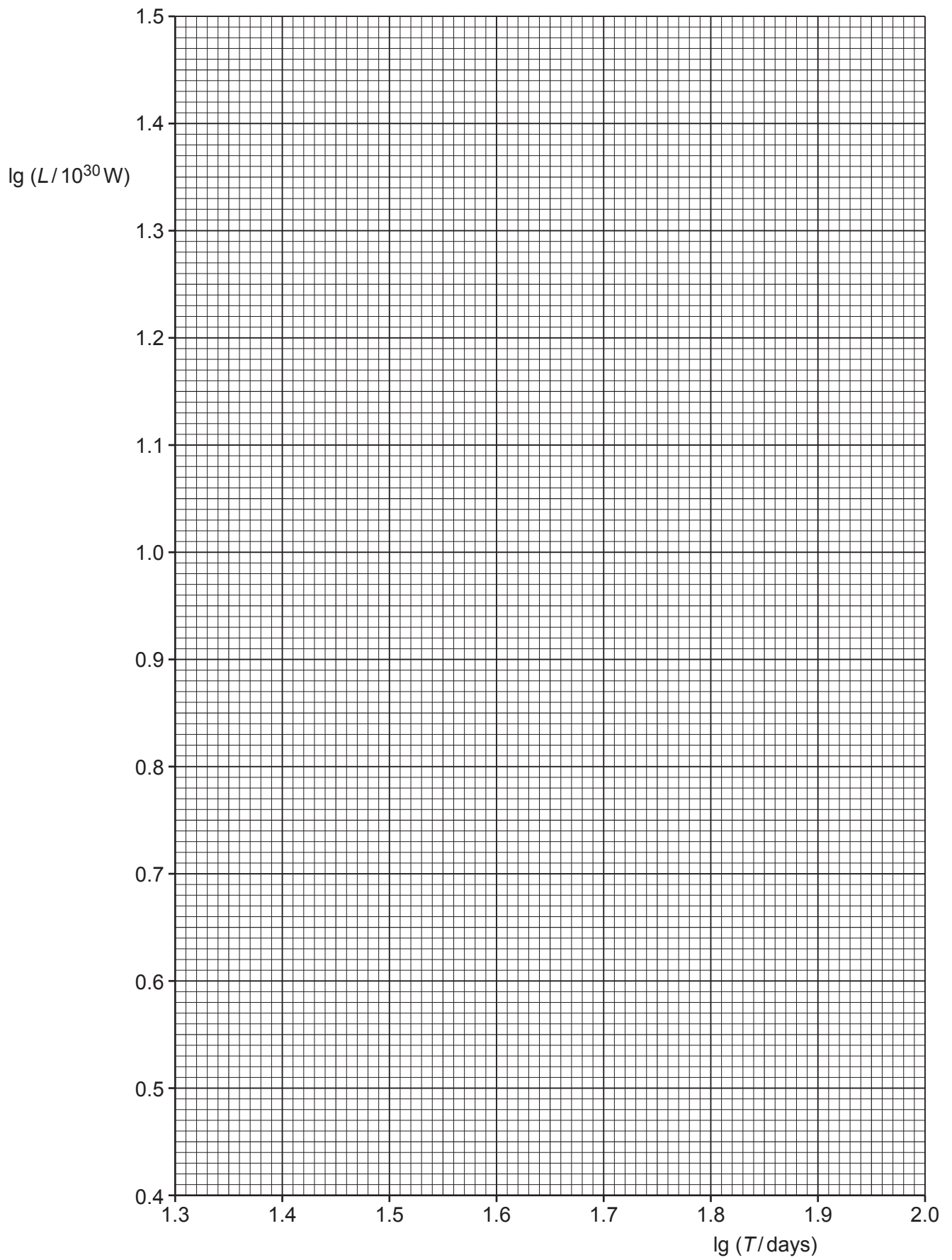
T/days	$L/10^{30}W$	$\lg(T/\text{days})$	$\lg(L/10^{30}W)$
22	2.9 ± 0.2		
32	4.9 ± 0.2		
42	6.9 ± 0.2		
54	9.8 ± 0.2		
78	16 ± 2		
97	21 ± 2		

Calculate and record values of $\lg(T/\text{days})$ and $\lg(L/10^{30}W)$ in Table 2.1.
Include the absolute uncertainties in $\lg(L/10^{30}W)$.

[2]

- (c) (i) Plot a graph of $\lg(L/10^{30}W)$ against $\lg(T/\text{days})$.
Include error bars for $\lg(L/10^{30}W)$. [2]
- (ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines. [2]
- (iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]



- (iv) Determine the y -intercept of the line of best fit. Include the absolute uncertainty in your answer.

y -intercept = [2]

- (d) Using your answers to (a), (c)(iii) and (c)(iv), determine the values of a and K . Include the absolute uncertainties in your values. You need not be concerned with units.

Data: $S = 3.85 \times 10^{26} \text{ W}$

$a =$

$K =$

[3]

- (e) A variable star has a period of 5.0 days.

Determine the luminosity L of this star.

$L =$ W [1]

[Total: 15]



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9702/53

May/June 2022

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
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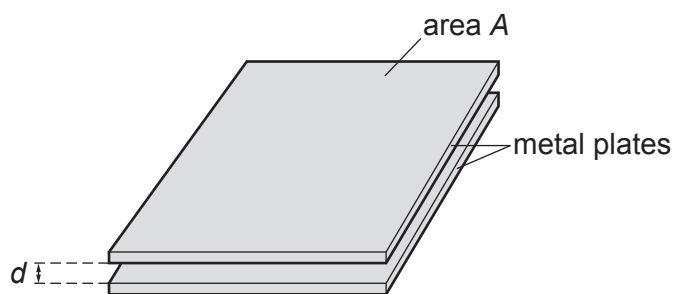


Fig. 1.1 (not to scale)

The plates are initially charged using a power supply.

The plates are then connected to an uncharged capacitor. The potential difference V across the capacitor is measured.

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where C is the capacitance of the capacitor, and K and W are constants.

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Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for K and W .

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- the analysis of the data
- any safety precautions to be taken.

Diagram

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It is suggested that L and M are related by the equation

$$L = SZM^n$$

where S is the luminosity of the Sun, and Z and n are constants.

- (a) A graph is plotted of $\lg L$ on the y -axis against $\lg M$ on the x -axis.

Determine expressions for the gradient and y -intercept.

gradient =

y -intercept =

[1]

(b) Values of M and L are given in Table 2.1.

Table 2.1

$M/10^{30} \text{ kg}$	$L/10^{28} \text{ W}$	$\lg (M/10^{30} \text{ kg})$	$\lg (L/10^{28} \text{ W})$
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Include the absolute uncertainties in $\lg (M/10^{30} \text{ kg})$.

[2]

(c) (i) Plot a graph of $\lg (L/10^{28} \text{ W})$ against $\lg (M/10^{30} \text{ kg})$.
Include error bars for $\lg (M/10^{30} \text{ kg})$.

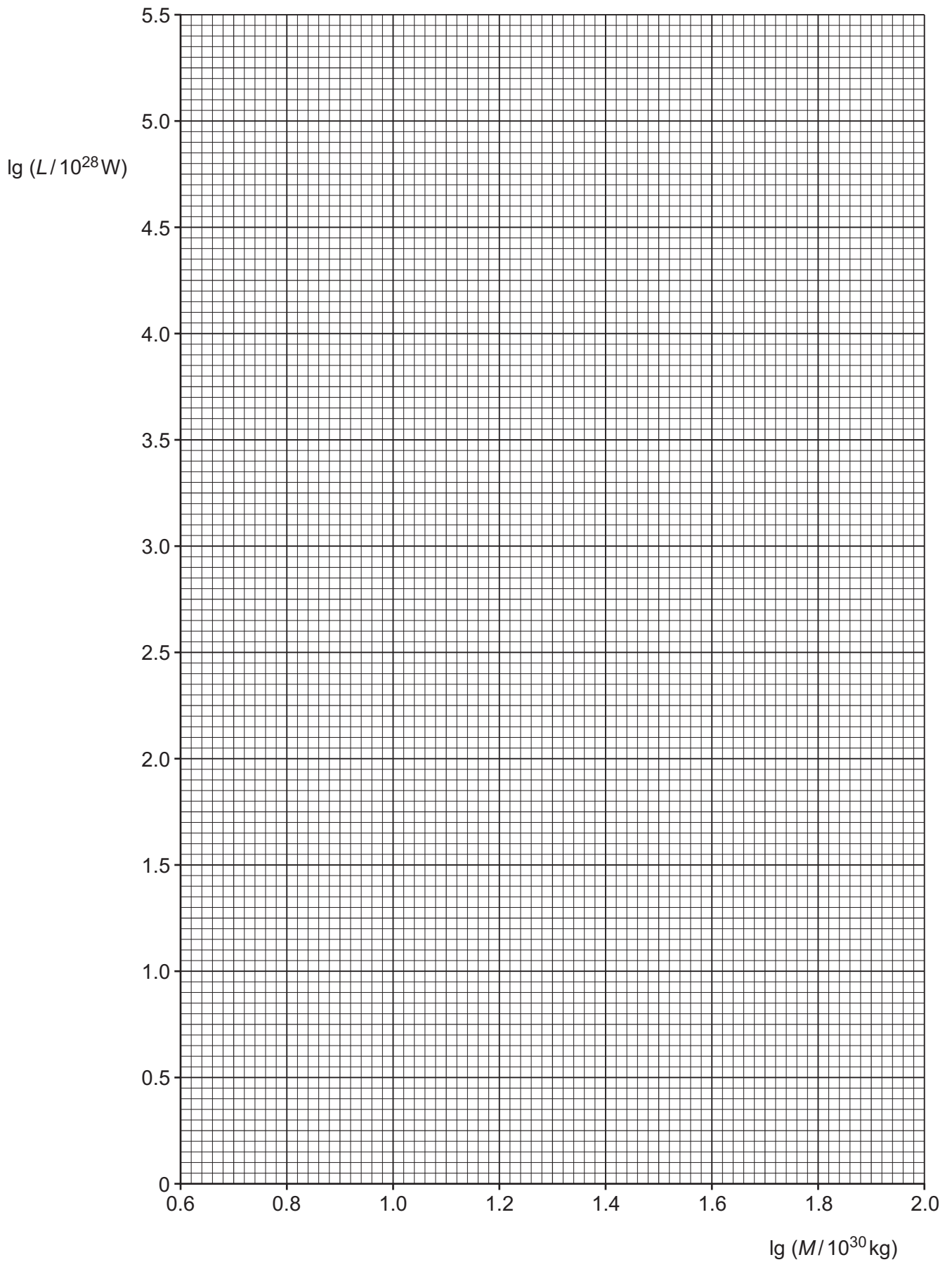
[2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines.

[2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]



- (iv) Determine the y -intercept of the line of best fit. Include the absolute uncertainty in your answer.

y -intercept = [2]

- (d) Using your answers to (a), (c)(iii) and (c)(iv), determine the values of n and Z . Include the absolute uncertainties in your values. You need not be concerned with units.

Data: $S = 3.85 \times 10^{26} \text{ W}$

$n = \dots\dots\dots$

$Z = \dots\dots\dots$

[3]

- (e) Another main-sequence star has a mass of $3.0 \times 10^{30} \text{ kg}$.

Determine the luminosity L of this star.

$L = \dots\dots\dots \text{ W}$ [1]

[Total: 15]



Cambridge International AS & A Level

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PHYSICS

9702/41

Paper 4 A Level Structured Questions

October/November 2022

2 hours

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [].

This document has **24** pages. Any blank pages are indicated.

Data

acceleration of free fall	$g = 9.81 \text{ m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
unified atomic mass unit	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27} \text{ kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$ $(\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ m F}^{-1})$
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_o = \frac{f_s v}{v \mp v_s}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \dots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$

gravitational potential	$\phi = -\frac{GM}{r}$
gravitational potential energy	$E_P = -\frac{GMm}{r}$
pressure of an ideal gas	$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$
simple harmonic motion	$a = -\omega^2 x$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t$ $v = \omega \sqrt{(x_0^2 - x^2)}$
electric potential	$V = \frac{Q}{4\pi\epsilon_0 r}$
electrical potential energy	$E_P = \frac{Qq}{4\pi\epsilon_0 r}$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$
capacitors in parallel	$C = C_1 + C_2 + \dots$
discharge of a capacitor	$x = x_0 e^{-\frac{t}{RC}}$
Hall voltage	$V_H = \frac{BI}{ntq}$
alternating current/voltage	$x = x_0 \sin \omega t$
radioactive decay	$x = x_0 e^{-\lambda t}$
decay constant	$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$
intensity reflection coefficient	$\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$
Stefan–Boltzmann law	$L = 4\pi\sigma r^2 T^4$
Doppler redshift	$\frac{\Delta\lambda}{\lambda} \hat{=} \frac{Df}{f} \hat{=} \frac{v}{c}$

- 1 (a) State the equation for the gravitational force F between two point masses m_1 and m_2 that are separated by a distance r . State the meaning of any other symbols you use.

[2]

- (b) A satellite is in a circular orbit of radius R around a planet of mass M .

Show that the period T of the orbit is given by

$$T^2 = kR^3$$

where k is a constant that depends on the value of M . Explain your reasoning.

[3]

- (c) A satellite is in a circular orbit around the Earth with a period of 24 hours. The mass of the Earth is 6.0×10^{24} kg.

- (i) Calculate the radius of the orbit.

radius = m [2]

(ii) State the **two** other conditions that must be met for the orbit to be geostationary.

1

.....

2

.....

[2]

[Total: 9]

- 2 Fig. 2.1 shows a laboratory thermometer that is calibrated to measure temperature in degrees Celsius.

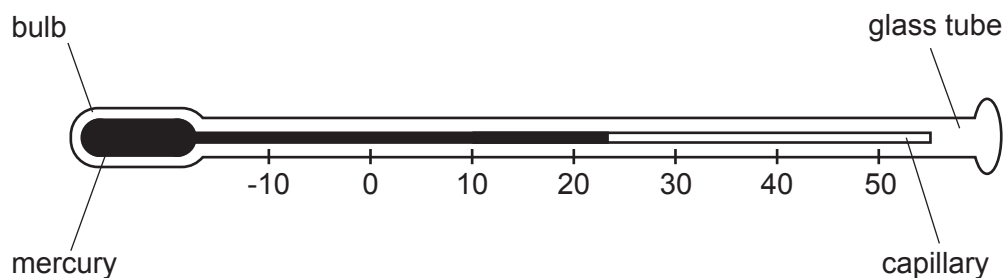


Fig. 2.1

The thermometer makes use of the fact that the density of mercury varies with temperature.

- (a) State **two** other physical properties of materials, apart from the density of a liquid, that can be used for measuring temperature.

1

2 [2]

- (b) The thermometer is initially at 23.0°C , as shown in Fig. 2.1. It is used to measure the temperature of an insulated beaker of water that is at 37.4°C . The bulb of the thermometer is inserted into the water, and the water is stirred until the reading on the thermometer becomes steady.

The mass of water in the beaker is 18.7 g.

The mass of mercury in the thermometer is 6.94 g.

The specific heat capacity of water is $4.18 \text{ J g}^{-1} \text{ K}^{-1}$.

The specific heat capacity of mercury is $0.140 \text{ J g}^{-1} \text{ K}^{-1}$.

The glass of the thermometer and the beaker containing the water can be considered to have negligible heat capacity.

- (i) Calculate, to three significant figures, the final steady temperature indicated by the thermometer in the water.

temperature = $^{\circ}\text{C}$ [4]

- (ii) Suggest **one** change that could be made to the design of the thermometer that would enable it to give a more accurate measurement of temperature.

.....
..... [1]

- (c) (i) Explain why the thermometer in Fig. 2.1 does **not** provide a direct measurement of thermodynamic temperature.

.....
.....
..... [2]

- (ii) Thermodynamic temperature T may be determined by the behaviour of a type of substance for which T is proportional to the product of pressure and volume.

State the name of this type of substance.

..... [1]

[Total: 10]

- 3 An object is suspended from a spring that is attached to a fixed point as shown in Fig. 3.1.

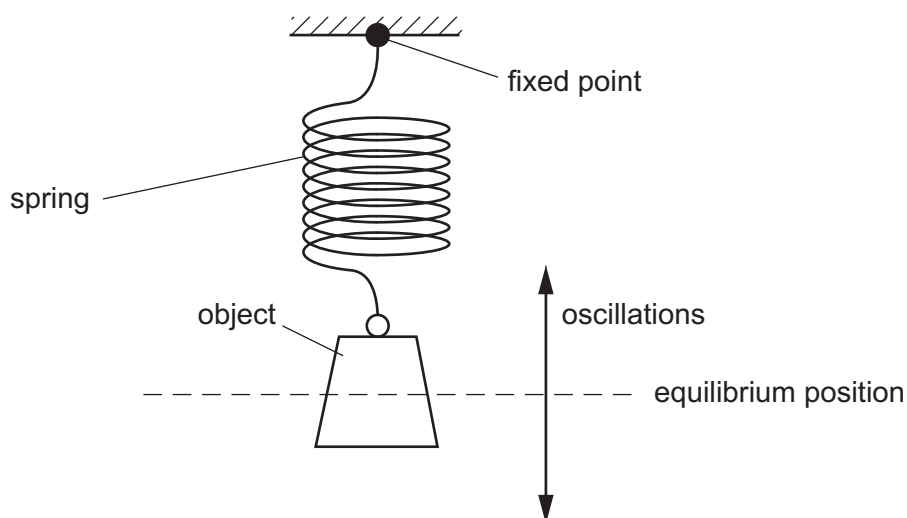


Fig. 3.1

The object oscillates vertically with simple harmonic motion about its equilibrium position.

- (a) State the defining equation for simple harmonic motion. Identify the meaning of each of the symbols used to represent physical quantities.

.....

 [2]

- (b) The variation with displacement x from the equilibrium position of the velocity v of the object is shown in Fig. 3.2.

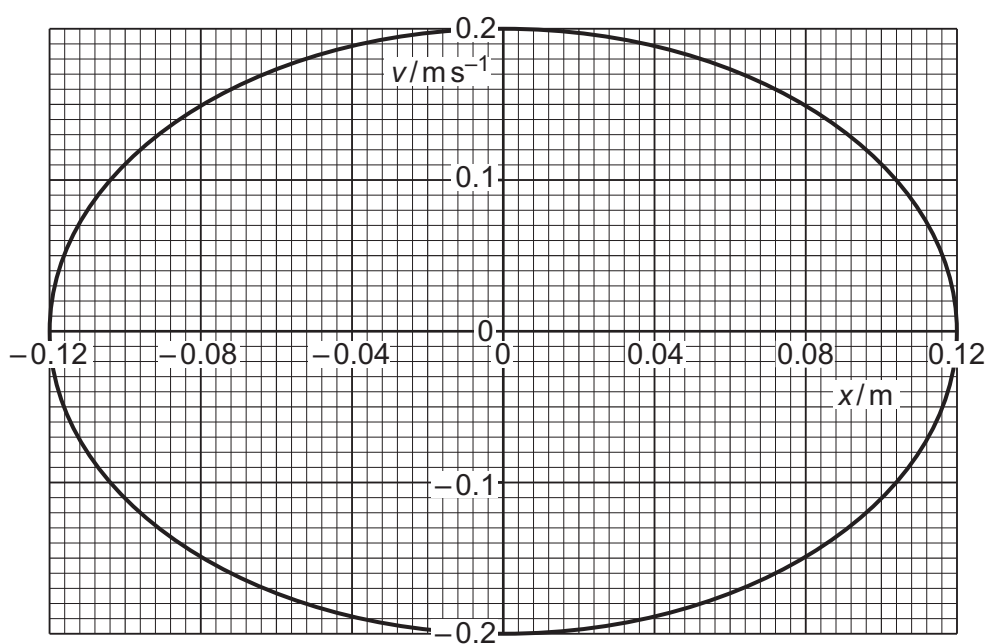


Fig. 3.2

The variation with x of the potential energy E_P of the oscillations of the object is shown in Fig. 3.3.

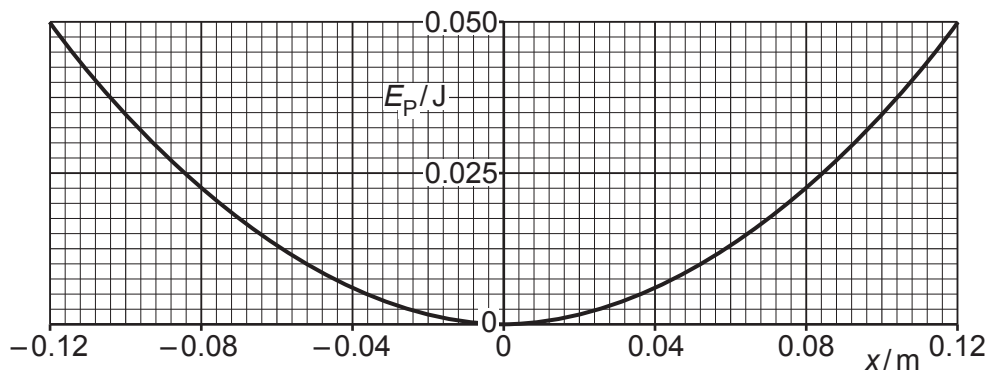


Fig. 3.3

Use Fig. 3.2 and Fig. 3.3 to:

- (i) determine the amplitude x_0 of the oscillations

$$x_0 = \dots\dots\dots \text{ m [1]}$$

- (ii) show that the angular frequency of the oscillations is 1.7 rad s^{-1}

[2]

- (iii) determine the mass M of the object.

$$M = \dots\dots\dots \text{ kg [2]}$$

(c) The oscillations of the object are now lightly damped.

(i) State what is meant by damping.

.....

.....

..... [2]

(ii) Assume that the damping does not change the angular frequency of the oscillations.

On Fig. 3.2, sketch the variation with x of v when the amplitude of the oscillations is 0.060 m. [2]

[Total: 11]

- 4 (a) State what is indicated by the direction of an electric field line.

.....
 [2]

- (b) Fig. 4.1 shows a pair of parallel metal plates with a potential difference (p.d.) of 2400 V between them.

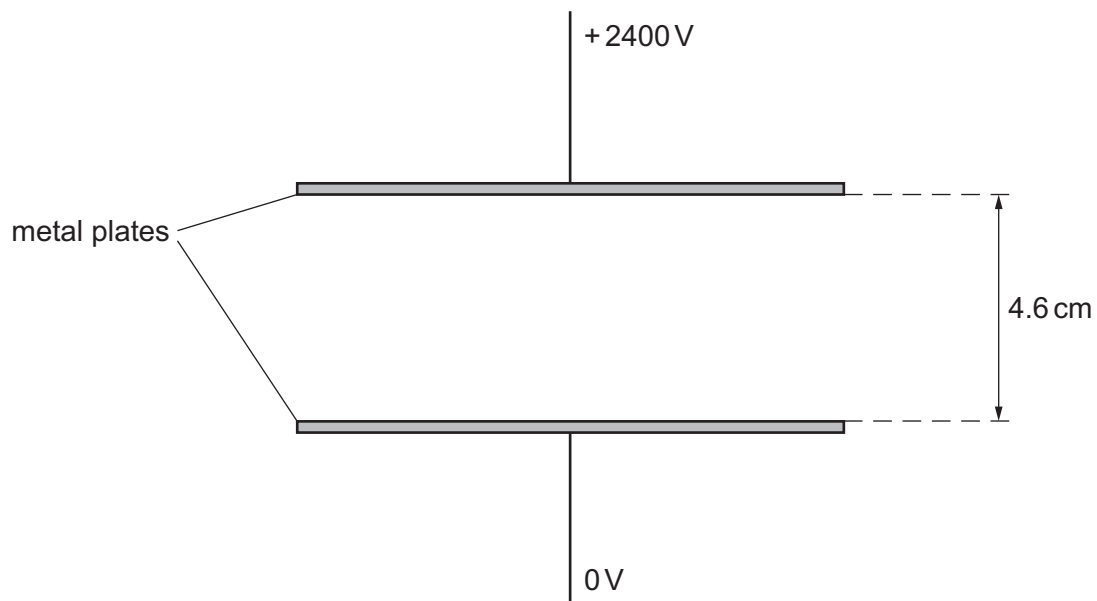


Fig. 4.1

The plates are separated by a distance of 4.6 cm. The plates are in a vacuum.

- (i) On Fig. 4.1, draw five lines to represent the electric field in the region between the plates. [3]
- (ii) Calculate the strength of the electric field between the plates.

electric field strength = NC^{-1} [2]

- (c) A moving proton enters the region between the plates from the left, as shown in Fig. 4.2.

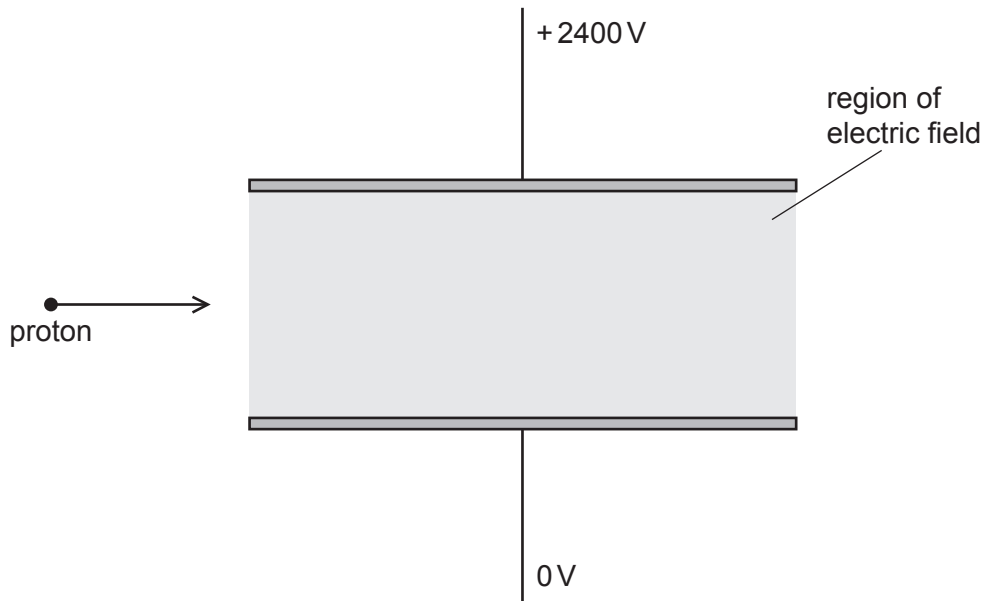


Fig. 4.2

- (i) The proton is deflected by the electric field.

On Fig. 4.2, draw a line to show the path of the proton as it moves through and out of the region of the electric field. [2]

- (ii) A helium nucleus (${}^4_2\text{He}$) now enters the region of the electric field along the same initial path as the proton and travelling at the same initial speed.

State and explain how the final speed of the helium nucleus compares with the final speed of the proton after leaving the region of the electric field.

.....

.....

.....

.....

..... [3]

[Total: 12]

- 5 A capacitor of capacitance $470\,\mu\text{F}$ is connected to a battery of electromotive force (e.m.f.) $24\,\text{V}$ in the circuit of Fig. 5.1.

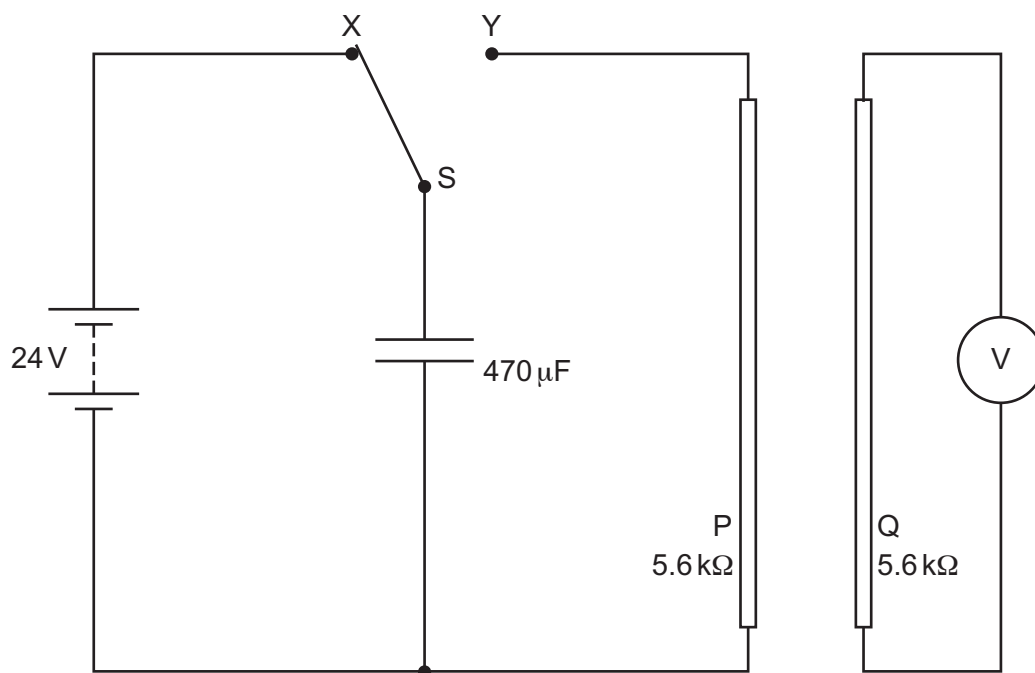


Fig. 5.1

The two-way switch S is initially at position X.

P and Q are identical long straight wires, each with a resistance of $5.6\,\text{k}\Omega$. These wires are placed near to, and parallel to, each other. Wire Q is connected to a voltmeter.

At time $t = 0$, switch S is moved to position Y so that the capacitor discharges through wire P.

- (a) (i) Calculate the charge Q_0 on the capacitor at time $t = 0$.

$$Q_0 = \dots\dots\dots \text{ C [2]}$$

- (ii) Calculate the current I_0 in wire P at time $t = 0$.

$$I_0 = \dots\dots\dots \text{ A [1]}$$

- (iii) Calculate the time constant τ of the discharge circuit.

$\tau = \dots\dots\dots$ s [2]

- (iv) On Fig. 5.2, sketch a line to show the variation with t of the current I in wire P as the capacitor discharges.



Fig. 5.2

[2]

- (b) (i) Explain why there is an induced e.m.f. across wire Q during the discharge of the capacitor.

.....

 [3]

- (ii) On Fig. 5.3, sketch a line to suggest the variation with t of the voltmeter reading V .



Fig. 5.3

[1]

[Total: 11]

- 6 Fig. 6.1 shows a thin slice of semiconducting material used in a Hall probe.

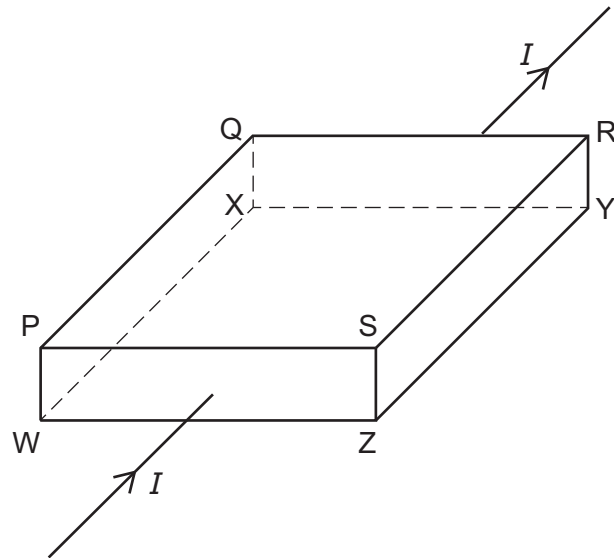


Fig. 6.1 (not to scale)

Current I passes through the slice in the direction shown.

The slice is placed in a uniform magnetic field of flux density B , so that two of its faces are perpendicular to the magnetic field.

A steady Hall voltage V_H is developed between face PQXW and face SRYZ.

- (a) (i) Use the letters in Fig. 6.1 to identify the faces that are perpendicular to the magnetic field.

..... and [1]

- (ii) Explain how the steady Hall voltage V_H is developed between faces PQXW and SRYZ.

.....

 [3]

(b) The magnitude of V_H is given by the equation

$$V_H = \frac{BI}{ntq}.$$

(i) State the meaning of the symbols n , t and q . You may refer to the letters in Fig. 6.1.

n :

t :

q : [3]

(ii) Suggest, with reference to the equation, why the slice of the material used in a Hall probe is thin.

.....

.....

..... [2]

[Total: 9]

- 7 (a) A sinusoidal alternating voltage has a root-mean-square (r.m.s.) potential difference (p.d.) of 4.2 V and a frequency of 50 kHz.

- (i) The alternating voltage is applied across a resistor of resistance $760\ \Omega$.

By considering the peak voltage, show that the maximum power dissipated by the resistor is 46 mW.

[2]

- (ii) On Fig. 7.1, draw a smooth curve to show how the power P dissipated in the resistor varies with time t between $t = 0$ and $t = 40\ \mu\text{s}$. Assume that $P = 0$ when $t = 0$.

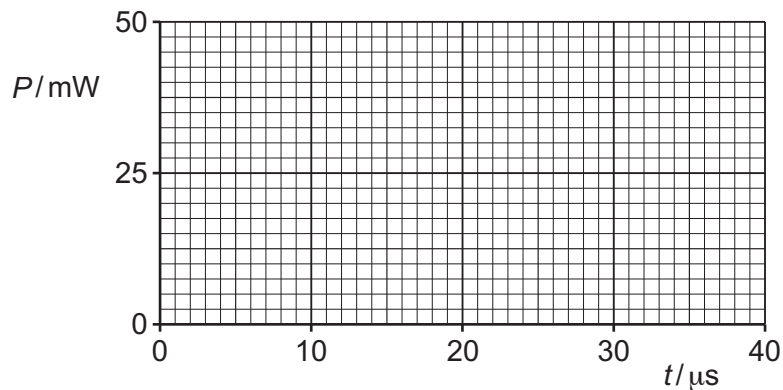


Fig. 7.1

[3]

- (iii) Use your line in (a)(ii) to explain why the mean power dissipated in the resistor is 23 mW.

.....

 [1]

(b) The alternating voltage in **(a)** is now applied to a piezoelectric crystal in air.

(i) Explain what happens to the air surrounding the crystal.

.....

.....

.....

..... [3]

(ii) A second piezoelectric crystal is placed in the air near to the first crystal.

Explain the effect of the surrounding air in **(b)(i)** on the second crystal.

..... [1]

[Total: 10]

- 8 (a) State what is meant by the work function energy of a metal.

.....

.....

..... [2]

- (b) Ultraviolet radiation of frequency $1.36 \times 10^{15} \text{ Hz}$ is incident, in a vacuum, on a metal surface. The power of the radiation incident on the surface is 8.36 mW . Photoelectrons are emitted with a maximum kinetic energy of $3.09 \times 10^{-19} \text{ J}$.

- (i) Determine the number of photons incident on the surface per unit time.

number per unit time = s^{-1} [2]

- (ii) Calculate the work function energy Φ of the metal.

$\Phi = \dots\dots\dots \text{ J}$ [2]

- (c) The frequency of the radiation incident on the surface in (b) is increased while the power remains constant.

State and explain the effect of this change on:

- (i) the maximum kinetic energy of the photoelectrons

.....

.....

..... [2]

- (ii) the rate of emission of photoelectrons.

.....

.....

..... [2]

[Total: 10]

- 9 (a) State what is meant by the luminosity of a star.

..... [1]

- (b) A star in the constellation Canis Major is a distance of 8.14×10^{16} m from the Earth and has a luminosity of 9.86×10^{27} W. The surface temperature of the star is 9830 K.

- (i) Calculate the radiant flux intensity of the radiation from the star observed from the Earth. Give a unit with your answer.

radiant flux intensity = unit [2]

- (ii) Determine the radius of the star.

radius = m [2]

- (c) Explain how the surface temperature of a distant star may be determined from the wavelength spectrum of the light from the star.

.....

 [3]

[Total: 8]

- 10 Carbon-15 ($^{15}_6\text{C}$) is an isotope of carbon that undergoes radioactive decay to nitrogen-15 ($^{15}_7\text{N}$), which is a stable isotope of nitrogen.

Radioactive decay is both a random and a spontaneous process.

(a) State what is meant by:

(i) random

.....
 [1]

(ii) spontaneous.

.....
 [1]

- (b) A small sample of carbon-15 decays. The mass M of carbon-15 in the sample decreases with time t .

Fig. 10.1 shows the variation with t of the value of $\ln (M/10^{-16}\text{g})$.

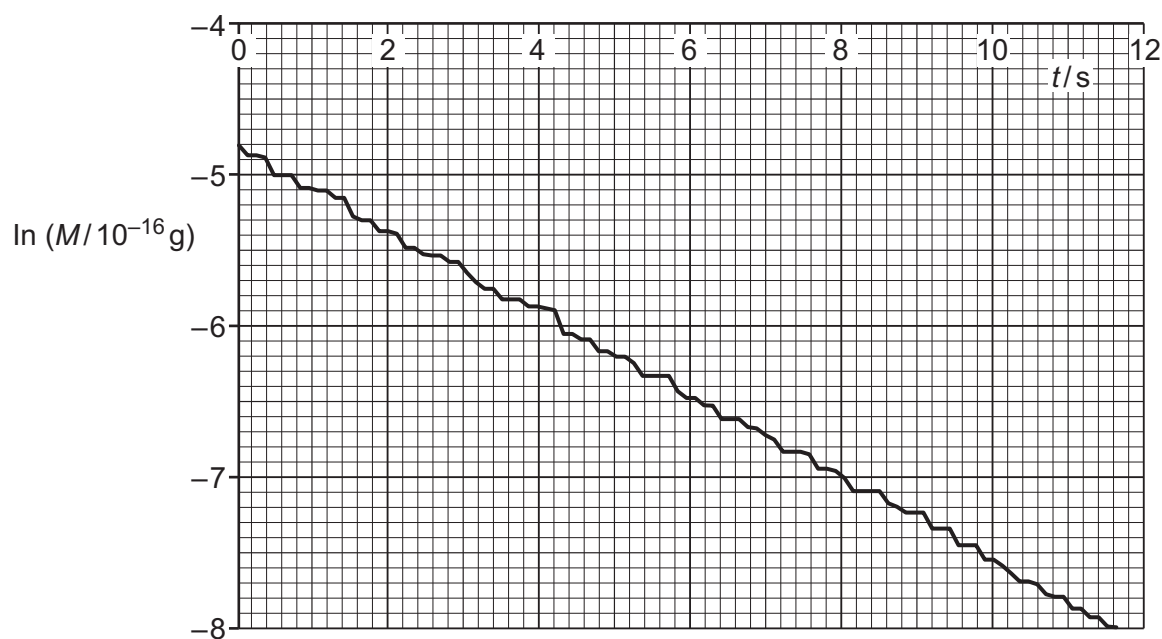


Fig. 10.1

- (i) State how Fig. 10.1 demonstrates that radioactive decay is random.

.....
 [1]

- (ii) On Fig. 10.1, draw the straight line of best fit.

[1]

- (iii) Show that the decay constant λ of carbon-15 is given by the magnitude of the gradient of your line in (b)(ii).

[1]

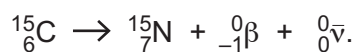
- (iv) Use your line in (b)(ii) to determine λ . Give a unit with your answer.

$\lambda = \dots\dots\dots$ unit $\dots\dots\dots$ [2]

- (v) Use your answer in (b)(iv) to calculate the half-life of carbon-15.

half-life = $\dots\dots\dots$ s [1]

- (c) The equation for the decay of carbon-15 can be written as



State and explain how the mass of the products of the decay must compare with the mass of the carbon-15 nucleus.

.....

 [2]

[Total: 10]

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9702/42

October/November 2022

2 hours

You must answer on the question paper.

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [].

This document has **24** pages. Any blank pages are indicated.

Data

acceleration of free fall	$g = 9.81 \text{ m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
unified atomic mass unit	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27} \text{ kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$ ($\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ m F}^{-1}$)
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_o = \frac{f_s v}{v \pm v_s}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \dots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$

gravitational potential	$\phi = -\frac{GM}{r}$
gravitational potential energy	$E_P = -\frac{GMm}{r}$
pressure of an ideal gas	$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$
simple harmonic motion	$a = -\omega^2 x$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t$ $v = \pm \omega \sqrt{(x_0^2 - x^2)}$
electric potential	$V = \frac{Q}{4\pi\epsilon_0 r}$
electrical potential energy	$E_P = \frac{Qq}{4\pi\epsilon_0 r}$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$
capacitors in parallel	$C = C_1 + C_2 + \dots$
discharge of a capacitor	$x = x_0 e^{-\frac{t}{RC}}$
Hall voltage	$V_H = \frac{BI}{ntq}$
alternating current/voltage	$x = x_0 \sin \omega t$
radioactive decay	$x = x_0 e^{-\lambda t}$
decay constant	$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$
intensity reflection coefficient	$\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$
Stefan–Boltzmann law	$L = 4\pi\sigma r^2 T^4$
Doppler redshift	$\frac{\Delta\lambda}{\lambda} \hat{=} \frac{Df}{f} \hat{=} \frac{v}{c}$

- 1 (a) Define gravitational field.

.....
 [1]

- (b) A spherical planet can be considered as a point mass at its centre.

- (i) On Fig. 1.1, draw gravitational field lines outside the planet to represent the gravitational field due to the planet.

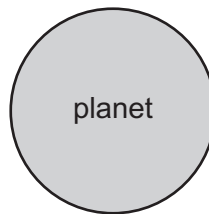


Fig. 1.1

[2]

- (ii) A satellite is in a circular orbit around the planet.

Explain, with reference to your answer in (b)(i), why the path of the satellite is circular.

.....

 [2]

- (c) An object rests on the surface of the Earth at the Equator.
The radius of the Earth is $6.4 \times 10^6 \text{ m}$.
- (i) Determine the centripetal acceleration of the object.

centripetal acceleration = ms^{-2} [3]

- (ii) Describe how the two forces acting on the object give rise to this centripetal acceleration.
You may draw a diagram if you wish.

.....
.....
..... [2]

[Total: 10]

- 2 (a) Define specific heat capacity.

.....

.....

..... [2]

- (b) A fixed mass of water in a beaker is at atmospheric pressure.

- (i) The initial temperature of the water is 0°C .

The water is supplied with thermal energy E , so that its temperature increases to 8°C . There is no net change in the volume of the water.

Use the first law of thermodynamics to complete Table 2.1 for this process.

Table 2.1

work done on water	thermal energy supplied to water	increase in internal energy of water
	$+ E$	

[2]

- (ii) The water is now heated so that its temperature increases by a further 8°C to a final temperature of 16°C . This process causes the volume of the water to increase so that work W is done.

Assume that the change in internal energy is the same as in (b)(i).

Use the first law of thermodynamics to complete Table 2.2 for this process.

Table 2.2

work done on water	thermal energy supplied to water	increase in internal energy of water

[2]

- (c) Use the information in (b) to suggest, with a reason, how the average specific heat capacity of water between 8°C and 16°C compares with its average value between 0°C and 8°C .

.....

..... [1]

[Total: 7]

- 3 (a) The equation of state for an ideal gas can be written as

$$pV = NkT.$$

State the meaning of each of the symbols in this equation.

p :

V :

N :

k :

T : [3]

- (b) Use the equation in (a) to show that the average translational kinetic energy E_K of a molecule of an ideal gas is given by

$$E_K = \frac{3}{2}kT.$$

[2]

- (c) The mass of an oxygen molecule is 5.31×10^{-26} kg. Assume that oxygen behaves as an ideal gas.

- (i) Use the equation in (b) to determine the root-mean-square (r.m.s.) speed u of an oxygen molecule at 23°C .

$u = \dots\dots\dots \text{ms}^{-1}$ [3]

- (ii) A fixed mass of oxygen gas at initial pressure P is sealed in a cylindrical container by a movable piston at one end, as shown in Fig. 3.1.

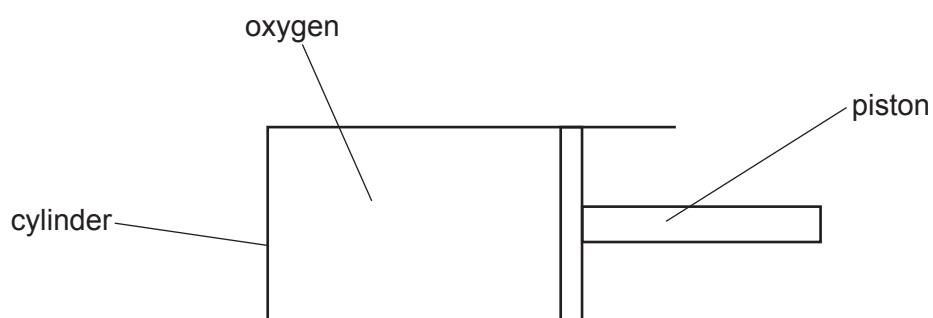


Fig. 3.1

The temperature of the gas is 23°C .

The piston is slowly moved into the cylinder so that the oxygen gas is compressed. At all times, the gas and the container remain in thermal equilibrium with the surroundings.

On Fig. 3.2, sketch the variation with pressure of the r.m.s. speed of the oxygen molecules as the pressure increases.



Fig. 3.2

[2]

[Total: 10]

- 4 Fig. 4.1 shows the variation with time t of the height h above the ground of an object of mass 36 kg that is undergoing vertical simple harmonic motion.

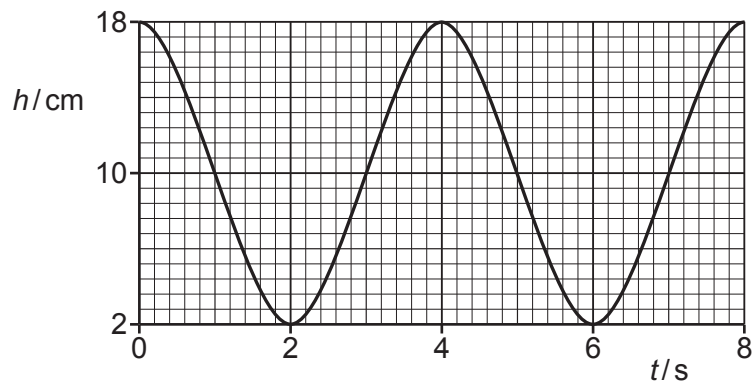


Fig. 4.1

- (a) For the oscillations of the object:

- (i) determine the amplitude x_0 , in cm

$$x_0 = \dots\dots\dots \text{ cm [1]}$$

- (ii) show that the angular frequency ω is 1.6 rad s^{-1}

[2]

- (iii) determine the total energy E .

$$E = \dots\dots\dots \text{ J [3]}$$

(b) On Fig. 4.2, sketch the variation with h of the kinetic energy E_K of the object.

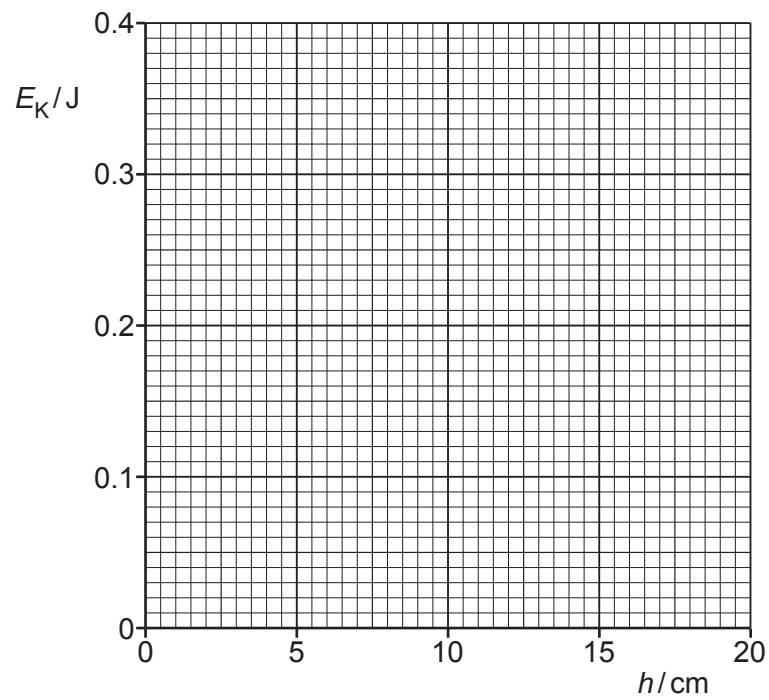


Fig. 4.2

[4]

[Total: 10]

- 5 (a) Define electric potential at a point.

.....

 [2]

- (b) An isolated conducting sphere is charged. Fig. 5.1 shows the variation of the potential V due to the sphere with displacement x from its centre.

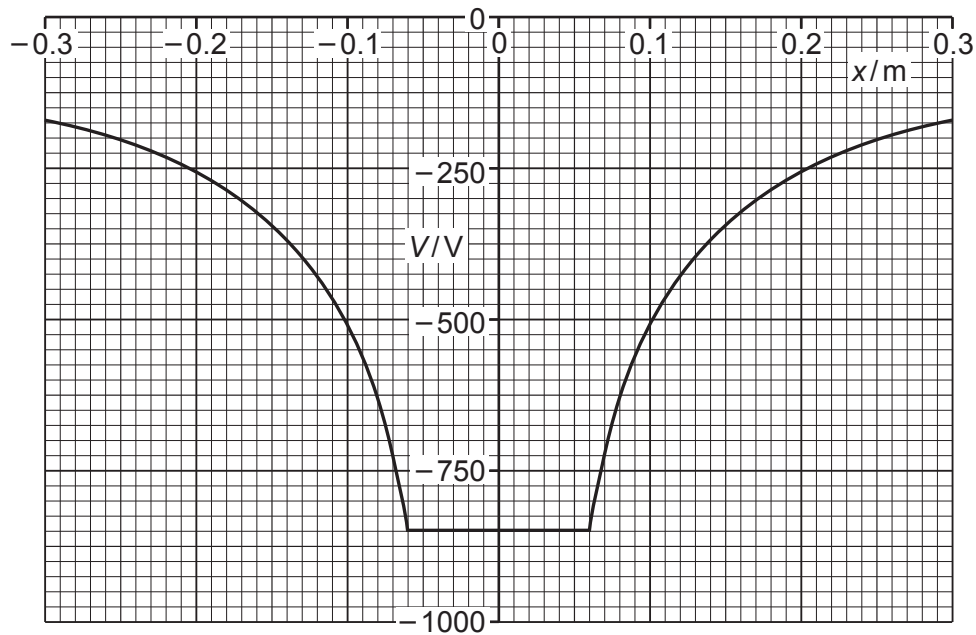


Fig. 5.1

Use Fig. 5.1 to determine:

- (i) the radius of the sphere

radius = m [1]

- (ii) the charge on the sphere.

charge = C [2]

- (c) Two spheres are identical to the sphere in (b). Each sphere has the same charge as the sphere in (b).

The spheres are held in a vacuum so that their centres are separated by a distance of 0.46 m. Assume that the charge on each sphere is a point charge at the centre of the sphere.

- (i) Calculate the electric potential energy E_p of the two spheres.

$$E_p = \dots\dots\dots \text{ J [2]}$$

- (ii) The two spheres are now released simultaneously so that they are free to move.

Describe and explain the subsequent motion of the spheres.

.....

.....

.....

..... [3]

[Total: 10]

- 6 A capacitor of capacitance C and a resistor of resistance R are connected as shown in Fig. 6.1.

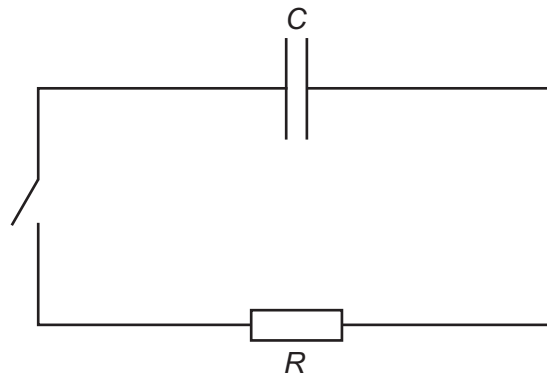


Fig. 6.1

Initially, the capacitor is charged and the switch is open.

The switch is closed at time $t = 0$.

Fig. 6.2 and Fig. 6.3 show, respectively, the variations with t of the charge Q on the capacitor and the potential difference (p.d.) V across the resistor.

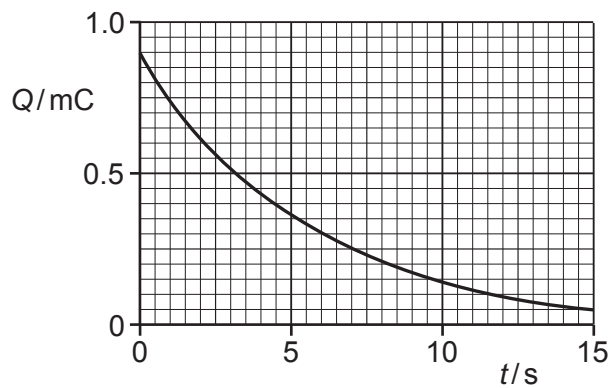


Fig. 6.2

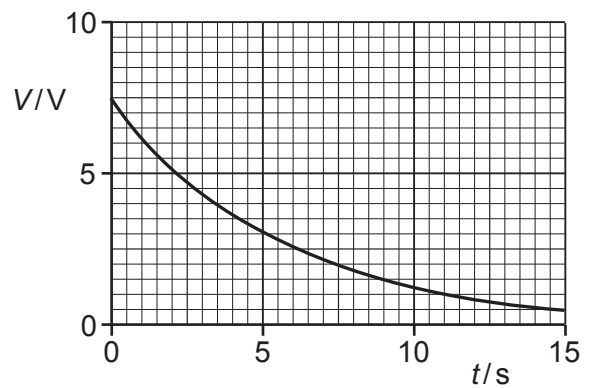


Fig. 6.3

- (a) Explain the shape of the line in Fig. 6.3 representing the variation of V with t .

.....

.....

.....

.....

..... [3]

(b) Use Fig. 6.2 to show that the time constant of the circuit in Fig. 6.1 is 5.5 s.

[3]

(c) Use Fig. 6.2, Fig. 6.3 and the information in (b) to determine:

(i) capacitance C , in μF

$C = \dots\dots\dots \mu\text{F}$ [2]

(ii) resistance R , in $\text{k}\Omega$.

$R = \dots\dots\dots \text{k}\Omega$ [2]

[Total: 10]

- 7 (a) Define magnetic flux density.

.....

.....

.....

..... [3]

- (b) An insulated rectangular coil of wire, consisting of 40 turns, is suspended in a cradle from a newton meter, as shown in Fig. 7.1.

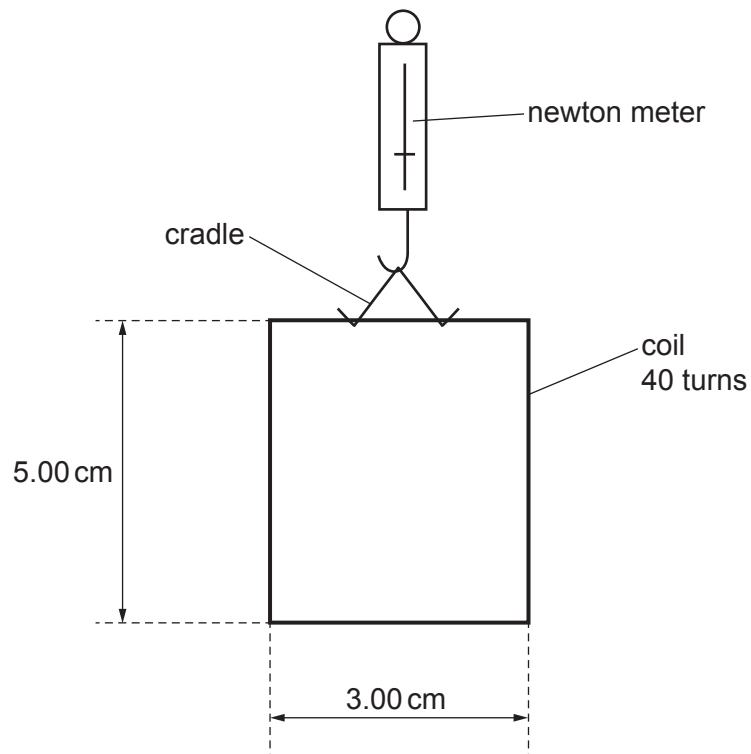


Fig. 7.1

The vertical sides of the coil have a length of 5.00 cm and the horizontal sides have a length of 3.00 cm. The initial reading on the newton meter is 0.563 N.

A U-shaped magnet rests on a top-pan balance that is set to a reading of 0.00 g. The lower edge of the coil is lowered into the region between the poles of the U-shaped magnet, as shown in the side view in Fig. 7.2.

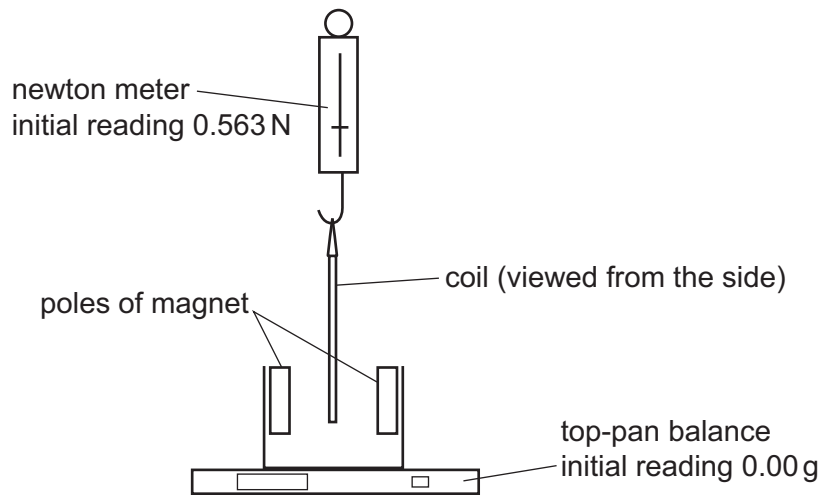


Fig. 7.2

The magnetic field in the region between the poles is uniform.
The lower edge of the coil is entirely within the uniform magnetic field.

A current of 3.94 A is now passed through the coil. This causes the reading on the top-pan balance to change to 2.16 g.

- (i) Explain why the current causes a vertical force to act on the coil.

.....

 [2]

- (ii) Determine, to three significant figures, the flux density B of the uniform magnetic field.

$B =$ T [3]

- (iii) Determine what is now the reading on the newton meter. Explain your reasoning.

reading = N [2]

[Total: 10]

[Turn over]

- 8 (a) State Lenz's law of electromagnetic induction.

.....

.....

..... [2]

- (b) Two coils of insulated wire are wound on an iron bar, as shown in Fig. 8.1.

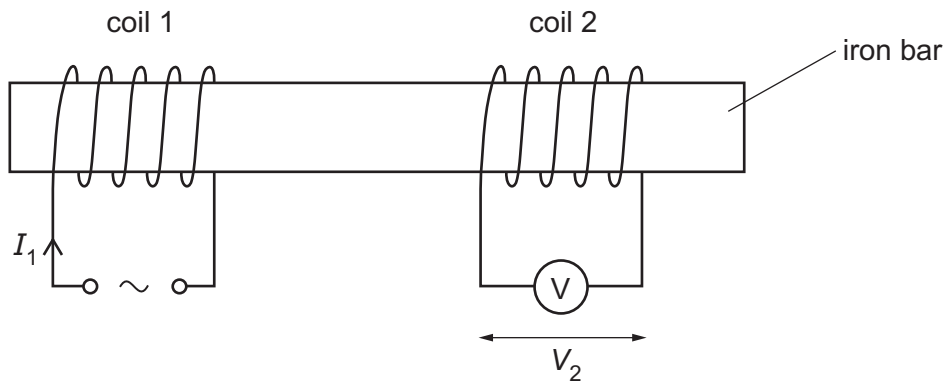


Fig. 8.1

There is a current I_1 in coil 1 that varies with time t as shown in Fig. 8.2.

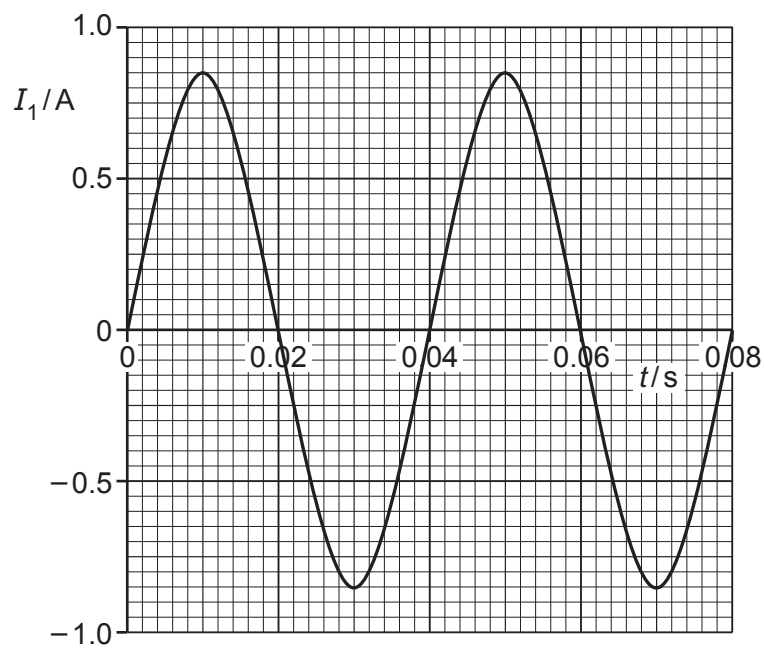


Fig. 8.2

- (i) The variation with t of I_1 can be represented by the equation

$$I_1 = X \sin Yt$$

where X and Y are constants.

Use Fig. 8.2 to determine the values of X and Y . Give units with your answers.

$X = \dots\dots\dots$ unit $\dots\dots\dots$

$Y = \dots\dots\dots$ unit $\dots\dots\dots$ [3]

- (ii) The current in coil 1 gives rise to a magnetic field in the iron bar.
Assume that the flux density of this magnetic field is proportional to I_1 .

An alternating electromotive force (e.m.f.) is induced across coil 2. The p.d. across coil 2 is measured using the voltmeter and has a root-mean-square (r.m.s.) value of 4.6 V.

On Fig. 8.3, sketch a line to show the variation with t of V_2 between $t = 0$ and $t = 0.08$ s.

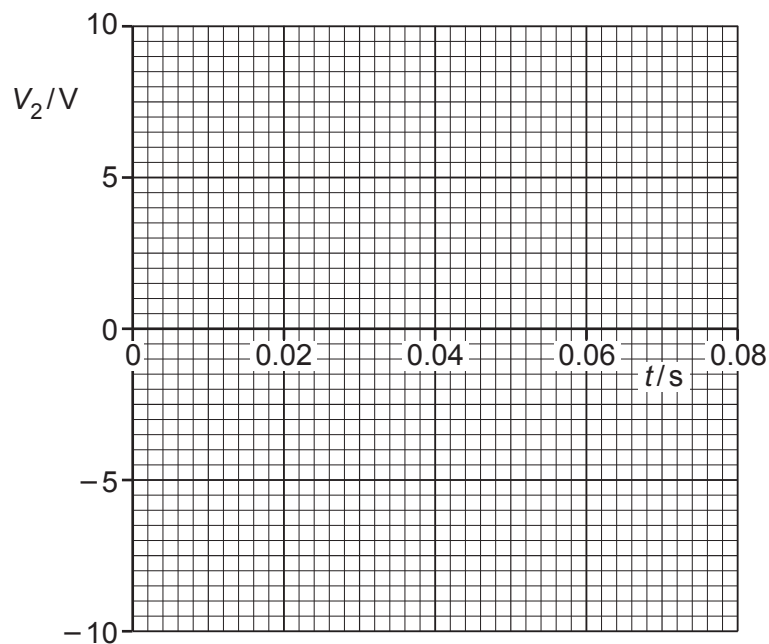


Fig. 8.3

[3]

- (iii) Use the laws of electromagnetic induction to explain the shape of your line in (b)(ii).

.....

 [3]

[Total: 11]

- 9 (a) Fig. 9.1 shows the visible part of the emission spectrum from hydrogen gas in a laboratory on the Earth. The numbers indicate the wavelength, in nm, represented by each line.

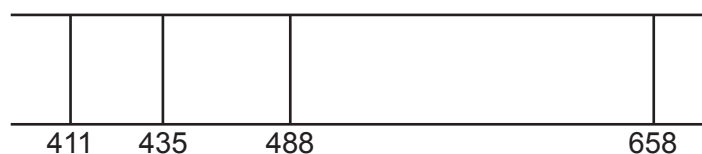


Fig. 9.1

- (i) Explain how the emission spectrum provides evidence for the existence of discrete energy levels for the electron in a hydrogen atom.

.....

 [3]

- (ii) Fig. 9.2 shows five of the energy levels in the hydrogen atom. The wavelengths of radiation shown in Fig. 9.1 relate to transitions to the -3.400 eV level in Fig. 9.2.

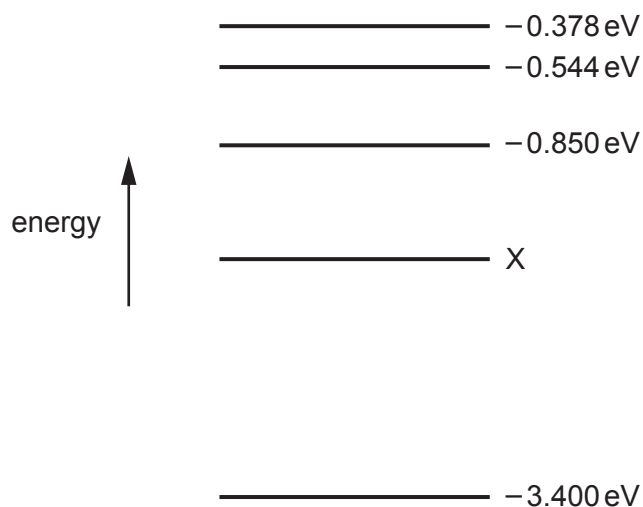


Fig. 9.2 (not to scale)

Show that the energy level X is -1.51 eV .

[3]

- (b) The same part of the emission spectrum from hydrogen as in (a), observed in light from stars in a distant galaxy, is shown in Fig. 9.3. The numbers indicate the wavelengths in nm.

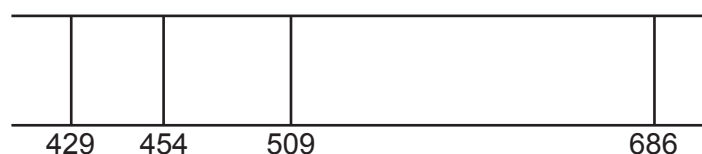


Fig. 9.3

The spectrum shows the same pattern as Fig. 9.1 but with different wavelengths.

- (i) State the name of the phenomenon that gives rise to the change in the wavelengths.
 [1]
- (ii) State what this phenomenon shows about the motion of the galaxy.
 [1]
- (iii) Use one of the lines in Fig. 9.1, and the corresponding line in Fig. 9.3, to determine the speed of the distant galaxy relative to the observer.

speed = ms^{-1} [3]

- (c) The galaxy in (b) is known to be a distance of $5.7 \times 10^{24} \text{ m}$ from the Earth.

Use your answer in (b)(iii) to determine a value for the Hubble constant H_0 .

$H_0 = \dots\dots\dots \text{s}^{-1}$ [2]

[Total: 13]

- 10** Positron emission tomography (PET scanning) involves the detection of gamma-radiation in order to identify the position of origin of positrons in the body.

(a) (i) Positrons are not naturally present in the body.

Explain how positrons come to be present in the body during PET scanning.

.....
.....
..... [2]

(ii) Explain how positrons cause the emission of gamma-radiation from the body during PET scanning.

.....
.....
.....
..... [3]

(b) Show that the wavelength of the gamma-radiation that is detected during PET scanning is approximately 2.4 pm. Explain your reasoning.

[4]

[Total: 9]

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PHYSICS

9702/43

Paper 4 A Level Structured Questions

October/November 2022

2 hours

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [].

This document has **24** pages. Any blank pages are indicated.

Data

acceleration of free fall	$g = 9.81 \text{ m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
unified atomic mass unit	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27} \text{ kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$ $(\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ m F}^{-1})$
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_o = \frac{f_s v}{v \mp v_s}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \dots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$

gravitational potential	$\phi = -\frac{GM}{r}$
gravitational potential energy	$E_P = -\frac{GMm}{r}$
pressure of an ideal gas	$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$
simple harmonic motion	$a = -\omega^2 x$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t$ $v = \omega \sqrt{(x_0^2 - x^2)}$
electric potential	$V = \frac{Q}{4\pi\epsilon_0 r}$
electrical potential energy	$E_P = \frac{Qq}{4\pi\epsilon_0 r}$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$
capacitors in parallel	$C = C_1 + C_2 + \dots$
discharge of a capacitor	$x = x_0 e^{-\frac{t}{RC}}$
Hall voltage	$V_H = \frac{BI}{ntq}$
alternating current/voltage	$x = x_0 \sin \omega t$
radioactive decay	$x = x_0 e^{-\lambda t}$
decay constant	$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$
intensity reflection coefficient	$\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$
Stefan–Boltzmann law	$L = 4\pi\sigma r^2 T^4$
Doppler redshift	$\frac{\Delta\lambda}{\lambda} \hat{=} \frac{Df}{f} \hat{=} \frac{v}{c}$

- 1 (a) State the equation for the gravitational force F between two point masses m_1 and m_2 that are separated by a distance r . State the meaning of any other symbols you use.

[2]

- (b) A satellite is in a circular orbit of radius R around a planet of mass M .

Show that the period T of the orbit is given by

$$T^2 = kR^3$$

where k is a constant that depends on the value of M . Explain your reasoning.

[3]

- (c) A satellite is in a circular orbit around the Earth with a period of 24 hours.
The mass of the Earth is 6.0×10^{24} kg.

- (i) Calculate the radius of the orbit.

radius = m [2]

(ii) State the **two** other conditions that must be met for the orbit to be geostationary.

1

.....

2

.....

[2]

[Total: 9]

- 2 Fig. 2.1 shows a laboratory thermometer that is calibrated to measure temperature in degrees Celsius.

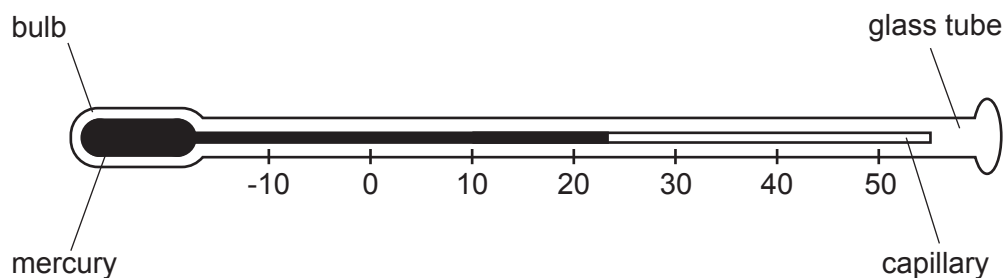


Fig. 2.1

The thermometer makes use of the fact that the density of mercury varies with temperature.

- (a) State **two** other physical properties of materials, apart from the density of a liquid, that can be used for measuring temperature.

1

2 [2]

- (b) The thermometer is initially at 23.0°C , as shown in Fig. 2.1. It is used to measure the temperature of an insulated beaker of water that is at 37.4°C . The bulb of the thermometer is inserted into the water, and the water is stirred until the reading on the thermometer becomes steady.

The mass of water in the beaker is 18.7 g.

The mass of mercury in the thermometer is 6.94 g.

The specific heat capacity of water is $4.18 \text{ J g}^{-1} \text{ K}^{-1}$.

The specific heat capacity of mercury is $0.140 \text{ J g}^{-1} \text{ K}^{-1}$.

The glass of the thermometer and the beaker containing the water can be considered to have negligible heat capacity.

- (i) Calculate, to three significant figures, the final steady temperature indicated by the thermometer in the water.

temperature = $^{\circ}\text{C}$ [4]

- (ii) Suggest **one** change that could be made to the design of the thermometer that would enable it to give a more accurate measurement of temperature.

.....
..... [1]

- (c) (i) Explain why the thermometer in Fig. 2.1 does **not** provide a direct measurement of thermodynamic temperature.

.....
.....
..... [2]

- (ii) Thermodynamic temperature T may be determined by the behaviour of a type of substance for which T is proportional to the product of pressure and volume.

State the name of this type of substance.

..... [1]

[Total: 10]

- 3 An object is suspended from a spring that is attached to a fixed point as shown in Fig. 3.1.

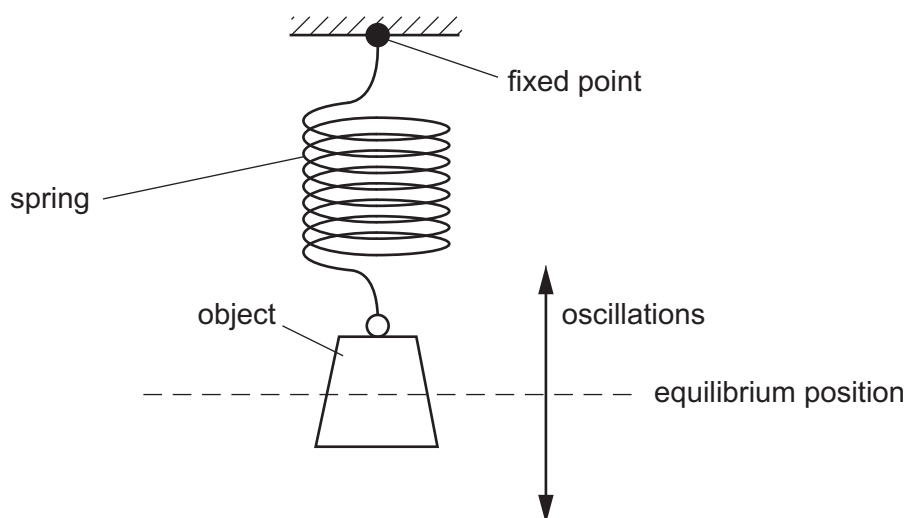


Fig. 3.1

The object oscillates vertically with simple harmonic motion about its equilibrium position.

- (a) State the defining equation for simple harmonic motion. Identify the meaning of each of the symbols used to represent physical quantities.

.....

 [2]

- (b) The variation with displacement x from the equilibrium position of the velocity v of the object is shown in Fig. 3.2.

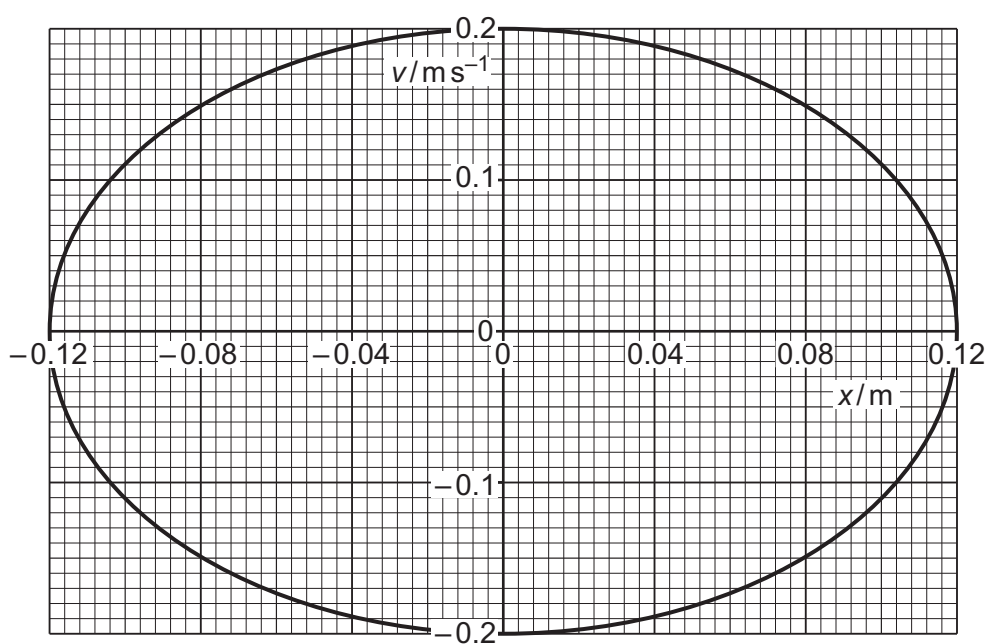


Fig. 3.2

The variation with x of the potential energy E_P of the oscillations of the object is shown in Fig. 3.3.

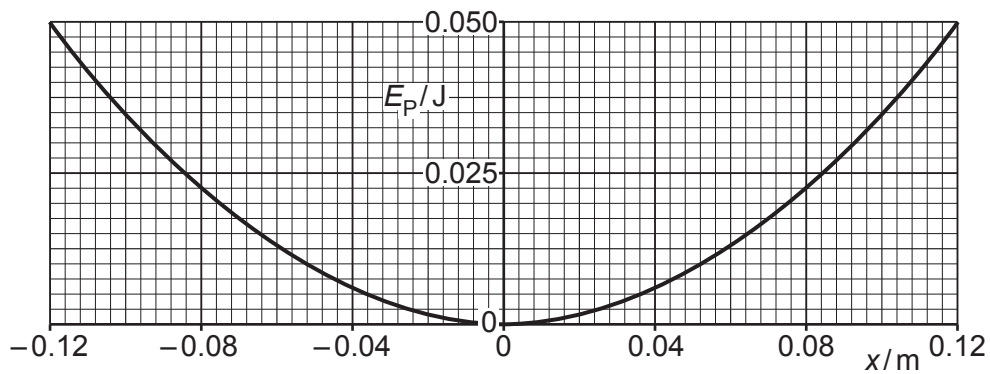


Fig. 3.3

Use Fig. 3.2 and Fig. 3.3 to:

- (i) determine the amplitude x_0 of the oscillations

$$x_0 = \dots\dots\dots \text{ m [1]}$$

- (ii) show that the angular frequency of the oscillations is 1.7 rad s^{-1}

[2]

- (iii) determine the mass M of the object.

$$M = \dots\dots\dots \text{ kg [2]}$$

(c) The oscillations of the object are now lightly damped.

(i) State what is meant by damping.

.....

.....

..... [2]

(ii) Assume that the damping does not change the angular frequency of the oscillations.

On Fig. 3.2, sketch the variation with x of v when the amplitude of the oscillations is 0.060 m. [2]

[Total: 11]

- 4 (a) State what is indicated by the direction of an electric field line.

.....
 [2]

- (b) Fig. 4.1 shows a pair of parallel metal plates with a potential difference (p.d.) of 2400 V between them.

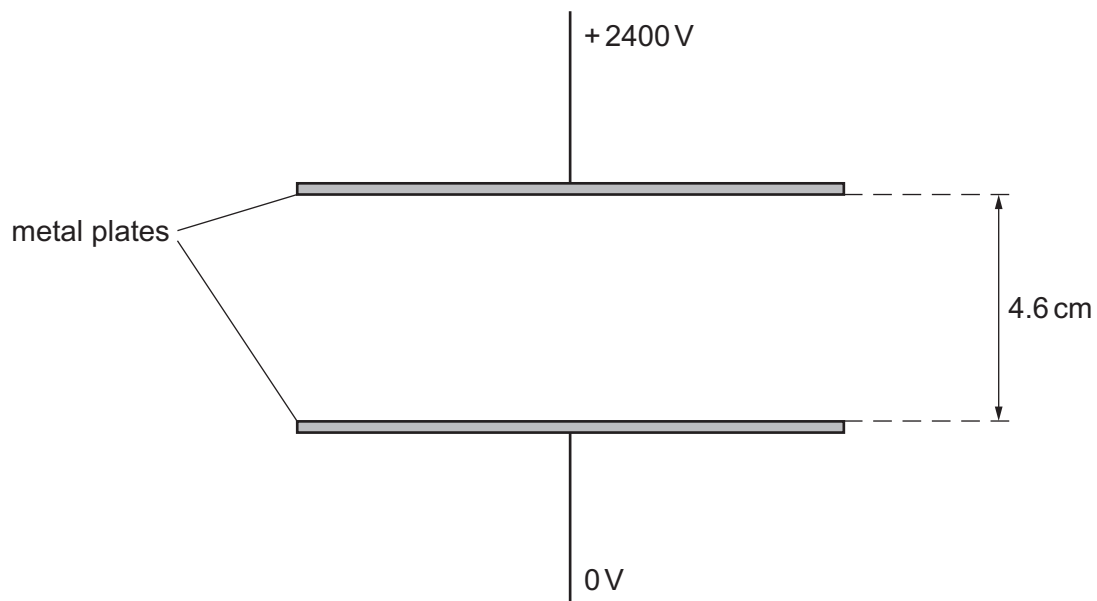


Fig. 4.1

The plates are separated by a distance of 4.6 cm. The plates are in a vacuum.

- (i) On Fig. 4.1, draw five lines to represent the electric field in the region between the plates. [3]
- (ii) Calculate the strength of the electric field between the plates.

electric field strength = NC^{-1} [2]

- (c) A moving proton enters the region between the plates from the left, as shown in Fig. 4.2.

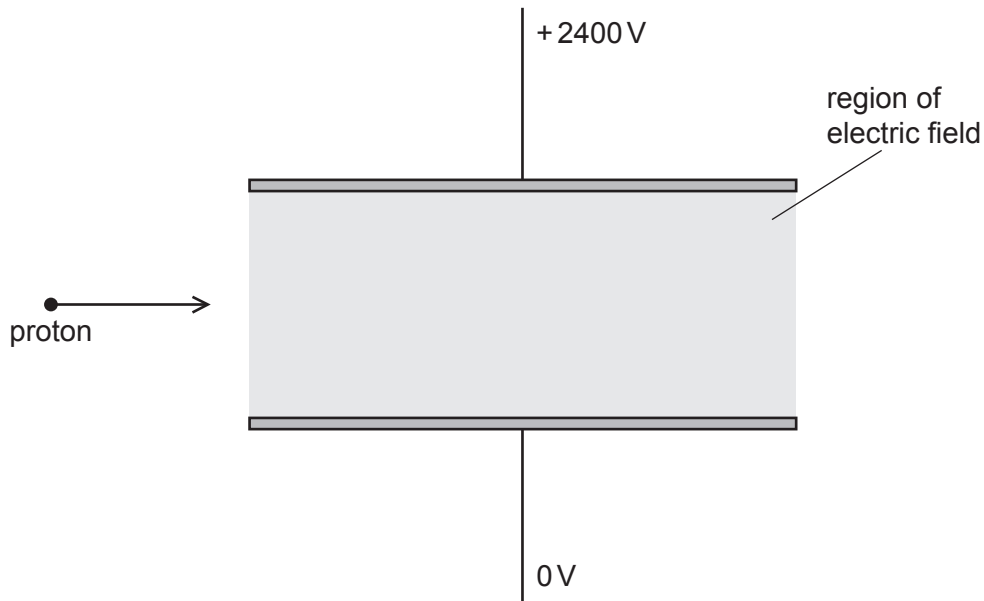


Fig. 4.2

- (i) The proton is deflected by the electric field.

On Fig. 4.2, draw a line to show the path of the proton as it moves through and out of the region of the electric field. [2]

- (ii) A helium nucleus (${}^4_2\text{He}$) now enters the region of the electric field along the same initial path as the proton and travelling at the same initial speed.

State and explain how the final speed of the helium nucleus compares with the final speed of the proton after leaving the region of the electric field.

.....

.....

.....

.....

..... [3]

[Total: 12]

- 5 A capacitor of capacitance $470\ \mu\text{F}$ is connected to a battery of electromotive force (e.m.f.) $24\ \text{V}$ in the circuit of Fig. 5.1.

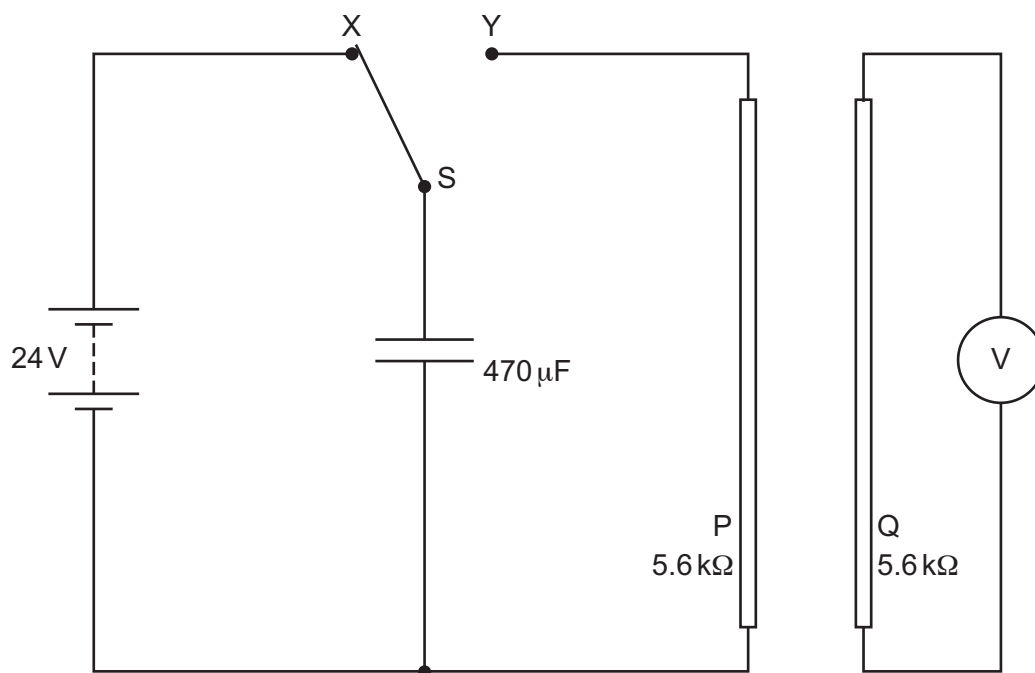


Fig. 5.1

The two-way switch S is initially at position X.

P and Q are identical long straight wires, each with a resistance of $5.6\ \text{k}\Omega$. These wires are placed near to, and parallel to, each other. Wire Q is connected to a voltmeter.

At time $t = 0$, switch S is moved to position Y so that the capacitor discharges through wire P.

- (a) (i) Calculate the charge Q_0 on the capacitor at time $t = 0$.

$$Q_0 = \dots\dots\dots \text{ C [2]}$$

- (ii) Calculate the current I_0 in wire P at time $t = 0$.

$$I_0 = \dots\dots\dots \text{ A [1]}$$

- (iii) Calculate the time constant τ of the discharge circuit.

$\tau = \dots\dots\dots$ s [2]

- (iv) On Fig. 5.2, sketch a line to show the variation with t of the current I in wire P as the capacitor discharges.



Fig. 5.2

[2]

- (b) (i) Explain why there is an induced e.m.f. across wire Q during the discharge of the capacitor.

.....

 [3]

- (ii) On Fig. 5.3, sketch a line to suggest the variation with t of the voltmeter reading V .



Fig. 5.3

[1]

[Total: 11]

- 6 Fig. 6.1 shows a thin slice of semiconducting material used in a Hall probe.

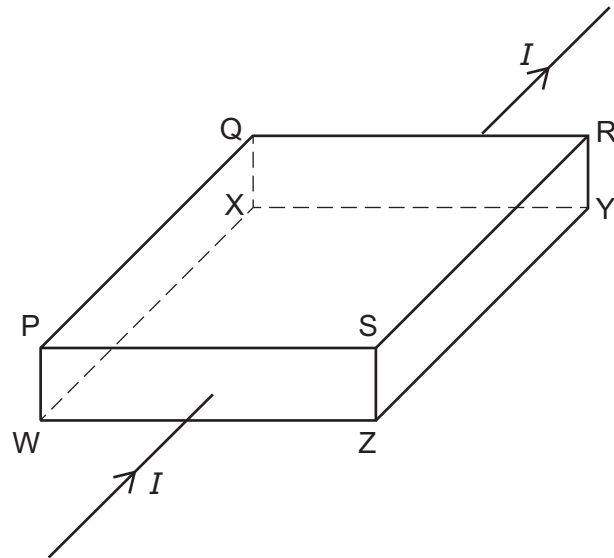


Fig. 6.1 (not to scale)

Current I passes through the slice in the direction shown.

The slice is placed in a uniform magnetic field of flux density B , so that two of its faces are perpendicular to the magnetic field.

A steady Hall voltage V_H is developed between face PQXW and face SRYZ.

- (a) (i) Use the letters in Fig. 6.1 to identify the faces that are perpendicular to the magnetic field.

..... and [1]

- (ii) Explain how the steady Hall voltage V_H is developed between faces PQXW and SRYZ.

.....

 [3]

(b) The magnitude of V_H is given by the equation

$$V_H = \frac{BI}{ntq}.$$

(i) State the meaning of the symbols n , t and q . You may refer to the letters in Fig. 6.1.

n :

t :

q : [3]

(ii) Suggest, with reference to the equation, why the slice of the material used in a Hall probe is thin.

.....

.....

..... [2]

[Total: 9]

- 7 (a) A sinusoidal alternating voltage has a root-mean-square (r.m.s.) potential difference (p.d.) of 4.2 V and a frequency of 50 kHz.

- (i) The alternating voltage is applied across a resistor of resistance $760\ \Omega$.

By considering the peak voltage, show that the maximum power dissipated by the resistor is 46 mW.

[2]

- (ii) On Fig. 7.1, draw a smooth curve to show how the power P dissipated in the resistor varies with time t between $t = 0$ and $t = 40\ \mu\text{s}$. Assume that $P = 0$ when $t = 0$.

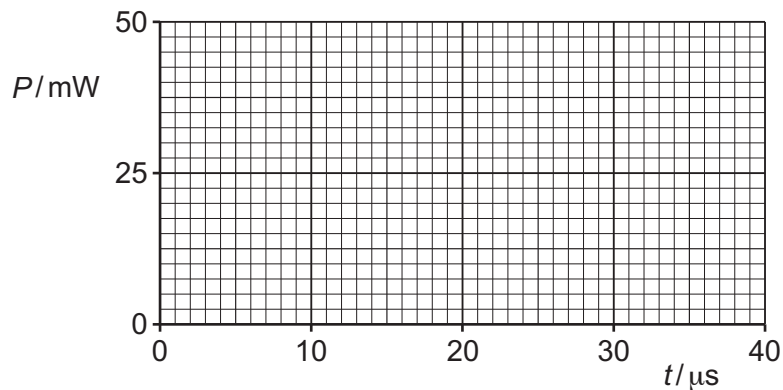


Fig. 7.1

[3]

- (iii) Use your line in (a)(ii) to explain why the mean power dissipated in the resistor is 23 mW.

.....

 [1]

(b) The alternating voltage in **(a)** is now applied to a piezoelectric crystal in air.

(i) Explain what happens to the air surrounding the crystal.

.....

.....

.....

..... [3]

(ii) A second piezoelectric crystal is placed in the air near to the first crystal.

Explain the effect of the surrounding air in **(b)(i)** on the second crystal.

..... [1]

[Total: 10]

- 8 (a) State what is meant by the work function energy of a metal.

.....

 [2]

- (b) Ultraviolet radiation of frequency $1.36 \times 10^{15} \text{ Hz}$ is incident, in a vacuum, on a metal surface. The power of the radiation incident on the surface is 8.36 mW . Photoelectrons are emitted with a maximum kinetic energy of $3.09 \times 10^{-19} \text{ J}$.

- (i) Determine the number of photons incident on the surface per unit time.

number per unit time = s^{-1} [2]

- (ii) Calculate the work function energy Φ of the metal.

$\Phi = \dots\dots\dots \text{ J}$ [2]

- (c) The frequency of the radiation incident on the surface in (b) is increased while the power remains constant.

State and explain the effect of this change on:

- (i) the maximum kinetic energy of the photoelectrons

.....

 [2]

- (ii) the rate of emission of photoelectrons.

.....

 [2]

[Total: 10]

- 9 (a) State what is meant by the luminosity of a star.

..... [1]

- (b) A star in the constellation Canis Major is a distance of 8.14×10^{16} m from the Earth and has a luminosity of 9.86×10^{27} W. The surface temperature of the star is 9830 K.

- (i) Calculate the radiant flux intensity of the radiation from the star observed from the Earth. Give a unit with your answer.

radiant flux intensity = unit [2]

- (ii) Determine the radius of the star.

radius = m [2]

- (c) Explain how the surface temperature of a distant star may be determined from the wavelength spectrum of the light from the star.

.....

 [3]

[Total: 8]

- 10 Carbon-15 ($^{15}_6\text{C}$) is an isotope of carbon that undergoes radioactive decay to nitrogen-15 ($^{15}_7\text{N}$), which is a stable isotope of nitrogen.

Radioactive decay is both a random and a spontaneous process.

(a) State what is meant by:

(i) random

.....
 [1]

(ii) spontaneous.

.....
 [1]

- (b) A small sample of carbon-15 decays. The mass M of carbon-15 in the sample decreases with time t .

Fig. 10.1 shows the variation with t of the value of $\ln (M/10^{-16}\text{g})$.

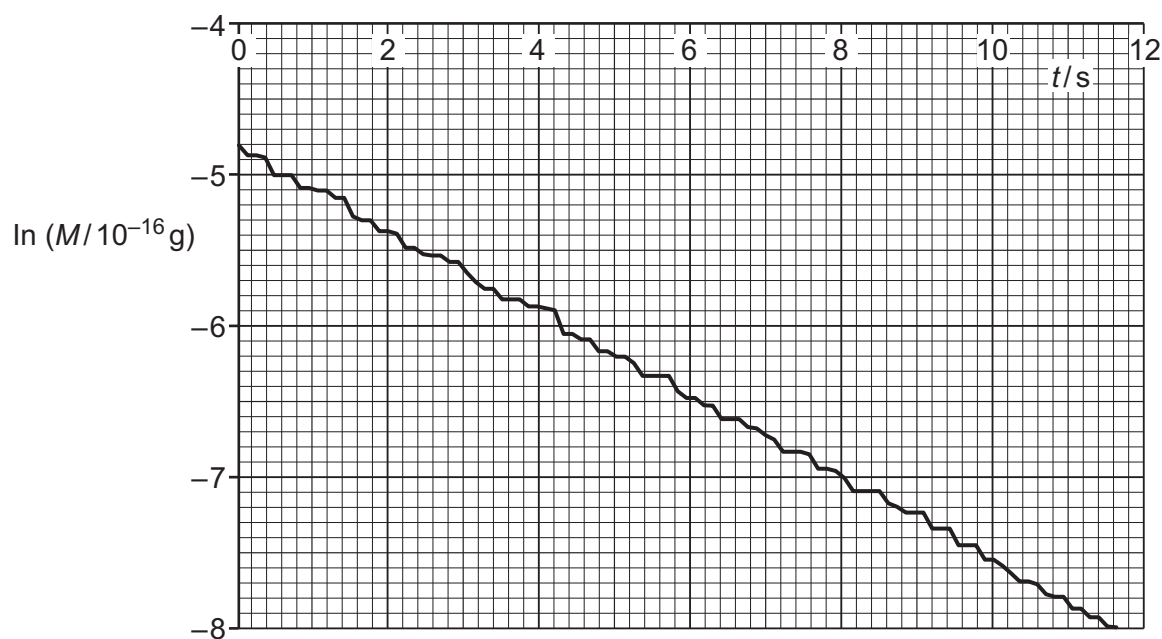


Fig. 10.1

- (i) State how Fig. 10.1 demonstrates that radioactive decay is random.

.....
 [1]

- (ii) On Fig. 10.1, draw the straight line of best fit.

[1]

- (iii) Show that the decay constant λ of carbon-15 is given by the magnitude of the gradient of your line in (b)(ii).

[1]

- (iv) Use your line in (b)(ii) to determine λ . Give a unit with your answer.

$\lambda = \dots\dots\dots$ unit $\dots\dots\dots$ [2]

- (v) Use your answer in (b)(iv) to calculate the half-life of carbon-15.

half-life = $\dots\dots\dots$ s [1]

- (c) The equation for the decay of carbon-15 can be written as



State and explain how the mass of the products of the decay must compare with the mass of the carbon-15 nucleus.

.....

 [2]

[Total: 10]

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9702/51

October/November 2022

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 30.
- The number of marks for each question or part question is shown in brackets [].

This document has 8 pages.

- 1 A thin copper sheet is suspended from a small hole near the top of the sheet and placed in a magnetic field, as shown in Fig. 1.1.

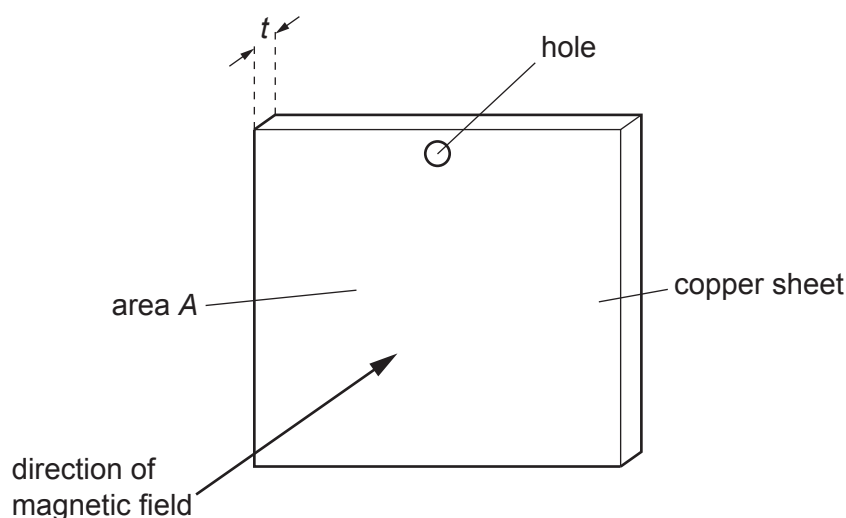


Fig. 1.1 (not to scale)

The sheet has area A and thickness t .

The sheet is displaced from its equilibrium position through a horizontal distance s_0 and then released so that it oscillates perpendicular to the direction of the magnetic field. The horizontal distance s of the sheet from its equilibrium position is measured after five complete oscillations.

It is suggested that s is related to A by the relationship

$$s = s_0 e^{-ABKt}$$

where B is the magnetic flux density of the field and K is a constant.

Plan a laboratory experiment to test the relationship between s and A .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine a value for K .

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

Diagram

.....

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- 2 A student investigates a circuit containing resistors and a metal wire as shown in Fig. 2.1.

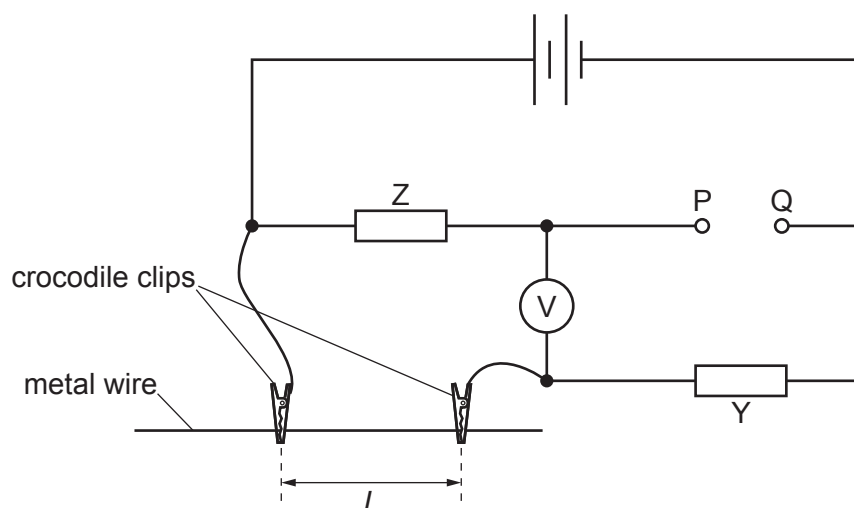


Fig. 2.1

Resistors Y and Z have resistances Y and Z respectively.

The student connects a resistor of resistance R between P and Q.

The student then adjusts the length of the wire between the crocodile clips until the voltmeter reads zero. The student measures the length L of wire between the crocodile clips.

The student repeats the experiment with different values of R .

It is suggested that L and R are related by the equation

$$\frac{Z}{R} = \frac{4\rho L}{\pi Y d^2}$$

where d is the diameter of the wire and ρ is the resistivity of the metal.

- (a) A graph is plotted of L on the y -axis against $\frac{1}{R}$ on the x -axis.

Determine an expression for the gradient.

gradient = [1]

(b) Values of R and L are given in Table 2.1.

Each resistance value R has a percentage uncertainty of $\pm 5\%$.

Table 2.1

R/Ω	$\frac{1}{R}/10^{-3}\Omega^{-1}$	L/cm
22		71.0
27		57.5
33		45.0
39		36.5
47		27.5
54		23.0

Calculate and record values of $\frac{1}{R}/10^{-3}\Omega^{-1}$ in Table 2.1.

Include the absolute uncertainties in $\frac{1}{R}$. [2]

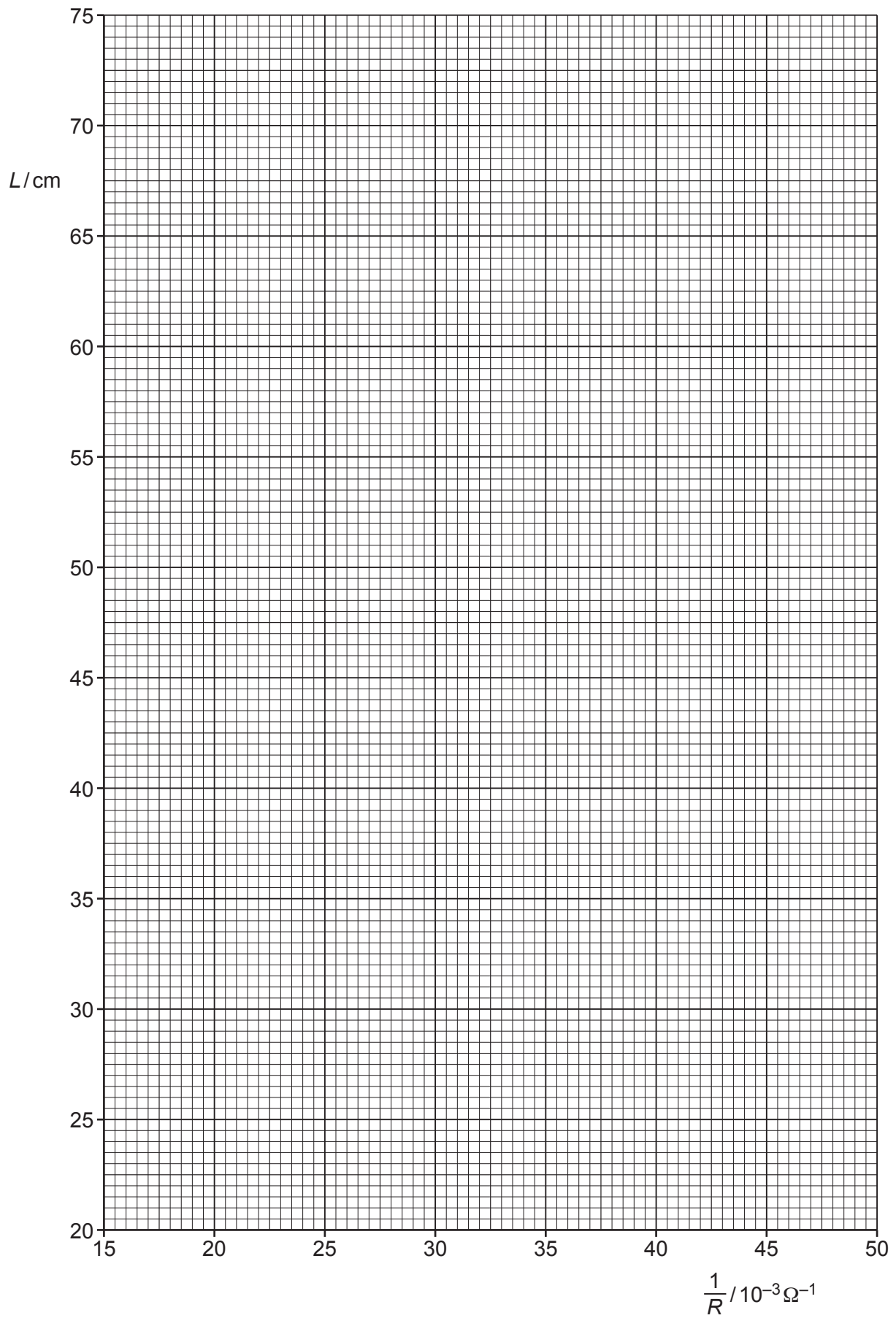
(c) (i) Plot a graph of L/cm against $\frac{1}{R}/10^{-3}\Omega^{-1}$.

Include error bars for $\frac{1}{R}$. [2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines. [2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]



- (d) The student measures the diameter of the wire. The student's values are:

0.263 mm 0.262 mm 0.263 mm 0.257 mm 0.262 mm 0.259 mm.

Determine the average diameter d . Include the absolute uncertainty in d .

$d = \dots\dots\dots$ mm [1]

- (e) (i) Resistors Y and Z each have a resistance of $22\,\Omega \pm 5\%$.

Using your answers to (a), (c)(iii) and (d), determine the value of ρ . Include an appropriate unit.

$\rho = \dots\dots\dots$ [2]

- (ii) Determine the percentage uncertainty in ρ .

percentage uncertainty in $\rho = \dots\dots\dots$ % [1]

- (f) Determine the resistance R that would give a value of L of 95.0 cm. Include the absolute uncertainty in your answer.

$R = \dots\dots\dots$ Ω [2]

[Total: 15]



CANDIDATE
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9702/52

October/November 2022

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
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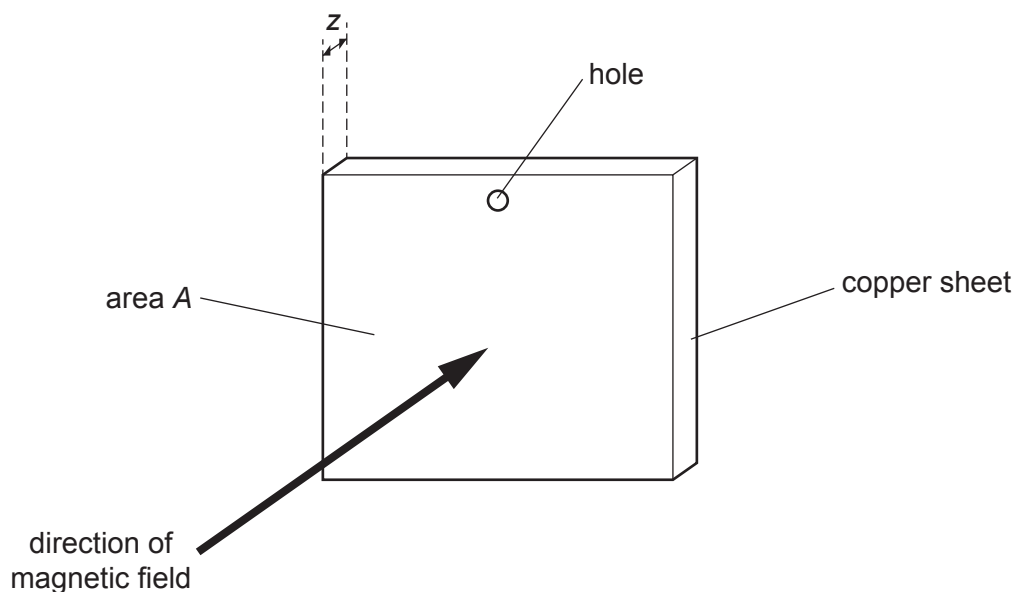


Fig. 1.1 (not to scale)

The sheet has area A and thickness z .

The sheet is displaced from its equilibrium position and then released so that it oscillates perpendicular to the direction of the magnetic field. The time t from when the sheet is released to when it becomes stationary is measured.

It is suggested that t is related to z by the relationship

$$t = \frac{Kz^q}{AB\rho}$$

where B is the magnetic flux density of the field, ρ is the density of copper, and K and q are constants.

Plan a laboratory experiment to test the relationship between t and z .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for K and q .

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

Diagram

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- 2 A student investigates stationary waves in a vertical tube using the apparatus shown in Fig. 2.1.

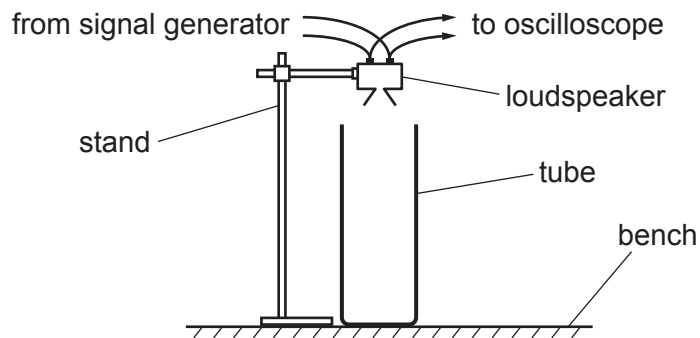


Fig. 2.1

The student slowly increases the frequency of the signal generator from zero and listens to the sound. The loudness of the sound varies several times between minimum and maximum as the frequency is increased. The lowest frequency giving maximum loudness is identified by $n = 1$. The next frequencies giving maximum loudness are identified by $n = 2, 3, 4, 5$ and 6 .

For each value of n , the student observes the trace on the oscilloscope screen. The student measures the distance d on the screen between two successive crests, as shown in Fig. 2.2.

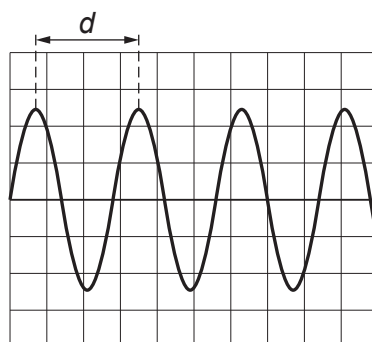


Fig. 2.2

The student then determines the period T and frequency f of the signal.

It is suggested that f and n are related by the equation

$$f = \frac{(2n - 1)c}{4h}$$

where c is the speed of sound in air and h is the height of the tube.

- (a) A graph is plotted of f on the y -axis against n on the x -axis.

Determine expressions for the gradient and y -intercept.

gradient =

y -intercept =

[1]

(b) The period T and frequency f are given by the equations

$$T = d \times \text{time-base} \quad \text{and} \quad f = \frac{1}{T}.$$

Values of n , d and the time-base of the oscilloscope are given in Table 2.1.

Table 2.1

n	d/cm	time-base / ms cm^{-1}	T/ms	f/Hz
1	1.4 ± 0.2	5		
2	2.9 ± 0.2	1		
3	3.6 ± 0.2	0.5		
4	2.7 ± 0.2	0.5		
5	2.1 ± 0.2	0.5		
6	8.8 ± 0.2	0.1		

Calculate and record values of T/ms and f/Hz in Table 2.1.

Include the absolute uncertainties in T and f .

[2]

(c) (i) Plot a graph of f/Hz against n . Include error bars for f .

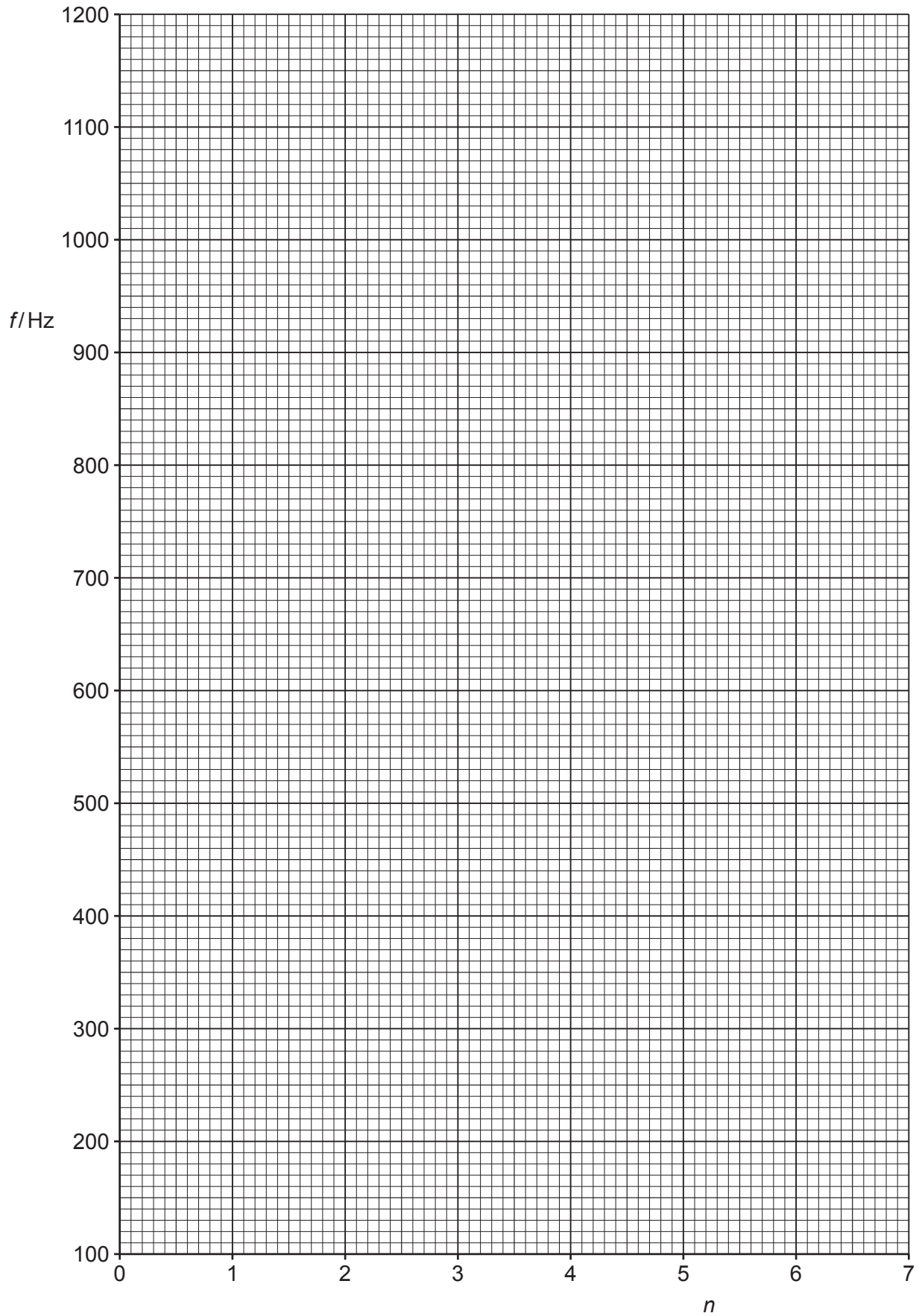
[2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines.

[2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]



- (d) The student measures the height of the tube. The student's values are:

83.5 cm 82.9 cm 83.3 cm 83.1 cm.

Determine the average height h . Include the absolute uncertainty in h .

$h = \dots\dots\dots$ cm [1]

- (e) (i) Using your answers to (a), (c)(iii) and (d), determine the value of c . Include an appropriate unit.

$c = \dots\dots\dots$ [2]

- (ii) Determine the percentage uncertainty in c .

percentage uncertainty in $c = \dots\dots\dots\%$ [1]

- (f) The experiment is repeated with a different tube.
When $n = 2$, the frequency is (130 ± 5) Hz.

Determine the height h of this tube. Include the absolute uncertainty in your answer.

$h = \dots\dots\dots$ m [2]

[Total: 15]



CANDIDATE
NAME

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CENTRE
NUMBER

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CANDIDATE
NUMBER

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9702/53

October/November 2022

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 30.
- The number of marks for each question or part question is shown in brackets [].

This document has 8 pages.

- 1 A thin copper sheet is suspended from a small hole near the top of the sheet and placed in a magnetic field, as shown in Fig. 1.1.

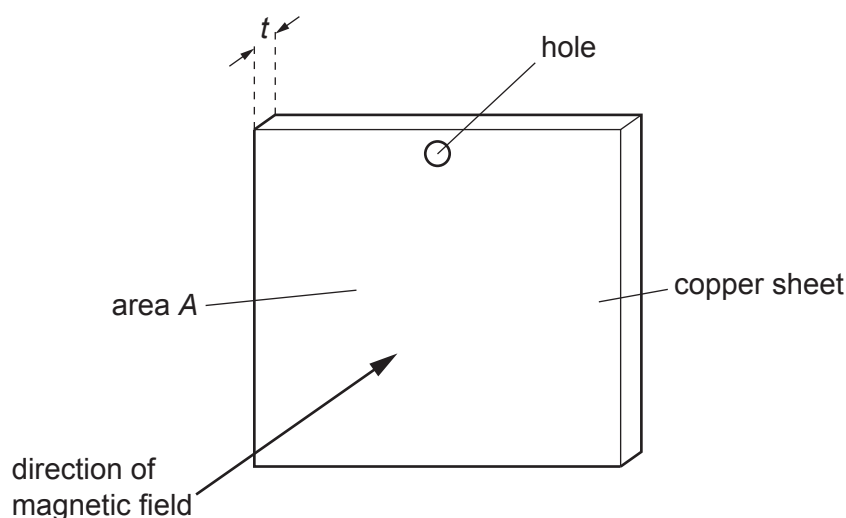


Fig. 1.1 (not to scale)

The sheet has area A and thickness t .

The sheet is displaced from its equilibrium position through a horizontal distance s_0 and then released so that it oscillates perpendicular to the direction of the magnetic field. The horizontal distance s of the sheet from its equilibrium position is measured after five complete oscillations.

It is suggested that s is related to A by the relationship

$$s = s_0 e^{-ABKt}$$

where B is the magnetic flux density of the field and K is a constant.

Plan a laboratory experiment to test the relationship between s and A .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine a value for K .

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

Diagram

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- 2 A student investigates a circuit containing resistors and a metal wire as shown in Fig. 2.1.

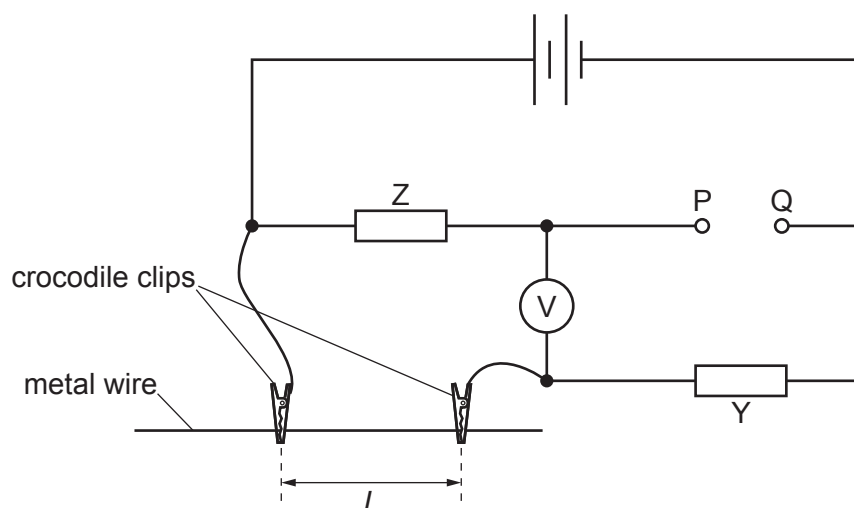


Fig. 2.1

Resistors Y and Z have resistances Y and Z respectively.

The student connects a resistor of resistance R between P and Q.

The student then adjusts the length of the wire between the crocodile clips until the voltmeter reads zero. The student measures the length L of wire between the crocodile clips.

The student repeats the experiment with different values of R .

It is suggested that L and R are related by the equation

$$\frac{Z}{R} = \frac{4\rho L}{\pi Y d^2}$$

where d is the diameter of the wire and ρ is the resistivity of the metal.

- (a) A graph is plotted of L on the y -axis against $\frac{1}{R}$ on the x -axis.

Determine an expression for the gradient.

gradient = [1]

(b) Values of R and L are given in Table 2.1.

Each resistance value R has a percentage uncertainty of $\pm 5\%$.

Table 2.1

R/Ω	$\frac{1}{R}/10^{-3}\Omega^{-1}$	L/cm
22		71.0
27		57.5
33		45.0
39		36.5
47		27.5
54		23.0

Calculate and record values of $\frac{1}{R}/10^{-3}\Omega^{-1}$ in Table 2.1.

Include the absolute uncertainties in $\frac{1}{R}$. [2]

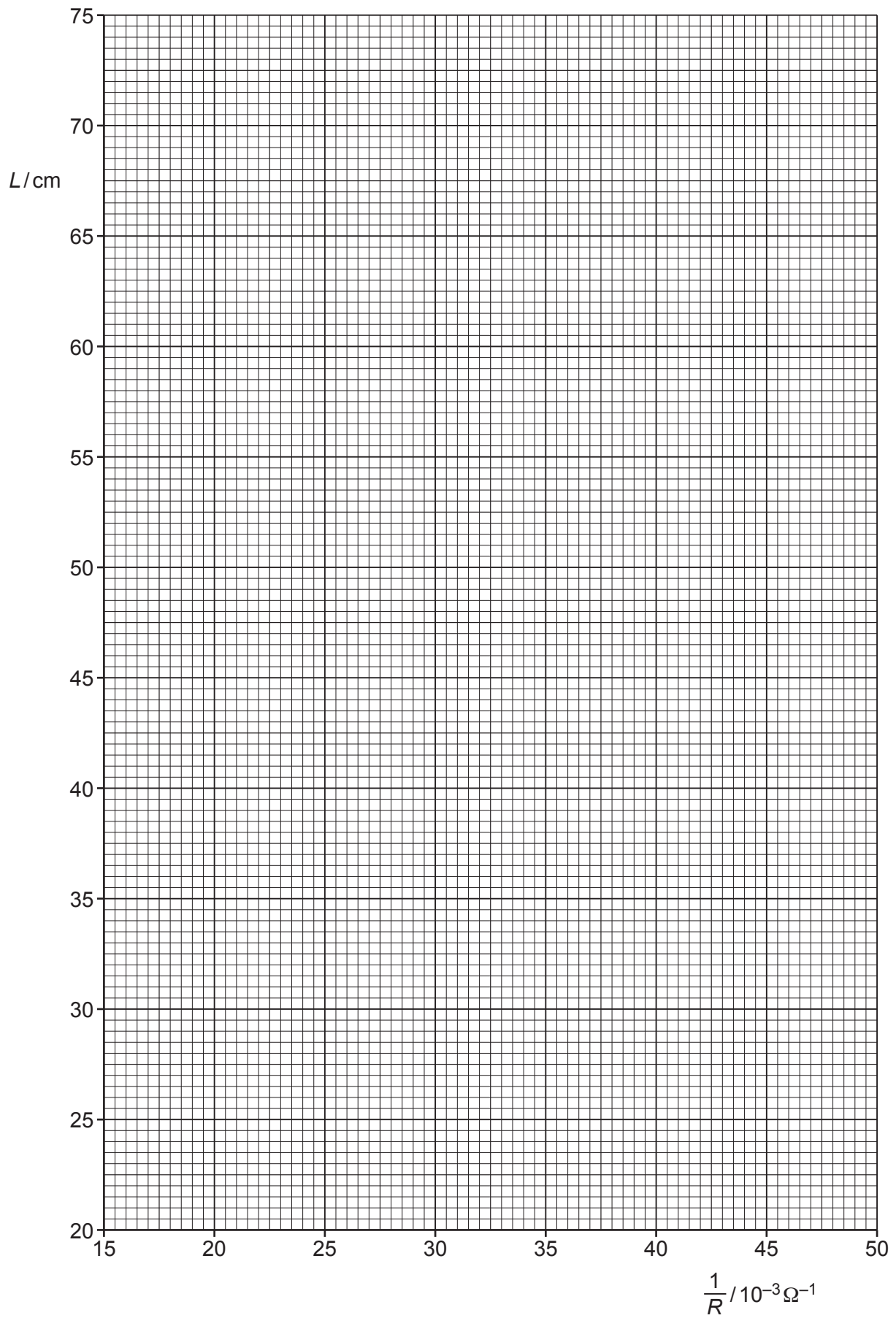
(c) (i) Plot a graph of L/cm against $\frac{1}{R}/10^{-3}\Omega^{-1}$.

Include error bars for $\frac{1}{R}$. [2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines. [2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]



- (d) The student measures the diameter of the wire. The student's values are:

0.263 mm 0.262 mm 0.263 mm 0.257 mm 0.262 mm 0.259 mm.

Determine the average diameter d . Include the absolute uncertainty in d .

$d = \dots\dots\dots$ mm [1]

- (e) (i) Resistors Y and Z each have a resistance of $22\,\Omega \pm 5\%$.

Using your answers to (a), (c)(iii) and (d), determine the value of ρ . Include an appropriate unit.

$\rho = \dots\dots\dots$ [2]

- (ii) Determine the percentage uncertainty in ρ .

percentage uncertainty in $\rho = \dots\dots\dots$ % [1]

- (f) Determine the resistance R that would give a value of L of 95.0 cm. Include the absolute uncertainty in your answer.

$R = \dots\dots\dots$ Ω [2]

[Total: 15]



CANDIDATE
NAME

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CENTRE
NUMBER

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CANDIDATE
NUMBER

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9702/41

May/June 2023

2 hours

You must answer on the question paper.

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [].

This document has **24** pages. Any blank pages are indicated.

Data

acceleration of free fall	$g = 9.81 \text{ m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
unified atomic mass unit	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27} \text{ kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$ ($\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ m F}^{-1}$)
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_o = \frac{f_s v}{v \pm v_s}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \dots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$

gravitational potential	$\phi = -\frac{GM}{r}$
gravitational potential energy	$E_P = -\frac{GMm}{r}$
pressure of an ideal gas	$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$
simple harmonic motion	$a = -\omega^2 x$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t$ $v = \omega \sqrt{(x_0^2 - x^2)}$
electric potential	$V = \frac{Q}{4\pi\epsilon_0 r}$
electrical potential energy	$E_P = \frac{Qq}{4\pi\epsilon_0 r}$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$
capacitors in parallel	$C = C_1 + C_2 + \dots$
discharge of a capacitor	$x = x_0 e^{-\frac{t}{RC}}$
Hall voltage	$V_H = \frac{BI}{ntq}$
alternating current/voltage	$x = x_0 \sin \omega t$
radioactive decay	$x = x_0 e^{-\lambda t}$
decay constant	$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$
intensity reflection coefficient	$\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$
Stefan–Boltzmann law	$L = 4\pi\sigma r^2 T^4$
Doppler redshift	$\frac{\Delta\lambda}{\lambda} \approx \frac{Df}{f} \approx \frac{v}{c}$

- 1 (a) (i) Define gravitational field.

.....
 [1]

- (ii) Define electric field.

.....
 [1]

- (iii) State **one** similarity and **one** difference between the gravitational potential due to a point mass and the electric potential due to a point charge.

similarity:

.....

difference:

.....

[2]

- (b) An isolated uniform conducting sphere has mass M and charge Q .

The gravitational field strength at the surface of the sphere is g .

The electric field strength at the surface of the sphere is E .

- (i) Show that

$$\frac{M}{Q} = \alpha \frac{g}{E}$$

where α is a constant.

[3]

- (ii) Show that the numerical value of α is $1.35 \times 10^{20} \text{ kg}^2 \text{ C}^{-2}$.

[1]

- (c) Assume that the Earth is a uniform conducting sphere of mass 5.98×10^{24} kg. The surface of the Earth carries a charge of -4.80×10^5 C that is evenly distributed.

- (i) Use the information in (b) to determine the electric field strength at the surface of the Earth. Give a unit with your answer.

electric field strength = unit [2]

- (ii) State how the direction of the electric field at the surface of the Earth compares with the direction of the gravitational field.

..... [1]

[Total: 11]

- 2 A steel sphere of mass 0.29 kg is suspended in equilibrium from a vertical spring. The centre of the sphere is 8.5 cm from the top of the spring, as shown in Fig. 2.1.

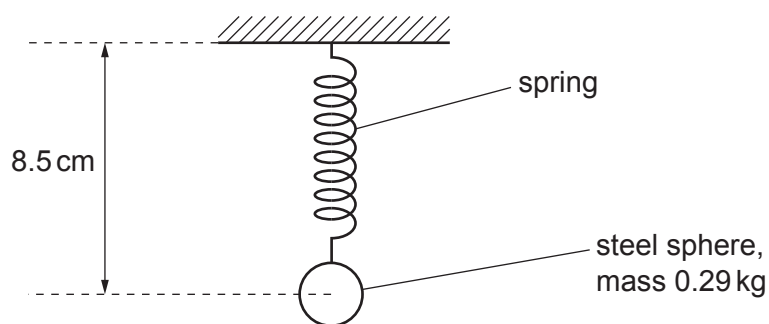


Fig. 2.1

The sphere is now set in motion so that it is moving in a horizontal circle at constant speed, as shown in Fig. 2.2.

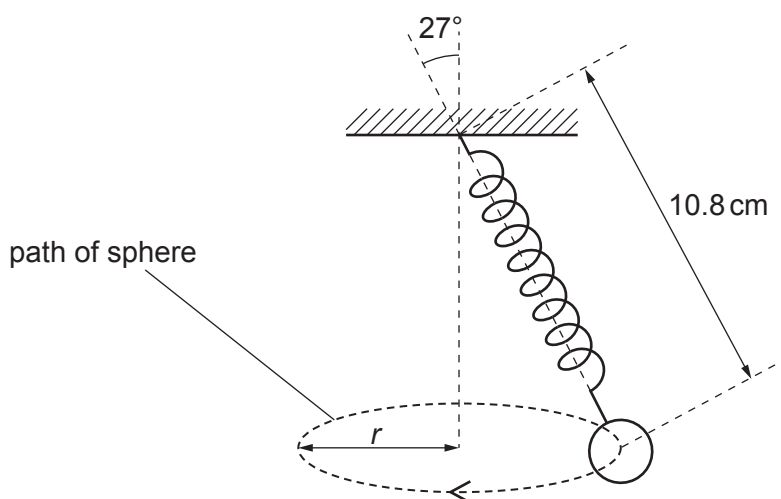


Fig. 2.2

The distance from the centre of the sphere to the top of the spring is now 10.8 cm .

- (a) Explain, with reference to the forces acting on the sphere, why the length of the spring in Fig. 2.2 is greater than in Fig. 2.1.

.....

.....

.....

.....

..... [3]

(b) The angle between the linear axis of the spring and the vertical is 27° .

(i) Show that the radius r of the circle is 4.9 cm.

[1]

(ii) Show that the tension in the spring is 3.2 N.

[2]

(iii) The spring obeys Hooke's law.

Calculate the spring constant, in N cm^{-1} , of the spring.

spring constant = N cm^{-1} [2]

(c) (i) Use the information in (b) to determine the centripetal acceleration of the sphere.

centripetal acceleration = ms^{-2} [2]

(ii) Calculate the period of the circular motion of the sphere.

period = s [2]

[Total: 12]

- 3 (a) State the reason why two objects that are at the same temperature are described as being in thermal equilibrium.

.....
 [1]

- (b) Fig. 3.1 shows the variations with temperature of the densities of mercury and of water between 0°C and 100°C .

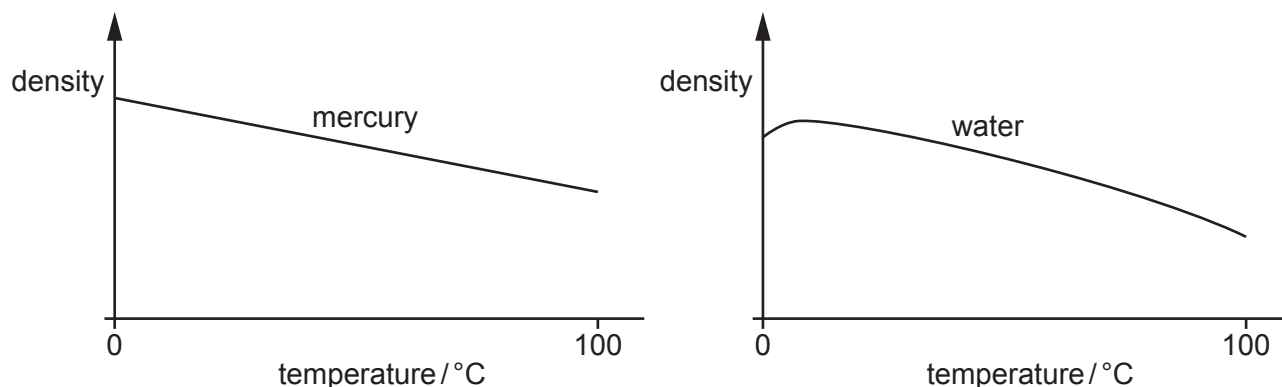


Fig. 3.1

Temperature may be measured using the variation with temperature of the density of a liquid.

Suggest why, for measuring temperature over this temperature range:

- (i) mercury is a suitable liquid

.....
 [1]

- (ii) water is not a suitable liquid.

.....

 [2]

- (c) A beaker contains a liquid of mass 120 g. The liquid is supplied with thermal energy at a rate of 810 W. The beaker has a mass of 42 g and a specific heat capacity of $0.84 \text{ J g}^{-1} \text{ K}^{-1}$. The beaker and the liquid are in thermal equilibrium with each other at all times and are insulated from the surroundings.

Fig. 3.2 shows the variation with time t of the temperature of the liquid.

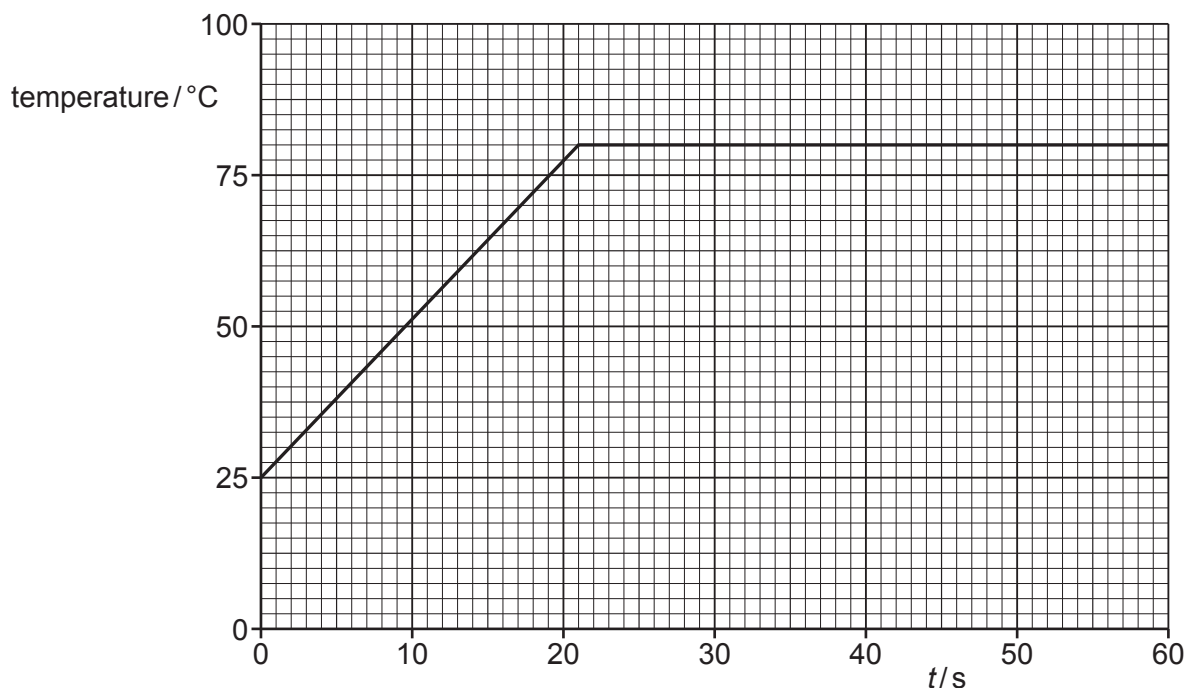


Fig. 3.2

- (i) State the boiling temperature, in °C, of the liquid.

temperature = °C [1]

- (ii) Determine the specific heat capacity, in $\text{J g}^{-1} \text{K}^{-1}$, of the liquid.

specific heat capacity = $\text{J g}^{-1} \text{K}^{-1}$ [4]

- (d) The experiment in (c) is repeated using water instead of the liquid in (c). The mass of liquid used, the power supplied, and the initial temperature are all unchanged.
The specific heat capacity of water is approximately twice that of the liquid in (c).
The boiling temperature of water is 100 °C.

On Fig. 3.2, sketch the variation with time t of the temperature of the water between $t = 0$ and $t = 60$ s. Numerical calculations are not required. [2]

[Total: 11]

- 4 (a) State **two** of the basic assumptions of the kinetic theory of gases.

1

.....

2

.....

[2]

- (b) An ideal gas has amount of substance n .

The gas is initially in state X, with pressure $2p$ and volume V .

The gas is cooled at constant volume to state Y, with pressure p .

The gas is then heated at constant pressure to state Z, with volume $2V$.

Finally, the gas returns at constant temperature to state X.

- (i) Determine an expression for the temperature T of the gas in state X, in terms of n , p and V .
Identify any other symbols that you use.

[2]

- (ii) On Fig. 4.1, sketch the variation with volume of pressure for the gas as the gas undergoes the three changes. The state X is labelled. Label states Y and Z.

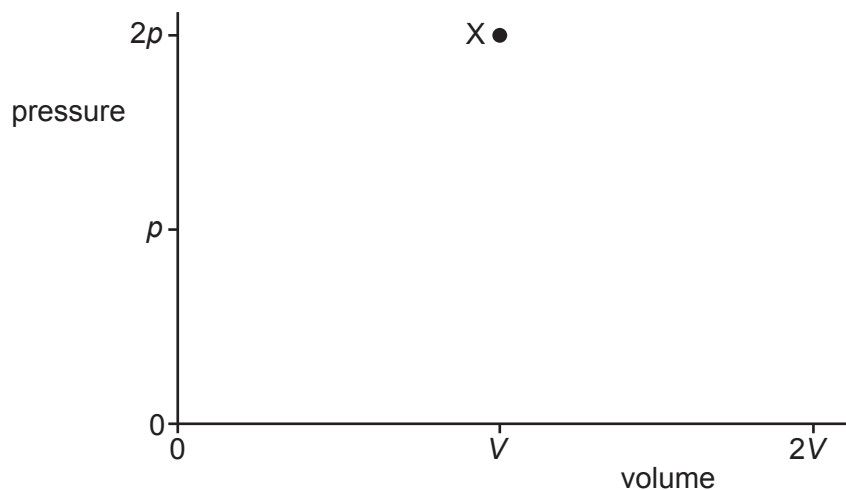


Fig. 4.1

[3]

- (iii) During the change of state from Y to Z, the increase in internal energy of the gas is U .
During the change of state from Z to X, the work done on the gas is W .

Complete Table 4.1 to indicate, for each of the three changes of state, the increase in internal energy of the gas, the thermal energy transferred to the gas and the work done on the gas, in terms of p , V , U and W .

Table 4.1

change	increase in internal energy of gas	thermal energy transferred to gas	work done on gas
X to Y			
Y to Z	$+U$		
Z to X			$+W$

[5]

[Total: 12]

5 Part of an electric circuit is shown in Fig. 5.1.

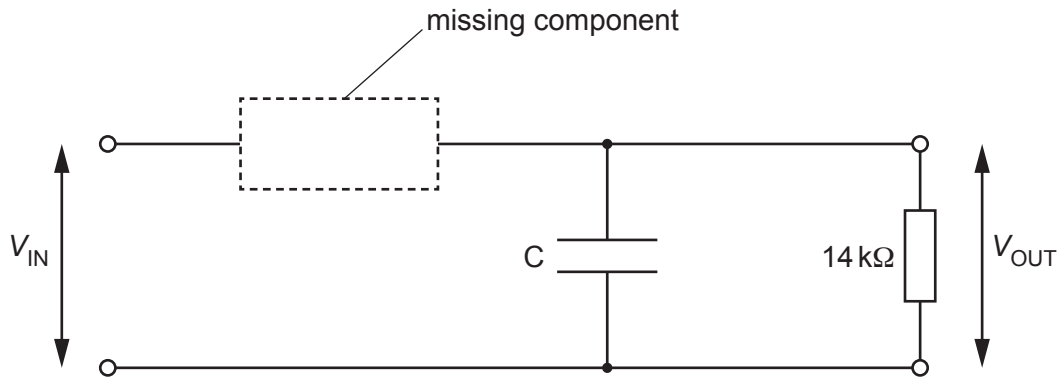


Fig. 5.1

The circuit is used to produce half-wave rectification of an alternating voltage of potential difference (p.d.) V_{IN} .

The output p.d. across the $14\text{ k}\Omega$ resistor is V_{OUT} .

(a) (i) A component is missing from the circuit of Fig. 5.1.

Complete the circuit diagram in Fig. 5.1 by adding the circuit symbol for the missing component, correctly connected. [1]

(ii) A capacitor C is shown in the circuit of Fig. 5.1.

State the effect on V_{OUT} of including the capacitor in the circuit.

..... [1]

(b) Fig. 5.2 shows the variation with time t of V_{IN} .

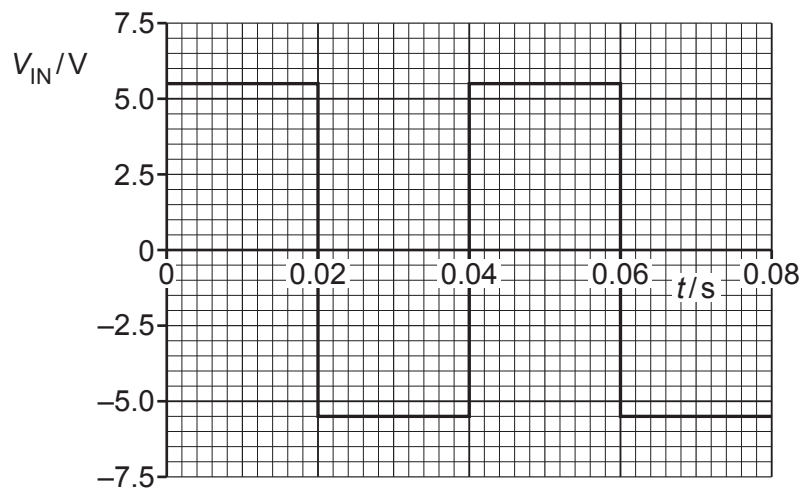


Fig. 5.2

Fig. 5.3 shows the variation with t of V_{OUT} .

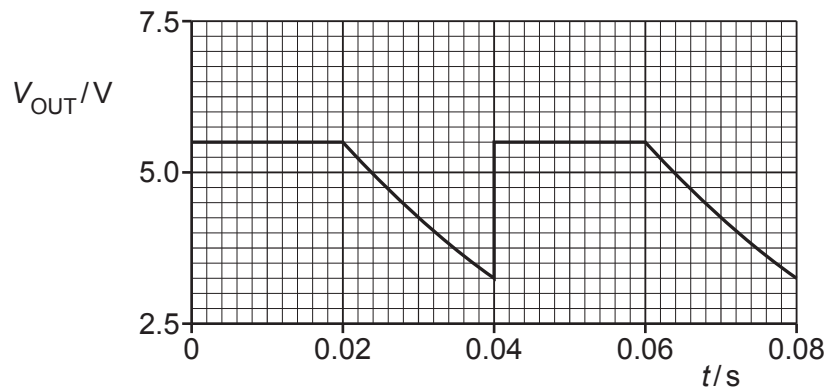


Fig. 5.3

- (i) Determine the frequency of V_{IN} .

frequency = Hz [1]

- (ii) Show that the time constant τ for the discharge of the capacitor through the resistor is 0.038 s.

[2]

- (iii) Calculate the capacitance of C. Give a unit with your answer.

capacitance = unit [2]

- (c) The circuit of Fig. 5.1 is modified so that it produces full-wave rectification of an input voltage.

Suggest, with a reason, how V_{OUT} now varies with time when V_{IN} is as shown in Fig. 5.2.

.....

 [2]

[Total: 9]

- 6 (a) State what is meant by a magnetic field.

.....

.....

..... [2]

- (b) A long, straight wire P carries a current into the page, as shown in Fig. 6.1.

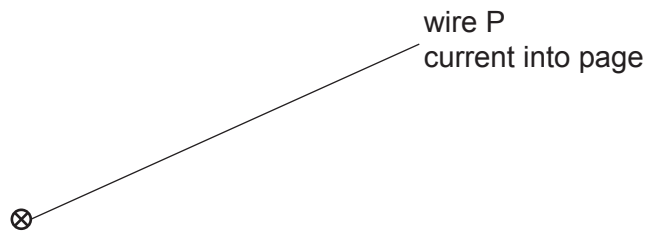


Fig. 6.1

On Fig. 6.1, draw four field lines to represent the magnetic field around wire P due to the current in the wire. [3]

- (c) A second long, straight wire Q, carrying a current of 5.0A out of the page, is placed parallel to wire P, as shown in Fig. 6.2.

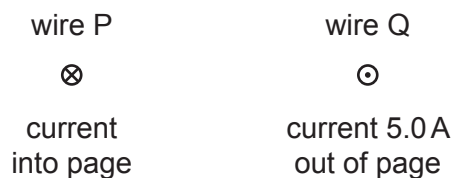


Fig. 6.2

The flux density of the magnetic field at wire Q due to the current in wire P is 2.6 mT.

- (i) Calculate the magnetic force per unit length exerted on wire Q by wire P.

force per unit length = Nm^{-1} [2]

- (ii) State the direction of the force exerted on wire Q by wire P.

..... [1]

- (iii) The flux density of the magnetic field at wire P due to the current in wire Q is 1.5 mT.

Determine the magnitude of the current in wire P. Explain your reasoning.

current = A [2]

[Total: 10]

- 7 (a) State what is meant by the de Broglie wavelength.

.....
 [1]

- (b) Fig. 7.1 shows a glass tube in which electrons are accelerated through a high p.d. to form a beam that is incident on a thin graphite crystal.

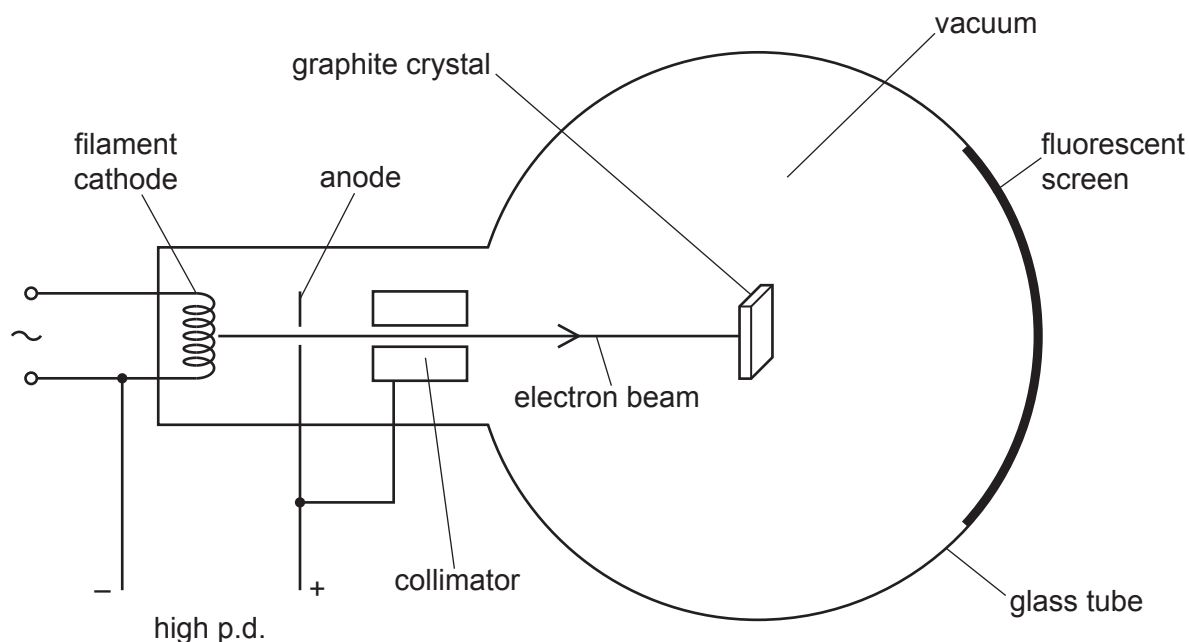


Fig. 7.1 (not to scale)

After passing through the graphite crystal, the electrons reach the fluorescent screen. The screen glows where the electrons strike it.

Fig. 7.2 shows the fluorescent screen viewed end-on, from the right-hand side of Fig. 7.1.

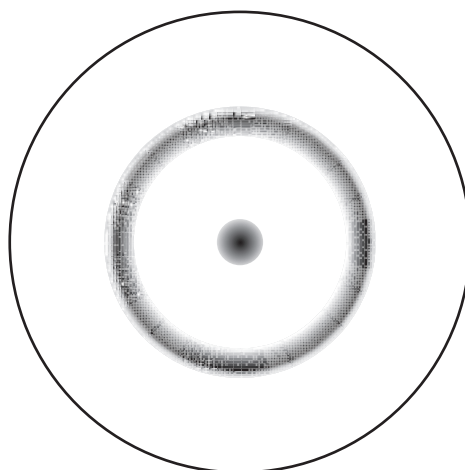


Fig. 7.2

- (i) State the name of the phenomenon demonstrated by the pattern shown in Fig. 7.2.

..... [1]

- (ii) Explain what can be concluded from the pattern in Fig. 7.2 about the nature of electrons.

.....

 [2]

- (c) The electrons in (b) are now accelerated through a greater potential difference between the cathode and the anode.

- (i) On Fig. 7.3, sketch the pattern that is now seen on the fluorescent screen in Fig. 7.1.

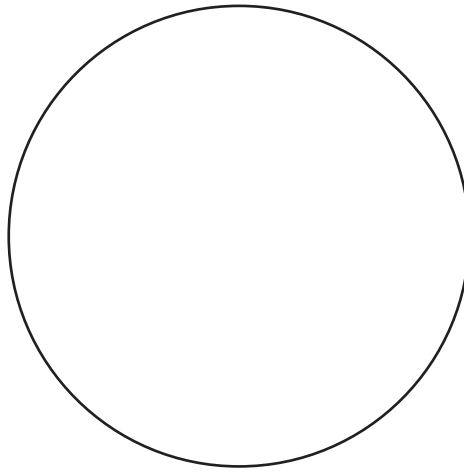


Fig. 7.3

[2]

- (ii) Explain, with reference to de Broglie wavelength, the change in the pattern on the fluorescent screen.

.....

 [3]

[Total: 9]

- 8 (a) Table 8.1 shows some data relating to the properties of air, gel and body tissue. The data are given to three significant figures.

Table 8.1

material	density / kg m^{-3}	speed of sound / m s^{-1}	specific acoustic impedance / $\text{kg m}^{-2} \text{s}^{-1}$
air		340	440
gel	1200	1400	
tissue	1090		1.68×10^6

- (i) Show that the specific acoustic impedance of gel is $1.68 \times 10^6 \text{ kg m}^{-2} \text{s}^{-1}$.

[1]

- (ii) Complete Table 8.1 by calculating the missing values to three significant figures. Use the space below for any working that you need.

[2]

- (b) Use the information in (a) to calculate the intensity reflection coefficient for:

- (i) an air–tissue boundary

intensity reflection coefficient = [2]

- (ii) a gel–tissue boundary.

intensity reflection coefficient = [1]

(c) Use your answers in (b) to explain why gel is applied to the skin during ultrasound scanning.

.....

.....

..... [2]

[Total: 8]

- 9 Carbon-11 is radioactive and decays by β^+ emission to form boron-11. Carbon-11 has a half-life of 20 minutes. Boron-11 is stable.

(a) Define half-life.

.....
 [1]

(b) A sample contains N_0 nuclei of carbon-11 and no other nuclei at time $t = 0$.

On Fig. 9.1, sketch the variation with t of the number of nuclei of **boron-11** in the sample.

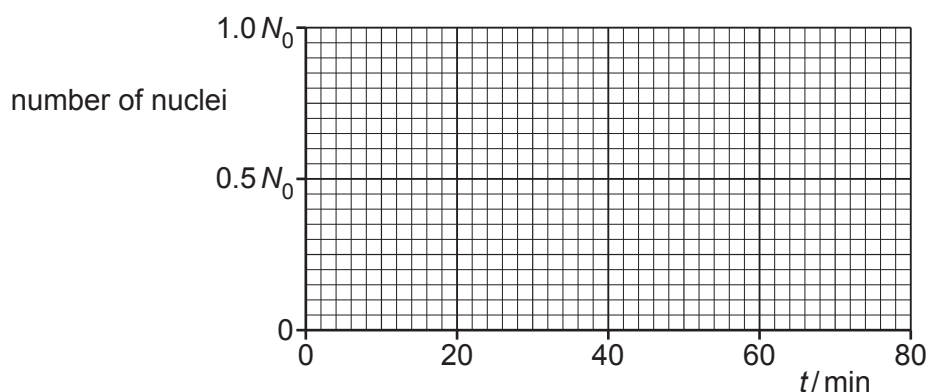


Fig. 9.1

[3]

(c) (i) Explain, with reference to the random nature of radioactive decay, why the activity of the carbon-11 sample in (b) decreases with time.

.....

 [2]

(ii) State, with reasons, whether a radiation detector placed near to the sample of carbon-11 indicates a measured count rate from the sample that is less than, the same as or greater than the activity of the sample.

.....

 [3]

[Total: 9]

- 10 (a) State Hubble's law. Identify any symbols that you use.

.....

.....

.....

..... [2]

- (b) A star of luminosity $3.8 \times 10^{31} \text{ W}$ is a distance of $1.8 \times 10^{24} \text{ m}$ from the Earth.

Calculate the radiant flux intensity at the Earth of the radiation emitted by the star.

radiant flux intensity = W m^{-2} [2]

- (c) The star in (b) is in a distant galaxy. A spectral line in the light from this galaxy is known to have a wavelength of 486 nm. This spectral line in the light from the galaxy observed on the Earth has a wavelength of 492 nm.

- (i) Explain why the wavelength observed on the Earth is different from the wavelength that the galaxy is known to have emitted.

.....

.....

..... [2]

- (ii) Determine a value for the Hubble constant H_0 .

$H_0 = \dots\dots\dots \text{s}^{-1}$ [3]

[Total: 9]

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9702/42

May/June 2023

2 hours

You must answer on the question paper.

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [].

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Data

acceleration of free fall	$g = 9.81 \text{ m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
unified atomic mass unit	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27} \text{ kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$ ($\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ m F}^{-1}$)
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_o = \frac{f_s v}{v \pm v_s}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \dots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$

gravitational potential	$\phi = -\frac{GM}{r}$
gravitational potential energy	$E_P = -\frac{GMm}{r}$
pressure of an ideal gas	$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$
simple harmonic motion	$a = -\omega^2 x$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t$ $v = \pm \omega \sqrt{(x_0^2 - x^2)}$
electric potential	$V = \frac{Q}{4\pi\epsilon_0 r}$
electrical potential energy	$E_P = \frac{Qq}{4\pi\epsilon_0 r}$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$
capacitors in parallel	$C = C_1 + C_2 + \dots$
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alternating current/voltage	$x = x_0 \sin \omega t$
radioactive decay	$x = x_0 e^{-\lambda t}$
decay constant	$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$
intensity reflection coefficient	$\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$
Stefan–Boltzmann law	$L = 4\pi\sigma r^2 T^4$
Doppler redshift	$\frac{\Delta\lambda}{\lambda} \hat{=} \frac{Df}{f} \hat{=} \frac{v}{c}$

- 1 (a) State Newton's law of gravitation.

.....

.....

..... [2]

- (b) A satellite is in a circular orbit around a planet. The radius of the orbit is R and the period of the orbit is T . The planet is a uniform sphere.

Use Newton's law of gravitation to show that R and T are related by

$$4\pi^2 R^3 = GMT^2$$

where M is the mass of the planet and G is the gravitational constant.

[2]

- (c) The Earth may be considered to be a uniform sphere of mass 5.98×10^{24} kg and radius 6.37×10^6 m.

A geostationary satellite is in orbit around the Earth.

Use the expression in (b) to determine the height of the satellite above the Earth's surface.

height = m [3]

(d) Another satellite is in a circular orbit around the Earth with the same orbital radius and period as the satellite in (c).

(i) Calculate the angular speed of the satellite in this orbit. Give a unit with your answer.

angular speed = unit [2]

(ii) Despite having the same orbital period, the orbit of this satellite is not geostationary.

Suggest **two** ways in which the orbit of this satellite could be different from the orbit of the satellite in (c).

1

.....

2

.....

[2]

[Total: 11]

- 2 (a) (i) State what is meant by an ideal gas.

.....

 [2]

- (ii) State the temperature, in degrees Celsius, of absolute zero.

temperature = °C [1]

- (b) A sealed vessel contains a mass of 0.0424 kg of an ideal gas at 227 °C.
 The pressure of the gas is 1.37×10^5 Pa and the volume of the gas is 0.640 m³.

Calculate:

- (i) the number of molecules of the gas in the vessel

number of molecules = [3]

- (ii) the mass of one molecule of the gas

mass = kg [1]

- (iii) the root-mean-square (r.m.s.) speed v of the molecules of the gas.

$v =$ ms⁻¹ [3]

- (c) The gas in (b) is now cooled gradually to absolute zero.

On Fig. 2.1, sketch the variation with thermodynamic temperature T of the r.m.s. speed of the molecules of the gas.

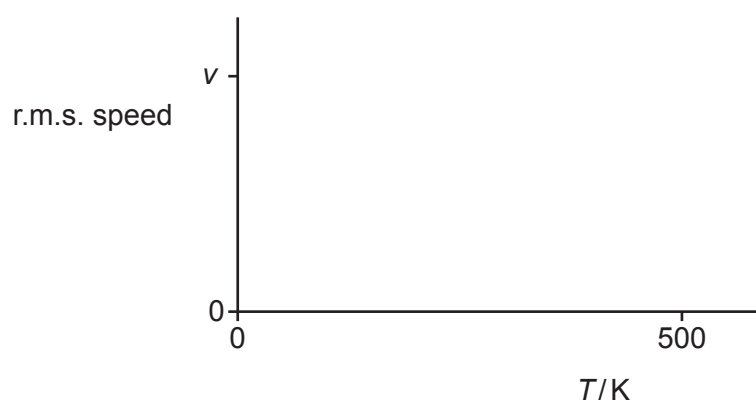


Fig. 2.1

[2]

[Total: 12]

- 3 (a) State the first law of thermodynamics. Identify the meaning of any symbols that you use.

.....

.....

..... [2]

- (b) The state of an ideal gas is continuously changed according to the cycle ABCDA shown in Fig. 3.1.

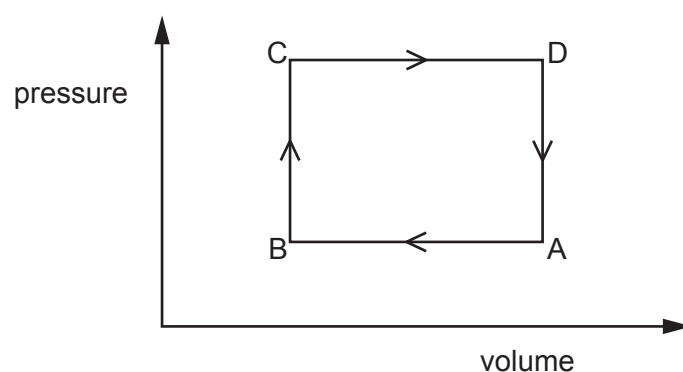


Fig. 3.1

- (i) Complete Table 3.1 for the changes A to B and B to C by placing two ticks (✓) in each row.

Table 3.1

change	change in internal energy			work done on gas		
	decrease	no change	increase	negative	zero	positive
A to B						
B to C						

[4]

- (ii) Use the first law of thermodynamics to describe and explain the energy transfers associated with one complete cycle ABCDA.

.....

.....

.....

.....

..... [3]

[Total: 9]

- 4 A small steel sphere is oscillating vertically on the end of a spring, as shown in Fig. 4.1.

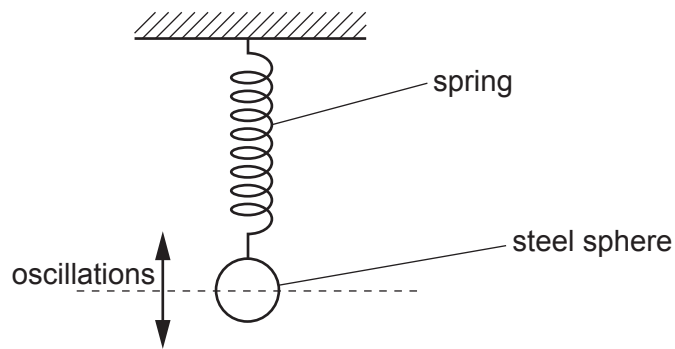


Fig. 4.1

The velocity v of the sphere varies with displacement x from its equilibrium position according to

$$v = \pm 9.7 \sqrt{(11.6 - x^2)}$$

where v is in cm s^{-1} and x is in cm.

- (a) (i) Calculate the frequency of the oscillations.

frequency = Hz [2]

- (ii) Show that the amplitude of the oscillations is 3.4 cm.

[1]

- (iii) Calculate the maximum acceleration a_0 of the sphere.

$a_0 = \dots \text{ms}^{-2}$ [2]

- (b) On Fig. 4.2, sketch the variation with x of the acceleration a of the sphere.

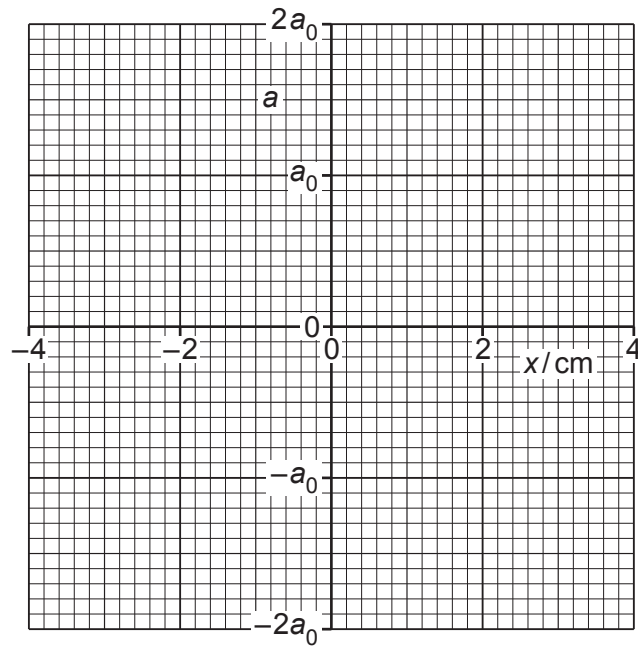


Fig. 4.2

[3]

- (c) Describe, without calculation, the interchange between the potential energy and the kinetic energy of the oscillations.

.....

.....

.....

.....

..... [3]

[Total: 11]

- 5 Two capacitors A and B are connected into the circuit shown in Fig. 5.1.

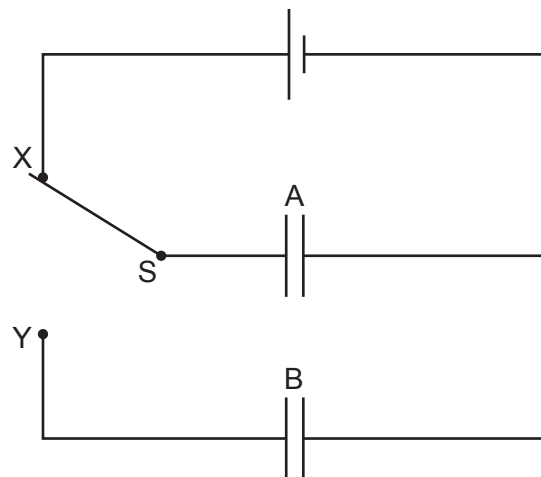


Fig. 5.1

Capacitor A has capacitance C and capacitor B has capacitance $3C$.

The electromotive force (e.m.f.) of the cell is V .

The two-way switch S is initially at position X, and capacitor B is initially uncharged.

- (a) State, in terms of V and C , expressions for:

- (i) the initial charge Q_A on the plates of capacitor A

$$Q_A = \dots\dots\dots [1]$$

- (ii) the initial energy E_A stored in capacitor A.

$$E_A = \dots\dots\dots [1]$$

- (b) The two-way switch S is now moved to position Y.

- (i) State and explain what happens to the charge that was initially on the plates of capacitor A.

.....

 [2]

- (ii) Show that the final potential difference (p.d.) V_B across capacitor B is given by

$$V_B = \frac{V}{4}.$$

Explain your reasoning.

[3]

- (iii) Determine an expression, in terms of V and C , for the decrease ΔE in the total energy that is stored in the capacitors as a result of the change of the position of the switch.

$$\Delta E = \dots\dots\dots [2]$$

[Total: 9]

- 6 A heavy aluminium disc has a radius of 0.36 m. The disc rotates with the wheels of a vehicle and forms part of an electromagnetic braking system on the vehicle.

In order to activate the braking system, a uniform magnetic field of flux density 0.17 T is switched on. This magnetic field is perpendicular to the plane of rotation of the disc, as shown in Fig. 6.1.

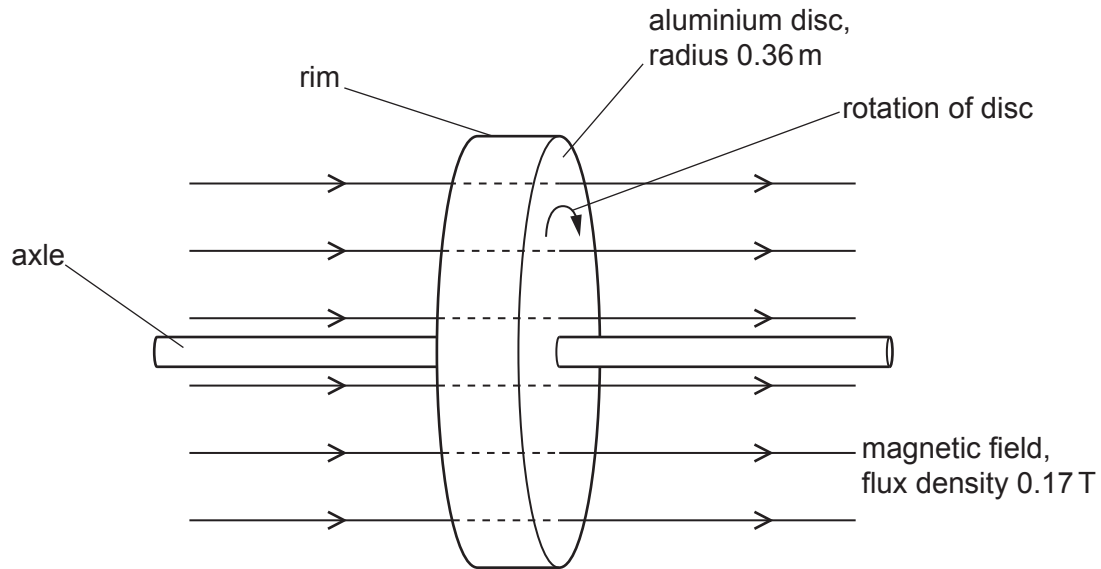


Fig. 6.1

- (a) (i) Define magnetic flux.

.....

.....

..... [2]

- (ii) Calculate the magnetic flux through the disc. Give a unit with your answer.

magnetic flux = unit [2]

- (b) The disc is rotating at a rate of 25 revolutions per second.

Calculate the magnitude of the electromotive force (e.m.f.) induced between the axle and the rim of the disc.

e.m.f. = V [3]

- (c) The axle and the rim are connected into an external circuit that enables the energy of the rotation of the disc to be stored for future use. The direction of rotation is shown in Fig. 6.1.

Use Lenz's law of electromagnetic induction to determine whether the current in the disc is from the rim to the axle or from the axle to the rim. Explain your reasoning.

.....

.....

.....

.....

..... [3]

[Total: 10]

- 7 Four diodes are used in a bridge rectifier circuit to produce rectification of a sinusoidal a.c. input voltage V_{IN} .
Fig. 7.1 shows part of the circuit, but three of the diodes are missing.

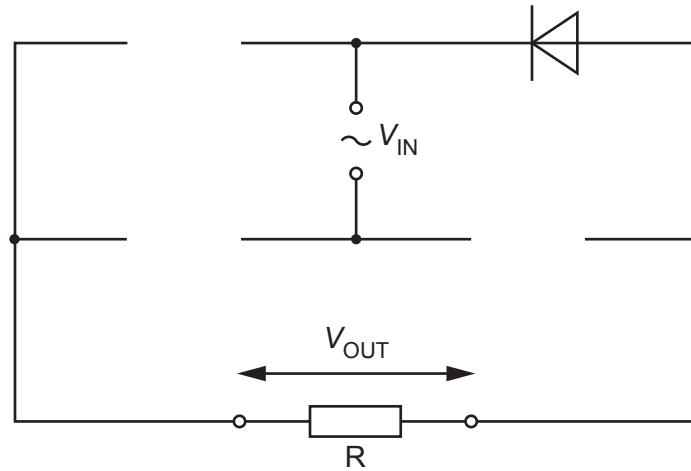


Fig. 7.1

The p.d. across the load resistor R is the output p.d. V_{OUT} of the bridge rectifier.

- (a) (i) State the name of the type of rectification produced by a bridge rectifier.
..... [1]
- (ii) Complete Fig. 7.1 by drawing the three missing diodes, correctly connected. [2]
- (iii) On Fig. 7.1, draw an arrow to indicate the direction of the current in resistor R . [1]
- (b) V_{IN} has amplitude V_0 and period T . Fig. 7.2 shows the variation with time t of V_{IN} .

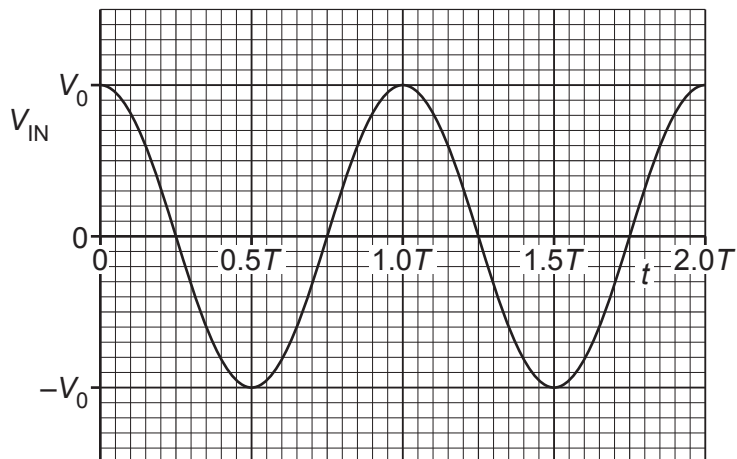


Fig. 7.2

- (i) On Fig. 7.3, sketch the variation of V_{OUT} with t between $t = 0$ and $t = 2.0T$.

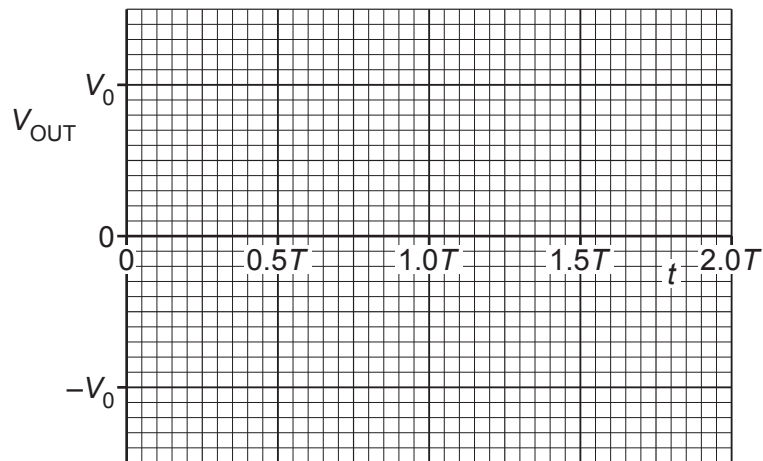


Fig. 7.3

[3]

- (ii) The power dissipated in the resistor is P .

On Fig. 7.4, sketch the variation of P with t between $t = 0$ and $t = 2.0T$.

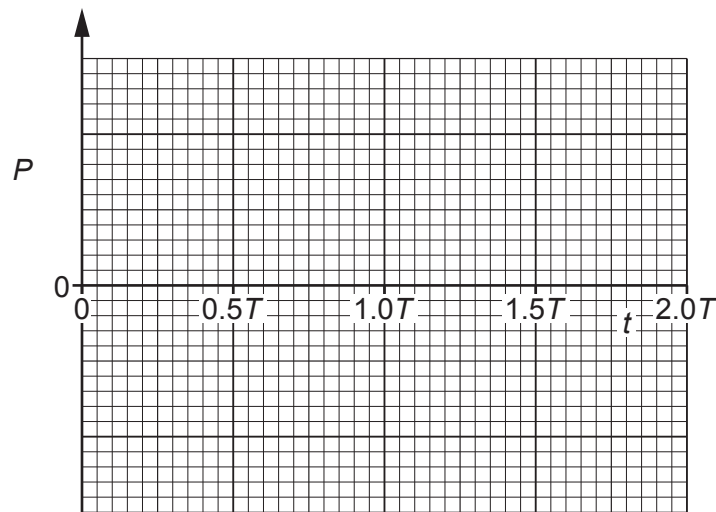


Fig. 7.4

[2]

- (iii) Suggest, with a reason, how the root-mean-square (r.m.s.) value of V_{OUT} compares with the r.m.s. value of V_{IN} .

.....

..... [1]

[Total: 10]

- 8 Fig. 8.1 shows the lowest four energy levels of an electron in an isolated atom.

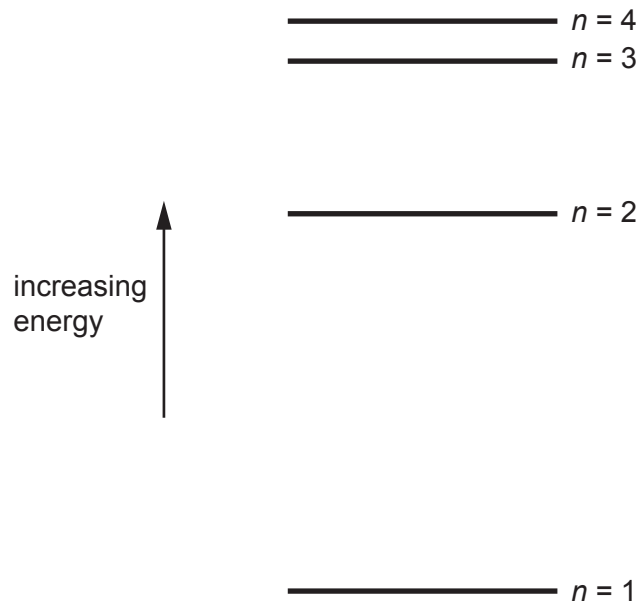


Fig. 8.1

Fig. 8.2 shows the lines in the emission spectrum of the atom that correspond to the transitions of the electron from $n = 3$ to $n = 1$ and from $n = 4$ to $n = 1$.

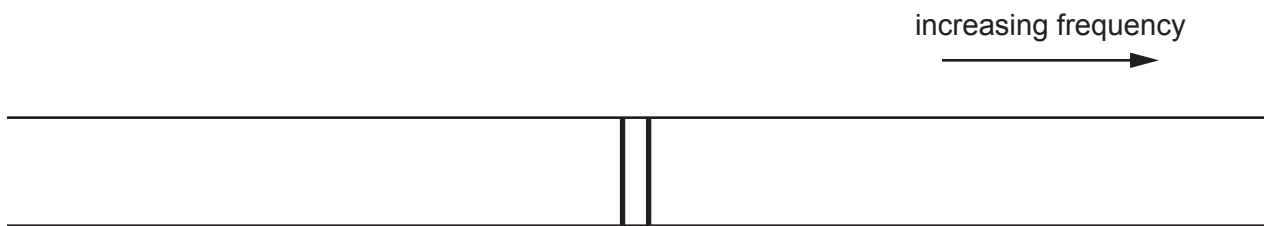


Fig. 8.2

- (a) Explain, with reference to photons, why there is a single frequency of electromagnetic radiation that corresponds to each of these transitions.

..... [2]

.....

.....

- (b) (i) On Fig. 8.2, draw a line that corresponds to the transition of the electron from $n = 2$ to $n = 1$. Label this line A. [2]
- (ii) On Fig. 8.2, draw a line that corresponds to the transition of the electron from $n = 3$ to $n = 2$. Label this line B. [2]

- (c) The frequency of radiation represented by line A is f_A .
The frequency of radiation represented by line B is f_B .
The energy of the ground state ($n = 1$) is E_1 .

Determine an expression, in terms of f_A , f_B , E_1 and the Planck constant h , for the energy E_3 of the energy level $n = 3$.

$$E_3 = \dots\dots\dots [2]$$

[Total: 8]

- 9 (a) Define mass defect.

.....

.....

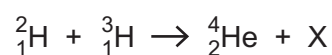
..... [2]

- (b) Table 9.1 shows the mass defects of three nuclei.

Table 9.1

nucleus	mass defect / u
${}^2_1\text{H}$	0.002 388
${}^3_1\text{H}$	0.009 105
${}^4_2\text{He}$	0.030 377

The nuclear fusion process in a particular star is described by



where X is a particle that has no mass defect.

- (i) State the name of particle X.

..... [1]

- (ii) Show that the energy released when one nucleus of ${}^4_2\text{He}$ is formed in this fusion reaction is $2.8 \times 10^{-12} \text{ J}$.

[3]

- (c) The star in (b) has a radius of $2.3 \times 10^9 \text{ m}$ and a luminosity of $1.4 \times 10^{28} \text{ W}$.
 All the energy released from the formation of ${}^4_2\text{He}$ is radiated away from the star.
 All the energy that is radiated from the star has been released in the formation of ${}^4_2\text{He}$.

Determine:

- (i) the mass of ${}^4_2\text{He}$ produced per unit time by the fusion process

mass per unit time = kg s^{-1} [3]

- (ii) the surface temperature of the star.

temperature = K [2]

[Total: 11]

- 10 (a) X-rays for use in medical diagnosis are produced in an X-ray tube. In the X-ray tube, charged particles are accelerated towards a metal target by an applied potential difference (p.d.).

(i) State the name of the charged particles that are accelerated by the applied p.d.

..... [1]

(ii) Explain how X-rays are produced at the metal target.

.....

.....

.....

..... [2]

(iii) Calculate the minimum wavelength of X-rays produced when the applied p.d. is 5.80 kV.

wavelength = m [3]

(b) X-rays pass through a medium that has an attenuation coefficient of 1.4 cm^{-1} .

Calculate the percentage of the X-ray energy that is **absorbed** by a 2.8 cm thickness of this medium.

percentage absorbed = % [3]

[Total: 9]

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9702/43

May/June 2023

2 hours

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Stefan–Boltzmann law	$L = 4\pi\sigma r^2 T^4$
Doppler redshift	$\frac{\Delta\lambda}{\lambda} \approx \frac{Df}{f} \approx \frac{v}{c}$

- 1 (a) (i) Define gravitational field.

.....
 [1]

- (ii) Define electric field.

.....
 [1]

- (iii) State **one** similarity and **one** difference between the gravitational potential due to a point mass and the electric potential due to a point charge.

similarity:

.....

difference:

.....

[2]

- (b) An isolated uniform conducting sphere has mass M and charge Q .
 The gravitational field strength at the surface of the sphere is g .
 The electric field strength at the surface of the sphere is E .

- (i) Show that

$$\frac{M}{Q} = \alpha \frac{g}{E}$$

where α is a constant.

[3]

- (ii) Show that the numerical value of α is $1.35 \times 10^{20} \text{ kg}^2 \text{ C}^{-2}$.

[1]

- (c) Assume that the Earth is a uniform conducting sphere of mass 5.98×10^{24} kg. The surface of the Earth carries a charge of -4.80×10^5 C that is evenly distributed.
- (i) Use the information in (b) to determine the electric field strength at the surface of the Earth. Give a unit with your answer.

electric field strength = unit [2]

- (ii) State how the direction of the electric field at the surface of the Earth compares with the direction of the gravitational field.

..... [1]

[Total: 11]

- 2 A steel sphere of mass 0.29 kg is suspended in equilibrium from a vertical spring. The centre of the sphere is 8.5 cm from the top of the spring, as shown in Fig. 2.1.

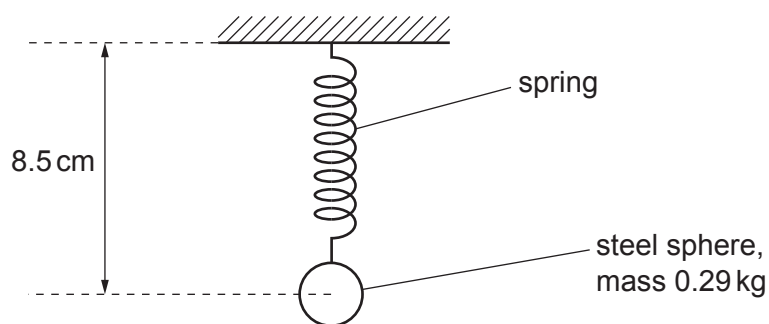


Fig. 2.1

The sphere is now set in motion so that it is moving in a horizontal circle at constant speed, as shown in Fig. 2.2.

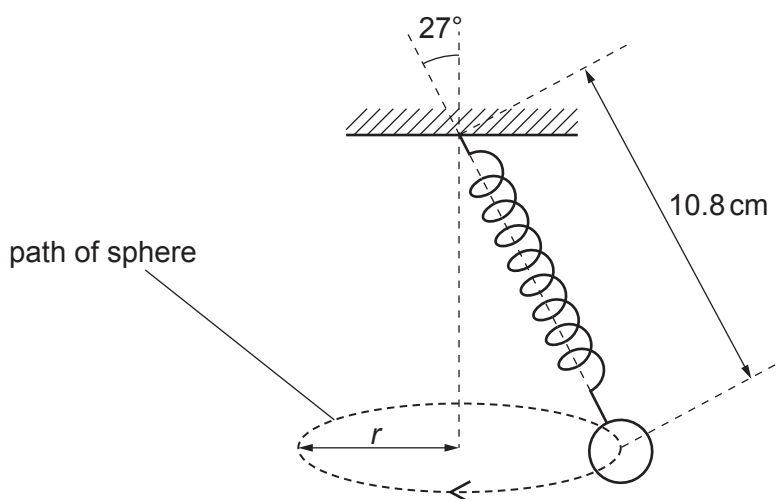


Fig. 2.2

The distance from the centre of the sphere to the top of the spring is now 10.8 cm .

- (a) Explain, with reference to the forces acting on the sphere, why the length of the spring in Fig. 2.2 is greater than in Fig. 2.1.

.....

.....

.....

.....

..... [3]

(b) The angle between the linear axis of the spring and the vertical is 27° .

(i) Show that the radius r of the circle is 4.9 cm.

[1]

(ii) Show that the tension in the spring is 3.2 N.

[2]

(iii) The spring obeys Hooke's law.

Calculate the spring constant, in N cm^{-1} , of the spring.

spring constant = N cm^{-1} [2]

(c) (i) Use the information in (b) to determine the centripetal acceleration of the sphere.

centripetal acceleration = ms^{-2} [2]

(ii) Calculate the period of the circular motion of the sphere.

period = s [2]

[Total: 12]

- 3 (a) State the reason why two objects that are at the same temperature are described as being in thermal equilibrium.

.....
 [1]

- (b) Fig. 3.1 shows the variations with temperature of the densities of mercury and of water between 0 °C and 100 °C.

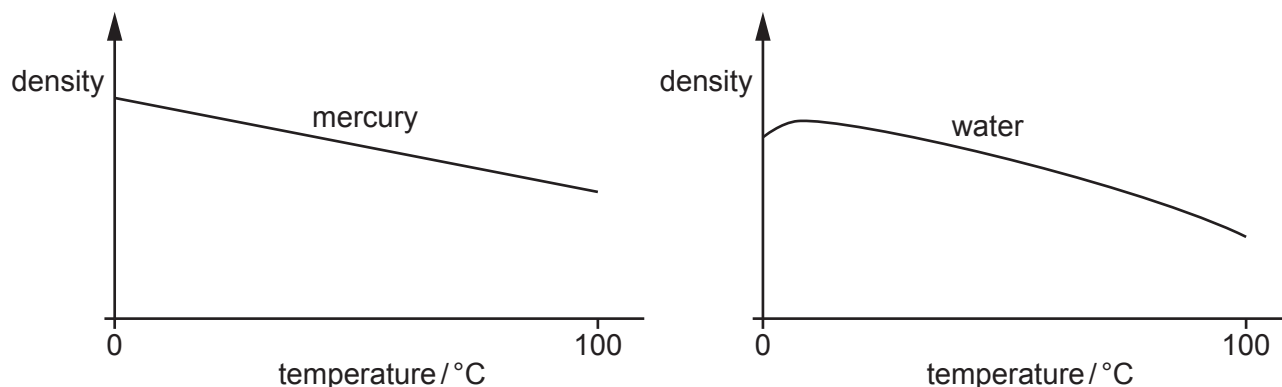


Fig. 3.1

Temperature may be measured using the variation with temperature of the density of a liquid.

Suggest why, for measuring temperature over this temperature range:

- (i) mercury is a suitable liquid

.....
 [1]

- (ii) water is not a suitable liquid.

.....

 [2]

- (c) A beaker contains a liquid of mass 120 g. The liquid is supplied with thermal energy at a rate of 810 W. The beaker has a mass of 42 g and a specific heat capacity of $0.84 \text{ J g}^{-1} \text{ K}^{-1}$. The beaker and the liquid are in thermal equilibrium with each other at all times and are insulated from the surroundings.

Fig. 3.2 shows the variation with time t of the temperature of the liquid.

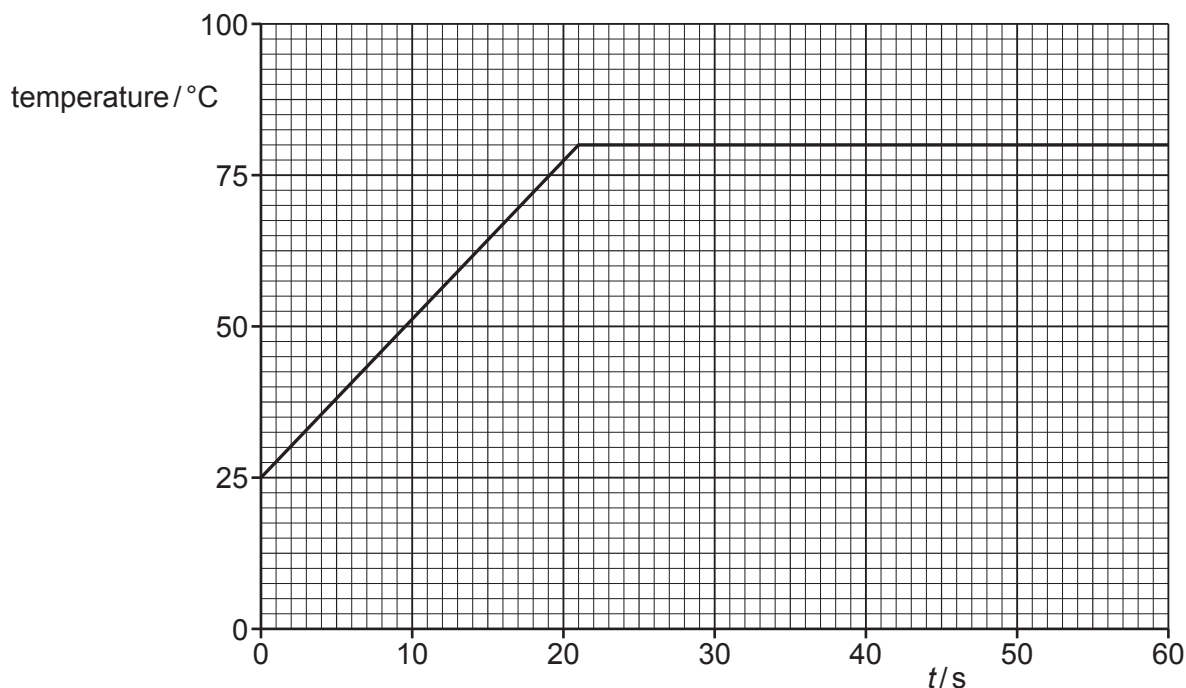


Fig. 3.2

- (i) State the boiling temperature, in °C, of the liquid.

temperature = °C [1]

- (ii) Determine the specific heat capacity, in $\text{J g}^{-1} \text{K}^{-1}$, of the liquid.

specific heat capacity = $\text{J g}^{-1} \text{K}^{-1}$ [4]

- (d) The experiment in (c) is repeated using water instead of the liquid in (c). The mass of liquid used, the power supplied, and the initial temperature are all unchanged.
The specific heat capacity of water is approximately twice that of the liquid in (c).
The boiling temperature of water is 100 °C.

On Fig. 3.2, sketch the variation with time t of the temperature of the water between $t = 0$ and $t = 60$ s. Numerical calculations are not required. [2]

[Total: 11]

- 4 (a) State **two** of the basic assumptions of the kinetic theory of gases.

1

.....

2

.....

[2]

- (b) An ideal gas has amount of substance n .

The gas is initially in state X, with pressure $2p$ and volume V .

The gas is cooled at constant volume to state Y, with pressure p .

The gas is then heated at constant pressure to state Z, with volume $2V$.

Finally, the gas returns at constant temperature to state X.

- (i) Determine an expression for the temperature T of the gas in state X, in terms of n , p and V .
Identify any other symbols that you use.

[2]

- (ii) On Fig. 4.1, sketch the variation with volume of pressure for the gas as the gas undergoes the three changes. The state X is labelled. Label states Y and Z.

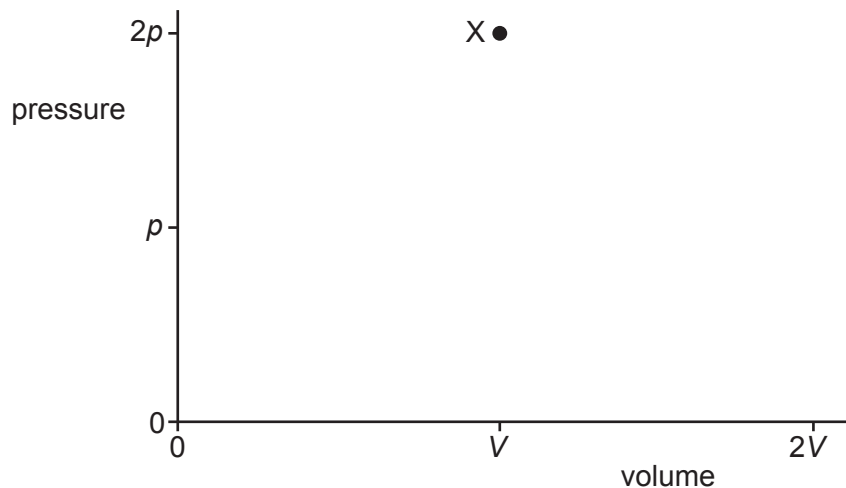


Fig. 4.1

[3]

- (iii) During the change of state from Y to Z, the increase in internal energy of the gas is U .
During the change of state from Z to X, the work done on the gas is W .

Complete Table 4.1 to indicate, for each of the three changes of state, the increase in internal energy of the gas, the thermal energy transferred to the gas and the work done on the gas, in terms of p , V , U and W .

Table 4.1

change	increase in internal energy of gas	thermal energy transferred to gas	work done on gas
X to Y			
Y to Z	$+U$		
Z to X			$+W$

[5]

[Total: 12]

5 Part of an electric circuit is shown in Fig. 5.1.

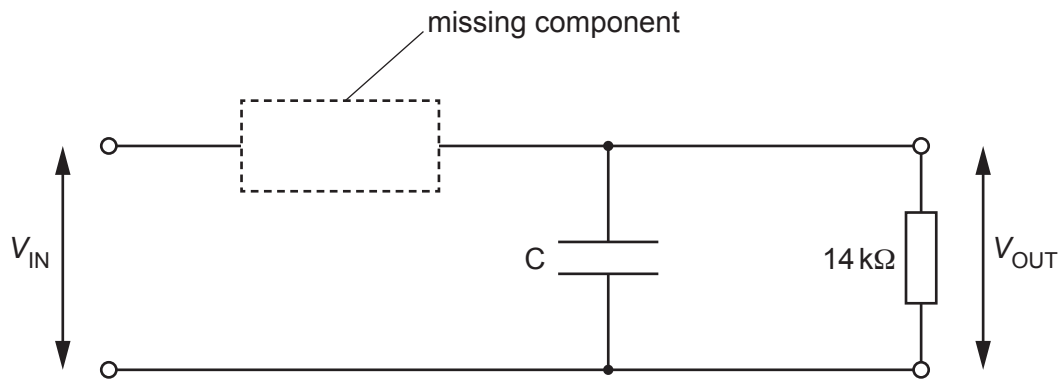


Fig. 5.1

The circuit is used to produce half-wave rectification of an alternating voltage of potential difference (p.d.) V_{IN} .

The output p.d. across the $14\text{ k}\Omega$ resistor is V_{OUT} .

(a) (i) A component is missing from the circuit of Fig. 5.1.

Complete the circuit diagram in Fig. 5.1 by adding the circuit symbol for the missing component, correctly connected. [1]

(ii) A capacitor C is shown in the circuit of Fig. 5.1.

State the effect on V_{OUT} of including the capacitor in the circuit.

..... [1]

(b) Fig. 5.2 shows the variation with time t of V_{IN} .

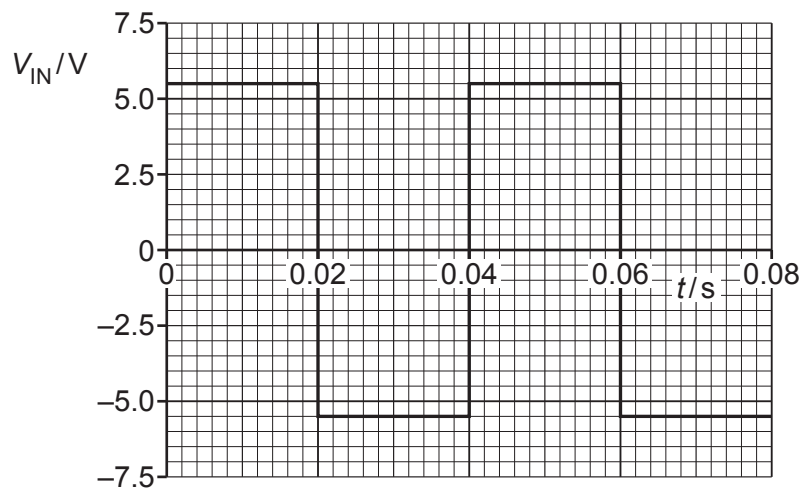


Fig. 5.2

Fig. 5.3 shows the variation with t of V_{OUT} .

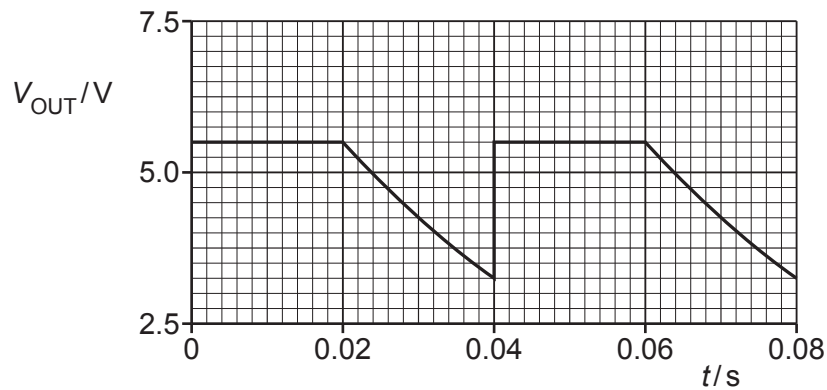


Fig. 5.3

- (i) Determine the frequency of V_{IN} .

frequency = Hz [1]

- (ii) Show that the time constant τ for the discharge of the capacitor through the resistor is 0.038 s.

[2]

- (iii) Calculate the capacitance of C. Give a unit with your answer.

capacitance = unit [2]

- (c) The circuit of Fig. 5.1 is modified so that it produces full-wave rectification of an input voltage.

Suggest, with a reason, how V_{OUT} now varies with time when V_{IN} is as shown in Fig. 5.2.

.....

 [2]

[Total: 9]

- 6 (a) State what is meant by a magnetic field.

.....

.....

..... [2]

- (b) A long, straight wire P carries a current into the page, as shown in Fig. 6.1.

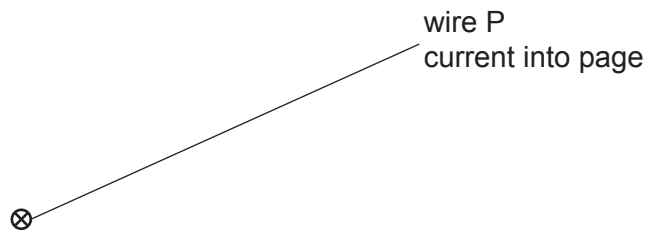


Fig. 6.1

On Fig. 6.1, draw four field lines to represent the magnetic field around wire P due to the current in the wire. [3]

- (c) A second long, straight wire Q, carrying a current of 5.0A out of the page, is placed parallel to wire P, as shown in Fig. 6.2.

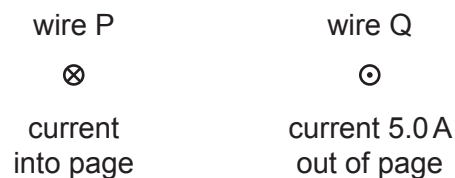


Fig. 6.2

The flux density of the magnetic field at wire Q due to the current in wire P is 2.6 mT.

- (i) Calculate the magnetic force per unit length exerted on wire Q by wire P.

force per unit length = Nm^{-1} [2]

- (ii) State the direction of the force exerted on wire Q by wire P.

..... [1]

- (iii) The flux density of the magnetic field at wire P due to the current in wire Q is 1.5 mT.

Determine the magnitude of the current in wire P. Explain your reasoning.

current = A [2]

[Total: 10]

- 7 (a) State what is meant by the de Broglie wavelength.

.....
 [1]

- (b) Fig. 7.1 shows a glass tube in which electrons are accelerated through a high p.d. to form a beam that is incident on a thin graphite crystal.

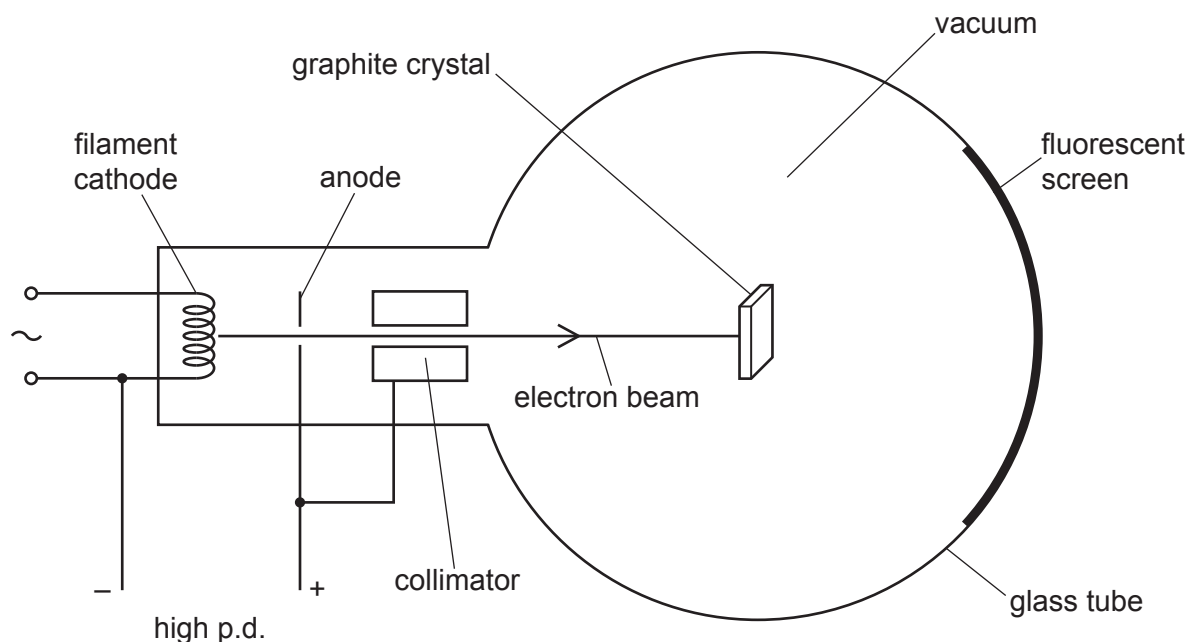


Fig. 7.1 (not to scale)

After passing through the graphite crystal, the electrons reach the fluorescent screen. The screen glows where the electrons strike it.

Fig. 7.2 shows the fluorescent screen viewed end-on, from the right-hand side of Fig. 7.1.

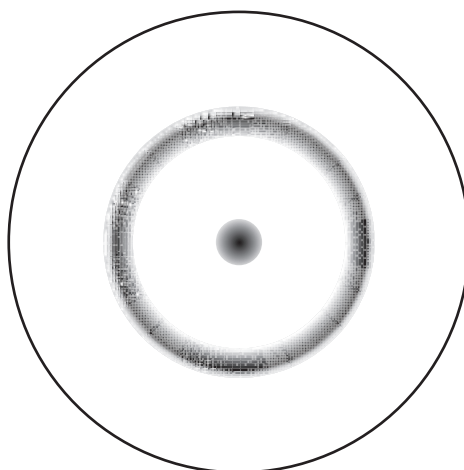


Fig. 7.2

- (i) State the name of the phenomenon demonstrated by the pattern shown in Fig. 7.2.

..... [1]

- (ii) Explain what can be concluded from the pattern in Fig. 7.2 about the nature of electrons.

.....

.....

..... [2]

- (c) The electrons in (b) are now accelerated through a greater potential difference between the cathode and the anode.

- (i) On Fig. 7.3, sketch the pattern that is now seen on the fluorescent screen in Fig. 7.1.

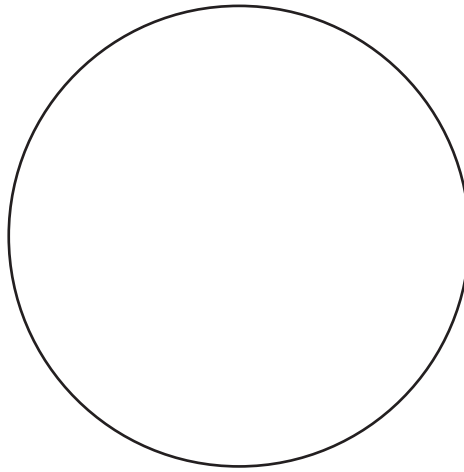


Fig. 7.3

[2]

- (ii) Explain, with reference to de Broglie wavelength, the change in the pattern on the fluorescent screen.

.....

.....

.....

.....

..... [3]

[Total: 9]

- 8 (a) Table 8.1 shows some data relating to the properties of air, gel and body tissue. The data are given to three significant figures.

Table 8.1

material	density / kg m^{-3}	speed of sound / m s^{-1}	specific acoustic impedance / $\text{kg m}^{-2} \text{s}^{-1}$
air		340	440
gel	1200	1400	
tissue	1090		1.68×10^6

- (i) Show that the specific acoustic impedance of gel is $1.68 \times 10^6 \text{ kg m}^{-2} \text{s}^{-1}$.

[1]

- (ii) Complete Table 8.1 by calculating the missing values to three significant figures. Use the space below for any working that you need.

[2]

- (b) Use the information in (a) to calculate the intensity reflection coefficient for:

- (i) an air–tissue boundary

intensity reflection coefficient = [2]

- (ii) a gel–tissue boundary.

intensity reflection coefficient = [1]

(c) Use your answers in (b) to explain why gel is applied to the skin during ultrasound scanning.

.....

.....

..... [2]

[Total: 8]

- 9 Carbon-11 is radioactive and decays by β^+ emission to form boron-11. Carbon-11 has a half-life of 20 minutes. Boron-11 is stable.

(a) Define half-life.

.....
 [1]

(b) A sample contains N_0 nuclei of carbon-11 and no other nuclei at time $t = 0$.

On Fig. 9.1, sketch the variation with t of the number of nuclei of **boron-11** in the sample.

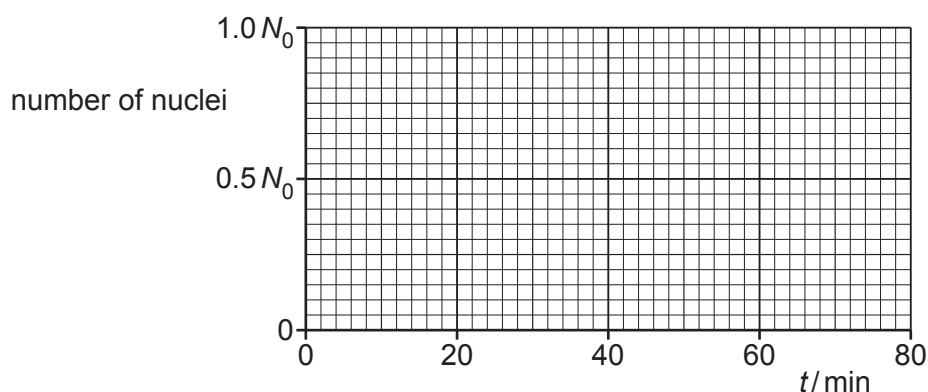


Fig. 9.1

[3]

(c) (i) Explain, with reference to the random nature of radioactive decay, why the activity of the carbon-11 sample in (b) decreases with time.

.....

 [2]

(ii) State, with reasons, whether a radiation detector placed near to the sample of carbon-11 indicates a measured count rate from the sample that is less than, the same as or greater than the activity of the sample.

.....

 [3]

[Total: 9]

- 10 (a) State Hubble's law. Identify any symbols that you use.

.....

.....

.....

..... [2]

- (b) A star of luminosity $3.8 \times 10^{31} \text{ W}$ is a distance of $1.8 \times 10^{24} \text{ m}$ from the Earth.

Calculate the radiant flux intensity at the Earth of the radiation emitted by the star.

radiant flux intensity = W m^{-2} [2]

- (c) The star in (b) is in a distant galaxy. A spectral line in the light from this galaxy is known to have a wavelength of 486 nm. This spectral line in the light from the galaxy observed on the Earth has a wavelength of 492 nm.

- (i) Explain why the wavelength observed on the Earth is different from the wavelength that the galaxy is known to have emitted.

.....

.....

..... [2]

- (ii) Determine a value for the Hubble constant H_0 .

$H_0 = \dots\dots\dots \text{ s}^{-1}$ [3]

[Total: 9]

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9702/51

May/June 2023

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 30.
- The number of marks for each question or part question is shown in brackets [].

This document has 8 pages.

- 1 A wooden cube of mass A is placed on an inclined plane. The cube is attached to a cylinder of mass B using string that passes over a pulley, as shown in Fig. 1.1.

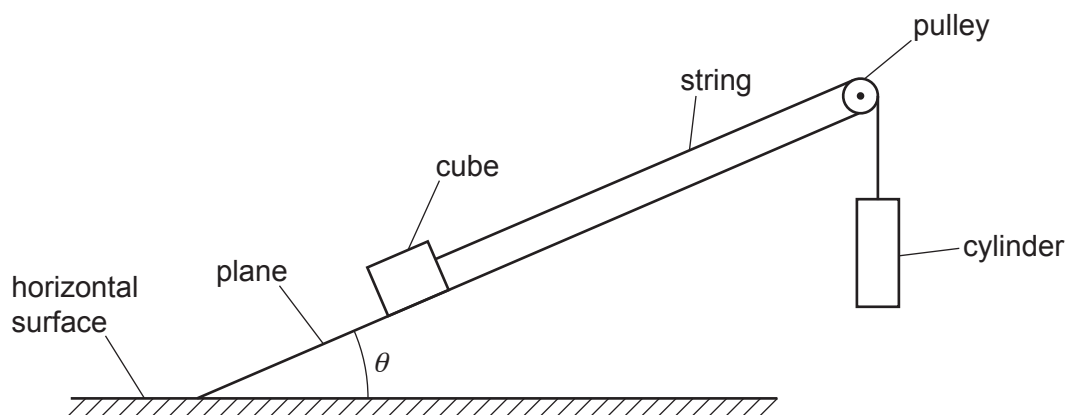


Fig. 1.1 (not to scale)

The angle between the plane and the horizontal surface is θ . Initially the cylinder is held at rest.

The cylinder is released. The time for the cylinder to fall a distance d is t .

It is suggested that t is related to θ by the relationship

$$\frac{2d}{t^2} = -\frac{AH \sin \theta}{(A + B)} - \frac{KA}{(A + B)}$$

where H and K are constants.

Plan a laboratory experiment to test the relationship between t and θ .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for H and K .

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

Diagram

[illegible]

- 2 A student investigates the discharge of capacitors in the circuit shown in Fig. 2.1.

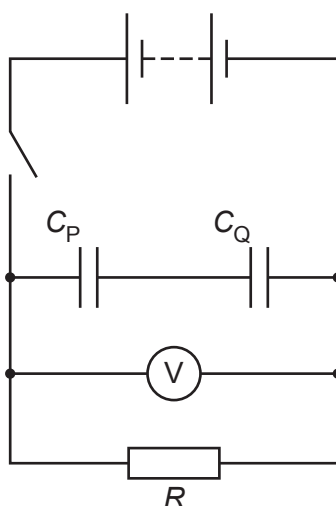


Fig. 2.1

The capacitors have capacitances C_P and C_Q .

The student closes the switch to charge the capacitors and then records the maximum reading V_0 on the voltmeter.

The switch is opened and a stop-watch is started. The capacitors discharge through the resistor and the reading on the voltmeter decreases. When the reading on the voltmeter is V the time t is recorded. The discharge of the capacitors is repeated and the mean time T is calculated.

The experiment is repeated for different values of C_P and C_Q .

For each combination of C_P and C_Q , the combined capacitance C is calculated.

It is suggested that C and T are related by the equation

$$\ln \left(\frac{V}{V_0} \right) = - \frac{T}{CR}$$

where R is the resistance of the resistor.

- (a) A graph is plotted of T on the y -axis against C on the x -axis.

Determine an expression for the gradient.

gradient = [1]

(b) Values of C_P , C_Q and t are given in Table 2.1.

Table 2.1

$C_P/10^{-4}\text{F}$	$C_Q/10^{-4}\text{F}$	$C/10^{-4}\text{F}$	t/s	t/s	T/s
2.2	1.5		12.9	14.5	
2.2	3.3		21.1	19.7	
2.2	5.6		23.7	24.9	
3.3	1.5		15.3	16.9	
5.6	1.5		19.0	17.6	
5.6	3.3		30.9	32.1	

The relationship between C , C_P and C_Q is

$$C = \frac{C_P C_Q}{C_P + C_Q}.$$

Calculate and record values of $C/10^{-4}\text{F}$ and T/s in Table 2.1.
Include the absolute uncertainties in T .

[2]

(c) (i) Plot a graph of T/s against $C/10^{-4}\text{F}$.
Include error bars for T .

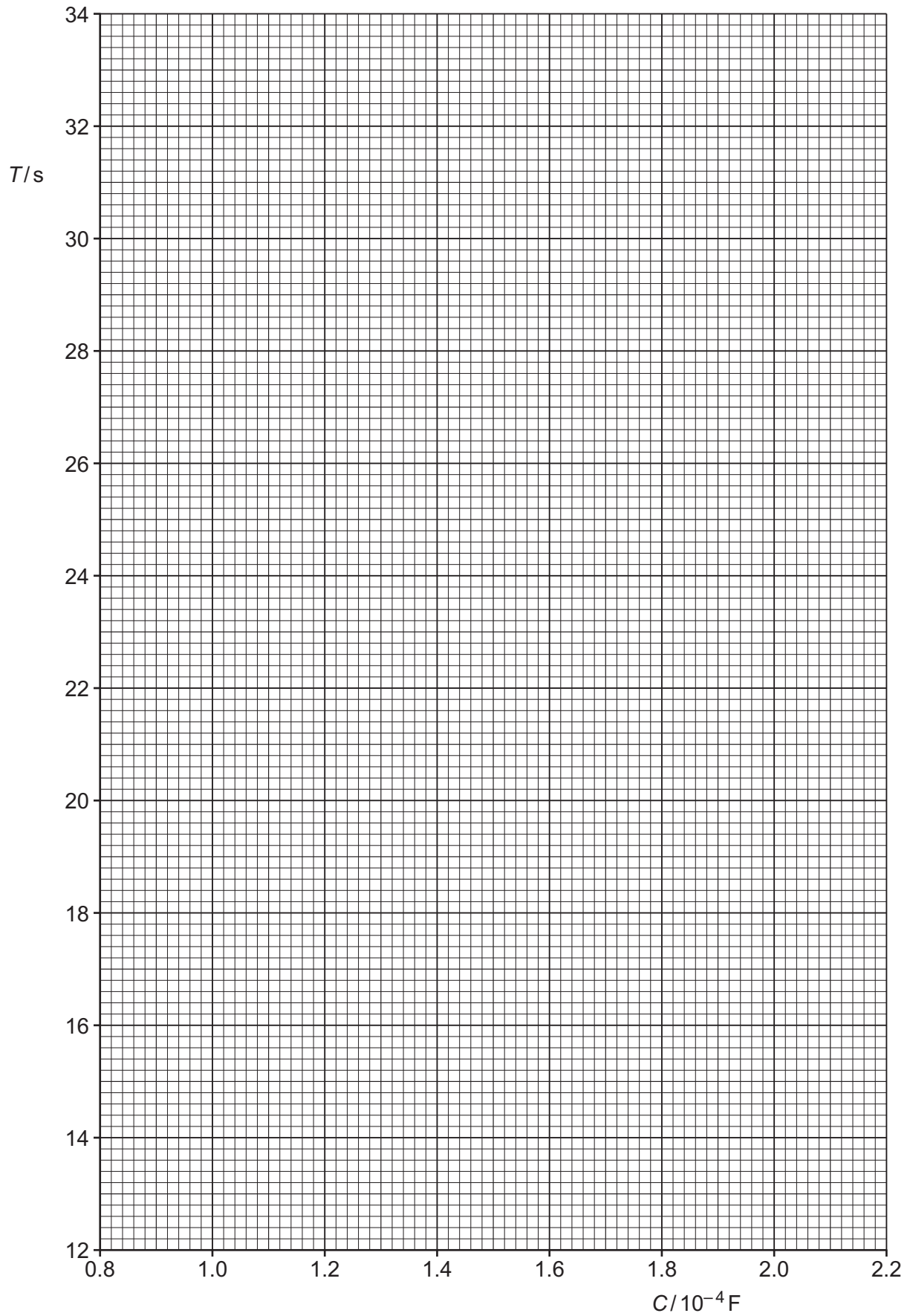
[2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines.

[2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]



- (d) The values of V_0 and V are:

$$V_0 = (4.8 \pm 0.1) \text{ V}$$

$$V = (2.4 \pm 0.1) \text{ V}$$

Calculate $\ln\left(\frac{V}{V_0}\right)$. Include the absolute uncertainty in $\ln\left(\frac{V}{V_0}\right)$.

$$\ln\left(\frac{V}{V_0}\right) = \dots\dots\dots [1]$$

- (e) (i) Using your answers to (a), (c)(iii) and (d), determine the value of R . Include an appropriate unit.

$$R = \dots\dots\dots [2]$$

- (ii) Determine the percentage uncertainty in R .

$$\text{percentage uncertainty in } R = \dots\dots\dots \% [1]$$

- (f) The experiment is repeated. Determine the value of C that gives a value of T of 60.0 s. Include the absolute uncertainty in your answer.

$$C = \dots\dots\dots \text{ F } [2]$$

[Total: 15]



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9702/52

May/June 2023

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
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- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 30.
- The number of marks for each question or part question is shown in brackets [].

This document has 8 pages.

- 1 Two coils, P and Q, are placed close to each other, as shown in Fig. 1.1.

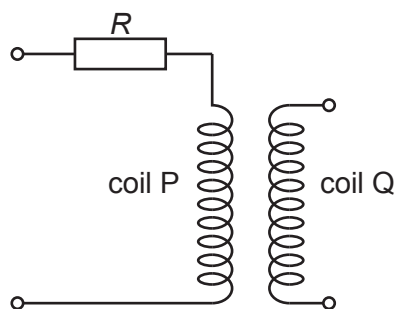


Fig. 1.1

A resistor of resistance R is connected in series with coil P.

A changing magnetic flux of frequency f in coil P causes an electromotive force (e.m.f.) E to be induced across the terminals of coil Q.

It is suggested that E is related to R by the relationship

$$E = 2\pi f M \left(\frac{V}{R + k} \right)$$

where V is the potential difference across the resistor and coil P, and k and M are constants.

Plan a laboratory experiment to test the relationship between E and R .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for k and M .

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

Diagram

[illegible]

- 2 A student investigates how the volume of a gas varies with its temperature. Air is trapped in a transparent cylinder of diameter d with a movable piston as shown in Fig. 2.1.

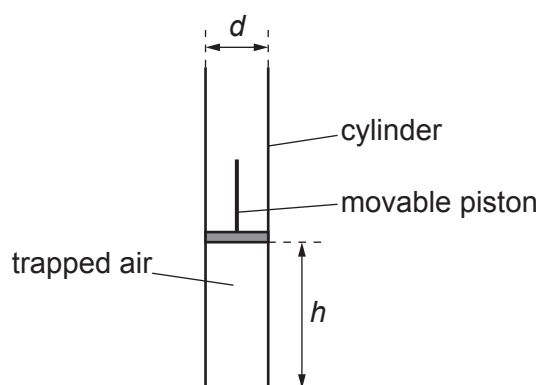


Fig. 2.1

The distance between the base of the cylinder and the bottom of the piston is h .

The trapped air is heated by placing the cylinder in water of temperature θ . The increase in temperature of the trapped air causes the piston to move. When the piston stops moving, the value of h is measured.

For each value of h , the volume V of the trapped air is calculated.

The experiment is repeated for different values of θ .

It is suggested that V and θ are related by the equation

$$pV = Yk(\theta + Z)$$

where k is the Boltzmann constant, p is the atmospheric pressure, and Y and Z are constants.

- (a) A graph is plotted of V on the y -axis against θ on the x -axis.

Determine expressions for the gradient and y -intercept.

gradient =

y -intercept =

[1]

(b) Values of θ and h are given in Table 2.1.

Table 2.1

$\theta / ^\circ\text{C}$	h / mm	$V / 10^{-5} \text{m}^3$
23	62.4 ± 0.1	
35	65.2 ± 0.1	
48	68.1 ± 0.1	
62	70.9 ± 0.1	
73	73.3 ± 0.1	
88	76.1 ± 0.1	

The value of d is $(27.9 \pm 0.1) \text{mm}$.

The volume V is calculated using the relationship

$$V = \frac{\pi d^2 h}{4}.$$

Calculate and record values of $V / 10^{-5} \text{m}^3$ in Table 2.1.

Include the absolute uncertainties in V .

[2]

(c) (i) Plot a graph of $V / 10^{-5} \text{m}^3$ against $\theta / ^\circ\text{C}$. Include error bars for V .

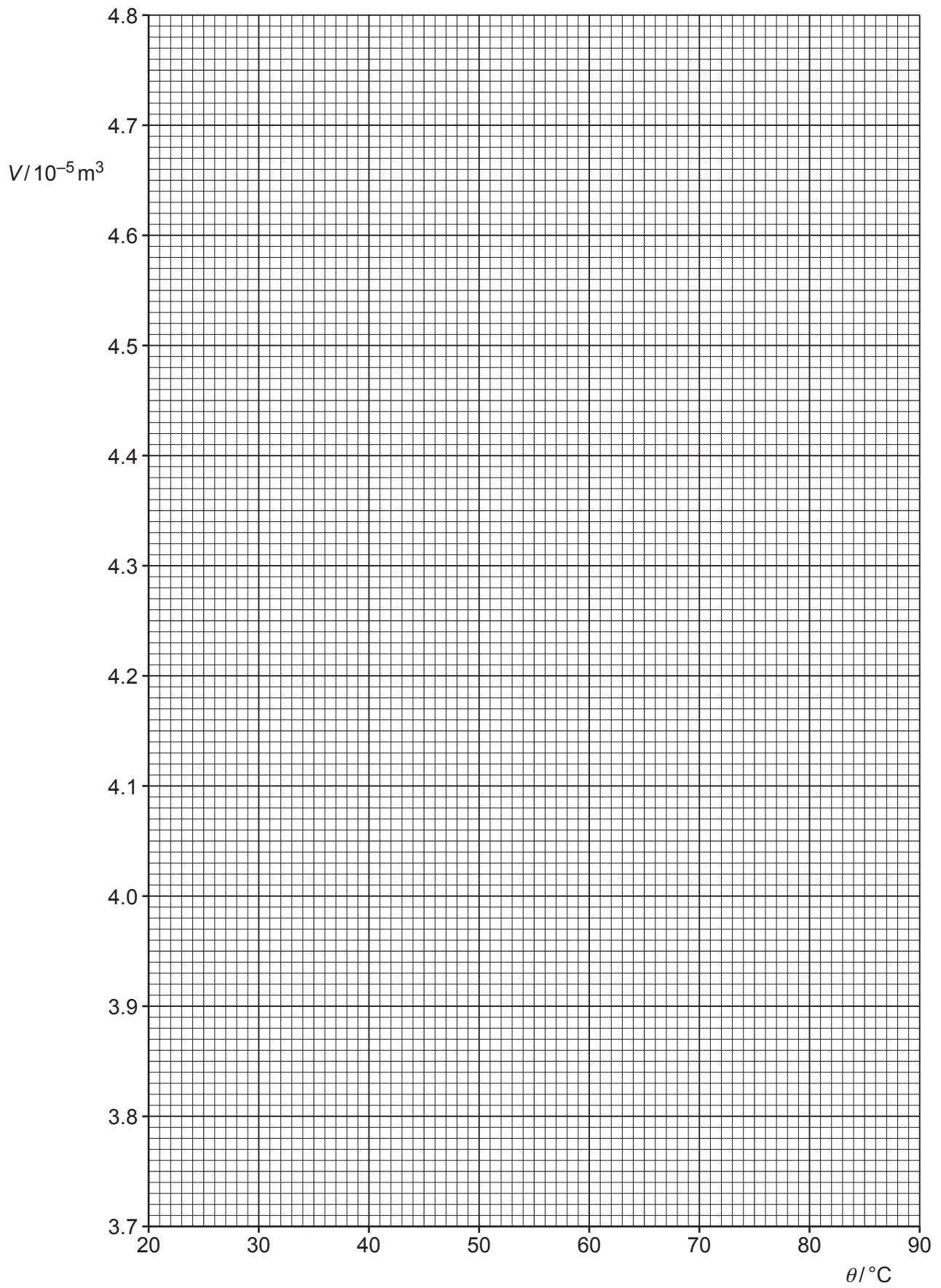
[2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines.

[2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]



- (iv) Determine the y -intercept of the line of best fit. Include the absolute uncertainty in your answer.

y -intercept = [2]

- (d) (i) Using your answers to (a), (c)(iii) and (c)(iv), determine the values of Y and Z . Include appropriate units.

Data: $p = (1.01 \pm 0.01) \times 10^5 \text{ Pa}$
 $k = 1.38 \times 10^{-23} \text{ J K}^{-1}$

$Y = \dots\dots\dots$

$Z = \dots\dots\dots$ [2]

- (ii) Determine the percentage uncertainty in Y .

percentage uncertainty in $Y = \dots\dots\dots\%$ [1]

- (e) The experiment is repeated. Determine the temperature θ that gives a value of h of 60.0 mm.

$\theta = \dots\dots\dots^\circ\text{C}$ [1]

[Total: 15]



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9702/53

May/June 2023

1 hour 15 minutes

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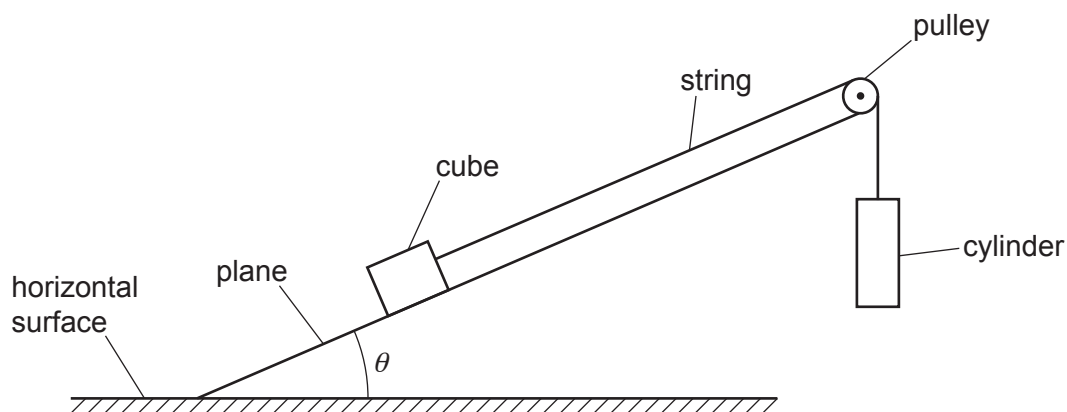


Fig. 1.1 (not to scale)

The angle between the plane and the horizontal surface is θ . Initially the cylinder is held at rest.

The cylinder is released. The time for the cylinder to fall a distance d is t .

It is suggested that t is related to θ by the relationship

$$\frac{2d}{t^2} = -\frac{AH \sin \theta}{(A + B)} - \frac{KA}{(A + B)}$$

where H and K are constants.

Plan a laboratory experiment to test the relationship between t and θ .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for H and K .

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

Diagram

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- 2 A student investigates the discharge of capacitors in the circuit shown in Fig. 2.1.

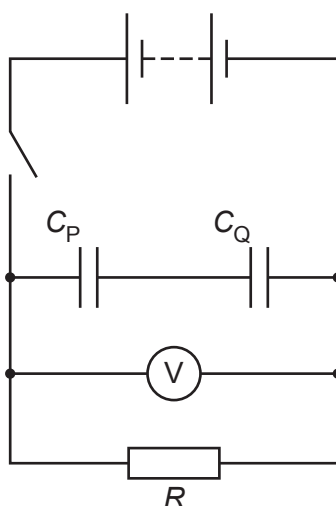


Fig. 2.1

The capacitors have capacitances C_P and C_Q .

The student closes the switch to charge the capacitors and then records the maximum reading V_0 on the voltmeter.

The switch is opened and a stop-watch is started. The capacitors discharge through the resistor and the reading on the voltmeter decreases. When the reading on the voltmeter is V the time t is recorded. The discharge of the capacitors is repeated and the mean time T is calculated.

The experiment is repeated for different values of C_P and C_Q .

For each combination of C_P and C_Q , the combined capacitance C is calculated.

It is suggested that C and T are related by the equation

$$\ln \left(\frac{V}{V_0} \right) = - \frac{T}{CR}$$

where R is the resistance of the resistor.

- (a) A graph is plotted of T on the y -axis against C on the x -axis.

Determine an expression for the gradient.

gradient = [1]

(b) Values of C_P , C_Q and t are given in Table 2.1.

Table 2.1

$C_P/10^{-4}\text{F}$	$C_Q/10^{-4}\text{F}$	$C/10^{-4}\text{F}$	t/s	t/s	T/s
2.2	1.5		12.9	14.5	
2.2	3.3		21.1	19.7	
2.2	5.6		23.7	24.9	
3.3	1.5		15.3	16.9	
5.6	1.5		19.0	17.6	
5.6	3.3		30.9	32.1	

The relationship between C , C_P and C_Q is

$$C = \frac{C_P C_Q}{C_P + C_Q}.$$

Calculate and record values of $C/10^{-4}\text{F}$ and T/s in Table 2.1.
Include the absolute uncertainties in T .

[2]

(c) (i) Plot a graph of T/s against $C/10^{-4}\text{F}$.
Include error bars for T .

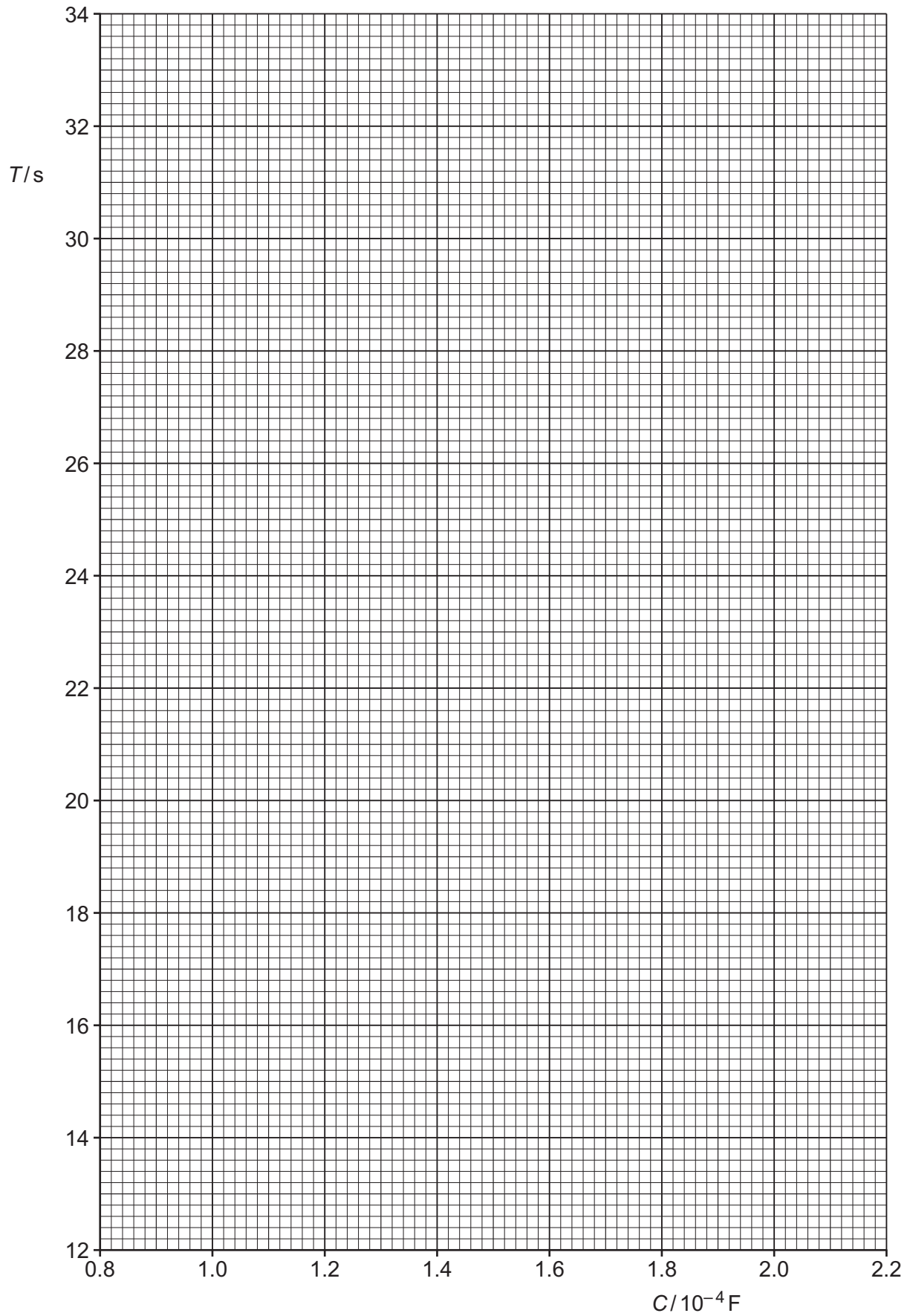
[2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines.

[2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]



- (d) The values of V_0 and V are:

$$V_0 = (4.8 \pm 0.1) \text{ V}$$

$$V = (2.4 \pm 0.1) \text{ V}$$

Calculate $\ln\left(\frac{V}{V_0}\right)$. Include the absolute uncertainty in $\ln\left(\frac{V}{V_0}\right)$.

$$\ln\left(\frac{V}{V_0}\right) = \dots\dots\dots [1]$$

- (e) (i) Using your answers to (a), (c)(iii) and (d), determine the value of R . Include an appropriate unit.

$$R = \dots\dots\dots [2]$$

- (ii) Determine the percentage uncertainty in R .

$$\text{percentage uncertainty in } R = \dots\dots\dots \% [1]$$

- (f) The experiment is repeated. Determine the value of C that gives a value of T of 60.0 s. Include the absolute uncertainty in your answer.

$$C = \dots\dots\dots \text{ F } [2]$$

[Total: 15]



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PHYSICS

9702/41

Paper 4 A Level Structured Questions

October/November 2023

2 hours

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [].

This document has **24** pages. Any blank pages are indicated.

Data

acceleration of free fall	$g = 9.81 \text{ m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
unified atomic mass unit	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27} \text{ kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$ $(\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ m F}^{-1})$
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_o = \frac{f_s v}{v \mp v_s}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \dots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$

gravitational potential	$\phi = -\frac{GM}{r}$
gravitational potential energy	$E_P = -\frac{GMm}{r}$
pressure of an ideal gas	$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$
simple harmonic motion	$a = -\omega^2 x$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t$ $v = \omega \sqrt{(x_0^2 - x^2)}$
electric potential	$V = \frac{Q}{4\pi\epsilon_0 r}$
electrical potential energy	$E_P = \frac{Qq}{4\pi\epsilon_0 r}$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$
capacitors in parallel	$C = C_1 + C_2 + \dots$
discharge of a capacitor	$x = x_0 e^{-\frac{t}{RC}}$
Hall voltage	$V_H = \frac{BI}{ntq}$
alternating current/voltage	$x = x_0 \sin \omega t$
radioactive decay	$x = x_0 e^{-\lambda t}$
decay constant	$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$
intensity reflection coefficient	$\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$
Stefan–Boltzmann law	$L = 4\pi\sigma r^2 T^4$
Doppler redshift	$\frac{\Delta\lambda}{\lambda} \hat{=} \frac{Df}{f} \hat{=} \frac{v}{c}$

- 1 (a) (i) State what is indicated by the direction of the gravitational field line at a point in a gravitational field.

.....
 [1]

- (ii) Explain, with reference to gravitational field lines, why the gravitational field near the surface of the Earth is approximately constant for small changes in height.

.....

 [2]

- (b) A large isolated uniform sphere has mass M and radius R .

Point P lies on a straight line passing through the centre of the sphere, at a variable displacement x from the centre, as shown in Fig. 1.1.

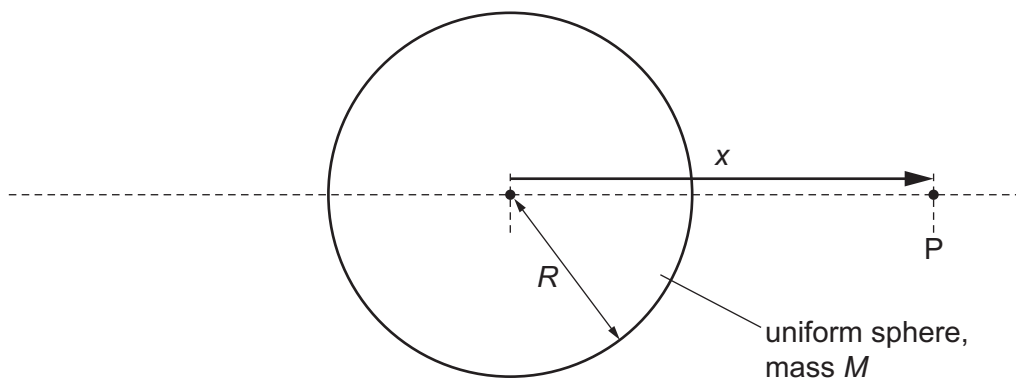


Fig. 1.1

Fig. 1.2 shows the variation with x of the gravitational field g at point P due to the sphere for the values of x for which P is inside the sphere.

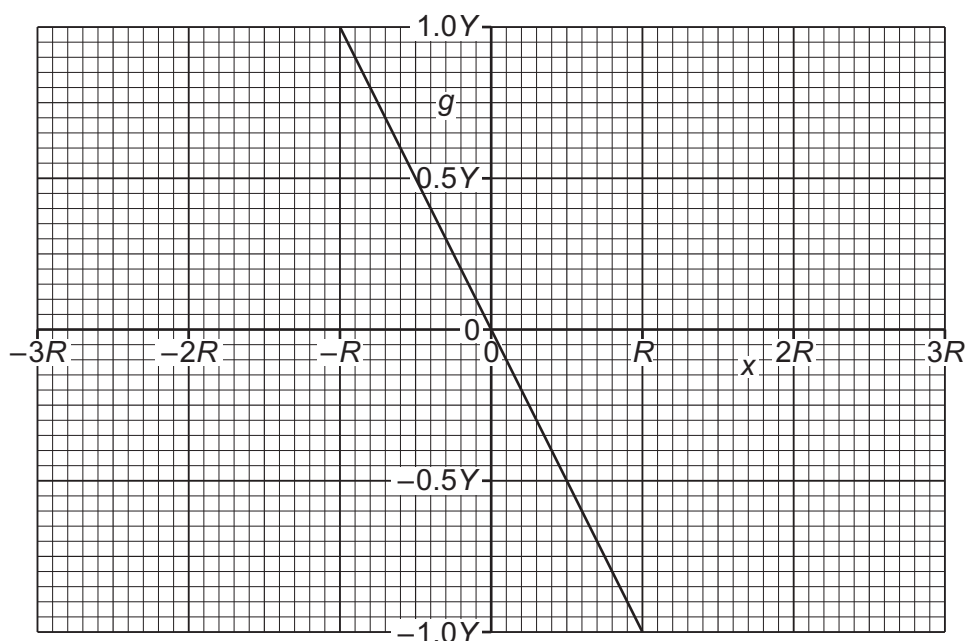


Fig. 1.2

The magnitude of the gravitational field at the surface of the sphere is Y .

- (i) Determine an expression for Y in terms of M and R . Identify any other symbols that you use.

[2]

- (ii) Explain why, at the surface of the sphere, g always has the opposite sign to x .

.....

 [2]

- (iii) Complete Fig. 1.2 to show the variation of g with x for values of x , up to $\pm 3R$, for which point P is outside the sphere. [3]

[Total: 10]

- 2 (a) Define specific heat capacity.

.....

.....

..... [2]

- (b) An ideal gas of mass 0.35 kg is heated at a constant pressure of $2.0 \times 10^5 \text{ Pa}$ so that its internal energy increases by 7600 J. During this process, the volume of the gas increases from 0.038 m^3 to 0.063 m^3 and the temperature increases by 56°C .

- (i) Show that the magnitude of the work done on the gas is 5000 J.

[1]

- (ii) Explain whether the work done on the gas is positive or negative.

.....

.....

..... [2]

- (iii) Determine the magnitude of the thermal energy q transferred to the gas.

$$q = \dots\dots\dots \text{ J [2]}$$

- (iv) Calculate the specific heat capacity of the gas for this process. Give a unit with your answer.

$$\text{specific heat capacity} = \dots\dots\dots \text{ unit } \dots\dots\dots [2]$$

- (c) The gas in (b) is now heated at constant volume rather than at constant pressure. The increase in internal energy of the gas is the same as in (b).

Use the first law of thermodynamics to explain whether the specific heat capacity of the gas for this process is less than, the same as, or greater than the answer in (b)(iv).

.....

.....

.....

.....

..... [3]

[Total: 12]

- 3 (a) The product pV for an ideal gas is given by

$$pV = \frac{1}{3} Nm \langle c^2 \rangle$$

where p is the pressure of the gas and V is the volume of the gas.

- (i) State the meaning of the symbols N , m and $\langle c^2 \rangle$ in this equation.

N :

m :

$\langle c^2 \rangle$:

[3]

- (ii) Use the equation of state for an ideal gas to show that the average translational kinetic energy E_K of a molecule of the gas at thermodynamic temperature T is given by

$$E_K = \frac{3}{2} kT.$$

[2]

- (b) The surface of a star consists mainly of a gas that may be assumed to be ideal. The molecules of the gas have a root-mean-square (r.m.s.) speed of 9300 m s^{-1} .

The mass of a molecule of the gas is $3.34 \times 10^{-27} \text{ kg}$.

Determine, to three significant figures, the temperature of the surface of the star.

temperature = K [2]

- (c) The radiant flux intensity of the radiation from the star in (b) is $2.52 \times 10^{-8} \text{ W m}^{-2}$ when observed at a distance of $4.16 \times 10^{16} \text{ m}$ from the star.

(i) Calculate the luminosity of the star. Give a unit with your answer.

luminosity = unit [2]

(ii) Determine the radius of the star.

radius = m [2]

- (d) The gas at the surface of a star has a very high pressure.

Use the basic assumptions of the kinetic theory to suggest why, in practice, a gas at the surface of a star is unlikely to behave as an ideal gas.

.....

 [2]

[Total: 13]

- 4 A heavy metal sphere of mass 0.81 kg is suspended from a string. The sphere is undergoing small oscillations from side to side, as shown in Fig. 4.1.

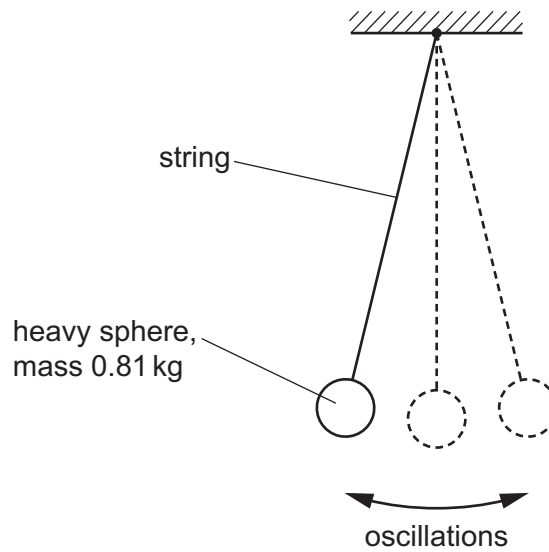


Fig. 4.1

The oscillations of the sphere may be considered to be simple harmonic with amplitude 0.036 m and period 3.0 s.

- (a) State what is meant by simple harmonic motion.

.....

.....

..... [2]

- (b) Calculate:

- (i) the angular frequency of the oscillations

angular frequency = rad s^{-1} [2]

(ii) the total energy of the oscillations.

total energy = J [2]

- (c) The suspended sphere is now lowered into water. The sphere is given a sideways displacement of $+0.036\text{ m}$ from its equilibrium position and is then released at time $t = 0$. The water causes the motion of the sphere to be critically damped.

On Fig. 4.2, sketch the variation of the displacement x of the sphere from its equilibrium position with t from $t = 0$ to $t = 6.0\text{ s}$.

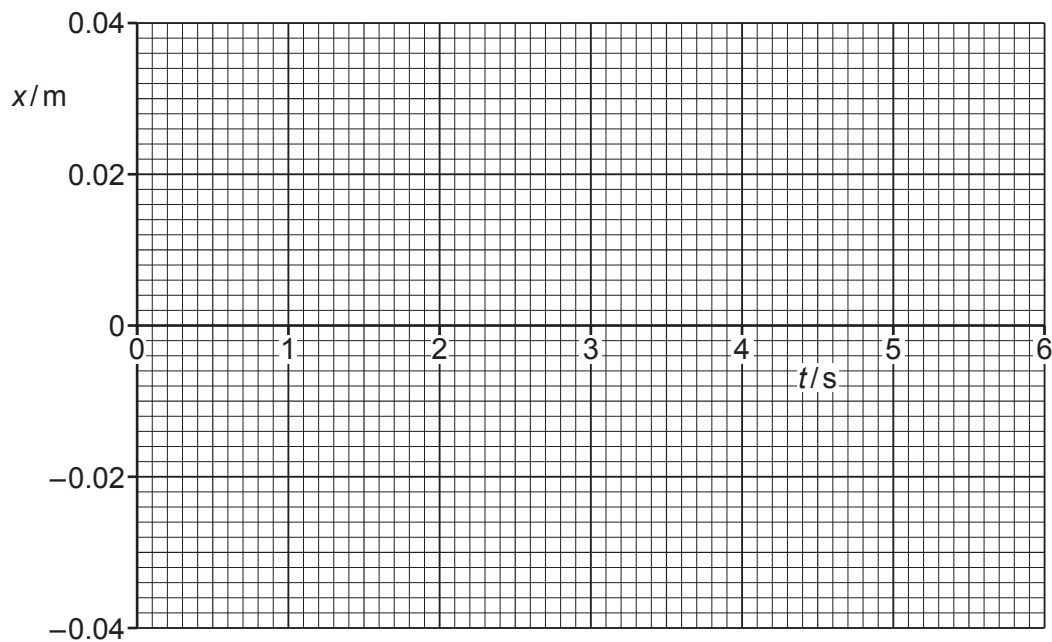


Fig. 4.2

[3]

[Total: 9]

- 5 (a) Define electric potential at a point.

.....

.....

..... [2]

- (b) Two isolated charged metal spheres X and Y are situated near to each other in a vacuum with their centres a distance of 24 m apart. Point P is at a variable distance x from the centre of sphere X on the line joining the centres of the spheres.

Fig. 5.1 shows the variation with x of the electric potential V due to the spheres at point P.

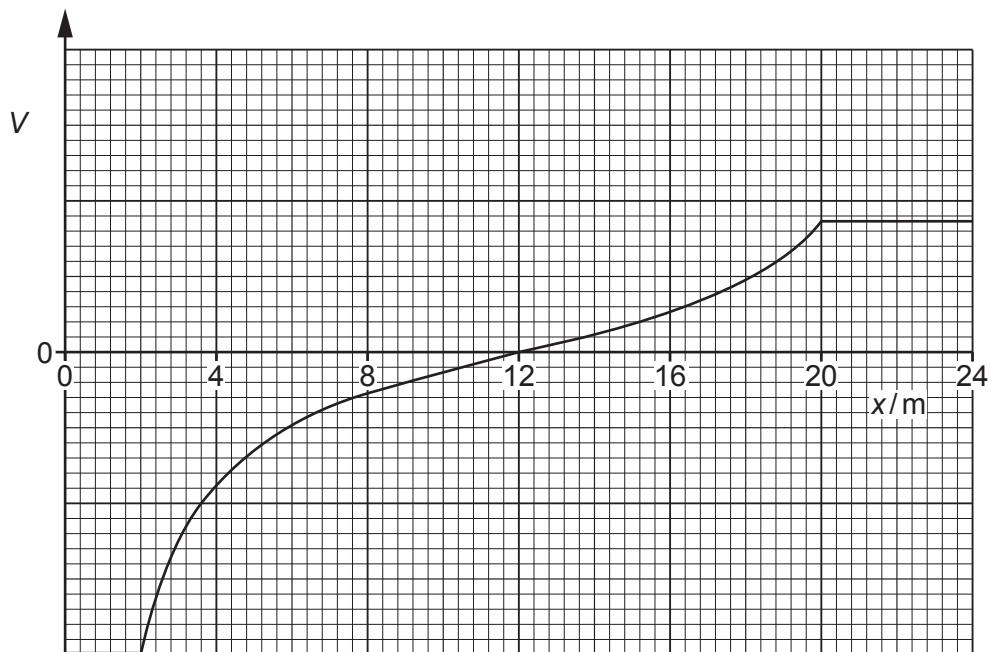


Fig. 5.1

State **three** conclusions that can be drawn about the spheres from Fig. 5.1. The conclusions may be qualitative or quantitative.

1

.....

2

.....

3

.....

[3]

- (c) A positively charged particle is placed at point P in (b), such that $x = 12\text{ m}$. The particle is released.

Describe and explain the subsequent motion of the particle.

.....

.....

.....

.....

..... [3]

[Total: 8]

- 6 (a) Define magnetic flux density.

.....

.....

.....

..... [2]

- (b) Electrons are moving in a vacuum with speed $1.7 \times 10^7 \text{ ms}^{-1}$. The electrons enter a uniform magnetic field of flux density 4.8 mT . Fig. 6.1 shows the path of the electrons.

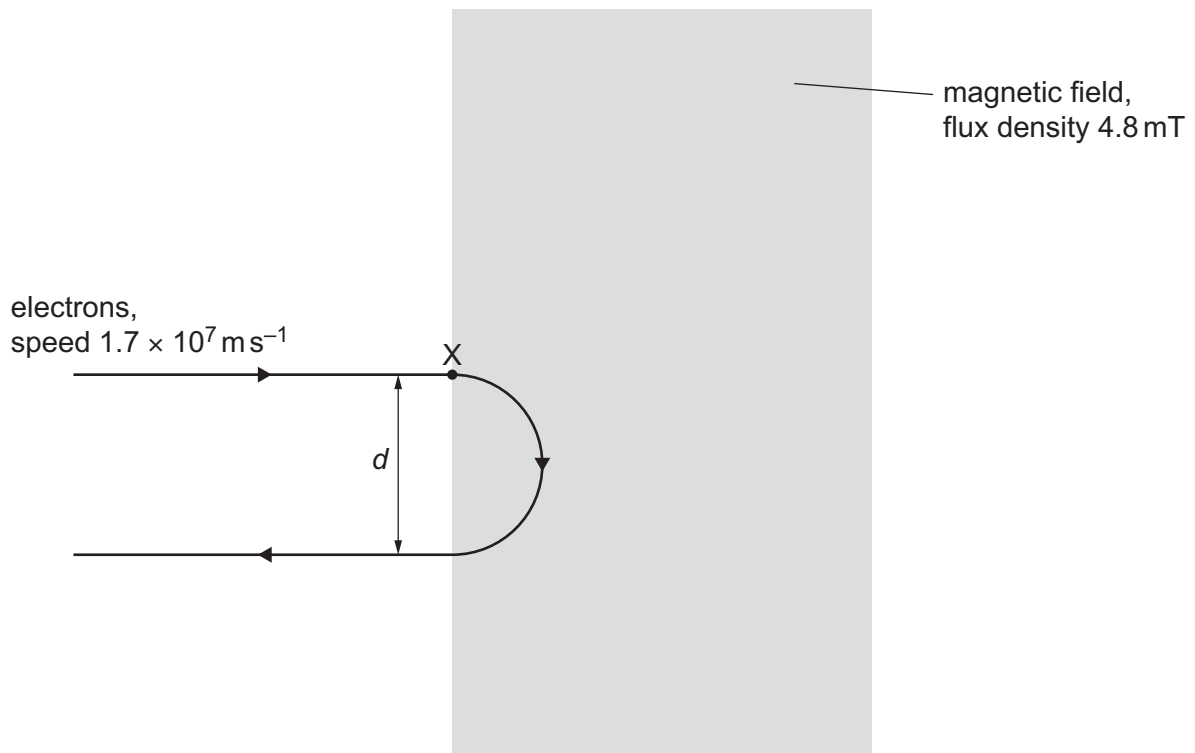


Fig. 6.1

The path of the electrons remains in the plane of the page.

- (i) State the direction of the magnetic field.

.....

..... [1]

- (ii) Show that the magnitude of the force exerted on each electron by the magnetic field is $1.3 \times 10^{-14} \text{ N}$.

[2]

- (iii) On Fig. 6.1, draw an arrow to indicate the direction of the centripetal acceleration of the electron where it enters the magnetic field at point X. [1]

- (iv) Use the information in **(b)(ii)** to calculate the distance d between the path of the electrons entering the magnetic field and the path of the electrons leaving it.

$d = \dots\dots\dots \text{ m}$ [3]

- (c) The electrons in **(b)** are replaced with positrons that are moving with speed $3.4 \times 10^7 \text{ ms}^{-1}$ along the same initial path as the electrons.

The positrons enter the magnetic field at point X on Fig. 6.1.

On Fig. 6.1, draw a line to show the path of the positrons through the magnetic field. [3]

[Total: 12]

- 7 A varying current I passes through a resistor of resistance R in the circuit shown in Fig. 7.1.

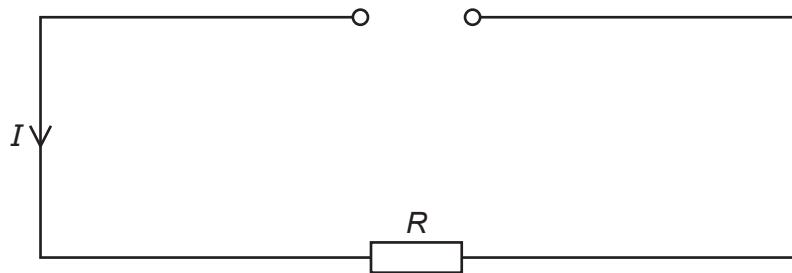


Fig. 7.1

Fig. 7.2 shows the variation with time t of I .

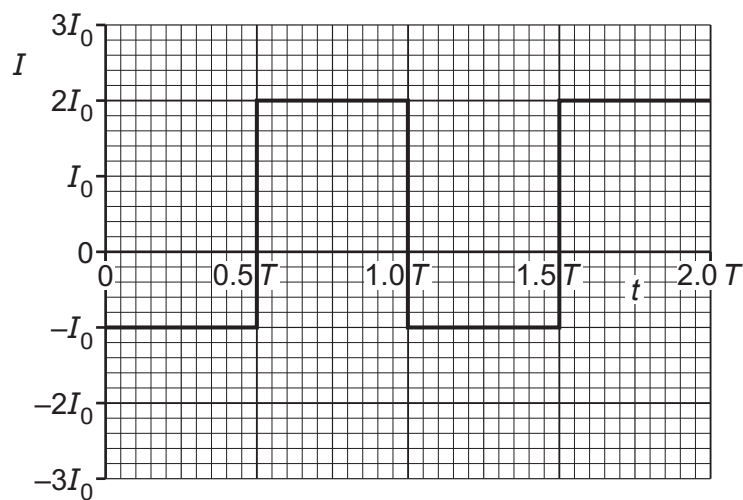


Fig. 7.2

The current has magnitude $2I_0$ when it is in the positive direction and I_0 when it is in the negative direction. The period of the variation of the current is T .

- (a) Determine expressions, in terms of I_0 and R , for the power P dissipated in the resistor for the times when:

- (i) the current is in the negative direction

$$P = \dots\dots\dots [1]$$

- (ii) the current is in the positive direction.

$$P = \dots\dots\dots [1]$$

- (b) On Fig. 7.3, sketch the variation of P with t between $t = 0$ and $t = 2.0T$. Label the power axis with an appropriate scale.

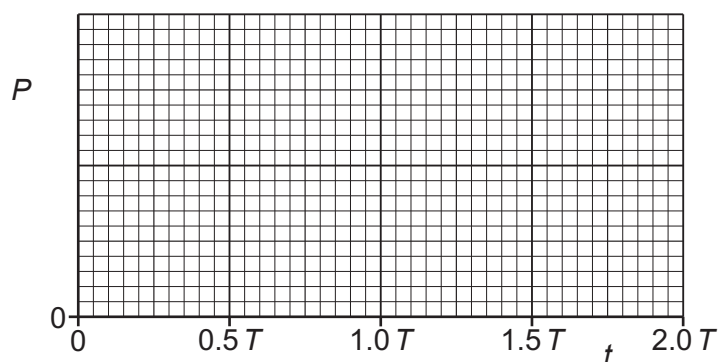


Fig. 7.3

[3]

- (c) Use your answer in (b) to determine an expression, in terms of I_0 and R , for:

- (i) the mean power $\langle P \rangle$ in the resistor

$$\langle P \rangle = \dots\dots\dots [1]$$

- (ii) the root-mean-square (r.m.s.) current $I_{\text{r.m.s.}}$ in the resistor.

$$I_{\text{r.m.s.}} = \dots\dots\dots [2]$$

[Total: 8]

- 8 (a) (i) Show that the momentum p of a photon of electromagnetic radiation with wavelength λ is given by

$$p = \frac{h}{\lambda}$$

where h is the Planck constant.

[2]

- (ii) Use the expression in (a)(i) to show that a photon in free space that has a momentum of $9.5 \times 10^{-28} \text{ N s}$ is a photon of red light.

[1]

- (b) A beam of red light of intensity 160 W m^{-2} is incident normally on a plane mirror, as shown in Fig. 8.1. The momentum of each photon in the beam is $9.5 \times 10^{-28} \text{ N s}$.

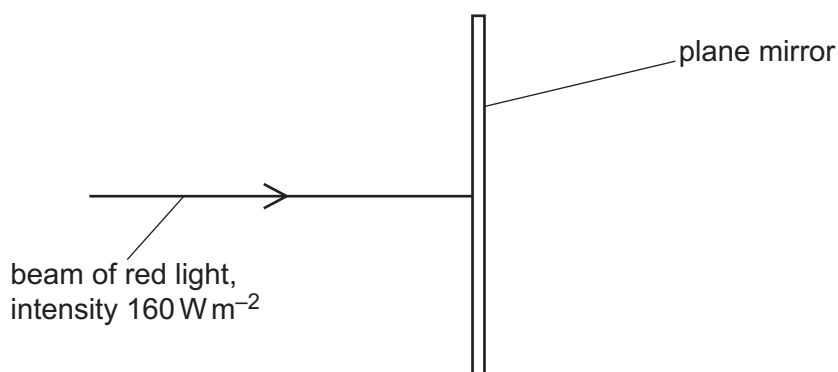


Fig. 8.1

All of the light is reflected by the mirror in the opposite direction to its original path. The cross-sectional area of the beam is $2.5 \times 10^{-6} \text{ m}^2$.

- (i) Show that the number of photons incident on the mirror per unit time is $1.4 \times 10^{15} \text{ s}^{-1}$.

[2]

- (ii) Use the information in **(b)(i)** to determine the pressure exerted by the light beam on the mirror.

pressure = Pa [3]

- (c) The beam of red light in **(b)** is now replaced with a beam of blue light of the same intensity.

Suggest and explain whether the pressure exerted on the mirror by the beam of blue light is less than, the same as, or greater than the pressure exerted by the beam of red light.

.....

 [2]

[Total: 10]

- 9 (a) State what is meant by nuclear fusion.

.....

.....

..... [2]

- (b) On Fig. 9.1, sketch the variation of binding energy per nucleon with nucleon number A for values of A between 1 and 250.

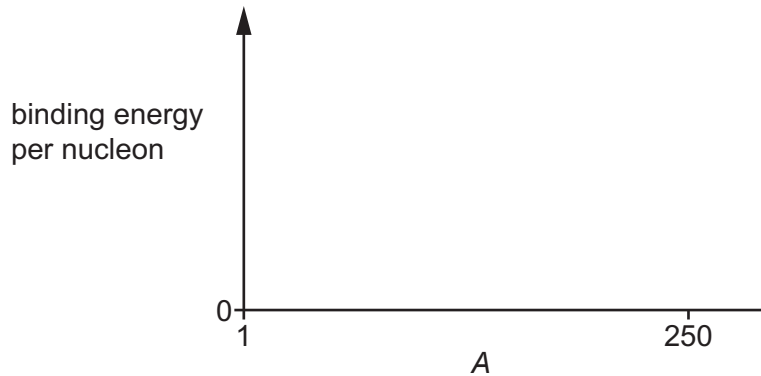


Fig. 9.1

[2]

- (c) On your line in Fig. 9.1, label:

- (i) a point X that could represent a nucleus that undergoes alpha-decay [1]
- (ii) a point Y that could represent a nucleus that undergoes nuclear fusion. [1]

- (d) A nucleus Z undergoes nuclear fission to form strontium-93 ($^{93}_{38}\text{Sr}$) and xenon-139 ($^{139}_{54}\text{Xe}$) according to

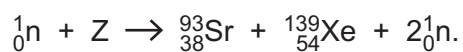


Table 9.1 shows the binding energies of the strontium-93 and xenon-139 nuclei.

Table 9.1

nucleus	binding energy/J
$^{93}_{38}\text{Sr}$	1.25×10^{-10}
$^{139}_{54}\text{Xe}$	1.81×10^{-10}

The fission of 1.00 mol of Z releases 1.77×10^{13} J of energy.

Determine the binding energy per nucleon, in MeV, of Z.

binding energy per nucleon = MeV [4]

[Total: 10]

- 10** Ultrasound and X-rays are both types of wave that are used in medical diagnosis to form images of internal body structures.

(a) Complete Table 10.1 to state, for each type of wave:

- the method of production of the wave
- whether the wave that is detected and used to form the image is the wave that has been absorbed, reflected or transmitted by the internal body structure.

Table 10.1

	ultrasound	X-rays
method of production		
detected wave (absorbed, reflected or transmitted)		

[4]

- (b) (i)** For one type of wave passing through tissue, the wave has 72% of its initial intensity after it has passed through 6.2 cm of the tissue.

Calculate the linear attenuation coefficient μ of the tissue for this wave.

$$\mu = \dots\dots\dots \text{ cm}^{-1} \quad [2]$$

- (ii)** Another wave of the same type as in **(b)(i)** passes through 9.3 cm of the same tissue.

Calculate the percentage of the initial intensity of the wave that is attenuated by the tissue.

$$\text{percentage attenuated} = \dots\dots\dots \% \quad [2]$$

[Total: 8]

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9702/42

October/November 2023

2 hours

You must answer on the question paper.

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molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
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simple harmonic motion	$a = -\omega^2 x$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t$ $v = \pm \omega \sqrt{(x_0^2 - x^2)}$
electric potential	$V = \frac{Q}{4\pi\epsilon_0 r}$
electrical potential energy	$E_P = \frac{Qq}{4\pi\epsilon_0 r}$
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discharge of a capacitor	$x = x_0 e^{-\frac{t}{RC}}$
Hall voltage	$V_H = \frac{BI}{ntq}$
alternating current/voltage	$x = x_0 \sin \omega t$
radioactive decay	$x = x_0 e^{-\lambda t}$
decay constant	$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$
intensity reflection coefficient	$\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$
Stefan–Boltzmann law	$L = 4\pi\sigma r^2 T^4$
Doppler redshift	$\frac{\Delta\lambda}{\lambda} \hat{=} \frac{Df}{f} \hat{=} \frac{v}{c}$

- 1 (a) Define the radian.

.....
 [1]

- (b) The minute hand of a clock revolves at constant angular speed around the face of the clock, completing one revolution every hour. A small piece of modelling clay is attached to the hand with its centre of gravity at a distance L from the fixed end of the hand, as shown in Fig. 1.1.

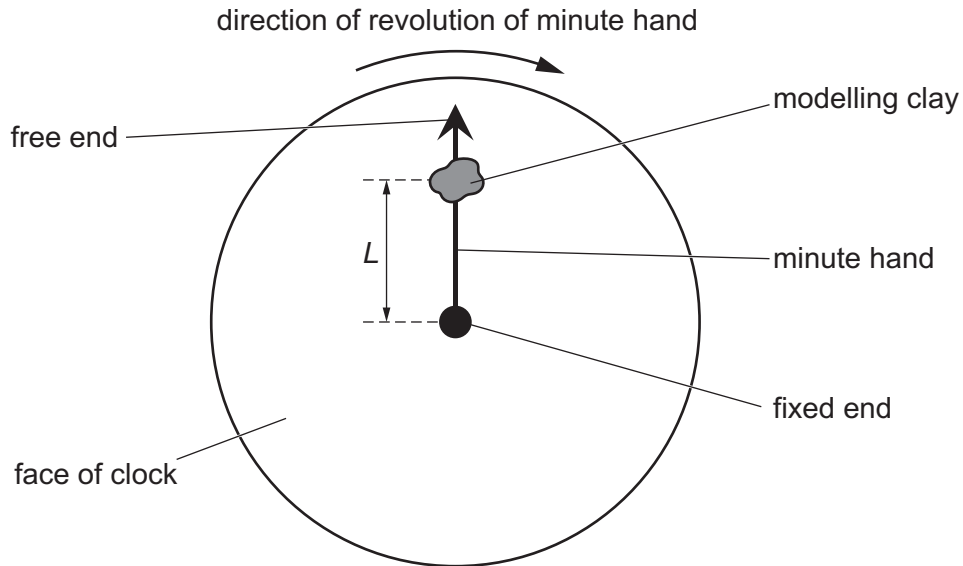


Fig. 1.1

Calculate the angular speed ω of the minute hand.

$$\omega = \dots\dots\dots \text{rad s}^{-1} \quad [2]$$

- (c) During a time interval of 1400 s, the centre of gravity of the piece of modelling clay in Fig. 1.1 moves through a total distance of 0.44 m.

- (i) Calculate the angle through which the minute hand moves in this time interval.

$$\text{angle} = \dots\dots\dots \text{rad} \quad [1]$$

(ii) Determine distance L .

$L = \dots\dots\dots$ m [2]

(iii) Calculate the magnitude of the centripetal acceleration of the piece of modelling clay.

centripetal acceleration = $\dots\dots\dots$ ms^{-2} [2]

(d) Use your answer in (c)(iii) to explain why the variation with time of the magnitude of the force exerted by the minute hand on the piece of modelling clay is negligible as the minute hand undergoes one full revolution.

$\dots\dots\dots$
 $\dots\dots\dots$
 $\dots\dots\dots$ [2]

[Total: 10]

- 2 (a) (i) Define gravitational potential at a point.

.....

 [2]

- (ii) The Moon may be considered to be an isolated uniform sphere of mass $7.3 \times 10^{22} \text{ kg}$ and radius $1.7 \times 10^6 \text{ m}$.

Calculate the gravitational potential at the surface of the Moon. Give a unit with your answer.

gravitational potential = unit [2]

- (b) An isolated uniform spherical planet has gravitational potential ϕ at its surface.

A particle of mass m is projected vertically upwards from the surface. The particle is given just enough kinetic energy to travel to an infinite distance away from the planet, escaping from the gravitational pull of the planet, without any additional work being done on it.

- (i) Determine an expression, in terms of m and ϕ , for the gravitational potential energy E_p of the particle at the surface of the planet.

$E_p = \dots\dots\dots$ [1]

- (ii) Show that the speed v at which the particle is projected upwards from the surface of the planet is given by

$$v = \sqrt{-2\phi}.$$

[2]

- (c) A particle is moving upwards at the surface of the Moon.

Use your answer in (a)(ii) and the expression in (b)(ii) to determine the minimum speed of this particle that will result in it escaping from the gravitational pull of the Moon.

speed = ms^{-1} [1]

- (d) Hydrogen may be assumed to be an ideal gas.
The mass of a hydrogen molecule is $3.34 \times 10^{-27} \text{ kg}$.

Calculate the root-mean-square (r.m.s.) speed of a hydrogen molecule in hydrogen gas that is at a temperature of 400 K.

r.m.s. speed = ms^{-1} [3]

- (e) The surface of the Moon reaches temperatures of approximately 400 K when in direct sunlight.

Use your answers in (c) and (d) to suggest a reason why the Moon does not have an atmosphere consisting of hydrogen.

.....
..... [1]

[Total: 12]

- 3 (a) State what is meant by the internal energy of a system.

.....

.....

..... [2]

- (b) Use the first law of thermodynamics to explain what happens to the internal energy:

- (i) of a spring when it is stretched at constant temperature within its elastic limit

.....

.....

.....

.....

..... [3]

- (ii) of a sample of water when it evaporates from a rain puddle on a hot day.

.....

.....

.....

.....

..... [3]

[Total: 8]

- 4 An electron in a metal rod moves randomly about a mean position. When an alternating voltage is applied to the ends of the rod, the mean position can be considered to oscillate with simple harmonic motion along the axis of the rod. Fig. 4.1 shows the variation with time t of the displacement x of the mean position from a fixed point on the axis of the rod.

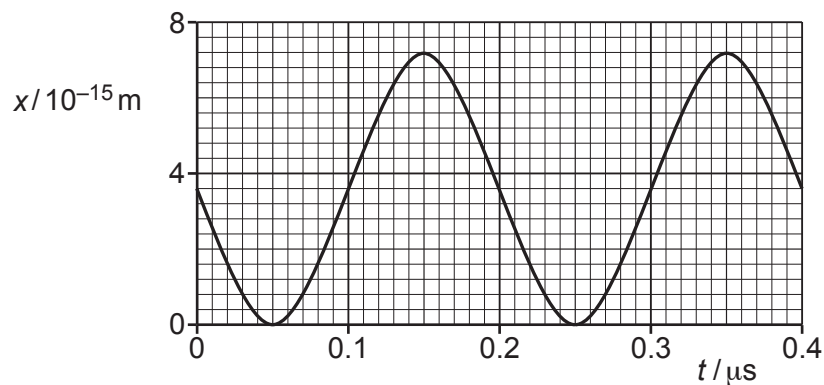


Fig. 4.1

- (a) (i) Determine the amplitude of the oscillations.

amplitude = m [1]

- (ii) Determine the angular frequency of the oscillations.

angular frequency = rad s^{-1} [1]

- (iii) Use your answers in (a)(i) and (a)(ii) to show that the maximum drift speed v_0 of the electron is $1.1 \times 10^{-7} \text{ m s}^{-1}$.

[2]

- (b) The rod has a cross-sectional area of 4.3 cm^2 and contains a number density of conduction electrons (charge carriers) of $8.5 \times 10^{28} \text{ m}^{-3}$.

All of the conduction electrons in the rod may be assumed to be oscillating in phase with, and with the same amplitude as, the oscillation shown in Fig. 4.1.

- (i) Use the information in (a)(iii) to calculate the magnitude I_0 of the maximum current in the rod.

$$I_0 = \dots\dots\dots \text{A} \quad [2]$$

- (ii) On Fig. 4.2, sketch the variation of the current I in the rod with time t between $t = 0$ and $t = 0.40 \mu\text{s}$.

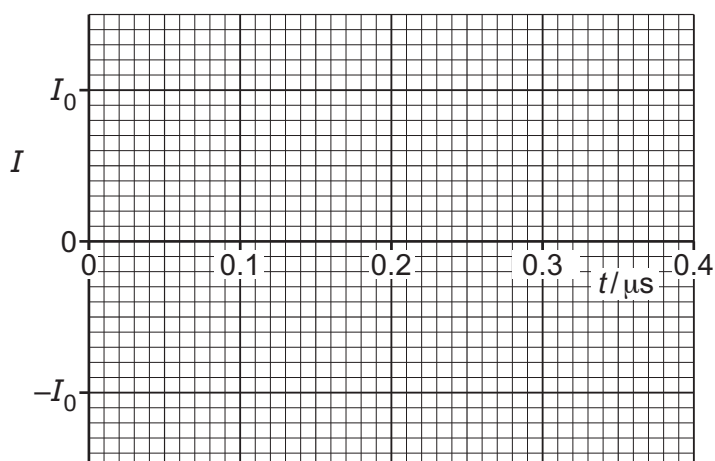


Fig. 4.2

[2]

- (iii) Use your answers in (a)(ii) and (b)(i) to determine an expression for I in terms of t , where I is in A and t is in s.

$$I = \dots\dots\dots [2]$$

- (iv) Determine the root-mean-square (r.m.s.) current in the rod.

$$\text{r.m.s. current} = \dots\dots\dots \text{A} \quad [1]$$

[Total: 11]

- 5 (a) State Coulomb's law.

.....

 [2]

- (b) Two identical oil droplets are in a vacuum. The centres of the droplets are a distance of $3.8 \times 10^{-6} \text{ m}$ apart. The droplets have equal charge and exert an electric force on each other of magnitude $6.3 \times 10^{-17} \text{ N}$.

Determine the magnitude of the charge on each droplet.

charge = C [2]

- (c) One of the oil droplets in (b) is now placed between two horizontal metal plates, as shown in Fig. 5.1.

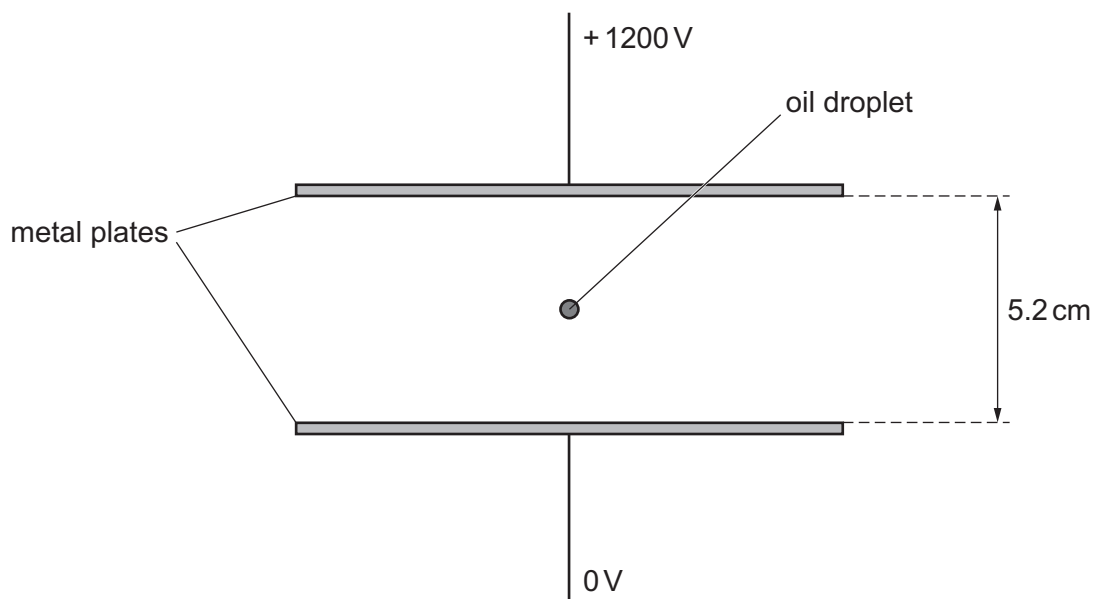


Fig. 5.1 (not to scale)

A potential difference (p.d.) of 1200 V is applied between the plates, with the top plate at the higher potential. The oil droplet is stationary and in equilibrium.

- (i) State the sign of the charge on the oil droplet.

..... [1]

- (ii) On Fig. 5.1, draw four lines to represent the electric field between the plates. [3]

- (iii) The distance between the plates is 5.2 cm.

Determine the mass of the oil droplet.

mass = kg [3]

[Total: 11]

- 6 A capacitor C is charged so that the potential difference (p.d.) V across its terminals is 8.0 V . The capacitor is connected into the circuit of Fig. 6.1.

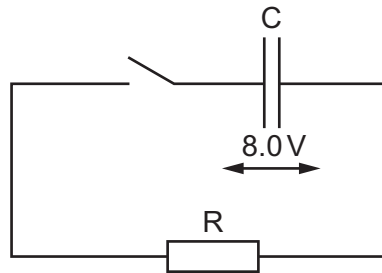


Fig. 6.1

The switch is initially open. The switch is closed at time $t = 0$.

- (a) Fig. 6.2 shows the variation of V with the charge Q on the plates of capacitor C as the capacitor discharges.

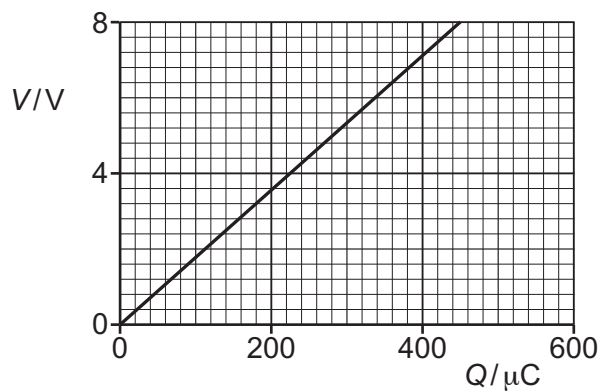


Fig. 6.2

- (i) Show that the energy stored in capacitor C at time $t = 0$ is 1.8 mJ .

[2]

- (ii) Determine the capacitance of capacitor C . Give a unit with your answer.

capacitance = unit [2]

- (b) Fig. 6.3 shows the variation with t of $-\ln \frac{V}{8.0\text{V}}$.

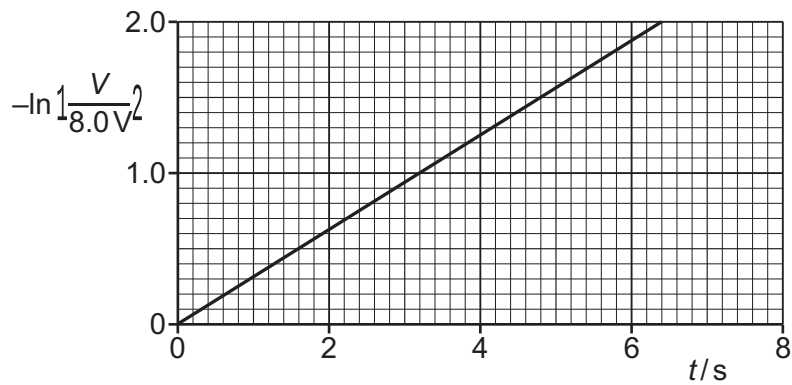


Fig. 6.3

- (i) Show that, when t is equal to one time constant, the value of $-\ln \frac{V}{8.0\text{V}}$ is equal to 1.0.

[2]

- (ii) Determine the time constant τ of the circuit in Fig. 6.1.

$\tau = \dots\dots\dots \text{s}$ [1]

- (iii) Calculate the resistance of resistor R.

resistance = $\dots\dots\dots \Omega$ [2]

[Total: 9]

- 7 (a) A Hall probe containing a thin slice of semiconducting material is placed in a uniform magnetic field of flux density B . The largest faces of the slice are perpendicular to the magnetic field, as shown in Fig. 7.1.

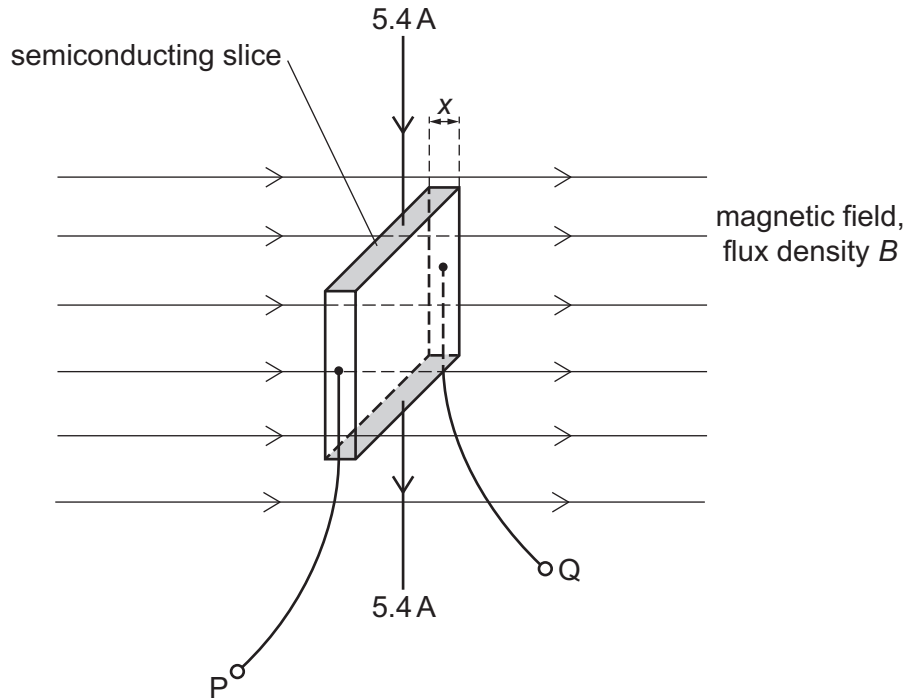


Fig. 7.1

The thickness x of the slice is 1.8 mm. The number density of charge carriers in the semiconducting material is $1.5 \times 10^{16} \text{ m}^{-3}$. A constant current of 5.4 A is passed through the slice between the shaded faces. The Hall voltage V_H that is developed between the terminals PQ is recorded.

Fig. 7.2 shows the variation with time t of B .

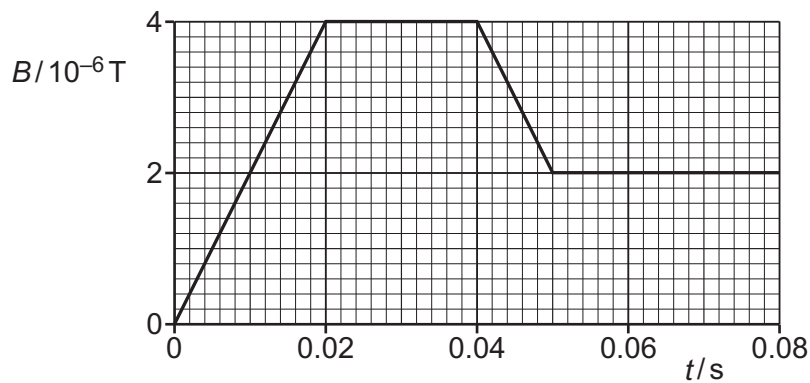


Fig. 7.2

- (i) Show that, when B is equal to $4.0 \times 10^{-6} \text{ T}$, the magnitude of V_H is 5.0 V.

- (ii) On Fig. 7.3, sketch the variation of V_H with t between $t = 0$ and $t = 0.080$ s.

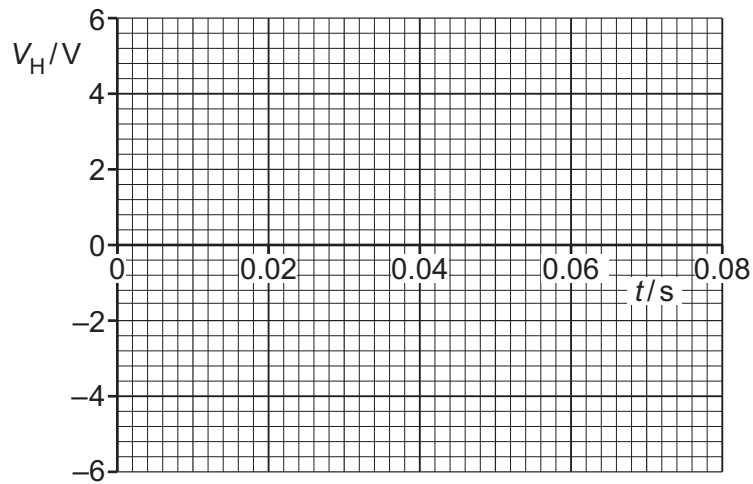


Fig. 7.3

[3]

- (b) The Hall probe in (a) is replaced with a small flat coil that has 3000 turns. The cross-sectional area of the coil is $3.4 \times 10^{-4} \text{ m}^2$. The plane of the coil is perpendicular to the magnetic field. The electromotive force (e.m.f.) E induced between the terminals of the coil is recorded as B varies as shown in Fig. 7.2.

- (i) Show that the magnitude of E at time $t = 0.010$ s is $2.0 \times 10^{-4} \text{ V}$.

[3]

- (ii) On Fig. 7.4, sketch the variation of E with t between $t = 0$ and $t = 0.080$ s.

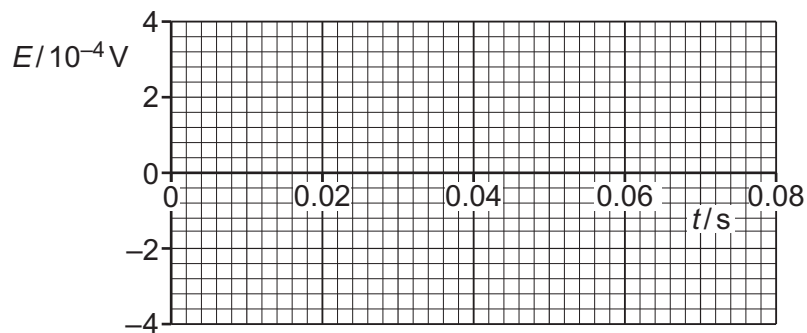


Fig. 7.4

[4]

[Total: 11]

- 8 (a) State what is meant by a photon.

.....

.....

..... [2]

- (b) When the surface of a metal plate is illuminated with electromagnetic radiation, electrons are sometimes emitted from the metal.

- (i) State the name of this phenomenon.

..... [1]

- (ii) It is observed that this phenomenon occurs only when the frequency of the electromagnetic radiation is greater than a certain minimum value, regardless of the intensity of the radiation.

Explain how this observation provides evidence for the existence of photons.

.....

.....

.....

.....

..... [3]

- (c) Fig. 8.1 shows the variation of the maximum kinetic energy of the emitted electrons in (b) with the frequency of the incident radiation.

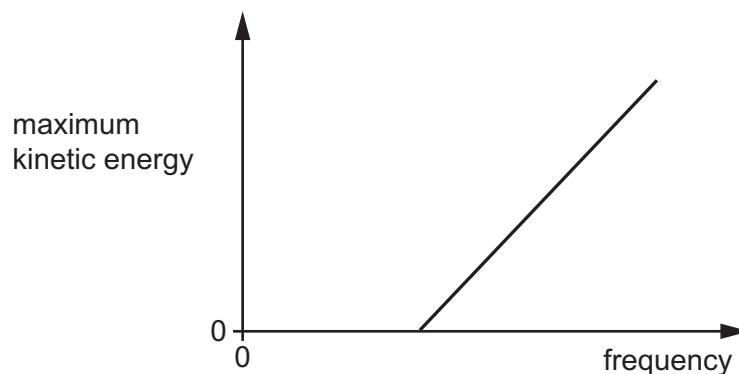


Fig. 8.1

State the name of the quantity represented by:

- (i) the gradient of the line in Fig. 8.1

..... [1]

- (ii) the y -intercept of the extrapolated line in Fig. 8.1.

..... [1]

[Total: 8]

- 9 Fluorine-18 ($^{18}_9\text{F}$) is a radioactive nuclide that is used as a tracer in positron emission tomography (PET scanning). Fluorine-18 decays to a nuclide of oxygen (O) according to



- (a) (i) State what is meant by a tracer.

.....
 [1]

- (ii) State the symbol of the particle that is represented by X and the values of P , Q and R .

X: P :

Q: R :

[2]

- (b) (i) Explain how the radioactive decay of fluorine-18 results in the emission from the body of the gamma-ray photons that are detected during a PET scan.

.....

 [2]

- (ii) Explain how the detection of the gamma-ray photons is used to produce an image of the tissue being examined.

.....

 [2]

- (c) The half-life of fluorine-18 is T .

A patient is injected with amount of substance n of fluorine-18.

- (i) Determine an expression for the initial value R_0 of the rate R of production of gamma-ray photons by the tracer, in terms of n , T and the Avogadro constant N_A .

$R_0 =$ [3]

- (ii) On Fig. 9.1, sketch the variation with time t of R .

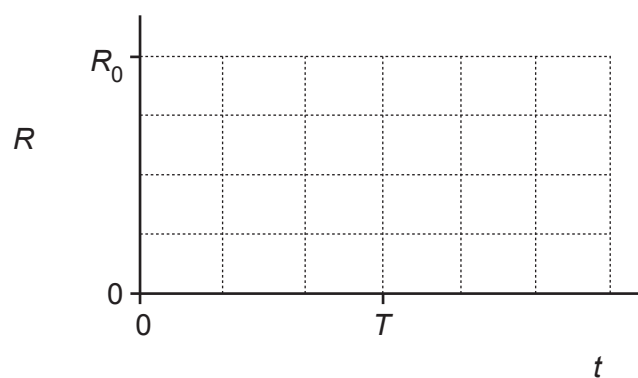


Fig. 9.1

[2]

[Total: 12]

- 10 (a) State Wien's displacement law. Identify any symbols that you use.

.....

.....

..... [2]

- (b) A cosmology student observes the electromagnetic radiation received from a star in a galaxy. The student uses Wien's law to estimate the surface temperature of the star, a standard candle to estimate the distance to the galaxy, and the Stefan–Boltzmann law to estimate the radius of the star.

The student observes that the radiation from the star is redshifted.

- (i) State what is meant by a standard candle.

..... [1]

- (ii) State the reason why the radiation from the star is redshifted.

..... [1]

- (iii) The true values of the quantities observed or estimated are those that are corrected to allow for redshift. However, the student does not correct for redshift.

By placing one tick (✓) in each row, complete Table 10.1 to indicate how the observations and estimates made by the student compare with the true values.

Table 10.1

	student's uncorrected value		
	too low	the same	too high
wavelength of radiation			
surface temperature of star			
distance to star			
radius of star			

[4]

[Total: 8]

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9702/43

October/November 2023

2 hours

You must answer on the question paper.

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [].

This document has **24** pages. Any blank pages are indicated.

Data

acceleration of free fall	$g = 9.81 \text{ m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
unified atomic mass unit	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27} \text{ kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$ $(\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ m F}^{-1})$
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_o = \frac{f_s v}{v \mp v_s}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \dots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$

gravitational potential	$\phi = -\frac{GM}{r}$
gravitational potential energy	$E_P = -\frac{GMm}{r}$
pressure of an ideal gas	$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$
simple harmonic motion	$a = -\omega^2 x$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t$ $v = \omega \sqrt{(x_0^2 - x^2)}$
electric potential	$V = \frac{Q}{4\pi\epsilon_0 r}$
electrical potential energy	$E_P = \frac{Qq}{4\pi\epsilon_0 r}$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$
capacitors in parallel	$C = C_1 + C_2 + \dots$
discharge of a capacitor	$x = x_0 e^{-\frac{t}{RC}}$
Hall voltage	$V_H = \frac{BI}{ntq}$
alternating current/voltage	$x = x_0 \sin \omega t$
radioactive decay	$x = x_0 e^{-\lambda t}$
decay constant	$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$
intensity reflection coefficient	$\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$
Stefan–Boltzmann law	$L = 4\pi\sigma r^2 T^4$
Doppler redshift	$\frac{\Delta\lambda}{\lambda} \hat{=} \frac{Df}{f} \hat{=} \frac{v}{c}$

- 1 (a) (i) State what is indicated by the direction of the gravitational field line at a point in a gravitational field.

.....
 [1]

- (ii) Explain, with reference to gravitational field lines, why the gravitational field near the surface of the Earth is approximately constant for small changes in height.

.....

 [2]

- (b) A large isolated uniform sphere has mass M and radius R .

Point P lies on a straight line passing through the centre of the sphere, at a variable displacement x from the centre, as shown in Fig. 1.1.

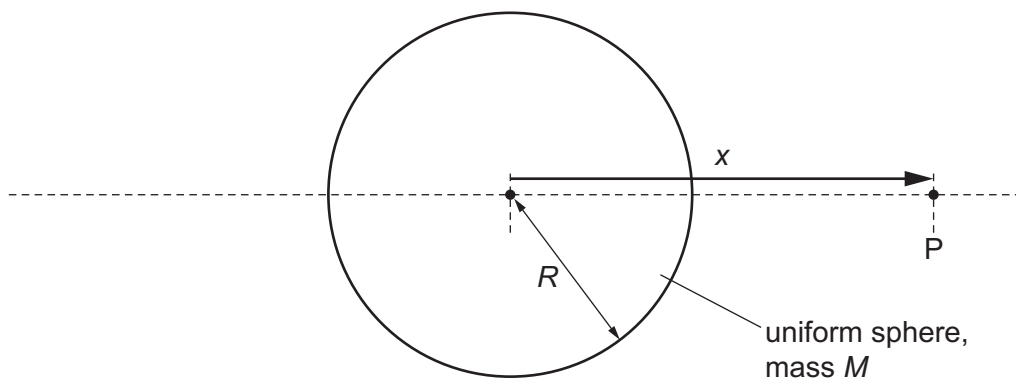


Fig. 1.1

Fig. 1.2 shows the variation with x of the gravitational field g at point P due to the sphere for the values of x for which P is inside the sphere.

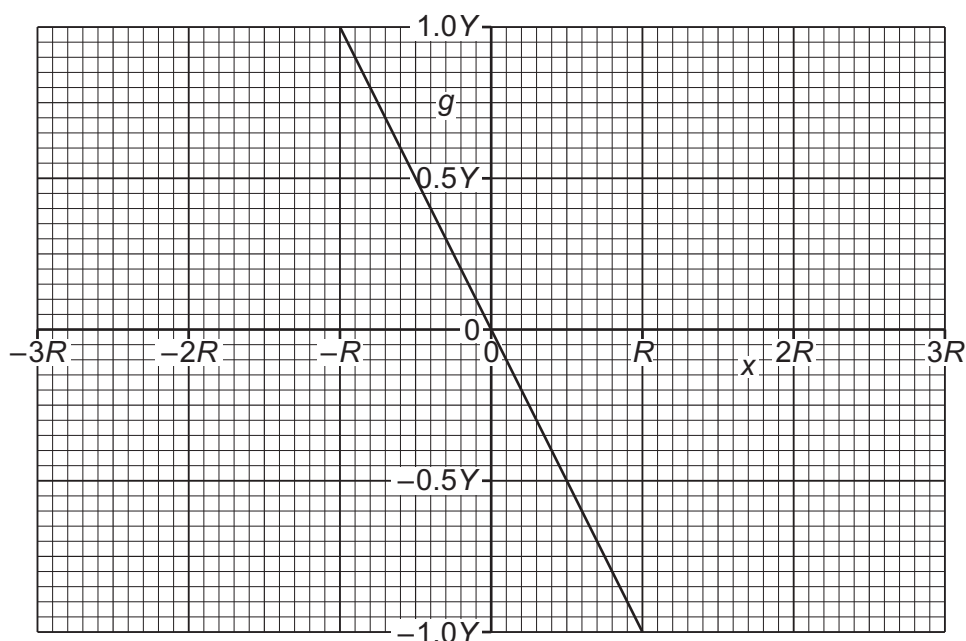


Fig. 1.2

The magnitude of the gravitational field at the surface of the sphere is Y .

- (i) Determine an expression for Y in terms of M and R . Identify any other symbols that you use.

[2]

- (ii) Explain why, at the surface of the sphere, g always has the opposite sign to x .

.....

 [2]

- (iii) Complete Fig. 1.2 to show the variation of g with x for values of x , up to $\pm 3R$, for which point P is outside the sphere. [3]

[Total: 10]

- 2 (a) Define specific heat capacity.

.....

.....

..... [2]

- (b) An ideal gas of mass 0.35 kg is heated at a constant pressure of $2.0 \times 10^5 \text{ Pa}$ so that its internal energy increases by 7600 J. During this process, the volume of the gas increases from 0.038 m^3 to 0.063 m^3 and the temperature increases by 56°C .

- (i) Show that the magnitude of the work done on the gas is 5000 J.

[1]

- (ii) Explain whether the work done on the gas is positive or negative.

.....

.....

..... [2]

- (iii) Determine the magnitude of the thermal energy q transferred to the gas.

$$q = \dots\dots\dots \text{ J [2]}$$

- (iv) Calculate the specific heat capacity of the gas for this process. Give a unit with your answer.

$$\text{specific heat capacity} = \dots\dots\dots \text{ unit } \dots\dots\dots [2]$$

- (c) The gas in (b) is now heated at constant volume rather than at constant pressure. The increase in internal energy of the gas is the same as in (b).

Use the first law of thermodynamics to explain whether the specific heat capacity of the gas for this process is less than, the same as, or greater than the answer in (b)(iv).

.....

.....

.....

.....

..... [3]

[Total: 12]

- 3 (a) The product pV for an ideal gas is given by

$$pV = \frac{1}{3} Nm \langle c^2 \rangle$$

where p is the pressure of the gas and V is the volume of the gas.

- (i) State the meaning of the symbols N , m and $\langle c^2 \rangle$ in this equation.

N :

m :

$\langle c^2 \rangle$:

[3]

- (ii) Use the equation of state for an ideal gas to show that the average translational kinetic energy E_K of a molecule of the gas at thermodynamic temperature T is given by

$$E_K = \frac{3}{2} kT.$$

[2]

- (b) The surface of a star consists mainly of a gas that may be assumed to be ideal. The molecules of the gas have a root-mean-square (r.m.s.) speed of 9300 m s^{-1} .

The mass of a molecule of the gas is $3.34 \times 10^{-27} \text{ kg}$.

Determine, to three significant figures, the temperature of the surface of the star.

temperature = K [2]

- (c) The radiant flux intensity of the radiation from the star in (b) is $2.52 \times 10^{-8} \text{ W m}^{-2}$ when observed at a distance of $4.16 \times 10^{16} \text{ m}$ from the star.

(i) Calculate the luminosity of the star. Give a unit with your answer.

luminosity = unit [2]

(ii) Determine the radius of the star.

radius = m [2]

- (d) The gas at the surface of a star has a very high pressure.

Use the basic assumptions of the kinetic theory to suggest why, in practice, a gas at the surface of a star is unlikely to behave as an ideal gas.

.....

 [2]

[Total: 13]

- 4 A heavy metal sphere of mass 0.81 kg is suspended from a string. The sphere is undergoing small oscillations from side to side, as shown in Fig. 4.1.

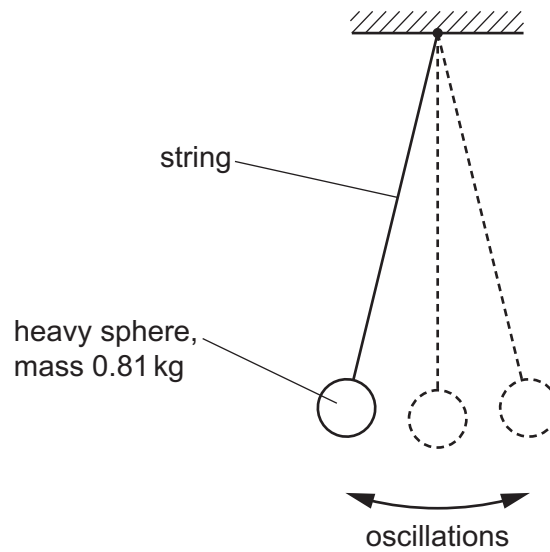


Fig. 4.1

The oscillations of the sphere may be considered to be simple harmonic with amplitude 0.036 m and period 3.0 s.

- (a) State what is meant by simple harmonic motion.

.....

.....

..... [2]

- (b) Calculate:

- (i) the angular frequency of the oscillations

angular frequency = rad s^{-1} [2]

(ii) the total energy of the oscillations.

total energy = J [2]

- (c) The suspended sphere is now lowered into water. The sphere is given a sideways displacement of $+0.036\text{ m}$ from its equilibrium position and is then released at time $t = 0$. The water causes the motion of the sphere to be critically damped.

On Fig. 4.2, sketch the variation of the displacement x of the sphere from its equilibrium position with t from $t = 0$ to $t = 6.0\text{ s}$.

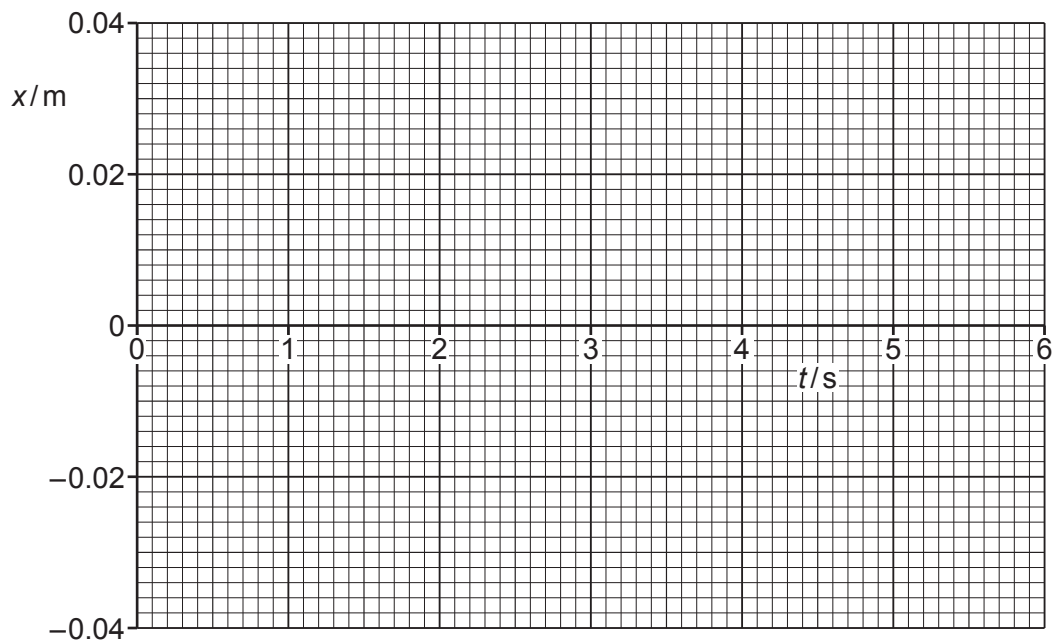


Fig. 4.2

[3]

[Total: 9]

- 5 (a) Define electric potential at a point.

.....

.....

..... [2]

- (b) Two isolated charged metal spheres X and Y are situated near to each other in a vacuum with their centres a distance of 24 m apart. Point P is at a variable distance x from the centre of sphere X on the line joining the centres of the spheres.

Fig. 5.1 shows the variation with x of the electric potential V due to the spheres at point P.

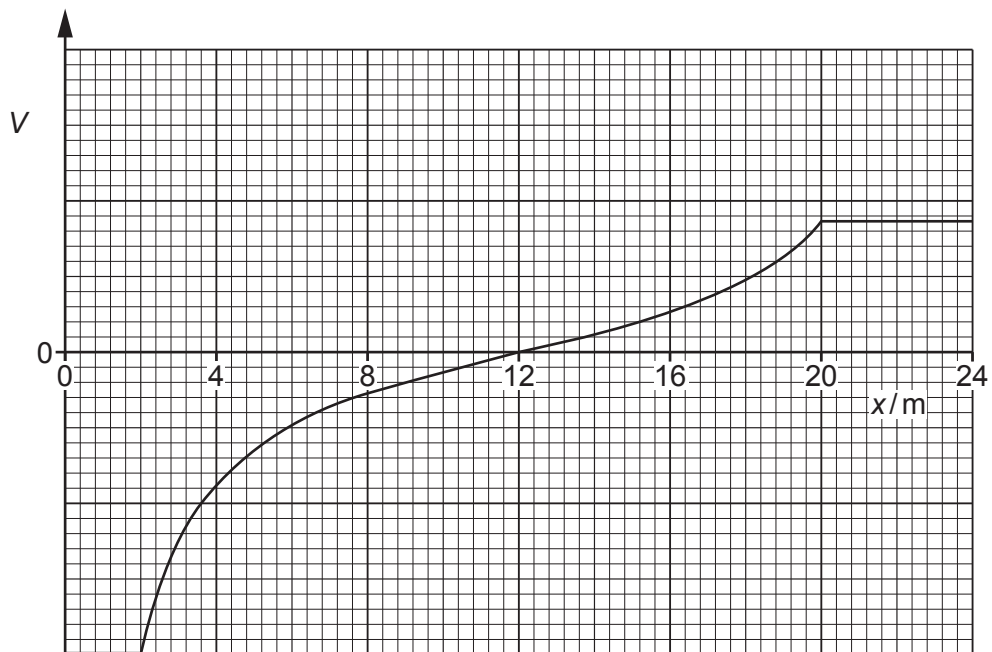


Fig. 5.1

State **three** conclusions that can be drawn about the spheres from Fig. 5.1. The conclusions may be qualitative or quantitative.

1

.....

2

.....

3

.....

[3]

- (c) A positively charged particle is placed at point P in (b), such that $x = 12\text{ m}$. The particle is released.

Describe and explain the subsequent motion of the particle.

.....

.....

.....

.....

..... [3]

[Total: 8]

- 6 (a) Define magnetic flux density.

.....

.....

..... [2]

- (b) Electrons are moving in a vacuum with speed $1.7 \times 10^7 \text{ ms}^{-1}$. The electrons enter a uniform magnetic field of flux density 4.8 mT . Fig. 6.1 shows the path of the electrons.

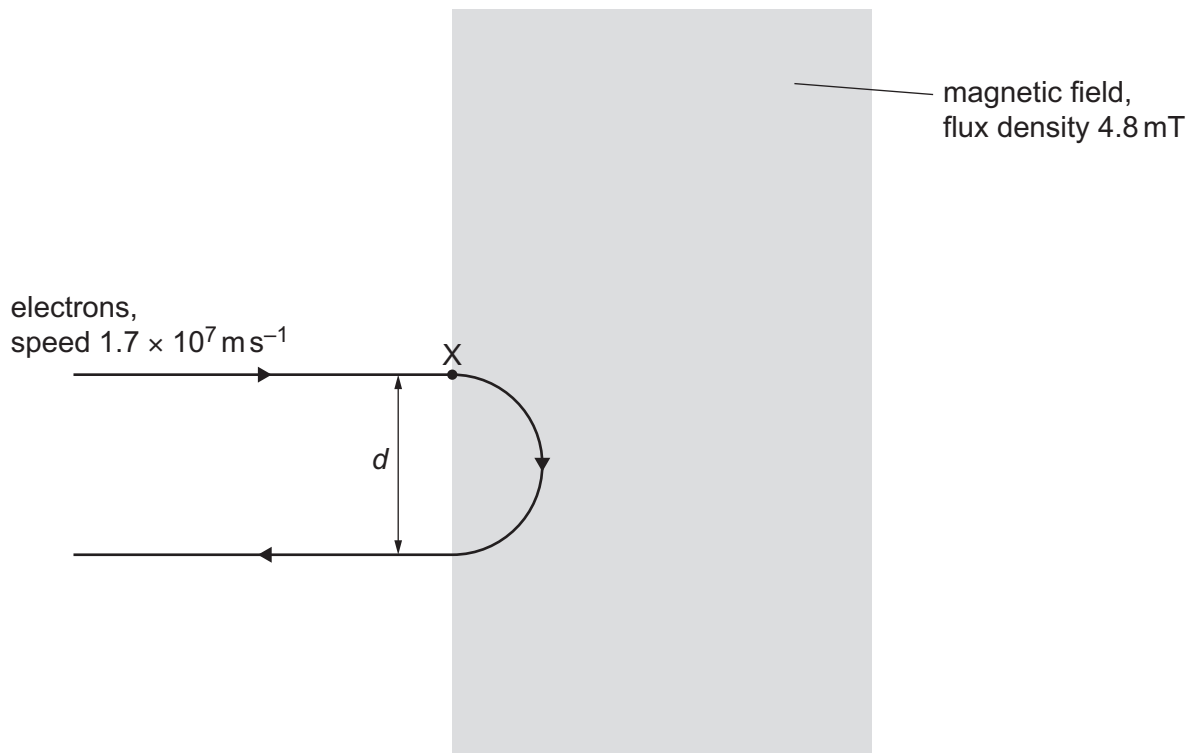


Fig. 6.1

The path of the electrons remains in the plane of the page.

- (i) State the direction of the magnetic field.

.....

..... [1]

- (ii) Show that the magnitude of the force exerted on each electron by the magnetic field is $1.3 \times 10^{-14} \text{ N}$.

[2]

- (iii) On Fig. 6.1, draw an arrow to indicate the direction of the centripetal acceleration of the electron where it enters the magnetic field at point X. [1]

- (iv) Use the information in **(b)(ii)** to calculate the distance d between the path of the electrons entering the magnetic field and the path of the electrons leaving it.

$d = \dots\dots\dots \text{ m}$ [3]

- (c) The electrons in **(b)** are replaced with positrons that are moving with speed $3.4 \times 10^7 \text{ ms}^{-1}$ along the same initial path as the electrons.

The positrons enter the magnetic field at point X on Fig. 6.1.

On Fig. 6.1, draw a line to show the path of the positrons through the magnetic field. [3]

[Total: 12]

- 7 A varying current I passes through a resistor of resistance R in the circuit shown in Fig. 7.1.

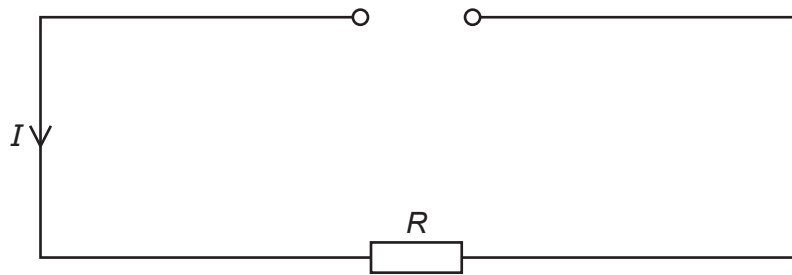


Fig. 7.1

Fig. 7.2 shows the variation with time t of I .

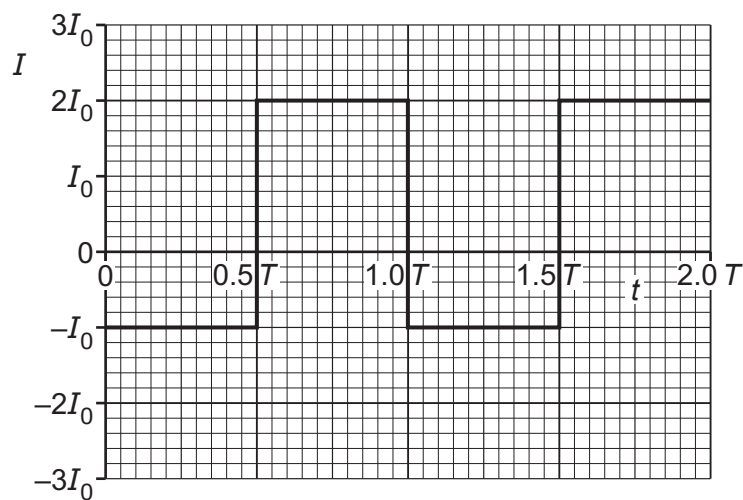


Fig. 7.2

The current has magnitude $2I_0$ when it is in the positive direction and I_0 when it is in the negative direction. The period of the variation of the current is T .

- (a) Determine expressions, in terms of I_0 and R , for the power P dissipated in the resistor for the times when:

- (i) the current is in the negative direction

$$P = \dots\dots\dots [1]$$

- (ii) the current is in the positive direction.

$$P = \dots\dots\dots [1]$$

- (b) On Fig. 7.3, sketch the variation of P with t between $t = 0$ and $t = 2.0T$. Label the power axis with an appropriate scale.

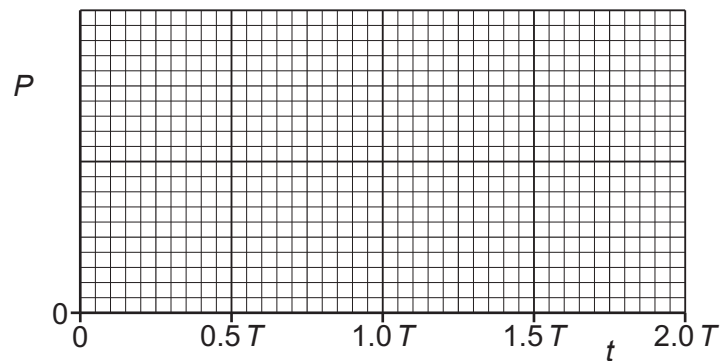


Fig. 7.3

[3]

- (c) Use your answer in (b) to determine an expression, in terms of I_0 and R , for:

- (i) the mean power $\langle P \rangle$ in the resistor

$$\langle P \rangle = \dots\dots\dots [1]$$

- (ii) the root-mean-square (r.m.s.) current $I_{\text{r.m.s.}}$ in the resistor.

$$I_{\text{r.m.s.}} = \dots\dots\dots [2]$$

[Total: 8]

- 8 (a) (i) Show that the momentum p of a photon of electromagnetic radiation with wavelength λ is given by

$$p = \frac{h}{\lambda}$$

where h is the Planck constant.

[2]

- (ii) Use the expression in (a)(i) to show that a photon in free space that has a momentum of $9.5 \times 10^{-28} \text{ N s}$ is a photon of red light.

[1]

- (b) A beam of red light of intensity 160 W m^{-2} is incident normally on a plane mirror, as shown in Fig. 8.1. The momentum of each photon in the beam is $9.5 \times 10^{-28} \text{ N s}$.

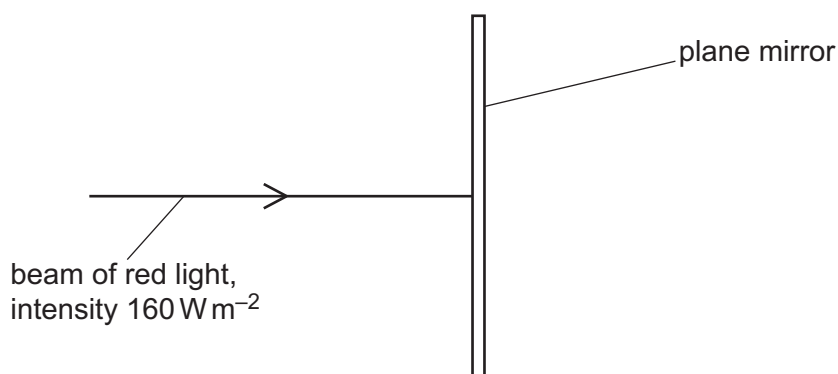


Fig. 8.1

All of the light is reflected by the mirror in the opposite direction to its original path. The cross-sectional area of the beam is $2.5 \times 10^{-6} \text{ m}^2$.

- (i) Show that the number of photons incident on the mirror per unit time is $1.4 \times 10^{15} \text{ s}^{-1}$.

[2]

- (ii) Use the information in **(b)(i)** to determine the pressure exerted by the light beam on the mirror.

pressure = Pa [3]

- (c) The beam of red light in **(b)** is now replaced with a beam of blue light of the same intensity.

Suggest and explain whether the pressure exerted on the mirror by the beam of blue light is less than, the same as, or greater than the pressure exerted by the beam of red light.

.....

 [2]

[Total: 10]

- 9 (a) State what is meant by nuclear fusion.

.....

.....

..... [2]

- (b) On Fig. 9.1, sketch the variation of binding energy per nucleon with nucleon number A for values of A between 1 and 250.

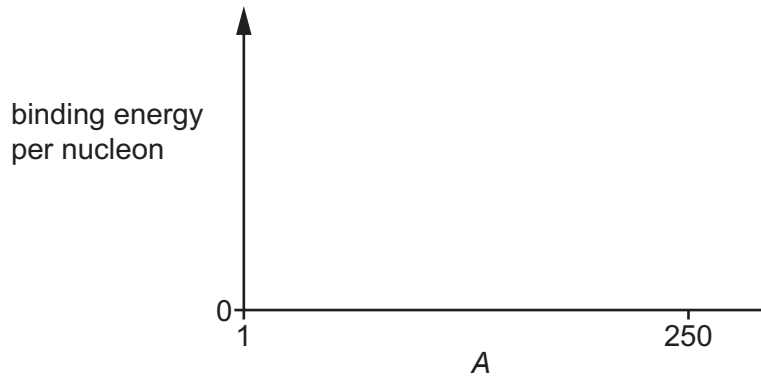


Fig. 9.1

[2]

- (c) On your line in Fig. 9.1, label:

- (i) a point X that could represent a nucleus that undergoes alpha-decay [1]
- (ii) a point Y that could represent a nucleus that undergoes nuclear fusion. [1]

- (d) A nucleus Z undergoes nuclear fission to form strontium-93 ($^{93}_{38}\text{Sr}$) and xenon-139 ($^{139}_{54}\text{Xe}$) according to

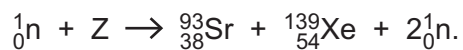


Table 9.1 shows the binding energies of the strontium-93 and xenon-139 nuclei.

Table 9.1

nucleus	binding energy/J
$^{93}_{38}\text{Sr}$	1.25×10^{-10}
$^{139}_{54}\text{Xe}$	1.81×10^{-10}

The fission of 1.00 mol of Z releases 1.77×10^{13} J of energy.

Determine the binding energy per nucleon, in MeV, of Z.

binding energy per nucleon = MeV [4]

[Total: 10]

- 10** Ultrasound and X-rays are both types of wave that are used in medical diagnosis to form images of internal body structures.

(a) Complete Table 10.1 to state, for each type of wave:

- the method of production of the wave
- whether the wave that is detected and used to form the image is the wave that has been absorbed, reflected or transmitted by the internal body structure.

Table 10.1

	ultrasound	X-rays
method of production		
detected wave (absorbed, reflected or transmitted)		

[4]

- (b) (i)** For one type of wave passing through tissue, the wave has 72% of its initial intensity after it has passed through 6.2 cm of the tissue.

Calculate the linear attenuation coefficient μ of the tissue for this wave.

$$\mu = \dots\dots\dots \text{ cm}^{-1} \quad [2]$$

- (ii)** Another wave of the same type as in **(b)(i)** passes through 9.3 cm of the same tissue.

Calculate the percentage of the initial intensity of the wave that is attenuated by the tissue.

$$\text{percentage attenuated} = \dots\dots\dots \% \quad [2]$$

[Total: 8]

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9702/51

October/November 2023

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 30.
- The number of marks for each question or part question is shown in brackets [].

This document has 8 pages.

- 1 Two coils, C and D, are placed with their axes on a straight line, as shown in Fig. 1.1.

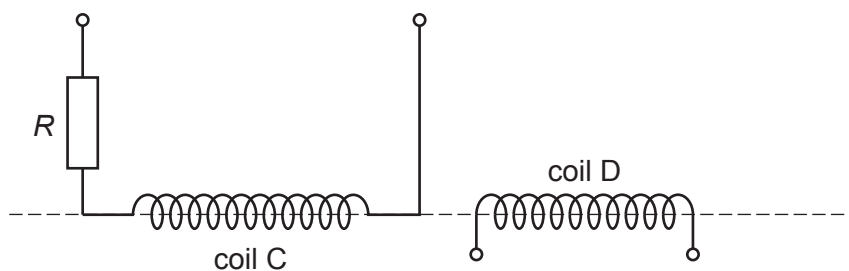


Fig. 1.1

A resistor of resistance R is connected in series with coil C.

A changing magnetic flux of frequency f in coil C causes an electromotive force (e.m.f.) E to be induced across the terminals of coil D.

It is suggested that E is related to f by the relationship

$$E = \frac{pf^qV}{R}$$

where V is the potential difference across the resistor and coil C, and p and q are constants.

Plan a laboratory experiment to test the relationship between E and f .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for p and q .

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

Diagram

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- 2 A block of modelling clay of mass M is attached to a string as shown in Fig. 2.1.

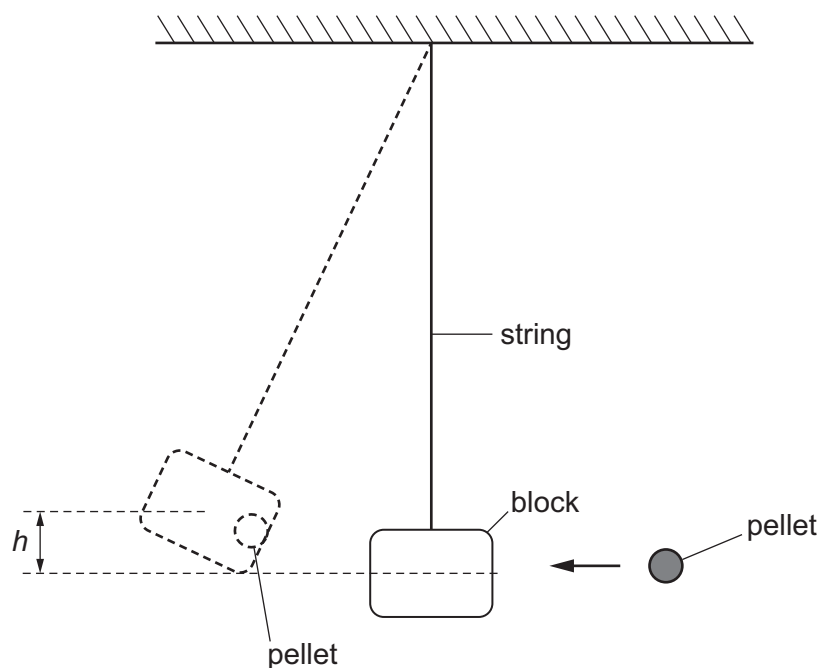


Fig. 2.1

A pellet travelling at speed u enters the block and causes the block to move through a vertical height h .

The experiment is repeated for different values of M .

It is suggested that h and M are related by the equation

$$\frac{1}{h} = 2gc \frac{M + Z}{uZ}^2$$

where g is the acceleration of free fall and Z is a constant.

- (a) A graph is plotted of $\sqrt{\frac{1}{h}}$ on the y -axis against M on the x -axis.

Determine expressions for the gradient and y -intercept.

gradient =

y -intercept =

[1]

(b) Values of M and h are given in Table 2.1.

Table 2.1

M/g	h/cm	$\sqrt{\frac{1}{h}}/\text{cm}^{-\frac{1}{2}}$
565	21.0 ± 0.2	
637	17.8 ± 0.2	
675	16.2 ± 0.2	
723	14.6 ± 0.2	
790	12.6 ± 0.2	
892	10.2 ± 0.2	

Calculate and record values of $\sqrt{\frac{1}{h}}/\text{cm}^{-\frac{1}{2}}$ in Table 2.1.

Include the absolute uncertainties in $\sqrt{\frac{1}{h}}$. [2]

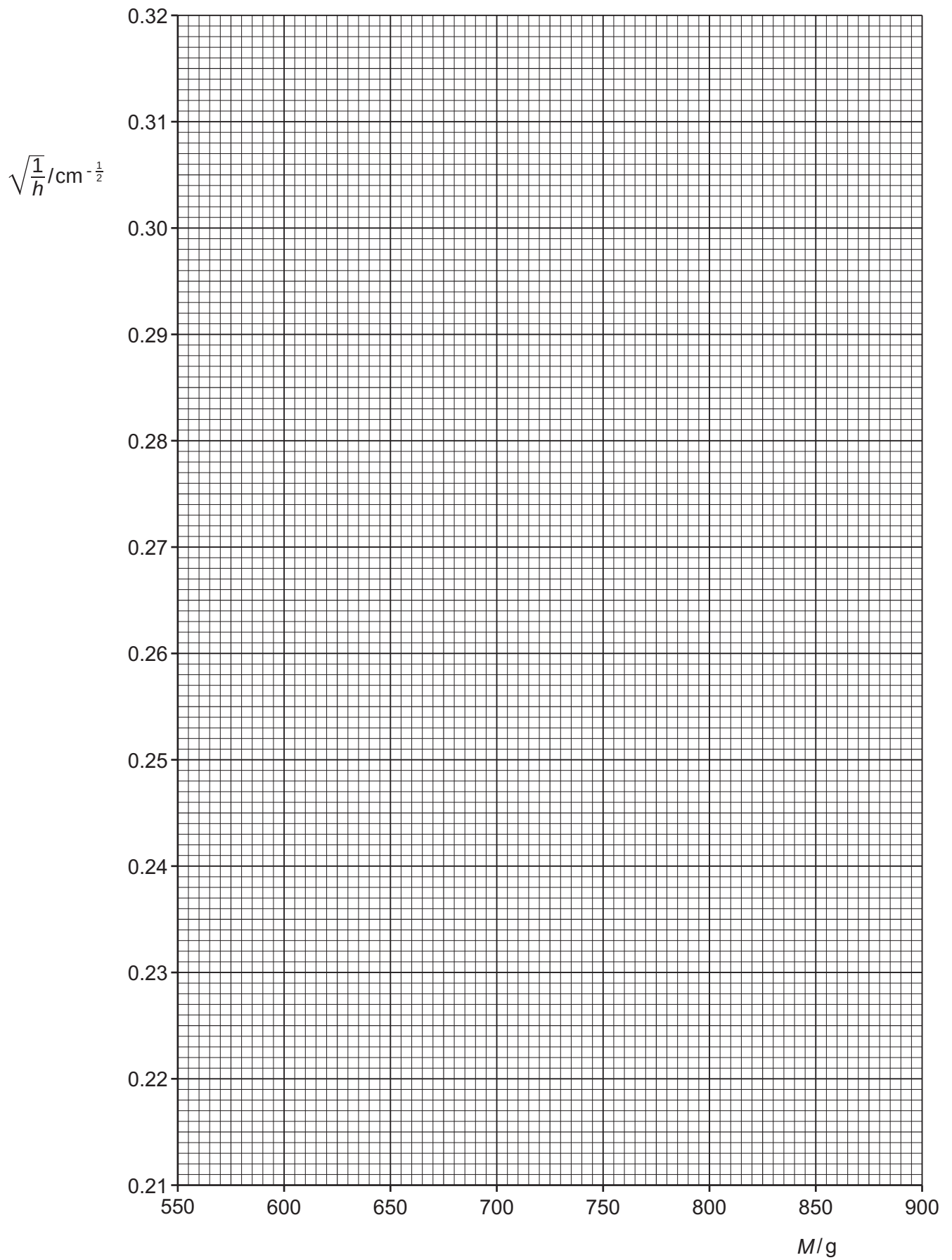
(c) (i) Plot a graph of $\sqrt{\frac{1}{h}}/\text{cm}^{-\frac{1}{2}}$ against M/g .

Include error bars for $\sqrt{\frac{1}{h}}$. [2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines. [2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]



- (iv) Determine the y -intercept of the line of best fit. Include the absolute uncertainty in your answer.

y -intercept = [2]

- (d) (i) Using your answers to (a), (c)(iii) and (c)(iv), determine the values of u and Z . Include appropriate units.

Data: $g = 981 \text{ cm s}^{-2}$

$u = \dots\dots\dots$

$Z = \dots\dots\dots$ [2]

- (ii) Determine the percentage uncertainty in Z .

percentage uncertainty in $Z = \dots\dots\dots\%$ [1]

- (e) The experiment is repeated. Determine the mass M that gives a value of h of 25.0 cm.

$M = \dots\dots\dots \text{ g}$ [1]

[Total: 15]



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9702/52

October/November 2023

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 30.
- The number of marks for each question or part question is shown in brackets [].

This document has 8 pages.

- 1 Two identical beakers, each of mass M , are attached to each other using string and suspended from a pulley, as shown in Fig. 1.1.

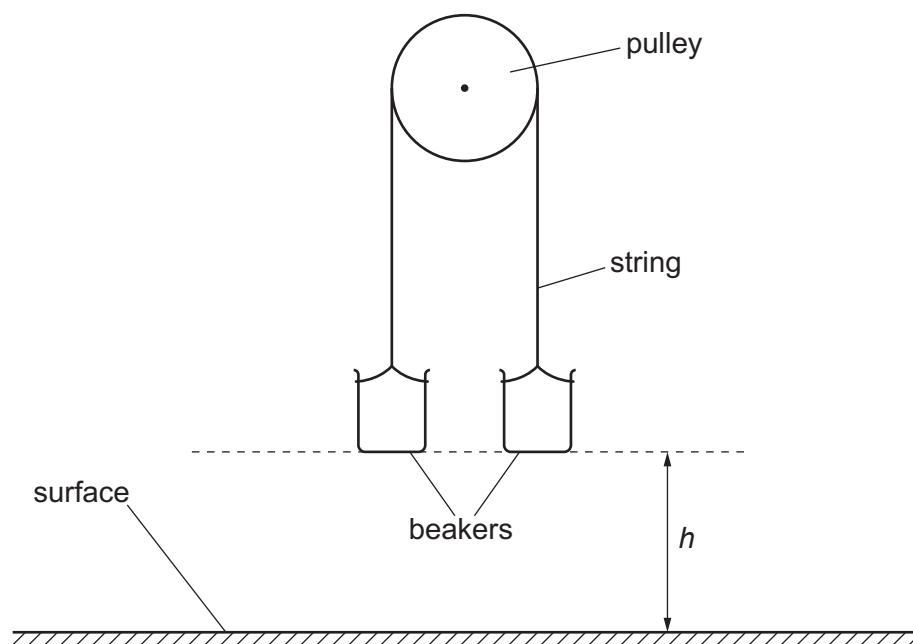


Fig. 1.1 (not to scale)

The beakers are held at rest at a height h above a surface. Cooking oil of volume V is added to one of the beakers.

The beakers are released so that the beaker with the oil begins to fall. The speed of the beaker as it reaches the surface is z .

It is suggested that z is related to V by the relationship

$$\frac{2h}{z^2} = \frac{2M}{abV} + \frac{1}{b}$$

where a and b are constants.

Plan a laboratory experiment to test the relationship between z and V .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for a and b .

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

Diagram

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[illegible]

- 2 A student investigates the discharge of capacitors in the circuit shown in Fig. 2.1.

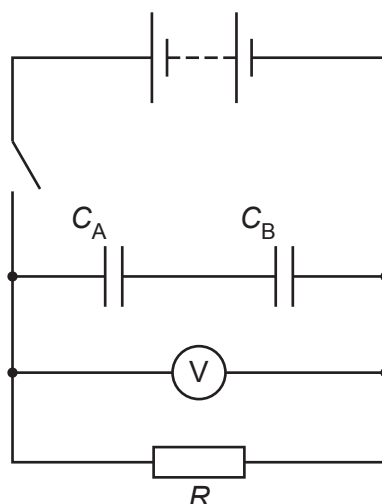


Fig. 2.1

The capacitors have capacitances C_A and C_B .

The student closes the switch to charge the capacitors.

The switch is opened and a stop-watch is started. The capacitors discharge through the resistor of resistance R . At a fixed time t the voltmeter reading V is recorded.

The experiment is repeated for different values of C_A and C_B .

For each combination of C_A and C_B , the combined capacitance C is calculated.

It is suggested that C and V are related by the equation

$$V = I_0 R e^{-\frac{t}{CR}}$$

where I_0 is the initial current in the resistor.

- (a) A graph is plotted of $\ln V$ on the y -axis against $\frac{1}{C}$ on the x -axis.

Determine expressions for the gradient and y -intercept.

gradient =

y -intercept =

[1]

(b) Values of C_A , C_B and V are given in Table 2.1.

Table 2.1

$C_A/10^{-4}\text{F}$	$C_B/10^{-4}\text{F}$	$\frac{1}{C}/10^4\text{F}^{-1}$	V/V	$\ln(V/V)$
2.2	2.2		2.45 ± 0.05	
2.2	3.3		2.75 ± 0.05	
2.2	5.6		3.05 ± 0.05	
3.3	3.3		3.10 ± 0.05	
3.3	5.6		3.50 ± 0.05	
5.6	5.6		3.85 ± 0.05	

The relationship between C , C_A and C_B is

$$\frac{1}{C} = \frac{C_A + C_B}{C_A C_B}.$$

Calculate and record values of $\frac{1}{C}/10^4\text{F}^{-1}$ and $\ln(V/V)$ in Table 2.1.

Include the absolute uncertainties in $\ln(V/V)$. [2]

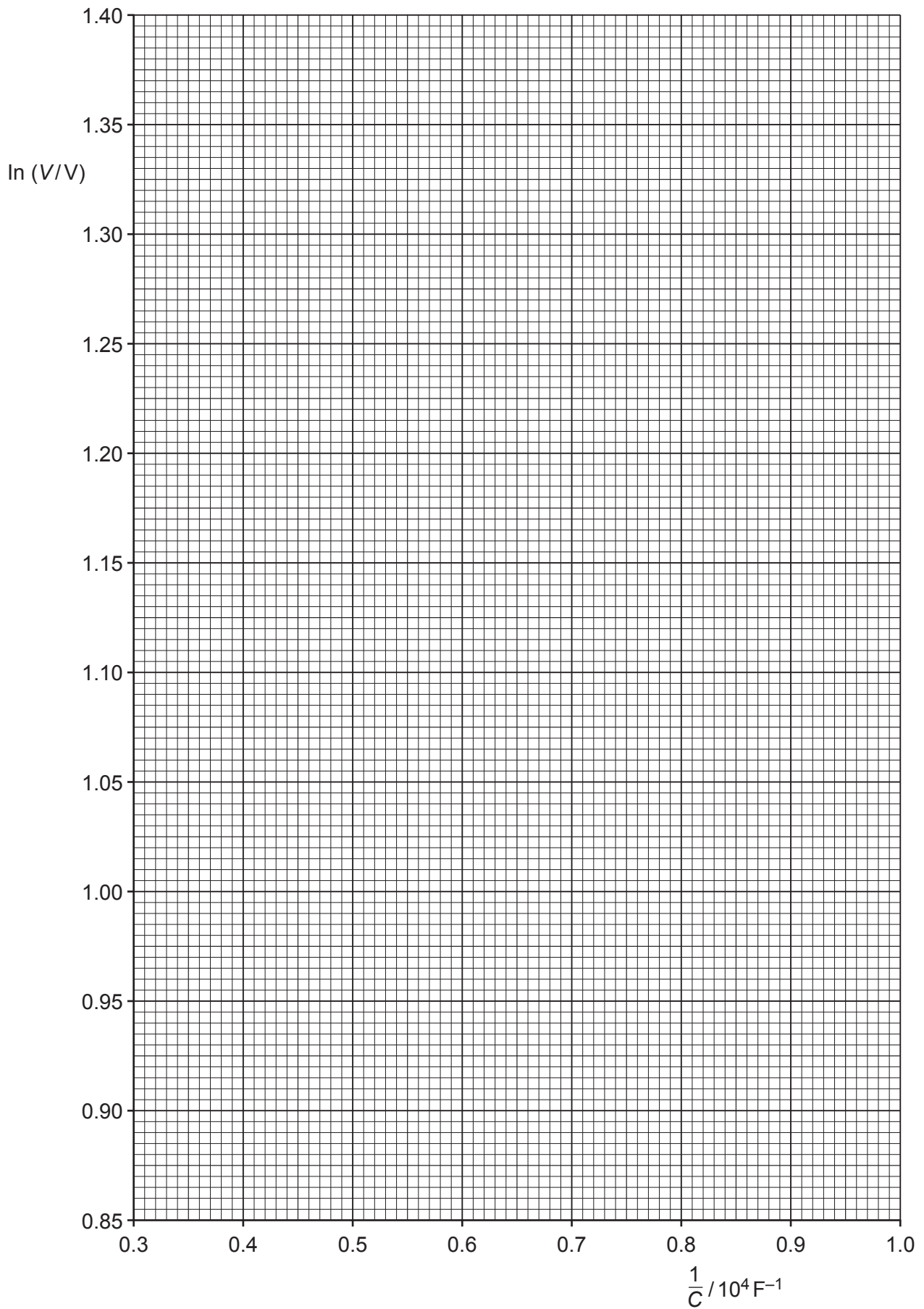
(c) (i) Plot a graph of $\ln(V/V)$ against $\frac{1}{C}/10^4\text{F}^{-1}$.

Include error bars for $\ln(V/V)$. [2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines. [2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]



- (iv) Determine the y -intercept of the line of best fit. Do **not** include the absolute uncertainty.

y -intercept = [1]

- (d) (i) Using your answers to (a), (c)(iii) and (c)(iv), determine the values of R and I_0 . Include appropriate units.

Data: $t = (30 \pm 1)\text{s}$

$R = \dots\dots\dots$

$I_0 = \dots\dots\dots$ [3]

- (ii) Determine the percentage uncertainty in R .

percentage uncertainty in $R = \dots\dots\dots\%$ [1]

- (e) The experiment is repeated with the same value of t . Determine the combined capacitance C that gives a value of V of 1.20 V.

$C = \dots\dots\dots\text{F}$ [1]

[Total: 15]



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9702/53

October/November 2023

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
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This document has 8 pages.

- 1 Two coils, C and D, are placed with their axes on a straight line, as shown in Fig. 1.1.

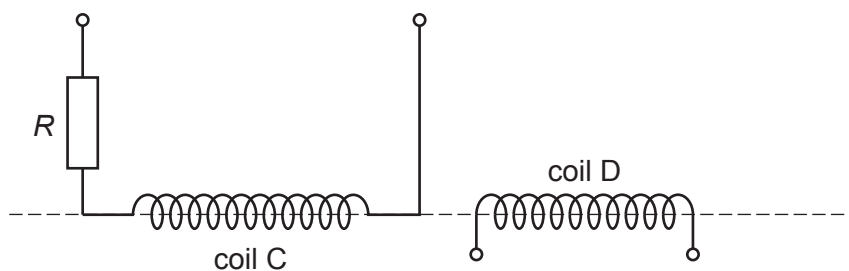


Fig. 1.1

A resistor of resistance R is connected in series with coil C.

A changing magnetic flux of frequency f in coil C causes an electromotive force (e.m.f.) E to be induced across the terminals of coil D.

It is suggested that E is related to f by the relationship

$$E = \frac{pf^qV}{R}$$

where V is the potential difference across the resistor and coil C, and p and q are constants.

Plan a laboratory experiment to test the relationship between E and f .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for p and q .

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

Diagram

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- 2 A block of modelling clay of mass M is attached to a string as shown in Fig. 2.1.

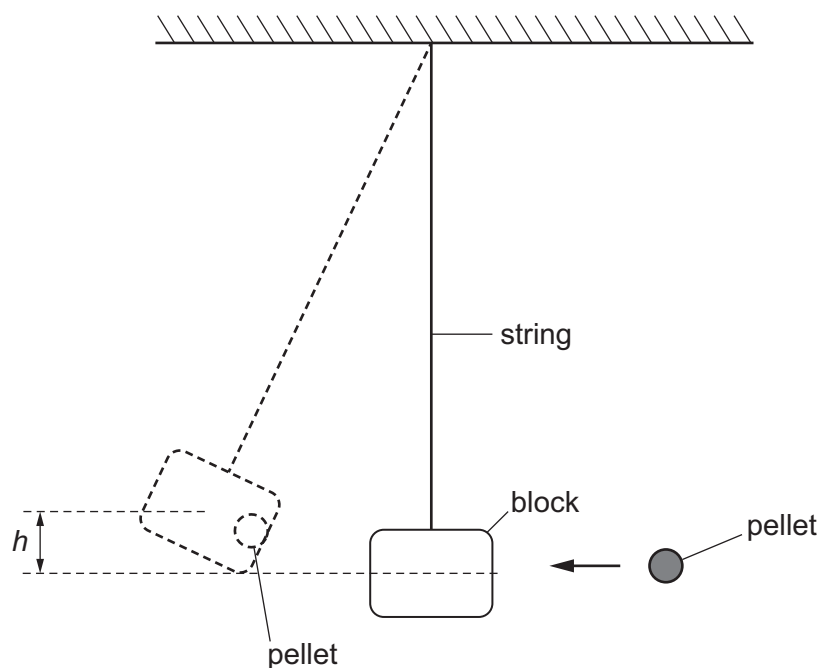


Fig. 2.1

A pellet travelling at speed u enters the block and causes the block to move through a vertical height h .

The experiment is repeated for different values of M .

It is suggested that h and M are related by the equation

$$\frac{1}{h} = 2gc \frac{M + Z}{uZ}^2$$

where g is the acceleration of free fall and Z is a constant.

- (a) A graph is plotted of $\sqrt{\frac{1}{h}}$ on the y -axis against M on the x -axis.

Determine expressions for the gradient and y -intercept.

gradient =

y -intercept =

[1]

(b) Values of M and h are given in Table 2.1.

Table 2.1

M/g	h/cm	$\sqrt{\frac{1}{h}}/\text{cm}^{-\frac{1}{2}}$
565	21.0 ± 0.2	
637	17.8 ± 0.2	
675	16.2 ± 0.2	
723	14.6 ± 0.2	
790	12.6 ± 0.2	
892	10.2 ± 0.2	

Calculate and record values of $\sqrt{\frac{1}{h}}/\text{cm}^{-\frac{1}{2}}$ in Table 2.1.

Include the absolute uncertainties in $\sqrt{\frac{1}{h}}$. [2]

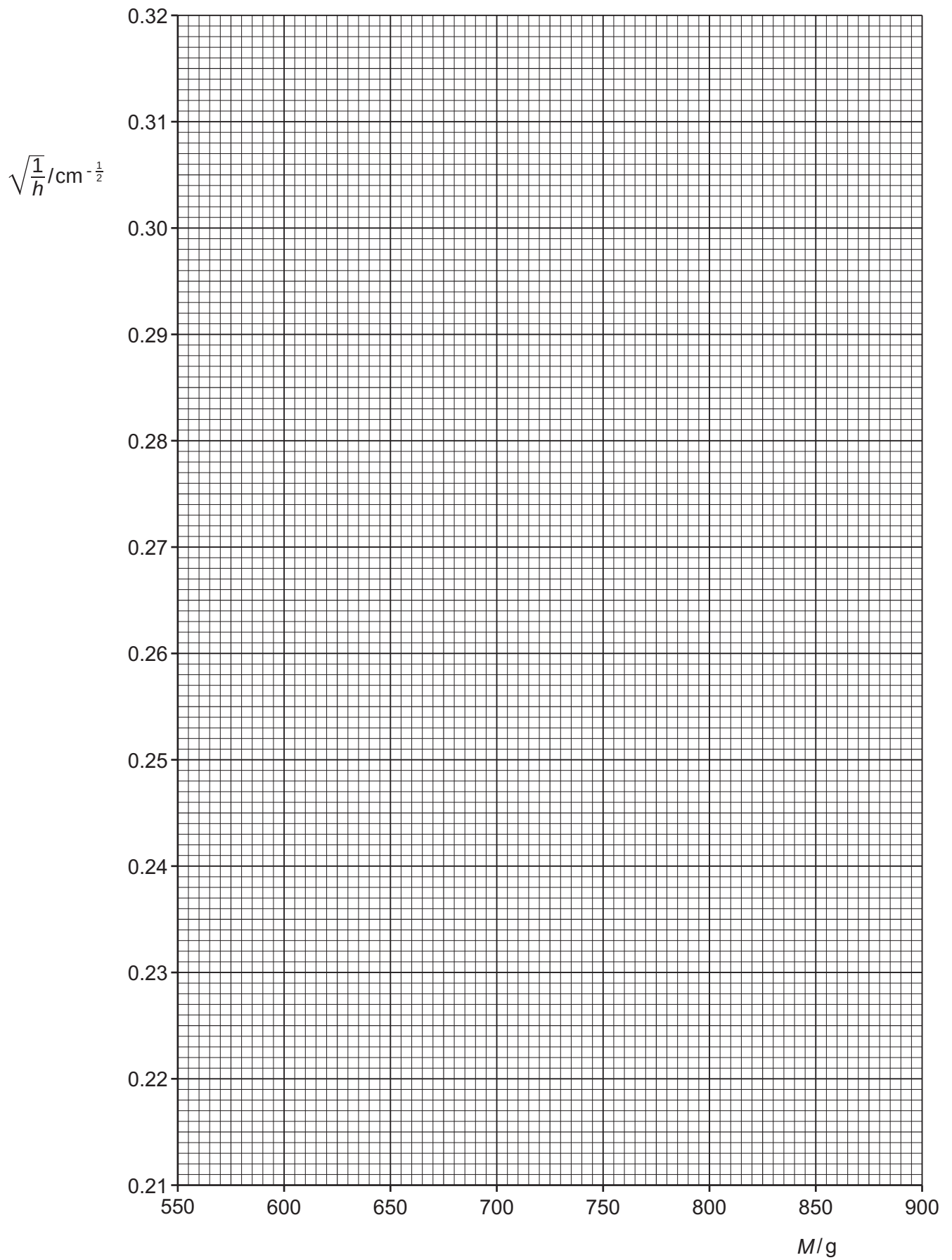
(c) (i) Plot a graph of $\sqrt{\frac{1}{h}}/\text{cm}^{-\frac{1}{2}}$ against M/g .

Include error bars for $\sqrt{\frac{1}{h}}$. [2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines. [2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]



- (iv) Determine the y -intercept of the line of best fit. Include the absolute uncertainty in your answer.

y -intercept = [2]

- (d) (i) Using your answers to (a), (c)(iii) and (c)(iv), determine the values of u and Z . Include appropriate units.

Data: $g = 981 \text{ cm s}^{-2}$

$u = \dots\dots\dots$

$Z = \dots\dots\dots$ [2]

- (ii) Determine the percentage uncertainty in Z .

percentage uncertainty in $Z = \dots\dots\dots\%$ [1]

- (e) The experiment is repeated. Determine the mass M that gives a value of h of 25.0 cm.

$M = \dots\dots\dots \text{ g}$ [1]

[Total: 15]



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9702/42

February/March 2024

2 hours

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [].

This document has **28** pages. Any blank pages are indicated.

Data

acceleration of free fall	$g = 9.81 \text{ m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
unified atomic mass unit	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27} \text{ kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$ $(\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ m F}^{-1})$
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_o = \frac{f_s v}{v \mp v_s}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \dots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$

gravitational potential	$\phi = -\frac{GM}{r}$
gravitational potential energy	$E_P = -\frac{GMm}{r}$
pressure of an ideal gas	$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$
simple harmonic motion	$a = -\omega^2 x$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t$ $v = \omega \sqrt{(x_0^2 - x^2)}$
electric potential	$V = \frac{Q}{4\pi\epsilon_0 r}$
electrical potential energy	$E_P = \frac{Qq}{4\pi\epsilon_0 r}$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$
capacitors in parallel	$C = C_1 + C_2 + \dots$
discharge of a capacitor	$x = x_0 e^{-\frac{t}{RC}}$
Hall voltage	$V_H = \frac{BI}{ntq}$
alternating current/voltage	$x = x_0 \sin \omega t$
radioactive decay	$x = x_0 e^{-\lambda t}$
decay constant	$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$
intensity reflection coefficient	$\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$
Stefan–Boltzmann law	$L = 4\pi\sigma r^2 T^4$
Doppler redshift	$\frac{\Delta\lambda}{\lambda} \hat{=} \frac{Df}{f} \hat{=} \frac{v}{c}$

- 1 (a) Explain why the gravitational potential near to a point mass is negative.

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..... [2]

- (b) A planet may be assumed to be a uniform sphere. It has gravitational potential ϕ at distance r from the centre of the planet.

The variation with $\frac{1}{r}$ of ϕ is shown in Fig. 1.1.

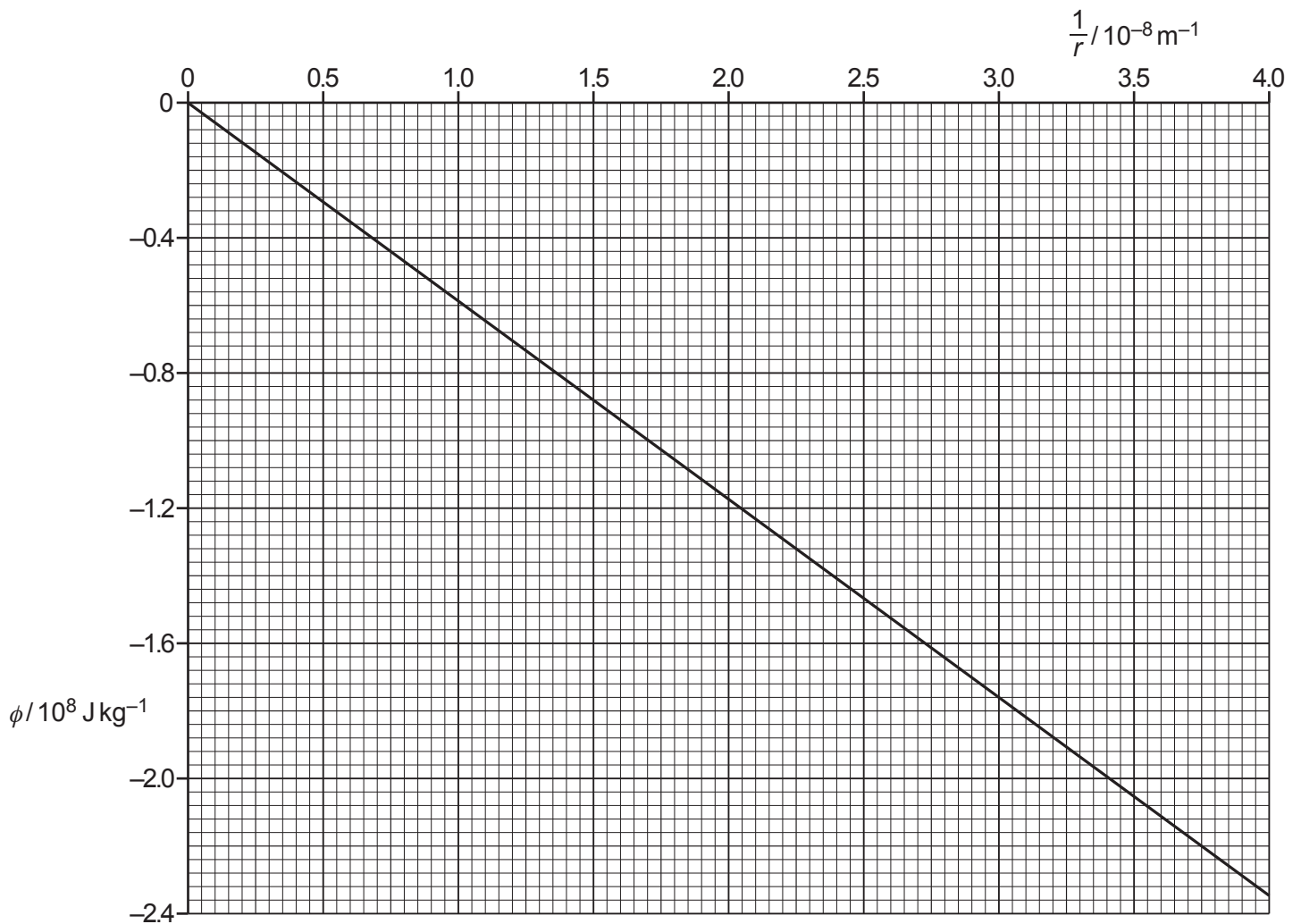


Fig. 1.1

- (i) Show that the mass of the planet is 8.8×10^{25} kg.

[2]

- (ii) The period of rotation of the planet is 0.72 Earth days.

A satellite in orbit around the planet remains above the same point on the surface of the planet.

Use the mass of the planet in **(b)(i)** to determine the radius R of the orbit of the satellite.

$R = \dots\dots\dots$ m [3]

- (iii) The speed of the satellite in **(b)(ii)** is 8400 ms^{-1} . The mass of the satellite is 1200 kg.

Determine the additional energy required to move the satellite from its orbit to infinity.

energy required = $\dots\dots\dots$ J [3]

[Total: 10]

- 2 (a) By referring to both kinetic energy and potential energy, explain what is meant by the internal energy of an ideal gas.

.....

.....

.....

..... [2]

- (b) A fixed mass of an ideal gas at a temperature of 20°C is sealed in a cylinder by a piston, as shown in Fig. 2.1.

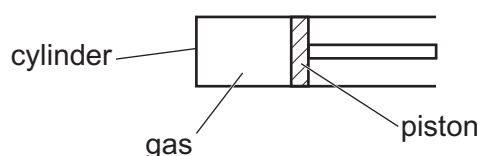


Fig. 2.1

The initial volume of the gas is $1.24 \times 10^{-4} \text{ m}^3$.

Thermal energy is supplied to the gas and its volume increases by $5.20 \times 10^{-5} \text{ m}^3$.

- (i) The piston is freely moving so that the gas is always at atmospheric pressure.

Atmospheric pressure is $1.01 \times 10^5 \text{ Pa}$.

Calculate the work done by the gas.

work done by gas = J [2]

- (ii) Calculate the final thermodynamic temperature T of the gas.

$$T = \dots\dots\dots \text{K} \quad [2]$$

- (iii) The mass of the gas is 16 g. For this expansion, there is a net transfer of 960 J of thermal energy to the gas.

Calculate the specific heat capacity c of the gas at this pressure.

$$c = \dots\dots\dots \text{J kg}^{-1} \text{K}^{-1} \quad [2]$$

- (c) The gas in (b) is allowed to return to its starting temperature. The piston is now fixed in position.

Thermal energy is supplied to increase the temperature to the same final temperature as in (b).

Use the first law of thermodynamics to suggest and explain how the specific heat capacity of the gas for this situation compares with the value in (b)(iii).

.....

 [3]

[Total: 11]

- 3 A small object of mass 24 g rests on a platform. The platform is attached to an oscillator, as shown in Fig. 3.1.

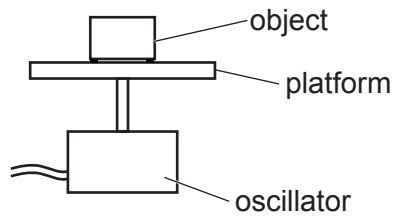


Fig. 3.1

The oscillator moves the platform up and down.

- (a) The total energy of the oscillations of the object is 2.2×10^{-4} J.
In one oscillation the object travels a total distance of 14 mm.

Calculate the angular frequency ω of the oscillations.

$$\omega = \dots\dots\dots \text{ rad s}^{-1} \quad [3]$$

- (b) The frequency of the oscillator is fixed, and the amplitude of the oscillations is gradually increased.
- (i) Calculate the maximum amplitude of the oscillations so the object does not lose contact with the platform.

$$\text{amplitude} = \dots\dots\dots \text{ m} \quad [2]$$

- (ii) The amplitude of the oscillations is increased so it is greater than the value in (b)(i).

State and explain the position in an oscillation where the object first loses contact with the platform.

.....

.....

.....

..... [2]

[Total: 7]

- 4 (a) Three capacitors are connected as shown in Fig. 4.1.

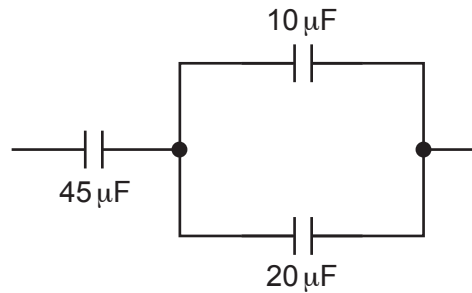


Fig. 4.1

Determine the total capacitance, in μF , of the network of three capacitors.

capacitance = μF [2]

- (b) A capacitor of capacitance $45\ \mu\text{F}$ is connected to a variable power supply initially set at $8.0\ \text{V}$. The output of the power supply increases so that the potential difference (p.d.) across the capacitor increases to $9.6\ \text{V}$.

Calculate the increase in energy ΔE stored in the capacitor.

$\Delta E = \dots\dots\dots\ \text{J}$ [2]

- (c) A sinusoidal a.c. power supply is connected to the input of a bridge rectifier. The output of the rectifier is connected to a load resistor.
- (i) Complete the circuit in Fig. 4.2 by adding a capacitor to smooth the p.d. across the load resistor.

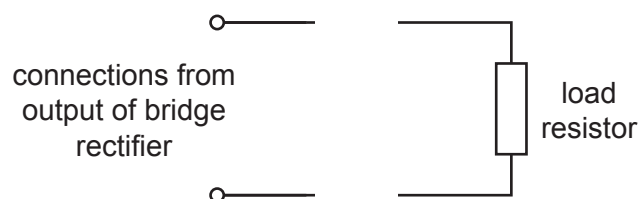


Fig. 4.2

[1]

(ii) The variation with time t of the p.d. V of the smoothed output is shown in Fig. 4.3.

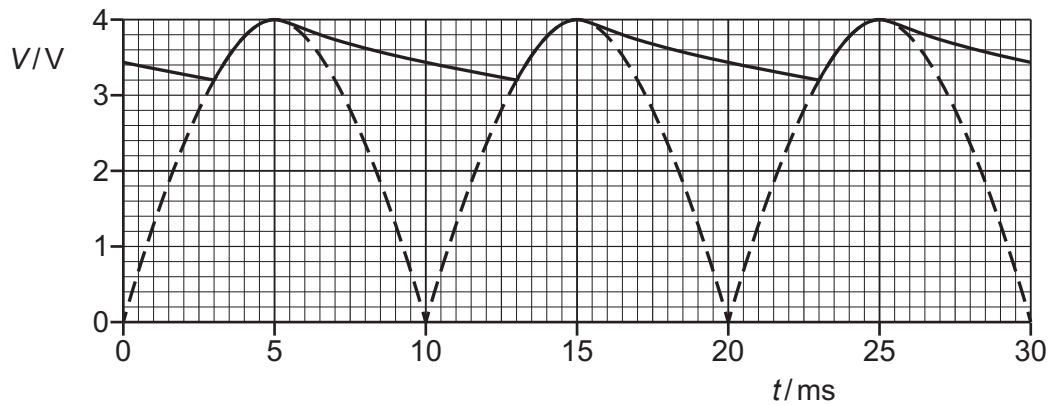


Fig. 4.3

Determine the time constant, in ms, of the smoothing circuit.

time constant = ms [3]

(d) A sinusoidal a.c. power supply has a maximum power of 16 W.

State the value of the mean power when the output of the power supply is:

(i) full-wave rectified

mean power = W [1]

(ii) half-wave rectified.

mean power = W [1]

[Total: 10]

- 5 (a) An object travels in a circle at constant speed.

State the names of **two** quantities that vary during the motion of the object.

1

2 [2]

- (b) A charged particle of mass m and with charge q enters a region of uniform magnetic field, perpendicular to the field lines. The magnetic flux density is B .

The particle travels in a circle with period T and radius r .

- (i) By considering the magnetic force acting on the particle, show that

$$B = \frac{2\pi m}{qT}.$$

[3]

- (ii) The particle is an alpha particle. The period of the circular motion is $2.5\mu\text{s}$.

Calculate B .

$$B = \dots\dots\dots \text{T} \quad [2]$$

- (iii) A second alpha particle is in the same uniform field. It travels in a circle of radius $2r$.

State and explain how the periods of the motion of the two particles compare.

.....

.....

..... [1]

- (iv) The speed of the alpha particle in (b)(ii) is $1.1 \times 10^6 \text{ m s}^{-1}$. An electric field is applied so that this particle now moves with constant velocity.

Use your answer in (b)(ii) to calculate the electric field strength E . Give the unit with your answer.

$E =$ unit [2]

[Total: 10]

- 6 (a) A small coil C has 64 turns and cross-sectional area 0.71 cm^2 . The coil is placed inside a solenoid as shown in Fig. 6.1.

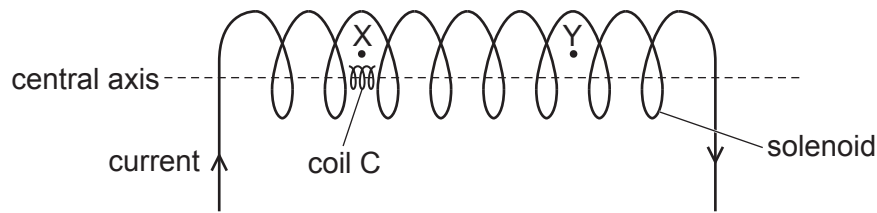


Fig. 6.1

The centre of coil C is on the central axis of the solenoid.

- (i) There is a constant current in the solenoid.
Coil C is moved through the solenoid from position X to position Y.

On Fig. 6.2, sketch a line to show the variation of the magnetic flux linkage in coil C with position as it moves from X to Y.

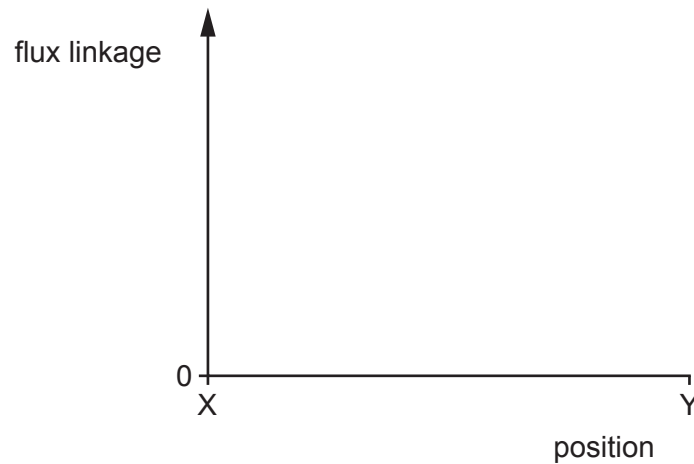


Fig. 6.2

[1]

- (ii) Explain the shape of your line in (a)(i).

.....

.....

.....

..... [2]

- (iii) Coil C is now held stationary at X. The current in the solenoid varies so that the magnetic flux density B at X varies from time 0 to time $4t$ as shown in Fig. 6.3.

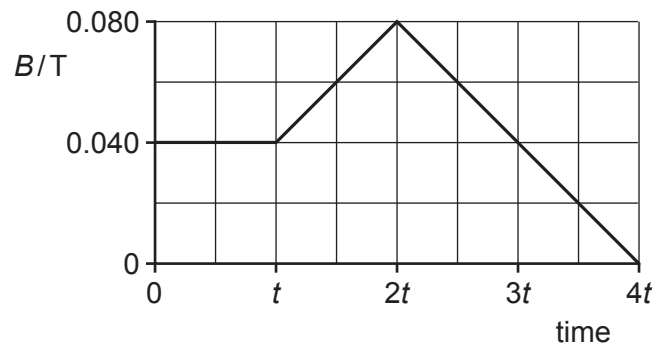


Fig. 6.3

Calculate the maximum magnetic flux linkage in coil C.

flux linkage = Wb [2]

- (iv) On Fig. 6.4, sketch a line to show the induced electromotive force (e.m.f.) E in coil C from time 0 to time $4t$.

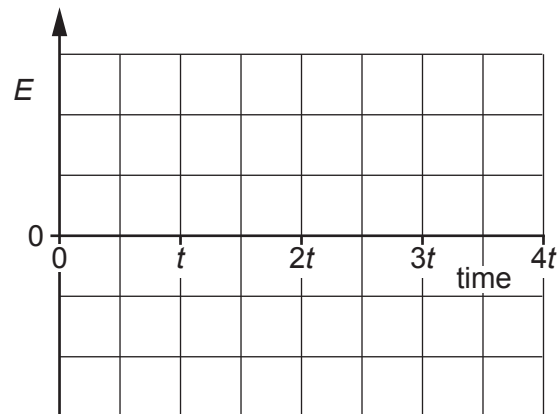


Fig. 6.4

[3]

- (b) A metal spring rests on a smooth table. The turns of the spring are equally spaced. The ends of the spring are connected to a d.c. power supply, as shown in Fig. 6.5.

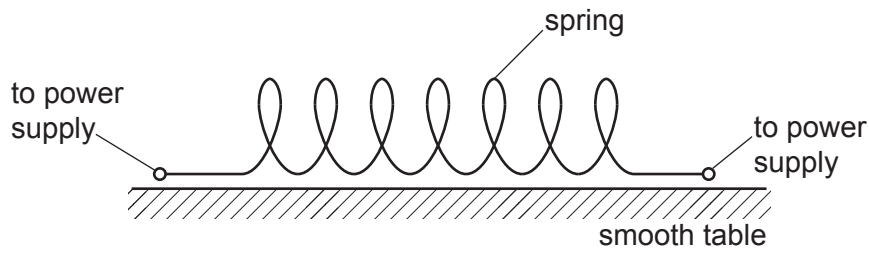


Fig. 6.5

The spring is connected to the d.c. power supply using flexible leads. The spring is not under tension.

With reference to magnetic fields, describe and explain the change in the distance between the turns of the spring when the power supply is first switched on.

.....

.....

.....

.....

..... [3]

[Total: 11]

- 7 (a) A photon has an energy of $3.11 \times 10^{-19} \text{ J}$.

Calculate the momentum of the photon.

momentum = N s [2]

- (b) A laser beam has a power of 350 mW. The light from the laser has a wavelength of 640 nm.

- (i) Determine the number of photons emitted by the laser in a time of 1.0 s.

number = [2]

- (ii) The laser beam is incident normally on a surface that absorbs all of the photons.

Show that the force F exerted on the surface by the laser beam is given by

$$F = \frac{P}{c}$$

where P is the power of the laser beam and c is the speed of light.

[2]

- (c) Light of a single wavelength is incident on the surface of different metals. The work function energy of the metals is given in Table 7.1.

Table 7.1

metal	work function energy / eV
tungsten	4.49
magnesium	3.68
potassium	2.26

- (i) Explain the term threshold wavelength.

.....

 [1]

- (ii) For the metals in Table 7.1, calculate the value of the largest threshold wavelength.

threshold wavelength = m [2]

[Total: 9]

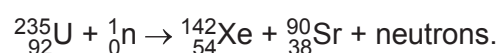
- 8 (a) State what is meant by the binding energy of a nucleus.

.....

.....

..... [2]

- (b) A nucleus of uranium-235 absorbs a neutron and becomes unstable. It then undergoes a fission reaction. One possible reaction is



- (i) Determine the number of neutrons produced in this fission reaction.

number = [1]

- (ii) Data for the binding energies per nucleon for this fission reaction are given in Table 8.1.

Table 8.1

isotope	binding energy per nucleon/MeV
uranium-235	7.59
xenon-142	8.37
strontium-90	8.72

Calculate the energy released, in MeV, from the fission of one nucleus of uranium-235.

energy = MeV [2]

- (iii) The isotope xenon-142 is unstable. The isotope xenon-132 is stable.

Suggest a reason why xenon-142 is unstable.

.....
..... [1]

- (iv) Xenon-142 decays into the isotope caesium-142.

A sample initially contains only nuclei of xenon-142. After a time equal to 6.0 s, the ratio

$$\frac{\text{number of decayed nuclei of xenon-142}}{\text{number of undecayed nuclei of xenon-142}}$$

is equal to 31.

Calculate the half-life of xenon-142. Show your working.

half-life = s [3]

[Total: 9]

- 9 (a) Electrons in a vacuum are accelerated through a potential difference of 84 kV. The electrons then strike a metal target and X-rays are produced.
- (i) Calculate the minimum wavelength of the X-rays that are produced.

wavelength = m [2]

- (ii) The melting points of two metals are given in Table 9.1.

Table 9.1

metal	melting point/°C
copper	1090
tungsten	3420

Suggest why the metal target is made from tungsten rather than copper.

.....

.....

..... [2]

- (b) An X-ray beam is incident normally on a sample of soft tissue and bone as shown in Fig. 9.1.

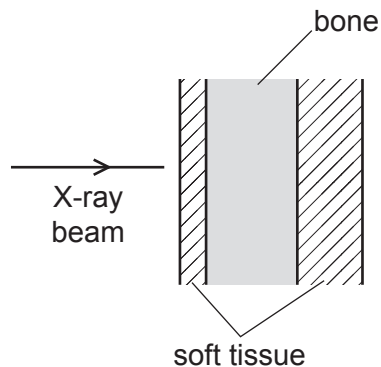


Fig. 9.1

Data for the two materials are given in Table 9.2.

Table 9.2

medium	linear attenuation coefficient μ/cm^{-1}	specific acoustic impedance $Z/10^6 \text{ kg m}^{-2} \text{ s}^{-1}$
soft tissue	0.22	1.7
bone	3.0	7.8

The total thickness of soft tissue is x . The total thickness of bone is also x .

The incident intensity of the X-ray beam is I_0 . The transmitted intensity of the X-ray beam is 13% of the incident intensity.

Determine x , in cm.

$x = \dots\dots\dots \text{ cm [3]}$

- (c) (i) Define the specific acoustic impedance of a medium.

.....

 [2]

- (ii) Use data from Table 9.2 to calculate the percentage of the intensity of ultrasound that is transmitted at a boundary between soft tissue and bone.

percentage transmitted =% [2]

- (iii) The ultrasound is now incident on the sample of soft tissue and bone shown in Fig. 9.1.

Suggest **two** reasons why the transmitted intensity through the sample is less than the answer in (c)(ii).

1

.....

2

.....

[2]

[Total: 13]

10 (a) The Sun has a surface temperature of 5780 K. The luminosity of the Sun is $3.85 \times 10^{26} \text{ W}$.

(i) Calculate the radius of the Sun.

radius = m [2]

(ii) The Earth is a distance of $1.50 \times 10^{11} \text{ m}$ from the Sun.

Calculate the radiant flux intensity F of the radiation from the Sun at a distance of $1.50 \times 10^{11} \text{ m}$. Give a unit with your answer.

$F = \dots\dots\dots$ unit [2]

(iii) The variation with wavelength of the intensity of radiation emitted from the Sun is shown in Fig. 10.1.

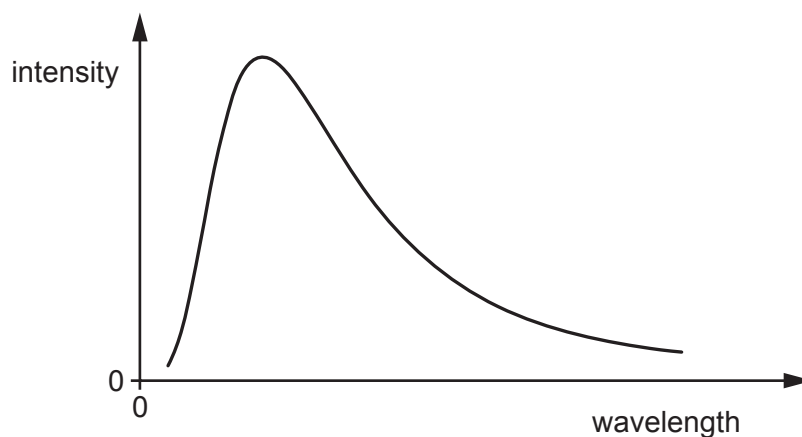


Fig. 10.1

Another star has the same radius as the Sun but has a lower surface temperature.

On Fig. 10.1, sketch a line to show the variation with wavelength of the intensity of the radiation emitted for this star. [2]

(b) A galaxy in the constellation Corona Borealis is moving away from the Earth.

(i) The visible emission spectrum for the Sun is shown in Fig. 10.2.

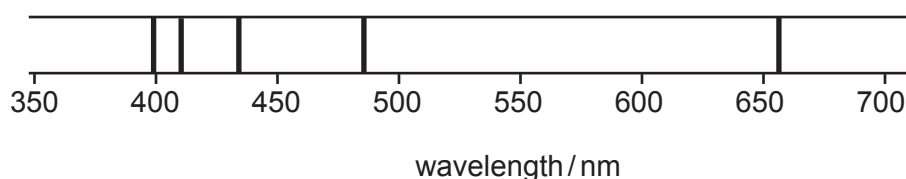


Fig. 10.2

The lines are at wavelengths of 397 nm, 410 nm, 434 nm, 486 nm and 656 nm. The compositions of the Sun and a star in the Corona Borealis galaxy are similar.

On Fig. 10.3, sketch the emission spectrum for the star in the Corona Borealis galaxy as observed from the Earth. No calculations are required.

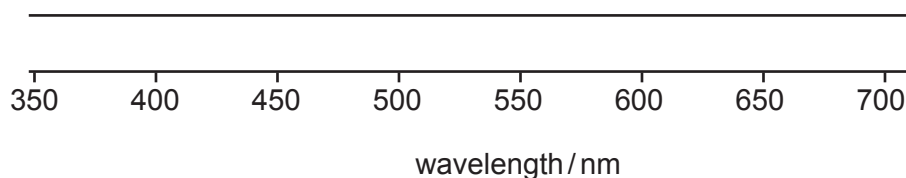


Fig. 10.3

[1]

(ii) The galaxy in Corona Borealis is moving away from the Earth at a speed of $21\,400\text{ km s}^{-1}$.

Use information from (b)(i) to calculate, in nm, the observed wavelength of the lowest visible energy emission for the star in the Corona Borealis galaxy.

wavelength = nm [2]

(iii) The wavelength in (b)(ii) is used to calculate a value for the surface temperature of the star in the Corona Borealis galaxy. The calculation does not give an accurate value.

State and explain whether this value of temperature is too high or too low.

.....

 [1]

[Total: 10]

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NUMBER

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9702/52

February/March 2024

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 30.
- The number of marks for each question or part question is shown in brackets [].

This document has 8 pages.

- 1 Fig. 1.1 shows a thin cylindrical metal rod of length L .



Fig. 1.1

One end of the rod is hit with a hammer. A stationary sound wave is set up within the rod. The rod vibrates at its resonant frequency f .

A microphone placed at the other end of the rod detects the sound wave emitted from the rod. The frequency of the detected sound is also f .

A number of rods of different length are available.

It is suggested that f is related to L by the relationship

$$2fL^n = \sqrt{\frac{E}{\rho}}$$

where ρ is the density of the metal, and E and n are constants.

Plan a laboratory experiment to test the relationship between f and L .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for E and n .

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

Diagram

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2 A student investigates an electrical circuit.

A power supply with negligible internal resistance is connected to six resistors, each of resistance Z , and a resistor of resistance R , as shown in Fig. 2.1.

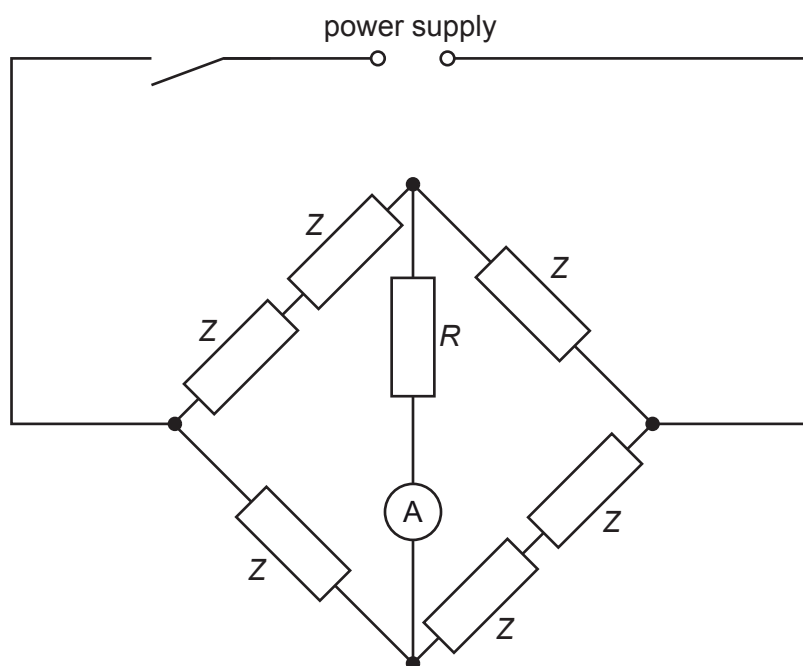


Fig. 2.1

The current measured by the ammeter is I .

The experiment is repeated for different values of R .

It is suggested that I and R are related by the equation

$$E = 3IR + 4IZ$$

where E is the electromotive force (e.m.f.) of the power supply.

(a) A graph is plotted of $\frac{1}{I}$ on the y -axis against R on the x -axis.

Determine expressions for the gradient and y -intercept.

gradient =

y -intercept =

[1]

(b) Values of R and I are given in Table 2.1.

Table 2.1

$R/\text{k}\Omega$	$I/\mu\text{A}$	$\frac{1}{I}/\text{A}^{-1}$
1.25	225 ± 5	
2.55	185 ± 5	
3.90	160 ± 5	
5.25	140 ± 5	
6.55	125 ± 5	
7.80	115 ± 5	

Calculate and record values of $\frac{1}{I}/\text{A}^{-1}$ in Table 2.1.

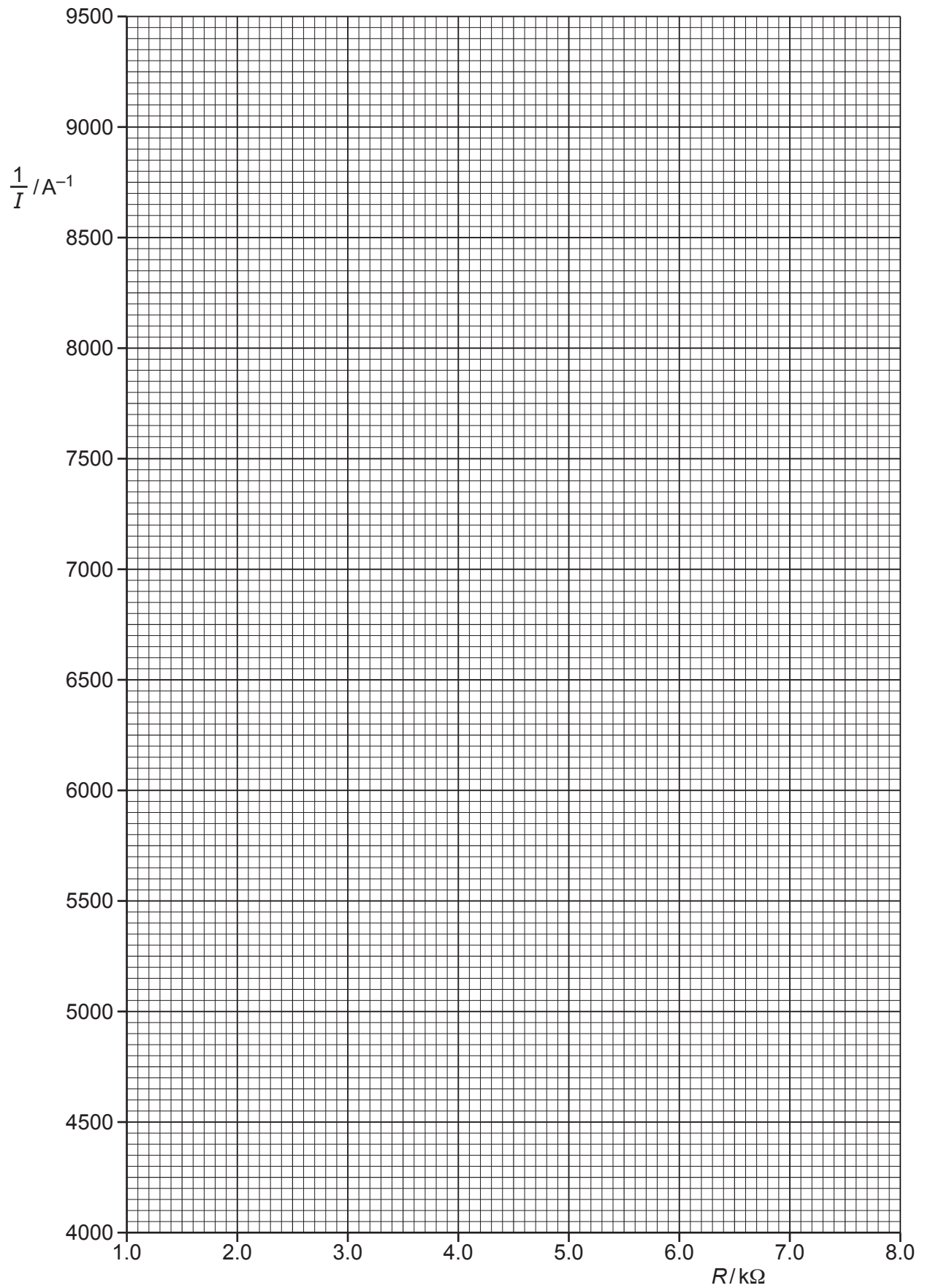
Include the absolute uncertainties in $\frac{1}{I}$. [2]

(c) (i) Plot a graph of $\frac{1}{I}/\text{A}^{-1}$ against $R/\text{k}\Omega$. Include error bars for $\frac{1}{I}$. [2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines. [2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]



- (iv) Determine the y -intercept of the line of best fit. Include the absolute uncertainty in your answer.

y -intercept = [2]

- (d) (i) Using your answers to (a), (c)(iii) and (c)(iv), determine the values of E and Z . Include appropriate units.

E =

Z = [2]

- (ii) Determine the percentage uncertainty in Z .

percentage uncertainty in Z = % [1]

- (e) The experiment is repeated. Determine the resistance R that gives a value of I of 0.10 mA.

R = Ω [1]

[Total: 15]



CANDIDATE
NAME

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9702/41

May/June 2024

2 hours

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [].

This document has **24** pages. Any blank pages are indicated.

Data

acceleration of free fall	$g = 9.81 \text{ m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
unified atomic mass unit	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27} \text{ kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$ ($\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ m F}^{-1}$)
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_o = \frac{f_s v}{v \pm v_s}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \dots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$

gravitational potential	$\phi = -\frac{GM}{r}$
gravitational potential energy	$E_P = -\frac{GMm}{r}$
pressure of an ideal gas	$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$
simple harmonic motion	$a = -\omega^2 x$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t$ $v = \pm \omega \sqrt{(x_0^2 - x^2)}$
electric potential	$V = \frac{Q}{4\pi\epsilon_0 r}$
electrical potential energy	$E_P = \frac{Qq}{4\pi\epsilon_0 r}$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$
capacitors in parallel	$C = C_1 + C_2 + \dots$
discharge of a capacitor	$x = x_0 e^{-\frac{t}{RC}}$
Hall voltage	$V_H = \frac{BI}{ntq}$
alternating current/voltage	$x = x_0 \sin \omega t$
radioactive decay	$x = x_0 e^{-\lambda t}$
decay constant	$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$
intensity reflection coefficient	$\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$
Stefan–Boltzmann law	$L = 4\pi\sigma r^2 T^4$
Doppler redshift	$\frac{\Delta\lambda}{\lambda} \approx \frac{Df}{f} \approx \frac{v}{c}$

- 1 (a) Define gravitational potential at a point.

.....

.....

..... [2]

- (b) A satellite X, of mass M , orbits a planet at a constant distance $4R$ from the centre of the planet, as shown in Fig. 1.1.

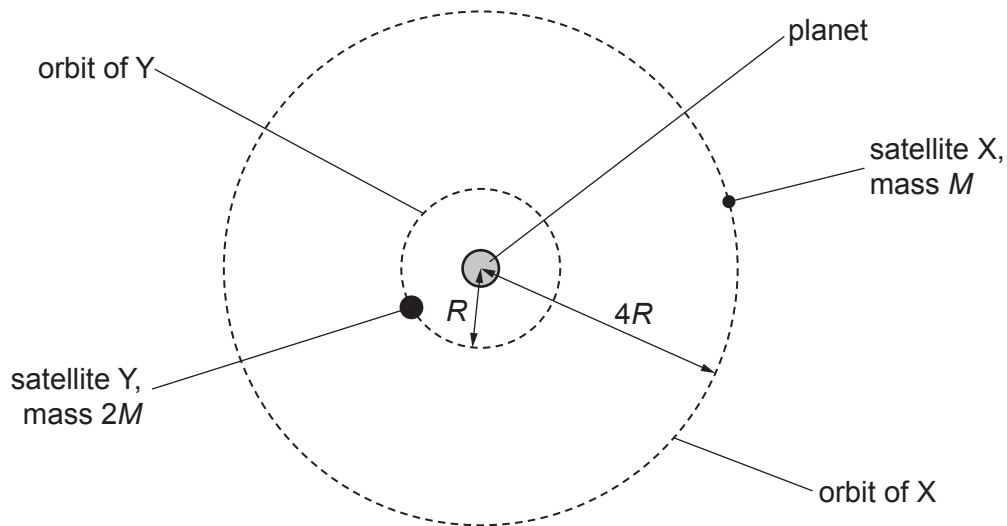


Fig. 1.1 (not to scale)

A second satellite Y, of mass $2M$, orbits the planet with orbital radius R .

The gravitational potential at X due to the planet is $-\Phi$. The planet is a uniform sphere.

- (i) Explain why the gravitational potential at X is negative.

.....

.....

..... [2]

- (ii) State an expression, in terms of Φ , for the gravitational potential at Y due to the planet.

gravitational potential = [2]

- (iii) Complete Table 1.1 by giving expressions, in terms of some or all of M , R and Φ , for the quantities indicated for each of the satellites X and Y.

Table 1.1

	satellite X	satellite Y
gravitational field strength at satellite due to planet		
gravitational potential energy of satellite		

[4]

[Total: 10]

- 2 (a) (i) State the magnitude and unit of absolute zero on the thermodynamic temperature scale.

..... [1]

- (ii) Explain why temperature measured using a laboratory liquid-in-glass thermometer does **not** give a measurement of thermodynamic temperature.

.....

..... [1]

- (b) Fig. 2.1 shows a simplified diagram of a type of thermometer called a platinum resistance thermometer.

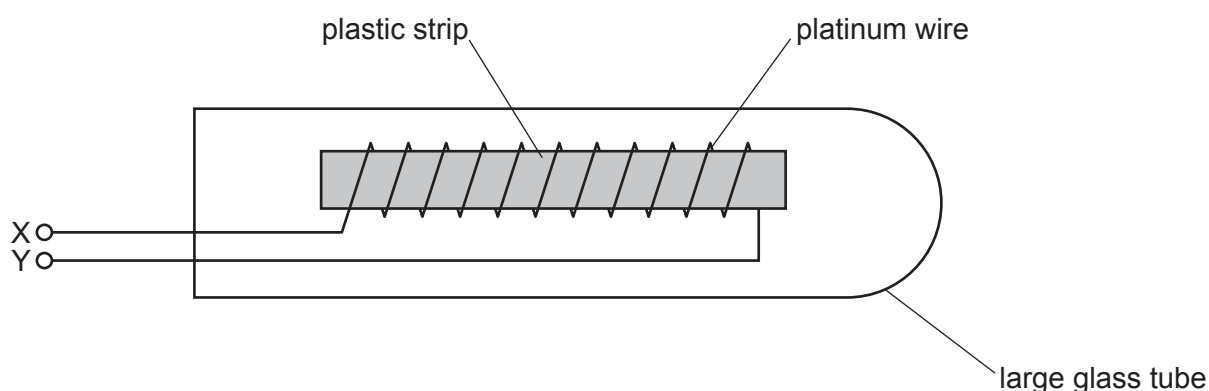


Fig. 2.1

The glass tube is immersed in the environment for which the temperature is to be determined. The resistance between the terminals X and Y is measured.

Fig. 2.2 shows the variation of the resistivity ρ of platinum with thermodynamic temperature T .

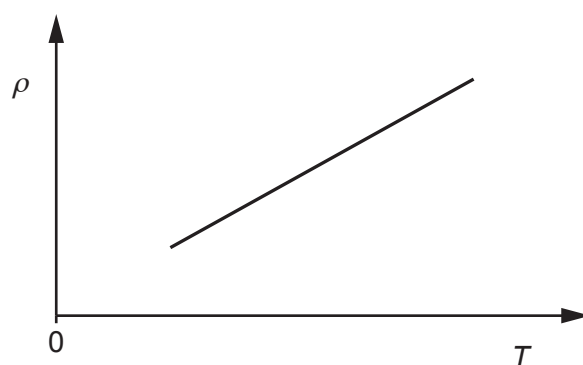


Fig. 2.2

- (i) Explain how Fig. 2.2 shows that platinum is a suitable metal for use in a resistance thermometer.

.....

 [2]

- (ii) Suggest a reason why a platinum resistance thermometer is **not** suitable for measuring a rapidly changing temperature.

.....

 [1]

- (iii) Suggest a type of thermometer that is suitable for measuring a rapidly changing temperature.

..... [1]

- (c) A negative temperature coefficient thermistor may be used as a type of resistance thermometer.

State **one** way in which the variation with temperature of the resistance of a thermistor differs from that of a platinum wire.

.....
 [1]

[Total: 7]

- 3 (a) (i) State what is meant by an ideal gas.

.....

 [2]

- (ii) Use one of the basic assumptions of the kinetic theory to explain what can be deduced about the potential energy associated with the random motion of molecules in an ideal gas.

.....

 [2]

- (b) A sample of 0.26 m^3 of an ideal gas is at pressure $2.0 \times 10^5 \text{ Pa}$ and temperature 290 K .

Determine:

- (i) the number N of molecules of the gas

$$N = \dots\dots\dots [2]$$

- (ii) the average translational kinetic energy E_K of one molecule of the gas

$$E_K = \dots\dots\dots \text{ J } [2]$$

- (iii) the internal energy of the gas. Explain your reasoning.

$$\text{internal energy} = \dots\dots\dots \text{ J } [2]$$

(c) The volume V of the gas in (b) is now varied, keeping its pressure constant.

On Fig. 3.1, sketch the variation with V of the internal energy U of the gas.



Fig. 3.1

[2]

[Total: 12]

- 4 (a) State what is meant by resonance.

.....

.....

..... [2]

- (b) A small ball is held in place using a stretched string. One end of the string is fixed to a wall and the other end is attached to a vibration generator, as shown in Fig. 4.1.

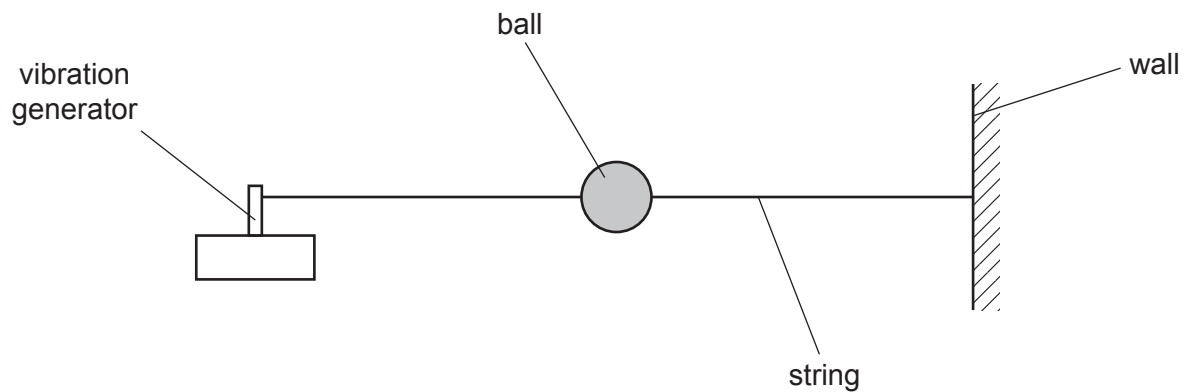


Fig. 4.1

Initially, the vibration generator is switched off.

A student displaces the ball vertically and then releases it. Fig. 4.2 shows the variation of the displacement of the ball with time after it is released.

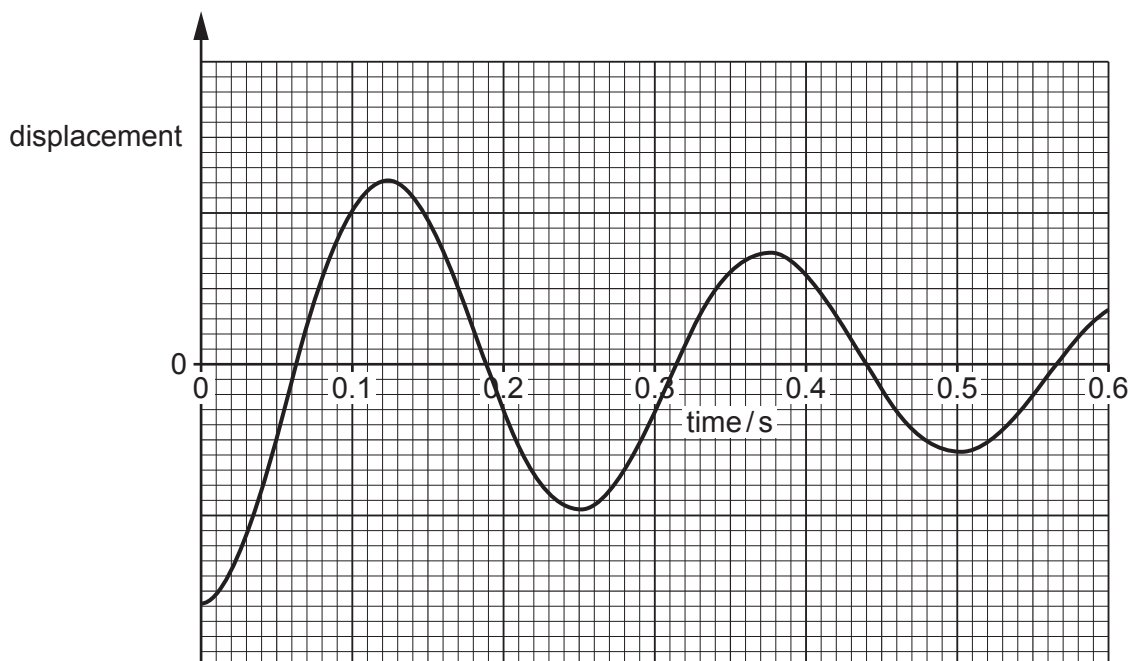


Fig. 4.2

- (i) State the name of the phenomenon illustrated by the decrease in the amplitude of the oscillations in Fig. 4.2.

..... [1]

- (ii) Explain the decrease with time of the amplitude of the oscillations of the ball.

.....

 [2]

- (iii) Determine the frequency of the oscillations of the ball.

frequency = Hz [1]

- (c) The vibration generator in (b) is switched on and its frequency f of vibration is gradually increased from 0 to 10 Hz.

On Fig. 4.3, sketch the variation with f of the amplitude of the oscillations of the ball.

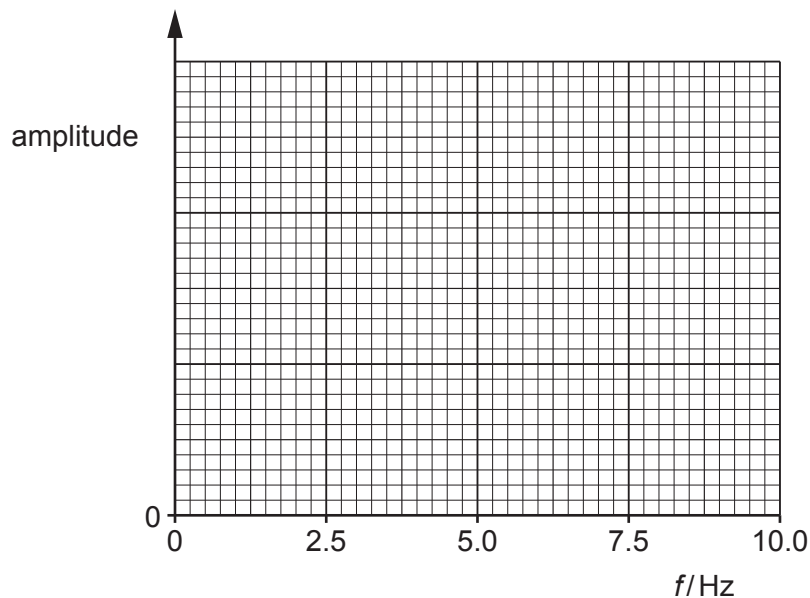


Fig. 4.3

[2]

[Total: 8]

- 5 (a) Define electric field.

.....

.....

..... [2]

- (b) Fig. 5.1 shows two parallel conducting plates that are in a vacuum. The plates are separated by a distance of 6.7 cm and have a potential difference (p.d.) of 430 V between them.

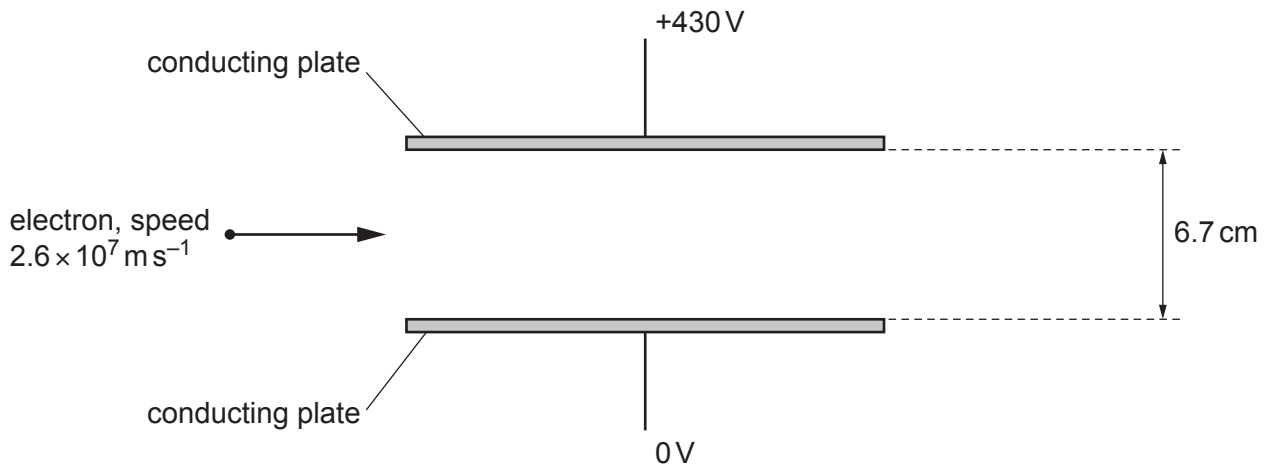


Fig. 5.1

- (i) On Fig. 5.1, draw four field lines to represent the electric field between the plates. [2]
- (ii) Determine the strength E of the electric field between the plates.

$$E = \dots\dots\dots \text{NC}^{-1} \text{ [2]}$$

- (iii) An electron travels at a speed of $2.6 \times 10^7 \text{ ms}^{-1}$ towards the region between the plates, as shown in Fig. 5.1.

On Fig. 5.1, draw the path of the electron as it moves between and beyond the plates. [2]

- (c) A uniform magnetic field is now applied in the region of the electric field in Fig. 5.1, so that the electron in (b)(iii) travels undeviated through the region.

- (i) Determine the direction of the uniform magnetic field.

..... [1]

- (ii) Explain, with reference to the forces exerted by the two fields on the electron, why the path of the electron is undeviated.

.....

.....

..... [2]

- (iii) Determine the flux density B of the uniform magnetic field. Give a unit with your answer.

$B =$ unit [2]

[Total: 13]

- 6 Fig. 6.1 shows a capacitor of capacitance C connected in series with a resistor of resistance R .

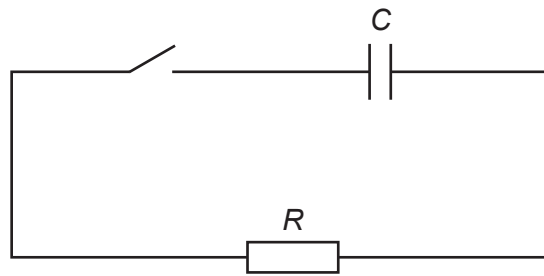


Fig. 6.1

Initially the switch is open and there is a p.d. of 12 V across the capacitor.

At time $t = 0$, the switch is closed so that there is a current I in the resistor.

Fig. 6.2 shows the variation of I with t .

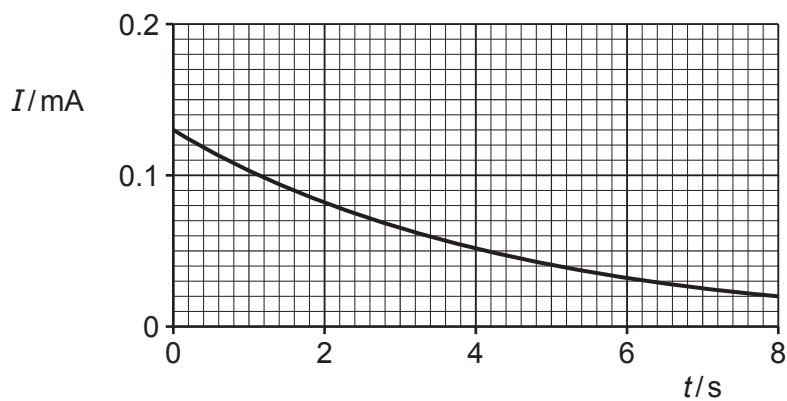


Fig. 6.2

- (a) Explain the shape of the line in Fig. 6.2.

.....

.....

.....

.....

..... [3]

(b) Use Fig. 6.2 to determine:

(i) resistance R

$$R = \dots\dots\dots \Omega \text{ [2]}$$

(ii) the time constant τ of the circuit in Fig. 6.1.

$$\tau = \dots\dots\dots \text{ s [3]}$$

(c) Use your answers in (b) to determine capacitance C .

$$C = \dots\dots\dots \text{ F [2]}$$

[Total: 10]

- 7 A circuit contains a power supply that provides a sinusoidal alternating input voltage V_{IN} . There is an output voltage V_{OUT} across a load resistor R , as shown in Fig. 7.1.

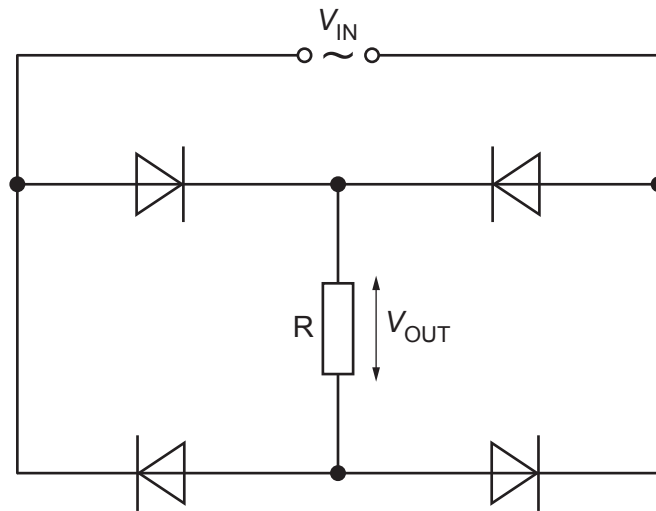


Fig. 7.1

- (a) State the purpose of the circuit in Fig. 7.1.

.....

.....

..... [2]

- (b) Fig. 7.2 shows the variation of V_{OUT} with time t .

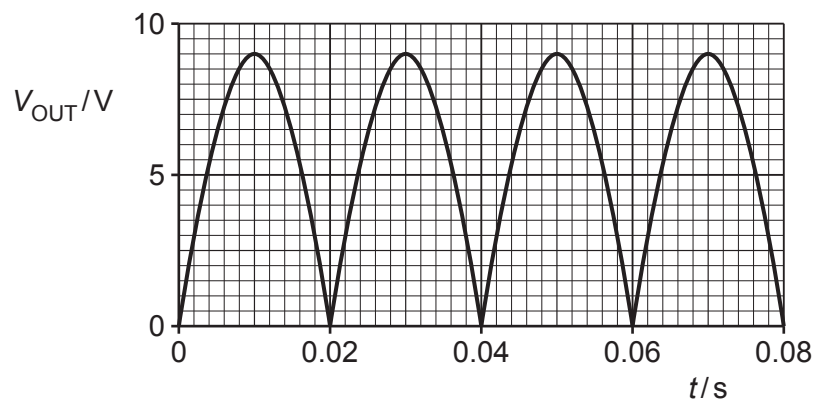


Fig. 7.2

- (i) The load resistor R has a resistance of $370\ \Omega$.

Show that the maximum power dissipated in R is 0.22 W .

[2]

- (ii) On Fig. 7.3, sketch the variation with t of the power P dissipated in R .

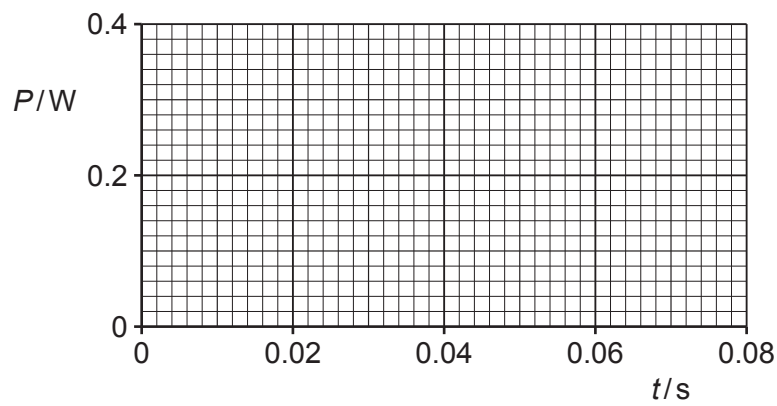


Fig. 7.3

[3]

- (iii) Calculate the mean power dissipated in R .

mean power = W [1]

- (c) The circuit of Fig. 7.1 is disconnected, and R is connected directly across the power supply.

Explain, without calculation, how the mean power now dissipated in R compares with the answer in (b)(iii).

.....

 [2]

[Total: 10]

- 8 (a) State what is meant by a photon.

.....

.....

..... [2]

- (b) Fig. 8.1 shows a tube in which X-rays are produced at a metal target.

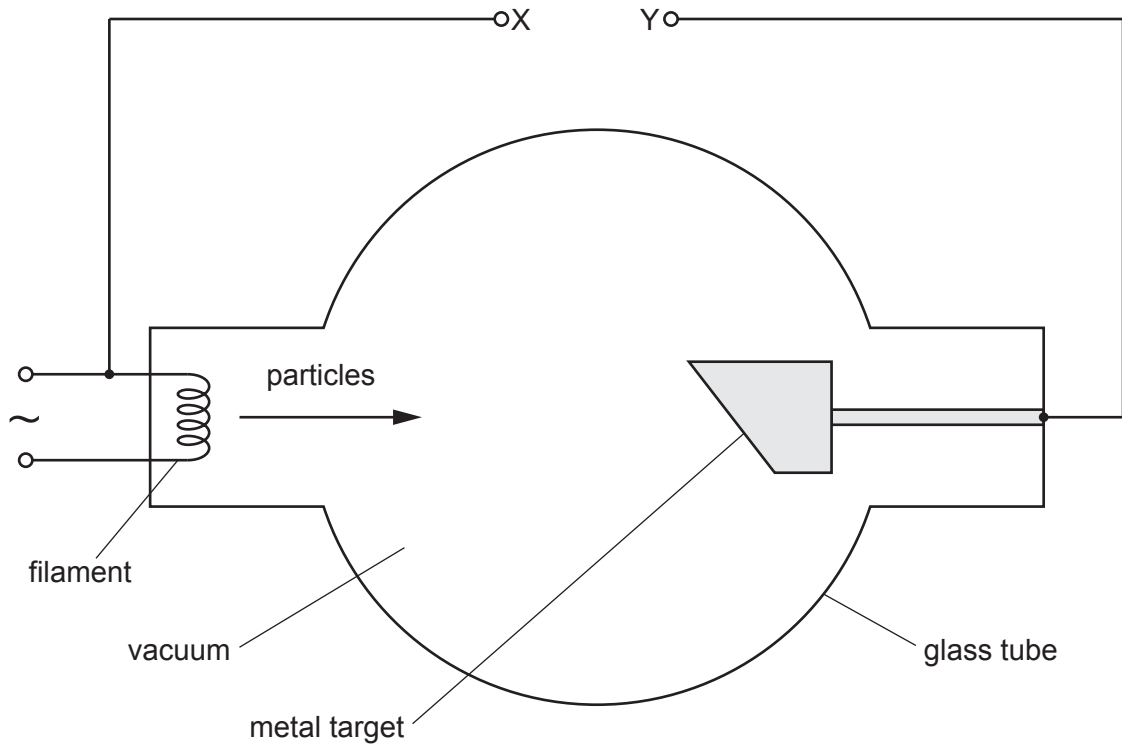


Fig. 8.1

Particles are accelerated from the filament to the target by a constant high voltage applied across the terminals X and Y.

- (i) State the name of the particles.

..... [1]

- (ii) On Fig. 8.1, use + and – signs to label terminals X and Y to indicate the polarity of the high voltage. [1]

(c) For an accelerating voltage of 32 kV in Fig. 8.1, determine:

(i) the maximum energy, in MeV, of an X-ray photon produced at the target

maximum photon energy = MeV [1]

(ii) the maximum momentum of an X-ray photon produced at the target

maximum photon momentum = N s [2]

(iii) the minimum wavelength of X-rays produced at the target.

minimum wavelength = m [3]

(d) Explain why X-rays can be used to produce images of internal body structures that have good contrast.

.....

.....

.....

.....

..... [3]

[Total: 13]

- 9 (a) Define half-life of a radioactive isotope.

.....

 [1]

- (b) Radioactive isotope X decays to isotope Y.

A sample contains only nuclei of X at time $t = 0$. Fig. 9.1 shows the variation with t of the numbers of nuclei of X and of Y as the sample decays.

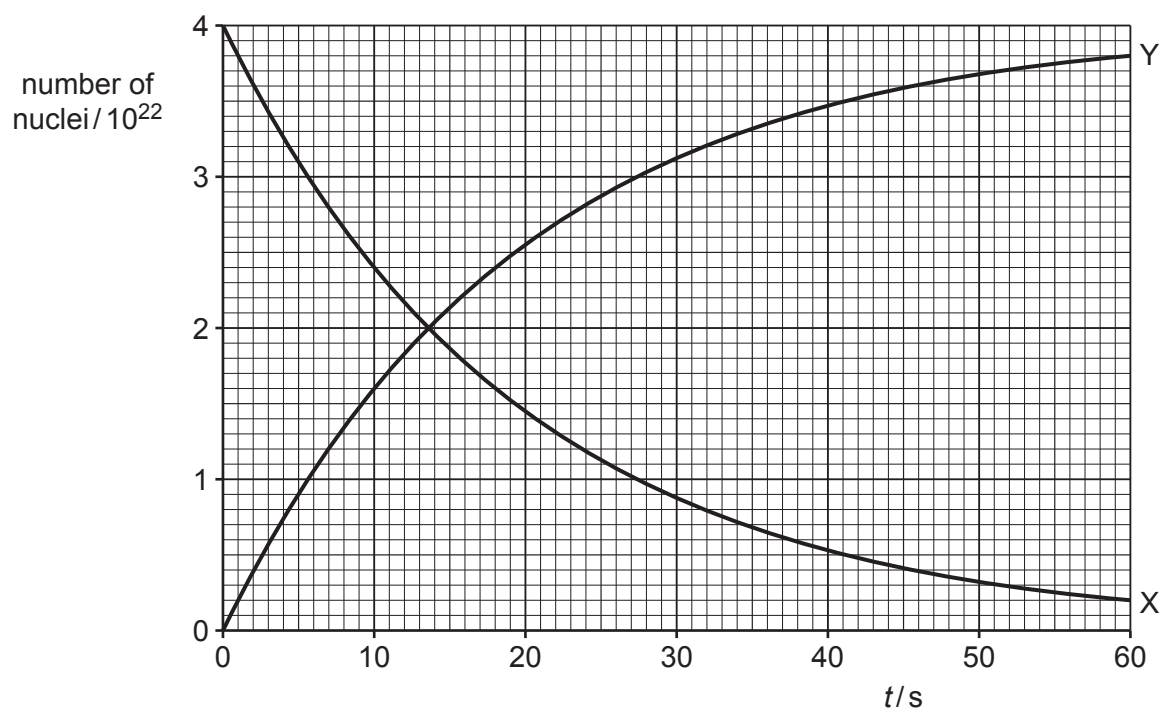


Fig. 9.1

- (i) State the name of the quantity represented by the magnitude of the gradient of line X in Fig. 9.1.

..... [1]

- (ii) State **three** conclusions about X or Y that may be drawn from Fig. 9.1. The conclusions may be qualitative or quantitative. Use the space below for any working that you need.

1
.....

2
.....

3
.....

[3]

- (c) The mass of radioactive isotope X in the sample in (b) is 7.3×10^{-4} kg at time $t = 0$.

Determine the nucleon number of isotope X.

nucleon number = [3]

[Total: 8]

- 10 (a) (i) State what is meant by the luminosity of a star.

.....

.....

..... [2]

- (ii) Explain how a standard candle in a distant galaxy can be used to determine the distance of the galaxy from an observer.

.....

.....

.....

.....

..... [3]

- (b) The Sun has a radius of 6.96×10^8 m and a surface temperature of 5780 K.
Light from the Sun is observed to have a peak intensity at a wavelength of 501 nm.

- (i) Calculate the luminosity of the Sun. Give a unit with your answer.

luminosity = unit [2]

- (ii) Another star emits radiation that has a peak intensity at a wavelength of 624 nm.

Determine the surface temperature of this star.

surface temperature =K [2]

[Total: 9]

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PHYSICS

9702/42

Paper 4 A Level Structured Questions

May/June 2024

2 hours

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [].

This document has **24** pages. Any blank pages are indicated.



Data

acceleration of free fall	$g = 9.81 \text{ m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
unified atomic mass unit	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27} \text{ kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$ $(\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ m F}^{-1})$
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_o = \frac{f_s v}{v \pm v_s}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \dots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$





gravitational potential

$$\phi = -\frac{GM}{r}$$

gravitational potential energy

$$E_P = -\frac{GMm}{r}$$

pressure of an ideal gas

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

simple harmonic motion

$$a = -\omega^2 x$$

velocity of particle in s.h.m.

$$v = v_0 \cos \omega t$$

$$v = \omega \sqrt{(x_0^2 - x^2)}$$

electric potential

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

electrical potential energy

$$E_P = \frac{Qq}{4\pi\epsilon_0 r}$$

capacitors in series

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$$

capacitors in parallel

$$C = C_1 + C_2 + \dots$$

discharge of a capacitor

$$x = x_0 e^{-\frac{t}{RC}}$$

Hall voltage

$$V_H = \frac{BI}{ntq}$$

alternating current/voltage

$$x = x_0 \sin \omega t$$

radioactive decay

$$x = x_0 e^{-\lambda t}$$

decay constant

$$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$$

intensity reflection coefficient

$$\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$$

Stefan–Boltzmann law

$$L = 4\pi\sigma r^2 T^4$$

Doppler redshift

$$\frac{\Delta\lambda}{\lambda} \approx \frac{Df}{f} \approx \frac{v}{c}$$





1 (a) Define the radian.

.....
 [1]

(b) A circular metal disc spins horizontally about a vertical axis, as shown in Fig. 1.1.

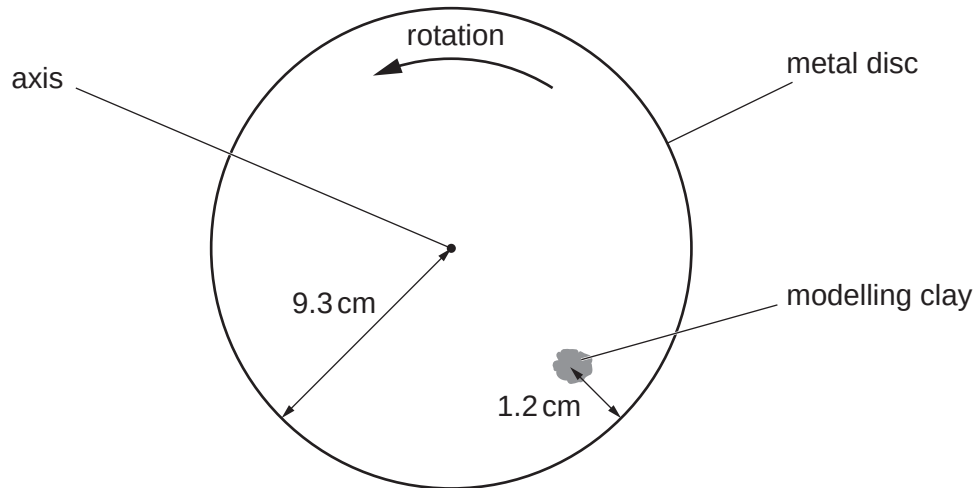


Fig. 1.1 (not to scale)

A piece of modelling clay is attached to the disc.

For the instant when the piece of modelling clay is in the position shown, draw on Fig. 1.1:

(i) an arrow, labelled V , showing the direction of the velocity of the modelling clay [1]

(ii) an arrow, labelled A , showing the direction of the acceleration of the modelling clay. [1]

(c) The metal disc in Fig. 1.1 has a radius of 9.3 cm.
 The centre of gravity of the modelling clay is 1.2 cm from the rim of the disc and moves with a speed of 0.68 m s^{-1} .

(i) Calculate the angular speed ω of the disc.

$\omega = \dots\dots\dots \text{ rad s}^{-1}$ [2]





(ii) Calculate the acceleration a of the centre of gravity of the modelling clay.

$a = \dots\dots\dots \text{ms}^{-2}$ [2]

(d) A second piece of modelling clay is attached to the disc in the position shown in Fig. 1.2.

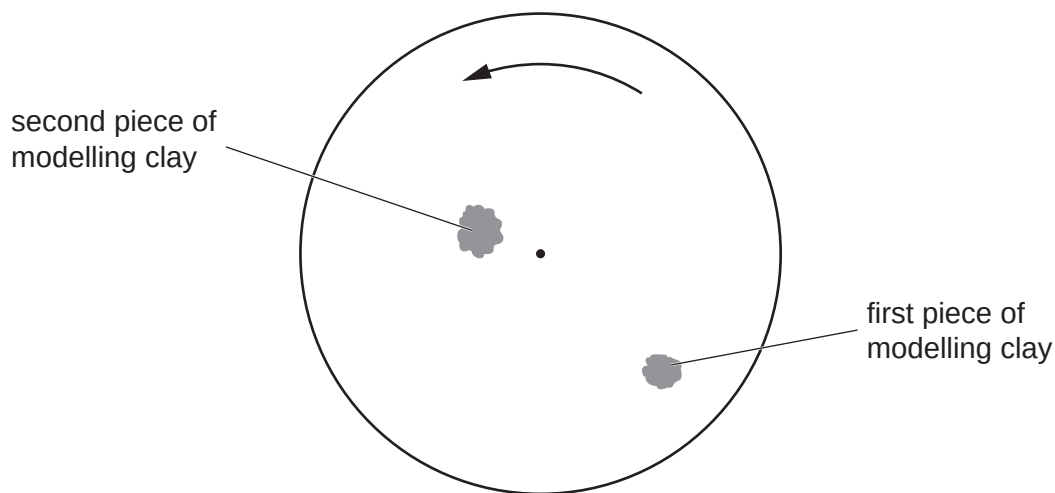


Fig. 1.2

The second piece of modelling clay has a larger mass than the first piece.

By placing **one** tick (3) in each row, complete Table 1.1 to show how the quantities indicated compare for the two pieces of modelling clay.

Table 1.1

quantity	less for second piece than first piece	same for both pieces	greater for second piece than first piece
angular speed			
linear speed			
acceleration			

[3]

[Total: 10]





- 2 (a) With reference to thermal energy, state what is meant by two objects being in thermal equilibrium.

.....

 [1]

- (b) Two cylinders X and Y each contain a sample of an ideal gas. The samples are in thermal equilibrium with each other.

X has a volume of 0.0260 m^3 and contains 0.740 mol of gas at a pressure of $1.20 \times 10^5 \text{ Pa}$. Y has a volume of 0.0430 m^3 and contains gas at a pressure of $2.90 \times 10^5 \text{ Pa}$. Data for the two cylinders are shown in Fig. 2.1.



Fig. 2.1

- (i) Show that the temperature of the gas in X is 234°C .

[3]

- (ii) Determine the number N of molecules of the gas in Y. Explain your reasoning.

$N = \dots\dots\dots$ [3]





- (iii) The gas in X consists of molecules that each have a mass that is four times the mass of a molecule of the gas in Y.

Explain how the root-mean-square (r.m.s.) speed of the molecules in X compares with the r.m.s. speed of the molecules in Y.

.....

.....

.....

.....

..... [3]

[Total: 10]







3 (a) State what is meant by the internal energy of a system.

.....

.....

..... [2]

(b) With reference to molecular kinetic and potential energies, describe and explain how the internal energy of the system changes when:

(i) a gas is heated at constant volume so that its temperature increases

.....

.....

.....

.....

..... [3]

(ii) a wire is stretched within its elastic limit at constant temperature.

.....

.....

.....

.....

..... [3]

[Total: 8]



- 4 A block of mass m oscillates vertically on a spring, as shown in Fig. 4.1.

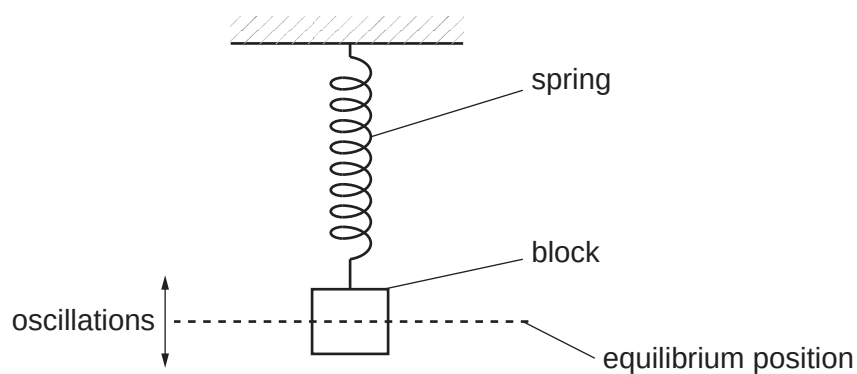


Fig. 4.1

The acceleration a of the block varies with displacement x from its equilibrium position, as shown in Fig. 4.2.

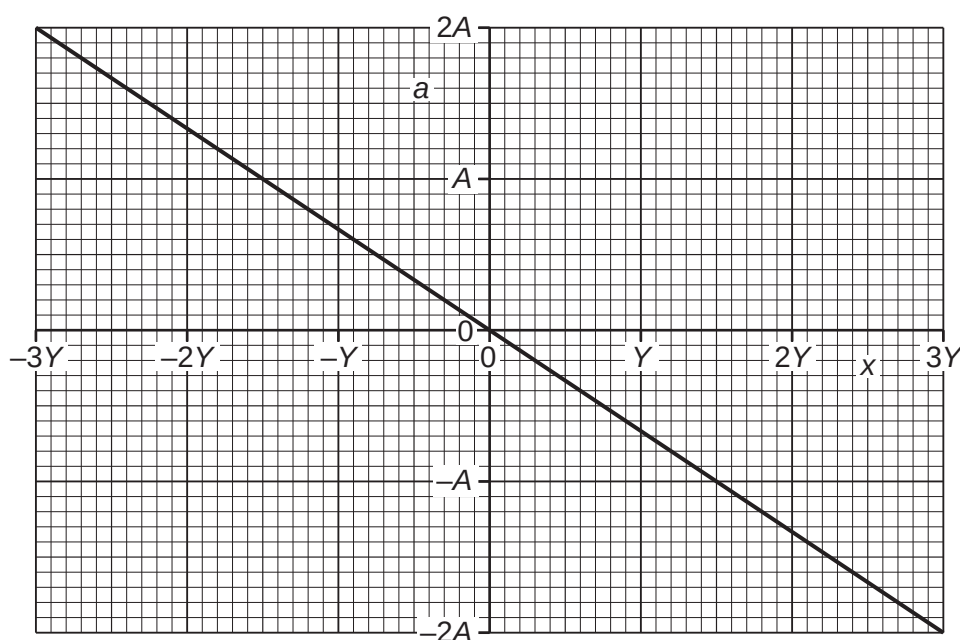


Fig. 4.2

The amplitude of the oscillations is $3Y$ and the maximum acceleration is $2A$.

- (a) Explain how Fig. 4.2 shows that the oscillations of the block are simple harmonic.

.....

.....

..... [2]



(b) Deduce expressions, in terms of some or all of m , A and Y , for:

(i) the angular frequency ω of the oscillations

$$\omega = \dots\dots\dots [1]$$

(ii) the maximum speed v_0 of the oscillations

$$v_0 = \dots\dots\dots [2]$$

(iii) the energy E of the oscillations.

$$E = \dots\dots\dots [2]$$

(c) The period of the oscillations is 0.75 s and the value of $3Y$ is 1.8 cm.

Determine an expression for x in terms of time t , where x is in cm and t is in seconds.

$$x = \dots\dots\dots [2]$$

[Total: 9]



- 5 (a) Define electric potential at a point.

.....

.....

..... [2]

- (b) Two isolated charged metal spheres X and Y are near to each other in a vacuum. The centres of the spheres are 1.2 m apart, as shown in Fig. 5.1.

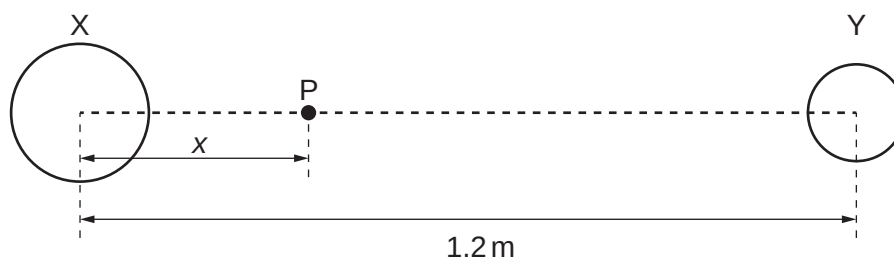


Fig. 5.1 (not to scale)

Point P is on the line joining the centres of spheres X and Y and is at a variable distance x from the centre of X.

Fig. 5.2 shows the variation with x of the total electric potential V due to the two spheres.

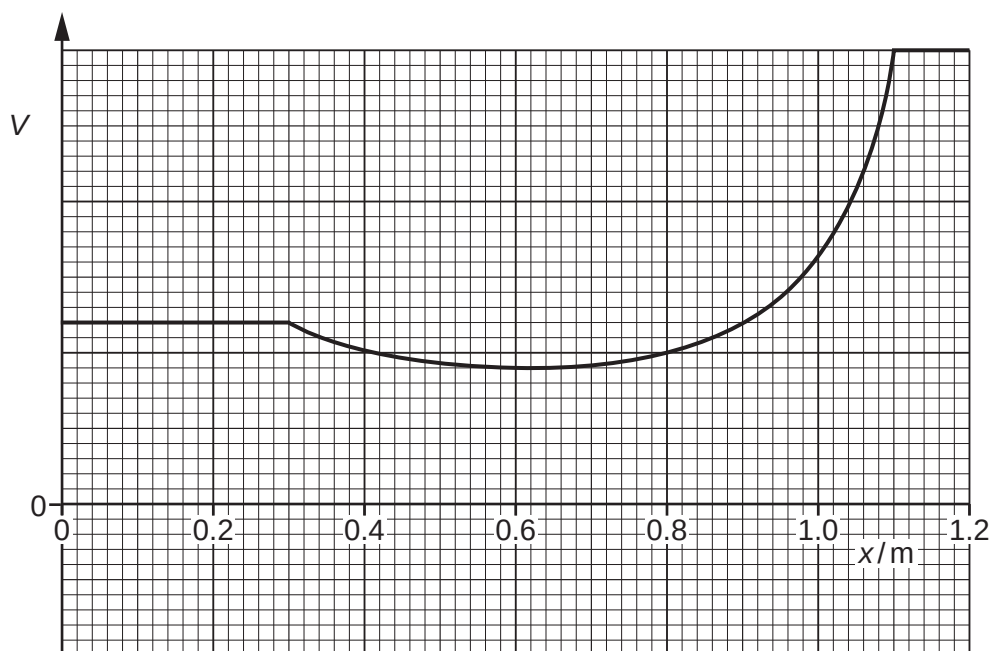


Fig. 5.2





State **three** conclusions that may be drawn about the spheres from Fig. 5.2. The conclusions may be qualitative or quantitative.

1

.....

2

.....

3

.....

[3]

- (c) A proton is held at rest on the line joining the centres of the spheres in (b) at the position where $x = 0.60$ m.

The proton is released.

Describe and explain, without calculation, the subsequent motion of the proton.

.....

.....

..... [2]

[Total: 7]





- 6 (a) Two capacitors X and Y are connected in series to a power supply of voltage V , as shown in Fig. 6.1.

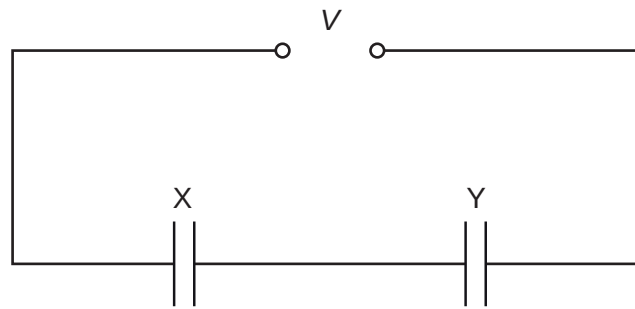


Fig. 6.1

The capacitance of X is C_X and the capacitance of Y is C_Y .

Derive an expression, in terms of C_X and C_Y , for the combined capacitance C_T of the capacitors in this circuit.
Explain your reasoning.

[3]

- (b) Two capacitors P and Q are connected in parallel to a power supply of voltage V .
The capacitance of P is $200\ \mu\text{F}$. The capacitance C_Q of Q can be varied between 0 and $400\ \mu\text{F}$.
When $C_Q = 0$, the total energy stored in the capacitors is $2.5\ \text{mJ}$.

- (i) Show that the supply voltage V is $5.0\ \text{V}$.

[2]



- (ii) Calculate the total energy, in mJ, stored in the capacitors when C_Q has its maximum value.

total energy = mJ [3]

- (iii) On Fig. 6.2, sketch the variation of the total energy E stored in the capacitors with C_Q , as C_Q varies from 0 to 400 μF .

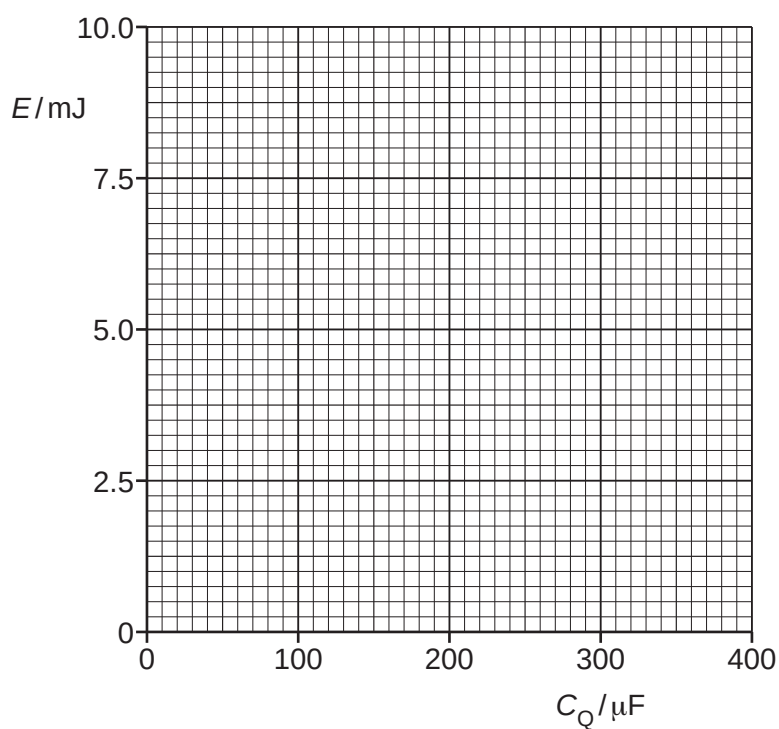


Fig. 6.2

[2]

[Total: 10]



7 (a) State Faraday's law of electromagnetic induction.

.....

.....

..... [2]

(b) Fig. 7.1 shows a coil at rest in a uniform magnetic field that is parallel to the axis of the coil.

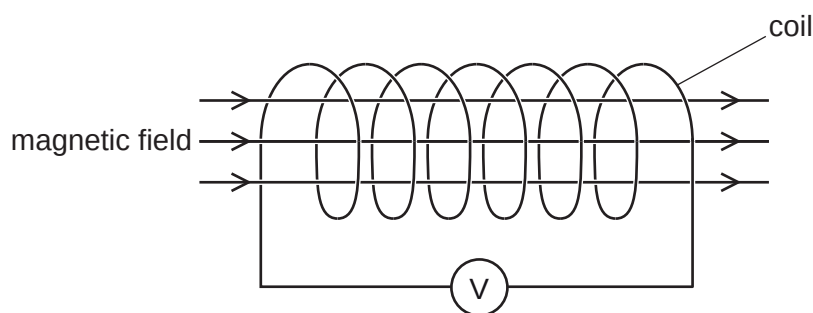


Fig. 7.1

The coil is connected to a centre-zero voltmeter.

The flux density B of the uniform magnetic field varies with time t as shown in Fig. 7.2.

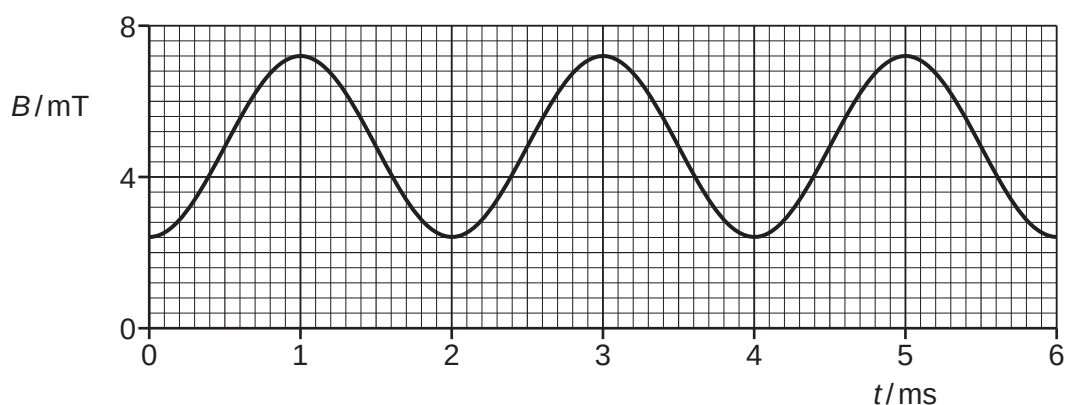


Fig. 7.2

The coil consists of 340 turns, each of cross-sectional area $3.2 \times 10^{-4} \text{ m}^2$.

(i) Calculate the maximum magnetic flux through **one** turn of the coil.

maximum magnetic flux = Wb [2]



- (ii) Determine the maximum rate of change of magnetic flux linkage in the coil.

maximum rate of change of flux linkage = Wb s^{-1} [3]

- (iii) State the maximum electromotive force (e.m.f.) V_0 induced across the coil.

$V_0 = \dots\dots\dots \text{V}$ [1]

- (iv) On Fig. 7.3, sketch the variation of the e.m.f. V induced across the coil with t from $t = 0$ to $t = 6.0 \text{ ms}$.

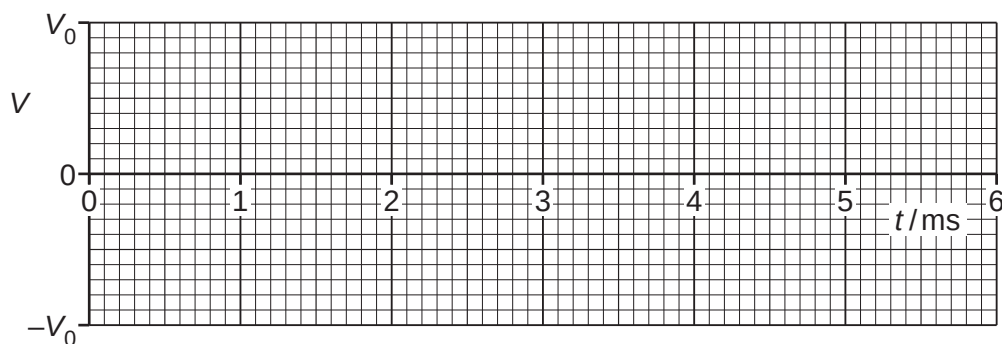


Fig. 7.3

[3]

- (v) The variation of V with t can be described by

$$V = A \sin Bt$$

where A and B are constants.

Determine the values of A and B . Give units with your answers.

$A = \dots\dots\dots \text{unit} \dots\dots\dots$

$B = \dots\dots\dots \text{unit} \dots\dots\dots$

[3]

[Total: 14]





- 8 Fig. 8.1 shows part of the emission spectrum of visible radiation emitted by hydrogen gas in a star in a distant galaxy.

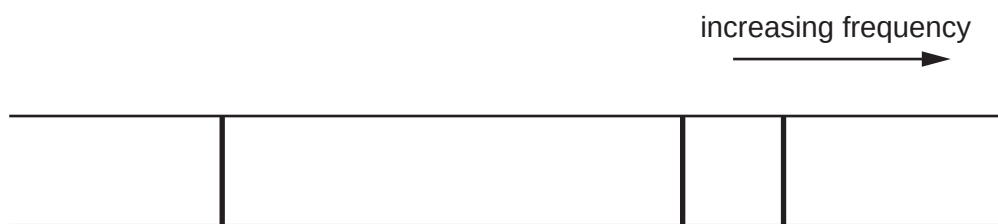


Fig. 8.1

The galaxy is moving away from the Earth at a speed of $6.2 \times 10^6 \text{ m s}^{-1}$.

- (a) (i) Explain how the positions of the lines in the emission spectrum seen by an observer on the Earth differ from the positions shown in Fig. 8.1.

.....

 [2]

- (ii) On Fig. 8.1, draw the three lines in possible positions in the spectrum seen by the observer. [2]

- (b) The lines in Fig. 8.1 correspond to electron transitions down to the energy level -3.40 eV . One of the lines represents emitted radiation of wavelength 488 nm .

- (i) Calculate the energy of a photon of this radiation.

photon energy = J [2]

- (ii) Determine the energy, in eV, of the energy level from which the electron transition originates to cause the emission of this radiation.

energy level = eV [2]





- (iii) Determine the wavelength, in nm, of this radiation as detected by the observer on the Earth.

wavelength = nm [2]

- (c) A value for the Hubble constant is $2.3 \times 10^{-18} \text{ s}^{-1}$.

Determine the distance of the galaxy from the Earth.

distance = m [2]

[Total: 12]





- 9 (a) State what is meant by the binding energy of a nucleus.

.....

 [2]

- (b) Table 9.1 shows the masses of two sub-atomic particles and a polonium-212 ($^{212}_{84}\text{Po}$) nucleus.

Table 9.1

	mass/u
proton	1.007 276
neutron	1.008 665
polonium-212 nucleus	211.942 749

For the polonium-212 nucleus, determine:

- (i) the mass defect Δm , in kg

$\Delta m = \dots\dots\dots$ kg [3]

- (ii) the binding energy

binding energy = $\dots\dots\dots$ J [2]

- (iii) the binding energy per nucleon.

binding energy per nucleon = $\dots\dots\dots$ J [1]



- (c) (i) On Fig. 9.1, sketch the variation with nucleon number A of binding energy per nucleon for values of A from 1 to 250.

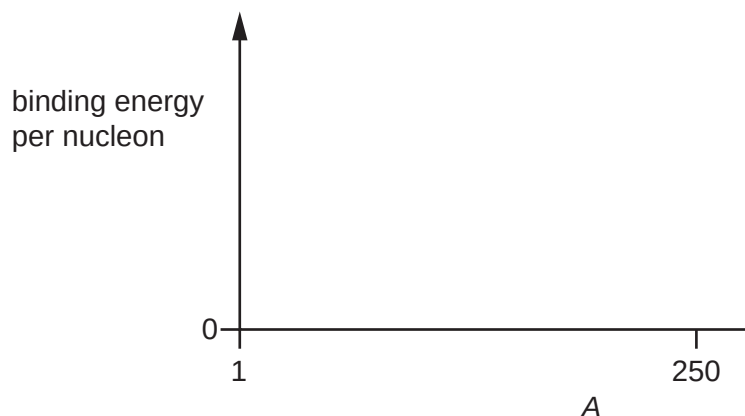


Fig. 9.1

- (ii) On your line in Fig. 9.1, draw an X to show the approximate position of polonium-212. [2]
- (iii) Polonium-212 is radioactive and undergoes alpha-decay. [1]

Suggest and explain, with reference to Fig. 9.1, why the alpha-decay of polonium-212 results in a release of energy.

.....

.....

..... [2]

[Total: 13]





- 10 (a) Describe how reflected ultrasound pulses may be used to obtain diagnostic information about internal structures.

.....

.....

.....

..... [2]

- (b) (i) Define specific acoustic impedance of a medium.

.....

.....

..... [2]

- (ii) Table 10.1 shows some data for water and for glass.

Table 10.1

	density / kg m ⁻³	speed of sound / m s ⁻¹
water	1000	1420
glass	2500	4560

Determine the intensity reflection coefficient for ultrasound that is incident on a water–glass boundary.

intensity reflection coefficient = [3]

[Total: 7]





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9702/43

May/June 2024

2 hours

You must answer on the question paper.

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [].

This document has **24** pages. Any blank pages are indicated.

Data

acceleration of free fall	$g = 9.81 \text{ m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
unified atomic mass unit	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27} \text{ kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$ ($\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ m F}^{-1}$)
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_o = \frac{f_s v}{v \pm v_s}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \dots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$

gravitational potential	$\phi = -\frac{GM}{r}$
gravitational potential energy	$E_P = -\frac{GMm}{r}$
pressure of an ideal gas	$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$
simple harmonic motion	$a = -\omega^2 x$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t$ $v = \omega \sqrt{(x_0^2 - x^2)}$
electric potential	$V = \frac{Q}{4\pi\epsilon_0 r}$
electrical potential energy	$E_P = \frac{Qq}{4\pi\epsilon_0 r}$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$
capacitors in parallel	$C = C_1 + C_2 + \dots$
discharge of a capacitor	$x = x_0 e^{-\frac{t}{RC}}$
Hall voltage	$V_H = \frac{BI}{ntq}$
alternating current/voltage	$x = x_0 \sin \omega t$
radioactive decay	$x = x_0 e^{-\lambda t}$
decay constant	$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$
intensity reflection coefficient	$\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$
Stefan–Boltzmann law	$L = 4\pi\sigma r^2 T^4$
Doppler redshift	$\frac{\Delta\lambda}{\lambda} \approx \frac{Df}{f} \approx \frac{v}{c}$

- 1 (a) Define gravitational potential at a point.

.....

.....

..... [2]

- (b) A satellite X, of mass M , orbits a planet at a constant distance $4R$ from the centre of the planet, as shown in Fig. 1.1.

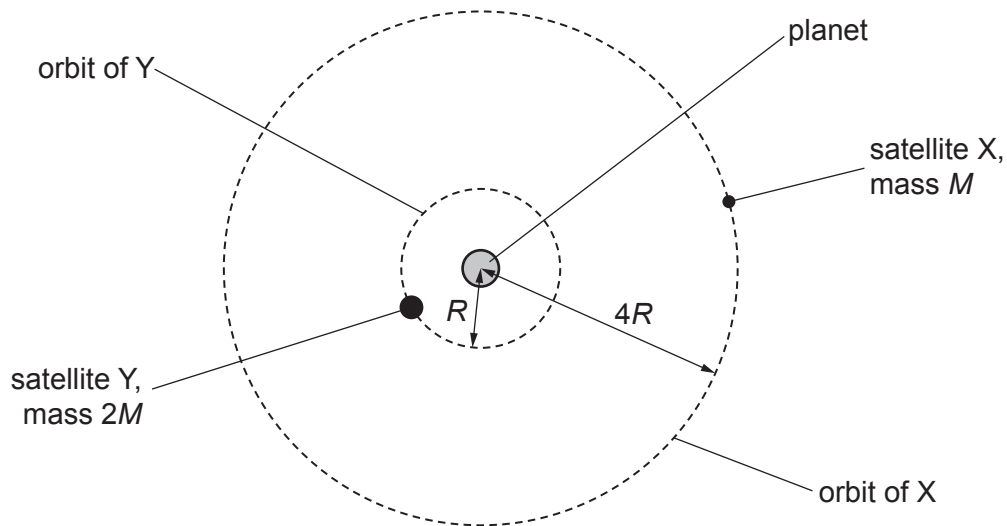


Fig. 1.1 (not to scale)

A second satellite Y, of mass $2M$, orbits the planet with orbital radius R .

The gravitational potential at X due to the planet is $-\Phi$. The planet is a uniform sphere.

- (i) Explain why the gravitational potential at X is negative.

.....

.....

..... [2]

- (ii) State an expression, in terms of Φ , for the gravitational potential at Y due to the planet.

gravitational potential = [2]

- (iii) Complete Table 1.1 by giving expressions, in terms of some or all of M , R and Φ , for the quantities indicated for each of the satellites X and Y.

Table 1.1

	satellite X	satellite Y
gravitational field strength at satellite due to planet		
gravitational potential energy of satellite		

[4]

[Total: 10]

- 2 (a) (i) State the magnitude and unit of absolute zero on the thermodynamic temperature scale.

..... [1]

- (ii) Explain why temperature measured using a laboratory liquid-in-glass thermometer does **not** give a measurement of thermodynamic temperature.

.....

..... [1]

- (b) Fig. 2.1 shows a simplified diagram of a type of thermometer called a platinum resistance thermometer.

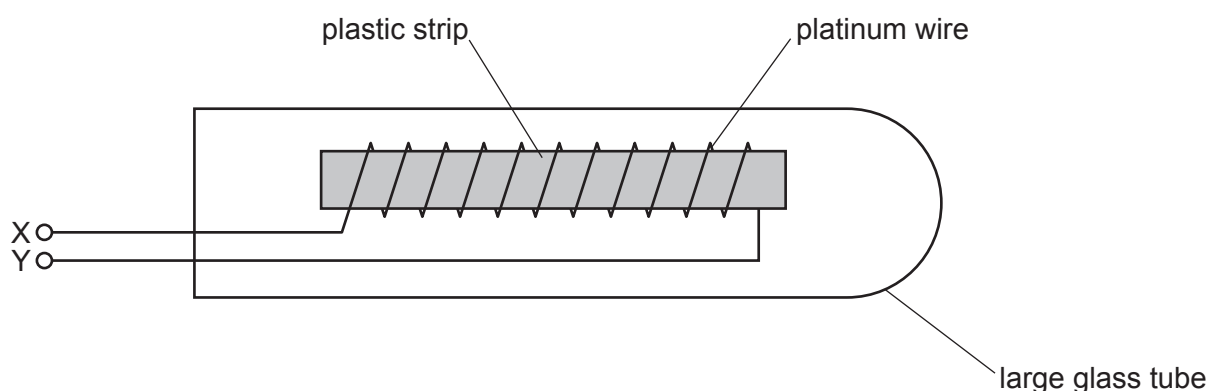


Fig. 2.1

The glass tube is immersed in the environment for which the temperature is to be determined. The resistance between the terminals X and Y is measured.

Fig. 2.2 shows the variation of the resistivity ρ of platinum with thermodynamic temperature T .

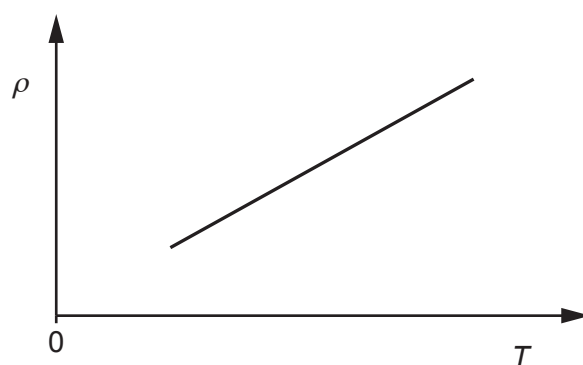


Fig. 2.2

- (i) Explain how Fig. 2.2 shows that platinum is a suitable metal for use in a resistance thermometer.

.....

 [2]

- (ii) Suggest a reason why a platinum resistance thermometer is **not** suitable for measuring a rapidly changing temperature.

.....

 [1]

- (iii) Suggest a type of thermometer that is suitable for measuring a rapidly changing temperature.

..... [1]

- (c) A negative temperature coefficient thermistor may be used as a type of resistance thermometer.

State **one** way in which the variation with temperature of the resistance of a thermistor differs from that of a platinum wire.

.....
 [1]

[Total: 7]

- 3 (a) (i) State what is meant by an ideal gas.

.....

 [2]

- (ii) Use one of the basic assumptions of the kinetic theory to explain what can be deduced about the potential energy associated with the random motion of molecules in an ideal gas.

.....

 [2]

- (b) A sample of 0.26 m^3 of an ideal gas is at pressure $2.0 \times 10^5 \text{ Pa}$ and temperature 290 K .

Determine:

- (i) the number N of molecules of the gas

$$N = \dots\dots\dots [2]$$

- (ii) the average translational kinetic energy E_K of one molecule of the gas

$$E_K = \dots\dots\dots \text{ J } [2]$$

- (iii) the internal energy of the gas. Explain your reasoning.

$$\text{internal energy} = \dots\dots\dots \text{ J } [2]$$

(c) The volume V of the gas in (b) is now varied, keeping its pressure constant.

On Fig. 3.1, sketch the variation with V of the internal energy U of the gas.



Fig. 3.1

[2]

[Total: 12]

- 4 (a) State what is meant by resonance.

.....

.....

..... [2]

- (b) A small ball is held in place using a stretched string. One end of the string is fixed to a wall and the other end is attached to a vibration generator, as shown in Fig. 4.1.

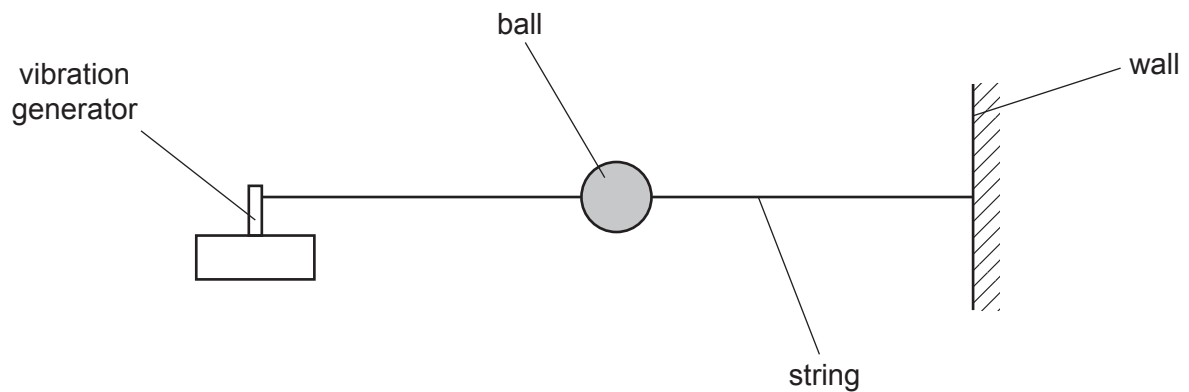


Fig. 4.1

Initially, the vibration generator is switched off.

A student displaces the ball vertically and then releases it. Fig. 4.2 shows the variation of the displacement of the ball with time after it is released.

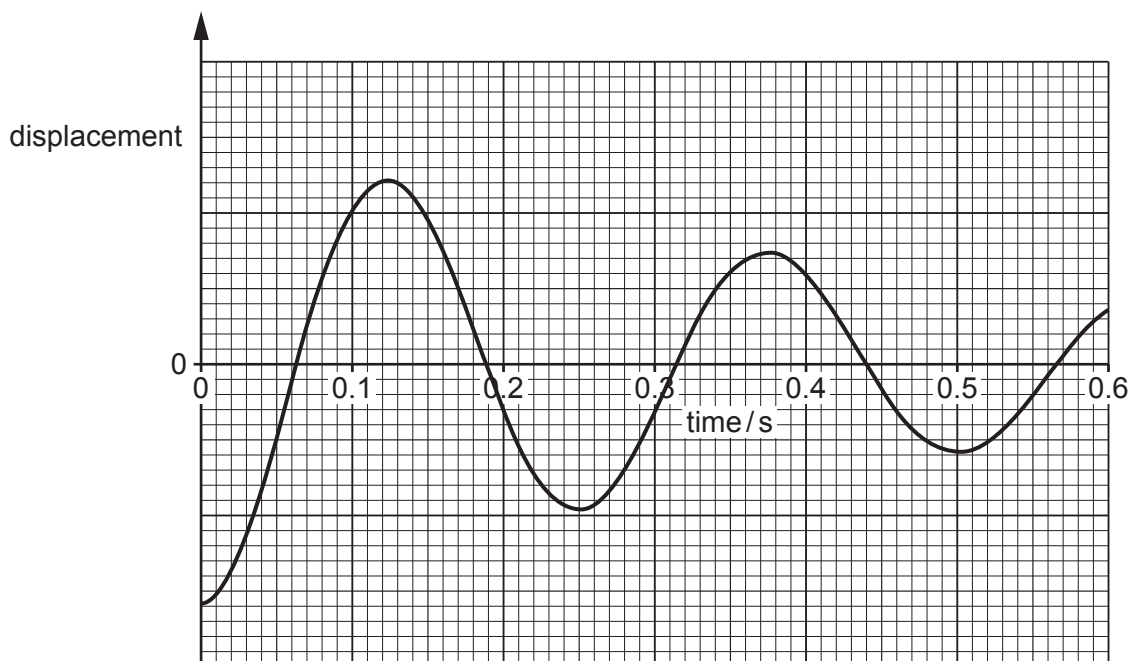


Fig. 4.2

- (i) State the name of the phenomenon illustrated by the decrease in the amplitude of the oscillations in Fig. 4.2.

..... [1]

- (ii) Explain the decrease with time of the amplitude of the oscillations of the ball.

.....

 [2]

- (iii) Determine the frequency of the oscillations of the ball.

frequency = Hz [1]

- (c) The vibration generator in (b) is switched on and its frequency f of vibration is gradually increased from 0 to 10 Hz.

On Fig. 4.3, sketch the variation with f of the amplitude of the oscillations of the ball.

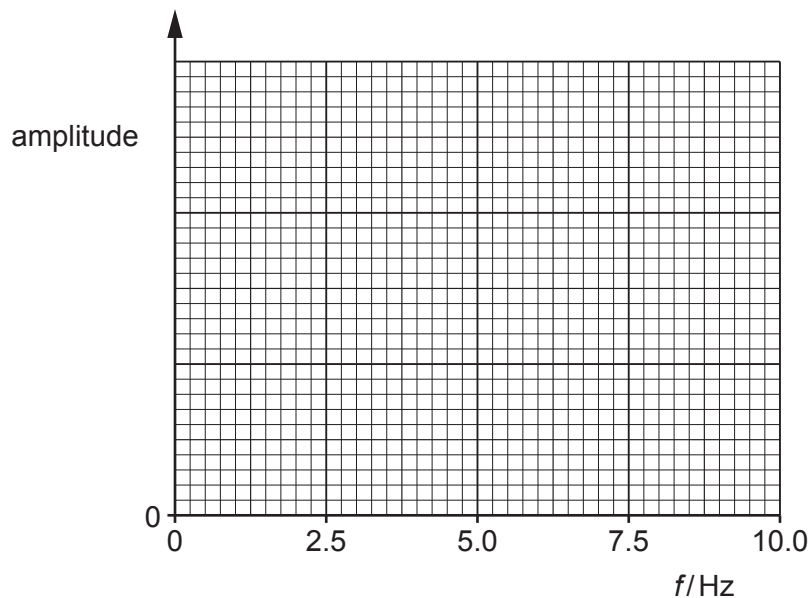


Fig. 4.3

[2]

[Total: 8]

- 5 (a) Define electric field.

.....

.....

..... [2]

- (b) Fig. 5.1 shows two parallel conducting plates that are in a vacuum. The plates are separated by a distance of 6.7 cm and have a potential difference (p.d.) of 430 V between them.

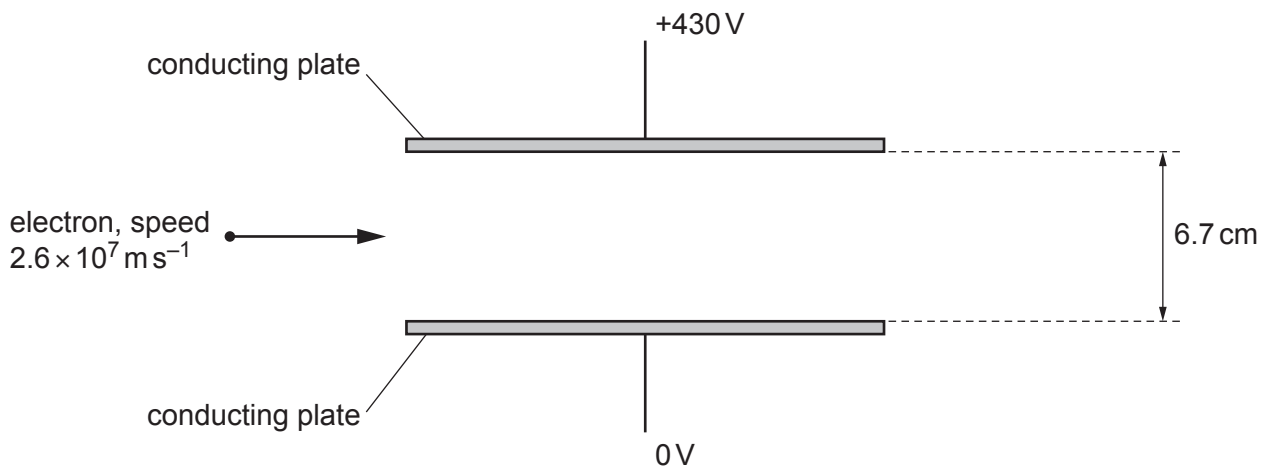


Fig. 5.1

- (i) On Fig. 5.1, draw four field lines to represent the electric field between the plates. [2]
- (ii) Determine the strength E of the electric field between the plates.

$$E = \dots\dots\dots \text{NC}^{-1} \text{ [2]}$$

- (iii) An electron travels at a speed of $2.6 \times 10^7 \text{ ms}^{-1}$ towards the region between the plates, as shown in Fig. 5.1.

On Fig. 5.1, draw the path of the electron as it moves between and beyond the plates. [2]

- (c) A uniform magnetic field is now applied in the region of the electric field in Fig. 5.1, so that the electron in (b)(iii) travels undeviated through the region.

- (i) Determine the direction of the uniform magnetic field.

..... [1]

- (ii) Explain, with reference to the forces exerted by the two fields on the electron, why the path of the electron is undeviated.

.....

.....

..... [2]

- (iii) Determine the flux density B of the uniform magnetic field. Give a unit with your answer.

$B =$ unit [2]

[Total: 13]

- 6 Fig. 6.1 shows a capacitor of capacitance C connected in series with a resistor of resistance R .

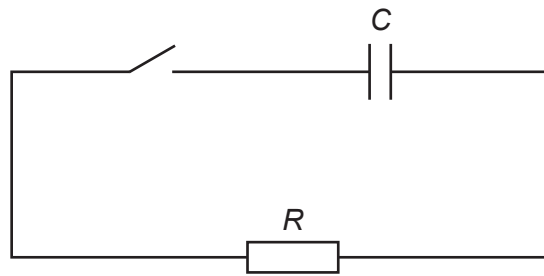


Fig. 6.1

Initially the switch is open and there is a p.d. of 12 V across the capacitor.

At time $t = 0$, the switch is closed so that there is a current I in the resistor.

Fig. 6.2 shows the variation of I with t .

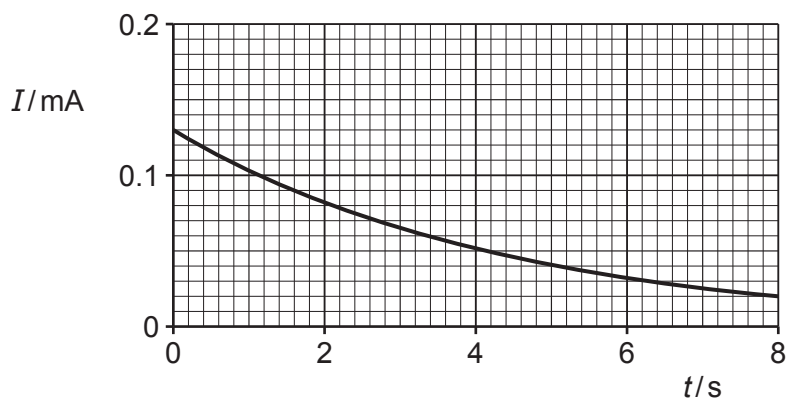


Fig. 6.2

- (a) Explain the shape of the line in Fig. 6.2.

.....

.....

.....

.....

..... [3]

(b) Use Fig. 6.2 to determine:

(i) resistance R

$$R = \dots\dots\dots \Omega \text{ [2]}$$

(ii) the time constant τ of the circuit in Fig. 6.1.

$$\tau = \dots\dots\dots \text{ s [3]}$$

(c) Use your answers in (b) to determine capacitance C .

$$C = \dots\dots\dots \text{ F [2]}$$

[Total: 10]

- 7 A circuit contains a power supply that provides a sinusoidal alternating input voltage V_{IN} . There is an output voltage V_{OUT} across a load resistor R , as shown in Fig. 7.1.

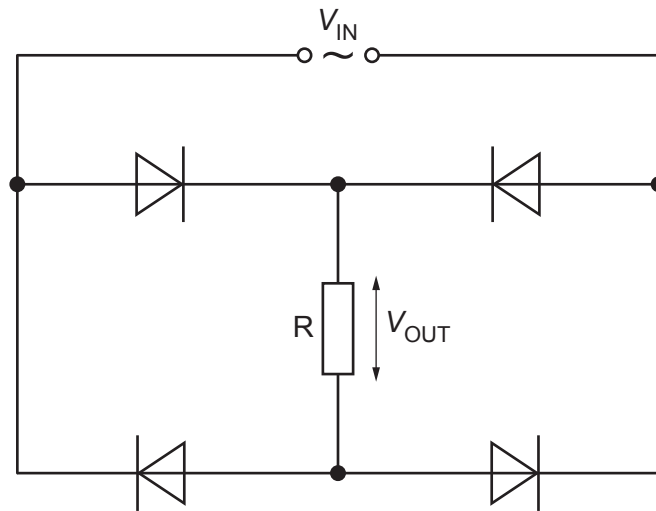


Fig. 7.1

- (a) State the purpose of the circuit in Fig. 7.1.

.....

.....

..... [2]

- (b) Fig. 7.2 shows the variation of V_{OUT} with time t .

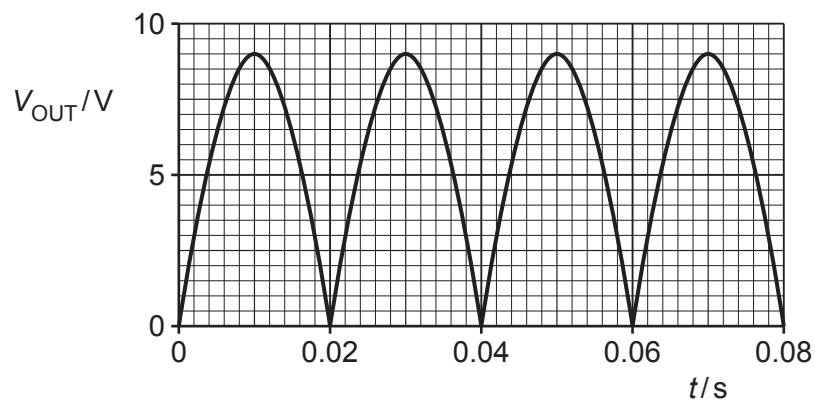


Fig. 7.2

- (i) The load resistor R has a resistance of $370\ \Omega$.

Show that the maximum power dissipated in R is 0.22 W .

[2]

- (ii) On Fig. 7.3, sketch the variation with t of the power P dissipated in R .

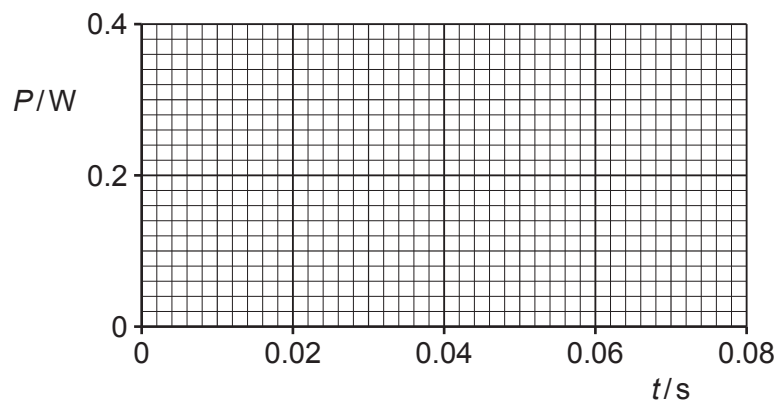


Fig. 7.3

[3]

- (iii) Calculate the mean power dissipated in R .

mean power = W [1]

- (c) The circuit of Fig. 7.1 is disconnected, and R is connected directly across the power supply.

Explain, without calculation, how the mean power now dissipated in R compares with the answer in (b)(iii).

.....

 [2]

[Total: 10]

- 8 (a) State what is meant by a photon.

.....

 [2]

- (b) Fig. 8.1 shows a tube in which X-rays are produced at a metal target.

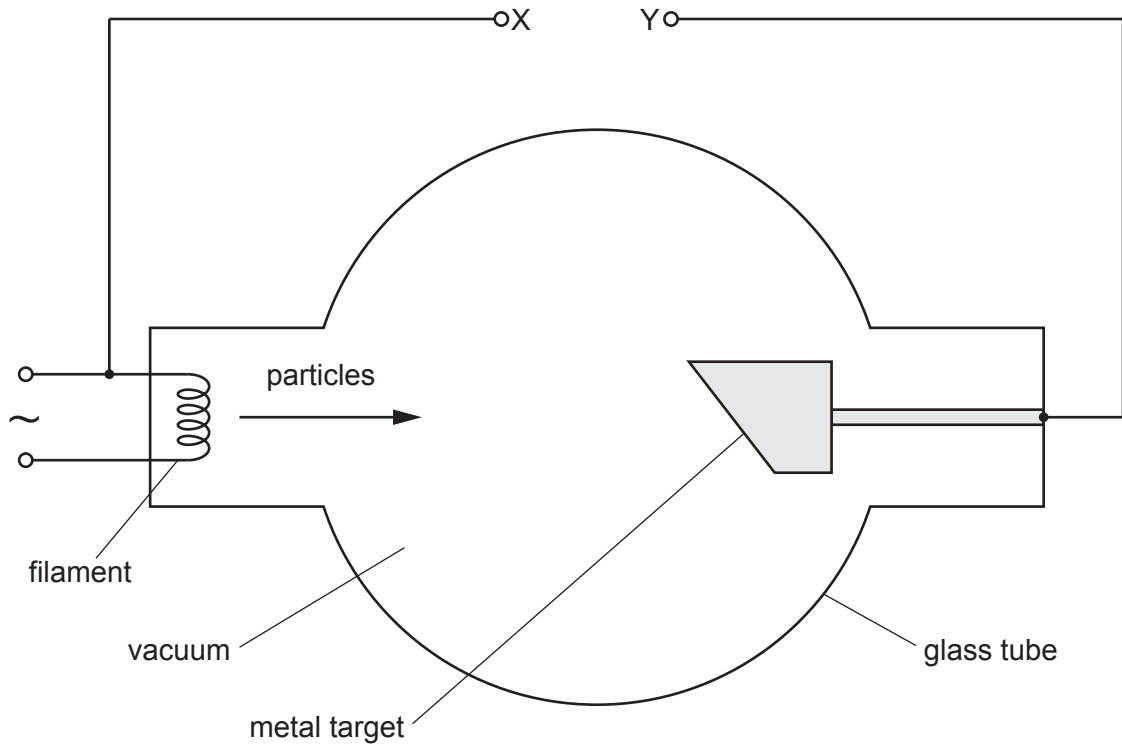


Fig. 8.1

Particles are accelerated from the filament to the target by a constant high voltage applied across the terminals X and Y.

- (i) State the name of the particles.

..... [1]

- (ii) On Fig. 8.1, use + and – signs to label terminals X and Y to indicate the polarity of the high voltage. [1]

(c) For an accelerating voltage of 32 kV in Fig. 8.1, determine:

(i) the maximum energy, in MeV, of an X-ray photon produced at the target

maximum photon energy = MeV [1]

(ii) the maximum momentum of an X-ray photon produced at the target

maximum photon momentum = N s [2]

(iii) the minimum wavelength of X-rays produced at the target.

minimum wavelength = m [3]

(d) Explain why X-rays can be used to produce images of internal body structures that have good contrast.

.....

.....

.....

.....

..... [3]

[Total: 13]

- 9 (a) Define half-life of a radioactive isotope.

.....

 [1]

- (b) Radioactive isotope X decays to isotope Y.

A sample contains only nuclei of X at time $t = 0$. Fig. 9.1 shows the variation with t of the numbers of nuclei of X and of Y as the sample decays.

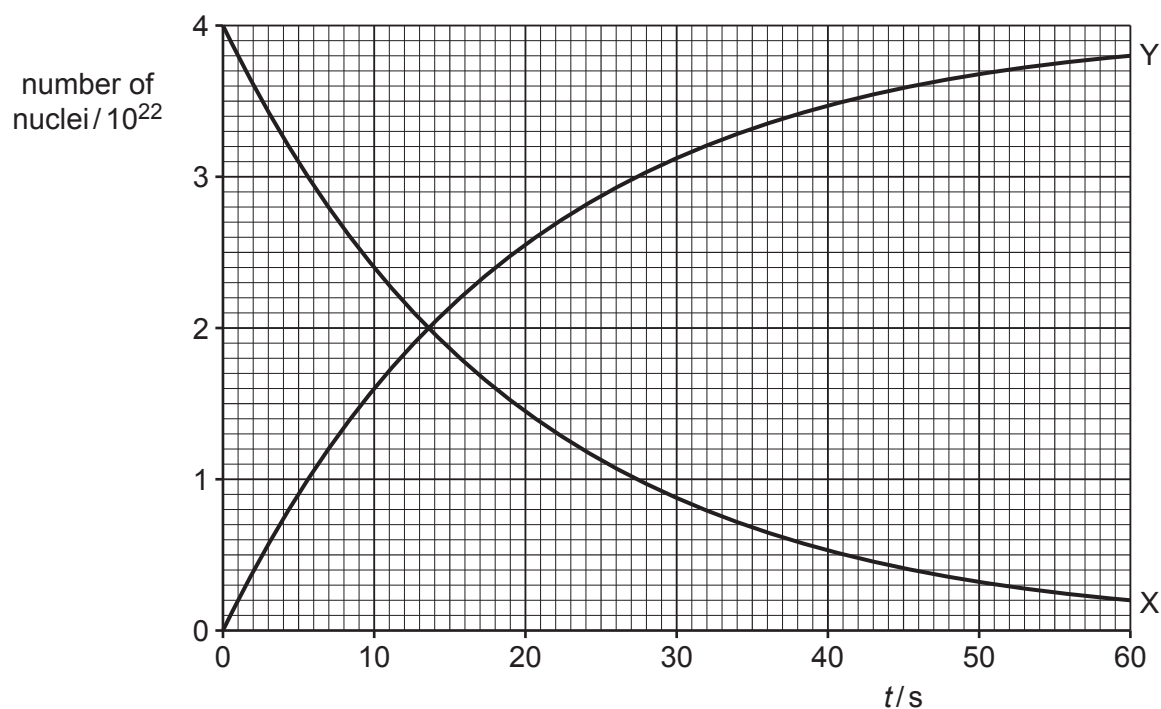


Fig. 9.1

- (i) State the name of the quantity represented by the magnitude of the gradient of line X in Fig. 9.1.

..... [1]

- (ii) State **three** conclusions about X or Y that may be drawn from Fig. 9.1. The conclusions may be qualitative or quantitative. Use the space below for any working that you need.

1

.....

2

.....

3

.....

[3]

- (c) The mass of radioactive isotope X in the sample in (b) is 7.3×10^{-4} kg at time $t = 0$.

Determine the nucleon number of isotope X.

nucleon number = [3]

[Total: 8]

- 10 (a) (i) State what is meant by the luminosity of a star.

.....

.....

..... [2]

- (ii) Explain how a standard candle in a distant galaxy can be used to determine the distance of the galaxy from an observer.

.....

.....

.....

.....

..... [3]

- (b) The Sun has a radius of 6.96×10^8 m and a surface temperature of 5780 K.
Light from the Sun is observed to have a peak intensity at a wavelength of 501 nm.

- (i) Calculate the luminosity of the Sun. Give a unit with your answer.

luminosity = unit [2]

- (ii) Another star emits radiation that has a peak intensity at a wavelength of 624 nm.

Determine the surface temperature of this star.

surface temperature =K [2]

[Total: 9]

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9702/51

May/June 2024

1 hour 15 minutes

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- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
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- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 30.
- The number of marks for each question or part question is shown in brackets [].

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[Turn over

- 1 Fig. 1.1 shows a small solid metal cylinder of mass m , length L and diameter d .

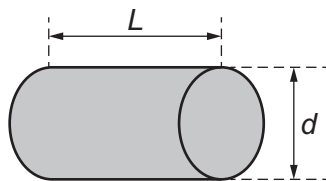


Fig. 1.1

The cylinder is heated to a uniform temperature. The cylinder is then removed from the heat source and the cylinder is wrapped in an insulating material.

The temperature of the room is T_R . At time t after the cylinder starts to cool, the surface temperature of the cylinder is T_C .

It is suggested that T_C is related to t by the relationship

$$(T_C - T_R) = Ze^{-\frac{UAt}{mc}}$$

where A is the total surface area of the cylinder, c is the specific heat capacity of the metal, and U and Z are constants.

Plan a laboratory experiment to test the relationship between T_C and t .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for U and Z .

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

Diagram

.....

.....

.....

.....

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- 2 A student investigates the sound from a horn attached to a car, as shown in Fig. 2.1.

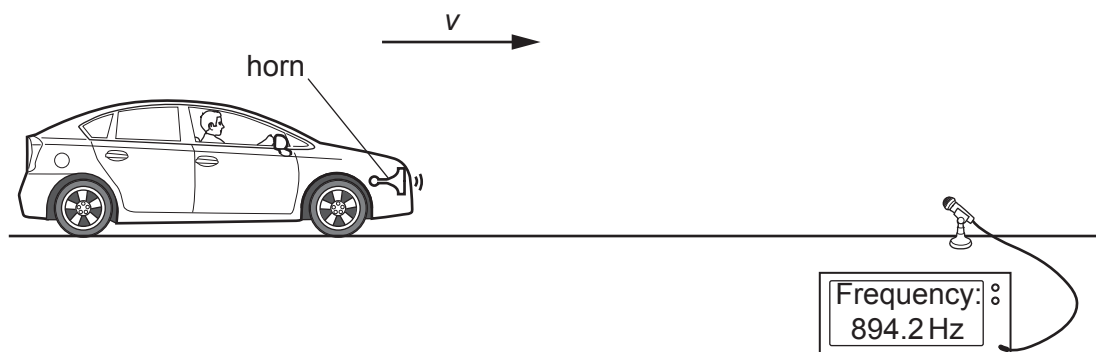


Fig. 2.1 (not to scale)

A microphone is placed at the side of the road and connected to a frequency meter. The car travels towards the microphone. The frequency f of the sound detected by the microphone is read from the frequency meter.

The speed of the car is measured by two speed detectors. The two measurements of speed are v_1 and v_2 . The average speed v of the car is determined from v_1 and v_2 .

The experiment is repeated for different speeds of the car.

It is suggested that f and v are related by the equation

$$f = \frac{f_s k}{k - v}$$

where f_s is the frequency of the sound emitted by the horn and k is a constant.

- (a) A graph is plotted of $\frac{1}{f}$ on the y -axis against v on the x -axis.

Determine expressions for the gradient and y -intercept.

gradient =

y -intercept =

[1]

(b) Values of v_1 , v_2 and f are given in Table 2.1.

Table 2.1

v_1/ms^{-1}	v_2/ms^{-1}	v/ms^{-1}	f/Hz	$\frac{1}{f}/10^{-3}\text{Hz}^{-1}$
3.1	3.9		894.2	
6.7	5.9		901.2	
9.2	8.2		908.0	
11.9	10.9		915.8	
13.3	14.5		923.6	
15.6	16.8		931.2	

Calculate and record values of v/ms^{-1} and $\frac{1}{f}/10^{-3}\text{Hz}^{-1}$ in Table 2.1.

Include the absolute uncertainties in v .

[2]

(c) (i) Plot a graph of $\frac{1}{f}/10^{-3}\text{Hz}^{-1}$ against v/ms^{-1} . Include error bars for v .

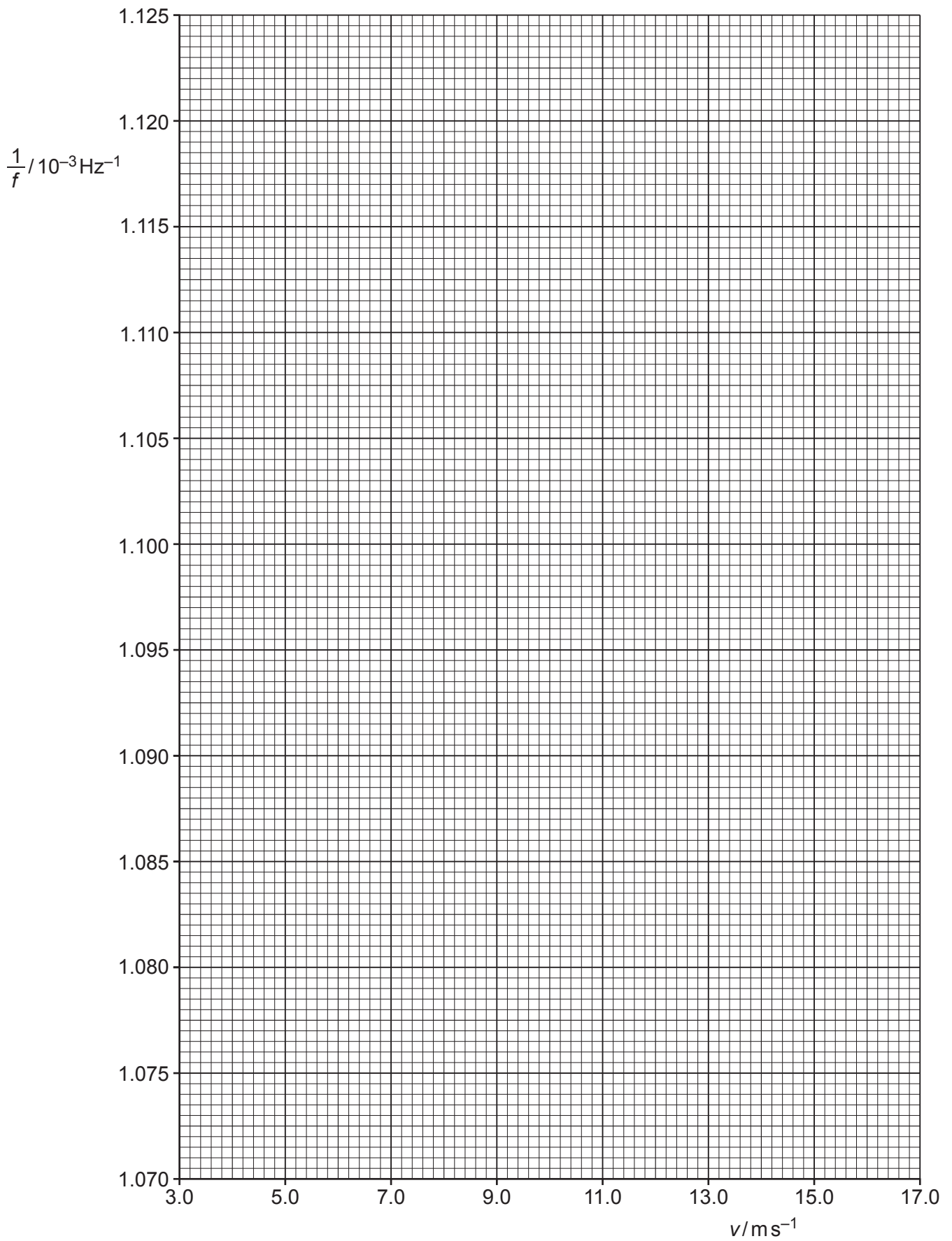
[2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines.

[2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]



- (iv) Determine the y -intercept of the line of best fit. Include the absolute uncertainty in your answer.

y -intercept = [2]

- (d) (i) Using your answers to (a), (c)(iii) and (c)(iv), determine the values of f_s and k . Include appropriate units.

f_s =

k = [2]

- (ii) Determine the percentage uncertainty in k .

percentage uncertainty in k = % [1]

- (e) The experiment is repeated. Determine the speed v that gives a value of f of 987.8 Hz.

v = ms^{-1} [1]

[Total: 15]



Cambridge International AS & A Level

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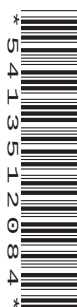


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PHYSICS

9702/52

Paper 5 Planning, Analysis and Evaluation

May/June 2024

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 30.
- The number of marks for each question or part question is shown in brackets [].

This document has 8 pages.

- 1 A spring is attached to a strong cylindrical magnet of length L and cross-sectional area A . The magnet is placed on thin card on top of a magnetic sheet on the bench, as shown in Fig. 1.1.

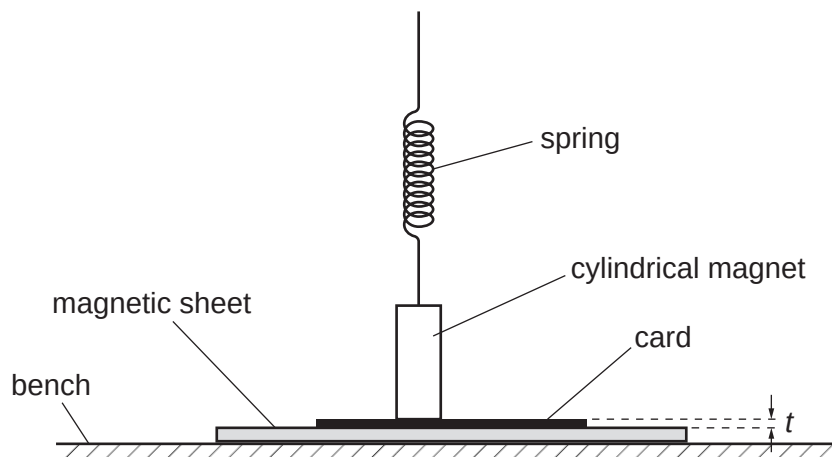


Fig. 1.1

The thickness of the card is t . The magnetic flux density at one of the poles of the magnet is B .

A force is applied upwards to the spring. The extension of the spring when the magnet just leaves the card is s .

It is suggested that s is related to t by the relationship

$$ks = \frac{ALBZ}{t}$$

where k is the spring constant of the spring and Z is a constant.

Plan a laboratory experiment to test the relationship between s and t .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine a value for Z .

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.





Diagram

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[illegible]

- 2 A student investigates the resonant frequency of a metal rod. The metal rod of length L is suspended from two rubber loops. A sensitive microphone with a cone is positioned at one end of the rod. The microphone is attached to an oscilloscope, as shown in Fig. 2.1.

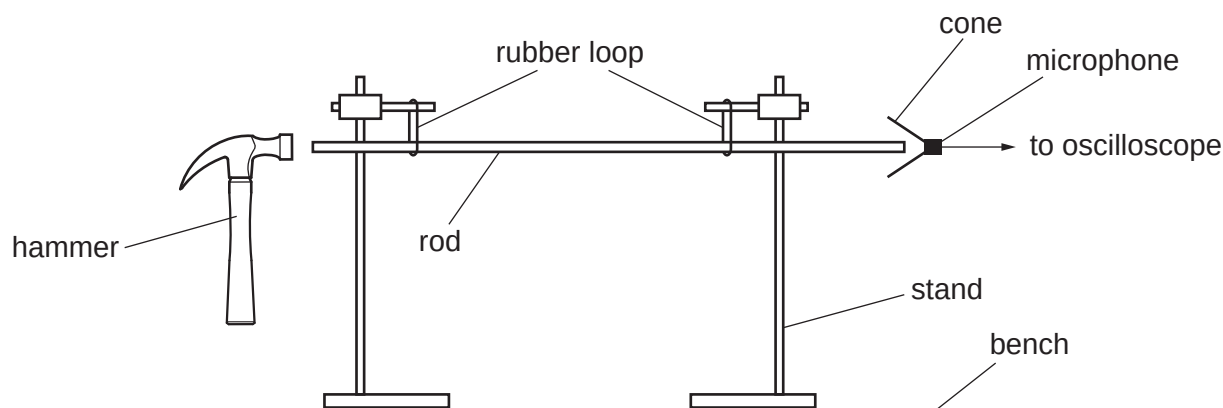


Fig. 2.1 (not to scale)

The rod is hit gently with a hammer.

The period T of the trace produced on the oscilloscope is determined.

The experiment is repeated for different values of L .

It is suggested that T and L are related by the equation

$$T = \frac{2L^n}{C}$$

where C and n are constants.

- (a) A graph is plotted of $\lg T$ on the y -axis against $\lg L$ on the x -axis.

Determine expressions for the gradient and y -intercept.

gradient =

y -intercept =

[1]





(b) Values of L and T are given in Table 2.1.

Table 2.1

L/cm	$T/10^{-5}\text{s}$	$\lg(L/\text{cm})$	$\lg(T/10^{-5}\text{s})$
54	24 ± 1		
70	32 ± 1		
86	39 ± 1		
108	49 ± 2		
140	64 ± 2		
167	74 ± 2		

Calculate and record values of $\lg(L/\text{cm})$ and $\lg(T/10^{-5}\text{s})$ in Table 2.1.
Include the absolute uncertainties in $\lg T$.

[2]

(c) (i) Plot a graph of $\lg(T/10^{-5}\text{s})$ against $\lg(L/\text{cm})$. Include error bars for $\lg T$. [2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines. [2]

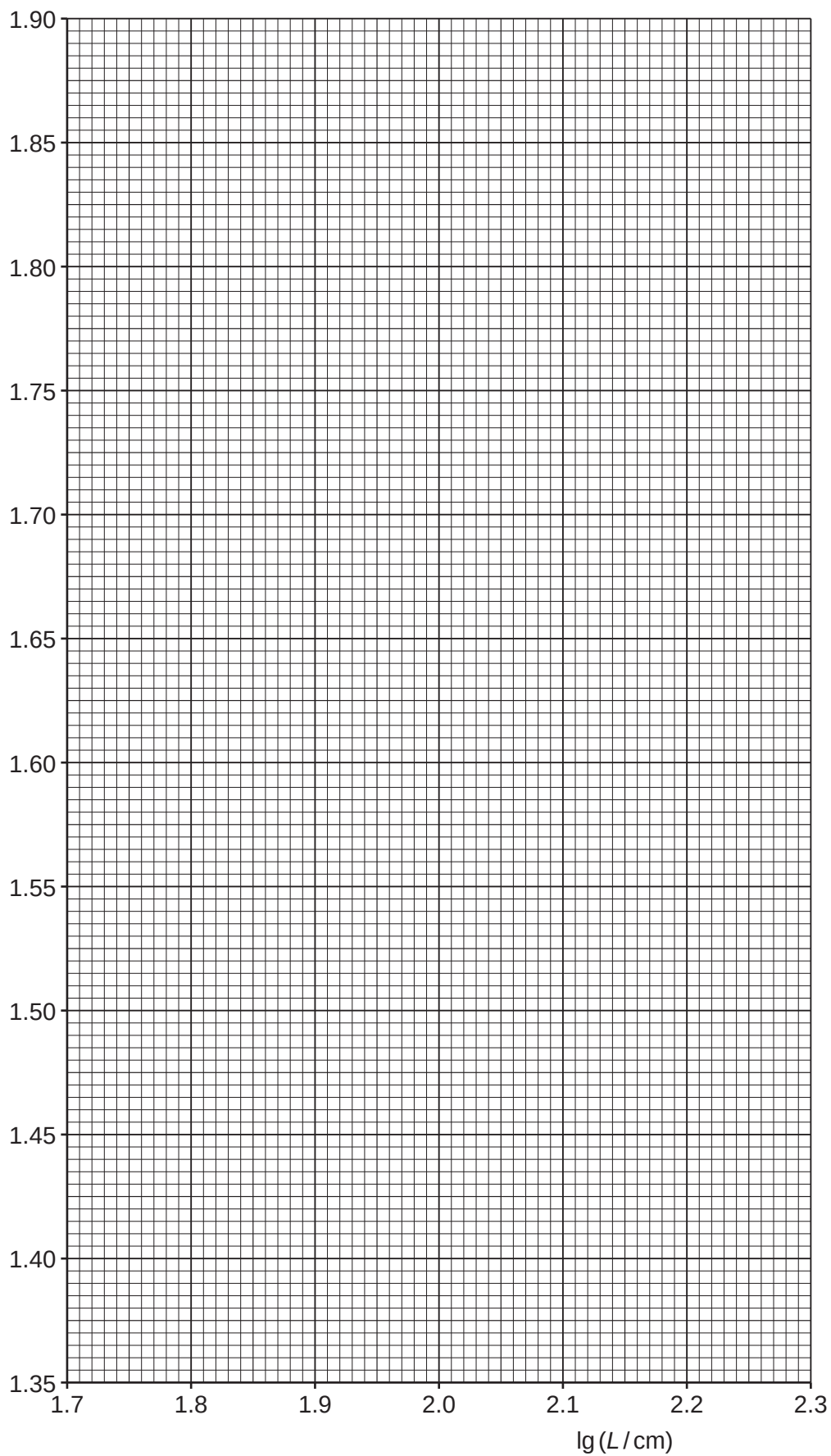
(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]





$\lg(T/10^{-5}\text{s})$





- (iv) Determine the y -intercept of the line of best fit. Include the absolute uncertainty in your answer.

y -intercept = [2]

- (d) Using your answers to (a), (c)(iii) and (c)(iv), determine the values of n and C . Include the absolute uncertainties in your values. You need not be concerned with units.

n =

C = [3]

- (e) The experiment is repeated. Determine the length L of the rod that gives a value of T of 0.10 ms.

L = cm [1]

[Total: 15]





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9702/53

May/June 2024

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 30.
- The number of marks for each question or part question is shown in brackets [].

This document has 8 pages.

- 1 Fig. 1.1 shows a small solid metal cylinder of mass m , length L and diameter d .

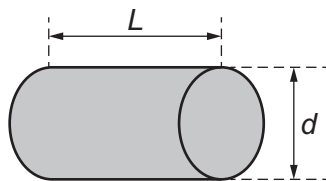


Fig. 1.1

The cylinder is heated to a uniform temperature. The cylinder is then removed from the heat source and the cylinder is wrapped in an insulating material.

The temperature of the room is T_R . At time t after the cylinder starts to cool, the surface temperature of the cylinder is T_C .

It is suggested that T_C is related to t by the relationship

$$(T_C - T_R) = Ze^{-\frac{UAt}{mc}}$$

where A is the total surface area of the cylinder, c is the specific heat capacity of the metal, and U and Z are constants.

Plan a laboratory experiment to test the relationship between T_C and t .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for U and Z .

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

Diagram

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- 2 A student investigates the sound from a horn attached to a car, as shown in Fig. 2.1.

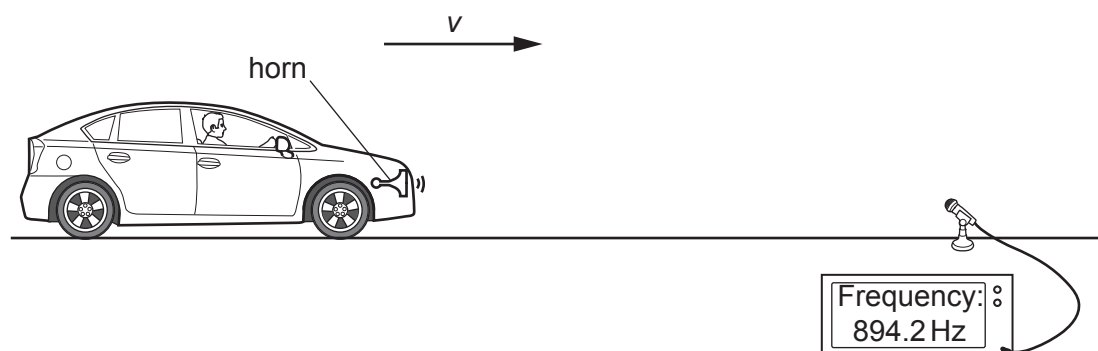


Fig. 2.1 (not to scale)

A microphone is placed at the side of the road and connected to a frequency meter. The car travels towards the microphone. The frequency f of the sound detected by the microphone is read from the frequency meter.

The speed of the car is measured by two speed detectors. The two measurements of speed are v_1 and v_2 . The average speed v of the car is determined from v_1 and v_2 .

The experiment is repeated for different speeds of the car.

It is suggested that f and v are related by the equation

$$f = \frac{f_s k}{k - v}$$

where f_s is the frequency of the sound emitted by the horn and k is a constant.

- (a) A graph is plotted of $\frac{1}{f}$ on the y -axis against v on the x -axis.

Determine expressions for the gradient and y -intercept.

gradient =

y -intercept =

[1]

(b) Values of v_1 , v_2 and f are given in Table 2.1.

Table 2.1

v_1/ms^{-1}	v_2/ms^{-1}	v/ms^{-1}	f/Hz	$\frac{1}{f}/10^{-3}\text{Hz}^{-1}$
3.1	3.9		894.2	
6.7	5.9		901.2	
9.2	8.2		908.0	
11.9	10.9		915.8	
13.3	14.5		923.6	
15.6	16.8		931.2	

Calculate and record values of v/ms^{-1} and $\frac{1}{f}/10^{-3}\text{Hz}^{-1}$ in Table 2.1.

Include the absolute uncertainties in v .

[2]

(c) (i) Plot a graph of $\frac{1}{f}/10^{-3}\text{Hz}^{-1}$ against v/ms^{-1} . Include error bars for v .

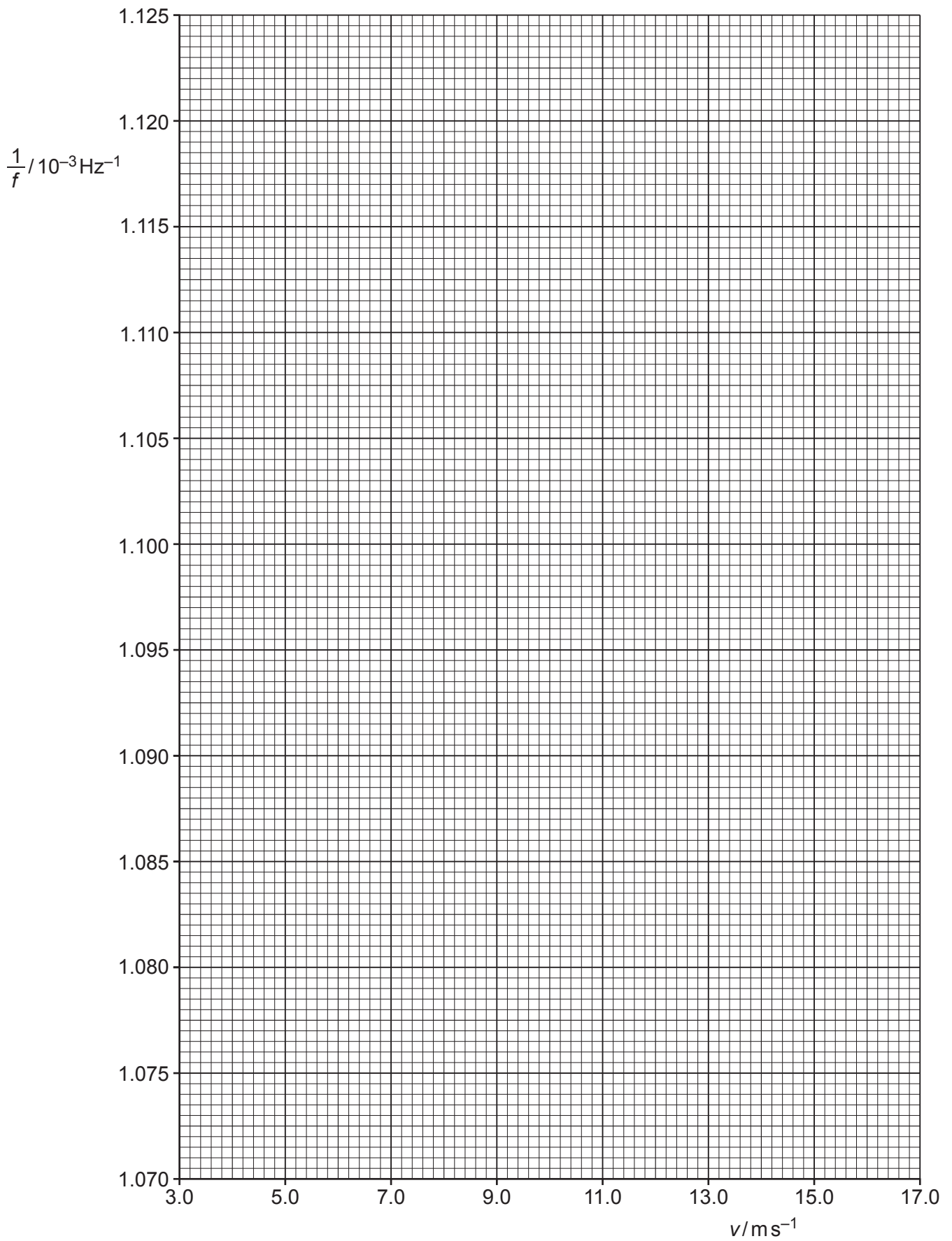
[2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines.

[2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]



- (iv) Determine the y -intercept of the line of best fit. Include the absolute uncertainty in your answer.

y -intercept = [2]

- (d) (i) Using your answers to (a), (c)(iii) and (c)(iv), determine the values of f_s and k . Include appropriate units.

f_s =

k = [2]

- (ii) Determine the percentage uncertainty in k .

percentage uncertainty in k = % [1]

- (e) The experiment is repeated. Determine the speed v that gives a value of f of 987.8 Hz.

v = ms^{-1} [1]

[Total: 15]



Cambridge International AS & A Level

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PHYSICS

9702/41

Paper 4 A Level Structured Questions

October/November 2024

2 hours

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [].

This document has **24** pages. Any blank pages are indicated.

**Data**

acceleration of free fall	$g = 9.81 \text{ m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
unified atomic mass unit	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27} \text{ kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$ $(\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ m F}^{-1})$
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_o = \frac{f_s v}{v \pm v_s}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \dots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$





gravitational potential

$$\phi = -\frac{GM}{r}$$

gravitational potential energy

$$E_P = -\frac{GMm}{r}$$

pressure of an ideal gas

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

simple harmonic motion

$$a = -\omega^2 x$$

velocity of particle in s.h.m.

$$v = v_0 \cos \omega t$$

$$v = \pm \omega \sqrt{(x_0^2 - x^2)}$$

electric potential

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

electrical potential energy

$$E_P = \frac{Qq}{4\pi\epsilon_0 r}$$

capacitors in series

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$$

capacitors in parallel

$$C = C_1 + C_2 + \dots$$

discharge of a capacitor

$$x = x_0 e^{-\frac{t}{RC}}$$

Hall voltage

$$V_H = \frac{BI}{ntq}$$

alternating current/voltage

$$x = x_0 \sin \omega t$$

radioactive decay

$$x = x_0 e^{-\lambda t}$$

decay constant

$$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$$

intensity reflection coefficient

$$\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$$

Stefan-Boltzmann law

$$L = 4\pi\sigma r^2 T^4$$

Doppler redshift

$$\frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$$





- 1 (a) State Newton's law of gravitation.

.....

.....

..... [2]

- (b) A planet may be considered as a uniform sphere.

A satellite is in circular orbit of period T around the planet at a height h above the surface. The height of the orbit can be adjusted by use of the satellite's rocket engines.

Fig. 1.1 shows the variation with h of $T^{\frac{2}{3}}$.

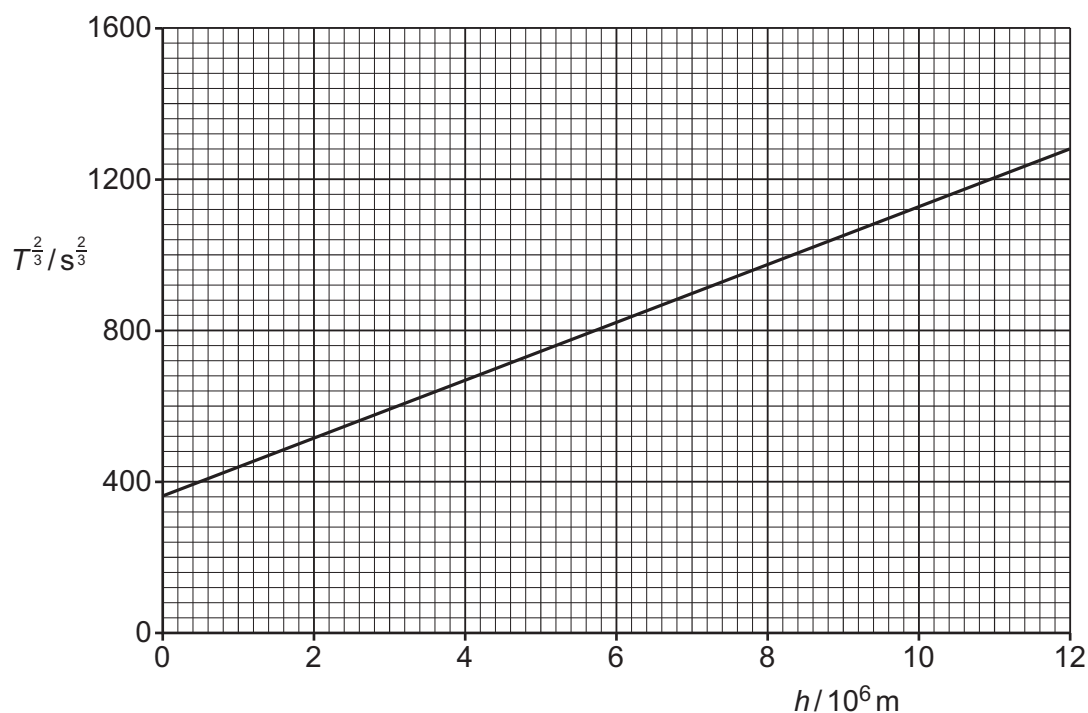


Fig. 1.1

- (i) By reference to forces, explain why the orbit of the satellite is circular.

.....

.....

..... [2]





- (ii) Use Newton's law of gravitation to show that h and T are related by

$$(h + B)^3 = \frac{GA}{4\pi^2} T^2$$

where G is the gravitational constant and A and B are constants that depend on the properties of the planet.

[3]

- (iii) Use the gradient and intercept of the line in Fig. 1.1 to determine values for A and B .
Give units with your answers.

$A = \dots\dots\dots$ unit $\dots\dots\dots$

$B = \dots\dots\dots$ unit $\dots\dots\dots$

[5]

[Total: 12]





2 (a) Define specific heat capacity.

.....

.....

..... [2]

(b) Two solid blocks X and Y are made from different metals. The blocks have different initial temperatures. Block Y is initially at room temperature.

The blocks are placed in direct thermal contact with each other at time $t = 0$. Fig. 2.1 shows the variation with t of the temperatures of the two blocks.

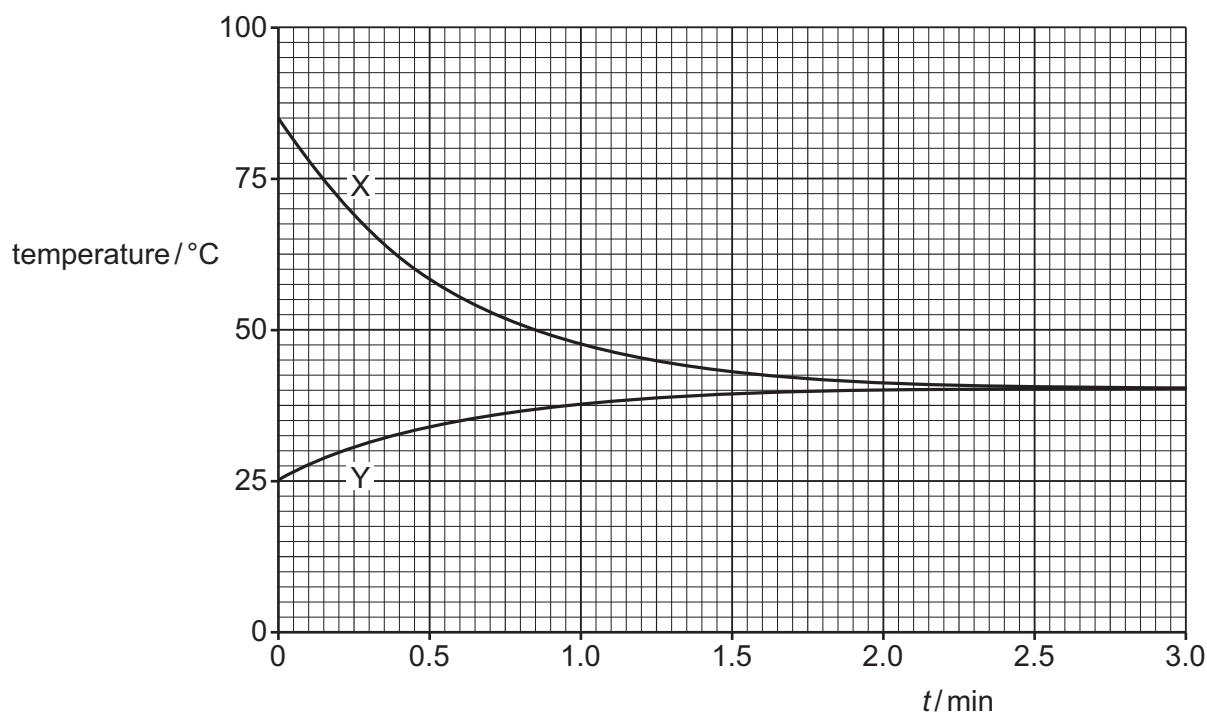


Fig. 2.1





- (i) State **three** conclusions that may be drawn from Fig. 2.1. The conclusions may be qualitative or quantitative.

1

.....

2

.....

3

.....

[3]

- (ii) The ratio $\frac{\text{mass of block Y}}{\text{mass of block X}}$ is equal to 1.3.

The metal in block Y has a specific heat capacity of $901 \text{ J kg}^{-1} \text{ K}^{-1}$.

Determine the specific heat capacity of the metal in block X.

specific heat capacity = $\text{J kg}^{-1} \text{ K}^{-1}$ [3]

[Total: 8]





3 (a) (i) State what is meant by the Avogadro constant.

.....

.....

..... [1]

(ii) State the relationship between the Avogadro constant N_A , the molar gas constant R and the Boltzmann constant k .

[1]

(b) Two samples X and Y of ideal gases are both at thermodynamic temperature T .

Sample X has volume V and consists of N molecules, each of mass m .

Sample Y has volume $2V$ and consists of $2N$ molecules, each of mass $2m$.

(i) Complete Table 3.1 by giving expressions, in terms of some or all of N , m , T , V and the constants in (a)(ii), for the quantities indicated.

Table 3.1

	sample X	sample Y
pressure		
amount of substance		
mean-square speed of molecules		
internal energy		

[4]





(ii) The temperature of sample X is now varied.

On Fig. 3.1, sketch the variation with thermodynamic temperature of the root-mean-square (r.m.s.) speed of the molecules of the gas.

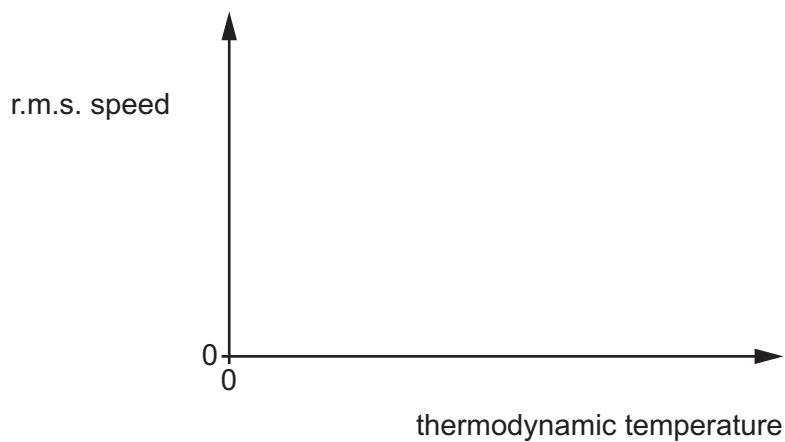


Fig. 3.1

[2]

[Total: 8]





- 4 (a) State what is meant by simple harmonic motion.

.....

.....

..... [2]

- (b) A block is suspended from a spring, as shown in Fig. 4.1.

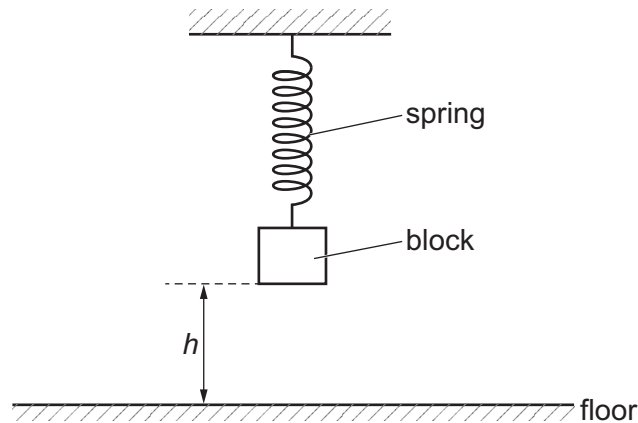


Fig. 4.1

The block is pulled down and released at time $t = 0$. It then oscillates vertically with simple harmonic motion.

Fig. 4.2 shows the variation of the velocity v of the block with height h of the base of the block above the floor.

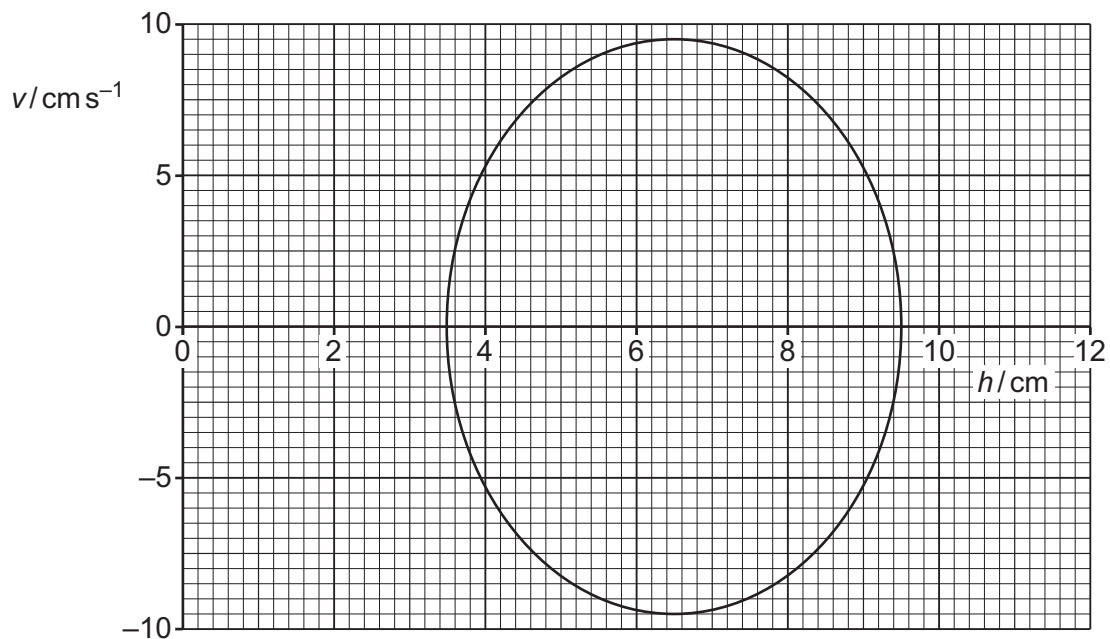


Fig. 4.2



- (i) Determine the amplitude, in cm, of the oscillations.

amplitude = cm [1]

- (ii) Show that the angular frequency of the oscillations is 3.2 rad s^{-1} .

[2]

- (iii) Calculate the period T of the oscillations.

$T = \dots\dots\dots$ s [2]

- (iv) On Fig. 4.3, sketch the variation of h with time t from $t = 0$ to $t = 6.0 \text{ s}$.

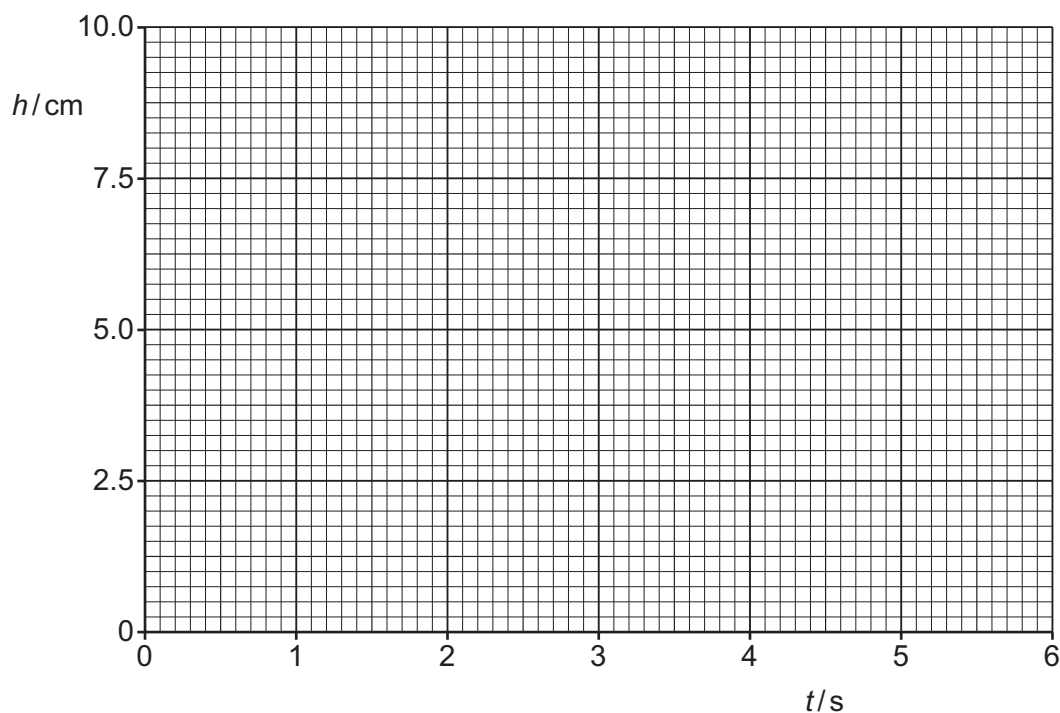


Fig. 4.3

[4]

[Total: 11]



- 5 (a) State the relationship between electric field and electric potential.

.....

.....

..... [2]

- (b) Two charged isolated insulating spheres X and Y are near to each other, as shown in Fig. 5.1.



Fig. 5.1

P is a point on the line joining the centres of the spheres.

Explain why it is **not** possible for the total electric potential and the resultant electric field to simultaneously be zero at point P.

.....

.....

.....

.....

..... [3]

- (c) The magnitudes of the charges on spheres X and Y in Fig. 5.1 are Q and $2Q$ respectively. The spheres may be considered as point charges at their centres.

Point P is a distance x from the centre of sphere X.

The electric potential at point P is zero.

- (i) Show that the distance y of point P from the centre of sphere Y is equal to $2x$.

[2]





- (ii) State an expression, in terms of Q , x and the permittivity of free space ϵ_0 , for the electric field strength E_x at P due to sphere X.

$$E_x = \dots\dots\dots [1]$$

- (iii) Determine an expression, in terms of Q , x and ϵ_0 , for the resultant electric field strength E at point P due to the two spheres.

$$E = \dots\dots\dots [2]$$

[Total: 10]





- 6 (a) (i) State what is meant by rectification of an alternating voltage.

.....
 [1]

- (ii) State the difference between half-wave rectification and full-wave rectification.

.....

 [2]

- (b) (i) Complete Fig. 6.1 to show a circuit that produces half-wave rectification of an alternating input voltage V_{IN} to produce output voltage V_{OUT} across the resistor R .

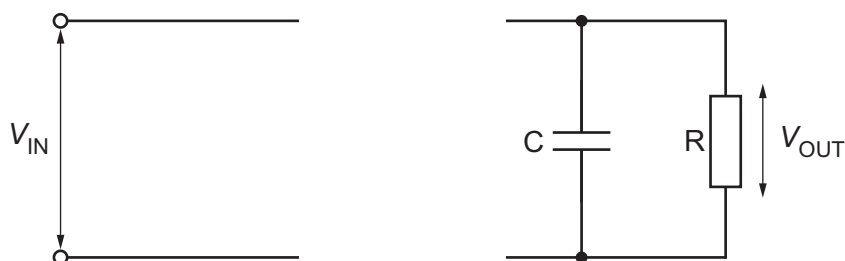


Fig. 6.1

[2]

- (ii) State the purpose of the capacitor C in the circuit of Fig. 6.1.

.....
 [1]

- (c) The input voltage V_{IN} in Fig. 6.1 is a square wave. Fig. 6.2 shows the variation of V_{IN} with time t .

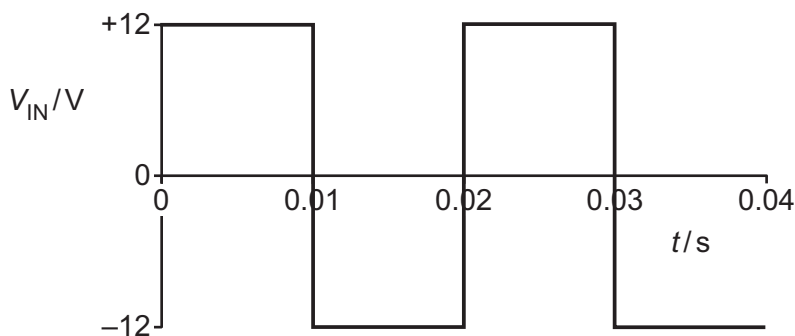


Fig. 6.2



Fig. 6.3 shows the variation of V_{OUT} with t .

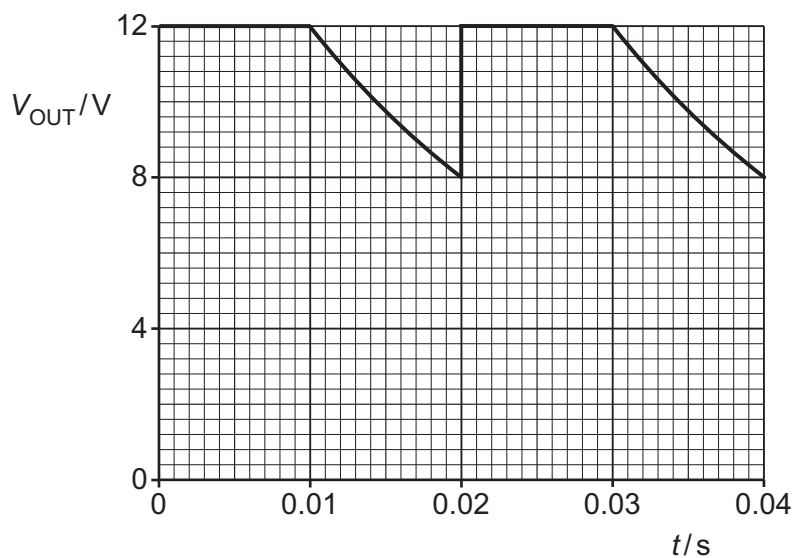


Fig. 6.3

The maximum energy stored in the capacitor is 0.041 J.

(i) Show that the capacitance of C is $570 \mu\text{F}$.

[2]

(ii) Determine the resistance of R .

resistance = Ω [3]

[Total: 11]





7 (a) Define magnetic flux density.

.....

.....

..... [2]

(b) A long, straight wire carries a current into the page, as shown in Fig. 7.1.



Fig. 7.1

On Fig. 7.1, draw four field lines to represent the magnetic field around the wire due to the current in it. [3]

(c) Two identical wires X and Y are placed parallel to each other. The wires both carry current into the page, as shown in Fig. 7.2.

X



Y



Fig. 7.2





- (i) Explain why the two wires exert a magnetic force on each other.

.....

.....

.....

..... [2]

- (ii) On Fig. 7.2, draw an arrow to show the direction of the magnetic force exerted on wire X. Label your arrow F. [1]

- (iii) The current in X is double the current in Y.

State how the magnetic force exerted on wire Y compares with the magnetic force exerted on wire X.

.....

.....

..... [2]

- (iv) The direction of the current in both wires is now reversed.

State, with a reason, the effect of this change on the direction of the force on wire X.

.....

..... [1]

[Total: 11]





- 8 A polished sheet of magnesium in a vacuum emits electrons when it is illuminated by ultraviolet radiation.

(a) State the name of this phenomenon.

..... [1]

- (b) For emission of electrons to occur, the frequency of the ultraviolet radiation must be at least 8.8×10^{14} Hz.

(i) Calculate the work function energy of magnesium.

work function energy = J [2]

- (ii) For ultraviolet radiation with a frequency of 11×10^{14} Hz, calculate the maximum speed of the emitted electrons.

maximum speed = ms^{-1} [3]



- (c) The frequency f of the ultraviolet radiation incident on the magnesium sheet is varied between $8.0 \times 10^{14} \text{ Hz}$ and $11 \times 10^{14} \text{ Hz}$.

On Fig. 8.1, sketch the variation with f of the maximum kinetic energy E_{MAX} of the emitted electrons. Use the space below for any working that you need.

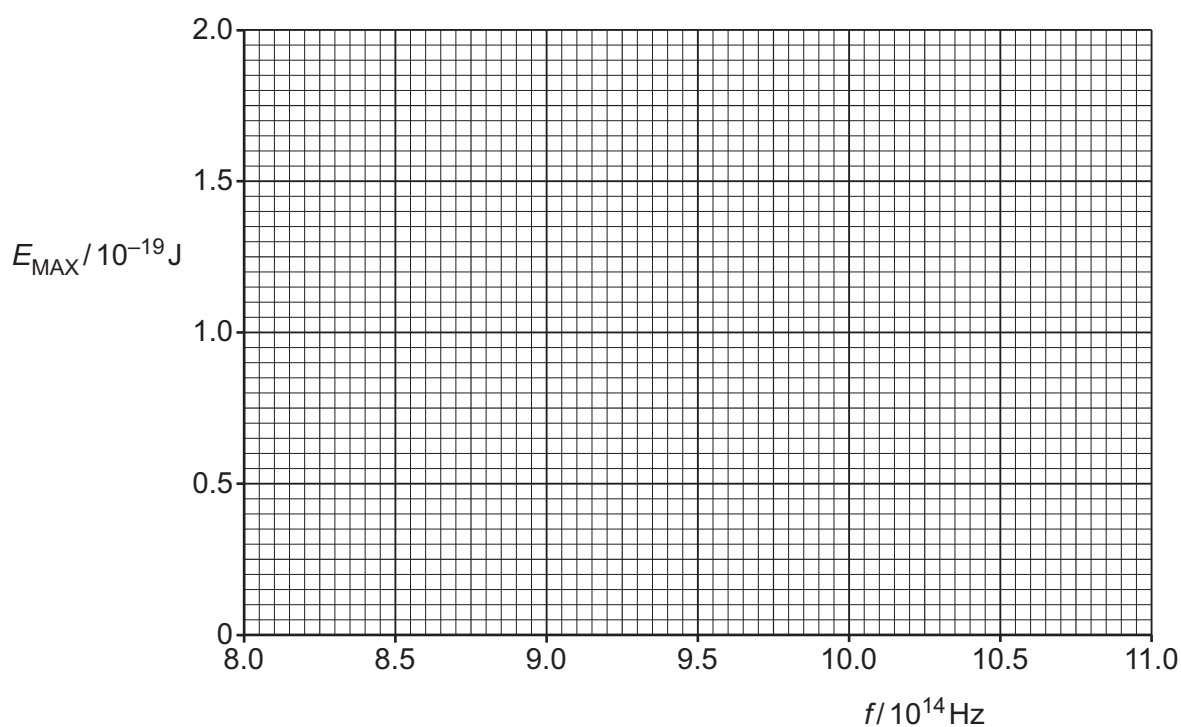


Fig. 8.1

[3]

[Total: 9]





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9 Fluorine-18 ($^{18}_9\text{F}$) decays by beta-plus (β^+) emission with a half-life of 110 minutes.

(a) (i) State the name of the beta-plus particle.

..... [1]

(ii) Show that the decay constant of fluorine-18 is $1.05 \times 10^{-4} \text{ s}^{-1}$.

[1]

(iii) Determine the activity of $2.1 \times 10^{-12} \text{ kg}$ of fluorine-18.

activity = Bq [3]

(b) A small sample of fluorine-18 injected into the body acts as a tracer for use in medical imaging.

(i) Describe how the interaction of a β^+ particle with an electron in the body enables the formation of an image.

.....

 [3]

(ii) Suggest why 110 minutes is a suitable half-life for a nuclide used as a tracer in medical diagnosis.

.....

 [2]

[Total: 10]





- 10 (a) Explain how redshift leads to the idea that the Universe is expanding.

.....

.....

.....

.....

..... [3]

- (b) Stars in a distant galaxy emit radiation. The total luminosity of the stars in the galaxy is $1.90 \times 10^{36} \text{ W}$.

The emission spectrum of the radiation contains a line X at a wavelength of 658 nm.

Radiation from the galaxy is observed on the Earth. The observed radiation has a radiant flux intensity of $8.42 \times 10^{-16} \text{ W m}^{-2}$. In the observed emission spectrum, line X is at a wavelength of 726 nm.

Determine:

- (i) the distance d of the galaxy from the Earth

$$d = \dots\dots\dots \text{ m [2]}$$

- (ii) the speed v of the galaxy relative to the Earth.

$$v = \dots\dots\dots \text{ ms}^{-1} [2]$$



- (c) Observations of many galaxies, such as the one in (b), lead to many pairs of values of d and v . Plotting these values reveals a trend.

- (i) On Fig. 10.1, sketch the variation of v with d .

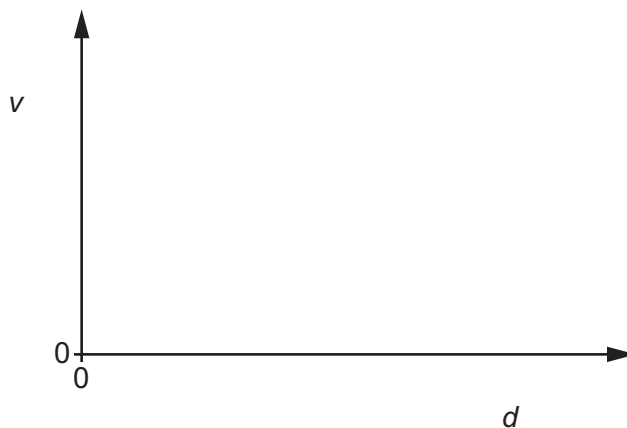


Fig. 10.1

[2]

- (ii) State the name of the quantity represented by the gradient of the line in Fig. 10.1.

..... [1]

[Total: 10]





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PHYSICS

9702/42

Paper 4 A Level Structured Questions

October/November 2024

2 hours

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

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- Write your name, centre number and candidate number in the boxes at the top of the page.
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- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [].

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**Data**

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elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
unified atomic mass unit	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27} \text{ kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$ $(\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ m F}^{-1})$
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_o = \frac{f_s v}{v \pm v_s}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \dots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$





gravitational potential

$$\phi = -\frac{GM}{r}$$

gravitational potential energy

$$E_P = -\frac{GMm}{r}$$

pressure of an ideal gas

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

simple harmonic motion

$$a = -\omega^2 x$$

velocity of particle in s.h.m.

$$v = v_0 \cos \omega t$$

$$v = \pm \omega \sqrt{(x_0^2 - x^2)}$$

electric potential

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

electrical potential energy

$$E_P = \frac{Qq}{4\pi\epsilon_0 r}$$

capacitors in series

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$$

capacitors in parallel

$$C = C_1 + C_2 + \dots$$

discharge of a capacitor

$$x = x_0 e^{-\frac{t}{RC}}$$

Hall voltage

$$V_H = \frac{BI}{ntq}$$

alternating current/voltage

$$x = x_0 \sin \omega t$$

radioactive decay

$$x = x_0 e^{-\lambda t}$$

decay constant

$$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$$

intensity reflection coefficient

$$\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$$

Stefan-Boltzmann law

$$L = 4\pi\sigma r^2 T^4$$

Doppler redshift

$$\frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$$





- 1 A metal wheel consists of an axle A, eight spokes and a rim, as shown in Fig. 1.1.

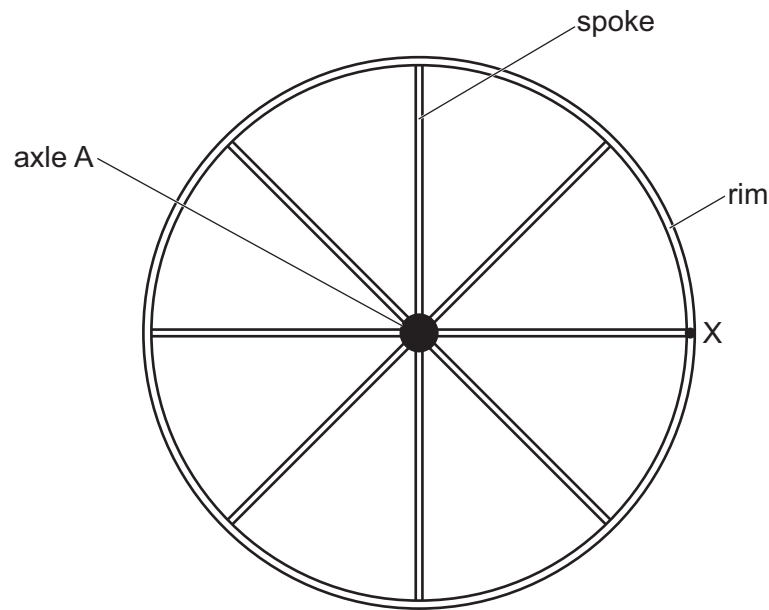


Fig. 1.1

Point X is on the rim at the end of one of the spokes.

The rim has a radius of 0.85 m.

The wheel is rotating clockwise with an angular speed of 140 rad s^{-1} .

- (a) For point X, determine:

- (i) the speed

speed = m s^{-1} [2]

- (ii) the centripetal acceleration.

acceleration = m s^{-2} [2]





(b) There is a uniform magnetic field of flux density 0.18 T into the plane of the page.

(i) State Lenz's law of electromagnetic induction.

.....

 [2]

(ii) Show that the time taken for point X to complete one revolution is 45 ms .

[1]

(iii) Calculate the magnetic flux cut by spoke AX during one revolution of the wheel.
 Give a unit with your answer.

magnetic flux = unit [3]

(iv) Determine the magnitude of the electromotive force (e.m.f.) induced across spoke AX.

induced e.m.f. = V [2]

(v) Use Lenz's law to explain whether the potential is higher at end A or end X of the spoke.

.....

 [1]

[Total: 13]





- 2 The Sun may be considered as a uniform sphere with a mass of $1.99 \times 10^{30} \text{ kg}$ and a surface temperature of 5780 K.

A probe with a mass of 2.63 kg moves in a straight line towards the Sun.

When it is at a distance x from the centre of the Sun, the probe measures the gravitational field strength g due to the Sun and the radiant flux intensity F of radiation from the Sun.

- (a) Define gravitational field.

.....
 [1]

- (b) For the position of the probe where $x = 1.47 \times 10^{11} \text{ m}$:

- (i) calculate g

$$g = \dots\dots\dots \text{N kg}^{-1} \quad [2]$$

- (ii) determine the gravitational potential energy E_p of the probe.

$$E_p = \dots\dots\dots \text{J} \quad [2]$$

- (c) (i) Show that, for any particular value of x , the numerical values of g and F are related by

$$g = \frac{4\pi GM}{L} F$$

where M is the mass of the Sun, L is the luminosity of the Sun and G is the gravitational constant.

[3]



(ii) Fig. 2.1 shows the variation of g with F .

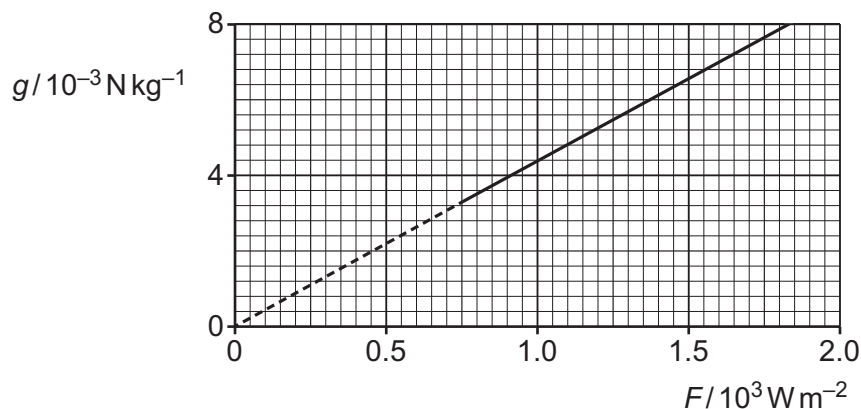


Fig. 2.1

Determine a value for the luminosity L of the Sun. Give a unit with your answer.

$L = \dots\dots\dots$ unit $\dots\dots\dots$ [2]

(iii) Use your answer in (c)(ii) to determine the radius r of the Sun.

$r = \dots\dots\dots$ m [2]

[Total: 12]





3 (a) Define specific latent heat.

.....

.....

..... [2]

(b) A dish containing $7.2 \times 10^{-5} \text{ m}^3$ of a substance rests on a laboratory bench. The substance is initially a liquid of density 710 kg m^{-3} . Atmospheric pressure is $1.0 \times 10^5 \text{ Pa}$.

The liquid is heated at its boiling point so that it completely vaporises. The increase in the internal energy of the substance during this process is 17.6 kJ . The final volume of the vapour is 0.017 m^3 .

(i) Show that the magnitude of the work done on the substance when it vaporises is 1.7 kJ .

[2]

(ii) Use the information in (b)(i) to calculate the thermal energy Q , in kJ , supplied to the substance to cause it to vaporise.

$Q = \dots\dots\dots \text{ kJ}$ [2]

(iii) Use your answer in (b)(ii) to determine a value for the specific latent heat of vaporisation L_v , in kJ kg^{-1} , of the substance.

$L_v = \dots\dots\dots \text{ kJ kg}^{-1}$ [2]





(c) The substance in (b) has a specific latent heat of fusion L_F .

Suggest and explain whether L_F is likely to be less than, the same as, or greater than the answer in (b)(iii).

.....

.....

.....

.....

..... [3]

[Total: 11]





4 (a) State **three** of the basic assumptions of the kinetic theory of gases.

- 1
-
- 2
-
- 3
-

[3]

(b) Explain how molecular movement causes the pressure exerted by a gas.

-
-
-
-
-
-

[3]

(c) Fig. 4.1 shows the variation with thermodynamic temperature T of the mean-square speeds $\langle c^2 \rangle$ for two gases X and Y.

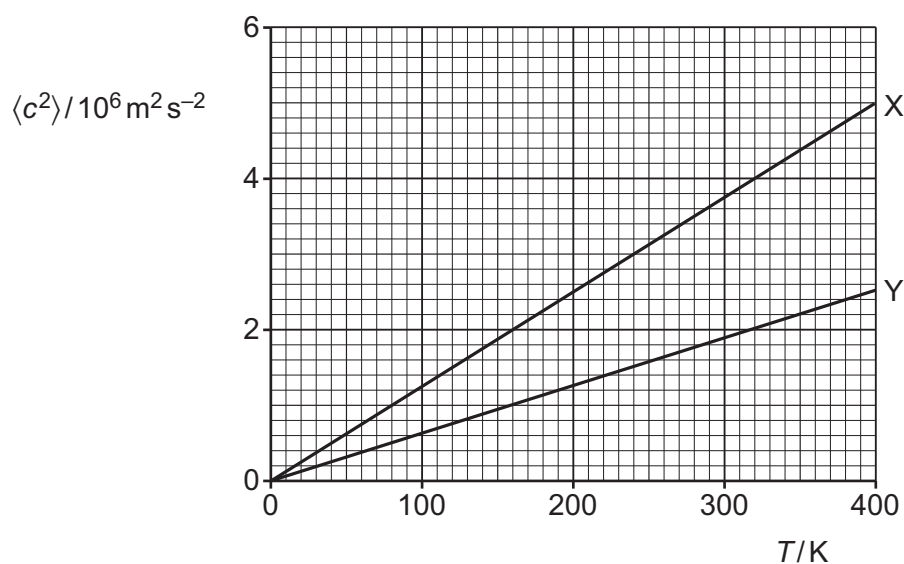


Fig. 4.1



Fig. 4.2 shows the variation with T of the product pV for samples of the two gases, where p is the pressure of the gas and V is the volume of the gas.

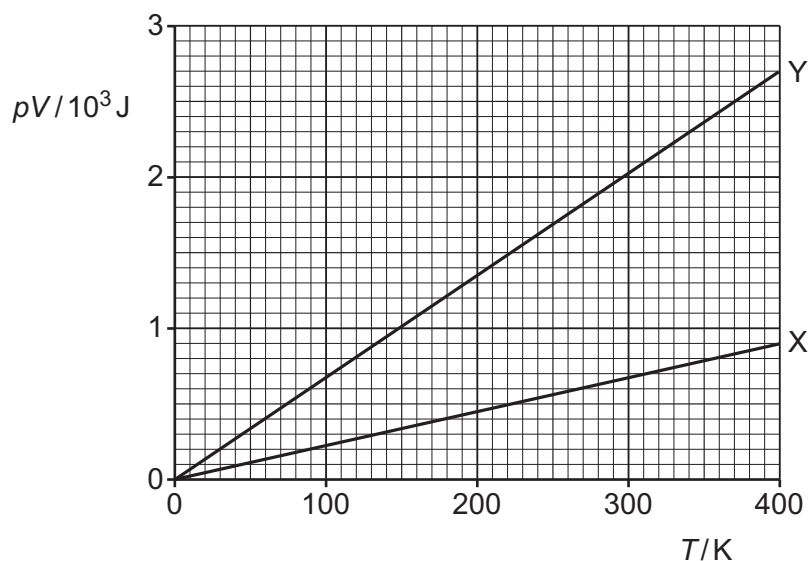


Fig. 4.2

State **three** conclusions about the gases and their samples that may be drawn from Fig. 4.1 and Fig. 4.2. The conclusions may be qualitative or quantitative. Use the space below for any working that you need.

- 1
- 2
- 3

[3]

[Total: 9]

- 5 Fig. 5.1 shows a pendulum consisting of a metal sphere suspended by a thin string.

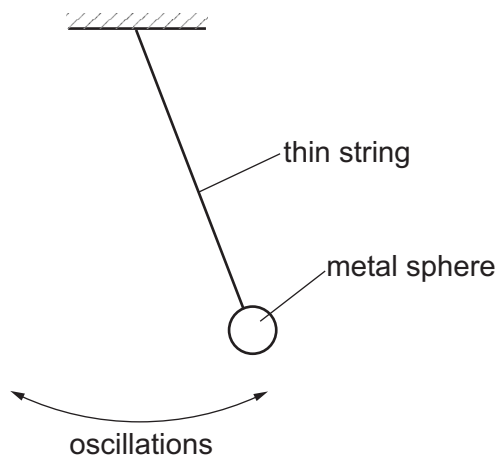


Fig. 5.1 (not to scale)

The sphere undergoes small oscillations about its equilibrium position. The oscillations may be considered to be simple harmonic.

Fig. 5.2 shows the variation with time t of the displacement x of the sphere from its equilibrium position.

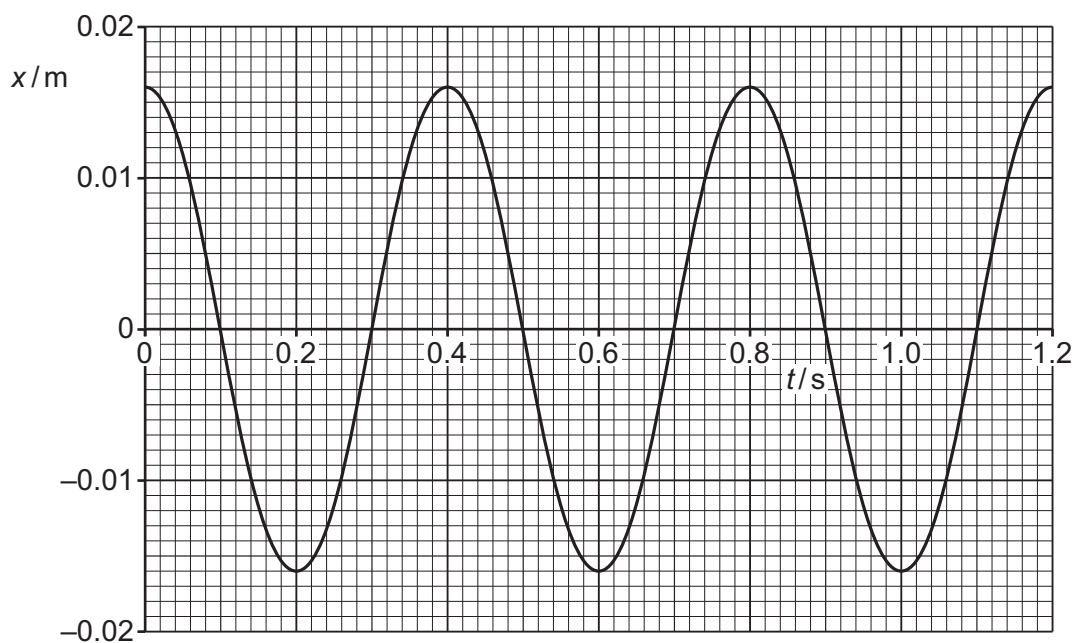


Fig. 5.2

- (a) On Fig. 5.1, draw an arrow, from the centre of the sphere, to represent the direction of the resultant force acting on the sphere when it is in the position shown. [1]



(b) The mass of the sphere is 0.15 kg.

(i) State the amplitude of the oscillations.

amplitude = m [1]

(ii) Determine the angular frequency of the oscillations.

angular frequency = rad s^{-1} [2]

(iii) Calculate the total energy of the oscillations.

total energy = J [2]

(c) On Fig. 5.3, sketch the variation with x of the kinetic energy E_K of the sphere.

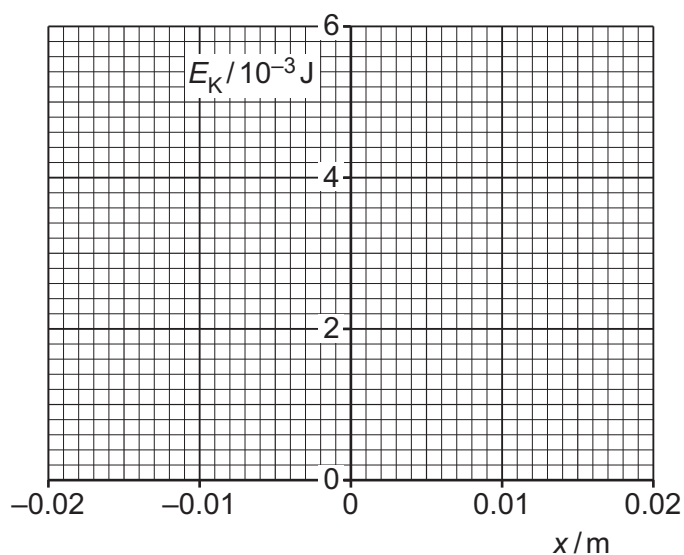


Fig. 5.3

[3]

[Total: 9]





6 (a) State Coulomb's law.

.....

.....

..... [2]

(b) Fig. 6.1 shows an isolated hollow conducting sphere that is positively charged.

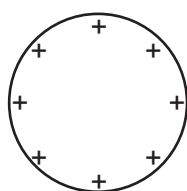


Fig. 6.1

On Fig. 6.1, draw field lines to represent the electric field outside the sphere. [3]

(c) Fig. 6.2 shows the variation of the electric field strength E with distance x from the centre of the sphere in (b).

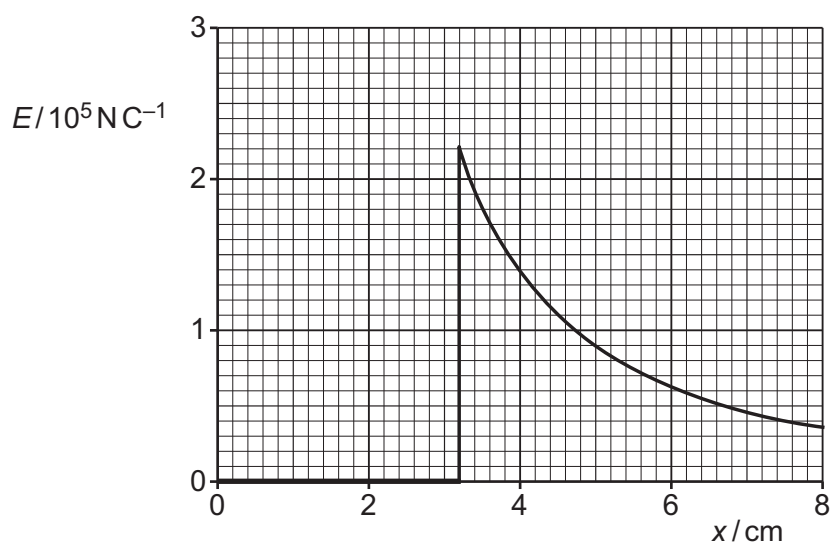


Fig. 6.2





(i) Determine the radius, in cm, of the sphere.

radius = cm [1]

(ii) Calculate the charge on the sphere.

charge = C [3]

(iii) Suggest an explanation for the fact that the electric field inside the sphere is zero.

.....
.....
..... [1]

[Total: 10]





- 7 (a) Define the capacitance of a parallel-plate capacitor.

.....

.....

..... [2]

- (b) An initially uncharged capacitor X, of capacitance C , is gradually charged so that the final potential difference (p.d.) between its plates is V and the final charge is Q .

- (i) On Fig. 7.1, sketch the variation of charge with p.d. for capacitor X as the p.d. increases from 0 to V .

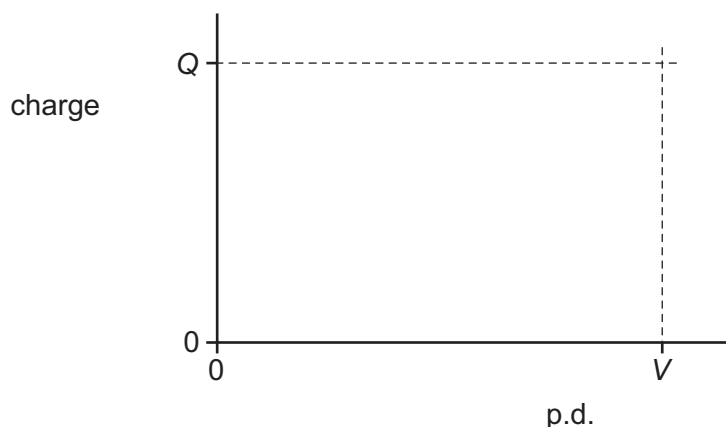


Fig. 7.1

- (ii) Determine an expression, in terms of Q and V , for the work W done on capacitor X during the charging process. Explain your reasoning.

$W =$ [2]





- (c) Another capacitor Y is initially uncharged. The fully charged capacitor X in (b) is now connected to capacitor Y, as shown in Fig. 7.2.

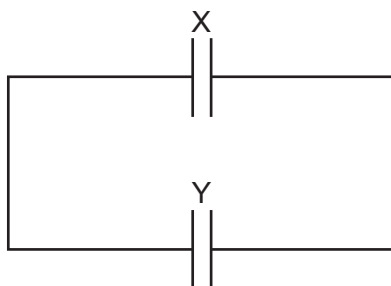


Fig. 7.2

The capacitance of capacitor Y is $3C$.

- (i) Complete Table 7.1 to show expressions, in terms of Q and V , for the final p.d.s across, and the final charges on, the two capacitors.
Use the space below for any working that you need.

Table 7.1

	X	Y
final p.d.		
final charge		

[3]

- (ii) State whether the total energy stored in the two capacitors is less than, the same as, or greater than the energy initially stored in capacitor X.

..... [1]

[Total: 10]





- 8 (a) State what is meant by the frequency of an alternating current.

.....
 [1]

- (b) An alternating current I in a resistor of resistance $680\ \Omega$ varies with time t according to

$$I = 3.5 \sin(40\pi t)$$

where I is in A and t is in s.

- (i) Show that the period of the alternating current is 50 ms.

[1]

- (ii) On Fig. 8.1, sketch the variation of I with t between $t = 0$ and $t = 100$ ms.

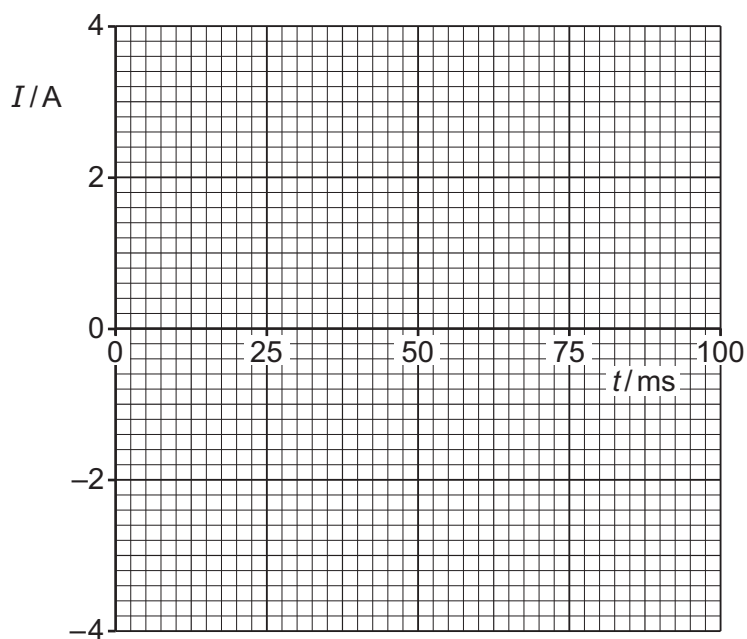


Fig. 8.1

[3]





(iii) Determine the root-mean-square (r.m.s.) current in the resistor.

r.m.s. current =A [1]

(c) Use data from (b), including your answer in (b)(iii), to show by calculation that the mean power in the 680Ω resistor is half of the peak power.

[3]

[Total: 9]





- 9 Electrons in a vacuum are accelerated from rest through a potential difference (p.d.) V to form a beam. The electrons each have mass m and charge q .

The beam is incident on a graphite crystal that acts as a diffraction grating. After passing through the crystal, the beam reaches a fluorescent screen. An interference pattern is observed on this screen.

- (a) Explain what this observation shows about the nature of electrons.

.....

 [1]

- (b) Determine an expression, in terms of m , q and V , for the momentum p of an electron in the beam.

$p =$ [3]

- (c) The p.d. through which the electrons are accelerated is now increased to a greater value.

Describe and explain the effect of this change on the interference pattern observed.

.....

 [2]



- (d) The electrons are now accelerated through different values of V , resulting in pairs of corresponding values for p and the de Broglie wavelength λ .

- (i) On Fig. 9.1, sketch the variation of p with $\frac{1}{\lambda}$.

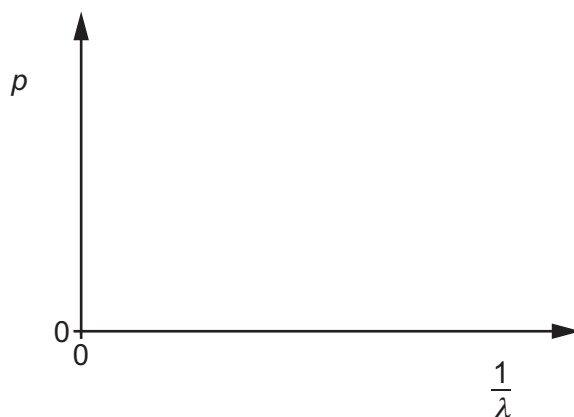


Fig. 9.1

[2]

- (ii) State the name of the quantity represented by the gradient of the line in Fig. 9.1.

..... [1]

[Total: 9]







10 (a) Radioactive decay is both random and spontaneous.

(i) State what is meant by random.

.....
 [1]

(ii) State what is meant by spontaneous.

.....
 [1]

(iii) State **one** piece of evidence for the random nature of decay.

.....
 [1]

(b) (i) Describe the differences between nuclear fission and nuclear fusion.

.....

 [3]

(ii) Explain, with reference to the variation of binding energy per nucleon with nucleon number, why the processes of nuclear fission and nuclear fusion both result in a release of energy.

.....

 [2]

[Total: 8]





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PHYSICS

9702/43

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October/November 2024

2 hours

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Stefan-Boltzmann law

$$L = 4\pi\sigma r^2 T^4$$

Doppler redshift

$$\frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$$





1 (a) State Newton's law of gravitation.

.....

.....

..... [2]

(b) A planet may be considered as a uniform sphere.

A satellite is in circular orbit of period T around the planet at a height h above the surface. The height of the orbit can be adjusted by use of the satellite's rocket engines.

Fig. 1.1 shows the variation with h of $T^{\frac{2}{3}}$.

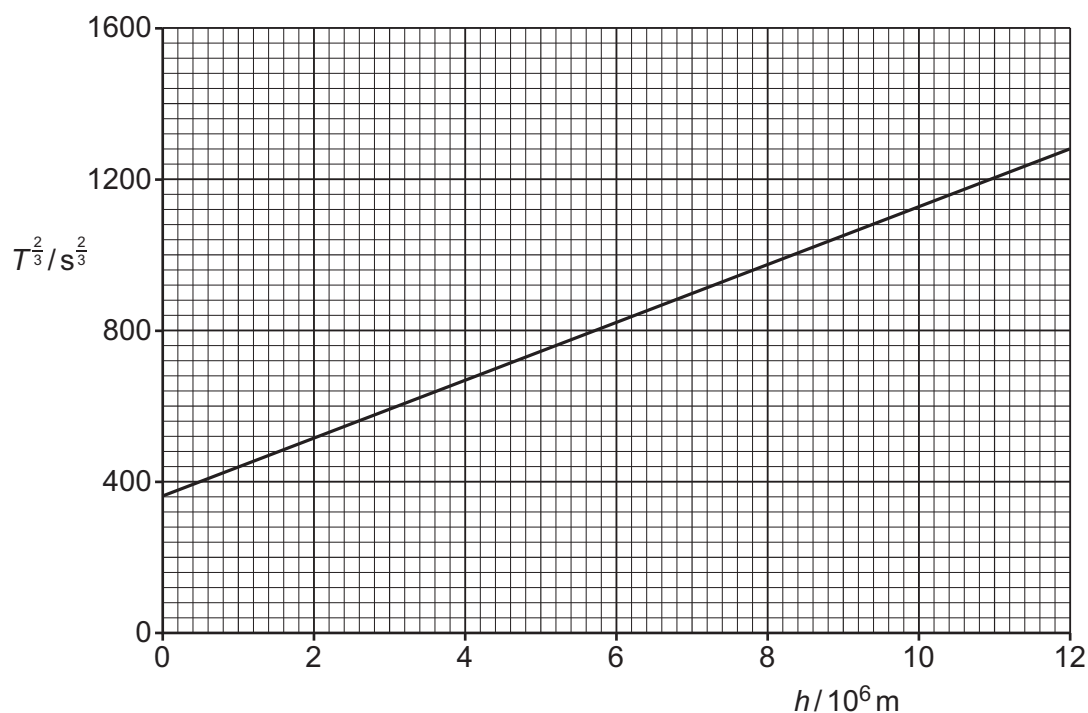


Fig. 1.1

(i) By reference to forces, explain why the orbit of the satellite is circular.

.....

.....

..... [2]





- (ii) Use Newton's law of gravitation to show that h and T are related by

$$(h + B)^3 = \frac{GA}{4\pi^2} T^2$$

where G is the gravitational constant and A and B are constants that depend on the properties of the planet.

[3]

- (iii) Use the gradient and intercept of the line in Fig. 1.1 to determine values for A and B .
Give units with your answers.

$A = \dots\dots\dots$ unit $\dots\dots\dots$

$B = \dots\dots\dots$ unit $\dots\dots\dots$

[5]

[Total: 12]





2 (a) Define specific heat capacity.

.....

.....

..... [2]

(b) Two solid blocks X and Y are made from different metals. The blocks have different initial temperatures. Block Y is initially at room temperature.

The blocks are placed in direct thermal contact with each other at time $t = 0$. Fig. 2.1 shows the variation with t of the temperatures of the two blocks.

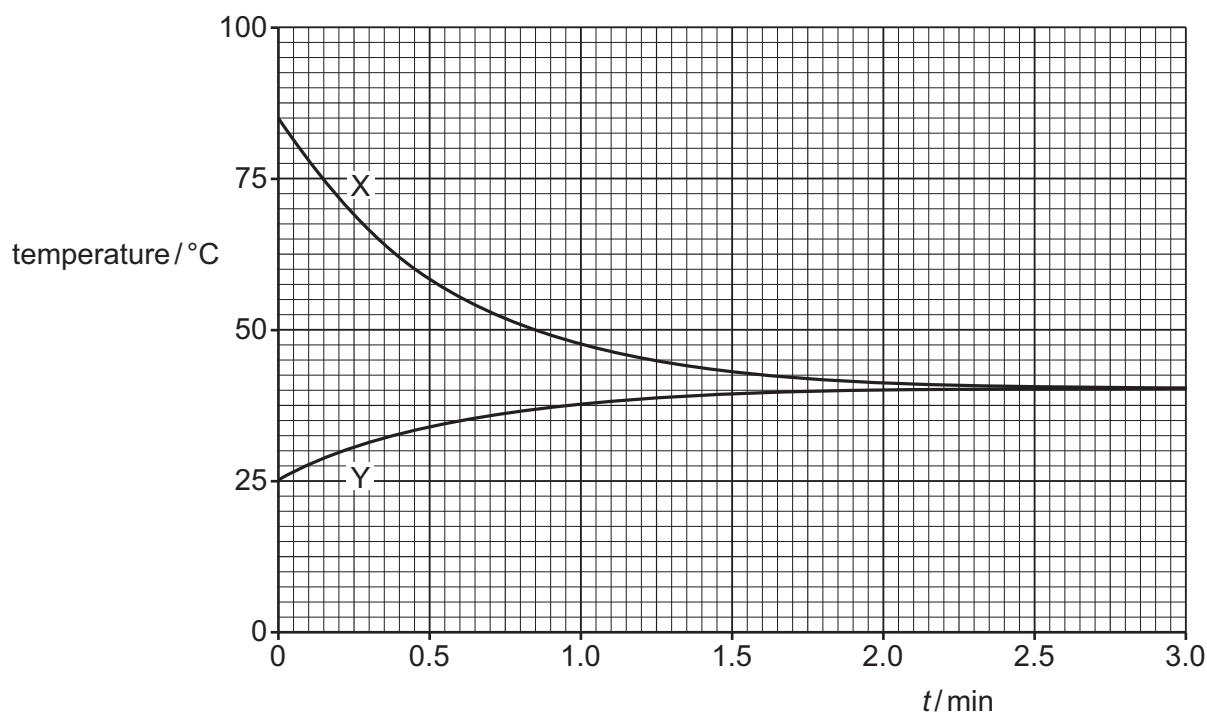


Fig. 2.1





- (i) State **three** conclusions that may be drawn from Fig. 2.1. The conclusions may be qualitative or quantitative.

1

.....

2

.....

3

.....

[3]

- (ii) The ratio $\frac{\text{mass of block Y}}{\text{mass of block X}}$ is equal to 1.3.

The metal in block Y has a specific heat capacity of $901 \text{ J kg}^{-1} \text{ K}^{-1}$.

Determine the specific heat capacity of the metal in block X.

specific heat capacity = $\text{J kg}^{-1} \text{ K}^{-1}$ [3]

[Total: 8]





- 3 (a) (i) State what is meant by the Avogadro constant.

.....

.....

..... [1]

- (ii) State the relationship between the Avogadro constant N_A , the molar gas constant R and the Boltzmann constant k .

[1]

- (b) Two samples X and Y of ideal gases are both at thermodynamic temperature T .

Sample X has volume V and consists of N molecules, each of mass m .

Sample Y has volume $2V$ and consists of $2N$ molecules, each of mass $2m$.

- (i) Complete Table 3.1 by giving expressions, in terms of some or all of N , m , T , V and the constants in (a)(ii), for the quantities indicated.

Table 3.1

	sample X	sample Y
pressure		
amount of substance		
mean-square speed of molecules		
internal energy		

[4]





(ii) The temperature of sample X is now varied.

On Fig. 3.1, sketch the variation with thermodynamic temperature of the root-mean-square (r.m.s.) speed of the molecules of the gas.

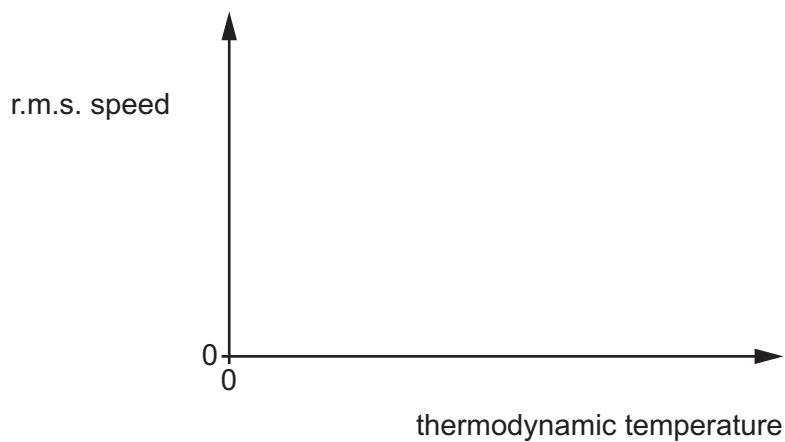


Fig. 3.1

[2]

[Total: 8]





- 4 (a) State what is meant by simple harmonic motion.

.....

.....

..... [2]

- (b) A block is suspended from a spring, as shown in Fig. 4.1.

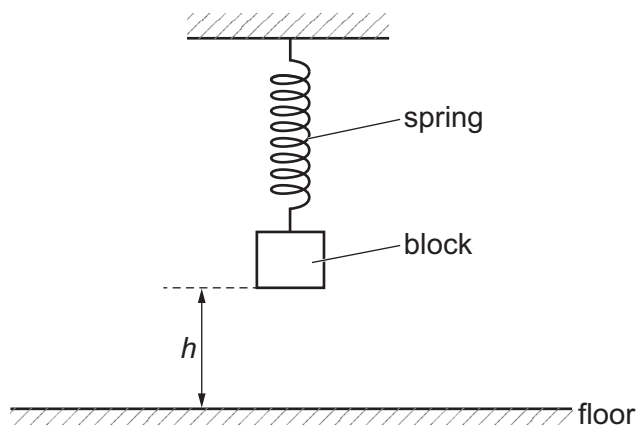


Fig. 4.1

The block is pulled down and released at time $t = 0$. It then oscillates vertically with simple harmonic motion.

Fig. 4.2 shows the variation of the velocity v of the block with height h of the base of the block above the floor.

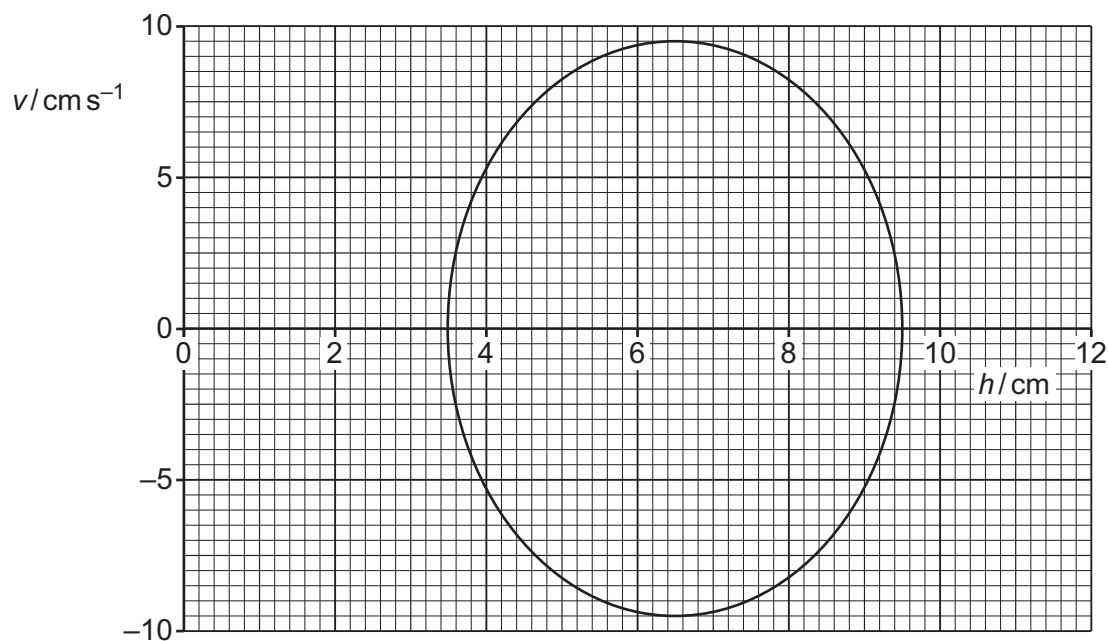


Fig. 4.2



- (i) Determine the amplitude, in cm, of the oscillations.

amplitude = cm [1]

- (ii) Show that the angular frequency of the oscillations is 3.2 rad s^{-1} .

[2]

- (iii) Calculate the period T of the oscillations.

$T = \dots\dots\dots$ s [2]

- (iv) On Fig. 4.3, sketch the variation of h with time t from $t = 0$ to $t = 6.0 \text{ s}$.

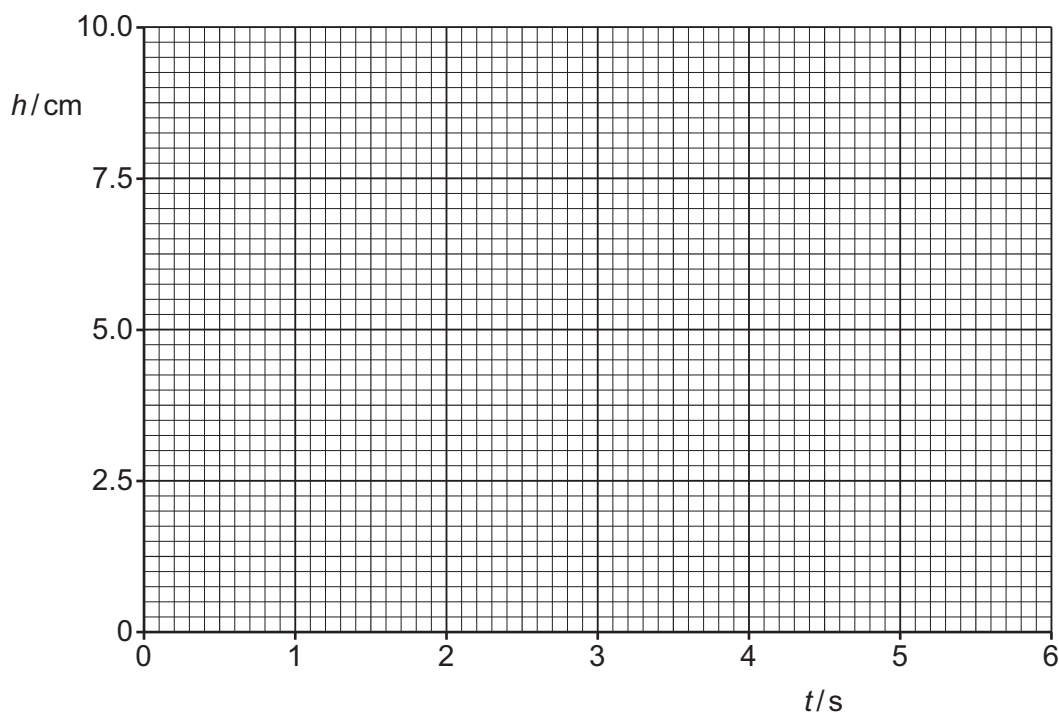


Fig. 4.3

[4]

[Total: 11]



- 5 (a) State the relationship between electric field and electric potential.

.....

.....

..... [2]

- (b) Two charged isolated insulating spheres X and Y are near to each other, as shown in Fig. 5.1.



Fig. 5.1

P is a point on the line joining the centres of the spheres.

Explain why it is **not** possible for the total electric potential and the resultant electric field to simultaneously be zero at point P.

.....

.....

.....

.....

..... [3]

- (c) The magnitudes of the charges on spheres X and Y in Fig. 5.1 are Q and $2Q$ respectively. The spheres may be considered as point charges at their centres.

Point P is a distance x from the centre of sphere X.

The electric potential at point P is zero.

- (i) Show that the distance y of point P from the centre of sphere Y is equal to $2x$.

[2]





- (ii) State an expression, in terms of Q , x and the permittivity of free space ϵ_0 , for the electric field strength E_x at P due to sphere X.

$$E_x = \dots\dots\dots [1]$$

- (iii) Determine an expression, in terms of Q , x and ϵ_0 , for the resultant electric field strength E at point P due to the two spheres.

$$E = \dots\dots\dots [2]$$

[Total: 10]





- 6 (a) (i) State what is meant by rectification of an alternating voltage.

.....
 [1]

- (ii) State the difference between half-wave rectification and full-wave rectification.

.....

 [2]

- (b) (i) Complete Fig. 6.1 to show a circuit that produces half-wave rectification of an alternating input voltage V_{IN} to produce output voltage V_{OUT} across the resistor R .

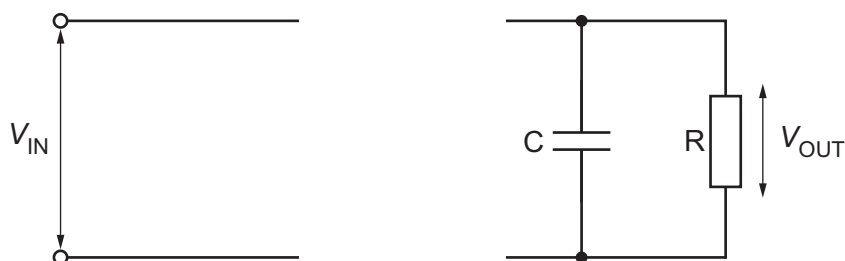


Fig. 6.1

[2]

- (ii) State the purpose of the capacitor C in the circuit of Fig. 6.1.

.....
 [1]

- (c) The input voltage V_{IN} in Fig. 6.1 is a square wave. Fig. 6.2 shows the variation of V_{IN} with time t .

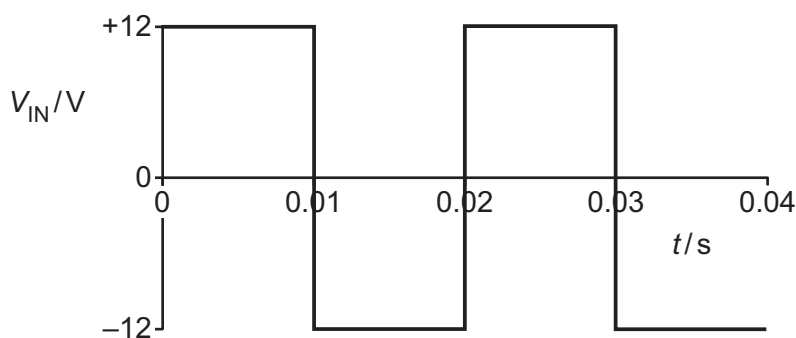


Fig. 6.2



Fig. 6.3 shows the variation of V_{OUT} with t .

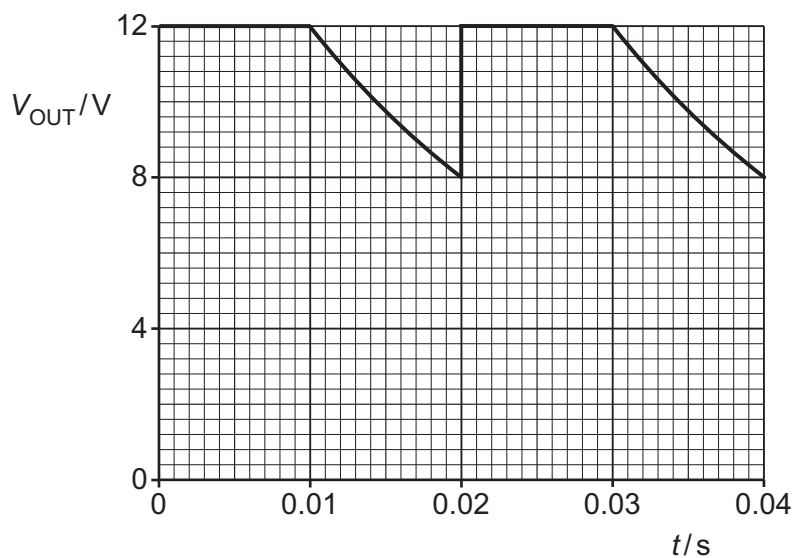


Fig. 6.3

The maximum energy stored in the capacitor is 0.041 J.

- (i) Show that the capacitance of C is $570 \mu\text{F}$.

[2]

- (ii) Determine the resistance of R .

resistance = Ω [3]

[Total: 11]





7 (a) Define magnetic flux density.

.....

.....

..... [2]

(b) A long, straight wire carries a current into the page, as shown in Fig. 7.1.



Fig. 7.1

On Fig. 7.1, draw four field lines to represent the magnetic field around the wire due to the current in it. [3]

(c) Two identical wires X and Y are placed parallel to each other. The wires both carry current into the page, as shown in Fig. 7.2.

X



Y



Fig. 7.2





- (i) Explain why the two wires exert a magnetic force on each other.

.....

.....

.....

..... [2]

- (ii) On Fig. 7.2, draw an arrow to show the direction of the magnetic force exerted on wire X. Label your arrow F. [1]

- (iii) The current in X is double the current in Y.

State how the magnetic force exerted on wire Y compares with the magnetic force exerted on wire X.

.....

.....

..... [2]

- (iv) The direction of the current in both wires is now reversed.

State, with a reason, the effect of this change on the direction of the force on wire X.

.....

..... [1]

[Total: 11]





- 8 A polished sheet of magnesium in a vacuum emits electrons when it is illuminated by ultraviolet radiation.

(a) State the name of this phenomenon.

..... [1]

- (b) For emission of electrons to occur, the frequency of the ultraviolet radiation must be at least 8.8×10^{14} Hz.

(i) Calculate the work function energy of magnesium.

work function energy = J [2]

- (ii) For ultraviolet radiation with a frequency of 11×10^{14} Hz, calculate the maximum speed of the emitted electrons.

maximum speed = ms^{-1} [3]



- (c) The frequency f of the ultraviolet radiation incident on the magnesium sheet is varied between $8.0 \times 10^{14} \text{ Hz}$ and $11 \times 10^{14} \text{ Hz}$.

On Fig. 8.1, sketch the variation with f of the maximum kinetic energy E_{MAX} of the emitted electrons. Use the space below for any working that you need.

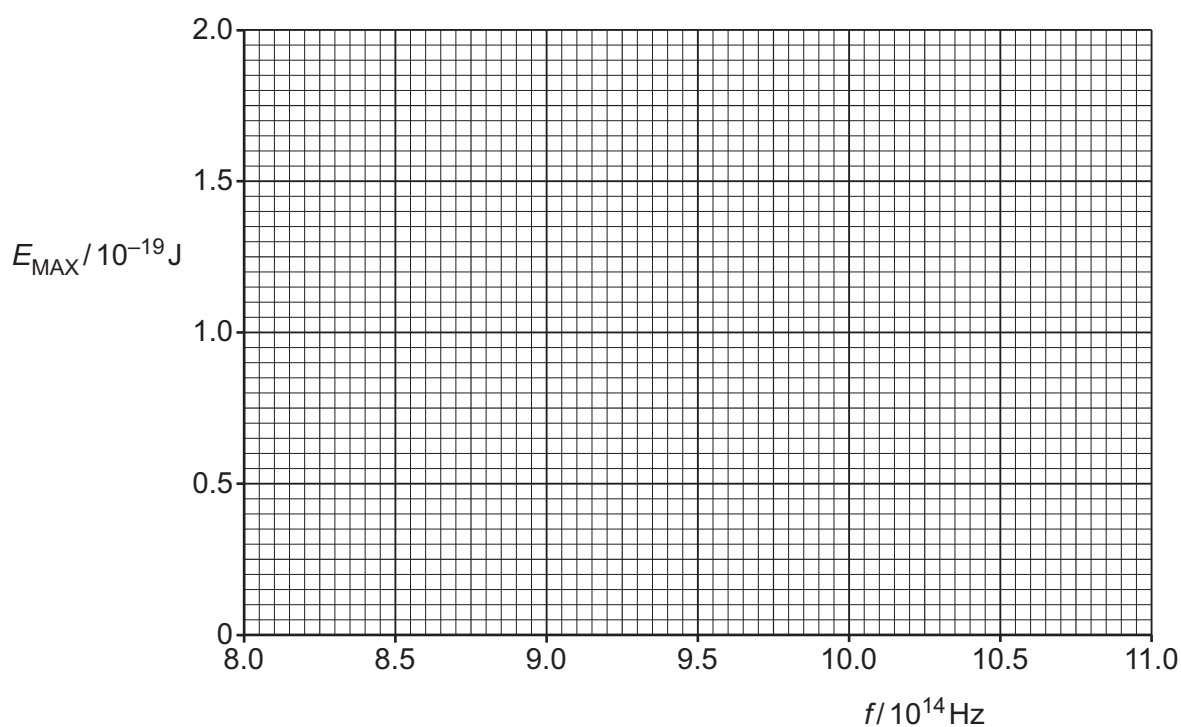


Fig. 8.1

[3]

[Total: 9]





9 Fluorine-18 ($^{18}_{9}\text{F}$) decays by beta-plus (β^+) emission with a half-life of 110 minutes.

(a) (i) State the name of the beta-plus particle.

..... [1]

(ii) Show that the decay constant of fluorine-18 is $1.05 \times 10^{-4} \text{ s}^{-1}$.

[1]

(iii) Determine the activity of $2.1 \times 10^{-12} \text{ kg}$ of fluorine-18.

activity = Bq [3]

(b) A small sample of fluorine-18 injected into the body acts as a tracer for use in medical imaging.

(i) Describe how the interaction of a β^+ particle with an electron in the body enables the formation of an image.

.....

 [3]

(ii) Suggest why 110 minutes is a suitable half-life for a nuclide used as a tracer in medical diagnosis.

.....

 [2]

[Total: 10]





- 10 (a) Explain how redshift leads to the idea that the Universe is expanding.

.....

.....

.....

.....

..... [3]

- (b) Stars in a distant galaxy emit radiation. The total luminosity of the stars in the galaxy is $1.90 \times 10^{36} \text{ W}$.

The emission spectrum of the radiation contains a line X at a wavelength of 658 nm.

Radiation from the galaxy is observed on the Earth. The observed radiation has a radiant flux intensity of $8.42 \times 10^{-16} \text{ W m}^{-2}$. In the observed emission spectrum, line X is at a wavelength of 726 nm.

Determine:

- (i) the distance d of the galaxy from the Earth

$$d = \dots\dots\dots \text{ m [2]}$$

- (ii) the speed v of the galaxy relative to the Earth.

$$v = \dots\dots\dots \text{ ms}^{-1} [2]$$



- (c) Observations of many galaxies, such as the one in (b), lead to many pairs of values of d and v . Plotting these values reveals a trend.

- (i) On Fig. 10.1, sketch the variation of v with d .

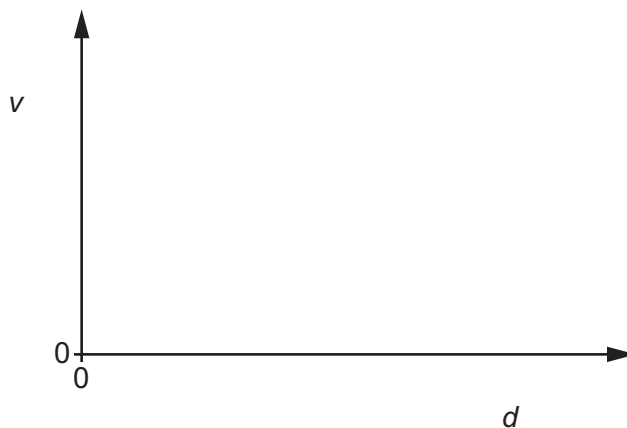


Fig. 10.1

[2]

- (ii) State the name of the quantity represented by the gradient of the line in Fig. 10.1.

..... [1]

[Total: 10]





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Cambridge International AS & A Level

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PHYSICS

9702/51

Paper 5 Planning, Analysis and Evaluation

October/November 2024

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 30.
- The number of marks for each question or part question is shown in brackets [].

This document has 8 pages.

- 1 A thin cylindrical bar magnet of length L and cross-sectional area A is attached to a block. An identical magnet is attached to a trolley, as shown in Fig. 1.1.

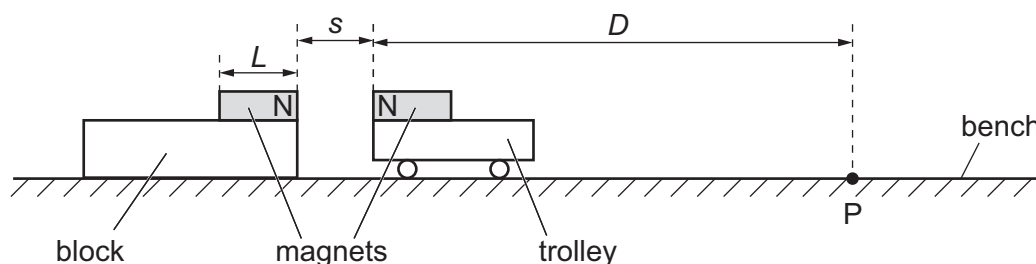


Fig. 1.1

The trolley is held so that the separation of the N poles of the two magnets is s .

Point P is a distance D from the N pole of the magnet on the stationary trolley.

The trolley is released. The speed v of the trolley at point P is determined using one light gate.

It is suggested that v is related to s by the relationship

$$\frac{mv^2}{2D} = \frac{KA^2B^2L^2}{s^4} - Q$$

where B is the magnetic flux density at the N pole of one of the magnets, m is the mass of the trolley, and K and Q are constants.

Plan a laboratory experiment to test the relationship between v and s .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for K and Q .

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.





Diagram

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[15]

- 2 A student investigates an electrical circuit. A power supply of electromotive force (e.m.f.) E_s and negligible internal resistance is connected in series to three resistors, each of resistance Z .

A cell, an ammeter and a resistor of resistance R are connected in parallel across one of these resistors, as shown in Fig. 2.1.

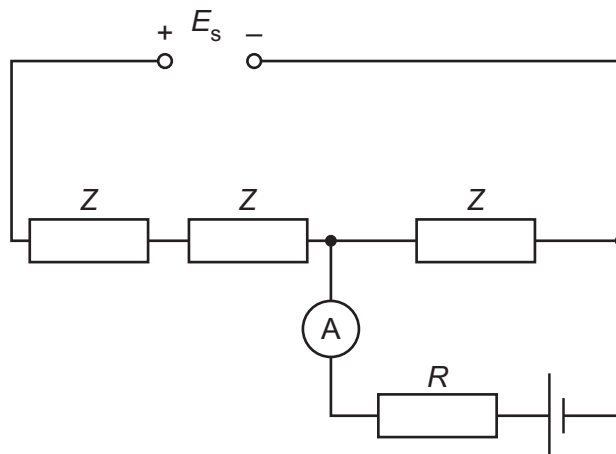


Fig. 2.1

The current I is measured by the ammeter for different values of R .

It is suggested that I and R are related by the equation

$$3E - E_s = I(3R + 2Z)$$

where E is the e.m.f. of the cell.

- (a) A graph is plotted of $\frac{1}{I}$ on the y -axis against R on the x -axis.

Determine expressions for the gradient and y -intercept.

gradient =

y -intercept =

[1]





(b) Values of R and I are given in Table 2.1.

Table 2.1

$R/\text{k}\Omega$	$I/\mu\text{A}$	$\frac{1}{I}/\text{A}^{-1}$
1.50	194 ± 2	
1.75	180 ± 2	
1.92	172 ± 2	
2.22	160 ± 2	
2.48	150 ± 2	
2.72	144 ± 2	

Calculate and record values of $\frac{1}{I}/\text{A}^{-1}$ in Table 2.1.

Include the absolute uncertainties in $\frac{1}{I}$.

[2]

(c) (i) Plot a graph of $\frac{1}{I}/\text{A}^{-1}$ against $R/\text{k}\Omega$. Include error bars for $\frac{1}{I}$.

[2]

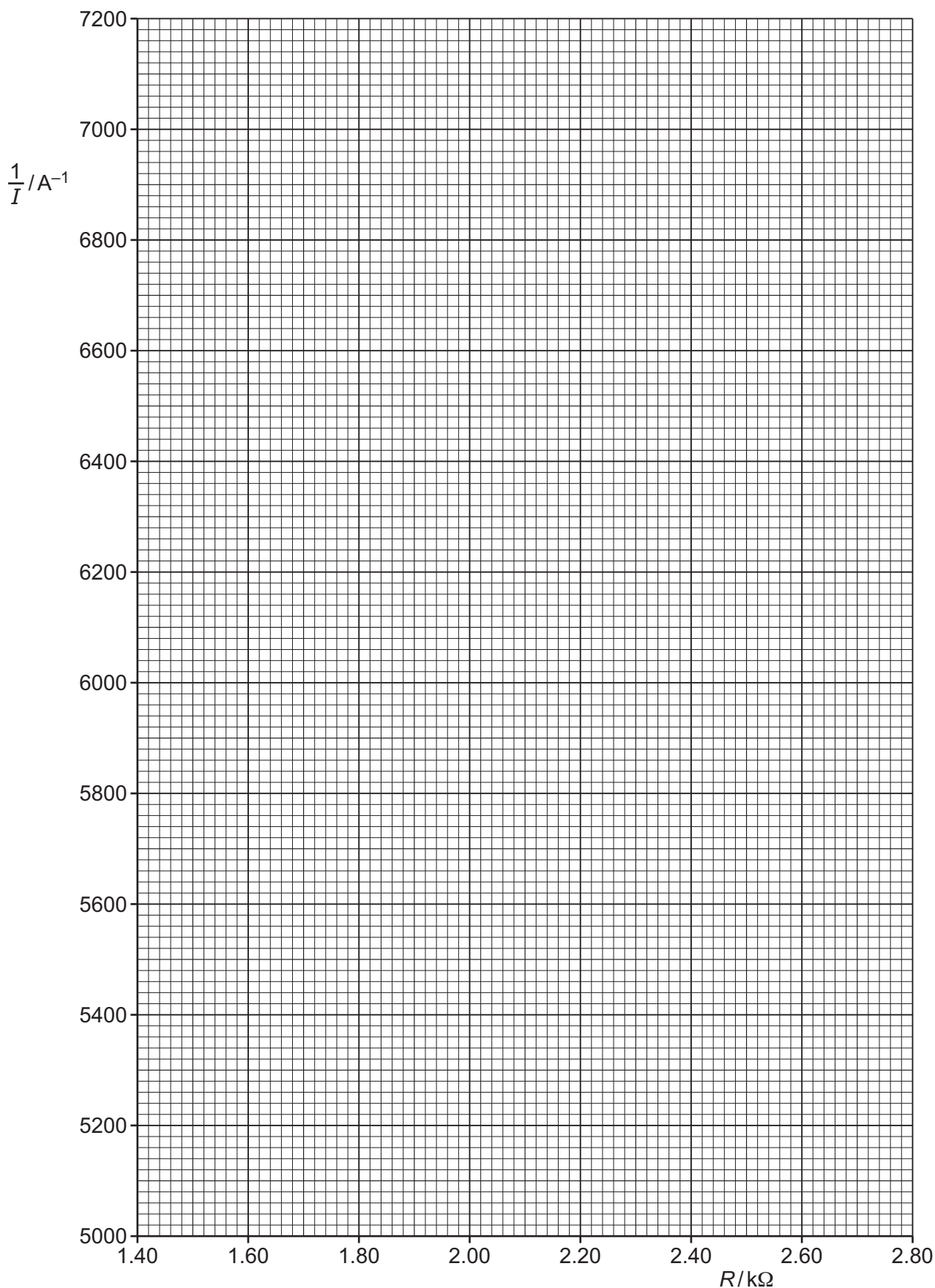
(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines.

[2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]







- (iv) Determine the y -intercept of the line of best fit. Include the absolute uncertainty in your answer.

y -intercept = [2]

- (d) (i) Using your answers to (a), (c)(iii) and (c)(iv), determine the values of E and Z . Include appropriate units.

Data: $E_s = (2.20 \pm 0.05) \text{ V}$

$E = \dots\dots\dots$

$Z = \dots\dots\dots$

[2]

- (ii) Determine the absolute uncertainty in E .

absolute uncertainty in $E = \dots\dots\dots$ [1]

- (e) The experiment is repeated. Determine the resistance R that gives a value of I of $250 \mu\text{A}$.

$R = \dots\dots\dots \Omega$ [1]

[Total: 15]





Cambridge International AS & A Level

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PHYSICS

9702/52

Paper 5 Planning, Analysis and Evaluation

October/November 2024

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
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- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

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- The number of marks for each question or part question is shown in brackets [].

This document has 8 pages.



1 Fig. 1.1 shows a coil made from resistance wire.

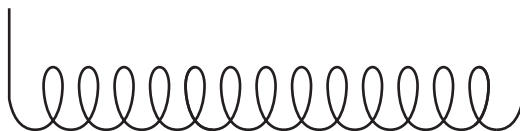


Fig. 1.1

The coil is placed in cooking oil of mass m . The total length of the resistance wire in the oil is L .

A potential difference V is applied to the coil. The temperature of the oil increases by $\Delta\theta$ in time t .

It is suggested that $\Delta\theta$ is related to L by the relationship

$$\frac{AtV^2}{L} = mK\Delta\theta + Z$$

where A is the cross-sectional area of the wire, and K and Z are constants.

Plan a laboratory experiment to test the relationship between $\Delta\theta$ and L .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for K and Z .

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.





Diagram

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[illegible]

- 2 A student investigates the refraction of white light entering a transparent rectangular block. A narrow beam of light enters the block at the midpoint of one of the shorter sides. The angle of incidence θ is measured, as shown in Fig. 2.1.

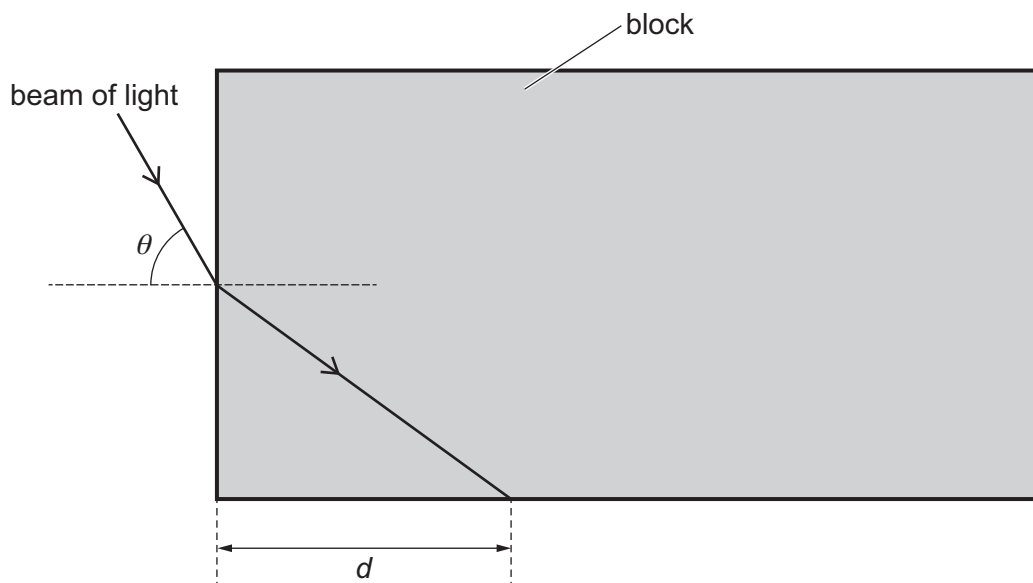


Fig. 2.1 (not to scale)

The distance d between the corner of the block and the point where the beam of light touches the boundary of the block is measured.

The experiment is repeated for different values of θ .

It is suggested that d and θ are related by the equation

$$\frac{B}{B + d^2} = \frac{\sin^2 \theta}{n^2}$$

where B and n are constants.

- (a) A graph is plotted of d^2 on the y -axis against $\frac{1}{\sin^2 \theta}$ on the x -axis.

Determine expressions for the gradient and y -intercept.

gradient =

y -intercept =

[1]





- (b) Values of θ , $\frac{1}{\sin^2 \theta}$ and d are given in Table 2.1.

Table 2.1

$\theta / ^\circ$	$\frac{1}{\sin^2 \theta}$	d / cm	d^2 / cm^2
28.5	4.39	24.8 ± 0.2	
33.5	3.28	21.4 ± 0.2	
42.5	2.19	17.1 ± 0.2	
50.0	1.70	14.7 ± 0.2	
57.5	1.41	12.9 ± 0.2	
63.5	1.25	11.8 ± 0.2	

Calculate and record values of d^2 / cm^2 in Table 2.1.
Include the absolute uncertainties in d^2 .

[2]

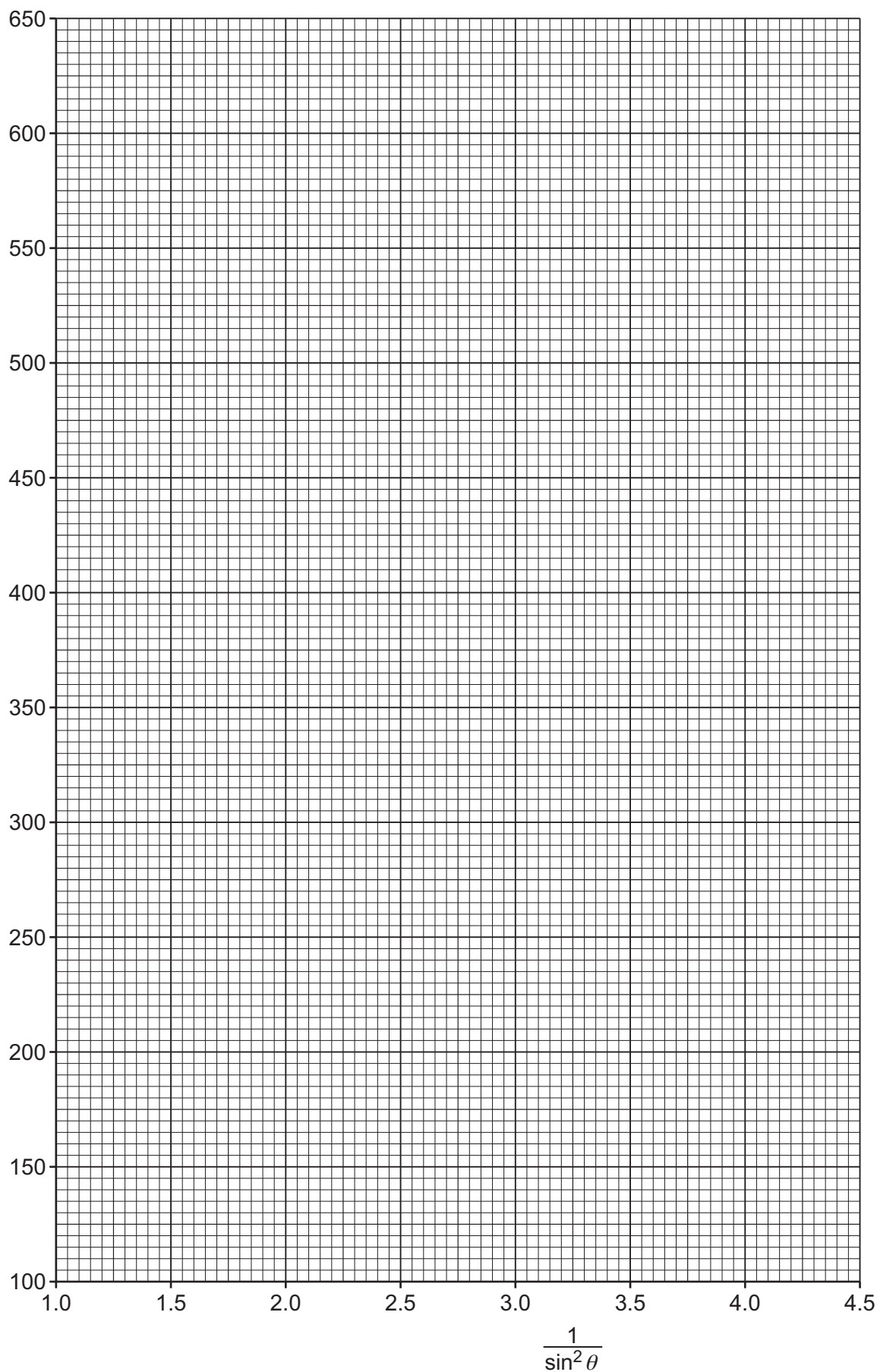
- (c) (i) Plot a graph of d^2 / cm^2 against $\frac{1}{\sin^2 \theta}$. Include error bars for d^2 . [2]
- (ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines. [2]
- (iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]





d^2/cm^2





- (iv) Determine the y -intercept of the line of best fit. Include the absolute uncertainty in your answer.

y -intercept = [2]

- (d) (i) Using your answers to (a), (c)(iii) and (c)(iv), determine the values of B and n . Include appropriate units.

B =

n = [2]

- (ii) Determine the percentage uncertainty in n .

percentage uncertainty in n = % [1]

- (e) The experiment is repeated. Determine the angle θ that gives a value of d of 30.0 cm.

θ = ° [1]

[Total: 15]





Cambridge International AS & A Level

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PHYSICS

9702/53

Paper 5 Planning, Analysis and Evaluation

October/November 2024

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
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- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

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This document has 8 pages.

- 1 A thin cylindrical bar magnet of length L and cross-sectional area A is attached to a block. An identical magnet is attached to a trolley, as shown in Fig. 1.1.

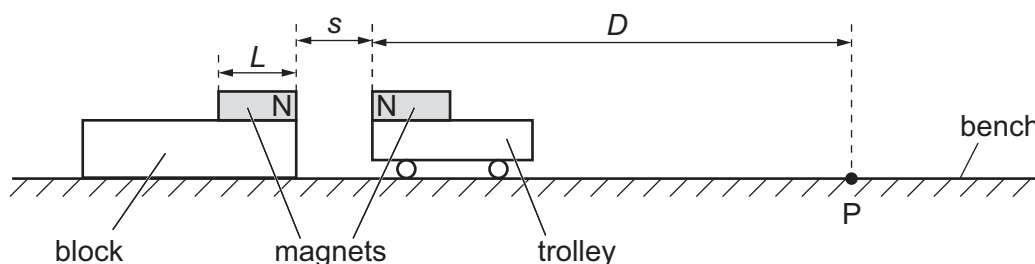


Fig. 1.1

The trolley is held so that the separation of the N poles of the two magnets is s .

Point P is a distance D from the N pole of the magnet on the stationary trolley.

The trolley is released. The speed v of the trolley at point P is determined using one light gate.

It is suggested that v is related to s by the relationship

$$\frac{mv^2}{2D} = \frac{KA^2B^2L^2}{s^4} - Q$$

where B is the magnetic flux density at the N pole of one of the magnets, m is the mass of the trolley, and K and Q are constants.

Plan a laboratory experiment to test the relationship between v and s .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for K and Q .

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.





Diagram

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[15]

- 2 A student investigates an electrical circuit. A power supply of electromotive force (e.m.f.) E_s and negligible internal resistance is connected in series to three resistors, each of resistance Z .

A cell, an ammeter and a resistor of resistance R are connected in parallel across one of these resistors, as shown in Fig. 2.1.

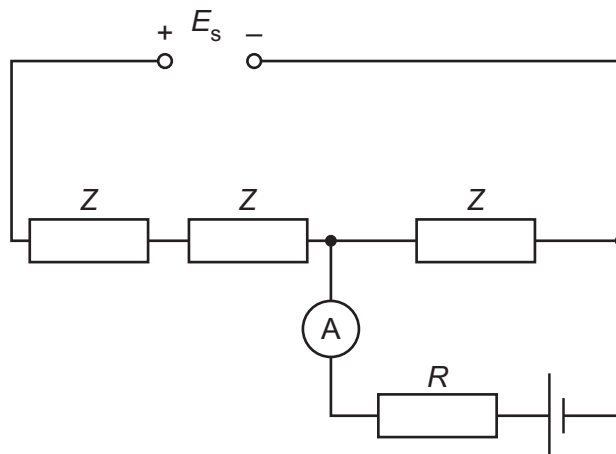


Fig. 2.1

The current I is measured by the ammeter for different values of R .

It is suggested that I and R are related by the equation

$$3E - E_s = I(3R + 2Z)$$

where E is the e.m.f. of the cell.

- (a) A graph is plotted of $\frac{1}{I}$ on the y -axis against R on the x -axis.

Determine expressions for the gradient and y -intercept.

gradient =

y -intercept =

[1]





(b) Values of R and I are given in Table 2.1.

Table 2.1

$R/\text{k}\Omega$	$I/\mu\text{A}$	$\frac{1}{I}/\text{A}^{-1}$
1.50	194 ± 2	
1.75	180 ± 2	
1.92	172 ± 2	
2.22	160 ± 2	
2.48	150 ± 2	
2.72	144 ± 2	

Calculate and record values of $\frac{1}{I}/\text{A}^{-1}$ in Table 2.1.

Include the absolute uncertainties in $\frac{1}{I}$.

[2]

(c) (i) Plot a graph of $\frac{1}{I}/\text{A}^{-1}$ against $R/\text{k}\Omega$. Include error bars for $\frac{1}{I}$.

[2]

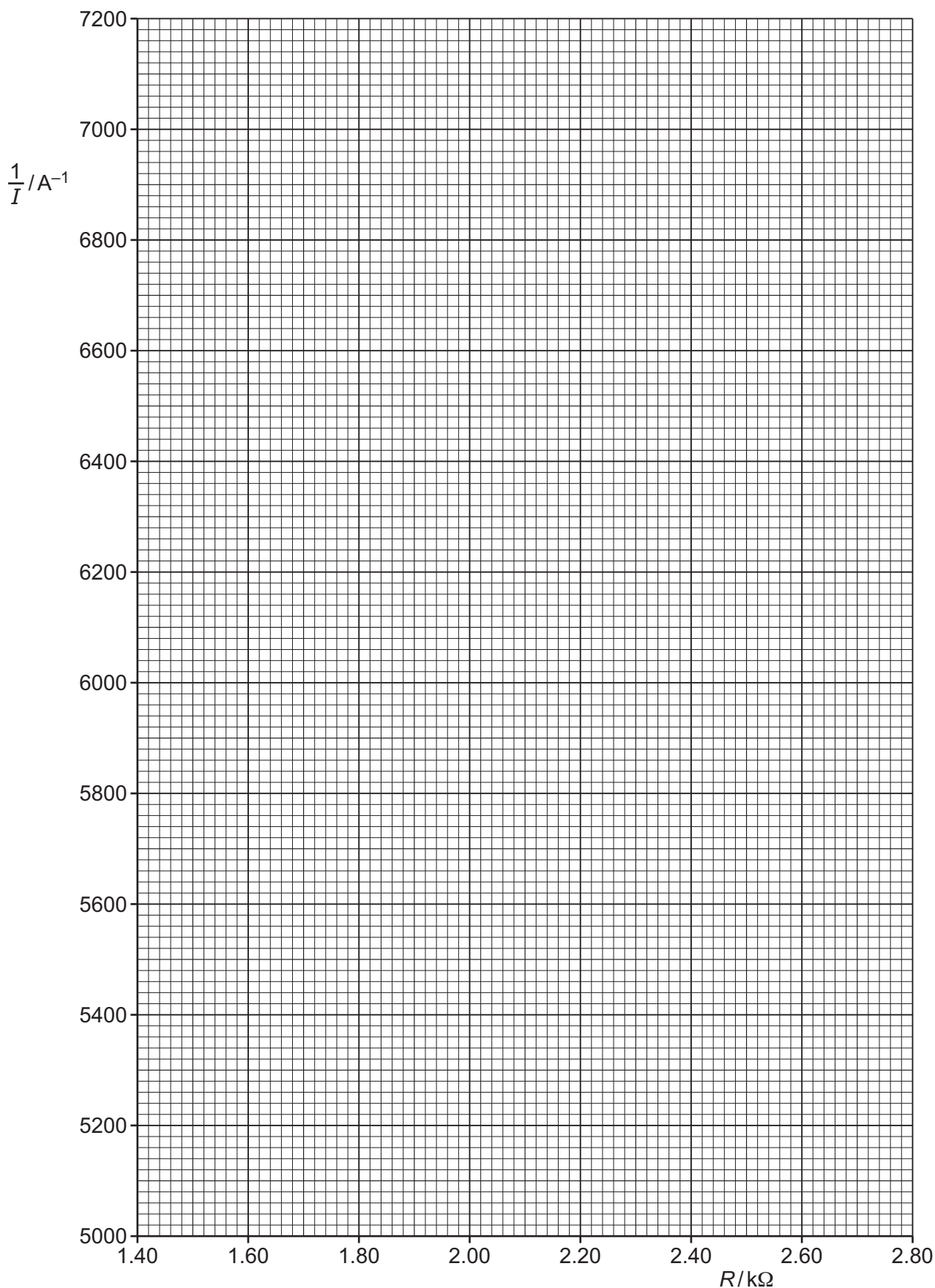
(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines.

[2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]







- (iv) Determine the y -intercept of the line of best fit. Include the absolute uncertainty in your answer.

y -intercept = [2]

- (d) (i) Using your answers to (a), (c)(iii) and (c)(iv), determine the values of E and Z . Include appropriate units.

Data: $E_s = (2.20 \pm 0.05) \text{ V}$

$E = \dots\dots\dots$

$Z = \dots\dots\dots$ [2]

- (ii) Determine the absolute uncertainty in E .

absolute uncertainty in $E = \dots\dots\dots$ [1]

- (e) The experiment is repeated. Determine the resistance R that gives a value of I of $250 \mu\text{A}$.

$R = \dots\dots\dots \Omega$ [1]

[Total: 15]





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PHYSICS

9702/41

Paper 4 A Level Structured Questions

May/June 2025

2 hours

You must answer on the question paper.

No additional materials are needed.

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- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

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This document has **24** pages. Any blank pages are indicated.



Data

acceleration of free fall	$g = 9.81 \text{ m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
unified atomic mass unit	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27} \text{ kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$ $(\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ m F}^{-1})$
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_o = \frac{f_s v}{v \pm v_s}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \dots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$





gravitational potential

$$\phi = -\frac{GM}{r}$$

gravitational potential energy

$$E_P = -\frac{GMm}{r}$$

pressure of an ideal gas

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

simple harmonic motion

$$a = -\omega^2 x$$

velocity of particle in s.h.m.

$$v = v_0 \cos \omega t$$

$$v = \pm \omega \sqrt{(x_0^2 - x^2)}$$

electric potential

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

electrical potential energy

$$E_P = \frac{Qq}{4\pi\epsilon_0 r}$$

capacitors in series

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$$

capacitors in parallel

$$C = C_1 + C_2 + \dots$$

discharge of a capacitor

$$x = x_0 e^{-\frac{t}{RC}}$$

Hall voltage

$$V_H = \frac{BI}{ntq}$$

alternating current/voltage

$$x = x_0 \sin \omega t$$

radioactive decay

$$x = x_0 e^{-\lambda t}$$

decay constant

$$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$$

intensity reflection coefficient

$$\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$$

Stefan–Boltzmann law

$$L = 4\pi\sigma r^2 T^4$$

Doppler redshift

$$\frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$$



- 1 (a) Define gravitational potential at a point.

.....

.....

..... [2]

- (b) Mars is a planet that may be considered to be an isolated uniform sphere of radius 3.4×10^6 m.

A satellite of mass 122 kg is in orbit around Mars at a constant height of 1.7×10^6 m above the surface of the planet.

The height of the orbit is increased to 6.8×10^6 m above the surface. This increases the gravitational potential energy of the satellite by 5.1×10^8 J.

- (i) Show that the mass of Mars is 6.4×10^{23} kg.

[3]

- (ii) Calculate the gravitational potential ϕ at the surface of Mars. Give a unit with your answer.

$\phi =$ unit [2]



(c) The satellite in (b) is moved to an orbit in which the satellite remains at the same point above the surface of Mars.

(i) The orbit has a period of 25 hours.

State what can be deduced from this about the rotation of Mars on its axis.

.....
..... [1]

(ii) State **one** other feature of this orbit.

.....
..... [1]

[Total: 9]



- 2 A helium atom may be modelled as a nucleus surrounded by two electrons in diametrically opposite circular orbits, each of radius 170 pm, as shown in Fig. 2.1.

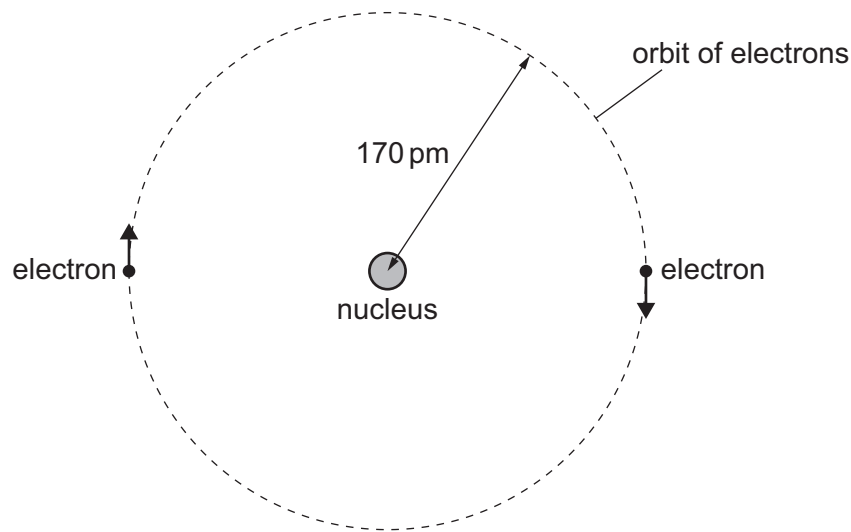


Fig. 2.1

- (a) State Coulomb's law.

.....

 [2]

- (b) (i) State the charge on the nucleus, in terms of the elementary charge e .

charge = e [1]

- (ii) Show that the electric force between the nucleus and one of the electrons is $1.6 \times 10^{-8} \text{ N}$.

[1]



(c) Assume that the force in (b)(ii) is the only force on the electrons.

(i) Calculate the speed of the orbiting electrons.

speed = ms^{-1} [2]

(ii) Calculate the period of the orbit of the electrons.

period = s [2]

(d) In practice, the orbit of each electron is affected by the presence of the other electron.

(i) For the position of one of the electrons, determine the ratio

$$\frac{\text{electric field strength due to the other electron}}{\text{electric field strength due to the nucleus}}.$$

ratio = [2]

(ii) Use your answer in (d)(i) to suggest and explain how the orbit of the electron is affected by the presence of the other electron.

.....
 [1]

[Total: 11]



- 3 (a) Define specific latent heat.

.....

 [2]

- (b) Explain why, for a substance, the specific latent heat of vaporisation is usually greater than the specific latent heat of fusion.

.....

 [3]

- (c) An ice cube of mass 37.0g at temperature 0.0°C is placed in a beaker containing water of mass 208g at temperature 26.4°C.

When all the ice has melted, and all the water in the beaker has reached thermal equilibrium, the final temperature of all the water is 10.3°C.

The specific heat capacity of water is $4.18 \text{ J g}^{-1} \text{ }^{\circ}\text{C}^{-1}$.

The beaker has negligible specific heat capacity and is perfectly insulated from the surroundings.

Determine a value, to three significant figures, for the specific latent heat of fusion of water.

specific latent heat of fusion = J g^{-1} [4]

[Total: 9]



4 (a) (i) State what is meant by the internal energy of a system.

.....

.....

..... [2]

(ii) Explain why the internal energy of an ideal gas is directly proportional to the thermodynamic temperature of the gas.

.....

.....

.....

..... [2]

(b) A sample of an ideal gas at thermodynamic temperature T has internal energy U .

The gas is compressed so that its temperature increases to $3T$.
During this compression, work W is done on the gas.

The gas is then cooled at constant volume so that its temperature decreases to $2T$.

Complete Table 4.1 to show, in terms of some or all of W , T and U , the work done on the gas, the thermal energy supplied to the gas and the increase in internal energy of the gas for each of the two processes.

Table 4.1

	work done on gas	thermal energy supplied to gas	increase in internal energy of gas
compression	$+W$		
cooling			

[4]

[Total: 8]



- 5 A cuboidal block floats in a liquid with its base horizontal, as shown in Fig. 5.1.

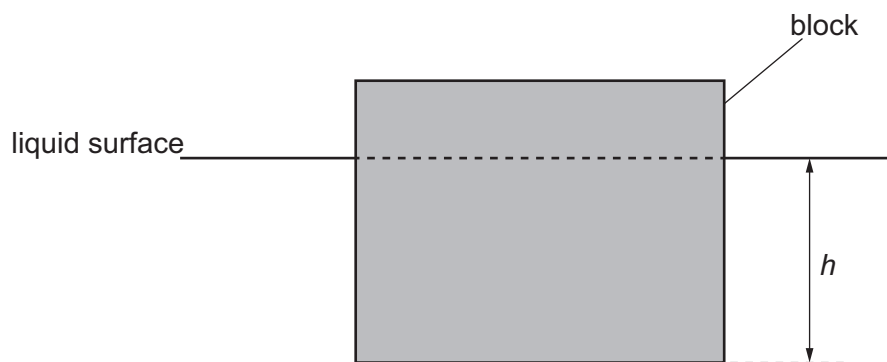


Fig. 5.1

The base of the block is at a depth h below the surface of the liquid.

The block is displaced downwards by a small distance and then released so that it oscillates.

Fig. 5.2 shows the variation with h of the acceleration a of the block.

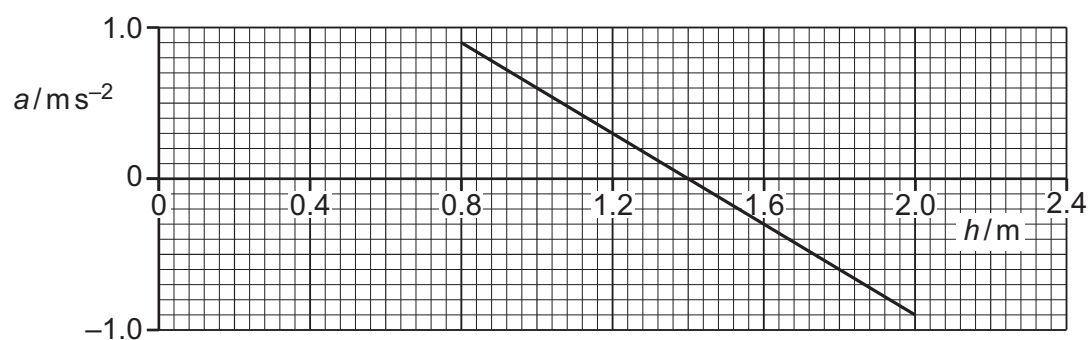


Fig. 5.2

Fig. 5.3 shows the variation with h of the kinetic energy E_K of the block.

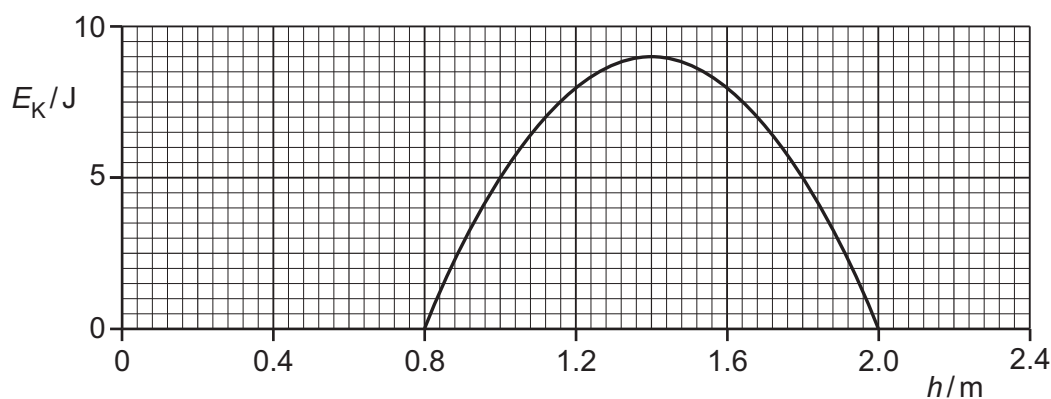


Fig. 5.3



- (a) (i) Determine the amplitude of the oscillations.

amplitude = m [1]

- (ii) State what the line in Fig. 5.2 shows about the nature of the oscillations.

..... [1]

- (b) State **three** other quantitative conclusions that can be drawn from Fig. 5.2 and Fig. 5.3 about the block and its oscillations. Use the space for any working.

1

.....

2

.....

3

.....

[3]

- (c) On Fig. 5.4, sketch the variation with h of the potential energy E_p of the oscillations.

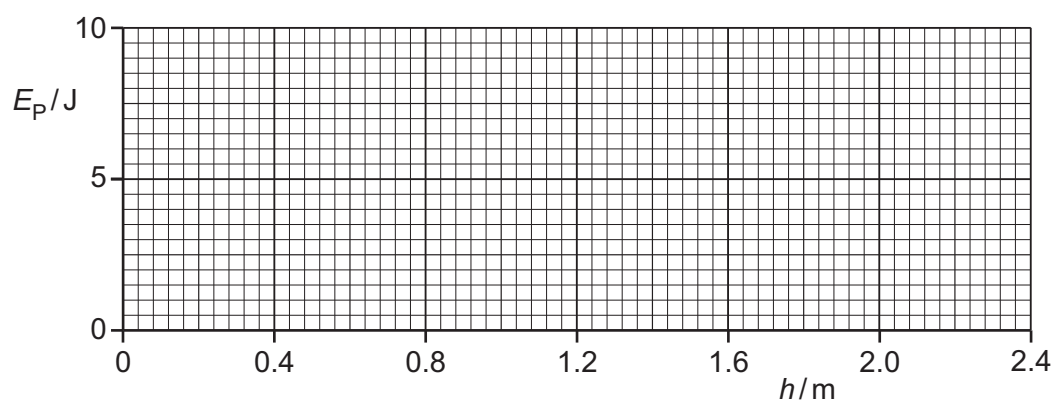


Fig. 5.4

[3]

[Total: 8]



- 6 Fig. 6.1 shows a circuit that rectifies an alternating input voltage V_{IN} and produces an output voltage V_{OUT} across a resistor R .

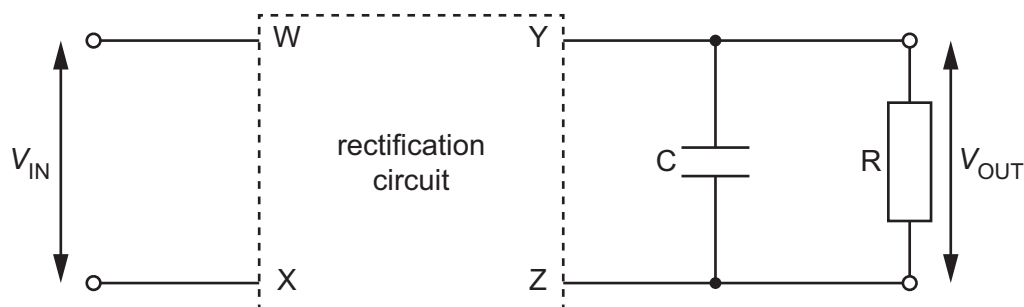


Fig. 6.1

The four terminals of the rectification circuit are labelled W, X, Y and Z.
A capacitor C is connected in parallel with resistor R .

- (a) (i) State what is meant by rectification.

.....
..... [1]

- (ii) State the purpose of capacitor C .

.....
..... [1]

- (b) Fig. 6.2 shows the variations with time t of the potential differences (p.d.s) V_{IN} and V_{OUT} .

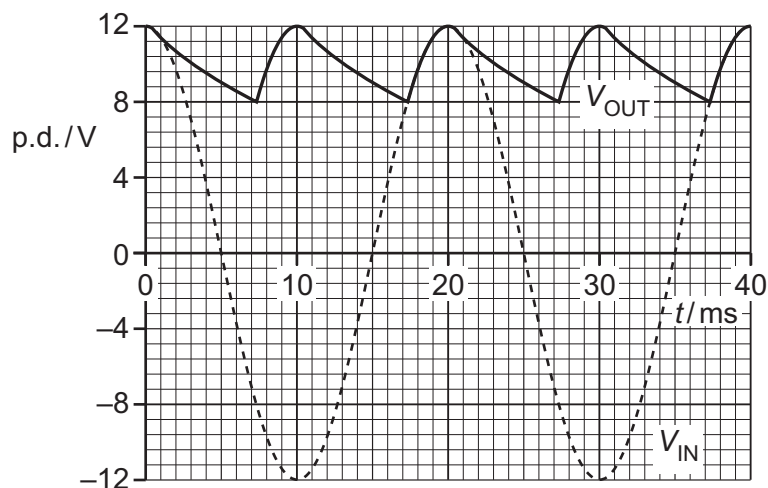


Fig. 6.2



- (i) The variation of V_{IN} with t can be represented by

$$V_{\text{IN}} = A \cos Bt$$

where A and B are constants.

Determine the values of A and B . Give a unit with your answer for A .

$$A = \dots\dots\dots \text{unit} \dots\dots\dots$$

$$B = \dots\dots\dots \text{rad s}^{-1} \quad [2]$$

- (ii) Determine the type of rectification produced by the circuit in Fig. 6.1.

..... [1]

- (iii) On Fig. 6.3, draw the circuit diagram for the components inside the rectification circuit.

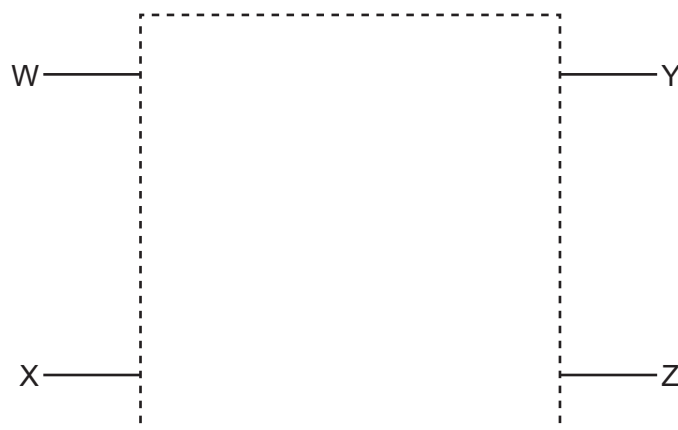


Fig. 6.3

[2]

- (iv) Determine a value for the time constant for the discharge of the capacitor C through the resistor R in Fig. 6.1.

time constant = s [3]



- (c) The capacitor C has a capacitance of $570\ \mu\text{F}$.

Use your answer in (b)(iv) to determine the resistance of resistor R.

resistance = Ω [2]

[Total: 12]

- 7 (a) Define magnetic flux density.

.....

 [2]

- (b) A particle of mass m and charge $+Q$ moves at speed v into a region where there is a uniform magnetic field, as shown in Fig. 7.1.

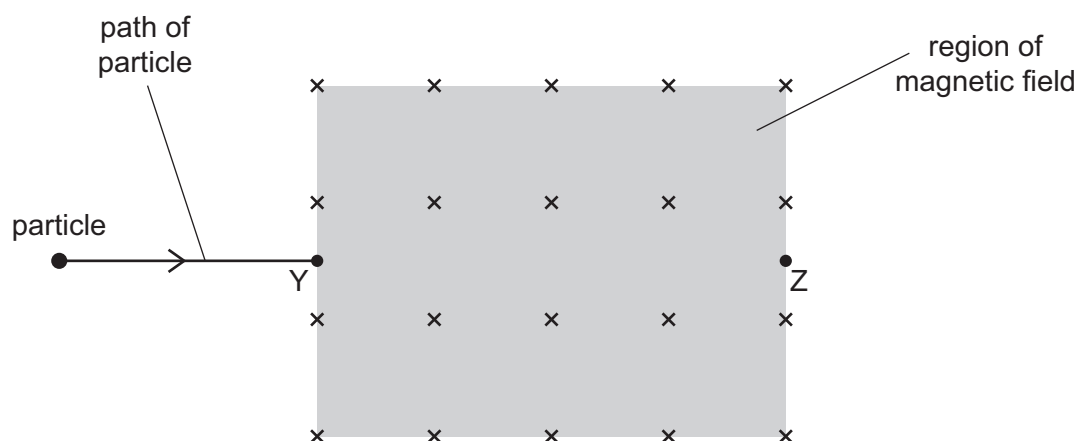


Fig. 7.1

The uniform magnetic field is into the page and has flux density B . The particle enters the region of the field at point Y.



- (i) State an expression, in terms of some or all of m , Q , B and v , for the magnetic force F that acts on the particle when it is at point Y.

$$F = \dots\dots\dots [1]$$

- (ii) On Fig. 7.1, draw an arrow at point Y to indicate the direction of the force in **(b)(i)**. [1]
- (iii) On Fig. 7.1, draw a line to show a possible path for the particle through the region of the magnetic field. [1]
- (c) (i) Explain how an electric field can be used with the magnetic field to ensure that the particle in **(b)** now passes through point Z.

.....

.....

.....

.....

..... [3]

- (ii) Derive an expression for v in terms of B and the electric field strength E .

$$v = \dots\dots\dots [2]$$

[Total: 10]



- 8 (a) State what is meant by the de Broglie wavelength.

.....
..... [1]

- (b) Calculate the de Broglie wavelength of an electron moving at a speed of $4.9 \times 10^7 \text{ m s}^{-1}$.

wavelength = m [2]

- (c) State **one** similarity and **one** difference between an electron and a positron.

similarity:
.....
difference:
..... [2]

- (d) An electron moving at a speed of $4.9 \times 10^7 \text{ m s}^{-1}$ collides with a positron that is travelling at the same speed in the opposite direction. As a result of the collision, two gamma-ray photons are produced.

- (i) State the name of this type of reaction.

..... [1]

- (ii) State what happens to the electron and to the positron.

.....
.....
..... [2]



- (iii) Explain why two gamma-ray photons are produced, rather than just one.

.....
..... [1]

- (iv) Show that the kinetic energy of the electron before the collision is $1.1 \times 10^{-15} \text{ J}$.

[1]

- (v) Use the information in (d)(iv) to determine, to three significant figures, the wavelength associated with the gamma radiation emitted in the collision.

wavelength = m [3]

[Total: 13]



- 9 (a) Define activity of a radioactive sample.

.....
 [1]

- (b) Explain why the variation with time of the activity of a radioactive sample is exponential in nature.

.....

 [3]

- (c) A sample contains a single radioactive isotope that decays to form a stable isotope.

The sample has an activity of 180 Bq at time $t = 0$.

At a time 8.4 minutes later, the activity is 120 Bq.

- (i) Determine the decay constant, in min^{-1} , of the radioactive isotope.

decay constant = min^{-1} [2]

- (ii) Use your answer in (c)(i) to determine the half-life, in min, of the radioactive isotope.

half-life = min [1]



- (iii) On Fig. 9.1, sketch the variation of the activity A of the sample with t for values of t between $t = 0$ and $t = 24$ min.

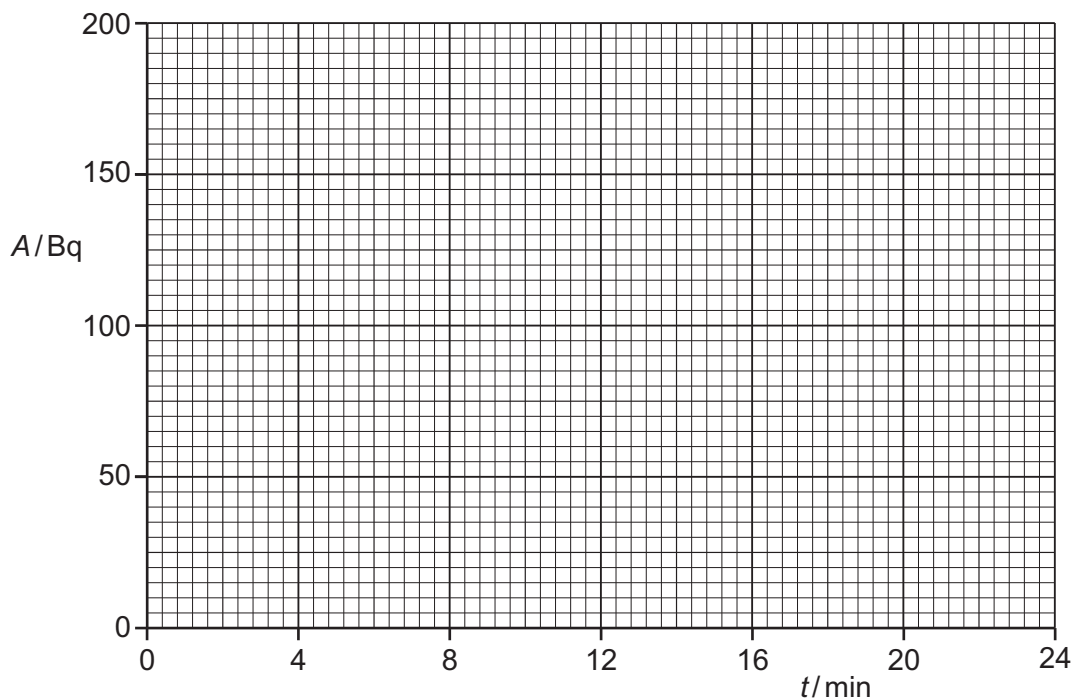


Fig. 9.1

[3]

[Total: 10]



10 (a) State Hubble's law.

.....

.....

..... [2]

(b) A star in a distant galaxy emits radiation that has a maximum intensity of emission at a wavelength of $4.62 \times 10^{-7} \text{ m}$.

Observations of the galaxy made on the Earth detect the maximum intensity of emission from the star at a wavelength of $4.91 \times 10^{-7} \text{ m}$.

(i) Explain why the observed wavelength and the emitted wavelength have different values.

.....

.....

..... [2]

(ii) Calculate the speed of the star relative to the Earth.

speed = m s^{-1} [2]

(iii) The wavelength of maximum intensity of emission is used to determine a value for the surface temperature of the star.

Explain how the temperature determined using the observed wavelength compares with the true value of temperature determined using the emitted wavelength.

.....

.....

..... [2]





(c) A value for the Hubble constant is $2.3 \times 10^{-18} \text{ s}^{-1}$.

Use your answer in (b)(ii) to determine the distance of the star in (b) from the Earth.

distance =m [2]

[Total: 10]









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PHYSICS

9702/42

Paper 4 A Level Structured Questions

May/June 2025

2 hours

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gravitational potential energy

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Stefan-Boltzmann law

$$L = 4\pi\sigma r^2 T^4$$

Doppler redshift

$$\frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$$



- 1 (a) Define the radian.

.....
 [1]

- (b) The rear wheel and the pedals of a bicycle are connected by a chain that passes around two cogs (toothed wheels), as shown in Fig. 1.1.

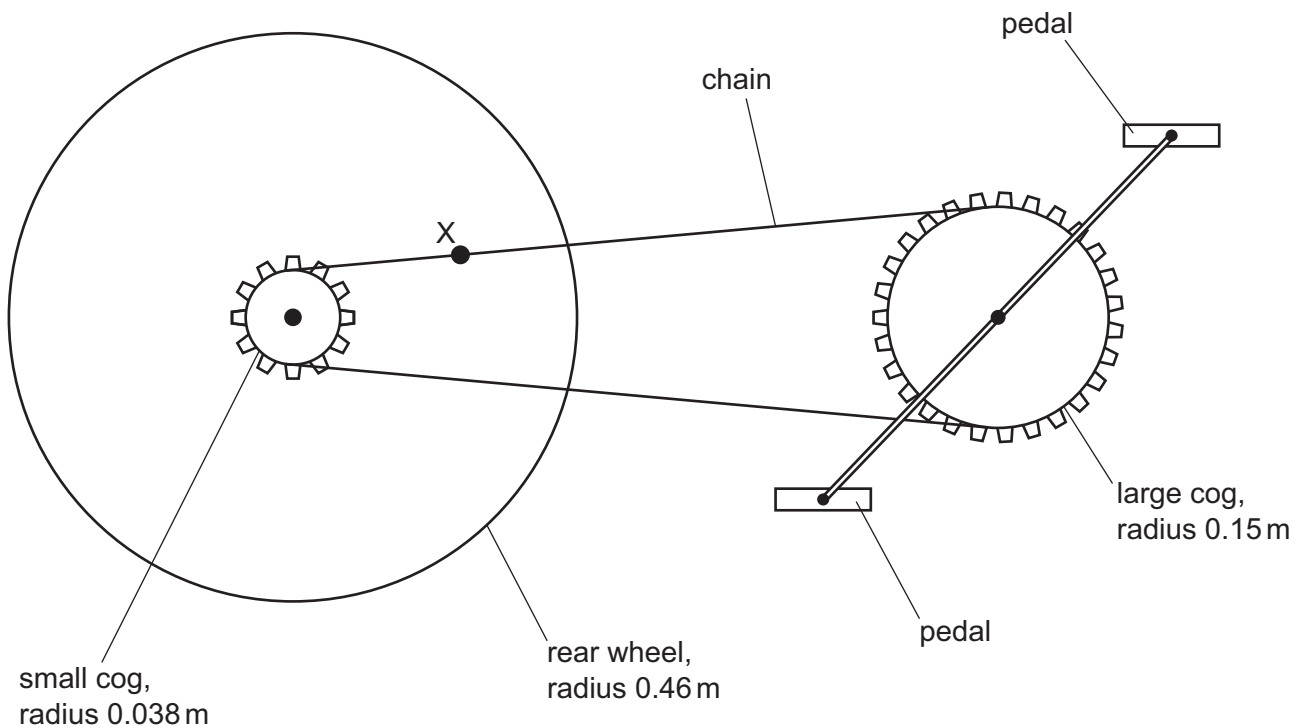


Fig. 1.1 (not to scale)

The small cog has a radius of 0.038 m and is fixed to the rear wheel so that it rotates with it. The large cog has a radius of 0.15 m and is fixed to the pedals so that it rotates with them. The rear wheel has a radius of 0.46 m.

The bicycle is being pedalled so that it moves in a straight line at a constant speed of 17 m s^{-1} .

- (i) Calculate the angular speed of the rear wheel.

angular speed = rad s^{-1} [2]



- (ii) Calculate the period of rotation of the small cog.

period = s [2]

- (iii) Show that the distance moved by point X on the chain during one full rotation of the small cog is 0.24 m.

[1]

- (iv) Use the information in (b)(iii) to determine the angle through which the large cog rotates during one full rotation of the small cog.

angle = rad [2]

- (c) The chain of the bicycle in (b) is moved onto a smaller cog fixed to the rear wheel. The speed of the bicycle does not change.

Explain, without calculation, the effect of this change on the angular speed of the pedals.

.....

 [2]

[Total: 10]



- 2 (a) (i) State what is represented by a gravitational field line.

.....

.....

..... [2]

- (ii) The Earth may be considered as a uniform sphere, as shown in Fig. 2.1.

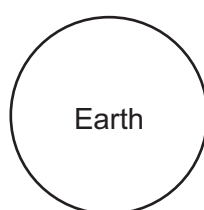


Fig. 2.1

On Fig. 2.1, draw field lines to represent the Earth's gravitational field outside the Earth. [2]

- (b) The Earth's magnetic field may be considered as being due to the Earth acting as a long solenoid, as shown in Fig. 2.2.

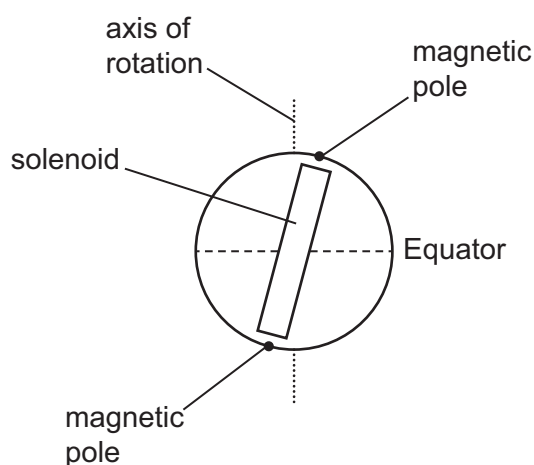


Fig. 2.2

The magnetic poles do not align with the geographic poles, which are on the axis of rotation.



Fig. 2.3 is a copy of Fig. 2.2 without the labels but with two magnetic field lines shown.

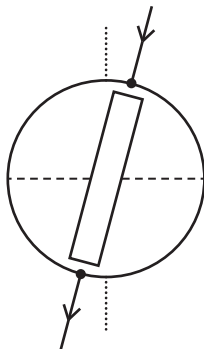


Fig. 2.3

- (i) On Fig. 2.3, label the magnetic poles with the letters N and S to indicate which one is the magnetic N pole and which one is the magnetic S pole. [1]
- (ii) On Fig. 2.3, draw field lines to represent the Earth's magnetic field outside the Earth. [2]
- (c) An observer moves around the surface of the Earth.

- (i) Use your answer in (a)(ii) to explain why the observed gravitational field of the Earth does not vary around the surface.

.....

.....

..... [2]

- (ii) With reference to your answer in (b)(ii), describe how the observed magnetic field of the Earth varies around the surface.

.....

.....

.....

.....

..... [3]

[Total: 12]



- 3 (a) Define specific heat capacity.

.....

.....

..... [2]

- (b) A block of aluminium has a volume of $3.612 \times 10^{-3} \text{ m}^3$ at a temperature of 0°C .

Aluminium has a density of $2.700 \times 10^3 \text{ kg m}^{-3}$ at 0°C .

It has a density of $2.620 \times 10^3 \text{ kg m}^{-3}$ at 500°C .

The block is heated so that its temperature increases from 0°C to 500°C at an atmospheric pressure of $1.01 \times 10^5 \text{ Pa}$.

The increase in internal energy of the block is 4.38 MJ .

- (i) Calculate the mass of the block.

mass = kg [2]

- (ii) Show that the volume of the block at a temperature of 500°C is $3.722 \times 10^{-3} \text{ m}^3$.

[1]

- (iii) Use the information in (b)(ii) to determine the magnitude of the work done on the block when its temperature is raised from 0°C to 500°C .

work done = J [2]



(iv) Explain whether the work done on the block is positive or negative.

.....

.....

..... [2]

(v) Use the first law of thermodynamics to determine, to three significant figures, a value for the specific heat capacity of aluminium. Explain your reasoning. Give a unit with your answer.

specific heat capacity = unit [3]

(c) Without further calculation, suggest with a reason how doubling the pressure in (b) is likely to affect the answer in (b)(v).

.....

.....

..... [1]

[Total: 13]



- 4 (a) The equation of state for an ideal gas may be written as

$$pVA = NBT$$

where p is the pressure of the gas, V is the volume of the gas, A is the Avogadro constant, B is another constant and N is the number of molecules of the gas.

- (i) State the meaning, in the equation, of the symbol T .

..... [1]

- (ii) Identify the constant B .

..... [1]

- (b) The product pV for an ideal gas is also given by

$$pV = \frac{1}{3} Nm \langle c^2 \rangle.$$

- (i) State the meanings, in this equation, of the symbols m and $\langle c^2 \rangle$.

m :

$\langle c^2 \rangle$:

[2]

- (ii) Use the equations in (a) and (b) to derive an expression, in terms of A , B and T , for the mean kinetic energy E_K of a molecule of the gas.

$$E_K = [2]$$



- (c) On Fig. 4.1, sketch the variation with T of the root-mean-square (r.m.s.) speed of the molecules of an ideal gas.

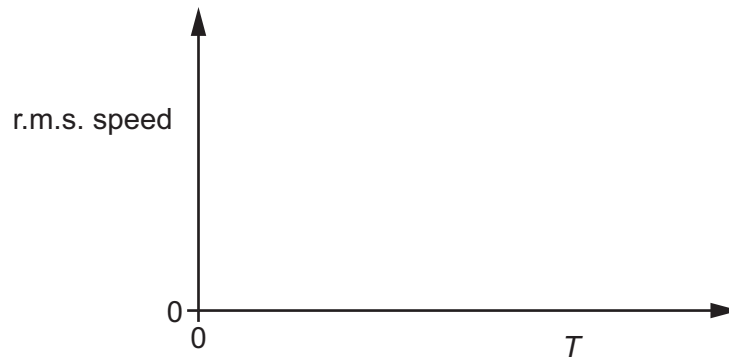


Fig. 4.1

[2]

[Total: 8]



- 5 (a) State what is meant by simple harmonic motion.

.....

 [2]

- (b) A block is suspended by a spring. The block oscillates vertically with simple harmonic motion.

The velocity v of the block varies with time t according to

$$v = 0.56 \cos 16t$$

where v is in m s^{-1} and t is in s.

- (i) Calculate the period of the oscillation.

period = s [1]

- (ii) Determine the amplitude x_0 of the oscillation.

$x_0 = \dots\dots\dots$ m [2]

- (iii) Use your answer in (b)(ii) to determine the equation for v in terms of the displacement x of the block, where v is in m s^{-1} and x is in m.

$v = \dots\dots\dots$ [1]



(iv) On Fig. 5.1, sketch the variation of v with x .

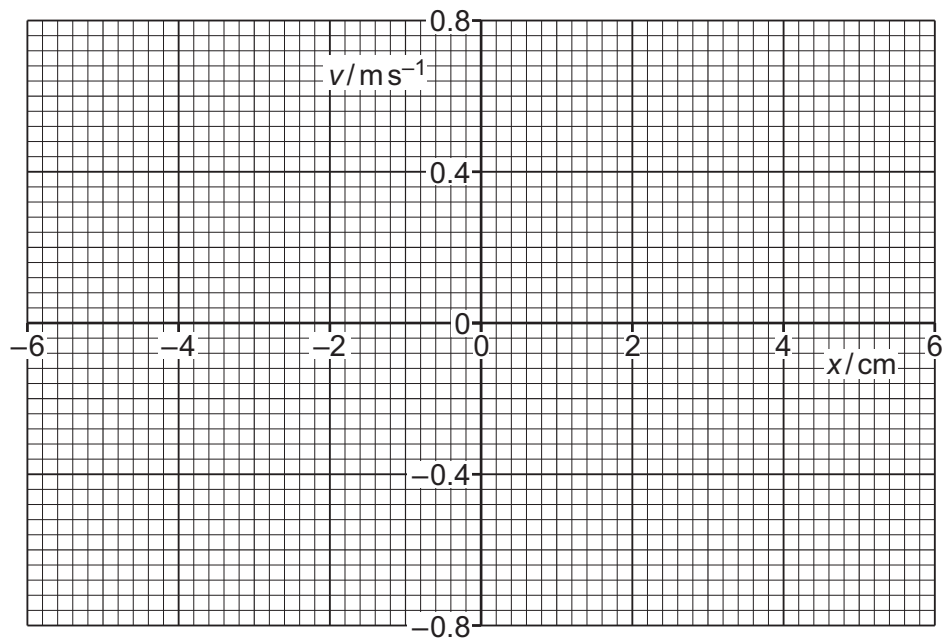


Fig. 5.1

[3]

[Total: 9]



- 6 Two parallel metal plates X and Y are separated by a distance of 0.041 m, as shown in Fig. 6.1.

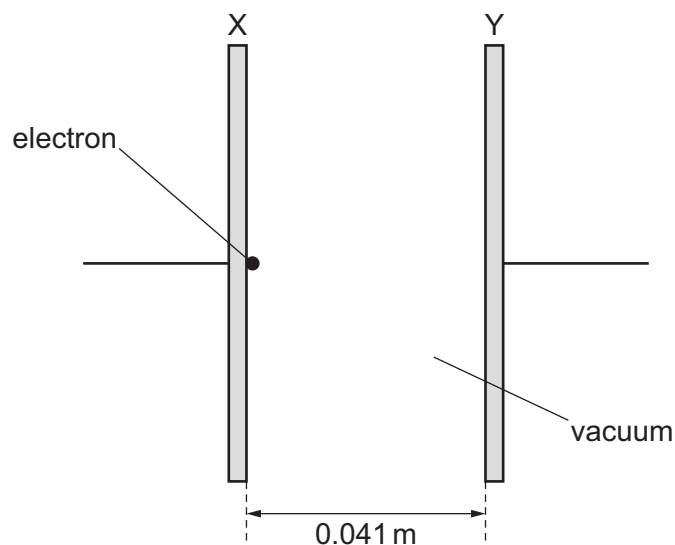


Fig. 6.1

There is a vacuum between the plates. An electron is at rest at the centre of plate X.

A potential difference (p.d.) of 58 kV is applied across the plates. This causes the electron to accelerate towards plate Y.

- (a) On Fig. 6.1, use the symbols + and – to indicate which of plates X and Y is the positive plate and which is the negative plate. [1]
- (b) (i) Calculate the electric field strength E between the plates. Give a unit with your answer.

$$E = \dots\dots\dots \text{unit} \dots\dots\dots [2]$$

- (ii) Determine the acceleration of the electron.

$$\text{acceleration} = \dots\dots\dots \text{ms}^{-2} [2]$$



(c) Many electrons are now accelerated from rest from plate X to plate Y in Fig. 6.1. When the electrons hit plate Y, the absorption of their kinetic energies results in the emission of electromagnetic waves.

(i) Show that the minimum wavelength of these electromagnetic waves is 21 pm.

[3]

(ii) State the region of the electromagnetic spectrum that contains these waves.

[1]

(iii) Explain how these electromagnetic waves may be used to form images of internal body structures.

[2]

[Total: 11]



- 7 Fig. 7.1 shows a circuit containing a capacitor of capacitance C and a resistor of resistance R .

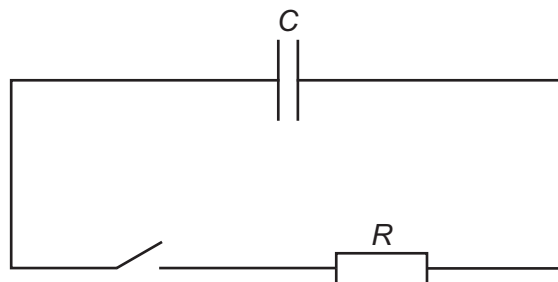


Fig. 7.1

Initially, the switch is open and the potential difference (p.d.) across the capacitor is 12 V.

The switch is closed at time $t = 0$ and the capacitor discharges through the resistor.

Fig. 7.2 shows the variation of the charge Q on the capacitor with the p.d. V_C across the capacitor as the capacitor discharges. Fig. 7.3 shows the variation of the current I in the resistor with the p.d. V_R across the resistor as the capacitor discharges.

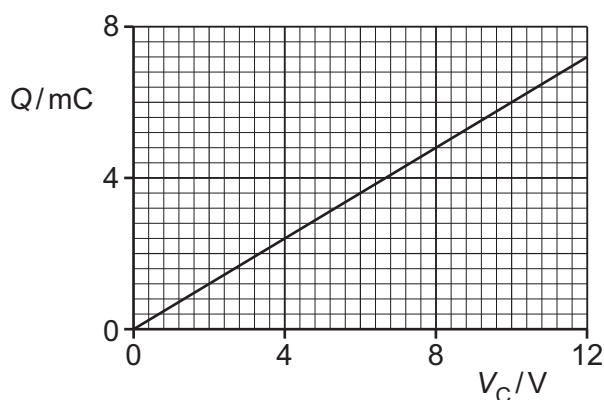


Fig. 7.2

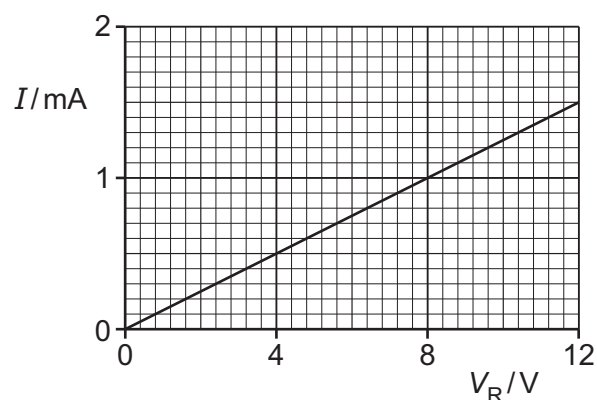


Fig. 7.3

- (a) State the relationship between V_C and V_R .

..... [1]

- (b) Determine:

- (i) the capacitance C , in μF

$C = \dots\dots\dots \mu\text{F}$ [2]





(ii) the resistance R , in $\text{k}\Omega$

$R = \dots\dots\dots \text{k}\Omega$ [2]

(iii) the time constant τ of the circuit.

$\tau = \dots\dots\dots \text{s}$ [2]

(c) Use Fig. 7.2, Fig. 7.3 and your answer in (a) to explain why the variation of Q with t is exponential in nature.

.....
.....
.....
.....
..... [3]

[Total: 10]



- 8 Fig. 8.1 shows a circuit that produces rectification of an alternating input voltage.

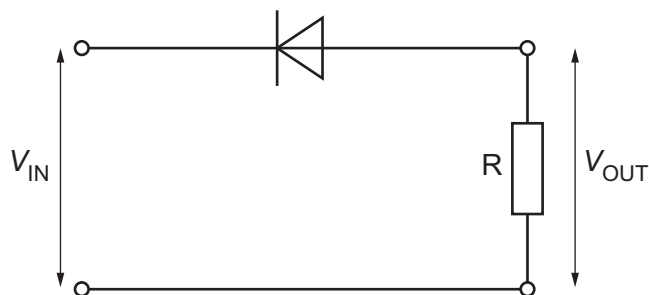


Fig. 8.1

The input voltage V_{IN} is sinusoidal. The rectified output voltage V_{OUT} is applied across resistor R .

The variation of V_{IN} with time t has amplitude V_0 and period T , as shown in Fig. 8.2.

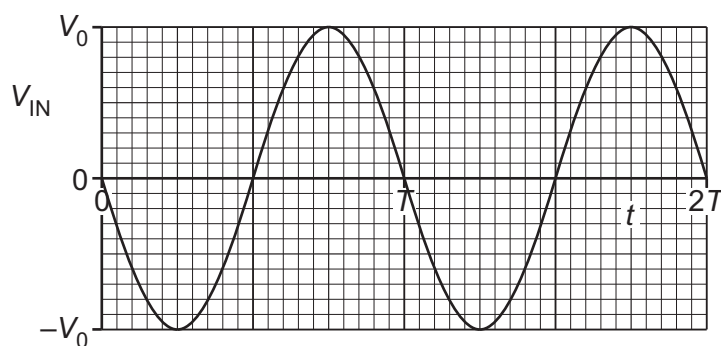


Fig. 8.2

The root-mean-square (r.m.s.) value of V_{IN} is 6.0 V.

- (a) (i) State the type of rectification produced by the circuit of Fig. 8.1.

..... [1]

- (ii) Calculate V_0 .

$V_0 = \dots\dots\dots$ V [1]



(b) Resistor R has resistance $45\ \Omega$.

Assume that there is no p.d. across the diode when it is conducting.

(i) Determine the peak power P_0 in the resistor.

$$P_0 = \dots\dots\dots \text{ W [2]}$$

(ii) On Fig. 8.3, sketch the variation of the power P in the resistor with t between $t = 0$ and $t = 2T$.

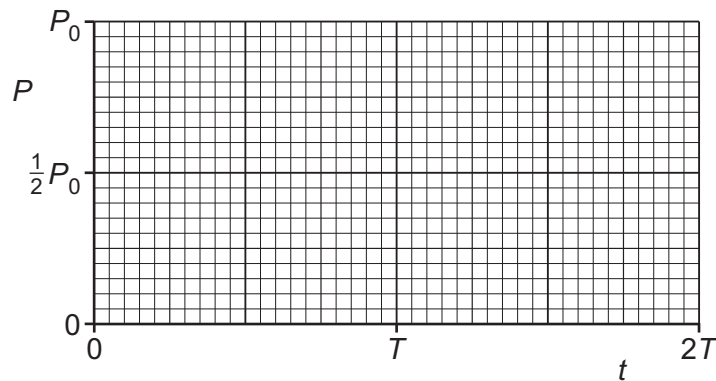


Fig. 8.3

[3]

(iii) Use the answer in (b)(ii) to explain why the mean power in the resistor is $\frac{1}{4}P_0$.

.....

 [2]

(iv) Use the information in (b)(iii) to determine the r.m.s. value of V_{OUT} .

$$\text{r.m.s. voltage} = \dots\dots\dots \text{ V [1]}$$

[Total: 10]



- 9 (a) State what is meant by the photoelectric effect.

.....

.....

..... [2]

- (b) The photoelectric effect is investigated in two stages using the circuit shown in Fig. 9.1.

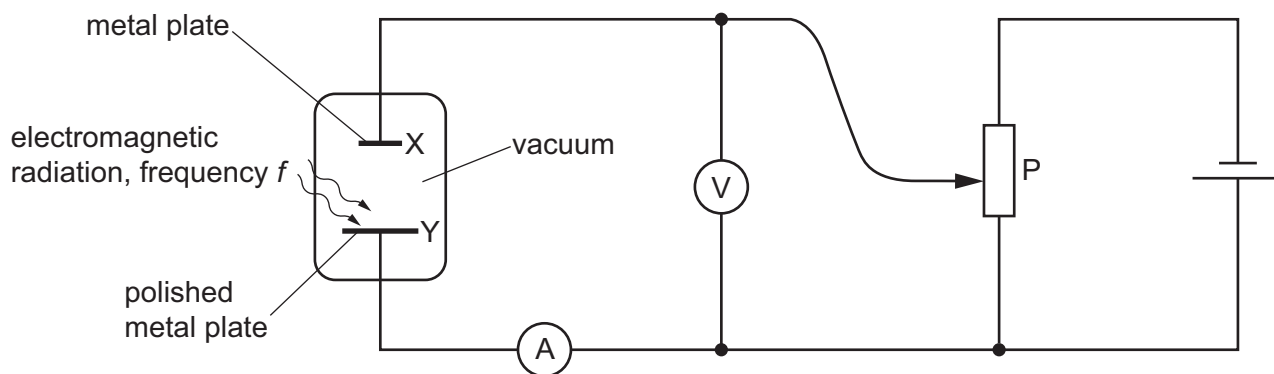


Fig. 9.1

The polished metal plate Y is illuminated with electromagnetic radiation of frequency f and constant power.

In stage 1 of the investigation, frequency f is set to a constant value of 2.5×10^{15} Hz. The current I in the ammeter is varied by adjusting the potentiometer P. Fig. 9.2 shows the variation of I with the voltmeter reading V . There is a value V_S of V at which the current just falls to zero.

In stage 2 of the investigation, stage 1 is repeated for different values of frequency. As frequency f is varied, the voltmeter reading V_S at which the current just falls to zero is measured. Fig. 9.3 shows the variation of V_S with f .

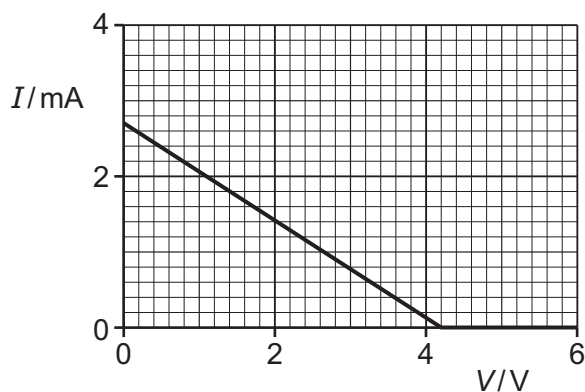


Fig. 9.2

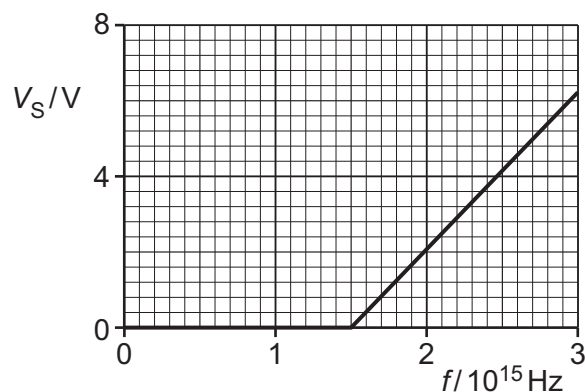


Fig. 9.3





- (i) Explain, with reference to photons, why V_S depends on the frequency of the incident electromagnetic radiation.

.....

.....

.....

.....

.....

..... [3]

- (ii) State **three** quantitative conclusions that can be drawn from the results in Fig. 9.2 and Fig. 9.3. Use the space for any working.

1

.....

2

.....

3

..... [3]

[Total: 8]



- 10 (a) Radioactive decay is a spontaneous process.

State the meaning, in this context, of the term spontaneous.

.....
 [1]

- (b) Two radioactive isotopes X and Y each decay to form a stable isotope.
 A sample initially contains only atoms of isotope X. At this time, its activity is $4A$.
 Another sample initially contains only atoms of Y. At this time, its activity is A .

Fig. 10.1 shows the variation of the activity of each sample with time t between $t = 0$ and $t = 6T$.

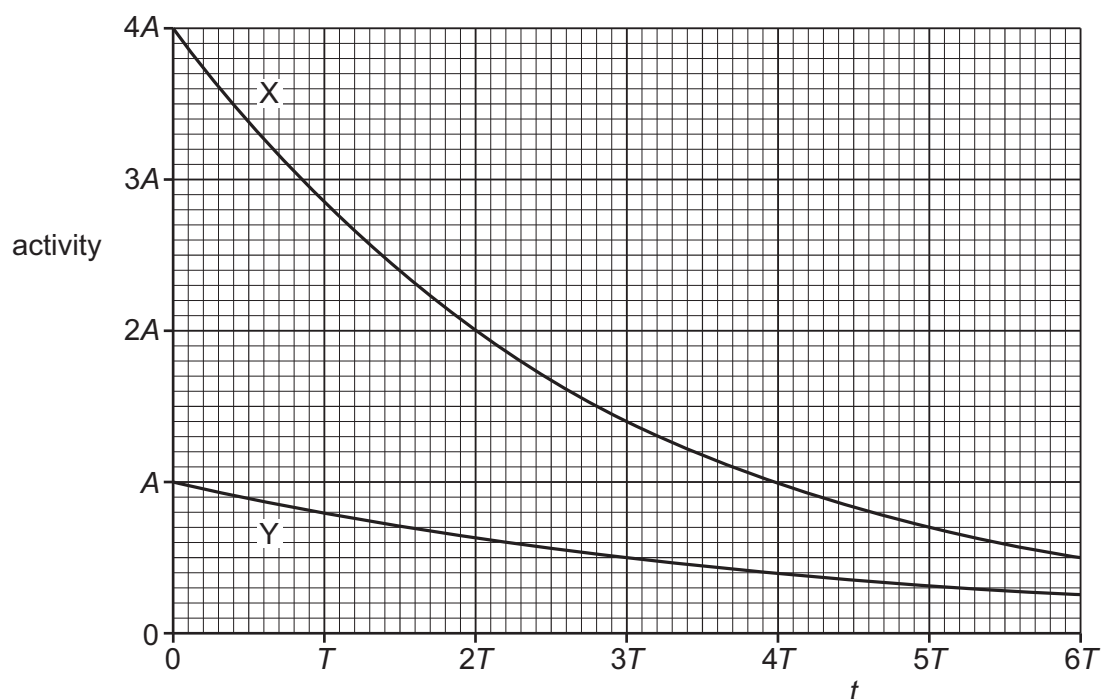


Fig. 10.1

- (i) Complete Table 10.1 to give expressions, in terms of either or both of A and T , for the quantities indicated for each of the samples.

Table 10.1

sample	half-life	decay constant	initial activity	initial number of nuclei
X			$4A$	
Y			A	

[3]



(ii) Determine, in terms of T , the time at which the two samples will have equal activities.

time = T [3]

(c) A radiation detector is placed near to one of the samples in (b).

Explain why the count rate measured by the detector is less than the activity of the sample.

.....
.....
..... [2]

[Total: 9]





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PHYSICS

9702/43

Paper 4 A Level Structured Questions

May/June 2025

2 hours

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [].

This document has **24** pages. Any blank pages are indicated.



Data

acceleration of free fall	$g = 9.81 \text{ m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
unified atomic mass unit	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27} \text{ kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$ $(\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ m F}^{-1})$
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_o = \frac{f_s v}{v \pm v_s}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \dots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$





gravitational potential

$$\phi = -\frac{GM}{r}$$

gravitational potential energy

$$E_P = -\frac{GMm}{r}$$

pressure of an ideal gas

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

simple harmonic motion

$$a = -\omega^2 x$$

velocity of particle in s.h.m.

$$v = v_0 \cos \omega t$$

$$v = \pm \omega \sqrt{(x_0^2 - x^2)}$$

electric potential

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

electrical potential energy

$$E_P = \frac{Qq}{4\pi\epsilon_0 r}$$

capacitors in series

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$$

capacitors in parallel

$$C = C_1 + C_2 + \dots$$

discharge of a capacitor

$$x = x_0 e^{-\frac{t}{RC}}$$

Hall voltage

$$V_H = \frac{BI}{ntq}$$

alternating current/voltage

$$x = x_0 \sin \omega t$$

radioactive decay

$$x = x_0 e^{-\lambda t}$$

decay constant

$$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$$

intensity reflection coefficient

$$\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$$

Stefan–Boltzmann law

$$L = 4\pi\sigma r^2 T^4$$

Doppler redshift

$$\frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$$



- 1 (a) Define gravitational potential at a point.

.....

.....

..... [2]

- (b) Mars is a planet that may be considered to be an isolated uniform sphere of radius 3.4×10^6 m.

A satellite of mass 122 kg is in orbit around Mars at a constant height of 1.7×10^6 m above the surface of the planet.

The height of the orbit is increased to 6.8×10^6 m above the surface. This increases the gravitational potential energy of the satellite by 5.1×10^8 J.

- (i) Show that the mass of Mars is 6.4×10^{23} kg.

[3]

- (ii) Calculate the gravitational potential ϕ at the surface of Mars. Give a unit with your answer.

$\phi =$ unit [2]



(c) The satellite in (b) is moved to an orbit in which the satellite remains at the same point above the surface of Mars.

(i) The orbit has a period of 25 hours.

State what can be deduced from this about the rotation of Mars on its axis.

.....
..... [1]

(ii) State **one** other feature of this orbit.

.....
..... [1]

[Total: 9]



- 2 A helium atom may be modelled as a nucleus surrounded by two electrons in diametrically opposite circular orbits, each of radius 170 pm, as shown in Fig. 2.1.

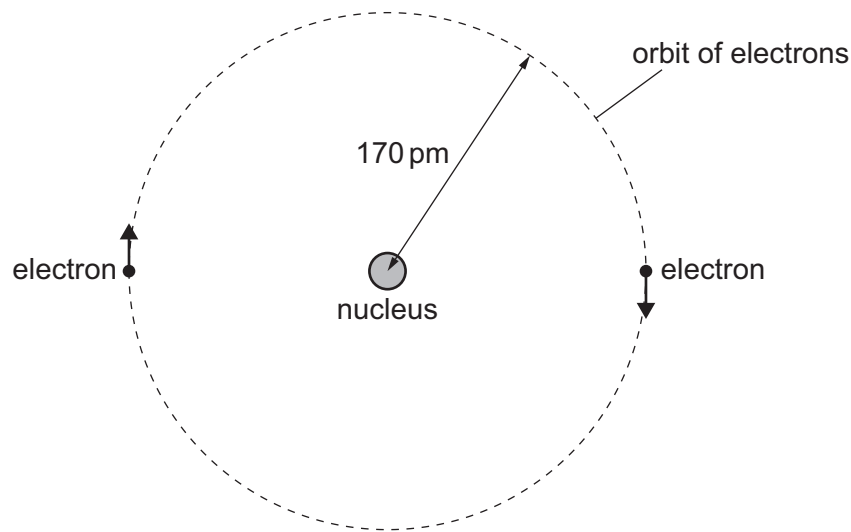


Fig. 2.1

- (a) State Coulomb's law.

.....

 [2]

- (b) (i) State the charge on the nucleus, in terms of the elementary charge e .

charge = e [1]

- (ii) Show that the electric force between the nucleus and one of the electrons is $1.6 \times 10^{-8} \text{ N}$.

[1]



(c) Assume that the force in (b)(ii) is the only force on the electrons.

(i) Calculate the speed of the orbiting electrons.

speed = ms^{-1} [2]

(ii) Calculate the period of the orbit of the electrons.

period = s [2]

(d) In practice, the orbit of each electron is affected by the presence of the other electron.

(i) For the position of one of the electrons, determine the ratio

$$\frac{\text{electric field strength due to the other electron}}{\text{electric field strength due to the nucleus}}.$$

ratio = [2]

(ii) Use your answer in (d)(i) to suggest and explain how the orbit of the electron is affected by the presence of the other electron.

.....
 [1]

[Total: 11]



- 3 (a) Define specific latent heat.

.....

 [2]

- (b) Explain why, for a substance, the specific latent heat of vaporisation is usually greater than the specific latent heat of fusion.

.....

 [3]

- (c) An ice cube of mass 37.0g at temperature 0.0°C is placed in a beaker containing water of mass 208g at temperature 26.4°C.

When all the ice has melted, and all the water in the beaker has reached thermal equilibrium, the final temperature of all the water is 10.3°C.

The specific heat capacity of water is $4.18 \text{ J g}^{-1} \text{ }^{\circ}\text{C}^{-1}$.

The beaker has negligible specific heat capacity and is perfectly insulated from the surroundings.

Determine a value, to three significant figures, for the specific latent heat of fusion of water.

specific latent heat of fusion = J g^{-1} [4]

[Total: 9]



4 (a) (i) State what is meant by the internal energy of a system.

.....

.....

..... [2]

(ii) Explain why the internal energy of an ideal gas is directly proportional to the thermodynamic temperature of the gas.

.....

.....

.....

..... [2]

(b) A sample of an ideal gas at thermodynamic temperature T has internal energy U .

The gas is compressed so that its temperature increases to $3T$.
During this compression, work W is done on the gas.

The gas is then cooled at constant volume so that its temperature decreases to $2T$.

Complete Table 4.1 to show, in terms of some or all of W , T and U , the work done on the gas, the thermal energy supplied to the gas and the increase in internal energy of the gas for each of the two processes.

Table 4.1

	work done on gas	thermal energy supplied to gas	increase in internal energy of gas
compression	$+W$		
cooling			

[4]

[Total: 8]



- 5 A cuboidal block floats in a liquid with its base horizontal, as shown in Fig. 5.1.

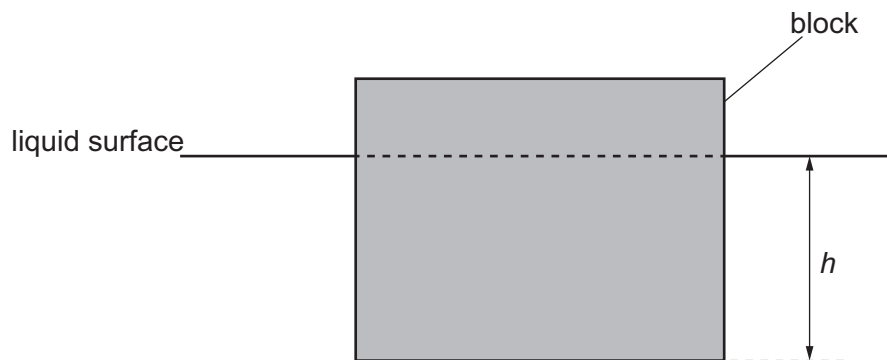


Fig. 5.1

The base of the block is at a depth h below the surface of the liquid.

The block is displaced downwards by a small distance and then released so that it oscillates.

Fig. 5.2 shows the variation with h of the acceleration a of the block.

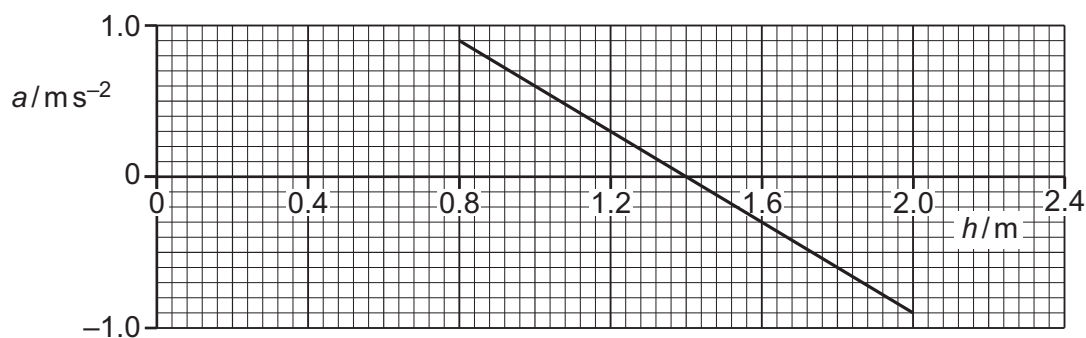


Fig. 5.2

Fig. 5.3 shows the variation with h of the kinetic energy E_K of the block.

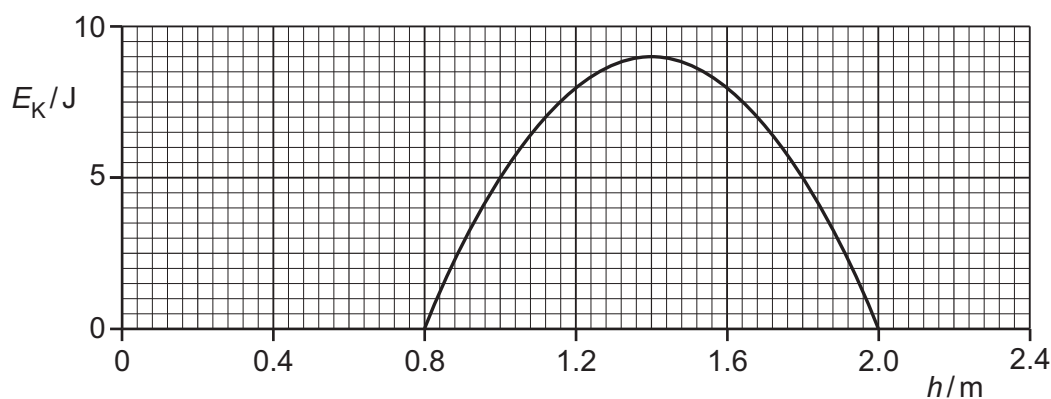


Fig. 5.3



- (a) (i) Determine the amplitude of the oscillations.

amplitude = m [1]

- (ii) State what the line in Fig. 5.2 shows about the nature of the oscillations.

..... [1]

- (b) State **three** other quantitative conclusions that can be drawn from Fig. 5.2 and Fig. 5.3 about the block and its oscillations. Use the space for any working.

1

.....

2

.....

3

.....

[3]

- (c) On Fig. 5.4, sketch the variation with h of the potential energy E_p of the oscillations.

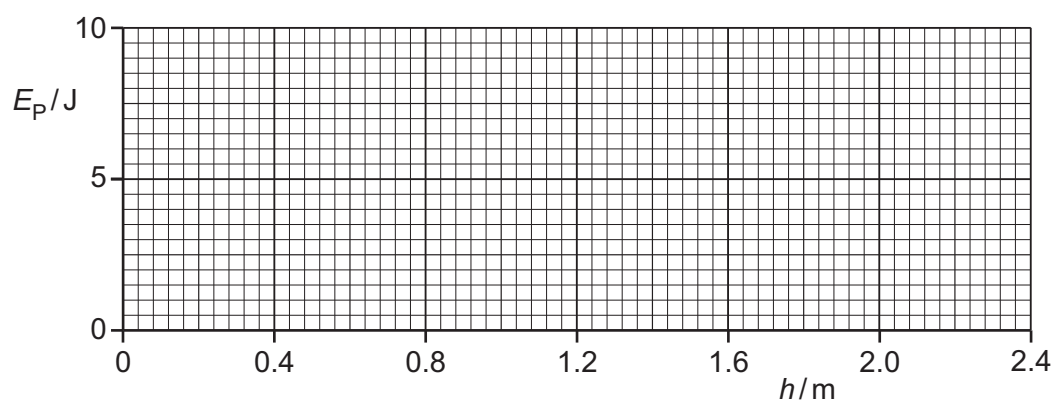


Fig. 5.4

[3]

[Total: 8]



- 6 Fig. 6.1 shows a circuit that rectifies an alternating input voltage V_{IN} and produces an output voltage V_{OUT} across a resistor R .

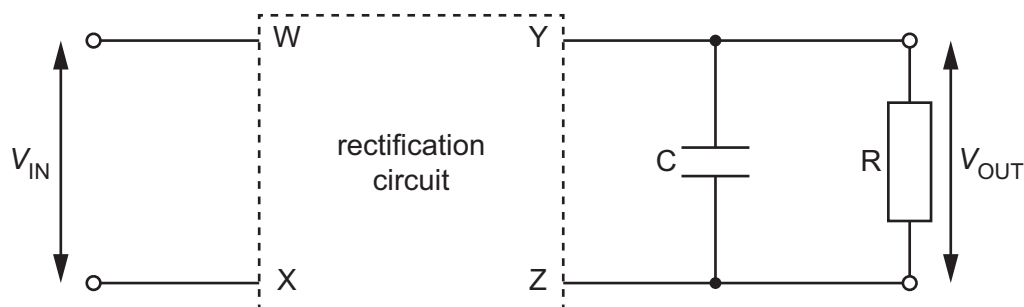


Fig. 6.1

The four terminals of the rectification circuit are labelled W, X, Y and Z.
A capacitor C is connected in parallel with resistor R .

- (a) (i) State what is meant by rectification.

.....
 [1]

- (ii) State the purpose of capacitor C .

.....
 [1]

- (b) Fig. 6.2 shows the variations with time t of the potential differences (p.d.s) V_{IN} and V_{OUT} .

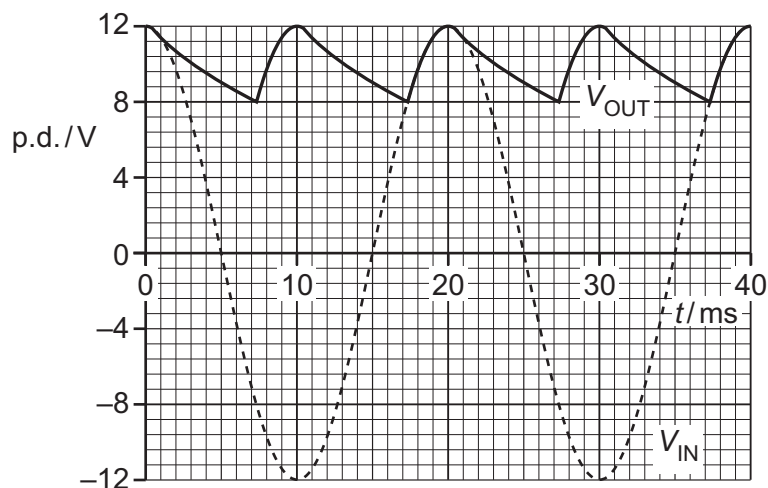


Fig. 6.2



- (i) The variation of V_{IN} with t can be represented by

$$V_{\text{IN}} = A \cos Bt$$

where A and B are constants.

Determine the values of A and B . Give a unit with your answer for A .

$$A = \dots\dots\dots \text{unit} \dots\dots\dots$$

$$B = \dots\dots\dots \text{rad s}^{-1} \quad [2]$$

- (ii) Determine the type of rectification produced by the circuit in Fig. 6.1.

..... [1]

- (iii) On Fig. 6.3, draw the circuit diagram for the components inside the rectification circuit.

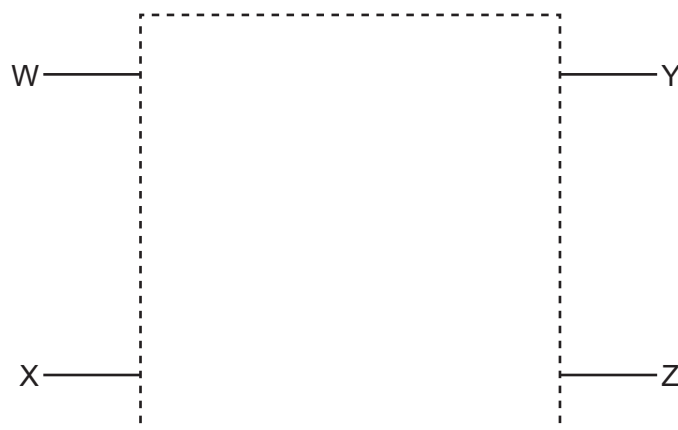


Fig. 6.3

[2]

- (iv) Determine a value for the time constant for the discharge of the capacitor C through the resistor R in Fig. 6.1.

time constant = s [3]



- (c) The capacitor C has a capacitance of $570\ \mu\text{F}$.

Use your answer in (b)(iv) to determine the resistance of resistor R.

resistance = Ω [2]

[Total: 12]

- 7 (a) Define magnetic flux density.

.....

 [2]

- (b) A particle of mass m and charge $+Q$ moves at speed v into a region where there is a uniform magnetic field, as shown in Fig. 7.1.

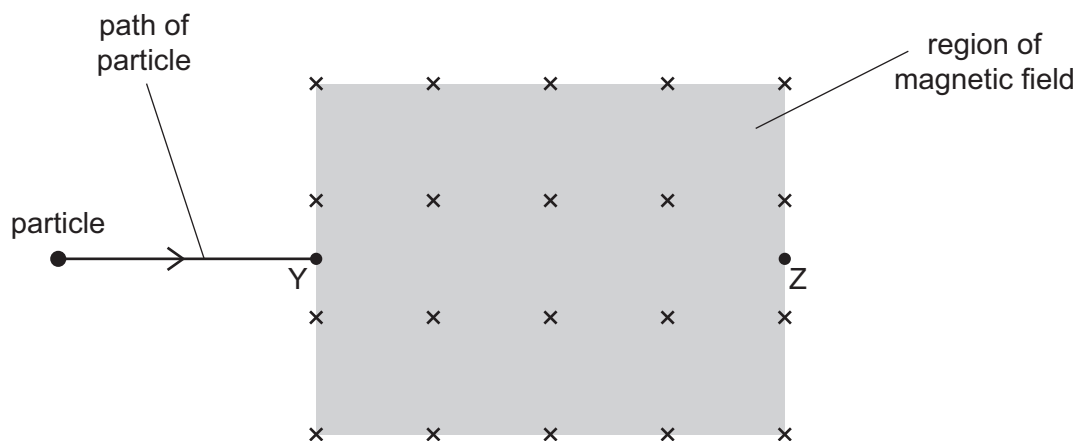


Fig. 7.1

The uniform magnetic field is into the page and has flux density B . The particle enters the region of the field at point Y.



- (i) State an expression, in terms of some or all of m , Q , B and v , for the magnetic force F that acts on the particle when it is at point Y.

$$F = \dots\dots\dots [1]$$

- (ii) On Fig. 7.1, draw an arrow at point Y to indicate the direction of the force in **(b)(i)**. [1]
- (iii) On Fig. 7.1, draw a line to show a possible path for the particle through the region of the magnetic field. [1]
- (c) (i) Explain how an electric field can be used with the magnetic field to ensure that the particle in **(b)** now passes through point Z.

.....

.....

.....

.....

..... [3]

- (ii) Derive an expression for v in terms of B and the electric field strength E .

$$v = \dots\dots\dots [2]$$

[Total: 10]



- 8 (a) State what is meant by the de Broglie wavelength.

.....
..... [1]

- (b) Calculate the de Broglie wavelength of an electron moving at a speed of $4.9 \times 10^7 \text{ m s}^{-1}$.

wavelength = m [2]

- (c) State **one** similarity and **one** difference between an electron and a positron.

similarity:
.....
difference:
..... [2]

- (d) An electron moving at a speed of $4.9 \times 10^7 \text{ m s}^{-1}$ collides with a positron that is travelling at the same speed in the opposite direction. As a result of the collision, two gamma-ray photons are produced.

- (i) State the name of this type of reaction.

..... [1]

- (ii) State what happens to the electron and to the positron.

.....
.....
..... [2]



- (iii) Explain why two gamma-ray photons are produced, rather than just one.

.....

..... [1]

- (iv) Show that the kinetic energy of the electron before the collision is $1.1 \times 10^{-15} \text{ J}$.

[1]

- (v) Use the information in (d)(iv) to determine, to three significant figures, the wavelength associated with the gamma radiation emitted in the collision.

wavelength = m [3]

[Total: 13]



- 9 (a) Define activity of a radioactive sample.

.....
 [1]

- (b) Explain why the variation with time of the activity of a radioactive sample is exponential in nature.

.....

 [3]

- (c) A sample contains a single radioactive isotope that decays to form a stable isotope.

The sample has an activity of 180 Bq at time $t = 0$.

At a time 8.4 minutes later, the activity is 120 Bq.

- (i) Determine the decay constant, in min^{-1} , of the radioactive isotope.

decay constant = min^{-1} [2]

- (ii) Use your answer in (c)(i) to determine the half-life, in min, of the radioactive isotope.

half-life = min [1]



- (iii) On Fig. 9.1, sketch the variation of the activity A of the sample with t for values of t between $t = 0$ and $t = 24$ min.

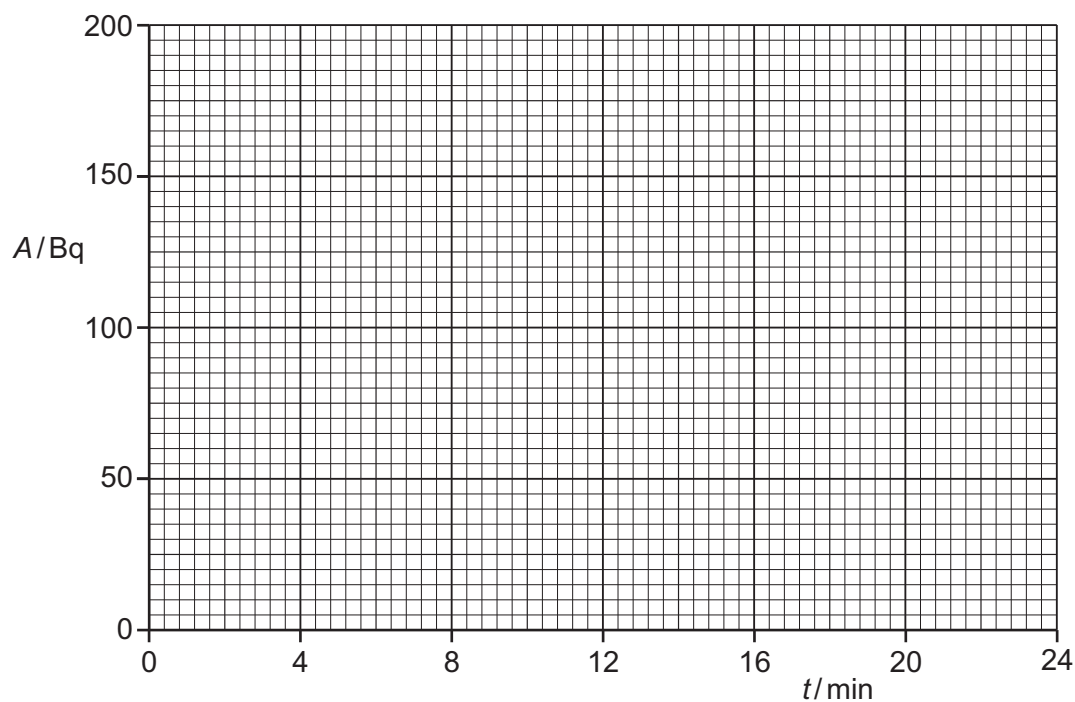


Fig. 9.1

[3]

[Total: 10]





10 (a) State Hubble's law.

.....

.....

..... [2]

(b) A star in a distant galaxy emits radiation that has a maximum intensity of emission at a wavelength of $4.62 \times 10^{-7} \text{ m}$.

Observations of the galaxy made on the Earth detect the maximum intensity of emission from the star at a wavelength of $4.91 \times 10^{-7} \text{ m}$.

(i) Explain why the observed wavelength and the emitted wavelength have different values.

.....

.....

..... [2]

(ii) Calculate the speed of the star relative to the Earth.

speed = m s^{-1} [2]

(iii) The wavelength of maximum intensity of emission is used to determine a value for the surface temperature of the star.

Explain how the temperature determined using the observed wavelength compares with the true value of temperature determined using the emitted wavelength.

.....

.....

..... [2]





(c) A value for the Hubble constant is $2.3 \times 10^{-18} \text{ s}^{-1}$.

Use your answer in (b)(ii) to determine the distance of the star in (b) from the Earth.

distance =m [2]

[Total: 10]









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Cambridge International AS & A Level

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PHYSICS

9702/51

Paper 5 Planning, Analysis and Evaluation

May/June 2025

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 30.
- The number of marks for each question or part question is shown in brackets [].

This document has 8 pages.

- 1 Fig. 1.1 shows a thin coil of cross-sectional area A and length l connected to a resistor of resistance S and two terminals.

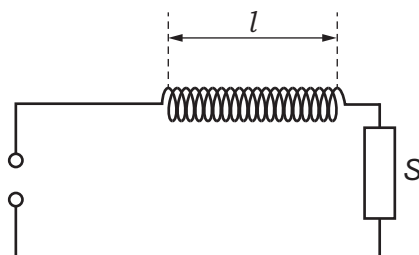


Fig. 1.1

An alternating voltage is applied to the terminals. The peak value of the alternating voltage is E and the frequency is f . The peak value of the potential difference V across the resistor is determined using an oscilloscope.

It is suggested that V is related to f by the relationship

$$\frac{ES}{V} = \frac{KAN^2f}{l}$$

where N is the number of turns on the coil and K is a constant.

Plan a laboratory experiment to test the relationship between V and f .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine a value for K .

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.



This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

..... [15]

- 2 A student investigates an electrical circuit.

The circuit is set up as shown in Fig. 2.1.

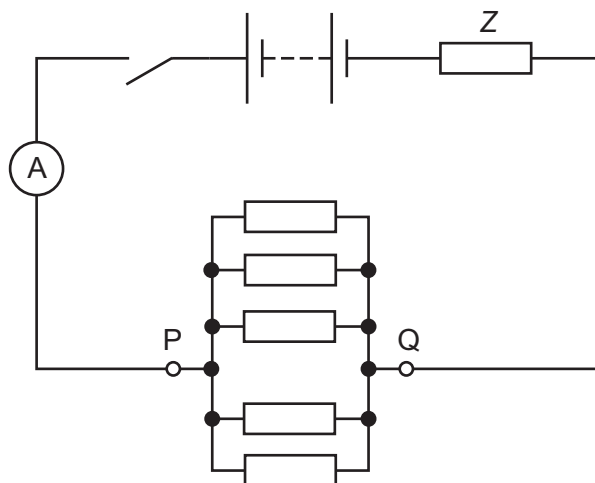


Fig. 2.1

A battery of negligible internal resistance is connected to a resistor of resistance Z . Five resistors, each of resistance R , are connected in parallel between P and Q.

The switch is closed. The total current I in the circuit is measured using the ammeter.

The experiment is then repeated by changing the number n of resistors, each of resistance R , connected in parallel between P and Q.

It is suggested that I and n are related by the equation

$$E = I \left(\frac{R}{n} + Z \right)$$

where E is the electromotive force (e.m.f.) of the battery.

- (a) A graph is plotted of $\frac{1}{I}$ on the y-axis against $\frac{1}{n}$ on the x-axis.

Determine expressions for the gradient and y-intercept.

gradient =

y-intercept =

[1]





(b) Values of n , $\frac{1}{n}$ and I are given in Table 2.1.

Table 2.1

n	$\frac{1}{n}$	$I/\mu\text{A}$	$\frac{1}{I}/10^3\text{A}^{-1}$
5	0.200	455 ± 5	
6	0.167	525 ± 5	
7	0.143	580 ± 5	
8	0.125	635 ± 5	
9	0.111	685 ± 5	
11	0.0909	765 ± 5	

Calculate and record values of $\frac{1}{I}/10^3\text{A}^{-1}$ in Table 2.1. Include the absolute uncertainties in $\frac{1}{I}$. [2]

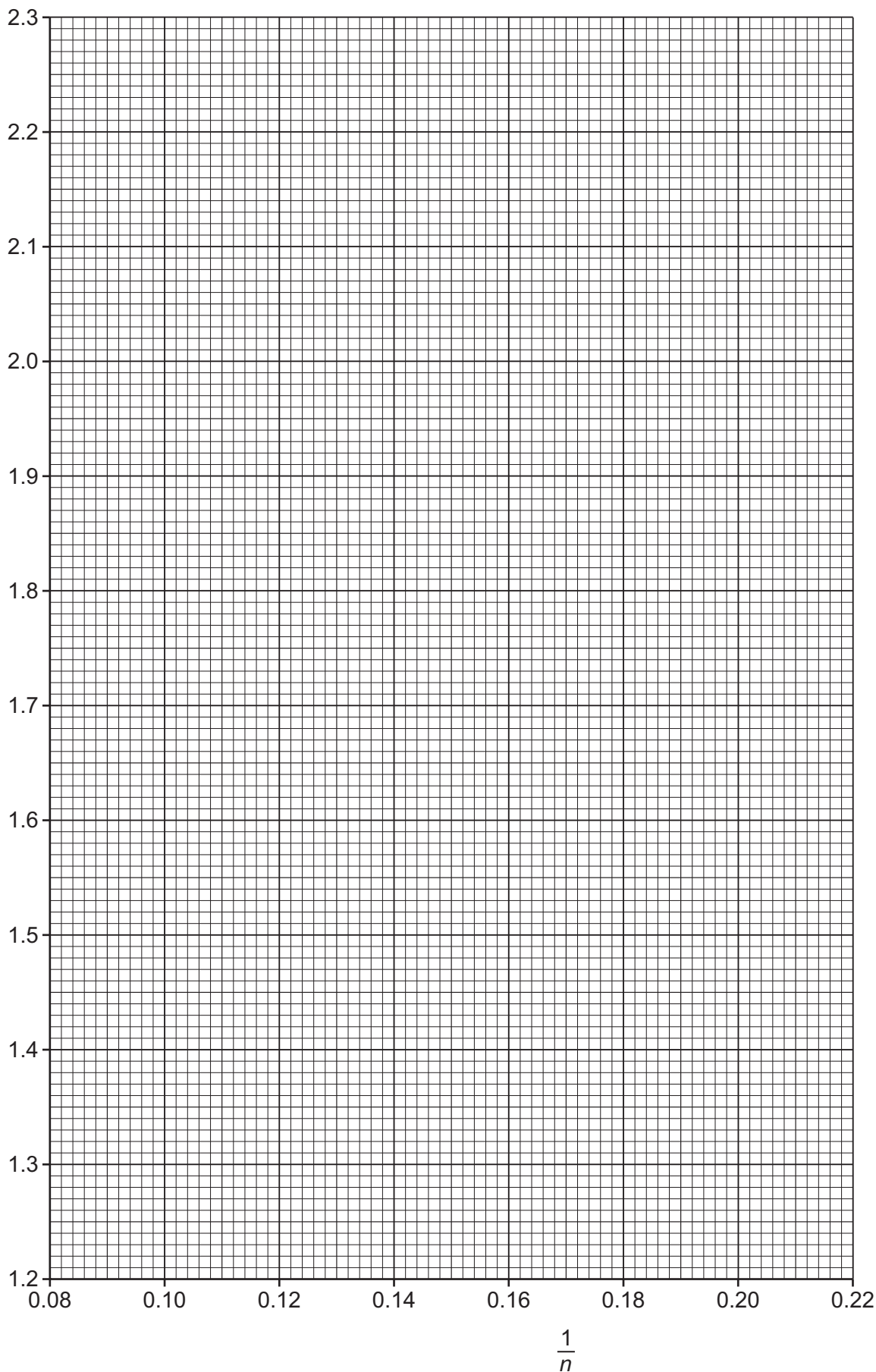
- (c) (i) Plot a graph of $\frac{1}{I}/10^3\text{A}^{-1}$ against $\frac{1}{n}$. Include error bars for $\frac{1}{I}$. [2]
- (ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines. [2]
- (iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]





$\frac{1}{I}/10^3\text{A}^{-1}$





- (iv) Determine the y -intercept of the line of best fit. Include the absolute uncertainty in your answer.

y -intercept = [2]

- (d) The e.m.f. E of the battery is determined twice during the experiment. The values obtained are 5.6 V and 6.0 V.

- (i) Using your answers to (a), (c)(iii) and (c)(iv), determine the values of R and Z . Include appropriate units.

R =

Z =

[2]

- (ii) Determine the percentage uncertainty in your value of R .

percentage uncertainty = % [1]

- (e) The experiment is repeated with 20 resistors, each of resistance R , connected in parallel between P and Q. Determine the total current I in the circuit.

I = A [1]

[Total: 15]





Cambridge International AS & A Level

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PHYSICS

9702/52

Paper 5 Planning, Analysis and Evaluation

May/June 2025

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 30.
- The number of marks for each question or part question is shown in brackets [].

This document has 8 pages.



- 1 A thin solid disc of radius r and thickness z is attached to a thin axle. String is wrapped around the axle, as shown in Fig. 1.1.

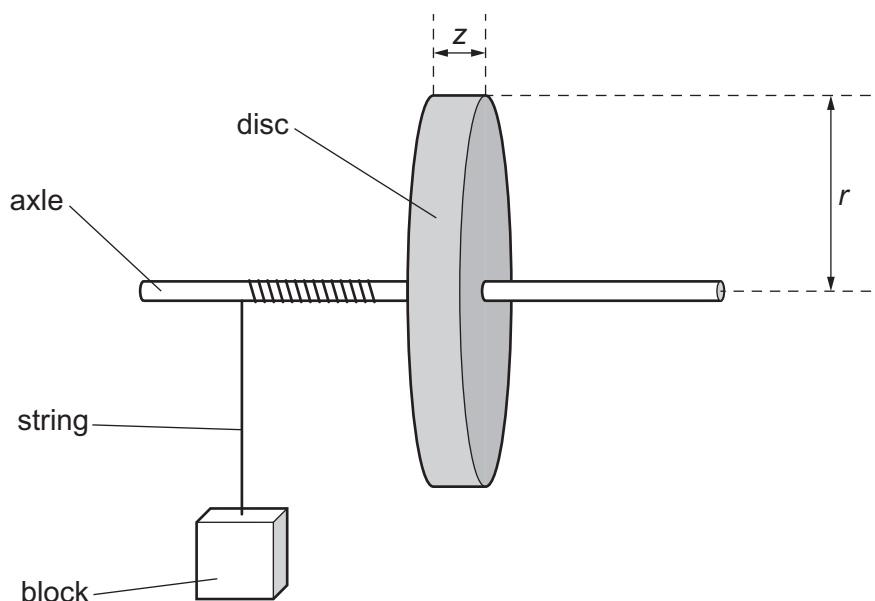


Fig. 1.1

A block of mass m is attached to the string.

The block is released from rest and falls downwards. The block has speed v when it has fallen through a distance h from the point of release. The value of v is determined using **one** light gate connected to a timer.

It is suggested that v is related to m by the relationship

$$\frac{h}{v^2} = \frac{\pi r^2 z}{2PQm} + \frac{1}{P}$$

where P and Q are constants.

Plan a laboratory experiment to test the relationship between v and m .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for P and Q .

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.





Diagram

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..... [15]

- 2 A student investigates a circuit containing capacitors. The circuit is connected with a capacitor of capacitance A , as shown in Fig. 2.1.

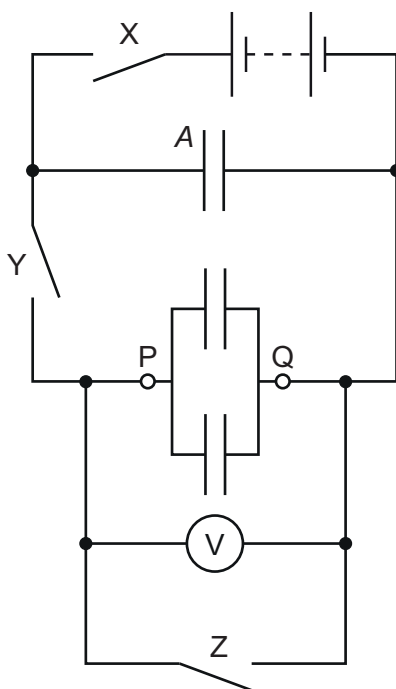


Fig. 2.1

Two capacitors, each of capacitance C , are connected in parallel between P and Q.

Initially, switch X and switch Z are closed and switch Y is open.

Switches X and Z are opened. Switch Y is then closed. The maximum potential difference between P and Q is measured using the voltmeter. This procedure is repeated and the mean maximum potential difference V between P and Q is determined.

The experiment is then repeated by changing the number n of capacitors, each of capacitance C , connected in parallel between P and Q.

It is suggested that V and n are related by the equation

$$EA = V(nC + A)$$

where E is the electromotive force (e.m.f.) of the battery.

- (a) A graph is plotted of $\frac{1}{V}$ on the y -axis against n on the x -axis.

Determine expressions for the gradient and y -intercept.

gradient =

y -intercept =

[1]





- (b) Values of n and the two measured values of the maximum potential difference V_1 and V_2 are given in Table 2.1.

Table 2.1

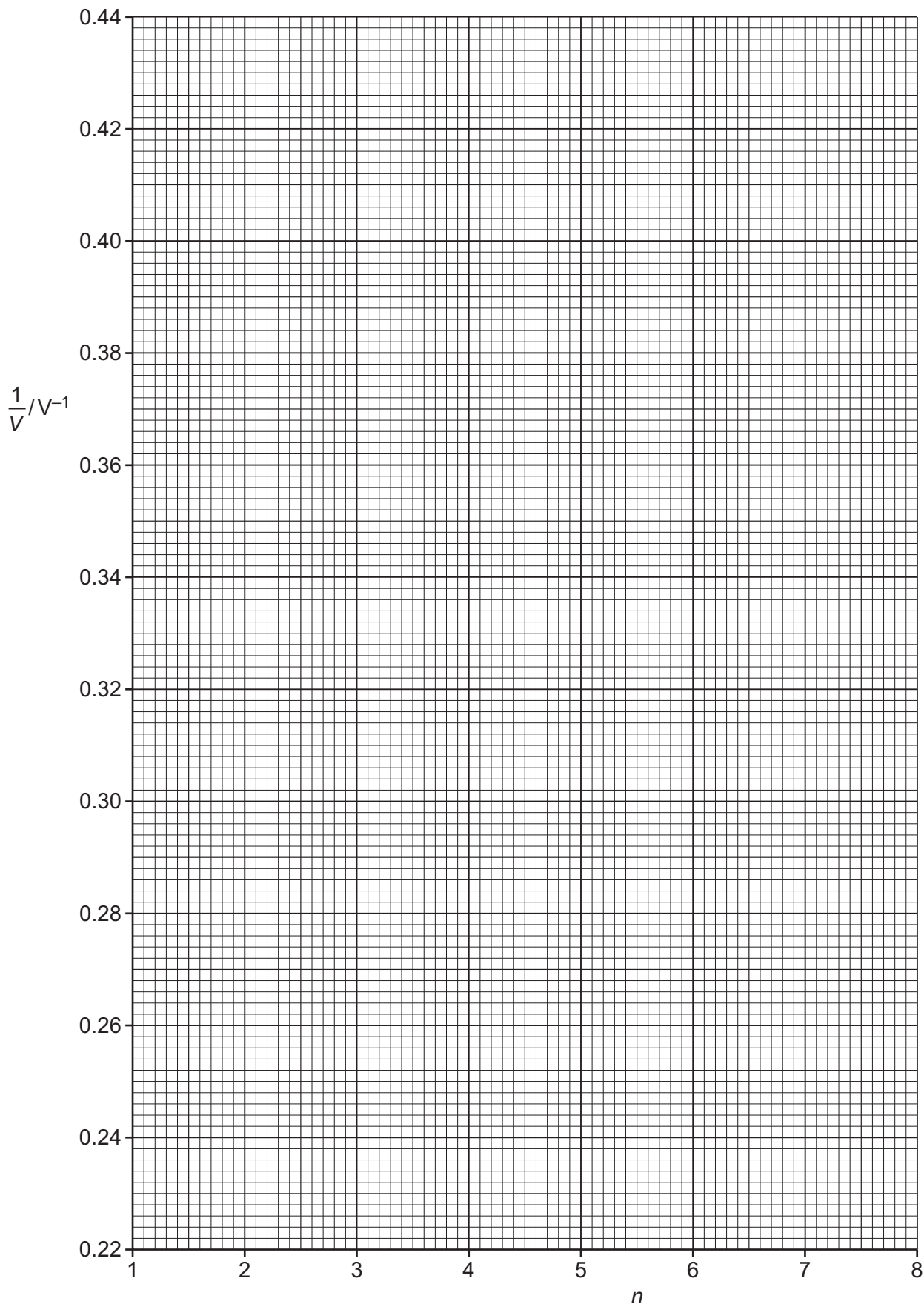
n	V_1/V	V_2/V	V/V	$\frac{1}{V}/V^{-1}$
2	4.30	4.20		
3	3.65	3.75		
4	3.30	3.20		
5	2.85	2.95		
6	2.65	2.55		
7	2.30	2.40		

Calculate and record values of V/V and $\frac{1}{V}/V^{-1}$ in Table 2.1. Include the absolute uncertainties in V and $\frac{1}{V}$. [2]

- (c) (i) Plot a graph of $\frac{1}{V}/V^{-1}$ against n . Include error bars for $\frac{1}{V}$. [2]
- (ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines. [2]
- (iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]







- (iv) Determine the y -intercept of the line of best fit. Include the absolute uncertainty in your answer.

y -intercept = [2]

- (d) (i) Using your answers to (a), (c)(iii) and (c)(iv), determine the values of E and C . Include appropriate units.

Data: $A = (2.2 \pm 0.2) \text{ mF}$

$E = \dots\dots\dots$

$C = \dots\dots\dots$

[2]

- (ii) Determine the percentage uncertainty in your value of C .

percentage uncertainty =% [1]

- (e) The experiment is repeated with 10 capacitors, each of capacitance C , connected in parallel between P and Q. Determine the maximum potential difference V between P and Q.

$V = \dots\dots\dots V$ [1]

[Total: 15]





Cambridge International AS & A Level

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PHYSICS

9702/53

Paper 5 Planning, Analysis and Evaluation

May/June 2025

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
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- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

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- The number of marks for each question or part question is shown in brackets [].

This document has 8 pages.

- 1 Fig. 1.1 shows a thin coil of cross-sectional area A and length l connected to a resistor of resistance S and two terminals.

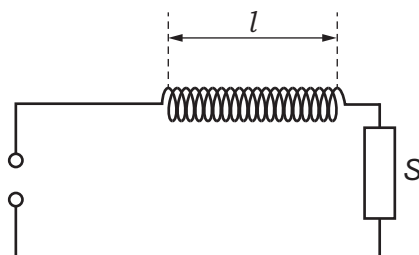


Fig. 1.1

An alternating voltage is applied to the terminals. The peak value of the alternating voltage is E and the frequency is f . The peak value of the potential difference V across the resistor is determined using an oscilloscope.

It is suggested that V is related to f by the relationship

$$\frac{ES}{V} = \frac{KAN^2f}{l}$$

where N is the number of turns on the coil and K is a constant.

Plan a laboratory experiment to test the relationship between V and f .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine a value for K .

In your plan you should include:

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- the control of variables
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[15]

- 2 A student investigates an electrical circuit.

The circuit is set up as shown in Fig. 2.1.

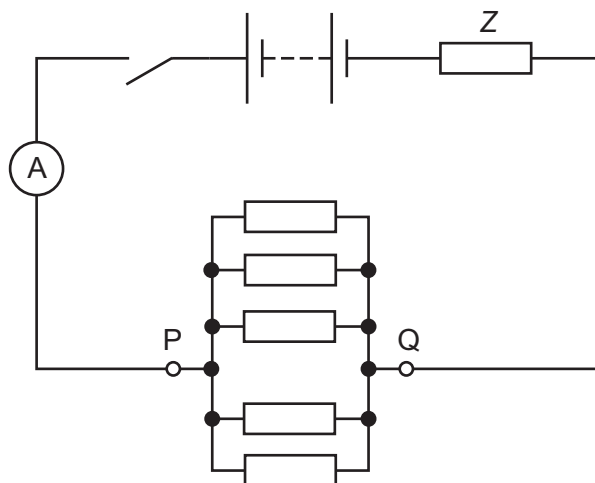


Fig. 2.1

A battery of negligible internal resistance is connected to a resistor of resistance Z . Five resistors, each of resistance R , are connected in parallel between P and Q.

The switch is closed. The total current I in the circuit is measured using the ammeter.

The experiment is then repeated by changing the number n of resistors, each of resistance R , connected in parallel between P and Q.

It is suggested that I and n are related by the equation

$$E = I \left(\frac{R}{n} + Z \right)$$

where E is the electromotive force (e.m.f.) of the battery.

- (a) A graph is plotted of $\frac{1}{I}$ on the y-axis against $\frac{1}{n}$ on the x-axis.

Determine expressions for the gradient and y-intercept.

gradient =

y-intercept =

[1]





(b) Values of n , $\frac{1}{n}$ and I are given in Table 2.1.

Table 2.1

n	$\frac{1}{n}$	$I/\mu\text{A}$	$\frac{1}{I}/10^3\text{A}^{-1}$
5	0.200	455 ± 5	
6	0.167	525 ± 5	
7	0.143	580 ± 5	
8	0.125	635 ± 5	
9	0.111	685 ± 5	
11	0.0909	765 ± 5	

Calculate and record values of $\frac{1}{I}/10^3\text{A}^{-1}$ in Table 2.1. Include the absolute uncertainties in $\frac{1}{I}$. [2]

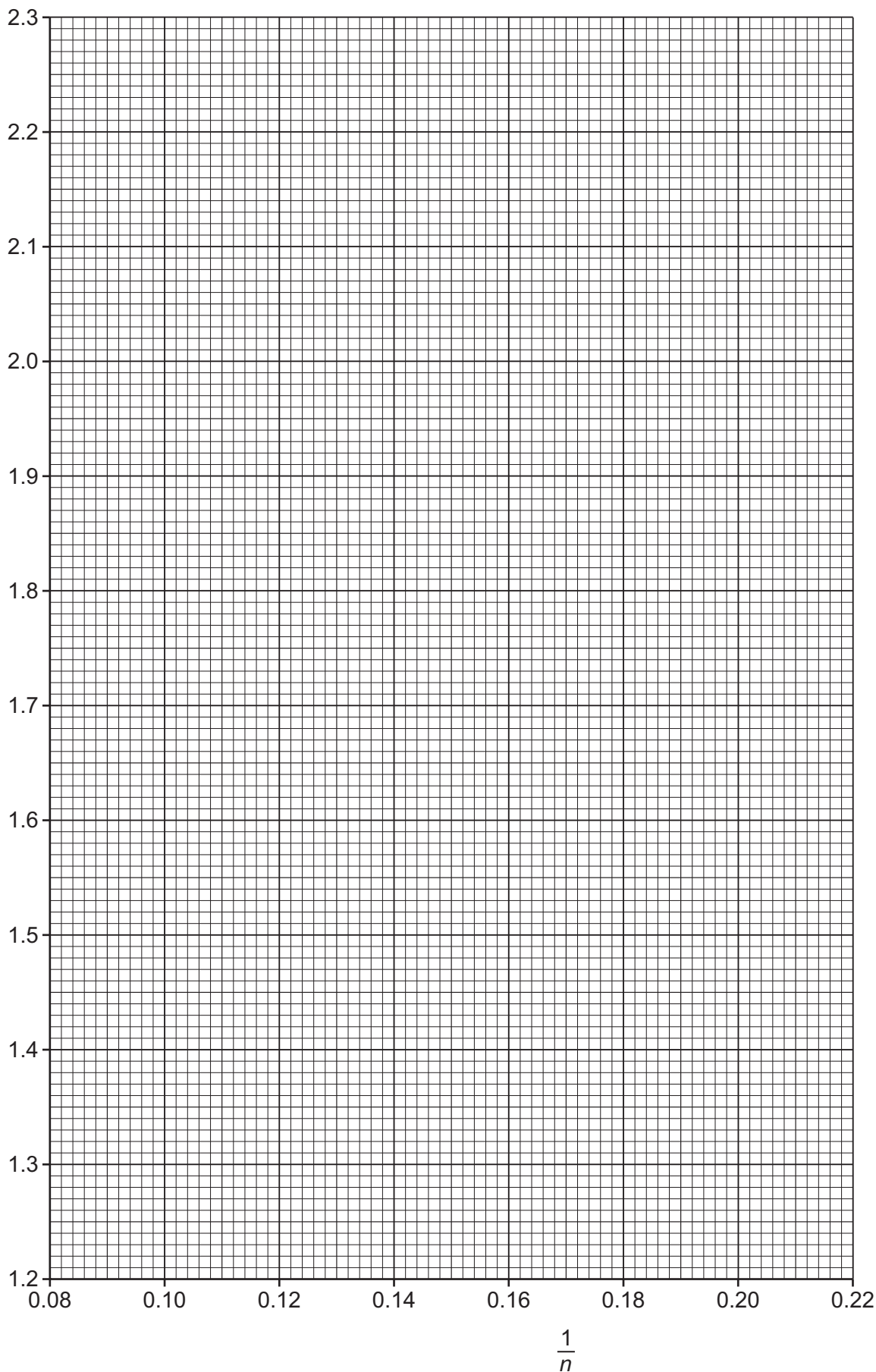
- (c) (i) Plot a graph of $\frac{1}{I}/10^3\text{A}^{-1}$ against $\frac{1}{n}$. Include error bars for $\frac{1}{I}$. [2]
- (ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines. [2]
- (iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]





$\frac{1}{I} / 10^3 \text{A}^{-1}$





- (iv) Determine the y -intercept of the line of best fit. Include the absolute uncertainty in your answer.

y -intercept = [2]

- (d) The e.m.f. E of the battery is determined twice during the experiment. The values obtained are 5.6 V and 6.0 V.

- (i) Using your answers to (a), (c)(iii) and (c)(iv), determine the values of R and Z . Include appropriate units.

R =

Z =

[2]

- (ii) Determine the percentage uncertainty in your value of R .

percentage uncertainty = % [1]

- (e) The experiment is repeated with 20 resistors, each of resistance R , connected in parallel between P and Q. Determine the total current I in the circuit.

I = A [1]

[Total: 15]





Cambridge International AS & A Level

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PHYSICS**9702/41**

Paper 4 A Level Structured Questions

October/November 2025**2 hours**

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [].

This document has **24** pages. Any blank pages are indicated.



Data

acceleration of free fall	$g = 9.81 \text{ m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
unified atomic mass unit	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27} \text{ kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$ $(\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ m F}^{-1})$
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_o = \frac{f_s v}{v \pm v_s}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \dots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$





3

gravitational potential

$$\phi = -\frac{GM}{r}$$

gravitational potential energy

$$E_P = -\frac{GMm}{r}$$

pressure of an ideal gas

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

simple harmonic motion

$$a = -\omega^2 x$$

velocity of particle in s.h.m.

$$v = v_0 \cos \omega t$$

$$v = \pm \omega \sqrt{(x_0^2 - x^2)}$$

electric potential

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

electrical potential energy

$$E_P = \frac{Qq}{4\pi\epsilon_0 r}$$

capacitors in series

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$$

capacitors in parallel

$$C = C_1 + C_2 + \dots$$

discharge of a capacitor

$$x = x_0 e^{-\frac{t}{RC}}$$

Hall voltage

$$V_H = \frac{BI}{ntq}$$

alternating current/voltage

$$x = x_0 \sin \omega t$$

radioactive decay

$$x = x_0 e^{-\lambda t}$$

decay constant

$$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$$

intensity reflection coefficient

$$\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$$

Stefan-Boltzmann law

$$L = 4\pi\sigma r^2 T^4$$

Doppler redshift

$$\frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$$



- 1 (a) In terms of velocity and acceleration, describe uniform circular motion of an object.

.....

.....

..... [2]

- (b) Fig. 1.1 shows the view from above of a polystyrene ball undergoing horizontal circular motion of radius R .

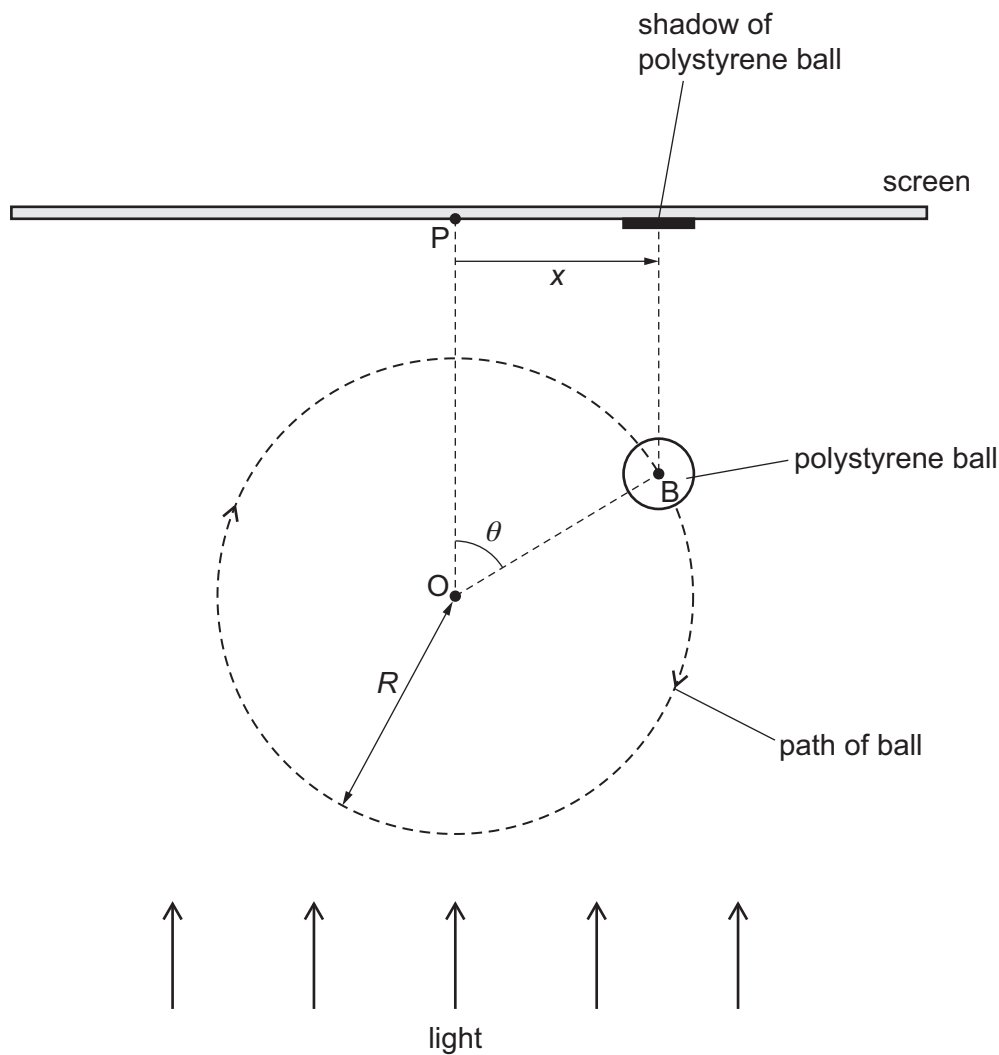


Fig. 1.1

The ball is illuminated by parallel light so that a shadow of the ball forms on a screen placed on the opposite side of the ball from the light source.

The line joining points O and P is perpendicular to the screen.

The angular speed of the circular motion is ω .



- (i) State an expression, in terms of R and ω , for the speed v of the ball.

$$v = \dots\dots\dots [1]$$

- (ii) Determine an expression, in terms of v and ω , for the centripetal acceleration of the ball.

$$\text{centripetal acceleration} = \dots\dots\dots [2]$$

- (c) The ball in (b) is in the position shown in Fig. 1.1, such that line OB is at an angle θ to the line OP.

- (i) Determine an expression, in terms of R and θ , for the displacement x of the shadow from P.

$$x = \dots\dots\dots [1]$$

- (ii) The value of θ is zero at time $t = 0$.

State an expression for θ in terms of ω and t .

$$\theta = \dots\dots\dots [1]$$

- (iii) Use your answers in (c)(i) and (c)(ii) to show that x is given by

$$x = R \sin \omega t.$$

[1]

- (iv) Explain, with reference to the equation in (c)(iii), why the motion of the shadow of the ball on the screen may be modelled as simple harmonic.

.....
 [1]



- (d) The circular motion of the ball in Fig. 1.1 has a diameter of 0.46 m and an angular speed of 1.9 rad s^{-1} .

For the simple harmonic motion of the shadow of the ball in Fig. 1.1, calculate:

- (i) the amplitude

amplitude = m [1]

- (ii) the period

period = s [2]

- (iii) the maximum acceleration.

maximum acceleration = m s^{-2} [2]

- (e) On Fig. 1.1, draw, and label with the letter A, the position of the shadow on the screen when the shadow has its maximum positive acceleration. [1]

[Total: 15]



- 2 (a) State **two** ways in which the first law of thermodynamics describes that the internal energy of a system may be changed.

1

.....

2

.....

[2]

- (b) (i) Use the first law of thermodynamics to explain why a bicycle pump gets hot when it is used to pump up a tyre quickly.

.....

.....

.....

.....

..... [3]

- (ii) With reference to molecular energies, explain why the temperature of water remains at 100 °C when it vaporises in a kettle, even though it is being heated.

.....

.....

.....

.....

..... [3]

[Total: 8]



- 3 (a) Define gravitational field at a point.

.....

..... [1]

- (b) Fig. 3.1 shows an isolated point mass of mass M .

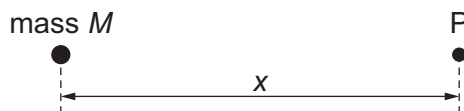


Fig. 3.1

Point P is at distance x from the point mass.

- (i) By considering the force exerted by the point mass on a test mass of mass m placed at P, derive an equation for the gravitational field strength g at P, in terms of M and x . Identify any other symbols you use.

[2]

- (ii) On Fig. 3.1, draw an arrow to indicate the direction of the gravitational field at P. [1]

- (iii) Point Q is at distance $\frac{x}{2}$ from the point mass, on the opposite side of the mass from P, as shown in Fig. 3.2.



Fig. 3.2

Compare the gravitational field at Q with that at P.

.....

.....

..... [2]



- (c) Two identical isolated uniform spheres X and Y each have radius R . The centres of the spheres are separated by distance L , as shown in Fig. 3.3.

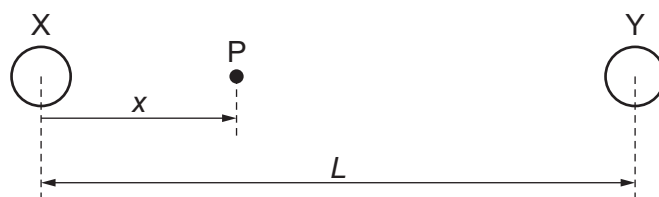


Fig. 3.3

Point P lies on the line joining the centres of X and Y, and is at a variable displacement x from the centre of sphere X.

The gravitational field strength at the surface of each sphere is g_0 .

On Fig. 3.4, sketch the variation with x of the gravitational field g at point P between $x = R$ and $x = L - R$.

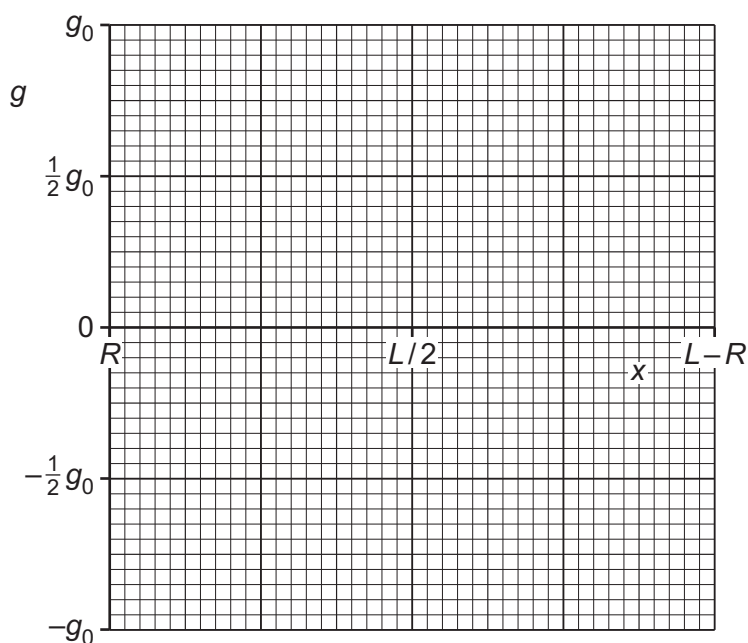


Fig. 3.4

[3]

[Total: 9]



4 (a) State the value of absolute zero on:

(i) the Celsius temperature scale

temperature = °C [1]

(ii) the thermodynamic temperature scale. Give a unit with your answer.

temperature = unit [1]

(b) A sample contains a fixed amount of gas. The gas has pressure p , volume V and thermodynamic temperature T .

Fig. 4.1 shows the variation of pV with kT for the sample, where k is the Boltzmann constant.

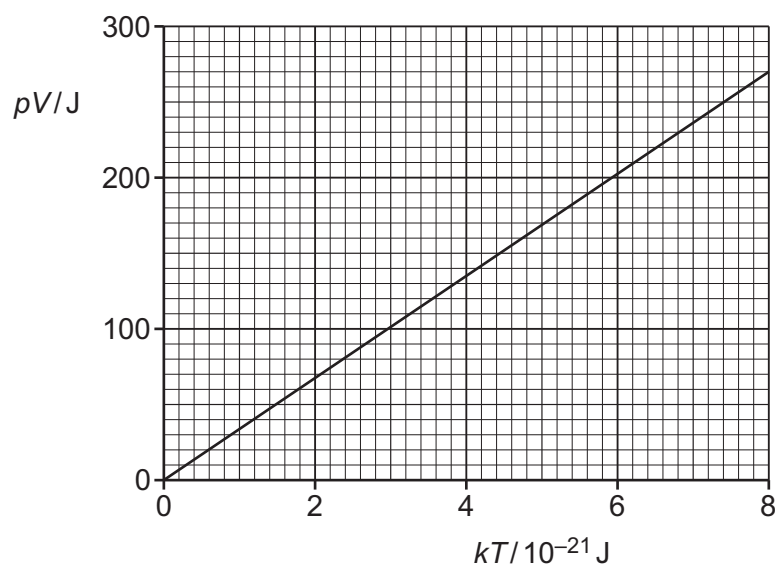


Fig. 4.1

(i) State what is indicated about the nature of the gas from the variation shown in Fig. 4.1.

..... [1]

(ii) Determine the number N of molecules of the gas in the sample.

$N =$ [2]



(iii) Use your answer in (b)(ii) to determine the amount n of gas in the sample.

$$n = \dots\dots\dots \text{mol} \quad [1]$$

- (c) The root-mean-square (r.m.s.) speed of the molecules of the gas is 1900 ms^{-1} when pV is equal to 270 J .

Determine the mass, in u , of one molecule of the gas, where u is the unified atomic mass unit.

$$\text{mass} = \dots\dots\dots u \quad [4]$$

[Total: 10]



- 5 (a) Define electric potential at a point.

.....

.....

..... [2]

- (b) A hydrogen atom may be considered to consist of a proton and an electron separated by a distance of 120 pm, as shown in Fig. 5.1.

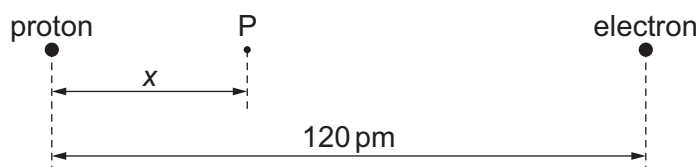


Fig. 5.1

The two particles may be considered as point charges.

Point P lies on the line joining the electron and the proton and is at a variable distance x from the proton.

- (i) Show that the electric potential V at point P when $x = 10$ pm is equal to 130 V.

[2]

- (ii) Calculate, to two significant figures, V when $x = 30$ pm.

$V = \dots\dots\dots V$ [2]

- (iii) On Fig. 5.1, draw a cross (x) at one position, other than infinity, where the electric potential is zero. [1]



(iv) On Fig. 5.2, sketch the variation of V with x between $x = 10\text{pm}$ and $x = 110\text{pm}$.

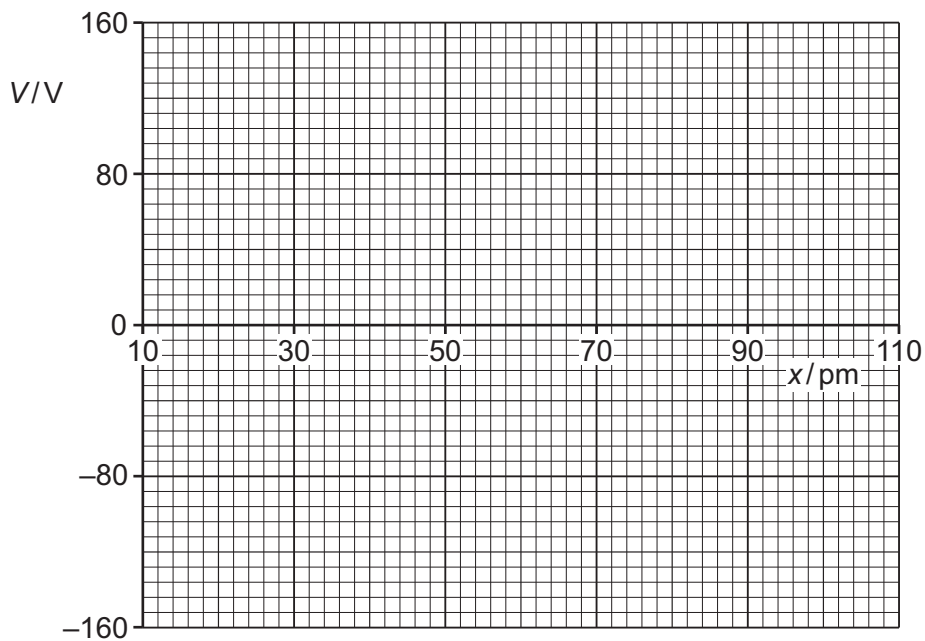


Fig. 5.2

[3]

[Total: 10]



- 6 (a) Two parallel plate capacitors C_1 and C_2 are connected to a supply that has a potential difference (p.d.) V_S . The capacitors may be connected in series or in parallel.

The supply provides charge Q_S and the plates of the two capacitors acquire charges Q_1 and Q_2 respectively. The p.d.s across the plates of the capacitors are V_1 and V_2 respectively.

Complete Table 6.1 to indicate how Q_S , Q_1 and Q_2 relate to each other, and how V_S , V_1 and V_2 relate to each other, for series and parallel connections of the capacitors to the supply.

Table 6.1

	relationship between charges	relationship between p.d.s
series		
parallel		

[4]

- (b) An isolated capacitor of capacitance $470\ \mu\text{F}$ stores 19 mJ of energy.

- (i) Calculate the p.d. across the capacitor.

p.d. =V [2]

- (ii) Calculate the charge on the capacitor.

charge = C [2]





- (iii) The capacitor is now connected in parallel with a capacitor of capacitance $180\mu\text{F}$ that is initially uncharged.

Determine the total energy, in mJ, now stored in the two capacitors.

energy = mJ [3]

[Total: 11]



- 7 (a) State Faraday's law of electromagnetic induction.

.....

.....

..... [2]

- (b) An aircraft is flying horizontally at constant speed v through the Earth's magnetic field, as shown in Fig. 7.1.

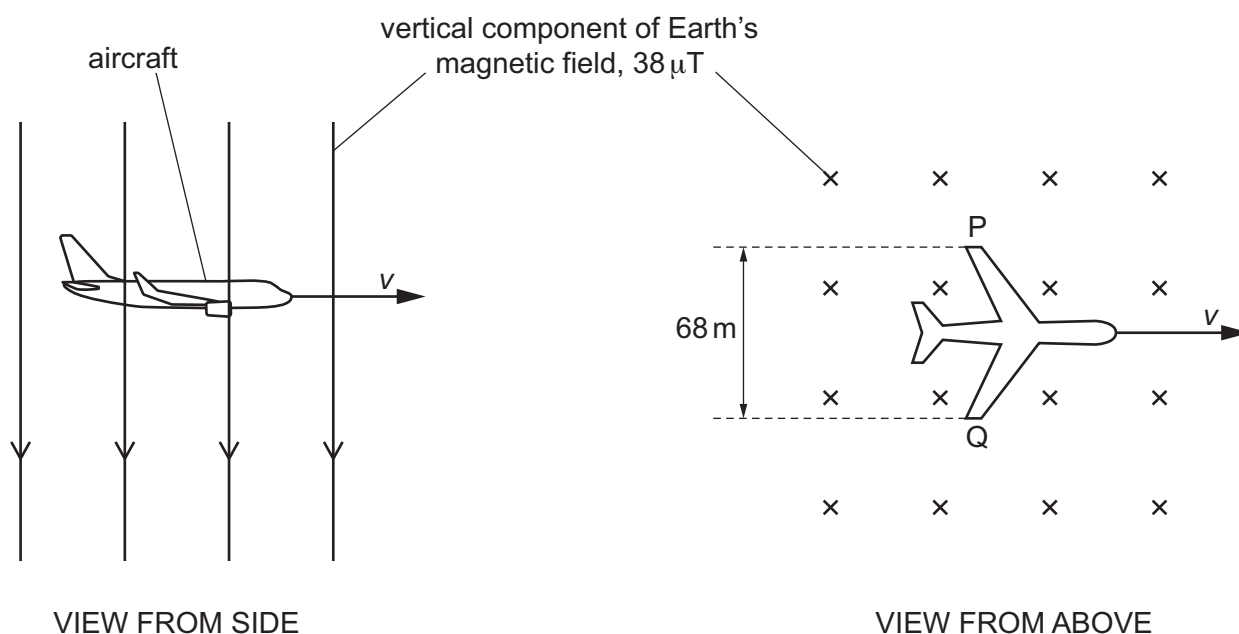


Fig. 7.1

At the location of the aircraft, the vertical component of the Earth's magnetic field is $38\mu\text{T}$ towards the ground.

The distance between the wingtips P and Q of the aircraft is 68 m.

As the aircraft moves through the magnetic field, an electromotive force (e.m.f.) of 0.54 V is induced between the wingtips P and Q.

- (i) Calculate the magnetic flux cut by the wings of the aircraft in a time of 15 s. Give a unit with your answer.

magnetic flux = unit [2]



- (ii) Determine the area of flux cut by the wings in a time of 15 s.

area =m² [2]

- (iii) Use your answer in (b)(ii) to determine the speed v of the aircraft.

v =ms⁻¹ [2]

- (iv) Use Lenz's law of electromagnetic induction to explain which of the wingtips P and Q is at the higher induced potential.

.....

.....

.....

.....

..... [3]

[Total: 11]



- 8 (a) State what is meant by a photon.

.....

.....

..... [2]

- (b) A stationary nucleus of uranium-238 ($^{238}_{92}\text{U}$) undergoes alpha decay to produce a nucleus of thorium-234 ($^{234}_{90}\text{Th}$). The kinetic energy of the emitted alpha particle is 4.200 MeV. A gamma-ray photon is also emitted during the decay.

Assume that the rebound kinetic energy of the thorium nucleus is negligible.

Table 8.1 shows the masses of the nuclides involved in the decay reaction. The mass of the uranium-238 nuclide is missing.

Table 8.1

nuclide	nuclide mass/u
$^4_2\alpha$	4.000 407
$^{234}_{90}\text{Th}$	233.915 174
$^{238}_{92}\text{U}$	

The total energy released in the decay of the nucleus of uranium-238 is 4.274 MeV.

- (i) Calculate the mass, in u, of the uranium-238 nuclide. Give your answer to five decimal places.

mass = u [3]



- (ii) Determine a value for the wavelength of the gamma radiation emitted during the decay of the uranium-238 nucleus.

wavelength = m [3]

- (iii) In practice, the rebound kinetic energy of the thorium nucleus is **not** negligible.

Explain, without further calculation, how your answer in **(b)(ii)** compares with the true wavelength of gamma radiation emitted during the decay of the uranium-238 nucleus.

.....

 [1]

- (c) Gamma radiation emitted during the decay of a sample of uranium-238 has a single wavelength.

Nuclei of cobalt-60 ($^{60}_{27}\text{Co}$) decay by beta emission, and also emit gamma radiation in the process.

Suggest why there is **not** a single wavelength for the gamma radiation emitted during the decay of a sample of cobalt-60.

.....

 [2]

[Total: 11]



9 (a) State Wien's displacement law.

.....

.....

..... [2]

(b) Fig. 9.1 shows the variation with d^{-2} of the radiant flux intensity F observed from a star X, where d is the distance of the observer from the star. Fig. 9.2 shows the variation with wavelength λ of the rates of emission P of radiation by star X and the Sun.

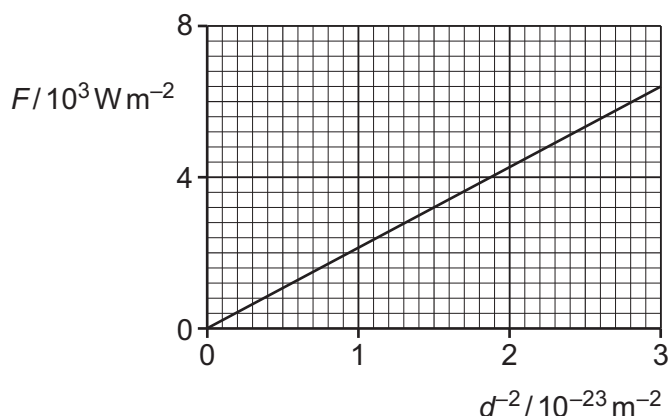


Fig. 9.1

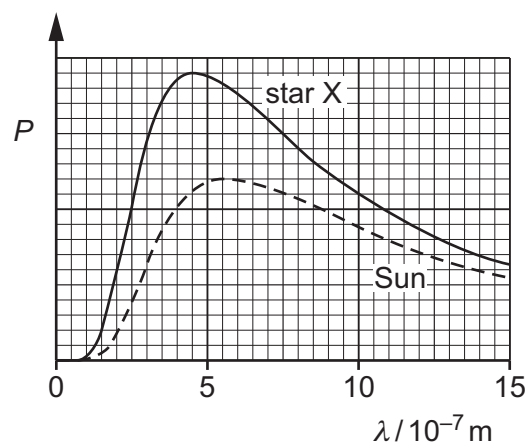


Fig. 9.2

The surface temperature of the Sun is 5770 K.

State **three** conclusions about star X that can be drawn from this data. The conclusions may be qualitative or quantitative. Use the space for any working.

1

.....

2

.....

3

.....



(c) Star X is in a galaxy that is moving away from the Earth.

Suggest, with a reason, how the line for star X in Fig. 9.2 would appear differently if it had been obtained from data measured on the Earth.

.....

.....

..... [2]

[Total: 7]



10 (a) Define specific acoustic impedance.

.....

.....

..... [2]

(b) Explain how ultrasound waves are detected by a piezoelectric crystal.

.....

.....

..... [2]

(c) Table 10.1 shows the specific acoustic impedance Z for body tissue, water and steel.

Table 10.1

material	$Z/\text{kg m}^{-2} \text{ s}^{-1}$
body tissue	1.38×10^6
water	1.48×10^6
steel	4.04×10^7

(i) Calculate the intensity reflection coefficient for ultrasound incident on a water–steel boundary.

intensity reflection coefficient = [2]

(ii) Explain, without calculation, what is likely to happen when ultrasound is incident on a body tissue–water boundary.

.....

.....

..... [2]

[Total: 8]







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PHYSICS

9702/42

Paper 4 A Level Structured Questions

October/November 2025

2 hours

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [].

This document has **24** pages. Any blank pages are indicated.



Data

acceleration of free fall	$g = 9.81 \text{ m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
unified atomic mass unit	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27} \text{ kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$ $(\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ m F}^{-1})$
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_o = \frac{f_s v}{v \pm v_s}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \dots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$





gravitational potential

$$\phi = -\frac{GM}{r}$$

gravitational potential energy

$$E_P = -\frac{GMm}{r}$$

pressure of an ideal gas

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

simple harmonic motion

$$a = -\omega^2 x$$

velocity of particle in s.h.m.

$$v = v_0 \cos \omega t$$

$$v = \pm \omega \sqrt{(x_0^2 - x^2)}$$

electric potential

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

electrical potential energy

$$E_P = \frac{Qq}{4\pi\epsilon_0 r}$$

capacitors in series

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$$

capacitors in parallel

$$C = C_1 + C_2 + \dots$$

discharge of a capacitor

$$x = x_0 e^{-\frac{t}{RC}}$$

Hall voltage

$$V_H = \frac{BI}{ntq}$$

alternating current/voltage

$$x = x_0 \sin \omega t$$

radioactive decay

$$x = x_0 e^{-\lambda t}$$

decay constant

$$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$$

intensity reflection coefficient

$$\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$$

Stefan-Boltzmann law

$$L = 4\pi\sigma r^2 T^4$$

Doppler redshift

$$\frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$$



- 1 The Earth may be considered as a uniform sphere of radius $6.37 \times 10^6 \text{ m}$.

Cambridge is at a point on the Earth's surface that has a latitude of 52.2° north of the Equator, as shown in Fig. 1.1.

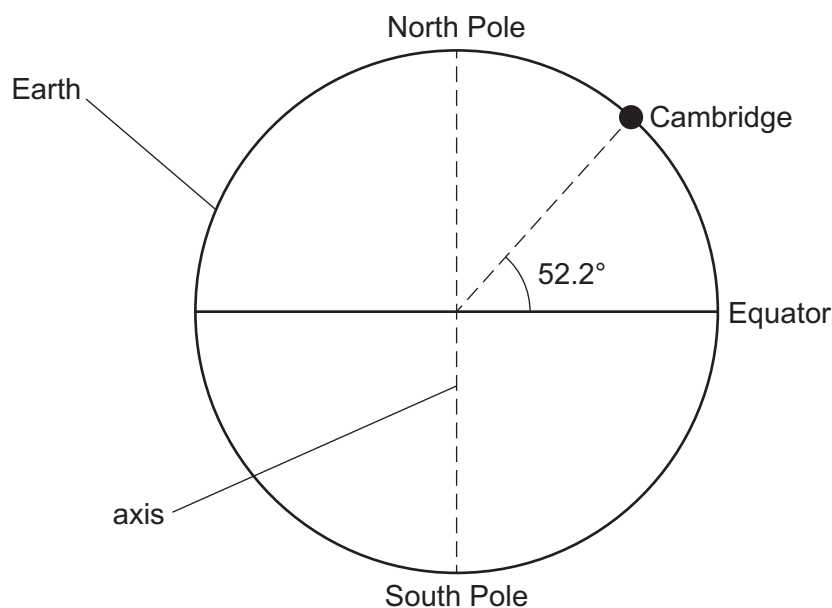


Fig. 1.1

As the Earth spins on its axis, Cambridge moves in a circle that is parallel to the Equator but with a smaller radius.

- (a) (i) Show that the radius of the circle around which Cambridge moves is $3.90 \times 10^6 \text{ m}$.

[1]

- (ii) Calculate the speed at which Cambridge moves around the circle.

speed = ms^{-1} [3]



(b) A student of mass 58.6 kg stands on horizontal ground in Cambridge.

- (i) Determine the magnitude of the resultant force that acts to cause the circular motion of the student.

resultant force = N [2]

- (ii) On Fig. 1.2, draw an arrow to show the direction of the resultant force that acts on the student.

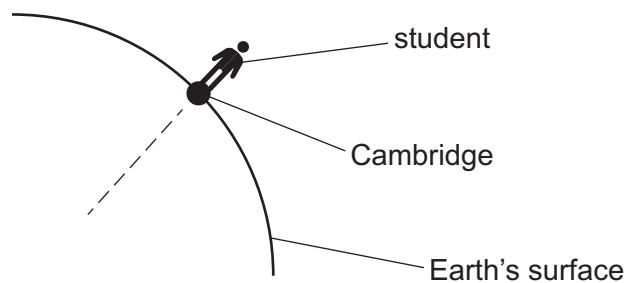


Fig. 1.2 (not to scale)

[1]

- (iii) On Fig. 1.3, draw labelled arrows from the student to show the directions of the forces that act on the student to cause the resultant force in (b)(ii).

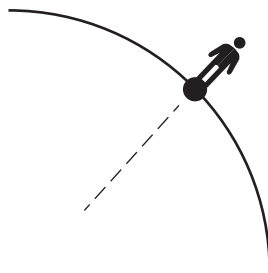


Fig. 1.3 (not to scale)

[2]

[Total: 9]



- 2 (a) State Newton's law of gravitation.

.....

 [2]

- (b) One of the basic assumptions of the kinetic theory of gases is that there are no forces exerted between the molecules of the gas except during collisions.

State **two** other basic assumptions of the kinetic theory of gases.

1

 2
 [2]

- (c) Hydrogen gas consists of molecules that each have a mass of 3.34×10^{-27} kg. Hydrogen may be considered to be an ideal gas.

A spherical balloon contains 0.0160 mol of hydrogen gas at a temperature of 282 K. At this temperature, the volume of gas in the balloon is $1.87 \times 10^{-4} \text{ m}^3$.

- (i) Determine the pressure of the gas.

pressure = Pa [2]

- (ii) Estimate the average separation of the hydrogen molecules in the gas.

average separation = m [2]



- (d) (i) Use your answer in (c)(ii) to calculate the average gravitational force between adjacent molecules in hydrogen gas.

average force = N [2]

- (ii) By considering the weight of a molecule, suggest with a reason whether your answer in (d)(i) is consistent with the assumption of the kinetic theory of gases that there are no forces exerted between molecules.

.....
.....
..... [1]

[Total: 11]



- 3 (a) State what is meant by two objects being in thermal equilibrium.

.....

.....

..... [2]

- (b) Fig. 3.1 shows a type of thermometer called a constant volume gas thermometer.

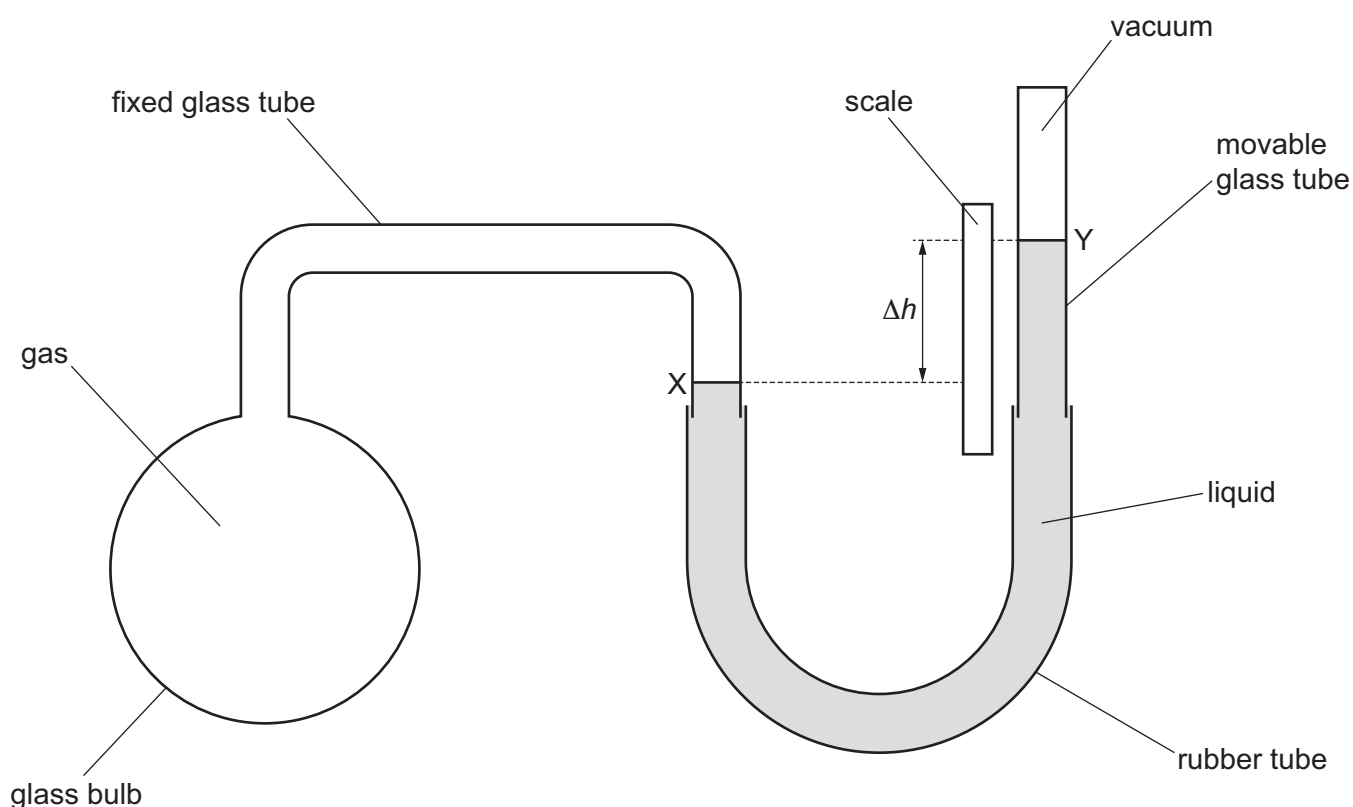


Fig. 3.1 (not to scale)

The thermometer is used to determine the thermodynamic temperature T of the gas in the glass bulb.

The glass bulb is immersed in the environment for which the temperature is to be measured. The height of the movable glass tube is then adjusted so that the level of the liquid on the left-hand side aligns with the reference line X marked on the fixed glass tube. The reference line Y is marked on the side of the movable glass tube. The level of the liquid at Y is higher than at X as a result of the pressure of the gas in the glass bulb.

The difference in height Δh between the liquid levels at X and Y is then measured using the scale. The thermodynamic temperature T of the gas is directly proportional to the pressure of the gas. This pressure is directly proportional to Δh .



- (i) The value of Δh can be used to calculate the pressure of the gas. In order to do this, the gravitational field strength is used, along with a property of the liquid.

State the property of the liquid that is used to calculate the pressure.

..... [1]

- (ii) Before the measurement of Δh can be made, the glass bulb needs to reach thermal equilibrium with the environment for which the temperature is to be measured.

State **two** disadvantages of using a constant volume gas thermometer to measure temperature.

1

.....

2

.....

[2]

- (iii) Suggest **one** situation in which a constant volume gas thermometer would be an appropriate type of thermometer to choose for measuring temperature.

.....

.....

..... [1]

- (iv) Level X aligns with 2.31 cm on the scale. At 0°C , level Y aligns with 8.69 cm.

At temperature θ , level Y aligns with 7.83 cm on the scale.

Determine a value for θ in $^\circ\text{C}$.

$\theta =$ $^\circ\text{C}$ [3]

[Total: 9]



- 4 A cylinder contains a fixed mass of an ideal gas at pressure $2Y$ and volume $6X$.

The gas undergoes a sequence of changes from its initial state A, through states B, C and D, then finally back to its initial state A, as shown in Fig. 4.1.

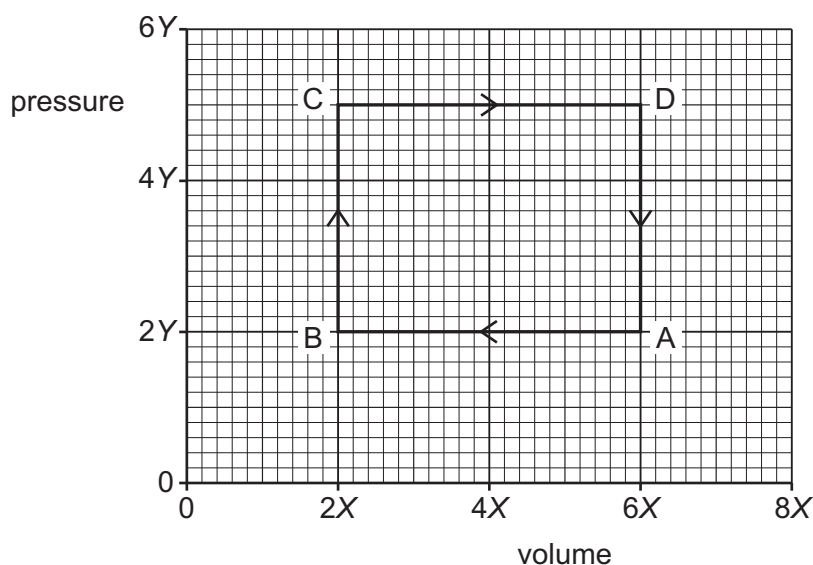


Fig. 4.1

Fig. 4.2 shows the variation with time of the internal energy of the gas.

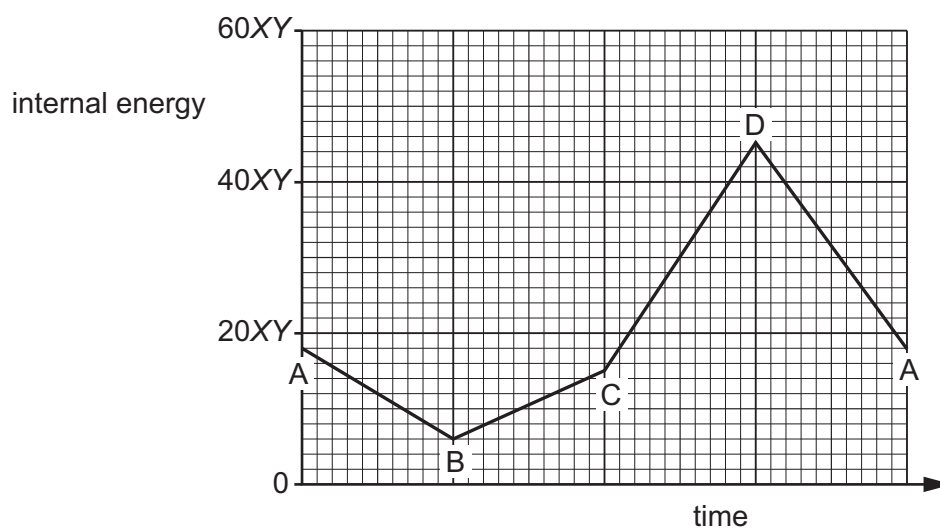


Fig. 4.2

- (a) State the first law of thermodynamics.

.....

.....

..... [2]



- (b) (i) Use Fig. 4.1 and Fig. 4.2 to determine the general expression for the internal energy U of the gas when it has pressure p and volume V .

$$U = \dots\dots\dots [1]$$

- (ii) An ideal gas at thermodynamic temperature T contains N molecules.

Use your answer in (b)(i) and the equation of state for an ideal gas to deduce an expression for U in terms of N and T . Identify any other symbols you use.

$$U = \dots\dots\dots [2]$$

- (c) Determine expressions, in terms of X and Y , for the work W done on the gas during:

- (i) change AB

$$W = \dots\dots\dots [1]$$

- (ii) change CD.

$$W = \dots\dots\dots [1]$$

- (d) Use your answers in (c) and the first law of thermodynamics to determine an expression, in terms of X and Y , for the net thermal energy Q supplied to the gas during one full cycle ABCDA. Explain your reasoning.

$$Q = \dots\dots\dots [3]$$

[Total: 10]



- 5 A steel ball on the end of a thin string oscillates with small oscillations, as shown in Fig. 5.1.

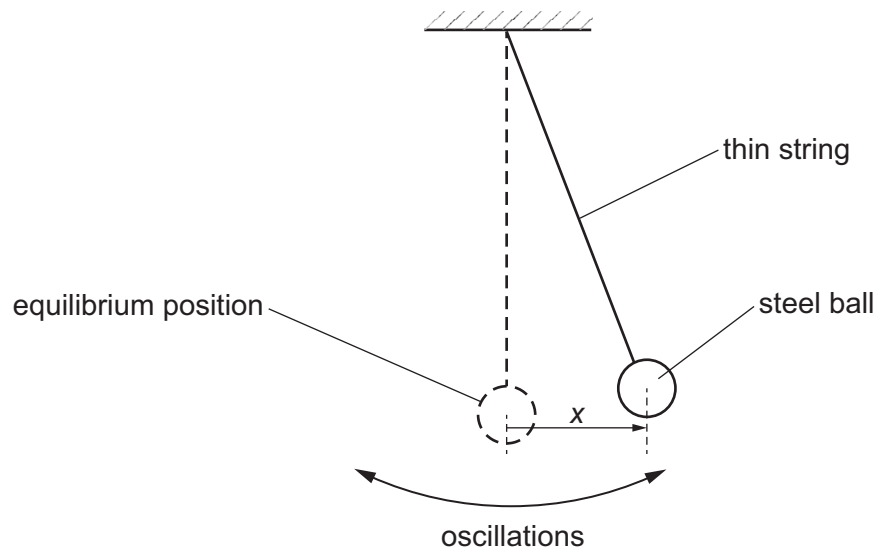


Fig. 5.1 (not to scale)

The displacement of the centre of the ball from its equilibrium position is x .

- (a) Fig. 5.2 shows the variation with x of the acceleration a of the ball.

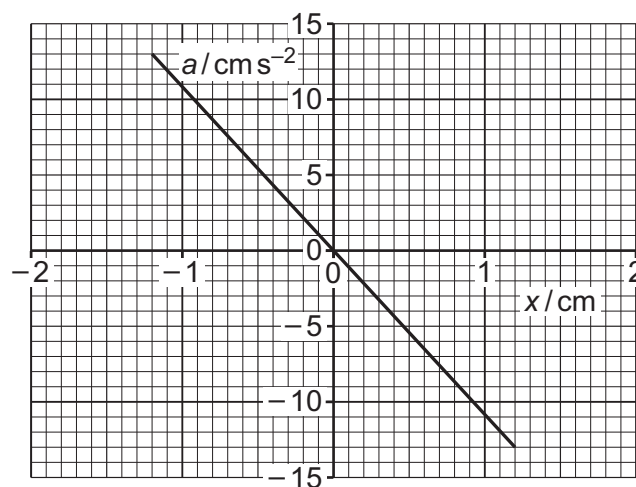


Fig. 5.2

- (i) Explain how Fig. 5.2 shows that the oscillations of the ball are simple harmonic.

.....

 [2]



(ii) Determine the period T of the oscillations.

$T = \dots\dots\dots$ s [3]

(b) At time $t = 0$, when the displacement of the ball has its maximum value, the ball is immersed in a trough containing thick oil so that the ball is just below the surface of the oil. This results in the subsequent motion of the ball being heavily damped.

(i) State what is meant by damping.

.....

 [2]

(ii) On Fig. 5.3, sketch a possible variation of the displacement x of the ball with t between $t = 0$ and $t = 2T$.

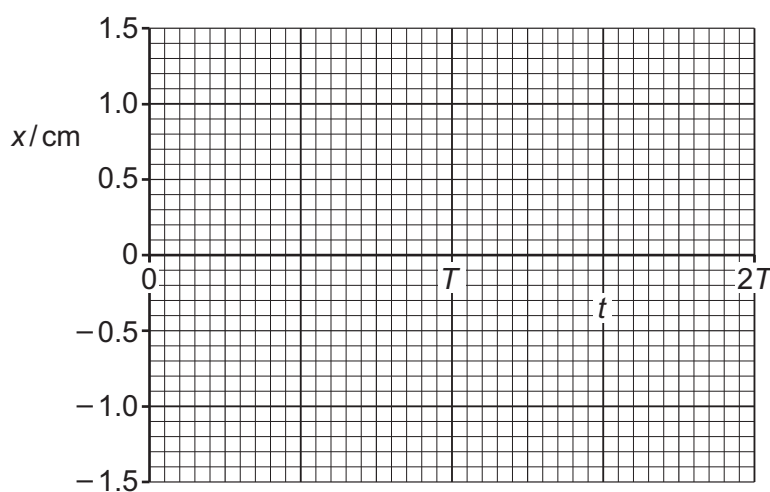


Fig. 5.3

[3]

[Total: 10]



- 6 (a) Define electric field at a point.

.....
 [1]

- (b) An isolated conducting sphere in a vacuum has a capacitance of 69 pF.
 The charge on the sphere is +83 pC.

- (i) On Fig. 6.1, draw field lines to represent the electric field outside the sphere due to the charge on the sphere.

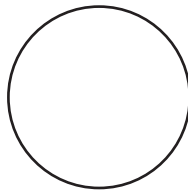


Fig. 6.1

[2]

- (ii) Calculate the electric potential at the surface of the sphere.

electric potential = V [2]

- (iii) Determine the radius of the sphere.

radius = m [2]



- (iv) Calculate the electric field strength E at the surface of the sphere. Give a unit with your answer.

$E =$ unit [2]

- (c) The sphere in (b) is discharged by connecting it to earth (0 V) through a resistor of resistance $120 \text{ M}\Omega$.

Calculate the time taken for the charge to fall to 26 pC .

time = s [2]

[Total: 11]



- 7 An alternating voltage V varies with time t according to

$$V = 18 \cos 40\pi t$$

where V is in V and t is in s.

- (a) For the alternating voltage:

- (i) show that the period is 0.050 s

[1]

- (ii) determine the root-mean-square (r.m.s.) voltage.

r.m.s. voltage = V [1]

- (b) On Fig. 7.1, sketch the variation of V with t for values of t from $t = 0$ to $t = 100$ ms.

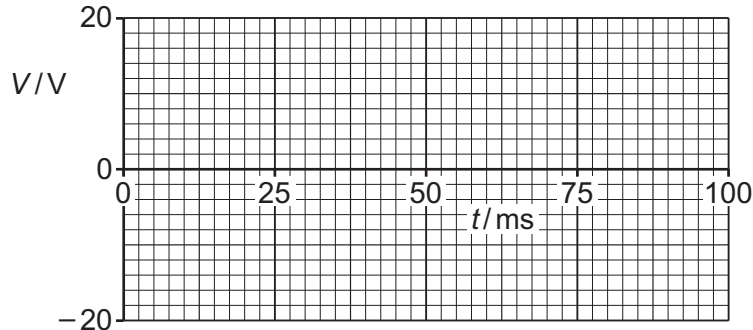


Fig. 7.1

[3]

- (c) The alternating voltage is rectified to produce an output voltage across a load resistor R , as shown in Fig. 7.2.

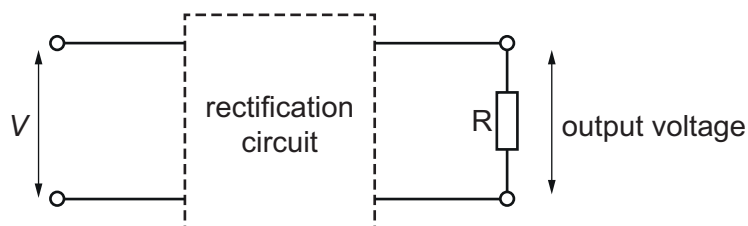


Fig. 7.2



Fig. 7.3 shows the variation with t of the power P in the load resistor.

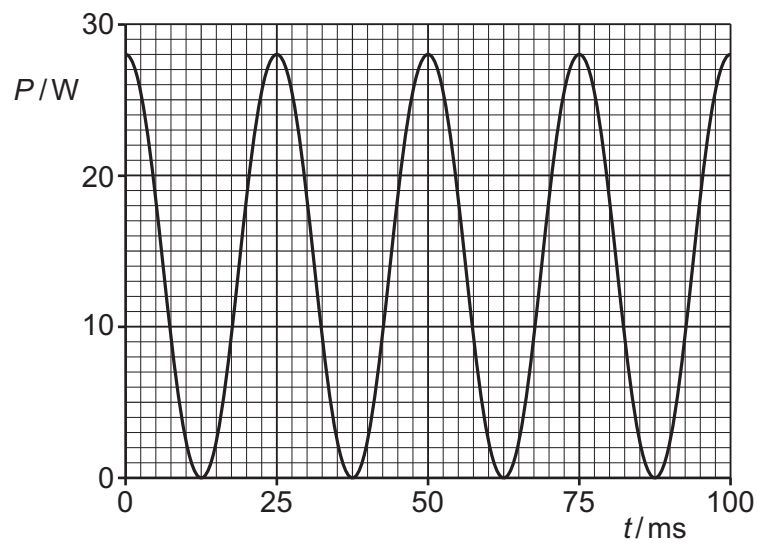


Fig. 7.3

State **three** conclusions that can be drawn from Fig. 7.3. The conclusions may be qualitative or quantitative. Use the space for any working.

1

.....

2

.....

3

.....

[3]

[Total: 8]



- 8 Fig. 8.1 shows the three lowest-frequency lines in the part of the emission spectrum for hydrogen that relates to electron transitions to the ground state (level $n = 1$).

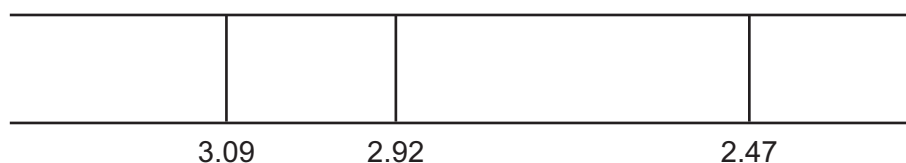


Fig. 8.1 (not to scale)

The numbers represent the frequencies, in 10^{15} Hz, associated with the spectral lines.

- (a) Use the photon model of electromagnetic radiation to explain how the existence of spectral lines in the emission spectrum provides evidence for discrete electron energy levels in the hydrogen atom.

.....

 [3]

- (b) The energy of the ground state (level $n = 1$) in a hydrogen atom is -13.6 eV.

- (i) Calculate the energy, in J, of the ground state.

energy = J [1]

- (ii) Show that the energy difference between levels $n = 1$ and $n = 2$ is 10.2 eV.

[2]



- (iii) Complete Table 8.1 to show the energy differences from the ground state, and the energies of the levels up to $n = 4$, in the hydrogen atom. Use the space for any working.

Table 8.1

level	(energy difference from $n = 1$)/eV	energy/eV
$n = 4$		
$n = 3$		
$n = 2$	10.2	
$n = 1$	0.0	-13.6

[4]

[Total: 10]



- 9 (a) State what is meant by the mass defect of a nucleus.

.....

.....

..... [2]

- (b) The nuclear fusion reaction for the formation of helium-4 from deuterium is represented by

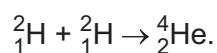


Table 9.1 shows the masses of the nuclides involved in this reaction.

Table 9.1

nuclide	nuclide mass/u
${}^2_1\text{H}$	2.013553
${}^4_2\text{He}$	4.001505

Calculate the energy released in the formation of 1.00 mol of helium-4.

energy = J [4]



- (c) The star Sirius has a radius of $1.19 \times 10^9 \text{ m}$ and loses mass due to nuclear fusion at a rate of $1.09 \times 10^{11} \text{ kg s}^{-1}$. Assume that the power of the radiation emitted by the star is equal to the power released by this process.

(i) Determine a value for the luminosity of Sirius. Give a unit with your answer.

luminosity = unit [2]

(ii) Use your answer in (c)(i) to determine the surface temperature of Sirius.

surface temperature = K [2]

- (d) Explain how cosmologists use standard candles to estimate the distance of a galaxy from the Earth.

.....

.....

.....

.....

..... [3]

[Total: 13]



- 10 (a) State what is meant by contrast in an X-ray image.

.....
 [1]

- (b) X-rays of intensity I_0 are incident normally on a structure, as shown in Fig. 10.1.

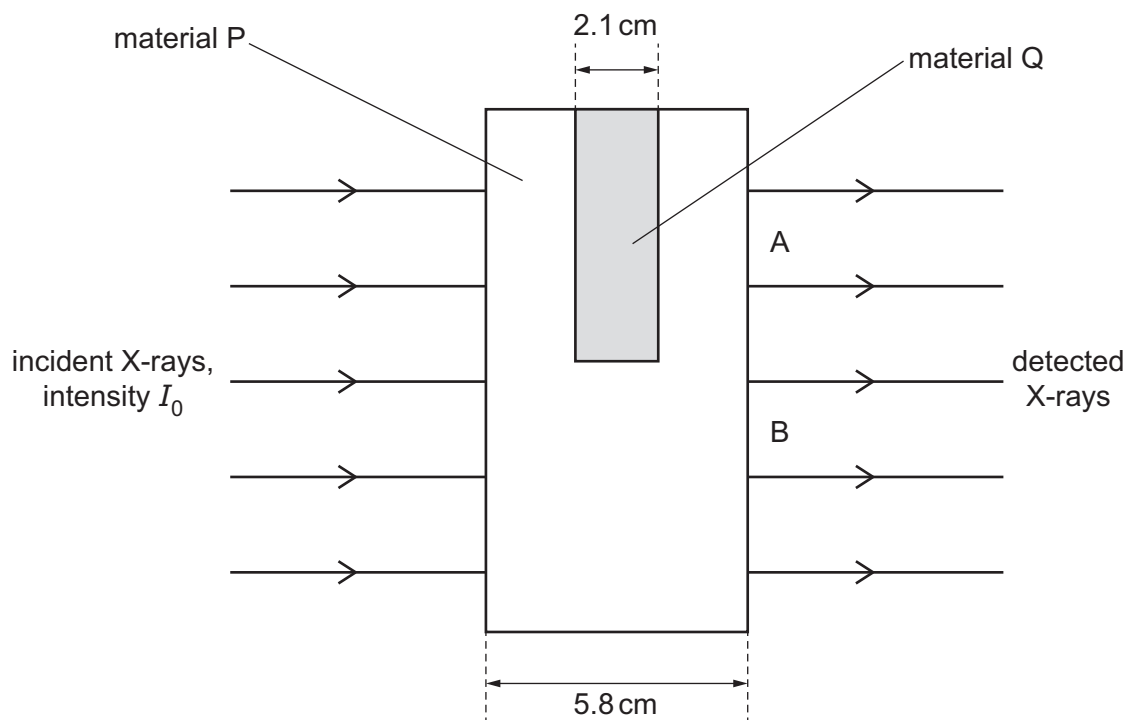


Fig. 10.1

Material P has a linear attenuation coefficient of 0.35 cm^{-1} .

The X-rays emerging from the structure in region A have an intensity of $0.053I_0$.

- (i) Show that the intensity of the X-rays emerging in region B is $0.13I_0$.

[1]





(ii) Determine the linear attenuation coefficient μ of material Q.

$\mu = \dots\dots\dots \text{cm}^{-1}$ [3]

(iii) Use the information in (b)(i) to suggest why the X-rays emerging from the structure form an image that has poor contrast.

.....
.....
..... [1]

(c) Explain how X-rays are used in computed tomography (CT) scanning to produce a three-dimensional image of an internal structure.

.....
.....
.....
.....
..... [3]

[Total: 9]





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Cambridge International AS & A Level

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PHYSICS**9702/43**

Paper 4 A Level Structured Questions

October/November 2025**2 hours**

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [].

This document has **24** pages. Any blank pages are indicated.



Data

acceleration of free fall	$g = 9.81 \text{ m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
unified atomic mass unit	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27} \text{ kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$ $(\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ m F}^{-1})$
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_o = \frac{f_s v}{v \pm v_s}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \dots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$





gravitational potential

$$\phi = -\frac{GM}{r}$$

gravitational potential energy

$$E_P = -\frac{GMm}{r}$$

pressure of an ideal gas

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

simple harmonic motion

$$a = -\omega^2 x$$

velocity of particle in s.h.m.

$$v = v_0 \cos \omega t$$

$$v = \pm \omega \sqrt{(x_0^2 - x^2)}$$

electric potential

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

electrical potential energy

$$E_P = \frac{Qq}{4\pi\epsilon_0 r}$$

capacitors in series

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$$

capacitors in parallel

$$C = C_1 + C_2 + \dots$$

discharge of a capacitor

$$x = x_0 e^{-\frac{t}{RC}}$$

Hall voltage

$$V_H = \frac{BI}{ntq}$$

alternating current/voltage

$$x = x_0 \sin \omega t$$

radioactive decay

$$x = x_0 e^{-\lambda t}$$

decay constant

$$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$$

intensity reflection coefficient

$$\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$$

Stefan-Boltzmann law

$$L = 4\pi\sigma r^2 T^4$$

Doppler redshift

$$\frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$$



- 1 (a) In terms of velocity and acceleration, describe uniform circular motion of an object.

.....

.....

..... [2]

- (b) Fig. 1.1 shows the view from above of a polystyrene ball undergoing horizontal circular motion of radius R .

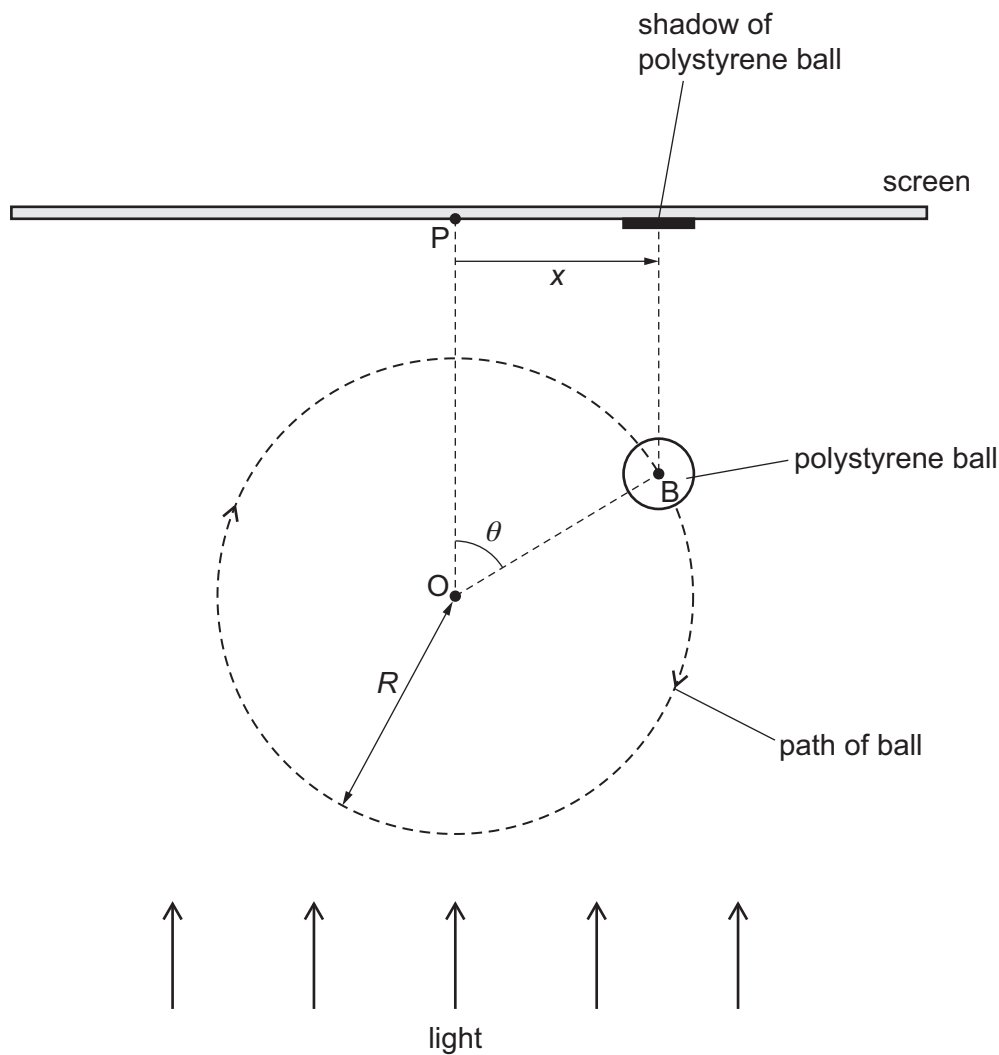


Fig. 1.1

The ball is illuminated by parallel light so that a shadow of the ball forms on a screen placed on the opposite side of the ball from the light source.

The line joining points O and P is perpendicular to the screen.

The angular speed of the circular motion is ω .



- (i) State an expression, in terms of R and ω , for the speed v of the ball.

$$v = \dots\dots\dots [1]$$

- (ii) Determine an expression, in terms of v and ω , for the centripetal acceleration of the ball.

$$\text{centripetal acceleration} = \dots\dots\dots [2]$$

- (c) The ball in (b) is in the position shown in Fig. 1.1, such that line OB is at an angle θ to the line OP.

- (i) Determine an expression, in terms of R and θ , for the displacement x of the shadow from P.

$$x = \dots\dots\dots [1]$$

- (ii) The value of θ is zero at time $t = 0$.

State an expression for θ in terms of ω and t .

$$\theta = \dots\dots\dots [1]$$

- (iii) Use your answers in (c)(i) and (c)(ii) to show that x is given by

$$x = R \sin \omega t.$$

[1]

- (iv) Explain, with reference to the equation in (c)(iii), why the motion of the shadow of the ball on the screen may be modelled as simple harmonic.

.....

..... [1]



- (d) The circular motion of the ball in Fig. 1.1 has a diameter of 0.46 m and an angular speed of 1.9 rad s^{-1} .

For the simple harmonic motion of the shadow of the ball in Fig. 1.1, calculate:

- (i) the amplitude

amplitude = m [1]

- (ii) the period

period = s [2]

- (iii) the maximum acceleration.

maximum acceleration = m s^{-2} [2]

- (e) On Fig. 1.1, draw, and label with the letter A, the position of the shadow on the screen when the shadow has its maximum positive acceleration. [1]

[Total: 15]



- 2 (a) State **two** ways in which the first law of thermodynamics describes that the internal energy of a system may be changed.

1

.....

2

.....

[2]

- (b) (i) Use the first law of thermodynamics to explain why a bicycle pump gets hot when it is used to pump up a tyre quickly.

.....

.....

.....

.....

..... [3]

- (ii) With reference to molecular energies, explain why the temperature of water remains at 100 °C when it vaporises in a kettle, even though it is being heated.

.....

.....

.....

.....

..... [3]

[Total: 8]



- 3 (a) Define gravitational field at a point.

.....

..... [1]

- (b) Fig. 3.1 shows an isolated point mass of mass M .

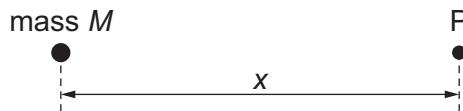


Fig. 3.1

Point P is at distance x from the point mass.

- (i) By considering the force exerted by the point mass on a test mass of mass m placed at P, derive an equation for the gravitational field strength g at P, in terms of M and x . Identify any other symbols you use.

[2]

- (ii) On Fig. 3.1, draw an arrow to indicate the direction of the gravitational field at P. [1]

- (iii) Point Q is at distance $\frac{x}{2}$ from the point mass, on the opposite side of the mass from P, as shown in Fig. 3.2.



Fig. 3.2

Compare the gravitational field at Q with that at P.

.....

.....

..... [2]



- (c) Two identical isolated uniform spheres X and Y each have radius R . The centres of the spheres are separated by distance L , as shown in Fig. 3.3.

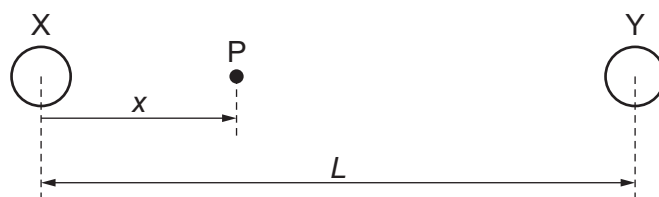


Fig. 3.3

Point P lies on the line joining the centres of X and Y, and is at a variable displacement x from the centre of sphere X.

The gravitational field strength at the surface of each sphere is g_0 .

On Fig. 3.4, sketch the variation with x of the gravitational field g at point P between $x = R$ and $x = L - R$.

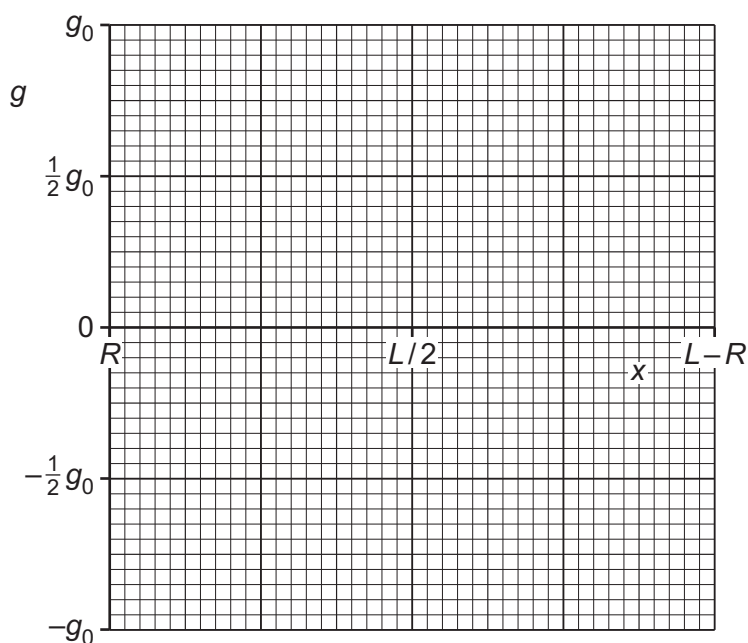


Fig. 3.4

[3]

[Total: 9]



4 (a) State the value of absolute zero on:

(i) the Celsius temperature scale

temperature = °C [1]

(ii) the thermodynamic temperature scale. Give a unit with your answer.

temperature = unit [1]

(b) A sample contains a fixed amount of gas. The gas has pressure p , volume V and thermodynamic temperature T .

Fig. 4.1 shows the variation of pV with kT for the sample, where k is the Boltzmann constant.

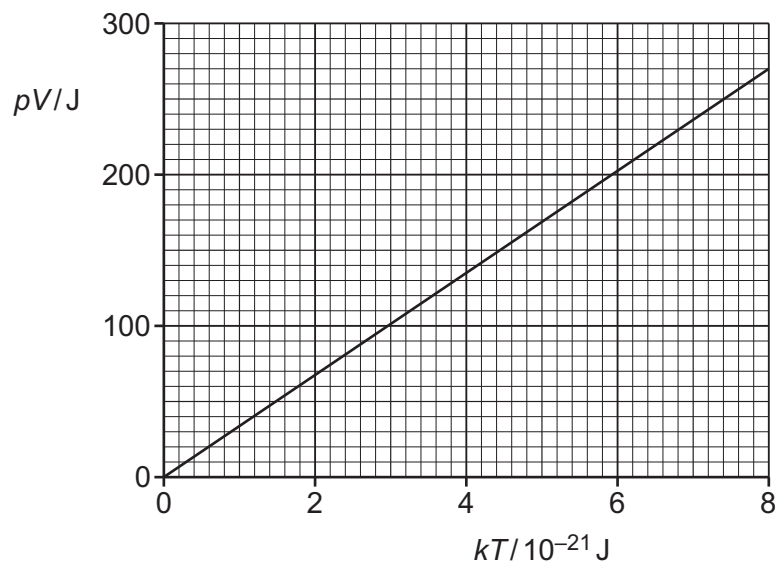


Fig. 4.1

(i) State what is indicated about the nature of the gas from the variation shown in Fig. 4.1.

..... [1]

(ii) Determine the number N of molecules of the gas in the sample.

$N =$ [2]



(iii) Use your answer in (b)(ii) to determine the amount n of gas in the sample.

$$n = \dots\dots\dots \text{mol} \quad [1]$$

- (c) The root-mean-square (r.m.s.) speed of the molecules of the gas is 1900 ms^{-1} when pV is equal to 270 J .

Determine the mass, in u , of one molecule of the gas, where u is the unified atomic mass unit.

$$\text{mass} = \dots\dots\dots u \quad [4]$$

[Total: 10]



- 5 (a) Define electric potential at a point.

.....

.....

..... [2]

- (b) A hydrogen atom may be considered to consist of a proton and an electron separated by a distance of 120 pm, as shown in Fig. 5.1.

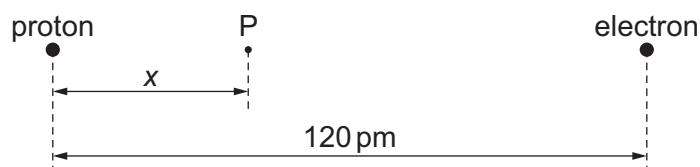


Fig. 5.1

The two particles may be considered as point charges.

Point P lies on the line joining the electron and the proton and is at a variable distance x from the proton.

- (i) Show that the electric potential V at point P when $x = 10$ pm is equal to 130 V.

[2]

- (ii) Calculate, to two significant figures, V when $x = 30$ pm.

$V = \dots\dots\dots V$ [2]

- (iii) On Fig. 5.1, draw a cross (x) at one position, other than infinity, where the electric potential is zero. [1]



(iv) On Fig. 5.2, sketch the variation of V with x between $x = 10\text{ pm}$ and $x = 110\text{ pm}$.

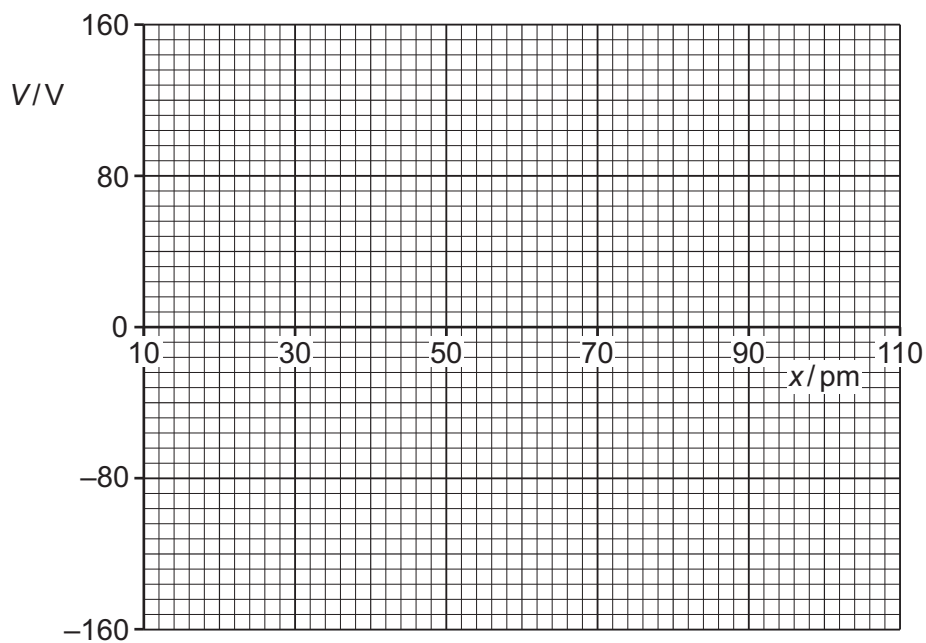


Fig. 5.2

[3]

[Total: 10]



- 6 (a) Two parallel plate capacitors C_1 and C_2 are connected to a supply that has a potential difference (p.d.) V_S . The capacitors may be connected in series or in parallel.

The supply provides charge Q_S and the plates of the two capacitors acquire charges Q_1 and Q_2 respectively. The p.d.s across the plates of the capacitors are V_1 and V_2 respectively.

Complete Table 6.1 to indicate how Q_S , Q_1 and Q_2 relate to each other, and how V_S , V_1 and V_2 relate to each other, for series and parallel connections of the capacitors to the supply.

Table 6.1

	relationship between charges	relationship between p.d.s
series		
parallel		

[4]

- (b) An isolated capacitor of capacitance $470\ \mu\text{F}$ stores 19 mJ of energy.

- (i) Calculate the p.d. across the capacitor.

p.d. =V [2]

- (ii) Calculate the charge on the capacitor.

charge = C [2]



- (iii) The capacitor is now connected in parallel with a capacitor of capacitance $180\mu\text{F}$ that is initially uncharged.

Determine the total energy, in mJ, now stored in the two capacitors.

energy = mJ [3]

[Total: 11]



- 7 (a) State Faraday's law of electromagnetic induction.

.....

.....

..... [2]

- (b) An aircraft is flying horizontally at constant speed v through the Earth's magnetic field, as shown in Fig. 7.1.

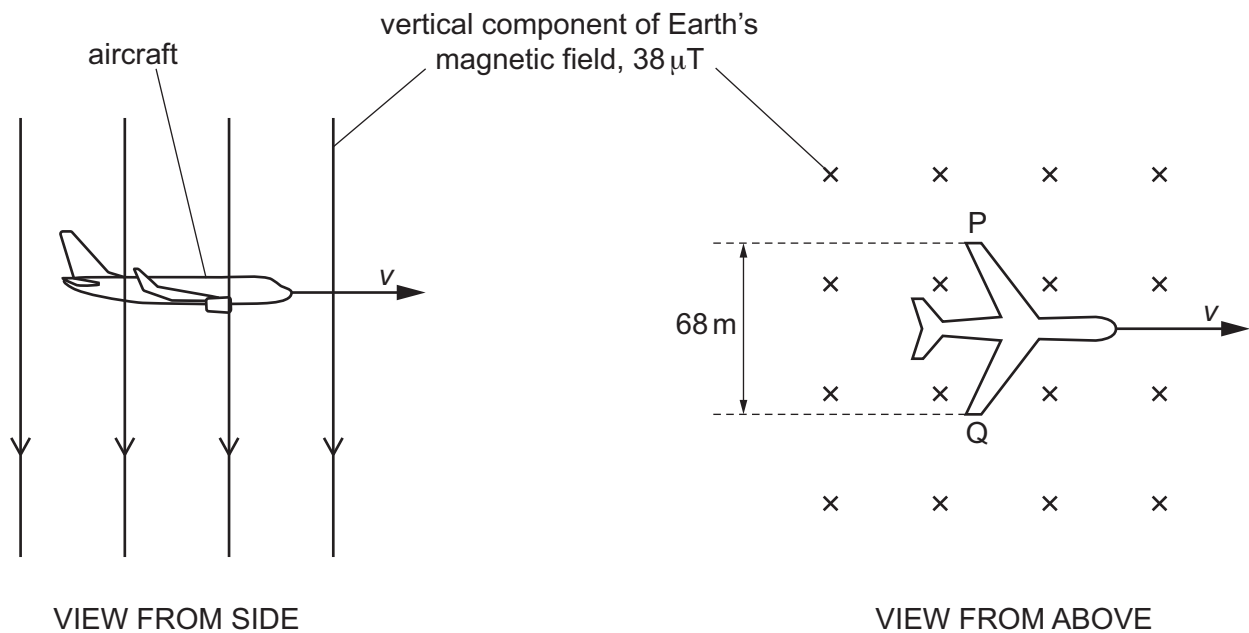


Fig. 7.1

At the location of the aircraft, the vertical component of the Earth's magnetic field is $38\mu\text{T}$ towards the ground.

The distance between the wingtips P and Q of the aircraft is 68 m.

As the aircraft moves through the magnetic field, an electromotive force (e.m.f.) of 0.54 V is induced between the wingtips P and Q.

- (i) Calculate the magnetic flux cut by the wings of the aircraft in a time of 15 s. Give a unit with your answer.

magnetic flux = unit [2]



- (ii) Determine the area of flux cut by the wings in a time of 15 s.

area =m² [2]

- (iii) Use your answer in (b)(ii) to determine the speed v of the aircraft.

v =ms⁻¹ [2]

- (iv) Use Lenz's law of electromagnetic induction to explain which of the wingtips P and Q is at the higher induced potential.

.....

.....

.....

.....

..... [3]

[Total: 11]



- 8 (a) State what is meant by a photon.

.....

.....

..... [2]

- (b) A stationary nucleus of uranium-238 ($^{238}_{92}\text{U}$) undergoes alpha decay to produce a nucleus of thorium-234 ($^{234}_{90}\text{Th}$). The kinetic energy of the emitted alpha particle is 4.200 MeV. A gamma-ray photon is also emitted during the decay.

Assume that the rebound kinetic energy of the thorium nucleus is negligible.

Table 8.1 shows the masses of the nuclides involved in the decay reaction. The mass of the uranium-238 nuclide is missing.

Table 8.1

nuclide	nuclide mass/u
$^4_2\alpha$	4.000 407
$^{234}_{90}\text{Th}$	233.915 174
$^{238}_{92}\text{U}$	

The total energy released in the decay of the nucleus of uranium-238 is 4.274 MeV.

- (i) Calculate the mass, in u, of the uranium-238 nuclide. Give your answer to five decimal places.

mass = u [3]



- (ii) Determine a value for the wavelength of the gamma radiation emitted during the decay of the uranium-238 nucleus.

wavelength = m [3]

- (iii) In practice, the rebound kinetic energy of the thorium nucleus is **not** negligible.

Explain, without further calculation, how your answer in **(b)(ii)** compares with the true wavelength of gamma radiation emitted during the decay of the uranium-238 nucleus.

.....

 [1]

- (c) Gamma radiation emitted during the decay of a sample of uranium-238 has a single wavelength.

Nuclei of cobalt-60 ($^{60}_{27}\text{Co}$) decay by beta emission, and also emit gamma radiation in the process.

Suggest why there is **not** a single wavelength for the gamma radiation emitted during the decay of a sample of cobalt-60.

.....

 [2]

[Total: 11]



9 (a) State Wien's displacement law.

.....

 [2]

(b) Fig. 9.1 shows the variation with d^{-2} of the radiant flux intensity F observed from a star X, where d is the distance of the observer from the star. Fig. 9.2 shows the variation with wavelength λ of the rates of emission P of radiation by star X and the Sun.

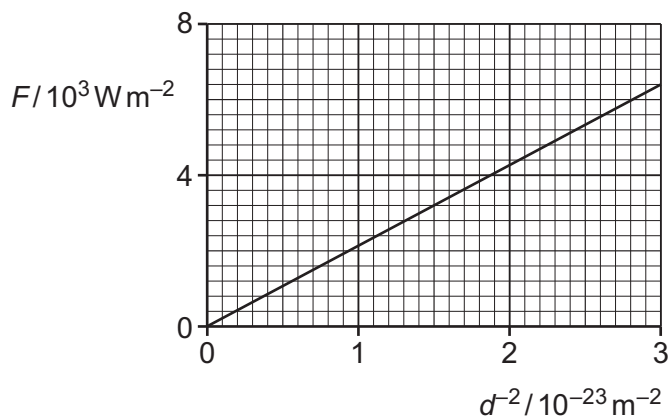


Fig. 9.1

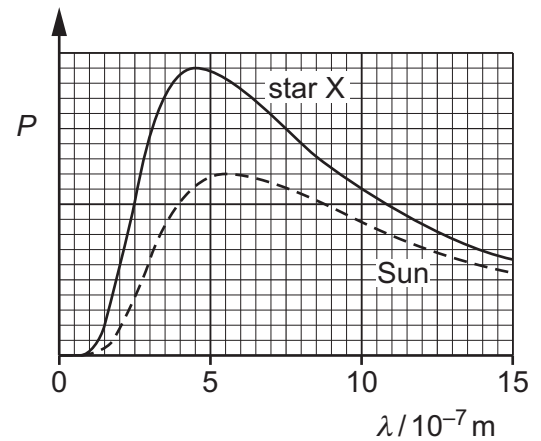


Fig. 9.2

The surface temperature of the Sun is 5770 K.

State **three** conclusions about star X that can be drawn from this data. The conclusions may be qualitative or quantitative. Use the space for any working.

1

 2

 3



(c) Star X is in a galaxy that is moving away from the Earth.

Suggest, with a reason, how the line for star X in Fig. 9.2 would appear differently if it had been obtained from data measured on the Earth.

.....

.....

..... [2]

[Total: 7]



10 (a) Define specific acoustic impedance.

.....

 [2]

(b) Explain how ultrasound waves are detected by a piezoelectric crystal.

.....

 [2]

(c) Table 10.1 shows the specific acoustic impedance Z for body tissue, water and steel.

Table 10.1

material	$Z/\text{kg m}^{-2} \text{s}^{-1}$
body tissue	1.38×10^6
water	1.48×10^6
steel	4.04×10^7

(i) Calculate the intensity reflection coefficient for ultrasound incident on a water–steel boundary.

intensity reflection coefficient = [2]

(ii) Explain, without calculation, what is likely to happen when ultrasound is incident on a body tissue–water boundary.

.....

 [2]

[Total: 8]







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PHYSICS

9702/51

Paper 5 Planning, Analysis and Evaluation

October/November 2025

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 30.
- The number of marks for each question or part question is shown in brackets [].

This document has 8 pages.

- 1 On a bench, a steel ball of radius r is used to compress a spring by a distance x . The ball is held at rest in this position, as shown in Fig. 1.1.

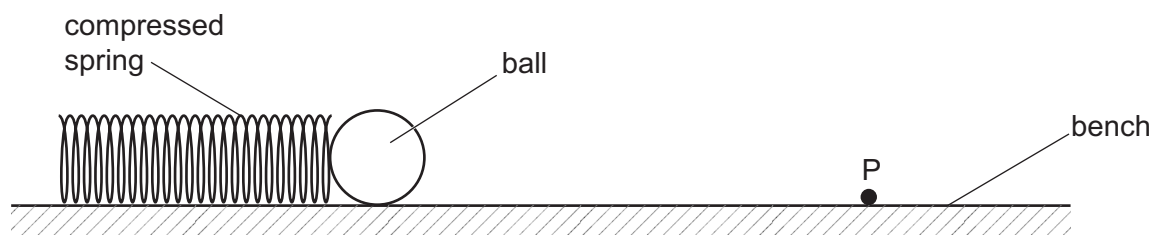


Fig. 1.1

The ball is released and rolls along the bench. At a fixed point P, the ball has speed v . The speed of the ball at P is determined using **one** light gate connected to a timer.

Several steel balls of different radii are available.

It is suggested that v is related to r by the relationship

$$v^2 = \frac{Ykx^2}{r^n \rho}$$

where k is the spring constant of the spring, ρ is the density of the steel, and Y and n are constants.

Plan a laboratory experiment to test the relationship between v and r .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for Y and n .

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.





Diagram

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[illegible]

- 2 A student investigates light from different galaxies.

Fig. 2.1 shows the lines in the absorption spectrum from a distant galaxy.

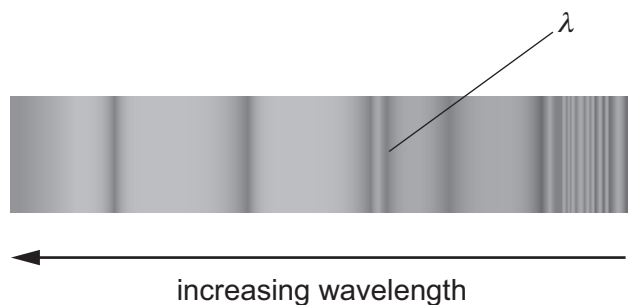


Fig. 2.1

The wavelength of one of the lines in the absorption spectrum is λ . The wavelength of this spectral line in the laboratory is λ_0 .

The observations of the same spectral line are repeated for different galaxies.

The student determines the distance d of each galaxy from the Earth.

It is suggested that λ and d are related by the equation

$$\frac{\lambda - \lambda_0}{\lambda_0} = \frac{Hd}{c}$$

where c is the speed of light in free space and H is the Hubble constant.

- (a) A graph is plotted of λ on the y -axis against $\frac{d}{c}$ on the x -axis.

Determine expressions for the gradient and y -intercept.

gradient =

y -intercept =

[1]



(b) Values of d and λ are given in Table 2.1.

Table 2.1

$d/10^{21} \text{ km}$	$\frac{d}{c}/10^{15} \text{ s}$	λ/nm
0.48 ± 0.12		658.4
1.04 ± 0.12		661.2
1.45 ± 0.12		664.2
1.80 ± 0.12		665.7
2.85 ± 0.12		672.4
3.75 ± 0.12		678.2

The value of c is $3.00 \times 10^5 \text{ km s}^{-1}$.

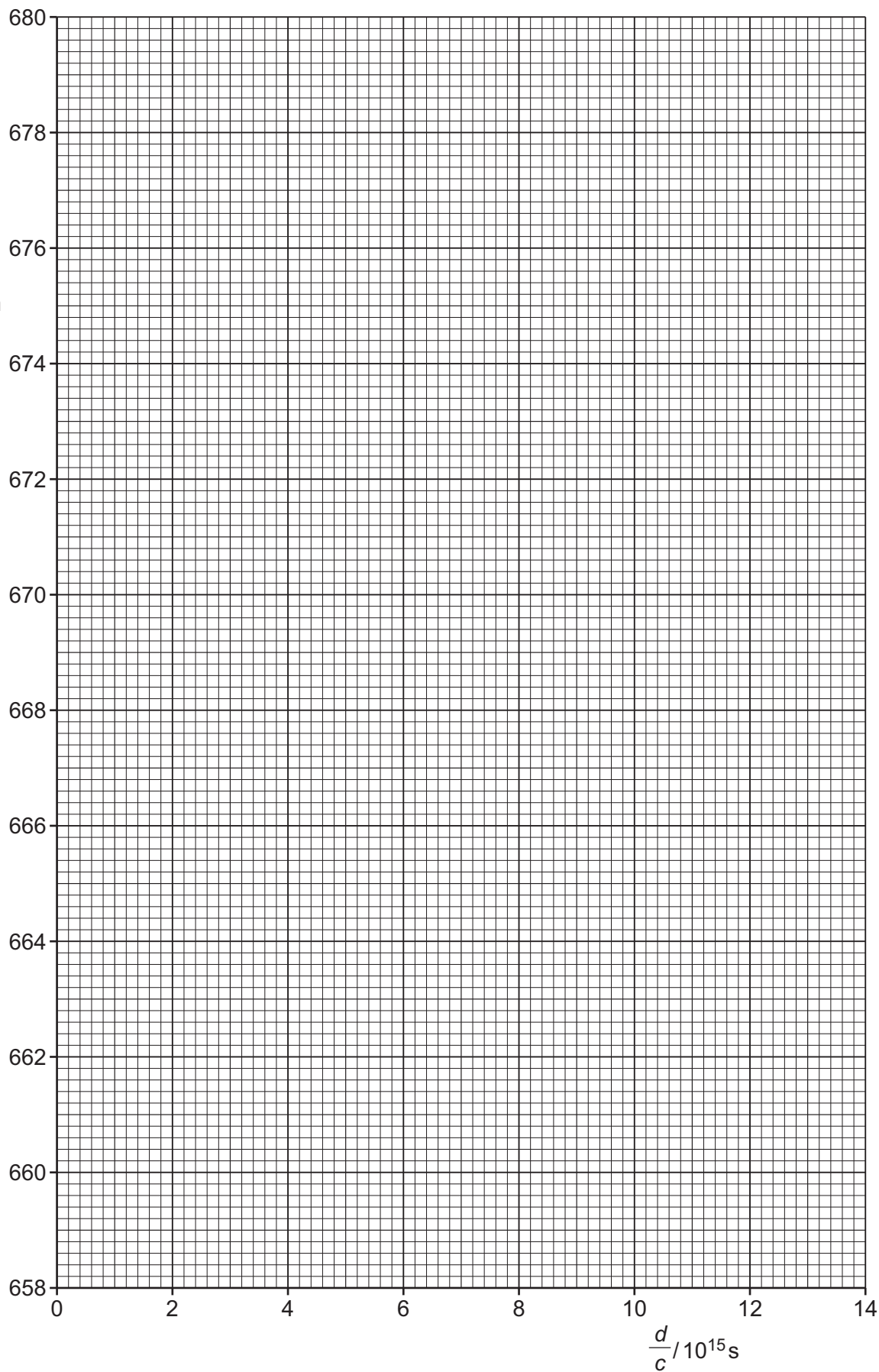
Calculate and record values of $\frac{d}{c}/10^{15} \text{ s}$ in Table 2.1. Include the absolute uncertainties in $\frac{d}{c}$. [2]

- (c) (i) Plot a graph of λ/nm against $\frac{d}{c}/10^{15} \text{ s}$. Include error bars for $\frac{d}{c}$. [2]
- (ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines. [2]
- (iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]



λ/nm





- (iv) Determine the y -intercept of the line of best fit. Include the absolute uncertainty in your answer.

y -intercept = [2]

- (d) Using your answers to (a), (c)(iii) and (c)(iv), determine the values of λ_0 and H . Include appropriate units.

λ_0 =

H = [2]

- (e) Hubble's law suggests that the age T of the universe is related to H by

$$T = \frac{1}{H}.$$

Determine a value for T . Include the absolute uncertainty in your answer.

T = s [2]

[Total: 15]





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CANDIDATE
NUMBER

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PHYSICS

9702/52

Paper 5 Planning, Analysis and Evaluation

October/November 2025

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 30.
- The number of marks for each question or part question is shown in brackets [].

This document has 8 pages.

1 Fig. 1.1 shows a model wind turbine with blades, each of length L , placed in moving air.

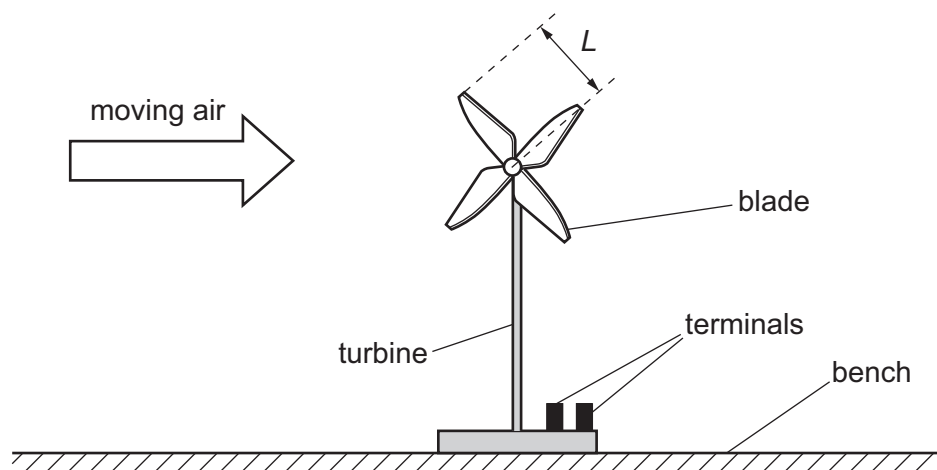


Fig. 1.1

The area of the circle swept by the blades of the turbine is A .

The output of the turbine has two terminals. The turbine is connected to a resistor of resistance R . At a speed v of the moving air, the current in the resistor is I .

The atmospheric pressure is P and the thermodynamic temperature of the air is T .

It is suggested that I is related to v by the relationship

$$\frac{I^2 R}{Q} = \frac{APv^3}{2T}$$

where Q is a constant.

Plan a laboratory experiment to test the relationship between I and v .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine a value for Q .

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.





Diagram

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.





..... [15]



- 2 A student observes the orbits of some of the moons around the planet Saturn, as shown in Fig. 2.1.

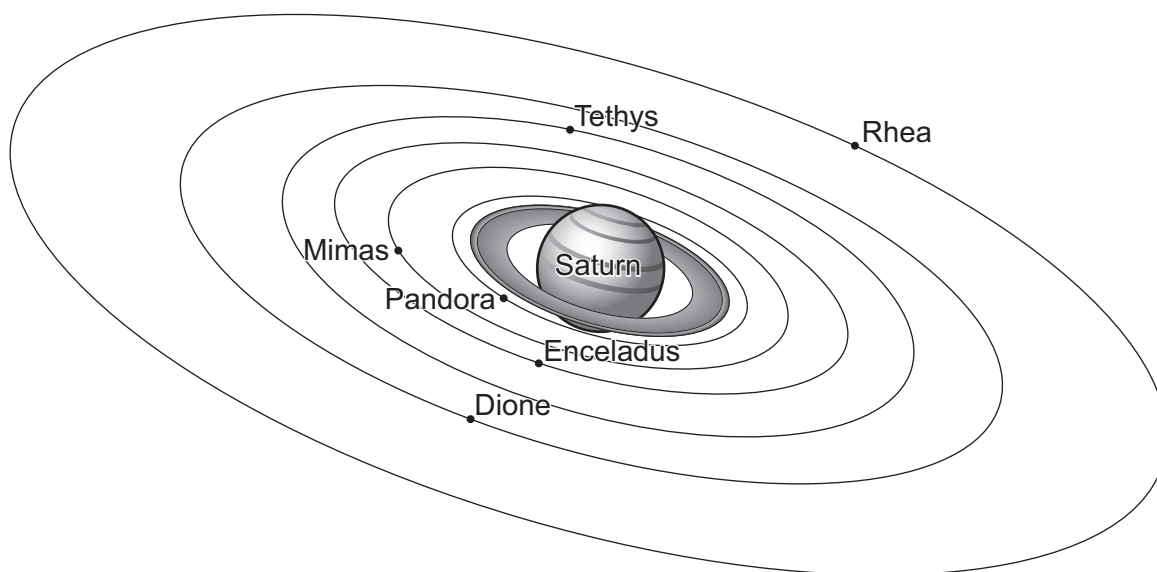


Fig. 2.1 (not to scale)

For the moon Pandora, the period of the orbit and the mean distance from the centre of Saturn are determined.

The measurements of period T and mean distance r are repeated for other moons.

It is suggested that T and r are related by the equation

$$T = \frac{2\pi r^n}{k}$$

where n and k are constants.

- (a) A graph is plotted of $\lg T$ on the y -axis against $\lg r$ on the x -axis.

Determine expressions for the gradient and y -intercept.

gradient =

y -intercept =

[1]



(b) Values of r and T are given for different moons in Table 2.1.

Table 2.1

moon	$r/10^8\text{ m}$	$T/10^3\text{ s}$	$\lg(r/10^8\text{ m})$	$\lg(T/10^3\text{ s})$
Pandora	1.42	52 ± 5		
Mimas	1.86	81 ± 5		
Enceladus	2.38	120 ± 10		
Tethys	2.95	170 ± 10		
Dione	3.77	240 ± 20		
Rhea	5.28	390 ± 30		

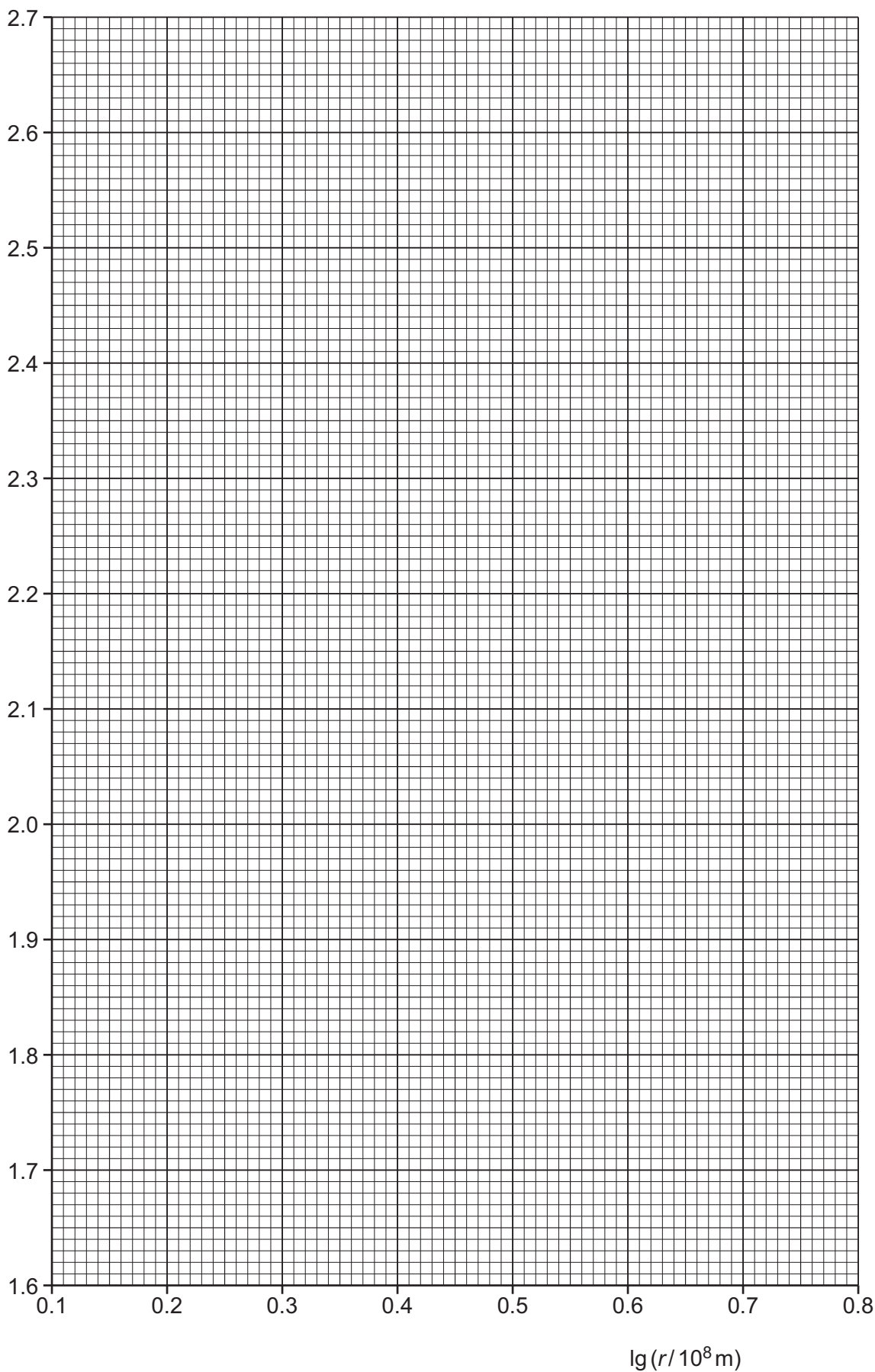
Calculate and record values of $\lg(r/10^8\text{ m})$ and $\lg(T/10^3\text{ s})$ in Table 2.1. Include the absolute uncertainties in $\lg(T/10^3\text{ s})$. [2]

- (c) (i) Plot a graph of $\lg(T/10^3\text{ s})$ against $\lg(r/10^8\text{ m})$. Include error bars for $\lg(T/10^3\text{ s})$. [2]
- (ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines. [2]
- (iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]



$\lg(T/10^3\text{s})$





- (iv) Determine the y -intercept of the line of best fit. Include the absolute uncertainty in your answer.

y -intercept = [2]

- (d) Using your answers to (a), (c)(iii) and (c)(iv), determine the values of n and k . Include the absolute uncertainties in n and k . You need not be concerned with units.

n =

k =

[3]

- (e) Titan is another moon of Saturn. The orbit of Titan has a period of 1.38×10^6 s.

Determine the value of r for Titan.

r = m [1]

[Total: 15]





Cambridge International AS & A Level

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- 1 On a bench, a steel ball of radius r is used to compress a spring by a distance x . The ball is held at rest in this position, as shown in Fig. 1.1.

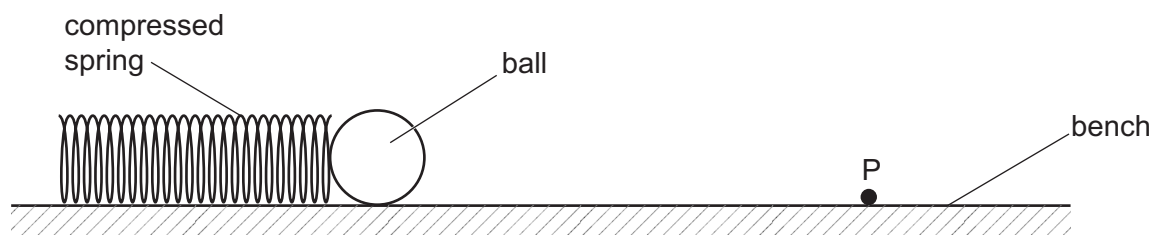


Fig. 1.1

The ball is released and rolls along the bench. At a fixed point P, the ball has speed v . The speed of the ball at P is determined using **one** light gate connected to a timer.

Several steel balls of different radii are available.

It is suggested that v is related to r by the relationship

$$v^2 = \frac{Ykx^2}{r^n \rho}$$

where k is the spring constant of the spring, ρ is the density of the steel, and Y and n are constants.

Plan a laboratory experiment to test the relationship between v and r .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for Y and n .

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.





Diagram

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[15]



- 2 A student investigates light from different galaxies.

Fig. 2.1 shows the lines in the absorption spectrum from a distant galaxy.

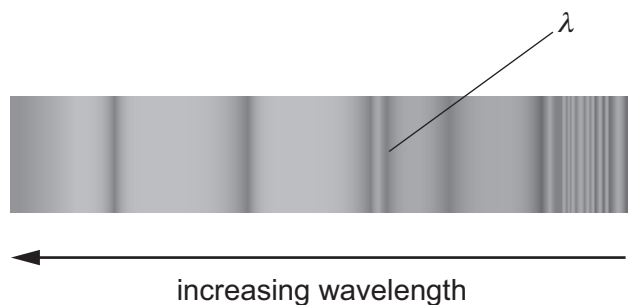


Fig. 2.1

The wavelength of one of the lines in the absorption spectrum is λ . The wavelength of this spectral line in the laboratory is λ_0 .

The observations of the same spectral line are repeated for different galaxies.

The student determines the distance d of each galaxy from the Earth.

It is suggested that λ and d are related by the equation

$$\frac{\lambda - \lambda_0}{\lambda_0} = \frac{Hd}{c}$$

where c is the speed of light in free space and H is the Hubble constant.

- (a) A graph is plotted of λ on the y -axis against $\frac{d}{c}$ on the x -axis.

Determine expressions for the gradient and y -intercept.

gradient =

y -intercept =

[1]



(b) Values of d and λ are given in Table 2.1.

Table 2.1

$d/10^{21} \text{ km}$	$\frac{d}{c}/10^{15} \text{ s}$	λ/nm
0.48 ± 0.12		658.4
1.04 ± 0.12		661.2
1.45 ± 0.12		664.2
1.80 ± 0.12		665.7
2.85 ± 0.12		672.4
3.75 ± 0.12		678.2

The value of c is $3.00 \times 10^5 \text{ km s}^{-1}$.

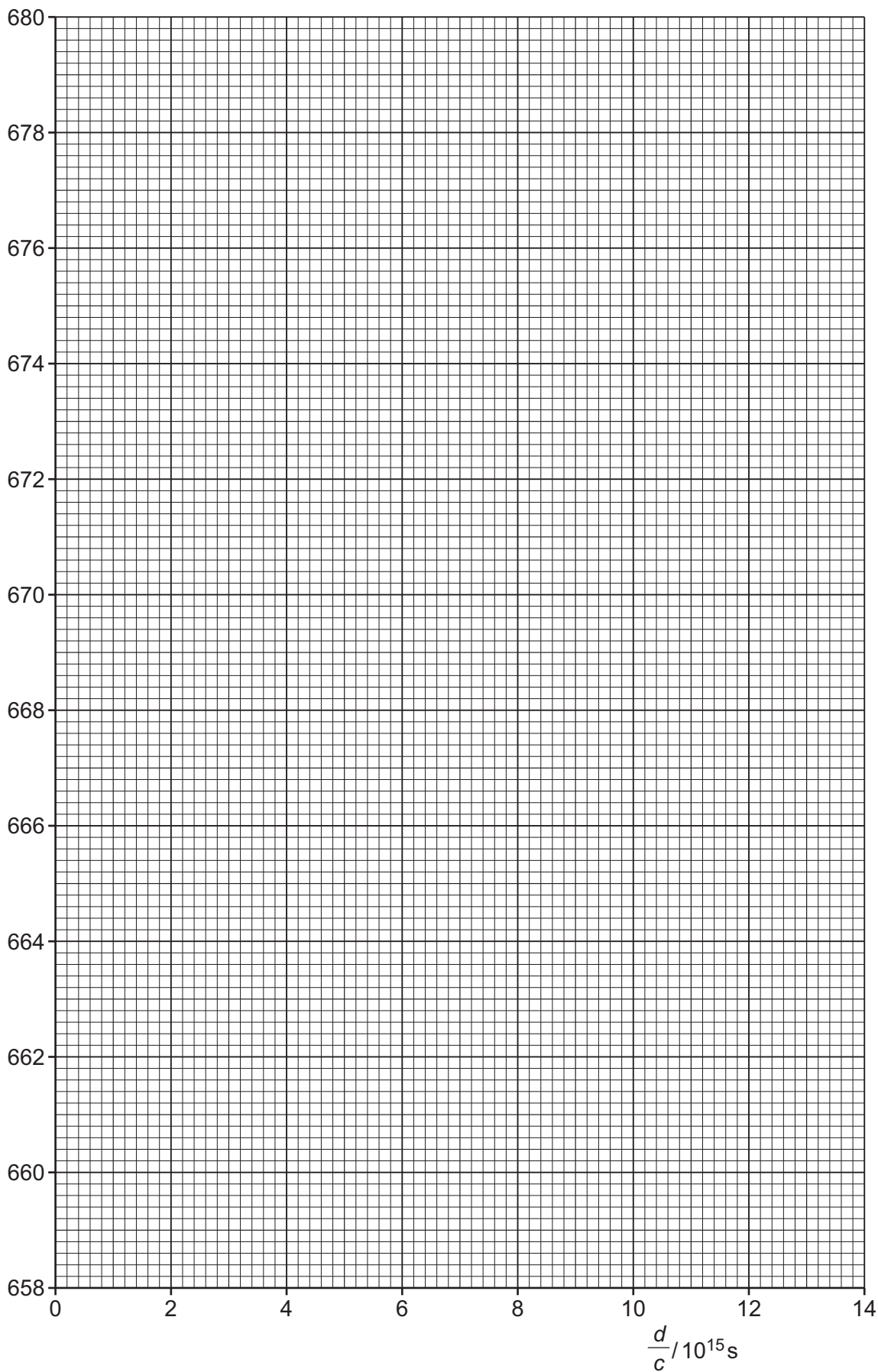
Calculate and record values of $\frac{d}{c}/10^{15} \text{ s}$ in Table 2.1. Include the absolute uncertainties in $\frac{d}{c}$. [2]

- (c) (i) Plot a graph of λ/nm against $\frac{d}{c}/10^{15} \text{ s}$. Include error bars for $\frac{d}{c}$. [2]
- (ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines. [2]
- (iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]



λ/nm





- (iv) Determine the y -intercept of the line of best fit. Include the absolute uncertainty in your answer.

y -intercept = [2]

- (d) Using your answers to (a), (c)(iii) and (c)(iv), determine the values of λ_0 and H . Include appropriate units.

λ_0 =

H = [2]

- (e) Hubble's law suggests that the age T of the universe is related to H by

$$T = \frac{1}{H}.$$

Determine a value for T . Include the absolute uncertainty in your answer.

T = s [2]

[Total: 15]

